Linear Algebra I/II

Lecture Notes & Transcript

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Contents

The Fibonacci Sequence (Chapter 1)

Introduction and Generalization

We begin our study with the classical Fibonacci sequence, a sequence of numbers that naturally emerges in various mathematical and biological contexts.

Definition 1.1 (*The Fibonacci Sequence*): The "*Fibonacci sequence*" is an infinite sequence of integers $a_0, a_1, a_2, a_3, \ldots, a_n, \ldots$ defined by the following recurrence relation:

$$\begin{cases} a_0 := 0 \\ a_1 := 1 \\ a_n := a_{n-1} + a_{n-2} \quad \text{for } n \geq 2 \end{cases} \tag{1.1}$$

Remark 1.1: This sequence was introduced to Western mathematics by Leonardo Fibonacci, a medieval mathematician from Pisa (1170–1250), to model the growth of rabbit populations. Computing the first few terms, we easily find:

$$a_0 = 0, \quad a_1 = 1, \quad a_2 = 1, \quad a_3 = 2, \quad a_4 = 3, \quad a_5 = 5, \quad a_6 = 8, \quad a_7 = 13, \quad a_8 = 21, \ldots$$

A natural, central question arises: Is there a closed, explicit formula for the general term a_n? To answer this, let us briefly look at other well-known mathematical sequences for comparison:

(a) **Arithmetic Sequence:** Defined by $a_0 := a$ and $a_n := a_{n-1} + d$ for $n \geq 1$. The explicit formula for the general term is given by $a_n = a + n \cdot d$.

(b) **Geometric Sequence:** Defined by $a_0 := a$ and $a_n := a_{n-1} \cdot q$ for $n \geq 1$. The explicit formula is given by $a_n = a \cdot q^n$.

Clearly, the recurrence relation for the Fibonacci sequence is fundamentally different. To find a formula for a_n, it is highly beneficial to step back and generalize the problem.

Definition 1.2 (*Generalized Fibonacci Sequence*): Let $a_0, a_1 \in \mathbb{R}$ be given initial real numbers. We define a sequence $\mathcal{A}$ to be a "*Fibonacci sequence*" if it satisfies the relation:

$$\begin{cases} \alpha_0 := a_0 \\ \alpha_1 := a_1 \\ \alpha_n := \alpha_{n-1} + \alpha_{n-2} \quad \text{for } n \geq 2 \end{cases} \tag{1.2}$$

We denote such a sequence uniquely determined by its starting values as $\mathcal{F}_{a_0,a_1}$. Furthermore, we denote the collection (or space) of all such generalized Fibonacci sequences by Fib.

Notice that our original, classical Fibonacci sequence is precisely $\mathcal{F}_{0,1}$.

The Algebraic Structure of Fib

The collection Fib possesses a very rich internal algebraic structure, which we will exploit to find our explicit formula.

Proposition 1.3 (*Closure Properties of* Fib): The collection of Fibonacci sequences, Fib, is closed under term-wise addition and scalar multiplication. Specifically,

(a) If $\mathcal{A}_1, \mathcal{A}_2 \in \mathrm{Fib}$, then $\mathcal{A}_1 + \mathcal{A}_2 \in \mathrm{Fib}$.

(b) If $\mathcal{A} \in \mathrm{Fib}$ and $\alpha \in \mathbb{R}$, then $\alpha\mathcal{A} \in \mathrm{Fib}$.

Proof:

We address each closure property:

(a) Let $\mathcal{A}_1 = (a_0, a_1, a_2, \dots)$ and $\mathcal{A}_2 = (b_0, b_1, b_2, \dots)$ be two sequences in Fib. We define their sum term-wise as $\mathcal{A}_1 + \mathcal{A}_2 := (c_0, c_1, c_2, \dots)$, where $c_n := a_n + b_n$. We must verify that the sequence (c_n) obeys the Fibonacci recurrence relation, namely $c_n = c_{n-1} + c_{n-2}$ for $n \geq 2$. By algebraic manipulation, we find:

$$c_{n-1} + c_{n-2} = (a_{n-1} + b_{n-1}) + (a_{n-2} + b_{n-2}) = (a_{n-1} + a_{n-2}) + (b_{n-1} + b_{n-2})$$

Because both $\mathcal{A}_1$ and $\mathcal{A}_2$ are Fibonacci sequences, we can substitute $a_{n-1} + a_{n-2} = a_n$ and $b_{n-1} + b_{n-2} = b_n$. Therefore,

$$c_{n-1} + c_{n-2} = a_n + b_n = c_n$$

This shows that $\mathcal{F}_{a_0,a_1} + \mathcal{F}_{b_0,b_1} = \mathcal{F}_{a_0+b_0,a_1+b_1}$, and therefore, Fib is closed under addition.

(b) Let $\mathcal{A} = (a_0, a_1, a_2, \dots) \in \mathrm{Fib}$ and $\alpha \in \mathbb{R}$. We define $\alpha\mathcal{A} := (\alpha a_0, \alpha a_1, \alpha a_2, \dots)$. We need to check that $\alpha a_n = \alpha a_{n-1} + \alpha a_{n-2}$. This is true because $a_n = a_{n-1} + a_{n-2}$, and we simply multiply both sides by α. In fact, we see that $\alpha\mathcal{F}_{a_0,a_1} = \mathcal{F}_{\alpha a_0,\alpha a_1}$, so Fib is closed under scalar multiplication.

Q.E.D.

Corollary 1.4: By Proposition 1.3, linear combinations of Fibonacci sequences are also Fibonacci sequences. That is, for any scalars $\alpha, \beta \in \mathbb{R}$, we have:

$$\alpha\mathcal{F}_{a_0,a_1} + \beta\mathcal{F}_{b_0,b_1} = \mathcal{F}_{\alpha a_0+\beta b_0,\alpha a_1+\beta b_1}$$

Remark 1.2: As a direct consequence of this structure, in order to find a general formula for *any* sequence $\mathcal{F}_{a_0,a_1}$, it is enough to find explicit formulas for the two foundational sequences $\mathcal{F}_{1,0}$ and $\mathcal{F}_{0,1}$. This is because any sequence can be decomposed as:

$$\mathcal{F}_{a_0,a_1} = a_0 \cdot \mathcal{F}_{1,0} + a_1 \cdot \mathcal{F}_{0,1}$$

We will see later in the course that Fib represents a 2-dimensional vector space. Every element in this space can be expressed by a linear combination of some two base elements (but 1 element alone is not enough). What precisely constitutes a *"vector space"* and *"dimension"* will be formally defined later in the course. The corresponding picture is drawn in Figure 1.1.

Deriving the Explicit Formula

Perhaps one of the sequences in Fib matches a form we already know?

First, we can rule out arithmetic sequences. Except for the trivial zero sequence $\mathcal{F}_{0,0}$, no Fibonacci sequence can be arithmetic. The reason is straightforward: for an arithmetic sequence, the difference between consecutive terms must be constant, meaning $a_n - a_{n-1} = d$. However, for a Fibonacci sequence, $a_n - a_{n-1} = a_{n-2}$. This difference grows and cannot remain constant. Thus, arithmetic sequences are unsuitable for this purpose.

Next, we ask: perhaps for some choice of a_0, a_1, the sequence $\mathcal{F}_{a_0,a_1}$ coincides with a geometric sequence? Let us try a geometric sequence of the form $\mathcal{G} = (1, q, q^2, q^3, \dots, q^n, \dots)$.

Lemma 1.5: A geometric sequence $\mathcal{G} = (1, q, q^2, \dots)$ with $q \neq 0$ is a Fibonacci sequence if and only if $q = \frac{1\pm\sqrt{5}}{2}$.

Proof:

For $\mathcal{G}$ to be a Fibonacci sequence, it must satisfy the recurrence relation for all $n \geq 2$:

$$q^n = q^{n-1} + q^{n-2}$$

Since $q \neq 0$, we can divide the entire equation by q^{n-2}, which simplifies the condition to a quadratic equation:

$$q^2 = q + 1 \implies q^2 - q - 1 = 0$$

Applying the standard quadratic formula yields two possible solutions:

$$q = \frac{1 \pm \sqrt{5}}{2}$$

We denote these two specific roots as $\varphi := \frac{1+\sqrt{5}}{2} \approx 1.618$ (the golden ratio) and $\psi := \frac{1-\sqrt{5}}{2} \approx -0.618$. Consequently, we have discovered two purely geometric sequences that belong to Fib:

$$\mathcal{F}_{1,\varphi} = (1, \varphi, \varphi^2, \varphi^3, \dots, \varphi^n, \dots)$$
$$\mathcal{F}_{1,\psi} = (1, \psi, \psi^2, \psi^3, \dots, \psi^n, \dots)$$

Q.E.D.

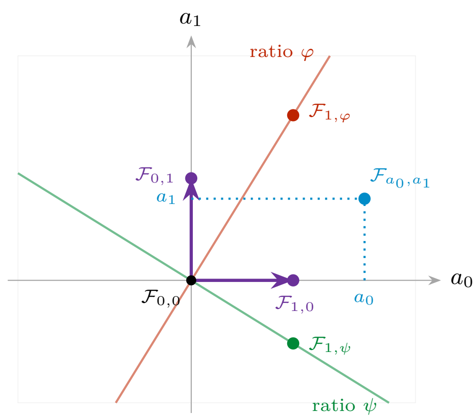


Figure 1.1: The space Fib drawn as a plane: a Fibonacci sequence is completely determined by its two initial values, so the point (a_0, a_1) *is* the sequence $\mathcal{F}_{a_0,a_1}$. Addition and rescaling of sequences become the usual addition and rescaling of points, which is the content of Proposition 1.3, and the decomposition $\mathcal{F}_{a_0,a_1} = a_0\mathcal{F}_{1,0} + a_1\mathcal{F}_{0,1}$ is just reading off coordinates. The two coloured lines carry the geometric Fibonacci sequences found in Lemma 1.5; they will turn out to be the natural axes of this plane.

Remark 1.3: The number φ is one of the most persistent constants in mathematics. The defining relation $\varphi^2 = \varphi + 1$ can be rewritten as $\varphi = 1 + \frac{1}{\varphi}$, which says that removing a unit square from a $\varphi \times 1$ rectangle leaves a rectangle of exactly the same proportions. Iterating this observation produces the classical subdivision of a rectangle into squares whose side lengths are consecutive Fibonacci numbers, drawn in Figure 1.2. The quarter circles inscribed into those squares fit together smoothly, and the resulting spiral grows by a factor of φ per quarter turn.

Theorem 1.6 (*Binet's Formula*): The explicit formula for the n-th term of the classical Fibonacci sequence $\mathcal{F}_{0,1}$ is given by "*Binet's Formula*":

$$a_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

Proof:

So far, by Lemma 1.5, we have found formulas for two specific sequences, $\mathcal{F}_{1,\varphi}$ and $\mathcal{F}_{1,\psi}$. Our original goal is to find a formula for the classical sequence $\mathcal{F}_{0,1}$.

We use the linearity principle established in Corollary 1.4. We seek scalars $\alpha, \beta \in \mathbb{R}$ such that:

$$\alpha \cdot \mathcal{F}_{1,\varphi} + \beta \cdot \mathcal{F}_{1,\psi} = \mathcal{F}_{0,1}$$

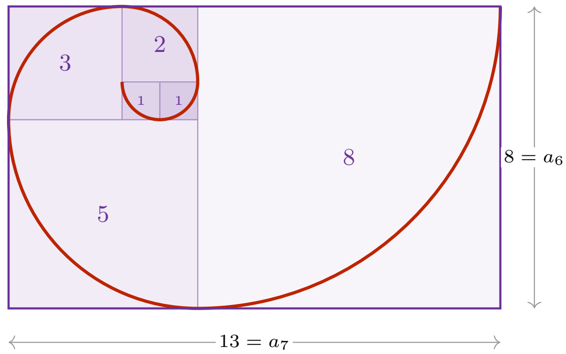


Figure 1.2: Squares with consecutive Fibonacci side lengths $1, 1, 2, 3, 5, 8$ tile a 13×8 rectangle, because $a_{n-1} + a_n = a_{n+1}$ is exactly the statement that a square of side a_n glued onto an $a_{n-1} \times a_n$ rectangle produces an $a_n \times a_{n+1}$ rectangle. Since $a_{n+1}/a_n \to \varphi$, these rectangles become ever closer to the golden rectangle.

Comparing the first two terms ($n = 0$ and $n = 1$) of these sequences yields a system of linear equations:

$$\begin{cases} \alpha \cdot 1 + \beta \cdot 1 = 0 \\ \alpha \cdot \varphi + \beta \cdot \psi = 1 \end{cases} \tag{1.3}$$

From the first equation, we immediately deduce that $\beta = -\alpha$. Substituting this into the second equation gives:

$$\alpha \cdot \varphi - \alpha \cdot \psi = 1 \implies \alpha(\varphi - \psi) = 1$$

Notice that the difference $\varphi - \psi = \sqrt{5}$. Therefore, $\alpha = 1/\sqrt{5}$ and $\beta = -1/\sqrt{5}$. This leads directly to:

$$a_n = \frac{1}{\sqrt{5}}(\varphi^n - \psi^n) = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

Q.E.D.

Where the Idea Came From: Symmetries of Fib

The derivation of Theorem 1.6 is complete, but it rests on a step that looks like a lucky guess: why should one ever think of hunting for a **geometric** sequence inside Fib? The answer is not luck, and uncovering it introduces the single most important theme of this course.

Let us look at maps

$$T\colon \mathrm{Fib} \longrightarrow \mathrm{Fib},$$

also called functions, transformations or operators. Such a T takes as input an arbitrary Fibonacci sequence $\mathcal{A}$ and returns as output another Fibonacci sequence $T(\mathcal{A})$. We have just seen in Proposition 1.3 that Fib carries an algebraic structure: it is closed under addition and under multiplication by scalars. It is therefore natural to restrict attention to those maps that **respect** this structure.

Definition 1.7 (*Linear Map on* Fib): A map $T\colon \mathrm{Fib} \to \mathrm{Fib}$ is called a "*linear map*" if

$$\begin{cases} T(\mathcal{A}+\mathcal{B}) = T(\mathcal{A}) + T(\mathcal{B}) & \text{for all } \mathcal{A}, \mathcal{B} \in \mathrm{Fib}, \\ T(\alpha\mathcal{A}) = \alpha\, T(\mathcal{A}) & \text{for all } \alpha \in \mathbb{R},\ \mathcal{A} \in \mathrm{Fib}\,. \end{cases} \tag{1.4}$$

There are many examples of such maps, though the first ones that come to mind are not very exciting:

(a) the identity $T := \mathrm{id}$, that is $T(\mathcal{A}) := \mathcal{A}$;

(b) the rescaling $T(\mathcal{A}) := \gamma \cdot \mathcal{A}$ for a fixed $\gamma \in \mathbb{R}$.

Neither of these takes any notice of the fact that the elements of Fib are *sequences*. Here is another one, which does:

Definition 1.8 (*The Shift Map*): The "*shift map*" $S\colon \mathrm{Fib} \to \mathrm{Fib}$ is defined by

$$S(a_0, a_1, a_2, \dots, a_n, \dots) := (a_1, a_2, \dots, a_n, \dots),$$

that is, it simply forgets the first term of the sequence.

Exercise 1.1: Check that S is linear.

Mathematicians like to probe a map $T\colon X \to X$ by looking for its special, or extreme, values. Two such notions will accompany us for the rest of the course.

Definition 1.9 (*Fixed Point and Eigenvector*): Let $T\colon X \to X$ be a map. An element $x_0 \in X$ is a "*fixed point*" of T if $T(x_0) = x_0$. An element $y_0 \in X$ is an "*eigenvector*" of T if $T(y_0)$ is proportional to y_0, that is, if

$$T(y_0) = \lambda \cdot y_0 \quad \text{for some } \lambda \in \mathbb{R}.$$

Let us test both notions on $X = \mathrm{Fib}$ with $T = S$ the shift map.

Proposition 1.10 (*Fixed Points of the Shift Map*): The only fixed point of the shift map S is the zero sequence $\mathcal{F}_{0,0}$.

Proof:

Suppose $\mathcal{A} = (a_0, a_1, a_2, \dots)$ satisfies $S(\mathcal{A}) = \mathcal{A}$. Comparing the two sequences term by term gives

$$a_0 = a_1, \quad a_1 = a_2, \quad a_2 = a_3, \quad \dots$$

and therefore $a_n = a_0$ for all n. Such a constant sequence is a Fibonacci sequence if and only if $a_0 = a_0 + a_0$, which forces $a_0 = 0$. Hence $\mathcal{A} = (0, 0, 0, \dots) = \mathcal{F}_{0,0}$. **Q.E.D.**

So the fixed points of S are not interesting. Let us weaken the requirement and ask for eigenvectors instead. This is exactly where the geometric sequences appear.

Proposition 1.11 (*Eigenvectors of the Shift Map*): A sequence $\mathcal{A} \neq \mathcal{F}_{0,0}$ is an eigenvector of the shift map S if and only if $\mathcal{A}$ is a geometric sequence $\mathcal{A} = (a_0, a_0\lambda, a_0\lambda^2, \dots)$ with $a_0 \neq 0$ whose ratio λ satisfies $\lambda^2 = \lambda + 1$. Consequently $\lambda \in \{\varphi, \psi\}$, while $a_0 \neq 0$ may be chosen freely.

Proof:

Assume $S(\mathcal{A}) = \lambda \cdot \mathcal{A}$ for some $\lambda \in \mathbb{R}$. Written out, the left-hand side is $(a_1, a_2, \dots, a_{n+1}, \dots)$ and the right-hand side is $(\lambda a_0, \lambda a_1, \dots, \lambda a_n, \dots)$. Comparing entries yields

$$a_n = a_{n-1} \cdot \lambda \quad \text{for all } n \geq 1,$$

and therefore

$$\mathcal{A} = (a_0,\, a_0\lambda,\, a_0\lambda^2,\, a_0\lambda^3,\, \dots,\, a_0\lambda^n,\, \dots),$$

which is precisely a geometric sequence. Note that $a_0 \neq 0$, since otherwise every term would vanish and $\mathcal{A}$ would be $\mathcal{F}_{0,0}$, which we excluded.

It remains to determine which ratios λ are admissible. Since $\mathcal{A}$ must in addition be a Fibonacci sequence, the relation $a_2 = a_1 + a_0$ has to hold, that is

$$\underbrace{a_0 \cdot \lambda^2}_{= a_2} = \underbrace{a_0 \cdot \lambda}_{= a_1} + a_0.$$

Dividing by $a_0 \neq 0$ gives $\lambda^2 = \lambda + 1$, the very same quadratic equation we met in Lemma 1.5. Its roots are φ and ψ, and a_0 is not constrained at all. Conversely, every geometric sequence with ratio φ or ψ satisfies $S(\mathcal{A}) = \lambda \cdot \mathcal{A}$ by the same computation read backwards. **Q.E.D.**

Conclusion 1.1: Taking $a_0 = 1$, the two geometric sequences

$$\mathcal{F}_{1,\varphi} = (1, \varphi, \varphi^2, \dots, \varphi^n, \dots) \qquad \text{and} \qquad \mathcal{F}_{1,\psi} = (1, \psi, \psi^2, \dots, \psi^n, \dots)$$

are Fibonacci sequences *and* they are eigenvectors of the shift operator S. The geometric sequences were therefore never a guess: they are forced upon us the moment we ask which sequences the shift map merely rescales.

It is worth restating this in the picture of Figure 1.1. Under the identification $\mathcal{F}_{a_0,a_1} \leftrightarrow (a_0, a_1)$, shifting a sequence discards a_0 and promotes a_1 and $a_2 = a_0 + a_1$ to the new initial pair, so

$$S(\mathcal{F}_{a_0,a_1}) = \mathcal{F}_{a_1,\, a_0+a_1}, \qquad \text{that is} \qquad \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} \longmapsto \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}. \tag{1.5}$$

The shift map thus acts on the plane Fib through a single fixed matrix, and Proposition 1.11 identifies the two lines that this matrix maps onto themselves, as drawn in Figure 1.3.

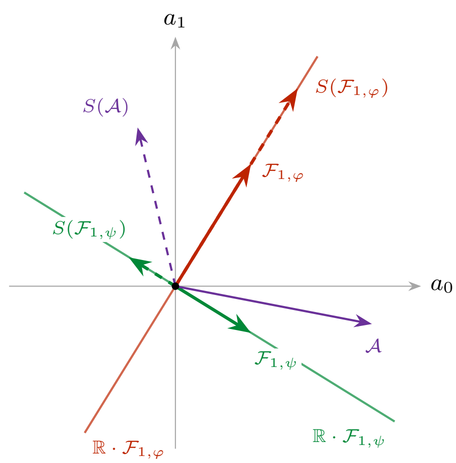


Figure 1.3: The shift map S acting on the plane Fib via (1.5); images are drawn dashed. A generic sequence $\mathcal{A}$ (purple) is thrown off its own direction. Only the two coloured lines are preserved: on $\mathbb{R} \cdot \mathcal{F}_{1,\varphi}$ the map stretches by $\varphi > 1$, while on $\mathbb{R} \cdot \mathcal{F}_{1,\psi}$ it reverses the direction and contracts by $|\psi| < 1$. These two lines are exactly the geometric Fibonacci sequences of Proposition 1.11.

Remark 1.4: The contraction along the second line is the geometric reason behind the asymptotics established below. Writing an arbitrary Fibonacci sequence in terms of the two eigen-directions, the ψ-component is crushed towards 0 under repeated shifting while the φ-component grows. Whatever sequence one starts from, its direction in the plane Fib converges to the line $\mathbb{R} \cdot \mathcal{F}_{1,\varphi}$.

The Upshot: Asymptotics and Matrix Form

AI-Note: This subsection has no counterpart in Prof. Biran's notes for Lecture 1A, which end with the exercises reproduced below. The closest-integer estimate, the matrix form and Cassini's identity are supplementary material added in this transcript. They are included because the matrix M reappears in the chapters on eigenvalues, but they should not be mistaken for lecture content.

The derivation above yields two profound observations regarding the growth and representation of the Fibonacci sequence.

Important remark 1.1: Clearly, as n grows, the term involving ψ becomes negligible. Since $|\psi| = |\frac{1-\sqrt{5}}{2}| \approx 0.618 < 1$, it follows that $\psi^n \to 0$ as $n \to \infty$. Therefore, for large n, we have the approximation $a_n \approx \frac{1}{\sqrt{5}}\varphi^n$. In fact, because $\left|\frac{\psi^n}{\sqrt{5}}\right| < \frac{1}{2}$ for all $n \geq 0$, we arrive at the following precise conclusion: the number a_n is the *closest integer* to

$$\frac{1}{\sqrt{5}} \cdot \left(\frac{1+\sqrt{5}}{2}\right)^n.$$

Furthermore, this implies that the ratio of consecutive terms satisfies:

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \varphi \tag{1.6}$$

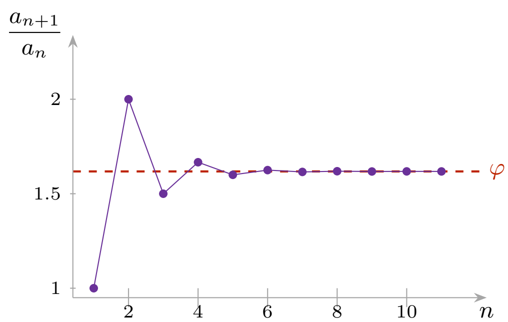


Figure 1.4: The quotients a_{n+1}/a_n of the classical Fibonacci sequence $\mathcal{F}_{0,1}$. They oscillate around the golden ratio φ, alternately overshooting and undershooting it, because the error term ψ^n changes sign at every step while its magnitude decays geometrically.

Finally, we can represent the recurrence relation using the language of matrices. Defining a vector $v_n := \binom{a_{n+1}}{a_n}$, the relation $a_{n+1} = a_n + a_{n-1}$ can be written as:

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_n \\ a_{n-1} \end{pmatrix} \implies v_n = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a_1 \\ a_0 \end{pmatrix} \tag{1.7}$$

This formulation shows that the behavior of the sequence is entirely governed by the powers of the matrix $M := \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. It is the shift map of (1.5) written in the reversed coordinates $\binom{a_{n+1}}{a_n}$, so by Proposition 1.11 its eigenvalues are again φ and ψ.

Exercise 1.2 (*Interesting Exercises*):

(a) General Formula: Find an explicit formula for the general Fibonacci sequence $\mathcal{F}_{a_0,a_1}$ that depends only on the initial values a_0, a_1 and the index n.

(b) The Shift Operator: Let $S : \text{Fib} \to \text{Fib}$ be the shift map defined by $S(a_0, a_1, \ldots) = (a_1, a_2, \ldots)$. Formally verify that S is a *linear map*.

(c) Golden Ratio Continued Fraction: Consider $\varphi = \frac{1+\sqrt{5}}{2}$.

$$\left.\varphi = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}}}\right\} \quad \text{what is actually the meaning of this?}$$

(d) Non-existence of Arithmetic Fibonacci Sequences: Complete the details to show that no Fibonacci sequence $\mathcal{F}_{a_0,a_1}$ (except for the trivial zero sequence) can be an arithmetic sequence.

(e) **Convergence:** For the classical Fibonacci sequence $\mathcal{F}_{0,1} = (a_0, a_1, a_2, \dots)$, calculate the limit

$$\lim_{n\to\infty} \frac{a_n}{a_{n-1}},$$

whose numerical behaviour is plotted in Figure 1.4.

(f) **Cassini's Identity:** Show that $a_{n+1}a_{n-1} - a_n^2 = (-1)^n$ for all $n \geq 1$. Hint: Use the determinant of the matrix M^n.

Logic and Mathematics (Chapter 2)

Before advancing into the formal theory of vector spaces, we must establish the rigorous grammar of mathematical reasoning. This chapter provides the logical foundation necessary for constructing and understanding proofs.

Mathematical Statements and Basic Operators

Definition 2.1 (*Mathematical Statement*): A "*mathematical statement*" (or logical statement) is a declarative sentence that is definitively either **True (T)** or **False (F)**. It is a fundamental axiom of logic that a statement cannot be both true and false simultaneously.

Example 2.1 (*Examples of Logical Statements*): Consider the following declarations:

- **A:** *"For every real number x, $x^2 \geq 0$."*
- **B:** *"Every prime number $p \geq 3$ is odd."*
- **C:** *"Every odd number $p \geq 3$ is prime."*

Statements A and B are True. Statement C is False (e.g., 9 is odd but not prime).

Negation, Conjunction, and Disjunction

Definition 2.2 (*Negation*): The "*negation*" of a statement A, denoted $\neg A$ (and sometimes $\overline{A}$), asserts that A does not hold.

A	$\neg A$
T	F
F	T

Definition 2.3 (*Conjunction / AND*): The statement $A \wedge B$ (read *"A and B"*, and sometimes written $A \& B$) is True if and only if both A and B are True.

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

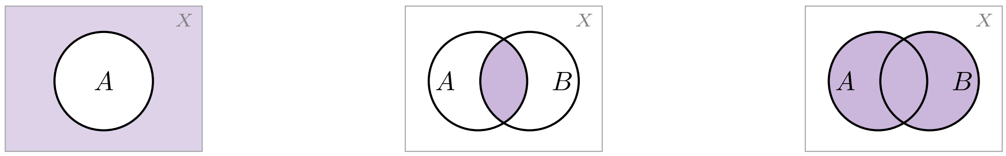
Example 2.2 (*Conjunction Example*): For $x \in \mathbb{R}$, the statement $(x^2 = 1) \wedge (x \geq 0)$ is logically equivalent to $x = 1$.

Definition 2.4 (*Disjunction / OR*): The statement $A \vee B$ (read *"A or B"*) is True if *at least one* of the statements A or B is True. This is an *inclusive* OR.

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Important remark 2.1: In everyday life, *"either A or B"* usually means one of the two but not both. In mathematics it means *"either A or B, or both"*: it could be only A, it could be only B, and it could be both A and B.

The three basic operators are conveniently visualised by shading the region of a Venn diagram on which the compound statement comes out True; see Figure 2.1.



(a) $\neg A$ (b) $A \wedge B$ (c) $A \vee B$

Figure 2.1: The three basic operators of Definitions 2.2 to 2.4, drawn on a universe X. The shaded region is where the compound statement is True. Note that in Figure 2.1c the overlap is shaded as well: this is the inclusive OR.

Example 2.3 (*Disjunction Example*): The statement $(x > 1) \vee (x < 2)$ is True for all $x \in \mathbb{R}$, since every real number satisfies at least one of the two conditions. Conversely, $(x < 1) \vee (x > 2)$ is precisely the negation of the statement $1 \leq x \leq 2$, as Figure 2.2 shows.

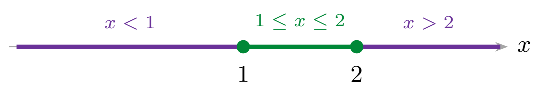


Figure 2.2: The statement $(x < 1) \vee (x > 2)$ holds exactly on the two purple rays, which together form the complement of the green interval. Hence it is the negation of $1 \leq x \leq 2$.

Implications and Equivalences

Definition 2.5 (*Implication*): The statement $A \implies B$ (*"A implies B"*) is defined to be False only when A is True and B is False. In all other cases, it is True.

A	B	$A \implies B$
T	T	T
T	F	F
F	T	T
F	F	T

Remark 2.1: Note that $A \implies B$ is logically equivalent to $\neg A \vee B$. Consequently, if the premise A is False, the implication is *vacuously true* regardless of B. As Prof. Biran noted:

$$(0 = 1) \implies \text{“The Earth is flat”}$$

is a mathematically **True** statement. Causality is irrelevant in formal logic.

Definition 2.6 (*Equivalence*): The statement $A \iff B$ (*"A if and only if B"*) holds when $A \implies B$ and $B \implies A$ are both True. This means A and B share the same truth value.

A	B	$A \iff B$
T	T	T
T	F	F
F	T	F
F	F	T

Example 2.4 (*Equivalences*): Both of the following statements are True:

$$(x^2 > 0) \iff (x \neq 0), \qquad (0 = 1) \iff \textit{“The Earth is flat”}.$$

The second one is True because both sides are False. Again, causality is irrelevant.

Proposition 2.7 (*Contrapositive*): Let A, B be statements. Then

$$(A \implies B) \iff (\neg B \implies \neg A).$$

Exercise 2.1 (*Double Negation and De Morgan's Laws*): Prove Proposition 2.7 via a truth table, and show in the same way that

(a) **Double Negation:** $\neg(\neg A) = A$;

(b) **De Morgan's Laws:** $\neg(A \wedge B) = \neg A \vee \neg B$ and $\neg(A \vee B) = \neg A \wedge \neg B$.

Spell the two De Morgan laws out in words.

Predicates and Quantifiers

A “*predicate*” $P(x)$ is a statement whose truth depends on a variable x. We use quantifiers to convert predicates into absolute statements.

Definition 2.8 (*Quantifiers*):

- **Universal ($\forall$):** *“For every x in X, $P(x)$ holds.”*
- **Existential ($\exists$):** *“There exists at least one x in X such that $P(x)$ holds.”*

Example 2.5 (*Quantifiers over Integers*): Let $P(n)$ be the predicate $n = n^2$ over $\mathbb{Z}$.

- $\forall n \in \mathbb{Z} : P(n)$ is False.
- $\exists n \in \mathbb{Z} : P(n)$ is True (take $n = 1$).
- $\forall n \in \mathbb{Z} : \big(n = n^2 \implies (n = 0) \vee (n = 1)\big)$ is True.

Quantifiers may of course be combined. For instance,

$$\forall y \in \mathbb{R} : \big(y \geq 0 \implies (\exists x \in \mathbb{R} : x^2 = y)\big)$$

is a True statement. In practice one writes this more compactly as

$$\forall\, 0 \leq y \in \mathbb{R},\ \exists x \in \mathbb{R} \text{ such that } x^2 = y. \tag{2.1}$$

Important remark 2.2 (*The Order of Quantifiers*): The order of quantifiers is non-commutative. Consider X (ETH students), Y (courses), and $A(x, y) :=$ *“student x takes course y”*.

- $\forall x \in X, \exists y \in Y : A(x, y)$ means every student takes at least one course (True).
- $\exists y \in Y, \forall x \in X : A(x, y)$ means there is one specific course taken by *every* student (False).

Figure 2.3 makes the asymmetry visible: reading the table row by row gives $\forall x\, \exists y$, whereas reading it column by column gives $\exists y\, \forall x$. Every row carries at least one mark, but no column is completely filled. The reader is encouraged to try the same interchange on (2.1) and to decide what the swapped statement asserts there.

Negation and Unique Existence

Proposition 2.9 (*Negating Quantifiers*):

$$\neg(\forall x \in X : A(x)) \iff \exists x \in X : \neg A(x)$$
$$\neg(\exists x \in X : A(x)) \iff \forall x \in X : \neg A(x)$$

Remark 2.2 (*Shorthand Notation*): In practice, we often condense multiple quantifiers of the same type:

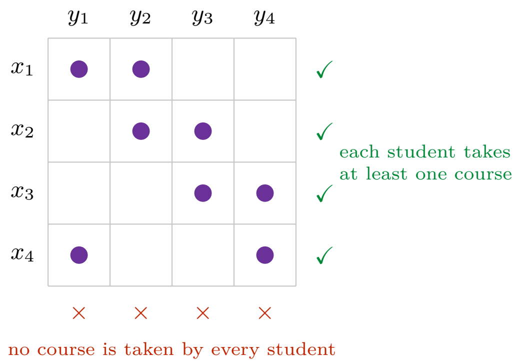


Figure 2.3: A mark in row x_i, column y_j records that student x_i takes course y_j. Every row contains a mark, so $\forall x \in X, \exists y \in Y : A(x, y)$ is True. No column is entirely filled, so $\exists y \in Y, \forall x \in X : A(x, y)$ is False. Swapping two quantifiers of different type therefore changes the meaning of a statement.

- $\exists x \in X, \exists y \in X \implies \exists x, y \in X$
- $\forall x \in X, \forall y \in X \implies \forall x, y \in X$

Definition 2.10 (*Unique Existence*): The notation $\exists! x \in X : P(x)$ means there exists exactly one x satisfying the predicate. Formally:

$$(\exists x \in X : P(x)) \wedge (\forall x, y \in X : (P(x) \wedge P(y) \implies x = y))$$

Example 2.6 (*Unique Existence of Roots*): $\exists! x \in \mathbb{R} : x^2 = 4$ is False (since $x = \pm 2$). However, $\exists! x \in \mathbb{R} : (x \geq 0) \wedge (x^2 = 4)$ is True.

Set Theory (Chapter 3)

In this chapter, we formalize the language of sets, which provides the universal framework for all modern mathematical structures, including the vector spaces we will investigate later.

Basic Definitions and Notation

Definition 3.1 (*Set*): A "*set*" is a collection of distinct objects, which are referred to as the *elements* of the set.

In our notation, we denote sets typically by capital letters. If an object x is an element of a set M, we write $x \in M$. If it is not, we write $x \notin M$.

Remark 3.1: Crucially, within a set, the order of elements is irrelevant, and repetitions do not contribute to the set's identity. For example:

$$M = \{1, 3, \text{moon}\} = \{3, \text{moon}, 1\}$$
$$M = \{1, 1, 1\} = \{1\}$$

We frequently define sets using a property or condition $A(x)$ that the elements must satisfy. The standard notation for this *"set-builder"* form is:

$$M = \{x \mid A(x)\} \quad \text{or} \quad M = \{x : A(x)\}$$

Example 3.1 (*Set-Builder Notation*): The set of integers whose square is less than or equal to 9 is written as:

$$\{x \in \mathbb{Z} \mid x^2 \leq 9\} = \{-3, -2, -1, 0, 1, 2, 3\}$$

Definition 3.2 (*The Empty Set*): The "*empty set*", denoted by $\emptyset$ or $\{\}$, is the unique set containing no elements.

It is important to note that a set is allowed to contain another set as one of its elements. For instance, consider the set $M = \{1, \text{elephant}, \{a, 5\}\}$. Here, the set $\{a, 5\}$ is a single element of M. Similarly, we can define a set containing only the empty set: $M = \{\emptyset\}$, which is a set containing exactly one element.

Subsets and Inclusion

We now define how sets relate to one another in terms of containment.

Definition 3.3 (*Subset and Proper Subset*): Let P and Q be sets.

(a) We say P is a "*subset*" of Q, denoted $P \subseteq Q$, if every element of P is also an element of Q. Formally: $\forall x \in P : x \in Q$.

(b) We say P is a "*proper subset*" of Q, denoted $P \subsetneq Q$, if $P \subseteq Q$ and $P \neq Q$.

(c) We write $P \nsubseteq Q$ to denote that P is not a subset of Q, meaning $\neg(P \subseteq Q)$.

Remark 3.2: Note that some authors use $P \subset Q$ to denote proper subsets, while others use it for generic subsets.

Example 3.2 (*Subsets and Non-Subsets*): We have

$$\{-1, 2, 5\} \subsetneq \{x \mid x \text{ is an integer and } x^2 \leq 30\},$$

the inclusion being proper because, for instance, 0 lies in the right-hand set but not in the left. On the other hand,

$$\{-1,2,5\} \not\subseteq \{x \mid x \text{ is an integer and } x^2 \leq 20\},$$

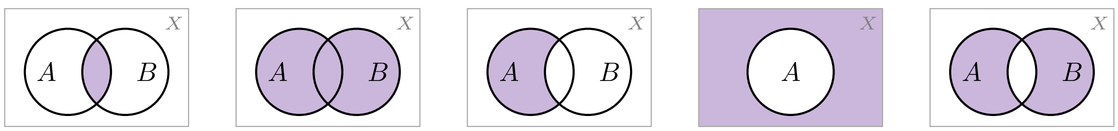
since $5^2 = 25 > 20$.

Claim: For any set Q, the empty set is a subset of Q (i.e., $\emptyset \subseteq Q$).

Set Operations

Just as we have logical operators for statements, we have operations to combine or modify sets. Let A and B be subsets of a universal set X.

(a) **Intersection:** $A \cap B = \{x \in X \mid x \in A \wedge x \in B\}$.
(b) **Union:** $A \cup B = \{x \in X \mid x \in A \vee x \in B\}$.
(c) **Difference:** $A \setminus B = \{x \in X \mid x \in A \wedge x \notin B\}$.
(d) **Complement:** The complement of A in X is $A^c = X \setminus A = \{x \in X \mid x \notin A\}$.



(a) $A \cap B$ (b) $A \cup B$ (c) $A \setminus B$ (d) A^c (e) $A \triangle B$

Figure 3.1: The set operations, drawn inside an ambient set X. In each panel the shaded region is the result of the operation. The last panel is the symmetric difference of Definition 3.4, which is exactly the union with the intersection removed.

Definition 3.4 (*Symmetric Difference*): The "*symmetric difference*" of A and B is defined by

$$A \triangle B := (A \cup B) \setminus (A \cap B).$$

Notation 3.1: The set X above is called the "*ambient set*", or the underlying set (in German, „*Grundmenge*"), of the discussion. The complement of A in X is sometimes also written $\overline{A}$ rather than A^c.

Unions and Intersections of Families of Sets

The operations $\cap$ and $\cup$ are not restricted to two sets; they extend to arbitrary collections.

Definition 3.5 (*Union and Intersection of a Family*): Let $\mathcal{A}$ be a family (that is, a set) of sets with $\mathcal{A} \neq \emptyset$. We define

$$\bigcup_{A \in \mathcal{A}} A := \{x \mid \exists A \in \mathcal{A} \text{ such that } x \in A\},$$

$$\bigcap_{A \in \mathcal{A}} A := \{x \mid \forall A \in \mathcal{A} : x \in A\}.$$

In practice, $\mathcal{A}$ sometimes merely parametrises, or indexes, a family of sets.

Example 3.3 (*An Indexed Family*): Let $\mathcal{A} = \{2,3,4,\dots\}$ and, for every $n \in \mathcal{A}$, set

$$A_n := \{x \mid x \in \mathbb{Z},\ x \geq 1, \text{ and } n^2 \text{ divides } x\}.$$

Thus $A_2 = \{4,8,12,\dots\}$, $A_3 = \{9,18,27,\dots\}$, and so on. Then

$$\bigcup_{n \in \mathcal{A}} A_n = \{a \in \mathbb{Z} \mid \exists k,r \in \mathbb{Z},\ k,r \geq 1, \text{ such that } a = k^2 \cdot r\}, \qquad \bigcap_{n \in \mathcal{A}} A_n = \emptyset.$$

The union is drawn in Figure 3.2. The intersection is empty because no integer $x \geq 1$ is divisible by n^2 for *every* n: any such x would exceed every bound.

AI-Note: As written in the notes, the condition describing the union allows $k = 1$, and with $k = 1$ the equation $a = k^2 \cdot r$ is satisfied by *every* $a \geq 1$. Taken literally the right-hand side would therefore be all of $\{a \in \mathbb{Z} \mid a \geq 1\}$, which is not the union of the A_n. Since the index set is $\mathcal{A} = \{2, 3, \dots\}$, the intended condition is $k \geq 2$, so that

$$\bigcup_{n \in \mathcal{A}} A_n = \{a \in \mathbb{Z} \mid a \geq 1 \text{ and } a \text{ is divisible by the square of some } k \geq 2\},$$

that is, the positive integers that are *not* squarefree. This is what Figure 3.2 shows.

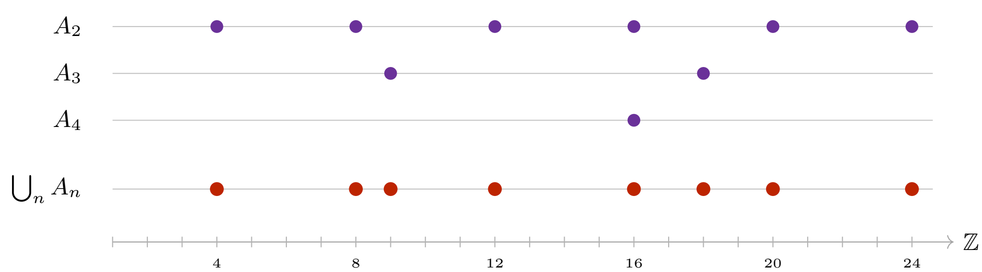


Figure 3.2: The family A_n of the preceding example, restricted to $1 \leq x \leq 24$. The set A_n collects the positive multiples of n^2, so $A_5, A_6, \dots$ contribute nothing in this range. The union (in red) consists of $4, 8, 9, 12, 16, 18, 20, 24$: precisely the integers divisible by a square greater than 1. Every row is eventually empty in any fixed range, which is why the intersection is empty.

Cartesian Products

To handle ordered pairs and higher-dimensional structures, we introduce the Cartesian product.

Definition 3.6 (*Cartesian Product*): The "*Cartesian product*" of two sets X and Y is the set of all ordered pairs (x, y) where $x \in X$ and $y \in Y$:

$$X \times Y := \{(x, y) \mid x \in X, y \in Y\}$$

Example 3.4 (*A Finite Cartesian Product*): Let $X = \{1, 2, \text{elephant}\}$ and $Y = \{4, \text{dog}\}$. Then

$$X \times Y = \{(1, 4), (1, \text{dog}), (2, 4), (2, \text{dog}), (\text{elephant}, 4), (\text{elephant}, \text{dog})\}.$$

The pairs are *ordered*, and this matters: $(1, 4) \in X \times Y$, whereas $(4, 1) \notin X \times Y$.

This construction generalizes to n sets. We denote the n-fold product of a set X with itself as X^n:

$$X^n := \underbrace{X \times \cdots \times X}_{n \text{ times}} = \{(x_1, \dots, x_n) \mid x_i \in X \text{ for } 1 \leq i \leq n\}$$

The elements of X^n are called n-tuples. For example, $X^2 := X \times X$, and $X^3 := X \times X \times X = \{(x_1, x_2, x_3) \mid x_1, x_2, x_3 \in X\}$, and so forth.

Example 3.5 (*Cartesian Products in the Plane*): If $X = \mathbb{R}$, then $X^2 = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ represents the standard Euclidean plane. If we take $X = [0, 2]$ and $Y = [1, 2]$, then $X \times Y$ represents a rectangle in the plane with corners at $(0, 1), (2, 1), (0, 2)$, and $(2, 2)$, as drawn in Figure 3.3.

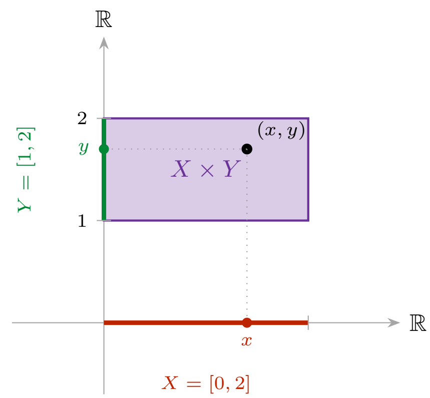


Figure 3.3: The Cartesian product $X \times Y$ for $X = [0, 2]$ and $Y = [1, 2]$. A point of the product is precisely a choice of one coordinate from each factor: dropping the point (x, y) onto the horizontal axis returns $x \in X$, and onto the vertical axis returns $y \in Y$.

Power Sets and Cardinality

Definition 3.7 (*Power Set*): Let X be a set. The "*power set*" of X, denoted by $\mathcal{P}(X)$ or 2^X, is the set of all subsets of X:

$$\mathcal{P}(X) := \{Q \mid Q \subseteq X\}$$

Example 3.6 (*Power Set of a Finite Set*): For $X = \{1, 2, 3\}$, the power set is:

$$\mathcal{P}(X) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Note that $\mathcal{P}(X)$ contains 8 elements.

Definition 3.8 (*Cardinality*): If X is a finite set, meaning a set with finitely many elements $X = \{x_1, \dots, x_n\}$, we say that the "*cardinality*" of X is the number of elements in X, denoted by $|X| := n$ or $\operatorname{card}(X) := n$.

Exercise 3.1 (*Cardinality of the Power Set*): Let A be a finite set. Prove by induction that the cardinality of its power set is given by:

$$|\mathcal{P}(A)| = 2^{|A|}$$

A Note on the Foundations: Russell's Paradox

In the early development of set theory, it was assumed that any property $A(x)$ could define a set (Naive Set Theory). However, Bertrand Russell showed that this leads to a contradiction.

Suppose there exists a *"set of all sets"*, denoted by S. We then define a subset $R \subseteq S$ as follows:

$$R := \{P \in S \mid P \notin P\},$$

that is, R is the set of all sets that do not contain themselves. For example, S itself is a set, so $S \in S$, and therefore $S \notin R$.

Now, we ask: is R an element of itself?

- If $R \in R$, then by the definition of R, it must be that $R \notin R$. This is a contradiction.
- If $R \notin R$, then by the definition of R, it must be that $R \in R$. This is also a contradiction.

The two implications close into the cycle drawn in Figure 3.4: each of the only two possible answers forces the other one.

Claim: Neither S nor R can be regarded as a set.

$$\begin{tikzcd} \boxed{R \in R} \arrow[rr, bend left, "\text{by definition of } R"] & & \boxed{R \notin R} \arrow[ll, bend left, "\text{by definition of } R"] \end{tikzcd}$$

Figure 3.4: Russell's paradox as a cycle. A statement and its negation imply one another, so neither can be assigned a truth value. The fault lies not in the reasoning but in the assumption that S is a set.

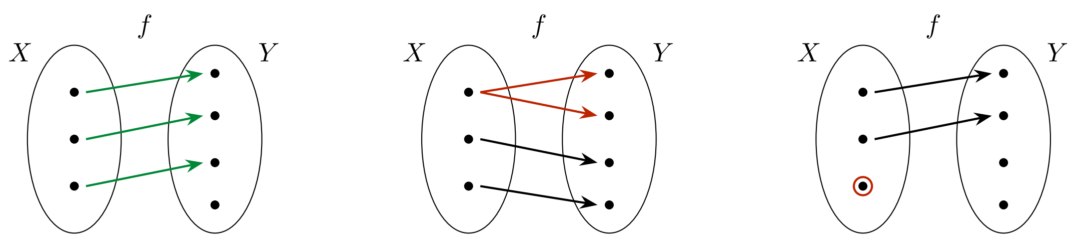
This paradox led to the development of Axiomatic Set Theory (e.g., Zermelo-Fraenkel), which places specific restrictions on how sets can be constructed to ensure mathematical consistency.

Maps (Chapter 4)

Functions and their Domains

Definition 4.1 (*Map / Function*): Let X and Y be sets. A map $f : X \to Y$ (also called a function or transformation) is an assignment that assigns to every element $x \in X$ a uniquely defined element $f(x) \in Y$.

In accordance with this definition, we can visually distinguish between valid and invalid assignments. A function must assign exactly one output to each input in its domain.



(a) O.K. (b) NOT O.K. (c) NOT O.K.

Figure 4.1: Which assignments are maps. In Figure 4.1a every element of X has exactly one arrow leaving it. In Figure 4.1b the top element of X is assigned two different values, so the value is not *uniquely defined*. In Figure 4.1c the circled element of X is assigned no value at all, so f is not defined on *every* element of X.

Informally, a map turns inputs into outputs:

$$\text{Input} \in X \xrightarrow{f} \text{output} \in Y.$$

X is called the *domain of definition* of f. Y is called the *target* (or target domain) of f.

Definition 4.2 (*Equality of Functions*): Two functions $f : X \to Y$ and $g : X' \to Y'$ are equal if they have the same domain of definition ($X' = X$), the same target ($Y' = Y$), and for all $x \in X$, we have $f(x) = g(x)$.

Sometimes a function is given by a formula, and sometimes it is not. Be careful: providing just a formula is often mathematically insufficient.

Example 4.1 (*Specifying Domain and Target*): **Bad notation:** $f(x) = x^2$. (What are the domain and target?)

O.K. notation: Let $X := \{x \in \mathbb{R} \mid 0 \leq x\}$. Define $f : X \to \mathbb{R}$ by $f(x) = x^2$.

Example 4.2 (*Student Database Function*): Let $X = \{x \in \mathbb{Z} \mid x \geq 0\}$ and let Y be the set of family names. We might try to define a map $g : X \to Y$ by setting $g(x)$ to be the family name of the student at ETH whose student number is x.

This g is **not** defined! What is $g(0)$? What is $g(10^{10})$? There are no students with these IDs.

To fix this, we could define $Y' = Y \cup \{\text{ERROR!}\}$ and introduce a rigorous function $h : X \to Y'$ given by:

$$h(x) = \begin{cases} \text{family name} & \text{if a student with student no. } x \text{ exists} \\ \text{ERROR!} & \text{otherwise} \end{cases}$$

Remark 4.1: Notation: We frequently use the barred arrow ($\mapsto$) to denote how individual elements are mapped. For example, $f : X \to Y, X \ni x \mapsto f(x) \in Y$. Concretely: $f : \mathbb{R} \to \mathbb{R}, \mathbb{R} \ni x \mapsto x^5 \in \mathbb{R}$.

Images and Restrictions

Definition 4.3 (*Image of a Map*): Let $f : X \to Y$ be a map. The *image* of f is the set:

$$f(X) := \{y \in Y \mid \exists x \in X \text{ s.t. } f(x) = y\}$$

(sometimes written as $\operatorname{image}(f)$). Clearly, $f(X) \subseteq Y$.

Definition 4.4 (*Restriction and Extension*): Let $f : X \to Y$ be a map, and let $A \subseteq X$ be a subset. We can define a new map $f|_A : A \to Y$, called the *restriction* of f to A, simply by $f|_A(x) := f(x)$ for all $x \in A$. Conversely, f is called an *extension* of $f|_A$.

Some Standard Maps

Example 4.3 (*Standard Maps*):

(a) **Identity Map:** $\mathrm{id} : X \to X$ (sometimes denoted id_X), defined by $\mathrm{id}(x) = x$ for all $x \in X$.
(b) **Constant Map:** $f : X \to Y$. Choose a fixed $y_0 \in Y$. Then $f(x) = y_0$ for all $x \in X$.
(c) **Characteristic Function:** Let $A \subseteq X$. The characteristic function on A, written $\mathbf{1}_A : X \to \mathbb{R}$ (and also χ_A), is defined as:

$$\mathbf{1}_A(x) := \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

(d) **Projections:** Given a Cartesian product $X_1 \times X_2$, the standard projection maps are:

$$\pi_1 : X_1 \times X_2 \to X_1, \quad (x_1, x_2) \mapsto x_1$$
$$\pi_2 : X_1 \times X_2 \to X_2, \quad (x_1, x_2) \mapsto x_2$$

Injectivity, Surjectivity, Bijectivity

Definition 4.5 (*Injectivity, Surjectivity, and Bijectivity*): Let $f : X \to Y$ be a map.

(a) We say f is **injective** if $\forall x_1, x_2 \in X$, $f(x_1) = f(x_2) \implies x_1 = x_2$. (Equivalently, by contrapositive: $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$).
(b) We say f is **surjective** if $\forall y \in Y, \exists x \in X$ such that $f(x) = y$. (Equivalently: $f(X) = Y$).
(c) We say f is **bijective** (or a *bijection*) if f is both injective and surjective. (Equivalently: $\forall y \in Y, \exists! x \in X : f(x) = y$).

Example 4.4 (*Injective and Surjective Squaring Functions*): Consider variations of the squaring function, which explicitly demonstrate the importance of the chosen domain and target:

- $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$. Not injective (e.g., $f(2) = f(-2)$). Not surjective (no real x maps to -1).
- $f : \mathbb{R}_{\geq 0} \to \mathbb{R}, f(x) = x^2$. Injective, but not surjective.
- $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}, f(x) = x^2$. Bijective.

Definition 4.6 (*Inverse Map*): Let $f : X \to Y$ be a bijection. The *inverse map* $f^{-1} : Y \to X$ is the map that assigns to every $y \in Y$ the unique $x \in X$ such that $f(x) = y$.

Example 4.5 (*Inverse of Squaring Function*): For the bijection $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ with $f(x) = x^2$, the inverse map is $f^{-1} : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$, strictly defined by $f^{-1}(y) = \sqrt{y}$.

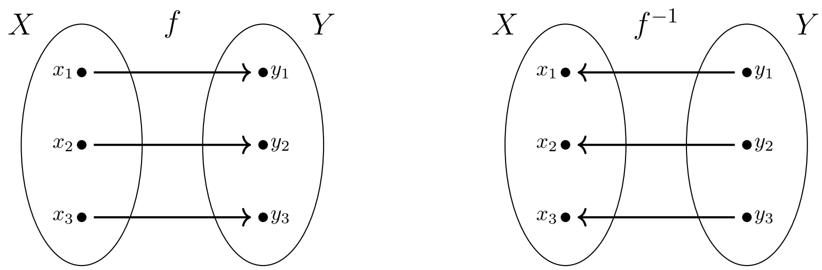


Figure 4.2: A bijection f and its inverse f^{-1}. Because f is bijective, every $y \in Y$ is hit by exactly one $x \in X$, so reversing every arrow again produces a map.

Composition of Maps

Definition 4.7 (*Composition*): Let $f : X \to Y$ and $g : Y \to Z$ be maps. The composition $g \circ f : X \to Z$ is defined by:

$$(g \circ f)(x) := g(f(x))$$

On elements, the composition is simply the two assignments performed one after the other:

$$x \xmapsto{f} f(x) \xmapsto{g} g(f(x))$$

Example 4.6 (*Composition with Inverse*): If $f : X \to Y$ is bijective and $f^{-1} : Y \to X$ is its inverse, then by definition:

$$f^{-1} \circ f = \mathrm{id}_X \quad \text{and} \quad f \circ f^{-1} = \mathrm{id}_Y$$

Remark 4.2: If $g \circ f$ is defined, $f \circ g$ is in general NOT defined (unless $Z = X$). But even when $Z = X$, in general $g \circ f \neq f \circ g$. Commutativity is not guaranteed.

Example 4.7 (*Non-Commutative Compositions*): Let $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$ and $g(x) = x + 1$.

$$\begin{aligned} (g \circ f)(x) &= g(x^2) = x^2 + 1 \\ (f \circ g)(x) &= f(x+1) = (x+1)^2 = x^2 + 2x + 1 \end{aligned}$$

Clearly, $g \circ f \neq f \circ g$.

Lemma 4.8 (*Composition of Injective and Surjective Maps*): Let $f : X \to Y$ and $g : Y \to Z$ be maps.

- **(a)** If f and g are injective, then $g \circ f$ is injective.
- **(b)** If f and g are surjective, then $g \circ f$ is surjective.
- **(c)** If f and g are bijective, then $g \circ f$ is bijective.

Proof:

- **(a)** Assume f and g are injective. Let $x_1, x_2 \in X$ such that $(g \circ f)(x_1) = (g \circ f)(x_2)$. This means $g(f(x_1)) = g(f(x_2))$. Because g is injective, this implies $f(x_1) = f(x_2)$. Because f is injective, this implies $x_1 = x_2$. Thus, $g \circ f$ is injective.
- **(b)** Assume f and g are surjective. Let $z \in Z$. Because g is surjective, there exists $y \in Y$ such that $g(y) = z$. Because f is surjective, there exists $x \in X$ such that $f(x) = y$. Therefore, $g(f(x)) = g(y) = z$, which means $(g \circ f)(x) = z$. Thus, $g \circ f$ is surjective.
- **(c)** This follows from **(a)** and **(b)**, because bijective means injective and surjective.

Q.E.D.

Graphs

Definition 4.9 (*Graph of a Map*): Let $f : X \to Y$ be a map. The "*graph*" of f is the following subset of $X \times Y$:

$$\mathrm{Graph}(f) := \{(x, y) \in X \times Y \mid y = f(x)\} \subseteq X \times Y.$$

For $X \subseteq \mathbb{R}$ and $Y = \mathbb{R}$ the graph is a curve in the plane, and the defining condition can be read off geometrically. Since f must assign to every $x \in X$ exactly one value, a subset of the plane is the graph of a map if and only if every vertical line meets it in exactly one point. This is the "*vertical line test*", illustrated in Figure 4.3.

(a) a graph (b) not a graph

Figure 4.3: The vertical line test. On the left every vertical line meets the curve exactly once, so the curve is the graph of a map. On the right the vertical line drawn meets the curve three times, so no single value can be assigned to that input and the curve is not the graph of a map.

We can often immediately determine the injectivity, surjectivity, and bijectivity of a function — or whether an assignment is a function at all — by examining its Cartesian graph using horizontal and vertical line tests.

Image and Inverse Image of Subsets

Definition 4.10 (*Image of a Subset*): Let $f : X \to Y$ be a map, and let $A \subseteq X$. The *image of A under f* is:

$$f(A) := \{y \in Y \mid \exists x \in A \text{ s.t. } f(x) = y\} = \{f(x) \mid x \in A\}$$

Definition 4.11 (*Inverse Image of a Subset*): Let $f : X \to Y$ be a map, and let $B \subseteq Y$. The *inverse image of B under f* is:

$$f^{-1}(B) := \{x \in X \mid f(x) \in B\} \subseteq X$$

Important remark 4.1: The notation $f^{-1}(B)$ is somewhat confusing, as this set exists and is perfectly well-defined even if the map f is not invertible!

Example 4.8 (*Preimages of Subsets*): Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$ (which is neither injective nor surjective). We can still compute inverse images of subsets without issue:

$$\begin{aligned} f^{-1}(\{4\}) &= \{-2, 2\} \\ f^{-1}(\{1, 9\}) &= \{-1, 1, -3, 3\} \\ f^{-1}(\{0\}) &= \{0\} \\ f^{-1}(\{-1\}) &= \emptyset \\ f^{-1}(\{-1, 0\}) &= \{0\} \end{aligned}$$

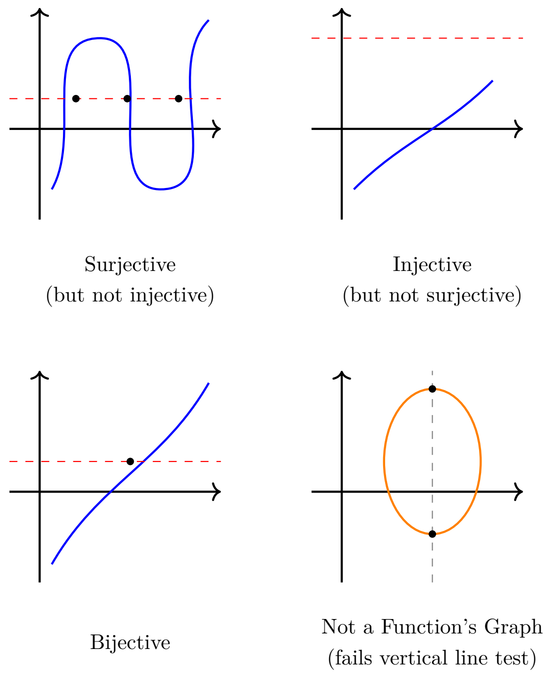


Figure 4.4: Reading off properties from a graph. A *horizontal* line at height y meets the graph once for each x with $f(x) = y$: meeting every horizontal line at least once means surjective, at most once means injective, and exactly once means bijective. A *vertical* line meeting the curve more than once means the curve is not a graph at all.

Partitions

Definition 4.12 (*Partition*): Let X be a set and $\mathcal{P}$ a family of subsets of X, none of which is empty. We say that $\mathcal{P}$ is a *partition* of X if the following hold:

(a) $X = \bigcup_{P \in \mathcal{P}} P$ (the subsets cover X).

(b) For any $P_1, P_2 \in \mathcal{P}$ with $P_1 \neq P_2$, we have $P_1 \cap P_2 = \emptyset$ (the subsets are mutually disjoint).

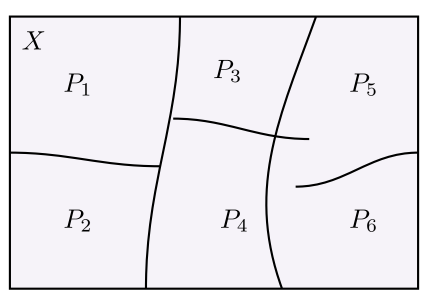


Figure 4.5: A partition of X. The pieces cover X and no two of them overlap, so every element of X lies in exactly one piece.

The last observation is what makes partitions useful, and it deserves to be stated on its own.

Lemma 4.13 (*Every Element Lies in Exactly One Piece*): Let $\mathcal{P}$ be a partition of X. Then for every $x \in X$ there exists exactly one $P \in \mathcal{P}$ with $x \in P$: at least one by condition **(a)** of Definition 4.12, and at most one by condition **(b)**.

Defining Functions by Cases

Partitions are precisely what licenses the case-by-case definitions we have already used, such as the characteristic function or the map h above.

Proposition 4.14 (*Gluing Functions along a Partition*): Let X and Y be sets and let $\mathcal{P}$ be a partition of X. Suppose that for every $P \in \mathcal{P}$ we are given a function $f_P : P \to Y$. Then there exists a unique function

$$f : X \to Y \quad \text{such that} \quad f|_P = f_P \ \text{ for all } P \in \mathcal{P}.$$

Proof:

Let $x \in X$. By Lemma 4.13, there exists exactly one $P \in \mathcal{P}$ with $x \in P$, so we may define

$$f(x) := f_P(x).$$

This assigns to every $x \in X$ a uniquely determined element of Y, so f is a well-defined map, and by construction $f|_P = f_P$ for every $P \in \mathcal{P}$.

For uniqueness, suppose $g : X \to Y$ also satisfies $g|_P = f_P$ for all $P \in \mathcal{P}$. Given $x \in X$, choose the piece P containing x; then $g(x) = g|_P(x) = f_P(x) = f(x)$. Hence $g = f$. **Q.E.D.**

Fields (Körper) (Chapter 5)

Before we can solve systems of linear equations or define abstract vector spaces, we must answer a more fundamental question: *Where do our numbers come from?*

In linear algebra, we do not restrict ourselves to the real numbers. We can perform our calculations over any algebraic structure where addition, subtraction, multiplication, and division behave exactly as we expect them to. Such a structure is called a **field** (in German: *Körper*).

The Field Axioms

To ensure that an algebraic structure behaves nicely, it must satisfy ten rigid rules. These are the Field Axioms.

Definition 5.1 (*Field*): A *field* is a tuple $(K, +, \cdot, 0, 1)$ consisting of a set K equipped with two binary operations:

$$+ : K \times K \to K, \quad (x, y) \mapsto x + y \quad \text{(Addition)}$$
$$\cdot : K \times K \to K, \quad (x, y) \mapsto x \cdot y \quad \text{(Multiplication)}$$

and two distinguished, unique elements $0, 1 \in K$, such that the following ten axioms hold:

Axioms of Addition:

- **(K1)** $\forall x, y, z \in K : x + (y + z) = (x + y) + z$ *(Associativity of Addition)*
- **(K2)** $\forall x, y \in K : x + y = y + x$ *(Commutativity of Addition)*
- **(K3)** $\forall x \in K : 0 + x = x$ *(Neutral Element of Addition)*
- **(K4)** $\forall x \in K, \exists x' \in K : x + x' = 0$ *(Inverse Element of Addition)*

Axioms of Multiplication:

- **(K5)** $\forall x, y, z \in K : x \cdot (y \cdot z) = (x \cdot y) \cdot z$ *(Associativity of Multiplication)*
- **(K6)** $\forall x \in K : 1 \cdot x = x$ *(Neutral Element of Multiplication)*
- **(K7)** $\forall x \in K \setminus \{0\}, \exists x' \in K : x' \cdot x = 1$ *(Inverse Element of Multiplication)*

Compatibility, Non-Triviality, and Commutativity:

- **(K8)** $\forall x, y, z \in K : x \cdot (y + z) = x \cdot y + x \cdot z$ and
  $\forall x, y, z \in K : (y + z) \cdot x = y \cdot x + z \cdot x$ *(Distributivity)*
- **(K9)** $1 \neq 0$ *(Non-triviality: the field has at least two elements)*
- **(K10)** $\forall x, y \in K : x \cdot y = y \cdot x$ *(Commutativity of Multiplication)*

Important remark 5.1: Axiom **(K10)** is not decorative. A structure satisfying **(K1)**–**(K9)** but not **(K10)** is called a "*skew field*" (or division ring); the quaternions are the standard example. Everything in this course is done over fields, so multiplication may always be reordered freely.

Remark 5.1 (*Notation*): We typically denote the additive inverse x' from (K4) as $-x$, allowing us to define subtraction: $x - y := x + (-y)$. Similarly, we denote the multiplicative inverse x' from (K7) as x^{-1} or $\frac{1}{x}$, allowing us to define division by non-zero elements: $x/y := x \cdot y^{-1}$.

AI-Note: The source material for this chapter is a single page, giving the definition above (and nothing else) in German, as *"Definition 1.3.9"*. Everything that follows — the examples, finite fields and modular arithmetic, $\mathbb{F}_p$, and the two proofs from the axioms — has no counterpart in the provided notes and was supplied by this transcript. It is standard material and consistent with the course, but it is not lecture content.

Infinite Fields

The most familiar fields are infinite. They form the backbone of calculus and classical geometry.

Example 5.1 (*Familiar Infinite Fields*):

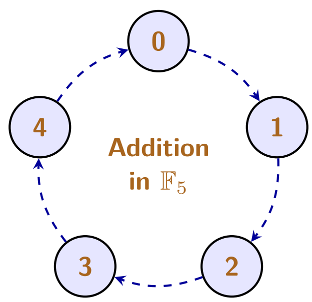
- $\mathbb{Q}$: The rational numbers (fractions). This is the smallest field containing the integers.
- $\mathbb{R}$: The real numbers.
- $\mathbb{C}$: The complex numbers, where we introduce $i^2 = -1$.

Important remark 5.2: The integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ do **not** form a field! While they satisfy addition and multiplication axioms perfectly, they completely fail (K7). For instance, there is no integer x' such that $2 \cdot x' = 1$. You cannot divide freely within $\mathbb{Z}$.

Finite Fields and Modular Arithmetic

A field does not have to be an infinite, continuous line of numbers. If we change our definitions of *"addition"* and *"multiplication"*, we can create perfectly valid fields that contain only a finite number of elements.

The secret to building a finite field is "*modular arithmetic*" (often called *"clock arithmetic"*). Instead of numbers growing infinitely on a number line, they wrap around a circle.



Definition 5.2 (*The Field $\mathbb{F}_p$*): Let p be a prime number. The set of integers modulo p, denoted as $\mathbb{F}_p$ (or sometimes $\mathbb{Z}/p\mathbb{Z}$), contains exactly p elements:

$$\mathbb{F}_p = \{0, 1, 2, \dots, p-1\}$$

Addition and multiplication are performed exactly as regular integers, but if the result exceeds $p-1$, we divide by p and keep only the **remainder**.

Theorem 5.3: $\mathbb{F}_p$ is a field if and only if p is a prime number.

Why must p be prime? Consider what happens if we try to build a field modulo 4 (which is not prime). In modulo 4, we have $2 \cdot 2 = 4 \equiv 0 \pmod 4$. We multiplied two non-zero numbers and got zero! In a true field, this is algebraically impossible, because it destroys the ability to divide (axiom **(K7)**). Prime numbers beautifully prevent this *"zero divisor"* problem, as Figure 5.1 makes visible.

Example 5.2 (*The Smallest Field: $\mathbb{F}_2$*): The absolute smallest field mathematically possible contains only two elements: $\{0, 1\}$. This is the field of Boolean logic, heavily used in computer science and cryptography. Operations are done modulo 2:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

$\cdot$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

(a) $\mathbb{F}_5$: no zero in the body

$\cdot$	1	2	3
1	1	2	3
2	2	$\boxed{0}$	2
3	3	2	1

(b) $\mathbb{Z}/4\mathbb{Z}$: $2 \cdot 2 = 0$

Figure 5.1: Multiplication tables of the non-zero elements. For the prime modulus 5 every entry is non-zero and every row is a rearrangement of $1, 2, 3, 4$, so each element has an inverse: the 1 in row x sits in column x^{-1}. For the composite modulus 4 the boxed entry $2 \cdot 2 = 0$ produces a zero divisor, and row 2 never reaches 1, so 2 has no inverse and **(K7)** fails.

Notice the fascinating property of $\mathbb{F}_2$: $1 + 1 = 0$. In this field, every element is its own additive inverse! Subtraction and addition are literally the same operation.

Example 5.3 (*Calculations in $\mathbb{F}_5$*): Let us explore $\mathbb{F}_5 = \{0, 1, 2, 3, 4\}$.

- **Addition:** $3 + 4 = 7$. But $7 \div 5 = 1$ with a remainder of 2. So, $3 + 4 = 2$ in $\mathbb{F}_5$.
- **Additive Inverses (K4):** What is -2? We need a number that adds to 2 to yield 0. Since $2 + 3 = 5 \equiv 0$, we know that $-2 = 3$.
- **Multiplicative Inverses (K7):** What is the inverse of 3 (i.e., $\frac{1}{3}$)? We need a number x such that $3 \cdot x = 1$. Let's test the options: $3 \cdot 2 = 6 \equiv 1$. Therefore, $3^{-1} = 2$.

Proving Basic Properties from the Axioms

The power of an axiomatic definition is that if we can prove a theorem using *only* axioms **(K1)** through **(K10)**, that theorem is universally true for **every field in existence** — whether it is the infinite space of real numbers $\mathbb{R}$, or the tiny binary world of $\mathbb{F}_2$.

Let us prove two seemingly *"obvious"* facts that actually require careful algebraic footwork to justify.

Lemma 5.4 (*Multiplication by Zero*): For any element x in a field K, we have $0 \cdot x = 0$.

Proof:

Let $x \in K$. By the neutral property of zero **(K3)**, we know that $0 = 0 + 0$. Let us multiply both sides of this equation by x:

$$(0 + 0) \cdot x = 0 \cdot x$$

By the distributivity axiom **(K8)**, we can expand the left side:

$$0 \cdot x + 0 \cdot x = 0 \cdot x$$

Now, by the additive inverse axiom **(K4)**, the element $(0 \cdot x)$ must have an inverse, $-(0 \cdot x)$. We add this inverse to both sides of our equation:

$$\big(0 \cdot x + 0 \cdot x\big) + \big(-(0 \cdot x)\big) = 0 \cdot x + \big(-(0 \cdot x)\big)$$

Using associativity of addition **(K1)**, we regroup the left side:

$$0 \cdot x + \Big(0 \cdot x + \big(-(0 \cdot x)\big)\Big) = 0$$

$$0 \cdot x + 0 = 0$$

Finally, applying **(K3)** once more, we drop the $+0$ to conclude:

$$0 \cdot x = 0$$

Q.E.D.

Lemma 5.5 (*Zero Divisors do not exist*): Let $a, b \in K$. If $a \cdot b = 0$, then either $a = 0$ or $b = 0$.

Proof:

Assume $a \cdot b = 0$. We want to show that if $a \neq 0$, then b must be exactly 0.

Suppose $a \neq 0$. Because a is a non-zero element in a field, axiom **(K7)** guarantees that it has a multiplicative inverse, a^{-1}. Let us multiply both sides of our equation $a \cdot b = 0$ by this inverse:

$$a^{-1} \cdot (a \cdot b) = a^{-1} \cdot 0$$

By Lemma 5.4, any element multiplied by 0 is 0, so the right side is 0:

$$a^{-1} \cdot (a \cdot b) = 0$$

By associativity of multiplication **(K5)**, we regroup the left side:

$$(a^{-1} \cdot a) \cdot b = 0$$

By the definition of the inverse **(K7)**, $a^{-1} \cdot a = 1$:

$$1 \cdot b = 0$$

Finally, by the neutral element of multiplication **(K6)**, $1 \cdot b = b$. Thus:

$$b = 0$$

We have proven that if the product is zero and a is not zero, b is forced to be zero. Q.E.D.

Systems of Linear Equations (Chapter 6)

Fields (Körper)

Before we solve systems of linear equations, we must fix the algebraic structure over which these equations are built. Throughout this course, we will work over a base "*field*" (in German: "*Körper*"), usually denoted by K.

A field is a set equipped with two operations (addition $+$ and multiplication $\cdot$) satisfying the ten axioms **(K1)**–**(K10)** of Definition 5.1: commutativity, associativity and distributivity, together with the existence of the identities 0 and 1 and of additive and multiplicative inverses. We briefly recall the examples we will use most often.

Example 6.1 (*Common Fields*):

- $\mathbb{Q} = \left\{a \mid a = \frac{m}{n},\ m, n \in \mathbb{Z},\ n \neq 0\right\}$ (the rational numbers)
- $\mathbb{R}$ (the real numbers)
- $\mathbb{C} = \{z \mid z = a + i \cdot b,\ a, b \in \mathbb{R}\}$, where $i \cdot i = -1$ (the complex numbers)

Example 6.2 (*Finite Fields*): Fields do not have to be infinite. We will frequently use finite fields, where arithmetic is performed modulo a prime number.

(a) $\mathbb{F}_2 = \{0, 1\}$. Operations are done modulo 2:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

(b) $\mathbb{F}_3 = \{0, 1, 2\}$. Operations are done modulo 3:

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

A Motivating Example

Let us solve a small system of linear equations over $\mathbb{R}$ by elimination:

$$\begin{cases} E_1: & x + 3y + 4z = -1 \\ E_2: & 2x - \ \ y + \ \ z = \ \ 5 \end{cases}$$

We operate on the equations so as to eliminate variables one at a time. First we remove x from the second equation by $-2 \cdot E_1 + E_2 \to E_2$:

$$\begin{cases} x + 3y + 4z = -1 \\ \quad\ - 7y - 7z = \ \ 7 \end{cases}$$

Next we normalise the second equation by $-\frac{1}{7} \cdot E_2 \to E_2$:

$$\begin{cases} x + 3y + 4z = -1 \\ \qquad\ \ y + \ \ z = -1 \end{cases}$$

Finally, instead of substituting backwards, we eliminate y **upwards** as well, using $-3 \cdot E_2 + E_1 \to E_1$:

$$\begin{cases} x \qquad\ + \ \ z = \ \ 2 \\ \qquad\ \ y + \ \ z = -1 \end{cases}$$

The system is now solved as far as it can be. There are three unknowns but only two equations, so one unknown remains free: let $z := c \in \mathbb{R}$ be an arbitrary real number. Then

$$y = -1 - c, \qquad x = 2 - c,$$

and the set of solutions is

$$\{(x, y, z) \mid x = 2 - c,\ y = -1 - c,\ z = c,\ c \in \mathbb{R}\}. \tag{6.1}$$

This is not a single point but an entire one-parameter family — geometrically, the line in which the two planes $x + 3y + 4z = -1$ and $2x - y + z = 5$ intersect, as sketched in Figure 6.1.

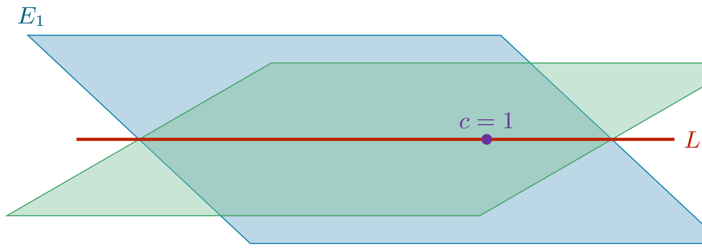


Figure 6.1: Schematic picture of the motivating example. Each equation cuts out a plane in $\mathbb{R}^3$, and the solution set L of the system is the line in which the two planes meet — one free parameter c, exactly as (6.1) records. Imposing a third independent equation adds a third plane, which meets L in a single point.

Example 6.3 (*One more equation removes the freedom*): Suppose we append a third equation $E_3 : 3x + 2y + z = 0$ to the system above. Substituting the parametrised solution (6.1) into it gives

$$3(2 - c) + 2(-1 - c) + c = 4 - 4c = 0,$$

so that $c = 1$ is forced. The freedom collapses and the system now has the unique solution

$$(x, y, z) = (1, -2, 1),$$

which is exactly the member of the family (6.1) with $c = 1$. A system may therefore have no solution, exactly one solution, or infinitely many; deciding which of these occurs is one of the goals of this chapter.

General Systems of Linear Equations

Definition 6.1 (*Linear Equation*): A "*linear equation*" in n variables $x_1, \dots, x_n$ over a field K is an equation of the form

$$a_1 x_1 + \cdots + a_n x_n = b,$$

where $a_1, \dots, a_n \in K$ are the "*coefficients*" and $b \in K$ is the "*constant term*".

- If $b = 0$, the equation is called "*homogeneous*".
- If $b \neq 0$, the equation is called "*inhomogeneous*".

Definition 6.2 (*System of Linear Equations*): A general system of m linear equations in n variables over a field K is written as

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n = b_2 \\ \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n = b_m \end{cases} \tag{6.2}$$

with $a_{i,j} \in K$ for $1 \leq i \leq m$, $1 \leq j \leq n$, and $b_i \in K$ for $1 \leq i \leq m$. Here i is the row index and j is the column index.

Definition 6.3 (*Solution*): A "*solution*" of the system is a tuple of scalars $s_1, \ldots, s_n \in K$ that satisfies all m equations simultaneously when we substitute $x_j = s_j$.

Definition 6.4 (*Equivalence of Systems*): Two systems of linear equations in the same variables are called "*equivalent*" if they have exactly the same set of solutions.

Matrices

To study systems of linear equations systematically, we strip away the variables and isolate the coefficients into a rectangular array.

Definition 6.5 (*Matrix*): An $m \times n$ "*matrix*" over K is a rectangular array consisting of $m \cdot n$ elements of K, arranged in m rows and n columns.

Notation 6.1: We denote a matrix A by its entries,

$$A = (a_{i,j})_{1 \le i \le m,\, 1 \le j \le n} \quad \text{or simply} \quad (a_{ij}),$$

where i denotes the row index and j the column index.

We define the set of all such matrices as

$$\mathcal{M}_{m \times n}(K) := \text{the set of all } m \times n \text{ matrices over } K.$$

Definition 6.6 (*Row and Column Vectors*): Special cases of matrices occur when one of the dimensions is 1:

(a) An $n \times 1$ matrix is called a "*column vector*". The set of all column vectors of length n is denoted by $K^n := \mathcal{M}_{n \times 1}(K)$:

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in K^n.$$

(b) A $1 \times n$ matrix is called a "*row vector*". The set of all row vectors of length n is denoted by $K^n_{\text{row}} := \mathcal{M}_{1 \times n}(K)$:

$$(x_1, x_2, \ldots, x_n) \in K^n_{\text{row}}.$$

Matrix-Vector Multiplication

Definition 6.7 (*Matrix-Vector Multiplication*): Let $A \in \mathcal{M}_{m \times n}(K)$ and $x \in K^n$. We define the product $A \cdot x$ to be the column vector in K^m obtained by pairing each row of A with the vector x:

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{1,1}x_1 + \cdots + a_{1,n}x_n \\ a_{2,1}x_1 + \cdots + a_{2,n}x_n \\ \vdots \\ a_{m,1}x_1 + \cdots + a_{m,n}x_n \end{pmatrix} \in K^m.$$

With this notation, the whole system (6.2) collapses into a single matrix equation. We write

$$(S): \quad Ax = b, \tag{6.3}$$

where $A \in \mathcal{M}_{m \times n}(K)$ is the "*coefficient matrix*", $x \in K^n$ is the vector of variables, and $b \in K^m$ is the "*constant vector*".

Notation 6.2: For a system (S) as in (6.3) we write

$$L(S) := \{(x_1, \ldots, x_n) \mid x_i \in K,\ Ax = b\} \subseteq K^n$$

for its "*set of solutions*". Two systems (S) and (S') in the same variables are equivalent (Definition 6.4) precisely when $L(S) = L(S')$.

Example 6.4 (*The motivating example in matrix form*): The system of the previous section can be rewritten first with indexed variables and then as a matrix equation:

$$\begin{cases} x + 3y + 4z = -1 \\ 2x - y + z = 5 \end{cases} \iff \begin{cases} x_1 + 3x_2 + 4x_3 = -1 \\ 2x_1 - x_2 + x_3 = 5 \end{cases} \iff \underbrace{\begin{pmatrix} 1 & 3 & 4 \\ 2 & -1 & 1 \end{pmatrix}}_{2\times 3} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_{3\times 1} = \underbrace{\begin{pmatrix} -1 \\ 5 \end{pmatrix}}_{2\times 1}.$$

Goals 6.1: Our aim for the rest of this chapter is to study the set of solutions $L(S)$ of a system $Ax = b$:

(a) Does the system have a solution at all, and if so, how many?
(b) How do we find **all** of the solutions, systematically?
(c) Does $L(S)$ carry a structure of its own?

Elementary Row Operations

Notation 6.3: We do not need to write down the unknowns $x_1, \ldots, x_n$ at every step: the coefficient matrix A and the constant vector b already carry all the information. We therefore record the system as its "*augmented matrix*" (or "*extended matrix*")

$$(A \mid b) \in \mathcal{M}_{m\times(n+1)}(K),$$

obtained by appending b to A as an extra column.

Example 6.5 (*An augmented matrix*): The system

$$\begin{cases} 2x_1 - x_2 + 3x_3 + 2x_4 = 0 \\ x_1 + 4x_2 - x_4 = 3 \\ 2x_1 + 6x_2 - x_3 + 5x_4 = 9 \end{cases} \qquad \text{has the augmented matrix} \qquad \left(\begin{array}{cccc|c} 2 & -1 & 3 & 2 & 0 \\ 1 & 4 & 0 & -1 & 3 \\ 2 & 6 & -1 & 5 & 9 \end{array}\right).$$

The three operations below act on the equations of a system, and equally on the rows of its augmented matrix.

Definition 6.8 (*Elementary Row Operations*): Let E_i denote the i-th equation of a system, equivalently the i-th row of its augmented matrix. The three "*elementary row operations*" (EROs) are:

(a) **Type 1 (scaling).** Choose $\lambda \in K \setminus \{0\}$ and multiply row i by λ.
Notation: $\lambda \cdot E_i \to E_i$.
(b) **Type 2 (addition).** Choose $\lambda \in K$ and $1 \le i, j \le m$ with $i \ne j$, and replace row j by row j **plus** λ times row i.
Notation: $\lambda \cdot E_i + E_j \to E_j$.
(c) **Type 3 (interchange).** Swap row i and row j.
Notation: $E_i \leftrightarrow E_j$.

Definition 6.9 (*Row Equivalence*): Let C and C' be $r \times s$ matrices with entries in K. We say that C' is "*row equivalent*" to C if C' can be obtained from C by a finite sequence of elementary row operations,

$$C = C_0 \longrightarrow C_1 \longrightarrow \cdots \longrightarrow C_k = C'.$$

Theorem 6.10 (*Row Operations Preserve the Solution Set*): Let $(S) : Ax = b$ and $(S') : A'x = b'$ be two systems of m linear equations in n unknowns. Suppose the augmented matrix $(A' \mid b')$ is row equivalent to $(A \mid b)$. Then

$$L(S') = L(S).$$

Proof:

We begin with three claims.

Claim (*Claim 1 — One Operation Gives One Inclusion*): If (S') is obtained from (S) by one elementary row operation, then $L(S) \subseteq L(S')$.

Proof of Claim 1. Let $(x_1, \ldots, x_n) \in L(S)$. We must show that $(x_1, \ldots, x_n) \in L(S')$. We treat the three types separately.

(a) Operation $\lambda \cdot E_i \to E_i$ with $\lambda \neq 0$. All equations of (S') coincide with those of (S) except for equation i, which changes as

$$a_{i,1}x_1 + \cdots + a_{i,n}x_n = b_i \qquad \longrightarrow \qquad \lambda a_{i,1}x_1 + \cdots + \lambda a_{i,n}x_n = \lambda b_i.$$

If $(x_1, \ldots, x_n)$ satisfies the left equation, then multiplying both sides by λ shows that it satisfies the right one as well.

(b) Operation $\lambda \cdot E_i + E_j \to E_j$. Again all equations of (S) and (S') coincide except for equation j:

$$\begin{aligned} &\text{row } j \text{ in } (S): \quad a_{j,1}x_1 + \cdots + a_{j,n}x_n = b_j \\ \longrightarrow \quad &\text{row } j \text{ in } (S'): \quad \big(a_{j,1} + \lambda a_{i,1}\big)x_1 + \cdots + \big(a_{j,n} + \lambda a_{i,n}\big)x_n = b_j + \lambda b_i \end{aligned}$$

If $(x_1, \ldots, x_n)$ satisfies rows i and j of (S), then adding λ times the i-th equation to the j-th shows that it satisfies row j of (S') too.

(c) Operation $E_i \leftrightarrow E_j$. Here we clearly have

$$(x_1, \ldots, x_n) \in L(S) \implies (x_1, \ldots, x_n) \in L(S'),$$

because the order in which the equations of a system are listed has no effect whatsoever on its set of solutions.

This concludes the proof of Claim 1. ◇

Claim (*Claim 2 — Every Operation is Reversible*): If (S') is obtained from (S) by one elementary row operation, then (S) can be obtained from (S') by one elementary row operation.

Proof of Claim 2. There are three cases, and in each of them we exhibit the inverse operation explicitly:

- **(a)** If $(S) \xrightarrow{\lambda \cdot E_i \to E_i} (S')$ with $\lambda \neq 0$, then $(S') \xrightarrow{\frac{1}{\lambda} \cdot E_i \to E_i} (S)$.
- **(b)** If $(S) \xrightarrow{\lambda \cdot E_i + E_j \to E_j} (S')$ with $i \neq j$, then $(S') \xrightarrow{-\lambda \cdot E_i + E_j \to E_j} (S)$.
- **(c)** If $(S) \xrightarrow{E_i \leftrightarrow E_j} (S')$, then $(S') \xrightarrow{E_j \leftrightarrow E_i} (S)$.

Note that case **(a)** is where the restriction $\lambda \neq 0$ in Definition 6.8 earns its keep: without it, $\frac{1}{\lambda}$ would not exist and the operation would not be reversible. This concludes the proof of Claim 2. ◇

Claim (*Claim 3 — One Operation Preserves the Solution Set*): If (S') is obtained from (S) by one elementary row operation, then $L(S) = L(S')$.

Proof of Claim 3. By Claim 1, we have $L(S) \subseteq L(S')$. By Claim 2, (S) is obtained from (S') by one elementary row operation, and hence Claim 1 applies again with the roles of (S) and (S') reversed, giving $L(S') \subseteq L(S)$. The two inclusions together give equality. This concludes the proof of Claim 3. ◇

We are now in a position to prove the theorem. By assumption, there is a finite sequence of elementary row operations

$$(S) = (S_0) \longrightarrow (S_1) \longrightarrow \cdots \longrightarrow (S_k) = (S').$$

Applying Claim 3 to each single step gives

$$L(S) = L(S_0) = L(S_1) = \cdots = L(S_k) = L(S').$$

Q.E.D.

Row-Reduced Matrices

Theorem 6.10 tells us that we may row-reduce as much as we like without changing the answer. It remains to decide **how far** the reduction can be pushed. The next two definitions describe the two successive normal forms we aim for.

Definition 6.11 (*Row-Reduced Matrix*): An $m \times n$ matrix A is called "*row-reduced*" if the following two conditions hold:

- **(a)** The first non-zero entry in each non-zero row of A is 1. This entry is called the "*pivot*" (or "*leading term*") of that row.
- **(b)** Each column of A which contains the leading non-zero entry of some row has all its other entries equal to 0.

Example 6.6 (*Row-reduced and not row-reduced*): Consider the following four matrices over $\mathbb{Q}$:

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 2 \\ 0 & 1 & -3 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & -3 \\ 0 & 0 & 0 \end{pmatrix} \qquad D = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 5 \end{pmatrix}$$

- A **is** row-reduced. The pivots sit in columns 4 and 2, and each of those columns is zero apart from its pivot.
- B is **not** row-reduced. Its pivots sit in columns 1, 2 and 3; but column 3 contains the pivot of row 3 and also the entry -1 in row 2, so condition **(b)** fails.
- C is **not** row-reduced. The first non-zero entry of row 1 is 2 and not 1, so condition **(a)** already fails.
- D **is** row-reduced. Its pivots sit in columns 3, 1 and 2, and each of those columns is zero apart from its pivot. Note that the pivots do **not** march to the right as we go down the rows — being row-reduced does not require this.

Theorem 6.12 (*Reduction to a Row-Reduced Matrix*): Every $m \times n$ matrix A is row equivalent to a row-reduced matrix.

Proof:

Let $A = (a_{ij})$ with $1 \leq i \leq m$, $1 \leq j \leq n$ and $a_{ij} \in K$. We proceed row by row.

Row $\#1$. If all entries in the first row of A are 0, then condition **(a)** of Definition 6.11 is satisfied for row $\#1$ and there is nothing to do. Otherwise, let k be the smallest index j for which $a_{1j} \neq 0$, that is

$$a_{1j} = 0 \quad \forall 1 \leq j < k, \qquad a_{1k} \neq 0,$$

so that the first row has the shape $(0,\ldots,0,a_{1k},*,\ldots,*)$. Multiply row #1 by $\frac{1}{a_{1k}}$; its leading entry is now 1, and condition **(a)** holds for row #1. Next, for each $2 \le i \le m$, add $(-a_{ik})$ times row #1 to row i, that is, perform

$$-a_{ik} \cdot E_1 + E_i \to E_i.$$

Schematically:

$$\begin{pmatrix} 0 & \ldots & 0 & 1 & \ldots \\ & \vdots & & \vdots & \\ a_{i1} & \ldots & \ldots & a_{ik} & \ldots \\ & \vdots & & \vdots & \end{pmatrix} \xrightarrow{-a_{ik}\cdot E_1+E_i\to E_i} \begin{pmatrix} 0 & \ldots & 0 & 1 & \ldots \\ & \vdots & & \vdots & \\ a_{i1} & \ldots & a_{i,k-1} & 0 & \ldots \\ & \vdots & & \vdots & \end{pmatrix}$$

After these operations, column k is zero apart from the 1 in row #1.

Row #2. If all entries in row #2 are 0, we do nothing to this row. If some entry in row #2 is non-zero, we multiply row #2 by a scalar so that its leading term becomes 1. Recall that row #1 has its leading non-zero entry in column k. Since we have just made all other entries of column k vanish, the leading non-zero term of row #2 can **not** lie in column k: it must appear in some other column, say column k' with $k' \neq k$. We now add suitable multiples of row #2 to all the other rows until we obtain a matrix in which all entries of column k' are 0, except the one in row #2. The two possible configurations are

$$\begin{pmatrix} 0 & \ldots & 0 & \ldots & 0 & 1 & * & \ldots & * \\ 0 & \ldots & 0 & 1 & *\cdots* & 0 & * & \ldots & * \\ & & 0 & & & 0 & & & \\ & & \vdots & & & \vdots & & & \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 & \ldots & 0 & 1 & *\cdots* & 0 & * & \ldots & * \\ 0 & \ldots & 0 & \ldots & 0 & 1 & * & \ldots & * \\ & & 0 & & & 0 & & & \\ & & \vdots & & & \vdots & & & \end{pmatrix}$$

according to whether $k' < k$ or $k < k'$.

Important remark 6.1: In the operations just performed using row #2 we did **not** change any of the entries of row #1 lying in columns $1,\ldots,k$; in particular, nothing was changed in column k. Also, in case row #1 was zero, the operations using row #2 do not affect row #1 at all. This is what guarantees that the work already done is never undone by the later steps.

We continue now to row #3 and repeat the same procedure, and so on. After a finite number of elementary row operations we arrive at a row-reduced matrix. **Q.E.D.**

Row-Reduced Echelon Matrices

The matrix D of the example above is row-reduced, and yet its pivots appear in the disorderly column order $3,1,2$. Sorting the rows costs nothing and yields a genuine normal form.

Definition 6.13 (*Row-Reduced Echelon Matrix*): An $m \times n$ matrix A is called "*row-reduced echelon*" if the following hold:

- **(a)** A is row-reduced.
- **(b)** Every row of A which has all its entries 0 appears below every row which has a non-zero entry.
- **(c)** If rows $1,\ldots,r$ are the non-zero rows of A, and if the leading non-zero entry of row i occurs in column k_i for $i = 1,\ldots,r$, then

$$k_1 < k_2 < \cdots < k_r.$$

Condition **(c)** is what produces the characteristic staircase shape shown in Figure 6.2.

$$
\begin{array}{c|c}
 & \begin{array}{ccccccc} & k_1 & & k_2 & & k_3 & \end{array} \\
\begin{array}{c} 1 \\ \\ r \\ r+1 \\ m \end{array} &
\begin{bmatrix}
0 & 1 & * & 0 & * & 0 & * \\
0 & 0 & 0 & 1 & * & 0 & * \\
0 & 0 & 0 & 0 & 0 & 1 & * \\
0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
\end{array}
$$

Figure 6.2: A row-reduced echelon matrix. The circled pivots are 1; everything below the green staircase is 0, and each pivot column is 0 apart from its pivot. Because $k_1 < k_2 < k_3$, the pivots move strictly to the right as we descend, and the $m-r$ zero rows are pushed to the bottom. The columns carrying a $*$ are the ones without a pivot; they will turn into the free variables when the matrix is read back as a system.

Theorem 6.14 (*Reduction to Row-Reduced Echelon Form*): Every $m \times n$ matrix A is row equivalent to a row-reduced echelon matrix.

Proof:

By Theorem 6.12 we already know that A is row equivalent to a row-reduced matrix A'. Performing a finite number of row interchanges on A' — an operation of Type 3, which preserves row equivalence — we may list the non-zero rows in increasing order of their pivot columns and move all zero rows to the bottom. The result satisfies conditions **(b)** and **(c)** of Definition 6.13, and interchanging rows does not disturb condition **(a)**. Hence we arrive at a row-reduced echelon matrix. **Q.E.D.**

Remark 6.1: Why is this useful for solving systems of equations? Let $Ax = b$ be a system of linear equations and consider its augmented matrix $(A \mid b)$. We may perform on $(A \mid b)$ the same elementary operations that we perform on A, and so obtain a row-equivalent matrix $(A' \mid b')$ in which A' is a row-reduced echelon matrix. By Theorem 6.10, the systems $Ax = b$ and $A'x = b'$ have the same solutions — but $A'x = b'$ is far easier to solve, since each pivot expresses one variable directly in terms of the pivot-free ones.

Worked Examples

Example 6.7 (*Solving a system by full reduction*): Consider the system over $\mathbb{R}$ whose augmented matrix is

$$\left(\begin{array}{cccc|c} 0 & -9 & 3 & 4 & 9 \\ 1 & 4 & 0 & -1 & 5 \\ 2 & 6 & -1 & 5 & -5 \end{array}\right).$$

We first normalise the leading entry of the first row, and then clear its column:

$$\xrightarrow{-\frac{1}{9}\cdot E_1 \to E_1} \left(\begin{array}{cccc|c} 0 & 1 & -\frac{1}{3} & -\frac{4}{9} & -1 \\ 1 & 4 & 0 & -1 & 5 \\ 2 & 6 & -1 & 5 & -5 \end{array}\right) \xrightarrow[-6\cdot E_1 + E_3 \to E_3]{-4\cdot E_1 + E_2 \to E_2} \left(\begin{array}{cccc|c} 0 & 1 & -\frac{1}{3} & -\frac{4}{9} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{9} & 9 \\ 2 & 0 & 1 & \frac{23}{3} & 1 \end{array}\right)$$

Next we clear the first column below row 2 and normalise the third row:

$$\xrightarrow{-2\cdot E_2 + E_3 \to E_3} \left(\begin{array}{cccc|c} 0 & 1 & -\frac{1}{3} & -\frac{4}{9} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{9} & 9 \\ 0 & 0 & -\frac{5}{3} & \frac{55}{9} & -17 \end{array}\right) \xrightarrow{-\frac{3}{5}\cdot E_3 \to E_3} \left(\begin{array}{cccc|c} 0 & 1 & -\frac{1}{3} & -\frac{4}{9} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{9} & 9 \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array}\right)$$

Now we clear the third column upwards, and finally sort the rows:

$$\xrightarrow[\frac{1}{3}\cdot E_3+E_1\to E_1]{-\frac{4}{3}\cdot E_3+E_2\to E_2}\left(\begin{array}{cccc|c} 0 & 1 & 0 & -\frac{5}{3} & \frac{12}{5} \\ 1 & 0 & 0 & \frac{17}{3} & -\frac{23}{5} \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array}\right) \xrightarrow{E_1\leftrightarrow E_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & \frac{17}{3} & -\frac{23}{5} \\ 0 & 1 & 0 & -\frac{5}{3} & \frac{12}{5} \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array}\right)$$

The last matrix is in row-reduced echelon form. Reading it back as a system gives

$$\begin{cases} x_1 \qquad\qquad + \frac{17}{3}x_4 = -\frac{23}{5} \\ \qquad x_2 \qquad - \frac{5}{3}x_4 = \frac{12}{5} \\ \qquad\qquad x_3 - \frac{11}{3}x_4 = \frac{51}{5} \end{cases}$$

Column 4 carries no pivot, so x_4 is free. Take $x_4 := a \in \mathbb{R}$ arbitrary. Then

$$x_3 = \frac{11}{3}a + \frac{51}{5}, \qquad x_2 = \frac{5}{3}a + \frac{12}{5}, \qquad x_1 = -\frac{17}{3}a - \frac{23}{5},$$

and these are **all** the solutions of the original system, by Theorem 6.10.

Example 6.8 (*Conditions for consistency*): Let us now see how the solutions behave for a general choice of the constants $b_1, b_2, b_3 \in \mathbb{R}$ in the system

$$\begin{pmatrix} 1 & -2 & 1 \\ 2 & 1 & 1 \\ 0 & 5 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

We construct the augmented matrix and reduce it:

$$\left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 2 & 1 & 1 & b_2 \\ 0 & 5 & -1 & b_3 \end{array}\right) \xrightarrow{-2\cdot E_1+E_2\to E_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 0 & 5 & -1 & b_2-2b_1 \\ 0 & 5 & -1 & b_3 \end{array}\right)$$

$$\xrightarrow{-1\cdot E_2+E_3\to E_3} \left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 0 & 5 & -1 & b_2-2b_1 \\ 0 & 0 & 0 & b_3-b_2+2b_1 \end{array}\right) \xrightarrow{\frac{1}{5}\cdot E_2\to E_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 0 & 1 & -\frac{1}{5} & \frac{1}{5}(b_2-2b_1) \\ 0 & 0 & 0 & b_3-b_2+2b_1 \end{array}\right)$$

$$\xrightarrow{2\cdot E_2+E_1\to E_1} \left(\begin{array}{ccc|c} 1 & 0 & \frac{3}{5} & \frac{1}{5}(b_1+2b_2) \\ 0 & 1 & -\frac{1}{5} & \frac{1}{5}(b_2-2b_1) \\ 0 & 0 & 0 & b_3-b_2+2b_1 \end{array}\right)$$

The last matrix is in row-reduced echelon form, and the equivalent system reads

$$\begin{cases} x_1 \qquad + \frac{3}{5}x_3 = \frac{1}{5}(b_1+2b_2) \\ \qquad x_2 - \frac{1}{5}x_3 = \frac{1}{5}(b_2-2b_1) \\ \qquad\qquad\quad 0 = b_3-b_2+2b_1 \end{cases}$$

Two cases arise from the last row.

- If $b_3 - b_2 + 2b_1 \neq 0$, the third equation reads $0 = \text{non-zero}$, and the system has **no** solutions.
- If $b_3 - b_2 + 2b_1 = 0$, the third equation is vacuous and the system is equivalent to

$$\begin{cases} x_1 + \frac{3}{5}x_3 = \frac{1}{5}(b_1+2b_2) \\ x_2 - \frac{1}{5}x_3 = \frac{1}{5}(b_2-2b_1) \end{cases}$$

  Column 3 carries no pivot, so x_3 is free. Take $x_3 := c \in \mathbb{R}$ arbitrary; then

$$x_2 = \frac{1}{5}c + \frac{1}{5}(b_2-2b_1), \qquad x_1 = -\frac{3}{5}c + \frac{1}{5}(b_1+2b_2).$$

So the constants b_1, b_2, b_3 cannot be chosen freely: the system is consistent if and only if $b_3 - b_2 + 2b_1 = 0$, and when it is, it has infinitely many solutions.

Example 6.9 (*A system with infinitely many solutions*): As a final illustration, consider

$$\begin{cases} x_1 - 2x_2 + \ \ x_3 + \ \ x_4 = 1 \\ 2x_1 - 4x_2 + \ \ x_3 + 3x_4 = 2 \\ x_1 - 2x_2 + 2x_3 - \ \ x_4 = 1 \end{cases}$$

We write the augmented matrix and apply EROs:

$$\left(\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 1 \\ 2 & -4 & 1 & 3 & 2 \\ 1 & -2 & 2 & -1 & 1 \end{array}\right) \xrightarrow[-1\cdot E_1 + E_3 \to E_3]{-2\cdot E_1 + E_2 \to E_2} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 \end{array}\right)$$

$$\xrightarrow{1\cdot E_2 + E_3 \to E_3} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 \end{array}\right)$$

From the third row, $-x_4 = 0$, and therefore, $x_4 = 0$. From the second row, $-x_3 + x_4 = 0$, and therefore, $x_3 = 0$. Substituting into the first row gives $x_1 - 2x_2 = 1$, that is, $x_1 = 1 + 2x_2$.

Here x_2 is the free variable. Let $x_2 := \alpha \in \mathbb{R}$. The general solution is

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1 + 2\alpha \\ \alpha \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Observe the shape of the answer: one particular solution plus an arbitrary multiple of a fixed vector. That the solution set of a system should always look like this is precisely the kind of structure our **(c)** question asked about, and we will return to it once we have the language of vector spaces.

AI-Note: The lecture notes for this chapter use F for the base field and R_i for the rows; this transcript writes K and E_i throughout, in line with the conventions used in the rest of the document. The two matrices in the schematic step of the proof of Theorem 6.12, and the final example above, are expanded slightly beyond what the notes show; the geometric picture in Figure 6.1 and the staircase in Figure 6.2 were added here. Everything else follows the notes directly.

Vector Spaces (Chapter 7)

Definition and Axioms

Definition 7.1 (*Vector Space*): A "*vector space*" over a field K is a set V, endowed with two operations:

$$+ : V \times V \longrightarrow V, \quad (v_1, v_2) \mapsto v_1 + v_2 \qquad \text{(Addition of vectors)}$$
$$\cdot : K \times V \longrightarrow V, \quad (\alpha, v) \mapsto \alpha \cdot v \qquad \text{(Scalar multiplication)}$$

such that the following conditions (axioms) hold:

- **(V1)** $\forall v_1, v_2, v_3 \in V\colon\ v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$.
- **(V2)** $\exists$ an element $0 \in V$ such that $0 + v = v + 0 = v$ for all $v \in V$. (Sometimes denoted 0_V).
- **(V3)** $\forall v \in V, \exists v' \in V$ such that $v + v' = 0$.
- **(V4)** $\forall v_1, v_2 \in V\colon\ v_1 + v_2 = v_2 + v_1$.
- **(V5)** $\forall a, b \in K, \forall v \in V\colon\ a \cdot (b \cdot v) = (a \cdot b) \cdot v$.
- **(V6)** $\forall v \in V\colon\ 1 \cdot v = v$.
- **(V7)** $\forall \alpha \in K, \forall v_1, v_2 \in V\colon\ \alpha \cdot (v_1 + v_2) = \alpha \cdot v_1 + \alpha \cdot v_2$.
- **(V8)** $\forall \alpha_1, \alpha_2 \in K, \forall v \in V\colon\ (\alpha_1 + \alpha_2) \cdot v = \alpha_1 \cdot v + \alpha_2 \cdot v$.

The elements of K are called "*scalars*" and the elements of V are called "*vectors*".

Lemma 7.2 (*Elementary Consequences of the Axioms*): Let V be a vector space over a field K. Then the following properties hold:

- **(a)** The 0 vector from (V2) is unique. To avoid confusion (e.g., with $0 \in K$), we will sometimes denote it by 0_V.
- **(b)** The element v' (corresponding to v) from (V3) is unique. We will denote it by $-v$. We will also write $v - w := v + (-w)$ for all $v, w \in V$.
- **(c)** For all $v \in V$, we have $0 \cdot v = 0_V$ (i.e., $0_K \cdot v = 0_V$).
- **(d)** For all $\alpha \in K$, we have $\alpha \cdot 0_V = 0_V$.
- **(e)** $(-1) \cdot v = -v$ for all $v \in V$.
- **(f)** $-(-v) = v$ for all $v \in V$.
- **(g)** If $a \cdot v = 0_V$ for some $a \in K$ and $v \in V$, then either $a = 0$ or $v = 0_V$.
- **(h)** Associativity for $+$ holds also for sums of n elements. That is, $v_1 + \cdots + v_n$ makes sense no matter how we place the brackets:

  $$v_1 + \big(v_2 + (\cdots + (v_{n-1} + v_n) \ldots)\big), \qquad (v_1 + v_2) + (\ldots), \qquad \text{and so on,}$$

  all give the same element of V.

Proof:

- **(a)** Assume there are two such zero elements, 0_1 and 0_2. Then, by applying (V2) to both, we get: $0_1 = 0_1 + 0_2 = 0_2 + 0_1 = 0_2$. Thus, $0_1 = 0_2$.
- **(b)** Assume there are two inverses v_1' and v_2' for an element v. Then we can write:

  $$v_1' = v_1' + 0_V = v_1' + (v + v_2') = (v_1' + v) + v_2' = 0_V + v_2' = v_2'$$

  Thus, the inverse is unique.
- **(c)** Consider $0 \cdot v = (0+0) \cdot v = 0 \cdot v + 0 \cdot v$, where the second equality is (V8). By adding $-(0 \cdot v)$ to both sides, we yield $0_V = 0 \cdot v$.
- **(d)** Similarly, $\alpha \cdot 0_V = \alpha \cdot (0_V + 0_V) = \alpha \cdot 0_V + \alpha \cdot 0_V$ by (V7). Adding $-(\alpha \cdot 0_V)$ yields $0_V = \alpha \cdot 0_V$.
- **(e)** Observe that $v + (-1) \cdot v = 1 \cdot v + (-1) \cdot v = (1 - 1) \cdot v = 0 \cdot v = 0_V$. Since the inverse is unique by part **(b)**, it must be that $(-1) \cdot v = -v$.

(f) By definition, $-(-v)$ is the unique element that added to $-v$ gives 0_V. But v itself satisfies $(-v)+v = v+(-v) = 0_V$ by (V4) and (V3), so uniqueness in part **(b)** forces $-(-v) = v$.

(g) Let $a \in K$ and $v \in V$ and assume that $a \cdot v = 0_V$. If $a = 0$ there is nothing to prove, so suppose $a \neq 0$. Because K is a field, there is an element $a^{-1} \in K$ with $a^{-1} \cdot a = 1$. Therefore,

$$v \overset{(\mathrm{V6})}{=} 1 \cdot v = (a^{-1} \cdot a) \cdot v \overset{(\mathrm{V5})}{=} a^{-1} \cdot (a \cdot v) = a^{-1} \cdot 0_V \overset{\textbf{(d)}}{=} 0_V.$$

So either $a = 0$ or $v = 0_V$, as claimed.

(h) This follows from (V1) by induction on n, exactly as for the associativity of addition in a field. From now on we will therefore write $v_1 + \cdots + v_n$ without brackets.

Q.E.D.

The Coordinate Space K^n

Example 7.1 (*The Coordinate Space K^n*): Let K be a field and $n \in \mathbb{N}$. Put

$$K^n := \{(a_1, \ldots, a_n) \mid a_i \in K,\ 1 \leq i \leq n\}.$$

The set K^n becomes a vector space over K if we endow it with the following operations. Let $v = (a_1, \ldots, a_n)$ and $w = (b_1, \ldots, b_n) \in K^n$. We define addition in K^n component-wise, using the addition of the field K in each slot:

$$v + w = (a_1, \ldots, a_n) + (b_1, \ldots, b_n) := (a_1 + b_1, \ldots, a_n + b_n)$$

For $\alpha \in K$, we define scalar multiplication as:

$$\alpha \cdot v = \alpha \cdot (a_1, \ldots, a_n) := (\alpha \cdot a_1, \ldots, \alpha \cdot a_n)$$

The zero vector is explicitly defined as $0 := (0, \ldots, 0) \in K^n$. With these operations K^n becomes a vector space over K; the proof rests entirely on the fact that K itself satisfies the field axioms. We call K^n also the "*n-dimensional coordinate space*" over K.

Exercise 7.1: Check the details: verify each of the eight axioms (V1)–(V8) for K^n.

Important remark 7.1: Sometimes we write the elements of K^n as row vectors and sometimes as column vectors, whichever is more convenient at the time. The two conventions describe the same vector space, and we will switch between them without further comment — but whenever a matrix product $A \cdot x$ is involved, the elements of K^n are always to be read as column vectors, as in Definition 6.7.

Linear Subspaces

Definition 7.3 (*Linear Subspace*): Let V be a vector space over K. A subset $U \subseteq V$ is called a "*linear subspace*" of V (or just a "*subspace*" of V) if the following holds:

(LSS1) $U \neq \emptyset$.

(LSS2) For all $u_1, u_2 \in U$, we have $u_1 + u_2 \in U$ (closed under addition).

(LSS3) For all $\alpha \in K$ and all $u \in U$, we have $\alpha \cdot u \in U$ (closed under scalar multiplication).

Lemma 7.4 (*A Subspace is a Vector Space*): Let V be a vector space over K and let $U \subseteq V$ be a subspace. Then U is a vector space in its own right, when endowed with the operations inherited from V.

Proof:

Conditions (LSS2) and (LSS3) say precisely that the operations $+$ and $\cdot$ of V restrict to operations

$$+\colon U \times U \longrightarrow U, \qquad \cdot\colon K \times U \longrightarrow U,$$

so that U carries the two operations required by Definition 7.1. That these operations satisfy the eight axioms of a vector space then follows by direct verification, since each axiom is an identity that already holds for all elements of V and therefore in particular for all elements of U.

The one axiom that deserves a word is (V2), because it asserts the **existence** of an element of U rather than an identity. Pick any element $u \in U$; this is possible because $U \neq \emptyset$ by (LSS1). By (LSS3),

$$0_V = 0_K \cdot u \in U,$$

where the first equality is Lemma 7.2 **(c)**. Now $0_V + w = w$ holds for every $w \in U$, because it already holds for every $w \in V$ by (V2) for V. Hence 0_V serves as the zero element of U.

Q.E.D.

Exercise 7.2: Carry out the direct verification of the remaining seven axioms for U.

In practice one rarely checks (LSS1)–(LSS3) separately. The following criterion compresses them into two conditions, one of which is a single membership test.

Lemma 7.5 (*Criterion for a Linear Subspace*): Let V be a vector space over K, and let $U \subseteq V$ be a subset. Then U is a linear subspace of V if and only if the following holds:

- **(a)** $0_V \in U$.
- **(b)** For all $\alpha_1, \alpha_2 \in K$ and all $u_1, u_2 \in U$, we have $\alpha_1 u_1 + \alpha_2 u_2 \in U$.

Proof:

Direction 1 ($\Longrightarrow$): Suppose $U \subseteq V$ is a subspace. Then $0_V \in U$ was established in the proof of Lemma 7.4, which gives **(a)**. For **(b)**, given $\alpha_1, \alpha_2 \in K$ and $u_1, u_2 \in U$, condition (LSS3) gives $\alpha_1 u_1 \in U$ and $\alpha_2 u_2 \in U$, and then (LSS2) gives $\alpha_1 u_1 + \alpha_2 u_2 \in U$.

Direction 2 ($\Longleftarrow$): Suppose **(a)** and **(b)** hold. Then $U \neq \emptyset$ by **(a)**, which is (LSS1). Let $u_1, u_2 \in U$. By **(b)** applied with $\alpha_1 = \alpha_2 = 1$,

$$\underbrace{1 \cdot u_1 + 1 \cdot u_2}_{= u_1 + u_2} \in U,$$

which is (LSS2). Finally, let $\alpha \in K$ and $u \in U$. By **(b)** applied with $\alpha_1 = \alpha$, $\alpha_2 = 0$ and $u_1 = u_2 = u$,

$$\underbrace{\alpha \cdot u + 0 \cdot u}_{= \alpha \cdot u} \in U,$$

which is (LSS3). Hence U is a linear subspace. **Q.E.D.**

Examples of Subspaces

Example 7.2 (*The Trivial Subspace*): Let V be a vector space over K. Then $\{0_V\} \subseteq V$ is a linear subspace: it is non-empty, and $0_V + 0_V = 0_V$ as well as $\alpha \cdot 0_V = 0_V$ for every $\alpha \in K$, by Lemma 7.2 **(d)**. At the other extreme, $V \subseteq V$ is of course a subspace of itself.

Example 7.3 (*A Plane Through the Origin, and One That Is Not*): Fix $b \in K$ and consider the subset

$$U_b := \{(x_1, x_2, x_3) \in K^3 \mid x_1 + x_2 + x_3 = b\} \subseteq K^3.$$

Claim (*U_b is a subspace exactly for $b = 0$*): $U_b \subseteq K^3$ is a linear subspace if and only if $b = 0$.

Proof of the direction $\Longrightarrow$. Suppose U_b is a subspace of K^3. Let (x_1, x_2, x_3) and $(y_1, y_2, y_3) \in U_b$. Then by definition,

$$x_1 + x_2 + x_3 = b, \qquad y_1 + y_2 + y_3 = b. \tag{7.1}$$

Since U_b is a subspace, it is closed under addition, and therefore, $(x_1+y_1, x_2+y_2, x_3+y_3) \in U_b$. By the definition of U_b, this implies

$$(x_1+y_1)+(x_2+y_2)+(x_3+y_3)=b.$$

But rearranging the terms and using (7.1), we also get

$$(x_1+y_1)+(x_2+y_2)+(x_3+y_3)=(x_1+x_2+x_3)+(y_1+y_2+y_3)=b+b.$$

Therefore, $b+b=b$, which implies $b=0$. ◇

Exercise 7.3: Prove the direction $\Longleftarrow$ of the claim on Page 47 that U_b is a subspace exactly for $b=0$: show that if $b=0$, then U_0 satisfies (LSS1), (LSS2) and (LSS3), and is therefore a linear subspace of K^3.

The Solution Set of a Linear System

The previous example is a special case of something much more important. It answers the third of the questions we set ourselves in the previous chapter, in the list of goals on Page 38: the solution set of a system of linear equations does carry a structure of its own — provided the system is homogeneous.

Let $A \in \mathcal{M}_{m\times n}(K)$ and fix $b \in K^m$, viewed as a column vector. Consider

$$L := \{x \in K^n \mid A\cdot x = b\} \subseteq K^n,$$

where the elements of K^n are viewed as column vectors as well, following the remark on row and column notation on Page 46. This is exactly the set $L(S)$ of solutions of the system $(S): Ax=b$ from (6.3).

Lemma 7.6 (*Linearity of Matrix-Vector Multiplication*): Let $A \in \mathcal{M}_{m\times n}(K)$. For all $u, v \in K^n$ and all $a \in K$ we have:

(a) $A\cdot(u+v) = A\cdot u + A\cdot v$.

(b) $A\cdot(a\cdot u) = a\cdot(A\cdot u)$.

Proof:

Write $A=(a_{ij})$ with $1\le i\le m$, $1\le j\le n$, and let

$$u=\begin{pmatrix}u_1\\ \vdots\\ u_n\end{pmatrix}, \qquad v=\begin{pmatrix}v_1\\ \vdots\\ v_n\end{pmatrix}.$$

We prove **(a)** by comparing the i-th entries of both sides. By Definition 6.7, entry number i of $A\cdot u$ is $a_{i1}u_1+\cdots+a_{in}u_n$, and entry number i of $A\cdot v$ is $a_{i1}v_1+\cdots+a_{in}v_n$. Adding these in K and regrouping by distributivity gives

$$\begin{aligned}\big(A\cdot u + A\cdot v\big)_i &= \big(a_{i1}u_1+\cdots+a_{in}u_n\big)+\big(a_{i1}v_1+\cdots+a_{in}v_n\big)\\ &= a_{i1}(u_1+v_1)+a_{i2}(u_2+v_2)+\cdots+a_{in}(u_n+v_n)\\ &= \big(A\cdot(u+v)\big)_i,\end{aligned}$$

the last equality again by Definition 6.7, since the j-th entry of $u+v$ is u_j+v_j. As this holds for every $1\le i\le m$, the two column vectors are equal. **Q.E.D.**

Exercise 7.4: Prove statement **(b)** of Lemma 7.6.

Theorem 7.7 (*When the Solution Set is a Subspace*): $L\subseteq K^n$ is a linear subspace if and only if $b=0\in K^m$.

Proof:

Direction 1 ($\Longleftarrow$): We assume $b = 0$ and have to show that $L \subseteq K^n$ is a subspace.

First, $x = 0 \in L$, since indeed $A \cdot 0 = 0$. Next, if $x = (x_1, \ldots, x_n) \in L$ and $y = (y_1, \ldots, y_n) \in L$, then by Lemma 7.6 **(a)**,

$$A \cdot (x + y) = A \cdot x + A \cdot y = 0 + 0 = 0,$$

and therefore, $x + y \in L$. This shows that $x, y \in L$ implies $x + y \in L$. Finally, let $x \in L$ and $a \in K$. By Lemma 7.6 **(b)**,

$$A \cdot (a \cdot x) = a \cdot (A \cdot x) = a \cdot 0 = 0,$$

so $a \cdot x \in L$. This shows L is a subspace.

Direction 2 ($\Longrightarrow$): Suppose $L \subseteq K^n$ is a subspace. We need to show that $b = 0$. Since L is a subspace, $0 \in L$ by Lemma 7.5 **(a)**. By the definition of L, this means $A \cdot 0 = b$. But $A \cdot 0 = 0$, and therefore, $b = 0$. **Q.E.D.**

Important remark 7.2: So the solution set of a **homogeneous** system is a linear subspace of K^n, while the solution set of an inhomogeneous one is never a subspace — it does not even contain 0. It is not, however, shapeless: recall the last example of the previous chapter, whose general solution had the form *"one particular solution plus an arbitrary multiple of a fixed vector"*. We will make that observation precise once we have the notion of dimension.

More Examples of Vector Spaces

Spaces of Sequences K^∞

Example 7.4 (*Spaces of Sequences*): Let K be a field. Define

$$K^\infty := \{(a_0, a_1, a_2, \ldots, a_k, \ldots) \mid a_i \in K \ \forall i\},$$

the set of all infinite sequences with entries in K. We define $+$ and $\cdot$ on K^∞ entry-wise:

$$\begin{aligned}(a_0, a_1, \ldots, a_k, \ldots) + (b_0, b_1, \ldots, b_k, \ldots) &:= (a_0 + b_0, a_1 + b_1, \ldots, a_k + b_k, \ldots), \\ a \cdot (a_0, a_1, \ldots, a_k, \ldots) &:= (a \cdot a_0, a \cdot a_1, \ldots, a \cdot a_k, \ldots) \quad \text{for } a \in K.\end{aligned}$$

Exercise 7.5: With these operations, K^∞ is a vector space over K.

Example 7.5 (*Fibonacci Sequences as a Subspace*): Recall the Fibonacci sequences from Proposition 1.3:

$$\mathrm{Fib} = \{(a_0, a_1, \ldots, a_k, \ldots) \in \mathbb{R}^\infty \mid a_k = a_{k-1} + a_{k-2} \ \forall k \geq 2\}.$$

Then $\mathrm{Fib} \subseteq \mathbb{R}^\infty$ is a linear subspace — which is precisely what we verified by hand in the very first chapter, before we had the word for it.

More generally, if we fix $\alpha, \beta \in \mathbb{R}$, then

$$U_{\alpha,\beta} := \{(a_0, a_1, \ldots, a_k, \ldots) \in \mathbb{R}^\infty \mid a_k = \alpha a_{k-1} + \beta a_{k-2} \ \forall k \geq 2\}$$

is also a linear subspace of $\mathbb{R}^\infty$, with $\mathrm{Fib} = U_{1,1}$.

Spaces of Functions K^S

Example 7.6 (*Function Spaces*): Fix a field K and a non-empty set S. We define the function space

$$K^S := \{f \colon S \to K \mid f \text{ is a function}\}.$$

We define addition $+$ and scalar multiplication $\cdot$ on K^S as follows. Let $f, g \in K^S$ and $a \in K$. We define $(f + g) \colon S \to K$ point-wise by $x \mapsto f(x) + g(x)$, where the addition on the right is that of K, and we define $(a \cdot f) \colon S \to K$ point-wise by $x \mapsto a \cdot f(x)$.

Exercise 7.6: K^S is a vector space over K.

Example 7.7 (*Continuous Functions*): Take $K = \mathbb{R}$ and let $S := [0,1] = \{x \in \mathbb{R} \mid 0 \le x \le 1\}$. Define

$$C(S) := \{f \in \mathbb{R}^S \mid f \text{ is continuous on } [0,1]\} \subseteq \mathbb{R}^S.$$

Then $C(S) \subseteq \mathbb{R}^S$ is a linear subspace.

Exercise 7.7: Prove that $C(S)$ is a linear subspace of $\mathbb{R}^S$: check that the constant function 0 is continuous, and that a sum of two continuous functions, as well as a scalar multiple of a continuous function, is again continuous on $[0,1]$.

Polynomials $K[x]$

Example 7.8 (*Polynomial Spaces*): Let K be a field and let x be a **formal** variable — we may think of x as a *"free element of K"*. A "*polynomial*" over K (or polynomial with coefficients in K) is an expression of the form

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n, \qquad n \in \mathbb{N},$$

where $a_i \in K$ for all $0 \le i \le n$. The a_i are called the "*coefficients*" of f. The "*zero polynomial*" is the one with $a_0 = a_1 = \cdots = 0$. Two polynomials are equal if and only if all their corresponding coefficients are equal.

To add two polynomials

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n, \qquad g(x) = b_0 + b_1 x + \cdots + b_m x^m,$$

we first pad the shorter one with zeros: if $n < m$ we set $a_{n+1} = \cdots = a_m = 0$, and if $n > m$ we set $b_{m+1} = \cdots = b_n = 0$. Putting $r := \max\{n, m\}$, both are then written with the same number of terms and we may define

$$f(x) + g(x) := (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \cdots + (a_r + b_r)x^r.$$

If $c \in K$, we define

$$c \cdot f(x) := c \cdot a_0 + c \cdot a_1 x + c \cdot a_2 x^2 + \cdots + c \cdot a_n x^n.$$

Notation 7.1: We write $K[x]$ for the set of all polynomials over K.

Exercise 7.8: With the operations above, $K[x]$ is a vector space over K.

If we view polynomials as functions $K \to K$, then the polynomials over K give us a linear subspace of K^K. This second point of view is however genuinely coarser than the first, and the following warning explains why.

Important remark 7.3: For some fields (e.g., $K = \mathbb{F}_2$, or $\mathbb{F}_p$ with p prime) two **different** polynomials may give the same function. For example, over $K = \mathbb{F}_2$ the polynomials $f(x) = x$ and $g(x) = x^2$ produce the same output value for every input in $\mathbb{F}_2$. But **as polynomials** $f(x)$ and $g(x)$ are strictly different objects, since their coefficients differ.

The Degree of a Polynomial

Let $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$ be a polynomial with $a_n \neq 0$. We say that the "*degree*" of f is n, and we write $\deg(f) := n \in \mathbb{Z}_{\ge 0}$. If $a_0 = a_1 = \cdots = a_n = 0$, meaning f is the zero polynomial, we set $\deg(f) := -\infty$. Consequently, $\deg(f) = -\infty$ if and only if $f = 0$.

Remark 7.1: Every polynomial has a well-defined degree: if $f \neq 0$ and $f(x) = a_0 + a_1 x + \cdots + a_r x^r$, then

$$\deg(f) := \max\{k \in \mathbb{Z}_{\ge 0} \mid a_k \neq 0\},$$

and this maximum is taken over a non-empty finite set, hence exists.

Example 7.9 (*Polynomials of Bounded Degree*): Fix $d \in \mathbb{Z}_{\geq 0} \cup \{-\infty\}$ and define

$$K[x]_d := \{f \in K[x] \mid \deg(f) \leq d\},$$

where we agree that $-\infty \leq k$ for all $k \in \mathbb{Z}$.

Claim (*$K[x]_d$ is a subspace*): $K[x]_d \subseteq K[x]$ is a linear subspace of $K[x]$.

Later, we will see that $K[x]_d$ is a subspace of dimension $d+1$.

Question 7.1:

(a) Is the set $\{f \in K[x] \mid \deg(f) = d\}$ also a subspace of $K[x]$?

(b) Consider the subset

$$\left\{ p \in K[x]_5 \,\middle|\, p(x) = a_0 + a_1 x + \cdots + a_5 x^5,\ \sum_{i=0}^{5} a_i = 0 \right\} \subseteq K[x]_5,$$

that is, all polynomials of degree ≤ 5 for which the sum of the coefficients is 0. Is this subset a linear subspace of $K[x]_5$?

Spaces of Matrices

Example 7.10 (*Matrix Spaces*): Consider the set $\mathcal{M}_{m\times n}(K)$ of all $m \times n$ matrices with entries in K. We define addition of matrices component-wise: if $A = (a_{ij})$ and $B = (b_{ij})$, then $A + B := (a_{ij} + b_{ij})$. Similarly, for $\alpha \in K$ we define scalar multiplication component-wise by $\alpha \cdot A := (\alpha\, a_{ij})$.

Claim (*$\mathcal{M}_{m\times n}(K)$ is a vector space*): With these operations, $\mathcal{M}_{m\times n}(K)$ is a vector space over K. Its zero vector is the $m \times n$ matrix all of whose entries are 0.

AI-Note: The lecture notes index sequences from 1, writing $K^\infty = \{(a_1, a_2, \dots)\}$ and Fib with the recursion starting at $k \geq 3$. This transcript indexes from 0 instead, so that Fib here is literally the same object as the one built in Proposition 1.3; nothing else changes. The notes also use W for a subspace, where this transcript writes U, matching the later chapters. The proofs of Lemma 7.2 **(e)**, **(f)** and **(h)** are left as exercises in the notes and are supplied here; the exercises marked as such above are the ones the notes leave open.

Span and Linear Combinations (Chapter 8)

Preparation

Let V be a vector space over a field K. Let $S \subseteq V$ be a subset (which is not necessarily a linear subspace). We aim to answer two fundamental and closely related questions:

(a) What is the **smallest** linear subspace of V that contains S?

(b) What structure do we obtain if we consider the subset of V formed by taking all possible linear combinations of elements from S?

To formalize the concept of a *"smallest"* subspace, we must first prove that the intersection of any collection of subspaces remains a subspace.

Lemma 8.1 (*Intersections of Subspaces*): Let V be a vector space over K. Let $\{W_i\}_{i\in I}$ be a collection of linear subspaces $W_i \subseteq V$ of V, indexed by an arbitrary set I of indices. Then their intersection

$$W := \bigcap_{i\in I} W_i$$

is a linear subspace of V.

Proof:

First, since every W_i is a subspace, the zero vector 0_V is contained in every W_i by Lemma 7.5 **(a)**. Therefore, $0_V \in \bigcap_{i\in I} W_i = W$.

Next, let $a_1, a_2 \in K$ and $w_1, w_2 \in W$. We must show that

$$a_1w_1 + a_2w_2 \in W. \tag{8.1}$$

Choose an arbitrary index $j \in I$. Since $w_1, w_2 \in \bigcap_{i\in I} W_i$, we in particular have $w_1, w_2 \in W_j$. Because $W_j \subseteq V$ is itself a linear subspace, Lemma 7.5 **(b)** applies to it and gives

$$a_1w_1 + a_2w_2 \in W_j.$$

We have proved that $a_1w_1 + a_2w_2 \in W_j$ holds for **every** $j \in I$, and therefore,

$$a_1w_1 + a_2w_2 \in \bigcap_{i\in I} W_i = W,$$

which is what we wanted to prove in (8.1). By Lemma 7.5, W is a linear subspace. **Q.E.D.**

The Span

Definition 8.2 (*Span*): Let $S \subseteq V$ be a subset. Let

$$\mathcal{N} := \{W \mid W \subseteq V \text{ is a linear subspace, and } S \subseteq W\}$$

be the collection of all linear subspaces of V that contain S. We define the "*span*" of S (in V) as

$$\operatorname{Sp}(S) := \bigcap_{W\in\mathcal{N}} W.$$

Definition 8.3 (*Linear Combination*): Let $n \in \mathbb{N}$, let $a_1, \ldots, a_n \in K$ be scalars and let $v_1, \ldots, v_n \in V$ be vectors. A "*linear combination*" of the vectors $v_1, \ldots, v_n$ with coefficients $a_1, \ldots, a_n$ is the vector

$$a_1v_1 + \cdots + a_nv_n \in V \quad \left(\text{or equivalently, } \sum_{i=1}^{n} a_iv_i\right).$$

Important remark 8.1: In this course, we consider **only finitely many elements** in the sums of linear combinations. Infinite sums require topological notions of limits and convergence, which fall outside the scope of general vector spaces.

The span deserves its name: it is a subspace, and it is the smallest one containing S.

Lemma 8.4 (*The Span is the Smallest Subspace Containing S*): Let $S \subseteq V$ be a subset. Then:

(a) $\mathrm{Sp}(S) \subseteq V$ is a linear subspace.

(b) If $W \subseteq V$ is a linear subspace with $S \subseteq W$, then $\mathrm{Sp}(S) \subseteq W$.

In other words, $\mathrm{Sp}(S)$ is a linear subspace of V, and moreover, among those subspaces that contain S, it is the smallest one.

Proof:

Statement **(a)** is immediate from Lemma 8.1: by Definition 8.2, $\mathrm{Sp}(S)$ is an intersection of a collection of linear subspaces of V, and any such intersection is again a linear subspace. Note that the collection $\mathcal{N}$ is never empty, since V itself is a subspace of V containing S.

For **(b)**, let $W \subseteq V$ be a linear subspace with $S \subseteq W$. Then W is one of the members of the collection $\mathcal{N}$ over which the intersection in Definition 8.2 is taken. An intersection is contained in each of the sets being intersected, and therefore,

$$\mathrm{Sp}(S) = \bigcap_{W' \in \mathcal{N}} W' \subseteq W.$$

Q.E.D.

The next lemma answers the second of our two opening questions, and shows that the two answers agree.

Lemma 8.5 (*Span as the Set of Linear Combinations*): Let $\emptyset \neq S \subseteq V$. Then $\mathrm{Sp}(S)$ is exactly the set of all possible linear combinations of vectors from S:

$$\begin{aligned}\mathrm{Sp}(S) &= \{a_1 v_1 + \cdots + a_n v_n \mid n \in \mathbb{N},\ a_i \in K,\ v_i \in S \text{ for all } 1 \leq i \leq n\} \\ &= \text{the subset of } V \text{ obtained by taking all possible linear combinations of vectors from } S.\end{aligned}$$

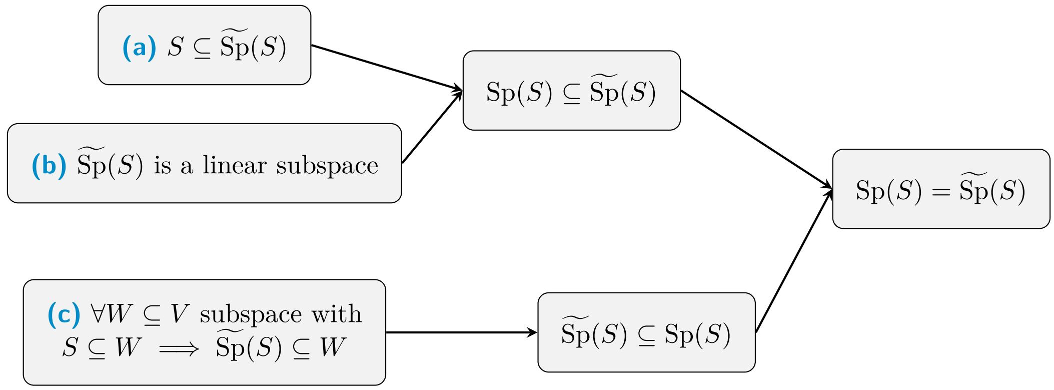
Remark 8.1: The set S itself can be finite or infinite. Regardless of the size of S, each individual linear combination uses only a finite number of elements from S.

Proof:

Let us denote the set of all linear combinations by

$$\widetilde{\mathrm{Sp}}(S) := \{a_1 v_1 + \cdots + a_n v_n \mid n \in \mathbb{N},\ a_i \in K,\ v_i \in S \text{ for all } 1 \leq i \leq n\}.$$

Before diving into the algebra, let us visualize the logical strategy of the proof. We will establish three facts, which together force $\mathrm{Sp}(S) = \widetilde{\mathrm{Sp}}(S)$:



Note that **(a)** and **(b)** imply that $\mathrm{Sp}(S) \subseteq \widetilde{\mathrm{Sp}}(S)$: together they say that $\widetilde{\mathrm{Sp}}(S)$ is one of the subspaces containing S, so Lemma 8.4 **(b)** applies to it. Furthermore, **(c)** implies that $\widetilde{\mathrm{Sp}}(S) \subseteq \mathrm{Sp}(S)$, because $\mathrm{Sp}(S)$ is itself a subspace W containing S by Lemma 8.4. Let us now prove the three statements.

(a) Let $v \in S$. Then we can write $v = 1 \cdot v \in \widetilde{\mathrm{Sp}}(S)$, which is a linear combination with $n = 1$. Thus $S \subseteq \widetilde{\mathrm{Sp}}(S)$.

(b) We verify the criterion of Lemma 7.5. First, $0_V \in \widetilde{\mathrm{Sp}}(S)$: since $S \neq \emptyset$ by assumption, we may pick some $u \in S$, and then $0_V = 0 \cdot u \in \widetilde{\mathrm{Sp}}(S)$ by Lemma 7.2 **(c)**.
Second, let $v, w \in \widetilde{\mathrm{Sp}}(S)$ and $\alpha, \beta \in K$. Write

$$v = a_1 v_1 + \cdots + a_n v_n, \qquad w = b_1 w_1 + \cdots + b_m w_m,$$

where $n, m \in \mathbb{N}$, $a_i, b_j \in K$ and $v_i, w_j \in S$ for all $1 \le i \le n$, $1 \le j \le m$. Then

$$\alpha v + \beta w = \alpha a_1 v_1 + \cdots + \alpha a_n v_n + \beta b_1 w_1 + \cdots + \beta b_m w_m,$$

which is a linear combination of the vectors $v_1, \ldots, v_n, w_1, \ldots, w_m \in S$ with coefficients $\alpha a_1, \ldots, \alpha a_n, \beta b_1, \ldots, \beta b_m \in K$. Hence $\alpha v + \beta w \in \widetilde{\mathrm{Sp}}(S)$, and $\widetilde{\mathrm{Sp}}(S)$ is a linear subspace.

(c) Let $W \subseteq V$ be a linear subspace such that $S \subseteq W$, and let $v \in \widetilde{\mathrm{Sp}}(S)$. We need to show that $v \in W$. Indeed, write $v = a_1 v_1 + \cdots + a_n v_n$ for some $n \in \mathbb{N}$, $v_i \in S$ and $a_i \in K$. By assumption $S \subseteq W$, and hence $v_i \in W$ for every i. Since W is a linear subspace it is closed under scalar multiplication and under finite sums, and therefore,

$$v = a_1 v_1 + \cdots + a_n v_n \in W.$$

Thus $\widetilde{\mathrm{Sp}}(S) \subseteq W$.

Q.E.D.

Summary 8.1: We now have two descriptions of the same object, and we may use whichever is more convenient:

$$\mathrm{Sp}(S) = \begin{cases} \bigcap_{W \in \mathcal{N}} W & \text{(by definition: the smallest subspace containing } S\text{)}, \\ \text{all linear combinations of vectors from } S & \text{(by Lemma 8.5)}. \end{cases}$$

They can be regarded as equivalent definitions of the span.

Notation 8.1: Let $n \in \mathbb{N}$ and $v_1, \ldots, v_n \in V$. We write

$$\mathrm{Sp}(v_1, \ldots, v_n) := \mathrm{Sp}(\{v_1, \ldots, v_n\}),$$

dropping the set brackets when the spanning set is given as an explicit finite list.

Lemma 8.6 (*The Span of a Finite List*): Let $n \in \mathbb{N}$ and $v_1, \ldots, v_n \in V$. Then

$$\mathrm{Sp}(v_1, \ldots, v_n) = \left\{ \sum_{i=1}^{n} \alpha_i v_i \;\middle|\; \alpha_i \in K \text{ for all } 1 \le i \le n \right\}.$$

Remark 8.2: Note that here the number of summands is fixed at n, whereas in Lemma 8.5 it was allowed to vary and the vectors were allowed to repeat. The point is that repetitions and omissions can always be absorbed into the coefficients: a repeated vector has its coefficients added together, and an omitted one is given the coefficient 0.

Exercise 8.1: Prove Lemma 8.6, using Lemma 8.5 together with the observation of the remark above about repetitions and omissions.

Remark 8.3: The span of the empty set is the zero subspace:

$$\operatorname{Sp}(\emptyset) = \{0_V\}.$$

This is not a convention but a consequence of Definition 8.2. Every subspace $W \subseteq V$ satisfies $\emptyset \subseteq W$, so the collection $\mathcal{N}$ consists of **all** subspaces of V; and since $\{0_V\}$ is one of them, the intersection of all of them is contained in $\{0_V\}$, while 0_V lies in every subspace. Using the other description, in terms of linear combinations, we just think of the sum of an empty collection of elements as 0.

Generating Sets and Finite-Dimensionality

Definition 8.7 (*Generating Set*): Let V be a vector space over K and let $S \subseteq V$. We say that S "*generates*" (or "*spans*") V if

$$\operatorname{Sp}(S) = V.$$

In this case S is called a "*generating set*" of V. More generally, if $\operatorname{Sp}(S) = U$, where $U \subseteq V$ is a subspace of V, we say that S generates (or spans) U.

Definition 8.8 (*Finite-Dimensional Vector Space*): A vector space V is called "*finite-dimensional*" if there exists a **finite** set $S \subseteq V$ such that $\operatorname{Sp}(S) = V$. Otherwise, V is called "*infinite-dimensional*".

Important remark 8.2: Note that this definition says nothing yet about **how many** vectors a generating set must contain — only that some finite generating set exists. Attaching an actual number to a finite-dimensional space requires the notion of a basis, and is the business of the next few chapters.

The Subspaces of the Plane

Before turning to computations, let us use the span to list **all** linear subspaces of a small vector space. Take $V := \mathbb{R}^2$ over $K := \mathbb{R}$.

Example 8.1 (*Lines Through the Origin*):

- The zero subspace: $\{0_{\mathbb{R}^2}\} = \{(0,0)\} = \operatorname{Sp}(\emptyset) = \operatorname{Sp}(0_{\mathbb{R}^2})$.
- Let $0 \neq v = (a,b) \in \mathbb{R}^2$. Then
  $$\operatorname{Sp}(v) = \{\alpha \cdot v \mid \alpha \in \mathbb{R}\} = \{(\alpha a, \alpha b) \mid \alpha \in \mathbb{R}\}.$$
  Geometrically, this is the line L_v that passes through $(0,0)$ and through $v = (a,b)$, as drawn in Figure 8.1.
- Let $0 \neq v, w \in \mathbb{R}^2$. Then
  $$\operatorname{Sp}(v) = \operatorname{Sp}(w) \iff \exists\, \alpha \neq 0 \text{ such that } w = \alpha \cdot v \iff L_v = L_w.$$

Exercise 8.2: Prove the two equivalences stated on Page 55: for $0 \neq v, w \in \mathbb{R}^2$, show that $\operatorname{Sp}(v) = \operatorname{Sp}(w)$ if and only if $w = \alpha \cdot v$ for some $\alpha \neq 0$, and that this holds if and only if $L_v = L_w$.

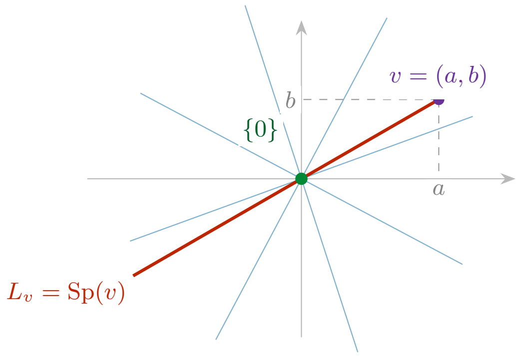


Figure 8.1: The linear subspaces of $\mathbb{R}^2$. There are exactly three kinds: the single point $\{0\}$ (green), the lines through the origin $L_v = \mathrm{Sp}(v)$ for $0 \neq v$ (one of them highlighted, a few others faint), and $\mathbb{R}^2$ itself. Two non-zero vectors span the same line precisely when each is a non-zero multiple of the other; as soon as w lies off L_v, the pair v, w already spans the whole plane.

Exercise 8.3 (*Important*): Let $W \subseteq \mathbb{R}^2$ be a linear subspace. Suppose $W = \mathrm{Sp}(v, w)$, where $v \neq 0$ and $w \notin \mathrm{Sp}(v)$. Show that $W = \mathbb{R}^2$.

Summary 8.2: $\mathbb{R}^2$ has the following linear subspaces, and no others:

(a) $\{0_{\mathbb{R}^2}\}$;
(b) $L_v = \mathrm{Sp}(v)$, where $0 \neq v \in \mathbb{R}^2$;
(c) $\mathbb{R}^2$ itself.

Checking for Span: The Bridge to Computations

Up to this point, our description of the span has been abstract. We now ask a practical, computational question.

Question 8.1: Let $v_1, \dots, v_n \in K^m$ and $w \in K^m$. How can we determine whether or not $w \in \mathrm{Sp}(v_1, \dots, v_n)$?

For example, take $v_1 := (1, 3)$ and $v_2 := (7, 73) \in \mathbb{R}^2$, together with $w := (11, 137)$. Here w is indeed in the span, because

$$-3 \cdot (1, 3) + 2 \cdot (7, 73) = (-3 + 14,\ -9 + 146) = (11, 137).$$

But how does one **find** the coefficients -3 and 2? Do they always exist? Are they unique?

Preparation: a matrix product is a linear combination of columns

Let

$$A = \begin{pmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & & \vdots \\ a_{m,1} & \dots & a_{m,n} \end{pmatrix} \in \mathcal{M}_{m \times n}(K), \qquad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in K^n,$$

and let v_j denote the j-th column of A, that is

$$v_1 := \begin{pmatrix} a_{1,1} \\ \vdots \\ a_{m,1} \end{pmatrix}, \qquad v_2 := \begin{pmatrix} a_{1,2} \\ \vdots \\ a_{m,2} \end{pmatrix}, \qquad \dots \qquad v_n := \begin{pmatrix} a_{1,n} \\ \vdots \\ a_{m,n} \end{pmatrix} \quad \in K^m,$$

so that A may be written in column-block form as

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{pmatrix}.$$

Carrying out the product $A \cdot x$ of Definition 6.7 and regrouping the terms by columns instead of by rows, we obtain

$$A \cdot x = x_1 \begin{pmatrix} | \\ v_1 \\ | \end{pmatrix} + x_2 \begin{pmatrix} | \\ v_2 \\ | \end{pmatrix} + \dots + x_n \begin{pmatrix} | \\ v_n \\ | \end{pmatrix} = \sum_{i=1}^{n} x_i \cdot v_i. \tag{8.2}$$

This reveals one of the most useful insights in all of linear algebra: **a matrix-vector product $A \cdot x$ is simply a linear combination of the columns of A, weighted by the entries of x.** Conversely, we can write any linear combination as a multiplication of a matrix A by a vector x.

The consequence for membership in a span

So the problem *"find coefficients $x_1, \dots, x_n \in K$ such that $x_1 v_1 + \dots + x_n v_n = w$"* is the very same problem as *"find a solution $x \in K^n$ of the system $A \cdot x = w$"*, where A is the matrix whose columns are $v_1, \dots, v_n$. We record this as a theorem.

Theorem 8.9 (*Span and Consistency of a Linear System*): Let $w, v_1, \dots, v_n \in K^m$. Let $A \in \mathcal{M}_{m \times n}(K)$ be the matrix whose columns are exactly the vectors $v_1, \dots, v_n$. Then the target vector w lies in the span of $v_1, \dots, v_n$ if and only if the system of linear equations $A \cdot x = w$ has at least one solution $x = (x_1, \dots, x_n)^\mathsf{T} \in K^n$:

$$A \cdot x = w \text{ has a solution } x \in K^n \iff x_1 v_1 + \dots + x_n v_n = w \iff w \in \mathrm{Sp}(v_1, \dots, v_n).$$

Proof:

The first equivalence is exactly (8.2), read from right to left and from left to right. The second is Lemma 8.6, which says that the elements of $\mathrm{Sp}(v_1, \dots, v_n)$ are precisely the vectors of the form $\sum_{i=1}^{n} x_i v_i$ with $x_i \in K$. **Q.E.D.**

Remark 8.4 (*How to check this in practice*): Because of this equivalence, whenever you are asked *"is w in the span of these vectors?"*, you do not need to guess the coefficients. You simply construct the augmented matrix $(A \mid w)$ and perform Gauss elimination, bringing it to row-reduced echelon form as in Theorem 6.14. If the resulting system yields a contradiction, the vector is **not** in the span. If the system is solvable, the solutions $x_1, \dots, x_n$ are exactly the coefficients needed to build w — and by Theorem 6.10, row reduction has not changed which coefficients those are. Whether the coefficients are **unique** is a separate question, and it is precisely the question of linear independence, which occupies the next chapter.

Some Spaces and How to Generate Them

To conclude, let us look at some standard vector spaces and identify natural sets of *"building block"* vectors that span them entirely.

Example 8.2 (*The Coordinate Space K^n*): Let $V := K^n$ and let

$$e_1 := (1, 0, \dots, 0), \quad e_2 := (0, 1, 0, \dots, 0), \quad \dots, \quad e_n := (0, \dots, 0, 1)$$

be the standard vectors. Every vector $(x_1, \dots, x_n) \in K^n$ can be written as $x_1 e_1 + \dots + x_n e_n$, and therefore,

$$K^n = \mathrm{Sp}(e_1, \dots, e_n).$$

Exercise 8.4: Check the identity $K^n = \mathrm{Sp}(e_1, \dots, e_n)$ in detail.

Example 8.3 (*Polynomials*): Let $W := K[x]_d$ be the space of polynomials over K of degree $\leq d$, introduced as a subspace of $K[x]$ on Page 51. We can generate this space using the standard monomials:

$$W = \mathrm{Sp}(1, x, x^2, \dots, x^d).$$

Let $V := K[x]$ be the space of **all** polynomials over K. Generating this space requires an infinite set, since there is no maximum degree:

$$V = \mathrm{Sp}\big(\{1, x, x^2, \dots, x^n, \dots\}\big).$$

Example 8.4 (*Spaces of Matrices*): Let $\mathcal{M} := \mathcal{M}_{m\times n}(K)$ be the space of all $m \times n$ matrices with entries in K. For $1 \leq k \leq m$ and $1 \leq \ell \leq n$, let $E_{k\ell} \in \mathcal{M}$ be the matrix that has all its entries 0 except the entry in position (k, ℓ), which is 1. In other words, $E_{k\ell} = (a_{ij})$, where

$$a_{ij} = \begin{cases} 1 & \text{if } i = k \text{ and } j = \ell, \\ 0 & \text{otherwise,} \end{cases} \qquad \text{so} \qquad E_{k\ell} = \begin{array}{cl} \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \dots & 1 & \dots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix} & \begin{array}{l} \\ \\ \leftarrow \text{row } k \\ \\ \\ \end{array} \end{array}$$

with the 1 sitting in column ℓ. There are exactly $m \cdot n$ such matrices.

For any matrix $A = (a_{ij}) \in \mathcal{M}_{m\times n}(K)$, we can pull it apart into a linear combination in which the scalar weights are simply the matrix entries themselves:

$$A = \sum_{k=1}^{m} \sum_{\ell=1}^{n} a_{k\ell} \cdot E_{k\ell},$$

and therefore,

$$\mathcal{M}_{m\times n}(K) = \mathrm{Sp}\big(\{E_{k\ell} \mid 1 \leq k \leq m,\ 1 \leq \ell \leq n\}\big).$$

For instance, in $\mathcal{M}_{2\times 2}(K)$ any matrix can be constructed like this:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

so that a matrix may be viewed as a *“vector”* built out of simpler matrices.

Exercise 8.5: Show that K^n, $\mathcal{M}_{m\times n}(K)$ and $K[x]_d$ are finite-dimensional in the sense of Definition 8.8. What about $K[x]$? Prove your answer.

AI-Note: The lecture notes write $\mathrm{Sp}(S)$ throughout, which this transcript keeps. The logical roadmap diagram inside the proof of Lemma 8.5, Figure 8.1, and the closing sentence about uniqueness in the practical remark on Page 57 were added here; everything else follows the notes. Lemma 8.4 is stated in the notes with the proof left as *“check the details”*, and is proved in full above.

Linear Independence (Chapter 9)

Linear Independence of a Finite List

Let V be a vector space over a field K. In the previous chapter, we studied how a set of vectors can *span* a subspace. We now ask whether a spanning set contains *"redundant"* vectors. This leads us to the core algebraic concept of linear independence.

Definition 9.1 (*Linear Independence*): Let $v_1, \dots, v_n$ be a finite list of vectors in V. We say that the list $v_1, \dots, v_n$ is **linearly independent** if the only linear combination that equals the zero vector 0_V is the trivial combination (where all coefficients are zero).

In formal notation, the vectors are linearly independent if, for all $a_1, \dots, a_n \in K$:

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n = 0_V \implies a_1 = 0, a_2 = 0, \dots, a_n = 0$$

Remark 9.1 (*The Empty Set*): By mathematical convention, the empty list of vectors $\emptyset$ is defined to be linearly independent. But *why* does this make sense?

Recall from Chapter 8 that we defined the span of the empty set as the zero subspace: $\mathrm{Sp}(\emptyset) = \{0_V\}$. If the empty set were to be linearly *dependent*, there would need to exist a linear combination of its elements that non-trivially yields 0_V. However, since $\emptyset$ contains literally no vectors, we cannot even form a linear combination with a non-zero coefficient! Because the condition for dependence is impossible to satisfy, the empty set is vacuously, and logically, independent.

Before looking at examples, we can immediately deduce several structural facts directly from the definition.

Lemma 9.2 (*Basic Properties of Linear Independence*): Let V be a vector space. The following properties hold for any finite list of vectors:

- **(a)** The zero vector 0_V can always be written as a trivial linear combination of any finite list of vectors: $0 \cdot v_1 + \cdots + 0 \cdot v_n = 0_V$.
- **(b)** The order of the vectors in a list does not affect their linear independence.
- **(c)** If a list $v_1, \dots, v_n$ contains two identical vectors (i.e., $v_k = v_l$ for some $k \neq l$), then the list is NOT linearly independent.
- **(d)** If the zero vector 0_V is an element of the list, then the list is NOT linearly independent.

Proof:

We address each property:

- **(a)** Follows trivially from the vector space axioms (specifically $0 \cdot v = 0_V$ for all $v \in V$).
- **(b)** Follows trivially from the commutativity of vector addition in V.
- **(c)** Assume $v_k = v_l$ for some $1 \leq k < l \leq n$. We can construct a specific linear combination where the coefficient for v_k is 1, the coefficient for v_l is -1, and all other coefficients are 0:

  $$0 \cdot v_1 + \cdots + 1 \cdot v_k + \cdots + (-1) \cdot v_l + \cdots + 0 \cdot v_n = v_k - v_l = 0_V$$

  Because we found a linear combination equal to 0_V where not all coefficients are zero (specifically, $a_k = 1 \neq 0$), the list fails the definition and is therefore not linearly independent.
- **(d)** Suppose $v_1 = 0_V$. We can assign a non-zero coefficient (e.g., 1) to v_1 and zero to all other vectors:

  $$1 \cdot 0_V + 0 \cdot v_2 + \cdots + 0 \cdot v_n = 0_V + 0_V + \cdots + 0_V = 0_V$$

  Again, since $a_1 = 1 \neq 0$, we have a non-trivial combination yielding the zero vector. The list is not linearly independent.

Q.E.D.

Property **(b)** above has a convenient consequence, which we will use constantly from here on.

Notation 9.1 (*From Lists to Sets*): Since the order of the vectors is irrelevant, whenever the list $v_1, \dots, v_n$ has **no repetitions** we can talk about the linear independence of the **set** $\{v_1, \dots, v_n\}$, and we shall do so freely. The proviso matters, because a set silently discards repetitions: $\{v, v\} = \{v\}$, whereas the two-term list v, v is never independent by Lemma 9.2 **(c)**. We return to this distinction on Page 61.

Examples of Linear Independence

Let us apply this definition to various vector spaces.

Example 9.1 (*In $\mathbb{R}^2$*): Consider the vectors $v_1 := \binom{1}{0}$ and $v_2 := \binom{0}{1}$ in $\mathbb{R}^2$.
Assume $a_1 v_1 + a_2 v_2 = 0_{\mathbb{R}^2}$. This means:

$$a_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus, $a_1 = 0$ and $a_2 = 0$. The vectors $\{v_1, v_2\}$ are linearly independent.

Example 9.2 (*Linearly dependent vectors in $\mathbb{R}^2$*): Consider $v_1 := \binom{1}{1}$ and $v_2 := \binom{2}{2}$.
We can easily spot that $2v_1 - v_2 = 0_{\mathbb{R}^2}$. Because we have coefficients $a_1 = 2$ and $a_2 = -1$ (which are non-zero) that yield the zero vector, these vectors are NOT linearly independent.

Example 9.3 (*The Standard Basis in K^n*): Consider the standard basis vectors in K^n:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots \quad , \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

Setting $a_1 e_1 + a_2 e_2 + \dots + a_n e_n = 0_{K^n}$ yields the column vector $(a_1, a_2, \dots, a_n)^\mathsf{T} = (0, 0, \dots, 0)^\mathsf{T}$. This immediately forces $a_i = 0$ for all i. Thus, the set $\{e_1, e_2, \dots, e_n\}$ is always linearly independent.

Example 9.4 (*Single Vectors*): Let V be a vector space and $v \in V$. The single-element set $\{v\}$ is linearly independent if and only if $v \neq 0_V$. (If $v = 0_V$, we know from Lemma 9.2 that it is not independent. If $v \neq 0_V$, then $a \cdot v = 0_V \implies a = 0$).

Arbitrary Families and Linear Dependence

As we advance in linear algebra, we frequently need to deal with infinite collections of vectors (such as the set of all polynomials). We must rigorously extend our definition to handle this. Before we do, we should say what a family actually is.

Question 9.1: What do we mean by a family of vectors in V, written $\mathcal{F} = \{v_i\}_{i \in I}$ and indexed by a set I?

Definition 9.3 (*Family*): The meaning of this is, by definition, that we have a set I (called the **set of indices**), and that we are given a function

$$\mathcal{F} : I \longrightarrow V, \qquad I \ni i \longmapsto v_i \in V.$$

So $\mathcal{F}$ assigns to each index $i \in I$ an element $v_i \in V$.

Example 9.5 (*Families and Their Index Sets*):

(1) $I = \{1, \ldots, n\}$ is a finite set. Then a family $\mathcal{F} = \{v_i\}_{i \in I}$ can be just written as $v_1, \ldots, v_n$; the finite lists of the previous subsections are thus a special case.

(2) $I = \mathbb{Z}_{\geq 1} = \{1, 2, 3, \ldots\}$. Then $\mathcal{F} = \{v_i\}_{i \in \mathbb{Z}_{\geq 1}}$ can be written as $v_1, v_2, v_3, \ldots$.

(3) We can have families $\mathcal{F} = \{v_i\}_{i \in I}$ indexed by other (in fact, any) sets I. For example $I = \mathbb{R}$, and so on.

Remark 9.2 (*Families of Elements of Any Set*): There is nothing special about V being a vector space as far as the concept of families goes. If X is a set and I is another set, we can talk about families $\mathcal{F} = \{x_i\}_{i \in I}$ of elements from X, indexed (or parametrized) by $i \in I$. Again, this means a function

$$\mathcal{F} \colon I \longrightarrow X, \qquad I \ni i \longmapsto x_i \in X.$$

Remark 9.3 (*Repetitions and Injectivity*): If this function $\mathcal{F}$ is **not** injective, then the family $\mathcal{F} = \{x_i\}_{i \in I}$ will have **repetitions**, i.e., there exist $i_1, i_2 \in I$ with $i_1 \neq i_2$ such that

$$x_{i_1} = x_{i_2}.$$

Conversely, an injective $\mathcal{F}$ is precisely a family without repetitions. This is the one feature of families that interacts with linear independence.

Back to linear independence.

Definition 9.4 (*Linear Independence of an Arbitrary Family*): Let $\mathcal{F} = \{v_i\}_{i \in I}$ be a *family of vectors* in V, indexed by an arbitrary set I (which may be infinite). We say that the family $\mathcal{F}$ is linearly independent if **every finite subset** of $\mathcal{F}$ is linearly independent.

Spelled out in terms of the indexing, this says: for every $n \in \mathbb{N}$ and any sequence of **distinct** indices $i_1, \ldots, i_n \in I$ (i.e., $i_k \neq i_\ell$ for all $k \neq \ell$), the list of vectors

$$v_{i_1},\ v_{i_2},\ \ldots,\ v_{i_n}$$

is linearly independent in the sense of Definition 9.1.

Remark 9.4 (*Repetitions Rule Out Independence*): Of course, if in $\mathcal{F}$ there is a repetition of some vector (i.e., there exist $i, j \in I$ with $i \neq j$ such that $v_i = v_j$), then the family $\mathcal{F}$ can **not** be linearly independent. Indeed, i, j is a choice of two distinct indices, so Definition 9.4 demands that the two-term list v_i, v_j be independent; but that list consists of the same vector twice, and is not independent by Lemma 9.2 **(c)**.

Since a family carries more information than the set of its values, it is worth asking how much of that extra information linear independence can actually see.

Question 9.2: What is the difference between a family $\mathcal{F} = \{v_i\}_{i \in I}$ and the set $S = \{v_i \mid i \in I\}$?

Answer 9.1:

(a) The set S does **not** contain repetitions (in case there are any in $\mathcal{F}$).

(b) The set S does **not** *"remember"* the indexing $i \in I$ of the elements v_i of S.

Now, **(b)** has no effect on linear independence, since by Lemma 9.2 **(b)** the order and the names of the indices are irrelevant. The effect of **(a)** is clear and has been discussed on Page 61: a repetition makes the family dependent while leaving the set unchanged.

Sometimes we consider a set of vectors $\emptyset \neq S \subseteq V$ directly, and it costs nothing to define independence for it.

Definition 9.5 (*Linear Independence of a Set*): Let $\emptyset \neq S \subseteq V$ be a set of vectors. We say that S is **linearly independent** if every finite list $v_1, \dots, v_n \in S$, $n \geq 1$, that has no repetitions (i.e., $v_i \neq v_j$ for all $i \neq j$) is linearly independent.

Example 9.6 (*Families of Polynomials*): In the space of bounded polynomials $K[x]_d$, the set of standard monomials $\{1, x, x^2, \dots, x^d\}$ is linearly independent. This remains true even for the **infinite family** of all monomials $\mathcal{F} = \{1, x, x^2, \dots\} \subseteq K[x]$. Any finite subset of this family will just be a finite collection of distinct monomials, which can only sum to the zero polynomial if all their coefficients are strictly zero.

Exercise 9.1: For each $k \geq 0$ let $e_k \in K^\infty$ be the sequence whose k-th entry is 1 and all of whose other entries are 0, where K^∞ is the space of sequences from Page 49. Show that the set

$$\{e_0, e_1, e_2, \dots, e_k, \dots\} \subseteq K^\infty$$

is linearly independent. (Note that this set does **not** span K^∞: a linear combination is a finite sum, so it can only produce sequences with finitely many non-zero entries.)

We now formally define the opposite of linear independence, specifically focusing on how it applies to an arbitrary family.

Definition 9.6 (*Linear Dependence*): A family $\mathcal{F} = \{v_i\}_{i \in I}$ of vectors in V is called **linearly dependent** if it is NOT linearly independent.

In other words, this means there exists some finite integer $n \in \mathbb{N}$, a selection of n distinct indices $i_1, \dots, i_n \in I$, and a set of scalars $a_1, \dots, a_n \in K$ **where not all a_j are zero**, such that:

$$a_1 v_{i_1} + a_2 v_{i_2} + \cdots + a_n v_{i_n} = 0_V$$

Such a combination, in which at least one coefficient is non-zero, is called a **non-trivial linear combination** of $v_{i_1}, \dots, v_{i_n}$. In other words, there is a non-trivial linear combination of elements from $\mathcal{F}$ that gives 0_V.

Remark 9.5 (*Two Ways to Be Dependent*): Let $\mathcal{F} = \{v_i\}_{i \in I}$ be a family of vectors in V.

(1) If there exists an $i \in I$ such that $v_i = 0_V$, then $\mathcal{F}$ is linearly dependent. Just take $n = 1$, the single index $i_1 := i$ with $v_{i_1} = 0_V$, and any $a_1 \neq 0$; then
$$a_1 \cdot v_{i_1} = a_1 \cdot 0_V = 0_V.$$

(2) If $\mathcal{F}$ has a repetition of some vector, then $\mathcal{F}$ is linearly dependent. This is the remark on Page 61, restated in the language of dependence.

Remark 9.6 (*Dependent Sets*): We can define a linearly dependent set S in a similar way to the above: $S \subseteq V$ is linearly dependent if it is not linearly independent in the sense of Definition 9.5, which spelled out gives the same description as in Definition 9.6, with the n distinct indices replaced by n pairwise distinct vectors $v_1, \dots, v_n \in S$.

Back to Linear Independence: Redundancy

This definition leads to an incredibly useful geometric intuition: a set of vectors is linearly dependent if and only if at least one vector is *"redundant"* — meaning it can be built entirely out of the other vectors in the set.

Algebraic Definition
$\{v_1, \dots, v_n\}$ is
Linearly Dependent
$(\sum a_i v_i = 0_V,\ a_i \neq 0)$

$\xleftrightarrow{\text{Proposition 9.7}}$

Geometric Redundancy
At least one vector v_k
is a linear combination
of the others.
$(v_k \in \mathrm{Sp}(v_1, \dots, \widehat{v_k}, \dots, v_n))$

Proposition 9.7 (*Equivalence of Dependence and Redundancy*)**:** A list of vectors $v_1, \ldots, v_n$ is linearly dependent if and only if at least one vector in the list can be written as a linear combination of the other vectors.

Proof:

We must prove both directions.

Direction 1 ($\implies$): Assume the list $v_1, \ldots, v_n$ is linearly dependent. By definition, there exist scalars $a_1, \ldots, a_n \in K$, not all zero, such that:

$$a_1v_1 + a_2v_2 + \cdots + a_nv_n = 0_V$$

Since not all coefficients are zero, we can choose an index k such that $a_k \neq 0$. We can isolate a_kv_k on one side of the equation by subtracting all other terms:

$$a_kv_k = -\sum_{\substack{i=1 \\ i\neq k}}^{n} a_iv_i$$

Because $a_k \neq 0$, its multiplicative inverse a_k^{-1} exists in the field K. Multiplying both sides by a_k^{-1} yields:

$$v_k = \sum_{\substack{i=1 \\ i\neq k}}^{n} \left(-\frac{a_i}{a_k}\right) v_i$$

Thus, v_k is a linear combination of the other vectors.

Direction 2 ($\impliedby$): Assume that at least one vector, say v_k, can be written as a linear combination of the others. This means there exist scalars b_i such that:

$$v_k = \sum_{\substack{i=1 \\ i\neq k}}^{n} b_iv_i$$

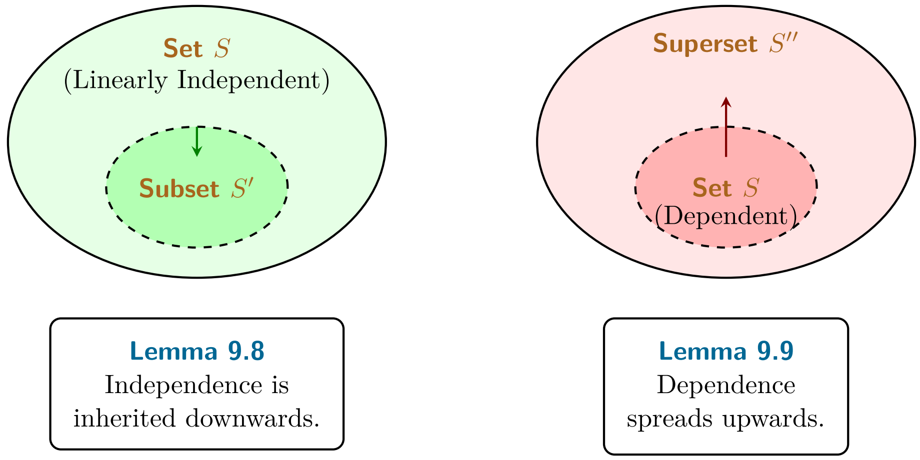
We can subtract v_k from both sides to obtain:

$$0_V = -1 \cdot v_k + \sum_{\substack{i=1 \\ i\neq k}}^{n} b_iv_i$$

This equation is a linear combination of the full list $v_1, \ldots, v_n$ that equals 0_V. Furthermore, the coefficient of v_k is -1, which is non-zero. Since we have found a non-trivial linear combination equaling 0_V, the list is linearly dependent by definition. **Q.E.D.**

Inheritance in Subsets and Supersets

How does linear independence behave when we add or remove vectors from a set? There is a strict, logical hierarchy to how these properties are inherited.



Lemma 9.8 (*Subsets of Independent Sets*): Let $S \subseteq V$ be a linearly independent set. If $S' \subseteq S$ is a subset, then S' is also linearly independent.

Proof:

We prove this by contradiction. Suppose S' is NOT linearly independent. Then S' must be linearly dependent. By definition, there exist vectors $v_1, \dots, v_k \in S'$ and scalars $a_1, \dots, a_k \in K$, not all zero, such that:

$$a_1 v_1 + \dots + a_k v_k = 0_V$$

However, since $S' \subseteq S$, every vector v_i is also an element of the larger set S. Therefore, this exact same equation represents a non-trivial linear combination of elements in S that equals 0_V. This would mean S is linearly dependent, which directly contradicts our initial assumption that S is linearly independent. Thus, S' must be linearly independent. **Q.E.D.**

Lemma 9.9 (*Supersets of Dependent Sets*): Let $S \subseteq V$ be a linearly dependent set. If S'' is a superset containing S (i.e., $S \subseteq S'' \subseteq V$), then S'' is also linearly dependent.

Proof:

Because S is linearly dependent, there exist vectors $v_1, \dots, v_k \in S$ and scalars $a_1, \dots, a_k \in K$, not all zero, such that:

$$a_1 v_1 + \dots + a_k v_k = 0_V$$

Since $S \subseteq S''$, the vectors $v_1, \dots, v_k$ also belong to S''. Therefore, this exact same non-trivial linear combination exists within S'', immediately satisfying the definition of linear dependence for S''. **Q.E.D.**

AI-Note: Definition 9.3 and the surrounding discussion of families — the index-set examples, the remark that nothing here needs V to be a vector space, and the link between repetitions and injectivity — are in the notes and were missing from this transcript; the same holds for Definition 9.5, for the comparison of a family with the set of its values on Page 61, for the exercise on Page 62, and for the remarks on Page 62. The notes write the two non-zero coefficients in the proof of Lemma 9.2 **(c)** as 1 and 1, which does not give $v_k - v_l$; the transcript already had the corrected 1 and -1, which is kept. Because this transcript indexes sequences from 0 (Page 49), the standard vectors of K^∞ in that exercise are $e_0, e_1, e_2, \dots$ rather than $e_1, e_2, \dots$ as in the notes. Proposition 9.7 and the two diagrams are not in the notes at this point; the proposition is kept here because the basis-extraction argument in the chapters on bases relies on it.

Bases of Vector Spaces (Part A) (Chapter 10)

Foundations of Bases and the Exchange Theorem (Section 10.a)

Definition and Unique Representation

We have previously explored two fundamental properties a set of vectors can possess: the ability to *span* the entire space, and the property of being *linearly independent* (having no redundancies). A basis is simply a set that achieves both simultaneously.

Definition 10.a.1 (*Basis*): Let V be a vector space over a field K. A subset $S \subseteq V$ is called a **basis** of V (plural: "*bases*") if the following two conditions hold:

(a) S is linearly independent.
(b) S generates (or spans) V, meaning $\mathrm{Sp}(S) = V$.

Why is a basis so mathematically desirable? Because it provides a perfect *"coordinate system"* for our abstract vector space.

Proposition 10.a.2 (*Unique Representation*): A subset $S \subseteq V$ is a basis of V if and only if every vector $v \in V$ can be written in a **unique way** as a linear combination of vectors in S.

To prove this rigorously—especially since S might be an infinite set, like the set of all polynomials—we must first formalize what we mean by *"a unique way as a linear combination."* In a vector space, we can only ever add a *finite* number of vectors together.

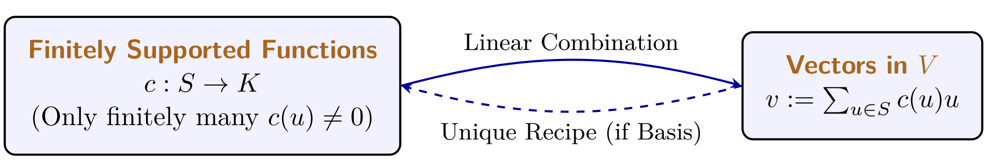
We formalize this using **finitely supported functions**. Let $c : S \to K$ be a function that assigns a scalar coefficient to every vector in S. We require that c is non-zero on only finitely many elements. In other words, the *"support"* of the function:

$$\mathrm{supp}(c) := \{u \in S \mid c(u) \neq 0\}$$

must be a finite set. We can then define our linear combination as the sum:

$$\sum_{u \in S} c(u) \cdot u$$

Because $\mathrm{supp}(c)$ is finite, the number of non-zero terms in this sum is strictly finite, making it a perfectly valid vector operation. (Furthermore, because vector addition is commutative, the order in which we sum these elements does not matter).



To say that every $v \in V$ can be written in a unique way means: if we have two finitely supported functions $c, c' : S \to K$ such that:

$$\sum_{u \in S} c(u) \cdot u = \sum_{u \in S} c'(u) \cdot u$$

then it must be true that $c(u) = c'(u)$ for all $u \in S$.

In case S is a finite set this has a very simple meaning. Write $S = \{v_1, \ldots, v_n\}$. Then if

$$v = a_1 v_1 + \cdots + a_n v_n \quad \text{with } a_i \in K, \qquad \text{and} \qquad v = b_1 v_1 + \cdots + b_n v_n \quad \text{with } b_i \in K,$$

then $a_1 = b_1, \ \ldots, \ a_n = b_n$.

Proof of Proposition 10.a.2:

We must prove both directions.

Direction 1 ($\implies$): Assume S is a basis. Because S spans V, every $v \in V$ can be written as a linear combination of elements in S. We must show this combination is unique. Suppose there are two combinations yielding the same vector: $\sum_{u\in S} c(u)u = \sum_{u\in S} c'(u)u$. Subtracting one from the other yields:

$$\sum_{u\in S} \big(c(u) - c'(u)\big)u = 0_V$$

Because S is a basis, it is linearly independent. The only way a linear combination of independent vectors can equal 0_V is if all coefficients are zero. Thus, $c(u) - c'(u) = 0 \implies c(u) = c'(u)$ for all $u \in S$. The representation is unique.

Direction 2 ($\impliedby$): Assume every $v \in V$ has a unique representation. Since every vector can be represented, S clearly spans V. We must show S is independent. Consider the linear combination $\sum_{u\in S} c(u)u = 0_V$. We already know one valid representation for the zero vector: assigning 0 to every coefficient. Since the hypothesis guarantees representations are *unique*, the all-zero assignment must be the *only* representation. Thus, S is linearly independent. **Q.E.D.**

Standard Examples of Bases

Let us identify the standard bases for the families of vector spaces we encounter most frequently.

Example 10.1 (*The Coordinate Space*): In $V = K^n$, the standard basis is $S = \{e_1, e_2, \dots, e_n\}$, where e_i is the column vector with a 1 in the i-th position and 0 elsewhere.

Example 10.2 (*Bounded Polynomials*): In $V = K[x]_d$, the space of polynomials of degree $\le d$, the standard basis is $S = \{1, x, x^2, \dots, x^d\}$.

Example 10.3 (*All Polynomials*): In $V = K[x]$, the space of all polynomials, the standard basis is the infinite set $S = \{1, x, x^2, \dots, x^n, \dots\}$. (Notice how beautifully our *"finitely supported function"* handles this: any specific polynomial has a finite degree, so it requires only a finite number of non-zero coefficients from this infinite basis).

Example 10.4 (*Matrices*): In $V = \mathcal{M}_{m\times n}(K)$, the standard basis is $S = \{E_{i,j} \mid 1 \le i \le m, 1 \le j \le n\}$, where $E_{i,j}$ is the matrix with a 1 in the (i,j) entry and 0 everywhere else. Written out, the 1 sits at the crossing of row i and column j:

$$E_{i,j} = \begin{array}{cl} \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} & \begin{array}{l} \\ \\ \leftarrow \text{row } i \\ \\ \\ \end{array} \end{array}$$

For example, $\mathcal{M}_{2\times 2}(K)$ has the following set as a basis:

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

The Size of a Basis and Exchange Lemmas

Looking at K^n, the standard basis has exactly n elements. Looking at $K[x]_d$, the standard basis has $d+1$ elements. This raises a monumental question: **Are all bases of a given vector space V exactly the same size?**

Yes, they are. Our next major goal is to prove this fact. To do so, we must understand how vectors can be swapped in and out of spanning sets without breaking the span. We develop this through a sequence of four powerful lemmas.

Lemma 10.a.3 (*Lemma A: Simple Exchange*): Let $v_1, \ldots, v_n$ be a basis of V. Let $u = a_1 v_1 + \cdots + a_n v_n \in V$ be a vector such that $a_1 \neq 0$. Then the modified list $u, v_2, \ldots, v_n$ is also a basis of V.

Proof:

We must show the new list is independent and spans V. **Independence:** Assume $b_1 u + b_2 v_2 + \cdots + b_n v_n = 0_V$. Substitute the definition of u:

$$b_1(a_1 v_1 + a_2 v_2 + \cdots + a_n v_n) + b_2 v_2 + \cdots + b_n v_n = 0_V$$

Regrouping the terms by the basis vectors v_i:

$$(b_1 a_1) v_1 + (b_1 a_2 + b_2) v_2 + \cdots + (b_1 a_n + b_n) v_n = 0_V$$

Since $v_1, \ldots, v_n$ are linearly independent, all coefficients must be zero. In particular, $b_1 a_1 = 0$. Because we assumed $a_1 \neq 0$, we are forced to conclude $b_1 = 0$. Once $b_1 = 0$, the equation reduces to $b_2 v_2 + \cdots + b_n v_n = 0_V$, which forces $b_2 = \cdots = b_n = 0$. Thus, the new list is independent.

Spanning: We can rearrange $u = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$ to solve for v_1:

$$v_1 = a_1^{-1} u - a_1^{-1} a_2 v_2 - \cdots - a_1^{-1} a_n v_n$$

This shows $v_1 \in \mathrm{Sp}(u, v_2, \ldots, v_n)$. Since all the other basis vectors $v_2, \ldots, v_n$ are trivially in this span, all vectors of the original basis are contained within it. Thus $V = \mathrm{Sp}(v_1, \ldots, v_n) \subseteq \mathrm{Sp}(u, v_2, \ldots, v_n)$. **Q.E.D.**

Lemma 10.a.4 (*Lemma B: Targeted Redundancy*): Let $v_1, \ldots, v_n$ be a list of vectors. Suppose it is linearly dependent. Then there exists an index $1 \leq j \leq n$ such that $v_j \in \mathrm{Sp}(v_1, \ldots, v_{j-1})$. Moreover, if we drop v_j from the list, the span of the remaining vectors is exactly the same as the original span.

Proof:

Because the list is dependent, there exist coefficients not all zero such that $\sum_{i=1}^{n} a_i v_i = 0_V$. Let j be the *largest* index such that $a_j \neq 0$. Then our equation is actually:

$$a_1 v_1 + a_2 v_2 + \cdots + a_j v_j = 0_V$$

Because $a_j \neq 0$, we can solve for v_j:

$$v_j = -\frac{a_1}{a_j} v_1 - \cdots - \frac{a_{j-1}}{a_j} v_{j-1}$$

This proves $v_j \in \mathrm{Sp}(v_1, \ldots, v_{j-1})$. Because v_j can be synthesized entirely from the vectors strictly preceding it, dropping v_j does not shrink the span.

Let us spell that last step out as the two inclusions it really is. Let $v \in \mathrm{Sp}(v_1, \ldots, v_n)$, so that

$$v = c_1 v_1 + \cdots + c_n v_n \qquad \text{for some } c_1, \ldots, c_n \in K.$$

Substituting the expression for v_j obtained above into this equation writes v as a linear combination of $v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n$, and therefore,

$$\mathrm{Sp}(v_1, \ldots, v_n) \subseteq \mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n).$$

The reverse inclusion

$$\mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n) \subseteq \mathrm{Sp}(v_1, \ldots, v_n)$$

is obvious, since every vector of the shorter list occurs in the longer one. The two inclusions give

$$\mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n) = \mathrm{Sp}(v_1, \ldots, v_n).$$

Q.E.D.

Exercise 10.1: What happens if $j = 1$ in Lemma 10.a.4?

The next lemma refines the previous one. It pins down **where** in the list the droppable vector can sit: if an initial segment of the list is independent, the droppable vector must lie strictly beyond it. This is what will later stop the exchange process from ever ejecting one of the vectors we are trying to insert.

Lemma 10.a.5 (*Lemma B′: The Drop Index Lies Beyond Any Independent Head*): Same assumptions as in Lemma 10.a.4, so let $v_1, \dots, v_n$ be a linearly dependent list and let j be the index it provides. Assume in addition that $v_1, \dots, v_k$ are linearly independent, where $1 \le k \le n$ is some index. Then $k < j$.

Proof:

Assume by contradiction that $j \le k$. By Lemma 10.a.4 we have $v_j \in \mathrm{Sp}(v_1, \dots, v_{j-1})$, that is,

$$v_j = a_1 v_1 + \cdots + a_{j-1} v_{j-1} \qquad \text{for some } a_1, \dots, a_{j-1} \in K.$$

Moving v_j to the other side gives

$$0_V = a_1 v_1 + \cdots + a_{j-1} v_{j-1} + (-1) v_j.$$

The coefficient of v_j is $-1 \neq 0$, so this is a non-trivial linear combination of $v_1, \dots, v_j$ that gives 0_V. Hence $v_1, \dots, v_j$ are linearly dependent. Since $j \le k$, the list $v_1, \dots, v_j$ sits inside the list $v_1, \dots, v_k$, so $v_1, \dots, v_k$ are linearly dependent as well. This contradicts the assumption that $v_1, \dots, v_k$ are linearly independent, and therefore, $k < j$. Q.E.D.

Lemma 10.a.6 (*Lemma C: Span Overcrowding*): If $\mathrm{Sp}(v_1, \dots, v_n) = V$ and $u \in V$, then the list $u, v_1, \dots, v_n$ is linearly dependent.

Proof:

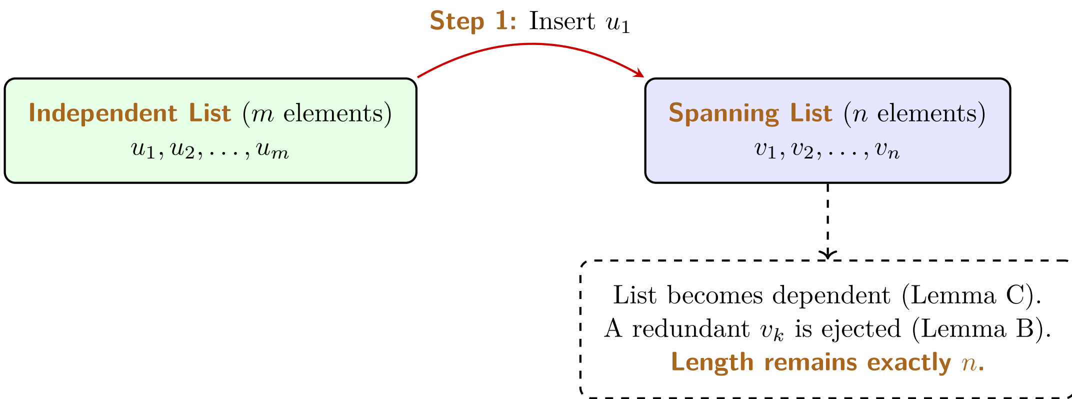
Because the v_i's span V, we can write $u = a_1 v_1 + \cdots + a_n v_n$. Rearranging this gives:

$$1 \cdot u - a_1 v_1 - \cdots - a_n v_n = 0_V$$

Since the coefficient of u is 1 (which is non-zero), this is a non-trivial linear combination that equals zero. The list is dependent. Q.E.D.

We now arrive at the crescendo of this chapter: The Steinitz Exchange Lemma. This theorem rigorously proves that an independent set can never outnumber a spanning set.

Theorem 10.a.7 (*Lemma D: Steinitz Exchange Theorem*): Let $u_1, \dots, u_m$ be a linearly independent list in V. Let $v_1, \dots, v_n$ be a list that spans V (i.e., $\mathrm{Sp}(v_1, \dots, v_n) = V$). Then $m \le n$.



Proof:

We proceed by injecting the independent u vectors into the spanning list one by one, forcefully replacing v vectors to maintain a constant list length of n.

Step 1: Add u_1 to the front of the spanning list: $u_1, v_1, \ldots, v_n$. Because the v_i's span V, Lemma C dictates this new list is dependent. By Lemma B, there is a redundant vector in this list that can be dropped without altering the span. Could the redundant vector be u_1? No. Lemma B specifies the redundant vector is in the span of the vectors *preceding* it. u_1 is at the very front (its preceding span is $\emptyset$), and $u_1 \neq 0_V$ because it is part of an independent set. Thus, the redundant vector must be one of the v_i's, say v_k. We drop v_k. Our list is now $u_1, v_1, \ldots, \widehat{v_k}, \ldots, v_n$. It still spans V, and it has exactly n elements.

Step 2: Add u_2 to the front: $u_2, u_1, \ldots, \widehat{v_k}, \ldots, v_n$. By Lemma C, this is dependent. By Lemma B, a redundant vector exists. It cannot be u_2 (front of the line). It cannot be u_1, because u_1 cannot be in the span of u_2 (since the u's are independent). Therefore, the redundant vector *must* be another v. We drop it. Length remains n.

The Contradiction: Assume, for the sake of contradiction, that $m > n$. We can continue the above process exactly n times. After step n, we will have systematically ejected every single v_i from the list, leaving us with a list consisting purely of u-vectors: $u_n, u_{n-1}, \ldots, u_1$. Crucially, because we only ever used Lemma B to drop vectors, this list of u-vectors **still spans V**.

Because $m > n$, we still have at least one vector left in our independent set: u_{n+1}. Let us add it to the front of our current spanning list:

$$u_{n+1}, u_n, u_{n-1}, \ldots, u_1$$

Because the set $u_n, \ldots, u_1$ spans V, Lemma C states that this new combined list must be linearly dependent. However, this list is just a subset of our original $u_1, \ldots, u_m$, which was explicitly assumed to be linearly independent! This is a fatal mathematical contradiction.

Therefore, our assumption that $m > n$ must be false. We conclude that $m \leq n$. **Q.E.D.**

Remark 10.1 (*The General Step*): At step j of the argument above, the list reads $u_1, \ldots, u_{j-1}, w_1, \ldots, w_{n-(j-1)}$, where the w's are the survivors of the original spanning list, and we insert u_j after u_{j-1}. Lemma 10.a.6 makes the resulting list of $n+1$ vectors dependent, and Lemma 10.a.4 supplies a vector that may be dropped without changing the span. That vector is never one of $u_1, \ldots, u_j$: those are linearly independent, being part of an independent list, so Lemma 10.a.5 forces the drop index to lie strictly beyond them. The ejected vector is therefore always one of the w's, which is what keeps the length at exactly n while the u's accumulate at the front.

Remark 10.2 (*We Have Proved More Than $m \leq n$*): The argument gives more than the inequality. It shows that $m \leq n$ **and** that one can remove m vectors from the list $v_1, \ldots, v_n$ in such a way that the remaining list $v_{i_1}, \ldots, v_{i_{n-m}}$, where $1 \leq i_1 < \cdots < i_{n-m} \leq n$, spans V when put together with $u_1, \ldots, u_m$:

$$\mathrm{Sp}(u_1, \ldots, u_m, v_{i_1}, \ldots, v_{i_{n-m}}) = V.$$

Indeed, we already know that $m \leq n$. Apply the procedure from the proof for the steps $j = 1, \ldots, m$. In each step we add another u_j to the list and drop one of the v_i's, while keeping the span equal to V. After $j = m$ steps we obtain the claimed list $u_1, \ldots, u_m, v_{i_1}, \ldots, v_{i_{n-m}}$.

AI-Note: The notes name their four lemmas A–D differently from this transcript: what is called Lemma 10.a.4 here is Lemma A in the notes, and the notes' Lemma B is Lemma 10.a.5, which was missing from the transcript altogether. The existing letters are left alone to keep the references in the later chapters valid, and the restored lemma is lettered B'. Lemma 10.a.3 does not appear in this part of the notes. Also restored: the finite-set reading of uniqueness before the proof of Proposition 10.a.2, the explicit form of $E_{i,j}$ and the basis of $\mathcal{M}_{2\times 2}(K)$ in Page 66, the second half of the proof of Lemma 10.a.4 (the notes prove the equality of spans by two inclusions; the transcript asserted it), Page 68, and Page 69, which the later chapters rely on. The proof of Proposition 10.a.2 is given in the notes under the simplifying assumption that S is finite, with the infinite case noted as similar; the transcript proves it for general S and that is kept.

Bases of a Vector Space (Part B) (Chapter 11)

The Dimension Theorem (Section 11.b)

The Strong Exchange Lemma

Before we establish the main theorem of this chapter, we formalize a stronger version of the Steinitz Exchange Theorem (Lemma D from the previous section). We proved that an independent list cannot be larger than a spanning list. However, the exact mechanism of the proof actually guarantees something even more powerful: we can precisely substitute the independent vectors into the spanning set.

Lemma 11.b.1 (*Lemma D': Strong Steinitz Exchange*): Let V be a vector space over K, and suppose that $V = \mathrm{Sp}(v_1, \dots, v_n)$ for some list of vectors $v_1, \dots, v_n \in V$. Let $u_1, \dots, u_m \in V$ be a linearly independent list.

Then $m \le n$, and moreover, one can delete exactly m vectors from the spanning list $v_1, \dots, v_n$ such that the remaining $n - m$ vectors, together with all the vectors $u_1, \dots, u_m$, still span V.

$$\mathrm{Sp}(u_1, \dots, u_m, v_1', \dots, v_{n-m}') = V$$

(where the v' vectors are the survivors of the original spanning list).

Proof:

The fact that $m \le n$ was already proven in Lemma D. To prove the *"moreover"* part, we simply apply the exact step-by-step substitution procedure described in the proof of Lemma D. For steps $j = 1, 2, \dots, m$, we sequentially add u_j to the front of the list. At each step, the list becomes dependent (Lemma C), and we forcefully eject a redundant v_k vector (Lemma B) to restore independence while perfectly preserving the span. After m steps, we have injected all m independent vectors and deleted exactly m of the original spanning vectors. The resulting mixed list still spans V. See also Page 69, where the same statement is drawn out of the proof of Lemma D directly. **Q.E.D.**

The Existence and Uniqueness of Dimension

We are now ready to state and prove the foundational theorem of finite-dimensional vector spaces. It guarantees that bases always exist and that they provide an absolute, rigid measure of the *"size"* of a vector space.

Theorem 11.b.2 (*The Basis Theorem*): Let V be a finite-dimensional vector space over K. Then V has a basis with finitely many elements.

Moreover, **every** basis of V is finite, and any two bases of V have exactly the same number of elements.

Remark 11.1 (*Infinite Dimensional Spaces*): This theorem is actually true even if V is not finite-dimensional (this is known as the *"Hamel Basis Theorem"* and requires Zorn's Lemma). However, in this course, we will only prove the theorem for finite-dimensional spaces. The infinite-dimensional case is rigorously covered in advanced algebra courses (e.g., *Grundstrukturen* in the 2nd semester).

Preparation: Two Lemmas

For the proof of the theorem we need some preparation. The first lemma is the converse reading of the redundancy lemma of the previous chapter: there, a dependent list was shown to contain a vector lying in the span of its predecessors; here, a list in which **no** vector lies in the span of its predecessors is shown to be independent.

Lemma 11.b.3 (*Lemma E: No Predecessor Span Means Independent*): Let $w_1, \dots, w_\ell \in V$ and assume that

$$w_j \notin \mathrm{Sp}(w_1, \dots, w_{j-1}) \qquad \text{for all } 1 \leq j \leq \ell.$$

Then the list $w_1, \dots, w_\ell$ is linearly independent.

Proof:

If $w_1, \dots, w_\ell$ were linearly dependent, then by Lemma 10.a.4 there would exist an index $1 \leq j \leq \ell$ with $w_j \in \mathrm{Sp}(w_1, \dots, w_{j-1})$, contradicting the assumption. **Q.E.D.**

The second lemma is the one that actually produces a basis. Note that it extracts the basis from **within** the given spanning list, rather than merely asserting that one exists somewhere.

Lemma 11.b.4 (*Lemma F: Every Finite Spanning List Contains a Basis*): Suppose that $\mathrm{Sp}(v_1, \dots, v_n) = V$. Then there exists a subset of $\{v_1, \dots, v_n\}$ which is a basis for V.

Proof:

The proof is algorithmic and is based on n steps. At each step we consider another vector from the list $v_1, \dots, v_n$ and decide whether or not to drop it from the list. At step number j we consider v_j:

Step 1. If $v_1 = 0_V$ we drop v_1 from the list. If $v_1 \neq 0_V$ we keep it.

Step j, for $2 \leq j \leq n$. If $v_j \in \mathrm{Sp}(v_1, \dots, v_{j-1})$ we drop v_j. Otherwise we keep it.

After performing the n steps above we obtain a new, possibly shorter, list

$$v_{i_1}, v_{i_2}, \dots, v_{i_m}, \qquad \text{where } 1 \leq i_1 < i_2 < \cdots < i_m \leq n.$$

We claim that $\mathrm{Sp}(v_{i_1}, \dots, v_{i_m}) = V$. Indeed, in any of the steps $1 \leq j \leq n$ we dropped the vector v_j only if $v_j \in \mathrm{Sp}(v_1, \dots, v_{j-1})$, and by Lemma 10.a.4 such a vector may be dropped without changing the span. Therefore,

$$\mathrm{Sp}(v_{i_1}, \dots, v_{i_m}) = \mathrm{Sp}(v_1, \dots, v_n) = V.$$

We also claim that $v_{i_1}, \dots, v_{i_m}$ is linearly independent. Indeed, for every $1 \leq k \leq m$ the vector v_{i_k} survived its step, which means

$$v_{i_k} \notin \mathrm{Sp}(v_1, v_2, \dots, v_{i_k - 1}),$$

and since $v_{i_1}, \dots, v_{i_{k-1}}$ are all taken from among $v_1, \dots, v_{i_k - 1}$, this gives in particular

$$v_{i_k} \notin \mathrm{Sp}(v_{i_1}, v_{i_2}, \dots, v_{i_{k-1}}).$$

By Lemma 11.b.3, the list $v_{i_1}, v_{i_2}, \dots, v_{i_m}$ is linearly independent.

Together with the fact that $\mathrm{Sp}(v_{i_1}, \dots, v_{i_m}) = V$, we get that $v_{i_1}, \dots, v_{i_m}$ is a basis for V.

Q.E.D.

Proof of the Basis Theorem

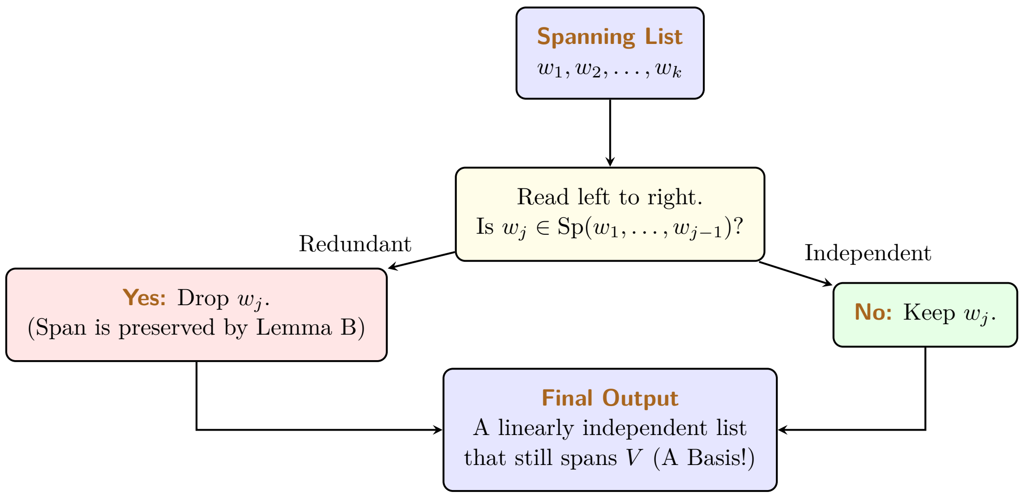
Because this theorem is so monumental, we will break its proof down into three distinct, manageable statements.

Statement 1: V has a finite basis.

Proof:

Because we assumed V is finite-dimensional, there must exist some finite list of vectors that spans V. Let us call this spanning set $S = \{w_1, \ldots, w_k\}$. If S is already linearly independent, it is a basis, and we are done.

If S is linearly dependent, we can extract a basis from it using the following systematic reduction algorithm (often called a *"Redundancy Filter"*):



Specifically:

(a) If the zero vector 0_V is in the list, drop it.

(b) Read the list from left to right. If a vector w_j is a linear combination of its strictly preceding vectors (i.e., $w_j \in \mathrm{Sp}(w_1, \ldots, w_{j-1})$), drop it.

By Lemma B, dropping this redundant vector does not change the span of the list. We repeat this until we have checked every vector. The remaining list still spans V. Furthermore, by Lemma 11.b.3, it must be linearly independent, because we systematically deleted every vector that could be written as a combination of the vectors preceding it. Thus, the remaining list is a finite basis. This is precisely the algorithm of Lemma 11.b.4, applied to the spanning list $w_1, \ldots, w_k$.

Q.E.D.

Statement 2: Every basis of V is finite.

Proof:

We prove this by contradiction. Suppose there exists a basis $\mathcal{A}$ of V that is *not* a finite set.

From Statement 1, we know that V definitely possesses at least one finite basis. Let us call this finite basis $S = \{v_1, \ldots, v_n\}$. The size of this finite basis is exactly n.

Because $\mathcal{A}$ is infinite, it contains infinitely many linearly independent vectors. Therefore, we can arbitrarily select $n + 100$ distinct vectors from $\mathcal{A}$. Let us call them $u_1, \ldots, u_{n+100}$. Because they belong to a basis, this chosen list of $n + 100$ vectors is strictly linearly independent.

We now have two lists:

- A spanning list S containing exactly n elements.
- An independent list containing $n + 100$ elements.

By the Steinitz Exchange Theorem (Lemma D), the number of elements in any independent list must be less than or equal to the number of elements in any spanning list. Therefore, we must have:

$$n + 100 \leq n$$

This implies $100 \leq 0$, which is a blatant mathematical contradiction! Thus, our initial assumption must be wrong. Every basis of V must be finite. **Q.E.D.**

Statement 3: Any two bases have the same cardinality.

Proof:

Let $\mathcal{A}$ and $\mathcal{B}$ be two arbitrary bases of V. By Statement 2, we know that both $\mathcal{A}$ and $\mathcal{B}$ must be finite sets. Let $|\mathcal{A}|$ denote the number of elements in $\mathcal{A}$, and $|\mathcal{B}|$ denote the number of elements in $\mathcal{B}$.

Because they are bases, both sets are simultaneously independent and spanning. This allows us to apply Lemma D twice in alternating directions:

Independent in Spanning

$$\begin{tikzcd} \boxed{\text{Basis } \mathcal{A}} \arrow[r, bend left, "|\mathcal{A}| \leq |\mathcal{B}|"] & \boxed{\text{Basis } \mathcal{B}} \arrow[l, bend left, "|\mathcal{B}| \leq |\mathcal{A}|"] \end{tikzcd}$$

Independent in Spanning

(a) Because $\mathcal{A}$ is linearly independent and $\mathcal{B}$ spans V, Lemma D dictates that $|\mathcal{A}| \leq |\mathcal{B}|$.

(b) Reversing their roles, because $\mathcal{B}$ is linearly independent and $\mathcal{A}$ spans V, Lemma D dictates that $|\mathcal{B}| \leq |\mathcal{A}|$.

Since $|\mathcal{A}| \leq |\mathcal{B}|$ and $|\mathcal{B}| \leq |\mathcal{A}|$, it must be that $|\mathcal{A}| = |\mathcal{B}|$. All bases contain exactly the same number of elements. **Q.E.D.**

The Concept of Dimension

Because the number of elements in a basis is a rigid, absolute invariant of the vector space, we can finally define the space's dimension.

Definition 11.b.5 (*Dimension*): Let V be a finite-dimensional vector space over K. We define the **dimension** of V to be the number of elements in any basis of V. We denote this integer by $\dim V := n \in \mathbb{Z}_{\geq 0}$.

If V is not finite-dimensional, we write $\dim V = \infty$.

Important remark 11.1: $\dim V$ is **well defined**: the definition does not depend on the specific choice of a basis for V. This is exactly what Statement 3 of Theorem 11.b.2 buys us. Without it, the phrase *"the number of elements in any basis"* would not name a single number, and the definition above would be empty.

Remark 11.2 (*The Zero Space*): Consider the zero vector space $V = \{0_V\}$. Its only spanning set is $\{0_V\}$, which is linearly dependent. The only linearly independent subset that spans it is the empty set $\emptyset$. Therefore, the basis of the zero space is $\emptyset$. Because the empty set contains 0 elements, we have:

$$\dim\{0_V\} = 0$$

Examples of Dimensions

Now that we have formally defined dimension, we can readily determine the dimensions of our familiar vector spaces by simply counting the elements in their standard bases.

Example 11.1 (*The Coordinate Space*): The standard basis for K^n is $\{e_1, \ldots, e_n\}$. Because this set contains exactly n elements, we have:

$$\dim K^n = n$$

Example 11.2 (*Bounded Polynomials*): The standard basis for the space of polynomials of degree $\leq d$ is $\{1, x, x^2, \ldots, x^d\}$. Be careful when counting here! Because we start at $x^0 = 1$, this set contains $d+1$ elements. Thus:

$$\dim K[x]_d = d + 1$$

Example 11.3 (*Matrices*): The standard basis for the matrix space $\mathcal{M}_{m\times n}(K)$ consists of the individual single-entry matrices $E_{i,j}$. Since there are m rows and n columns, there are exactly $m \cdot n$ such matrices. Thus:

$$\dim \mathcal{M}_{m\times n}(K) = m \cdot n$$

Example 11.4 (*Function Spaces*): Let S be an arbitrary set. Recall that the space of all functions from S to K, denoted as $K^S := \{f \mid f : S \to K\}$, forms a vector space over K.

The dimension of this space matches the cardinality of the underlying set S:

$$\dim K^S = |S|$$

Exercise 11.1 (*Basis for the Space of Functions on a Finite Set*): If S is a finite set, say $S = \{s_1, \ldots, s_n\}$, can you construct an explicit basis for K^S?
(Hint: Try to define a family of functions f_i where each function outputs 1 for exactly one specific element s_i, and 0 for all other elements).

Remark 11.3: When the domain S is an infinite set, the space K^S is infinite-dimensional. In this case, we unfortunately do not have a *"nice"* or easily describable basis for it.

AI-Note: The notes prepare the Basis Theorem with two lemmas that the transcript had dropped: Lemma 11.b.3 and Lemma 11.b.4, both restored above with their proofs. Statement 1 of the theorem is proved in the notes by citing Lemma 11.b.4; the transcript inlines the algorithm instead, which is kept, with the citation corrected. That citation had pointed at Proposition 9.7, which is about a vector lying in the span of **all** the others, whereas the filter only ever tests a vector against the ones preceding it. Lemma 11.b.3 is the statement that closes that gap. Also restored: Page 74, which the notes flag explicitly. The notes state Lemma 11.b.1 here and refer back to the closing remark of the previous lecture for its proof; both are kept, and the cross-reference is made explicit.

Dimension and Subspaces (Chapter 12)

In this chapter, we synthesize the fundamental results regarding finite-dimensional vector spaces and explore the relationship between a space and its subspaces through the lens of dimensionality.

Summary 12.1 (*Properties of Finite-Dimensional Spaces*): Let V be a finite-dimensional vector space over a field K. The following structural principles hold:

(a) Every finite list of vectors that spans V contains a sublist which constitutes a "*basis*" of V. Formally, if $(v_1, \ldots, v_n)$ spans V, then there exists a sublist $(v_{i_1}, \ldots, v_{i_d})$ with $1 \leq i_1 < i_2 < \cdots < i_d \leq n$ that is a basis for V, where $d = \dim V$.

(b) Every linearly independent list of vectors in V can be extended to a basis of V. If the list $(u_1, \ldots, u_m)$ is linearly independent, there exist vectors $w_1, \ldots, w_{d-m}$ such that

$$(u_1, \ldots, u_m, w_1, \ldots, w_{d-m})$$

is a basis of V, where $d = \dim V$.

Theorem 12.1 (*Equivalent Characterizations of a Basis*): Let V be a vector space with $\dim V = n \in \mathbb{Z}_{\geq 0}$. Let $v_1, \ldots, v_n \in V$ be a list of n vectors. Then the following statements are equivalent:

(a) The vectors $v_1, \ldots, v_n$ are linearly independent.

(b) The vectors $v_1, \ldots, v_n$ generate V, i.e., $\mathrm{Sp}(v_1, \ldots, v_n) = V$.

(c) The vectors $v_1, \ldots, v_n$ constitute a basis for V.

Proof:

The equivalence is established through the following implications:

(a) $\Longrightarrow$ (c): Suppose that **(a)** holds, so the list $v_1, \ldots, v_n$ is linearly independent. According to the property that every linearly independent list can be extended to a basis, and observing that the list already possesses length $n = \dim V$, the extension requires no additional vectors. Thus, the list is already a basis for V, satisfying **(c)**.

(c) $\Longrightarrow$ (a) & (b): This follows directly from the definition of a basis, which by requirement must satisfy both the condition of linear independence in **(a)** and the spanning property in **(b)**.

(b) $\Longrightarrow$ (c): If the spanning property in **(b)** holds, the list $v_1, \ldots, v_n$ generates V. By the properties of finite-dimensional spaces, this list must contain a sublist that is a basis. Since $\dim V = n$, any basis must have exactly length n. Therefore, the sublist must be the entire list itself, making $v_1, \ldots, v_n$ a basis as required by **(c)**. **Q.E.D.**

The previous theorem settles what happens for a list whose length is exactly $\dim V$. The next one records what happens when the length is wrong in either direction: too short a list cannot reach every vector, and too long a list must contain a redundancy.

Theorem 12.2 (*Lists of the Wrong Length*): Suppose that $n = \dim V < \infty$, and let $v_1, \ldots, v_k$ be a list of vectors in V. Then:

(1) If $k < n$, the list $v_1, \ldots, v_k$ can **not** span V.

(2) If $k > n$, the list $v_1, \ldots, v_k$ can **not** be linearly independent.

Proof:

(1) Assume that $k < n$, and suppose by contradiction that $\mathrm{Sp}(v_1, \ldots, v_k) = V$. By item **(1)** of the summary on Page 76 we can delete some of the vectors from the list $v_1, \ldots, v_k$ and obtain a basis $v_{i_1}, \ldots, v_{i_{k'}}$ of V. Then

$$\dim V = k' \leq k < n = \dim V,$$

a contradiction.

(2) Assume that $k > n$, and suppose by contradiction that $v_1, \ldots, v_k$ are linearly independent. By item **(2)** of the summary on Page 76 we can extend this list to a basis of V. Then

$$\dim V \geq k > n = \dim V,$$

a contradiction. **Q.E.D.**

Proposition 12.3 (*Dimensional Constraints on Subspaces*): Let V be a finite-dimensional vector space over K. Then every linear subspace $U \subseteq V$ is also finite-dimensional. Moreover, $\dim U \leq \dim V$, with the equality $\dim U = \dim V$ holding if and only if $U = V$.

Proof:

Let $n := \dim V$. We first establish that U possesses a finite basis. If $U = \{0\}$, the empty set serves as the basis and we are done. Assume $U \neq \{0\}$. We select a non-zero vector $v_1 \in U$. If $\mathrm{Sp}(v_1) = U$, we have found our basis.

If $\mathrm{Sp}(v_1) \neq U$, we choose a second vector $v_2 \in U \setminus \mathrm{Sp}(v_1)$. It follows from the properties of linear independence (Lemma 11.b.3) that the list $\{v_1, v_2\}$ is linearly independent. We proceed iteratively: after j steps, we obtain a linearly independent list $\{v_1, \ldots, v_j\} \subseteq U$. If $\mathrm{Sp}(v_1, \ldots, v_j) = U$, the process terminates. Otherwise, we choose $v_{j+1} \in U \setminus \mathrm{Sp}(v_1, \ldots, v_j)$. By the principle that adding a vector outside the span of a linearly independent set preserves independence (again Lemma 11.b.3), the expanded list $\{v_1, \ldots, v_{j+1}\}$ remains linearly independent.

Crucially, in a space V of dimension n, no linearly independent list can exceed length n. Consequently, this algorithmic procedure must terminate in k steps, where $k \leq n$. At this stage, we have a list $\{v_1, \ldots, v_k\}$ that is linearly independent and spans U, thereby constituting a basis for U. This implies U is finite-dimensional and $\dim U = k \leq n = \dim V$.

It remains to settle when equality holds. One direction is immediate: if $U = V$, then clearly $\dim U = \dim V$. The converse is the substantial one.

Furthermore, if $\dim U = \dim V = n$, then U contains a linearly independent list of length n. By Theorem 12.1, such a list satisfies property **(a)** and is therefore also a basis for V. This implies $\mathrm{Sp}(v_1, \ldots, v_n) = V$. Since $U = \mathrm{Sp}(v_1, \ldots, v_n)$ by the construction of the basis, we conclude $U = V$. **Q.E.D.**

AI-Note: Theorem 12.2 is stated and proved in the notes and was missing from this transcript; it is restored above. The notes justify the two independence steps in the proof of Proposition 12.3 by citing Lemma E of the previous chapter (Lemma 11.b.3), which the transcript appealed to without naming; the citations are put back, leaving the surrounding wording as it was. The trivial direction of the equality claim in Proposition 12.3, that $U = V$ gives $\dim U = \dim V$, was also missing and is supplied. The notes restate Lemma 11.b.3 in full at this point as a recollection from the earlier lecture; it is only cross-referenced here.

Row and Column Spaces (Chapter 13)

In this chapter, we bridge the abstract theory of vector spaces with the algorithmic power of matrix theory. We use Gauss elimination as a scalpel to dissect the structure of coordinate spaces.

Linear Combinations and Matrix Equations

Consider the vector space K^m over a field K. Let $w_1, \dots, w_n \in K^m$ be a collection of vectors. We begin by addressing three fundamental questions that govern our understanding of these spaces.

Question 13.1 (*The Three Fundamental Questions*):

(a) How can we systematically determine if a given vector $w \in K^m$ belongs to the span $\mathrm{Sp}(w_1, \dots, w_n)$?

(b) Can we describe the subspace $\mathrm{Sp}(w_1, \dots, w_n)$ precisely using algebraic conditions?

(c) How can we determine whether the vectors $w_1, \dots, w_n$ are linearly independent?

Answer 13.1 (*To Questions 1 and 2*): By definition, $w \in \mathrm{Sp}(w_1, \dots, w_n)$ if and only if there exist scalars $x_1, \dots, x_n \in K$ such that $w = x_1 w_1 + \cdots + x_n w_n$. We can assemble the vectors w_j as columns of an $m \times n$ matrix A. This abstract linear combination is equivalent to the matrix equation:

$$\underbrace{\begin{pmatrix} | & & | \\ w_1 & \dots & w_n \\ | & & | \end{pmatrix}}_{A} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} | \\ w \\ | \end{pmatrix}$$

Thus, w is in the span if and only if the linear system $Ax = w$ is *"consistent"* (i.e., it possesses at least one valid solution vector x). We analyze this by applying Gauss elimination on the *"augmented matrix"* $(A \mid w)$.

To answer the second question and describe the entire span precisely, we replace the specific target w with a general, variable vector $b = (b_1, \dots, b_m)^{\mathsf{T}}$ and reduce the augmented matrix $(A \mid b)$ to a row-reduced echelon form $(A' \mid b')$.

Two cases have to be distinguished. Suppose first that A' has no zero rows at all, so that every row of A' carries a pivot. Then the system $A'x = b'$ can be solved no matter what the right-hand side b' is, and by Theorem 6.10 the original system $Ax = b$ is then solvable as well. Since this argument applies to every $b \in K^m$, we conclude that in this case

$$\mathrm{Sp}(w_1, \dots, w_n) = K^m.$$

In the remaining case, A' does have zero rows. This row operation procedure transforms our original problem into a much simpler, equivalent linear system $A'x = b'$. As shown in Prof. Biran's notes, the resulting matrix generally has the following characteristic block structure:

$$(A' \mid b') = \left(\begin{array}{ccc|c} * & \dots & * & b'_1 \\ \vdots & \text{Rows with pivots} & \vdots & \vdots \\ * & \dots & * & b'_r \\ \hline 0 & \dots & 0 & b'_{r+1} \\ \vdots & \text{Zero rows} & \vdots & \vdots \\ 0 & \dots & 0 & b'_m \end{array}\right) \tag{13.1}$$

In this reduced block structure, any row of A' consisting entirely of zeros translates to an algebraic equation of the form $0 = b'_i$. For the overall system to be consistent (meaning a solution actually exists), it is an absolute requirement that the right-hand side of these specific equations also evaluates to zero.

In other words, the system is consistent if and only if the entries in b' corresponding to the zero-rows of A' strictly vanish. Therefore, we can concisely describe the span as the set of all vectors satisfying these constraints:

$$\operatorname{Sp}(w_1, \ldots, w_n) = \left\{ b \in K^m \;\middle|\; b'_{r+1} = \cdots = b'_m = 0 \right\}.$$

Example 13.1 (*Calculating a Specific Span*): Let $v_1 = (1,1,1)^\mathsf{T}$ and $v_2 = (1,0,-1)^\mathsf{T} \in K^3$. To describe $\operatorname{Sp}(v_1, v_2)$, we set up the augmented matrix with a general vector b and row-reduce:

$$\left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 1 & 0 & b_2 \\ 1 & -1 & b_3 \end{array}\right) \xrightarrow[-1\cdot E_1 + E_3 \to E_3]{-1\cdot E_1 + E_2 \to E_2} \left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & -1 & b_2 - b_1 \\ 0 & -2 & b_3 - b_1 \end{array}\right)$$

$$\xrightarrow{-1\cdot E_2 \to E_2} \left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & 1 & b_1 - b_2 \\ 0 & -2 & b_3 - b_1 \end{array}\right)$$

$$\xrightarrow{2\cdot E_2 + E_3 \to E_3} \left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & 1 & b_1 - b_2 \\ \hline 0 & 0 & b_1 - 2b_2 + b_3 \end{array}\right)$$

The system is consistent if and only if the expression corresponding to the zero-row evaluates to zero. Thus, the span is perfectly described by that single plane equation:

$$\operatorname{Sp}(v_1, v_2) = \{(b_1, b_2, b_3)^\mathsf{T} \in K^3 \mid b_1 - 2b_2 + b_3 = 0\}.$$

Answer 13.2 (*To Question 3*): The vectors are linearly independent if and only if the homogeneous equation $Ax = 0$ has only the trivial solution (i.e., $x_1 = \cdots = x_n = 0$).

This occurs if and only if the row-reduced echelon form A' has a pivot in every column. If even a single column lacks a pivot, the corresponding variable becomes *"free"* (e.g., it can be parameterized to take on any scalar value). The existence of a free variable immediately allows for non-trivial solutions, which by definition means the vectors are linearly dependent.

The machinery just developed gives a second, entirely computational proof of the two counting statements of Theorem 12.2, now specialised to the concrete space K^m. It is instructive to compare the two arguments: the earlier one was extracted from the abstract theory of bases, whereas the one below simply counts pivots.

Theorem 13.1 (*Fundamental Constraints on Coordinate Vectors*): Let $v_1, \ldots, v_n \in K^m$.

(a) If $n > m$, then the vectors $v_1, \ldots, v_n$ are linearly dependent.

(b) If $n < m$, then the vectors $v_1, \ldots, v_n$ cannot span K^m.

Proof:

(a) Let $A = (v_1 \mid \cdots \mid v_n)$. The number of pivots in A' is bounded by the number of rows m. If $n > m$, at least $n - m$ columns are forced to be without pivots. These free variables guarantee non-trivial solutions to $Ax = 0$, hence dependence.

(b) The number of pivots is bounded by the number of columns n. If $n < m$, there are at least $m - n$ zero-rows in A'. By the logic of equation (13.1), we can easily choose a vector b such that the resulting b' has non-zero entries in those bottom rows, making the system inconsistent.

It is worth recording what the elimination produces here in slightly more detail. Applying Gauss elimination to the extended matrix $(v_1 \mid \cdots \mid v_n \mid b)$ yields a matrix whose left block is in row-reduced echelon form and whose last column splits into two parts: an upper part b', sitting next to the rows with pivots, and a lower part b'', sitting next to the zero rows. Every entry of b'' is an expression of the form $\alpha_1 b_1 + \cdots + \alpha_m b_m$ with $\alpha_1, \ldots, \alpha_m \in K$, where the coefficients α_i may of course differ from entry to entry. The span is therefore

$$\mathrm{Sp}(v_1, \ldots, v_n) = \{b \in K^m \mid b'' = 0\} \subsetneq K^m,$$

The inclusion is strict: b'' has at least one entry, and the linear form $\alpha_1 b_1 + \cdots + \alpha_m b_m$ computing it cannot have all its coefficients equal to zero, since the row operations leading to it are reversible and hence cannot annihilate a whole row.

Q.E.D.

Row and Column Spaces

Definition 13.2 (*Subspaces of a Matrix*): Let $A \in \mathcal{M}_{m\times n}(K)$, and denote by $u_1, \ldots, u_m \in K^n$ the rows of A and by $v_1, \ldots, v_n \in K^m$ the columns of A, so that

$$A = \begin{pmatrix} — & u_1 & — \\ & \vdots & \\ — & u_m & — \end{pmatrix} = \begin{pmatrix} | & & | \\ v_1 & \ldots & v_n \\ | & & | \end{pmatrix}.$$

(a) The "*column space*" $\mathrm{Cols}(A) \subseteq K^m$ is the span of the columns of A, that is, $\mathrm{Cols}(A) := \mathrm{Sp}(v_1, \ldots, v_n)$.

(b) The "*row space*" $\mathrm{Rows}(A) \subseteq K^n$ is the span of the rows of A, that is, $\mathrm{Rows}(A) := \mathrm{Sp}(u_1, \ldots, u_m)$.

Note the difference in the ambient spaces: the rows of A have n entries and are viewed as row vectors, so that $\mathrm{Rows}(A)$ is a subspace of K^n, while the columns have m entries and are viewed as column vectors, so that $\mathrm{Cols}(A)$ is a subspace of K^m.

Proposition 13.3 (*Invariance of Row Space*): Elementary row operations do not change the row space: $\mathrm{Rows}(A) = \mathrm{Rows}(A')$. More generally, if $A, B \in \mathcal{M}_{m\times n}(K)$ are row-equivalent matrices in the sense of Definition 6.9, then $\mathrm{Rows}(A) = \mathrm{Rows}(B)$.

Proof:

New rows are, simply, linear combinations of the old rows, and because elementary operations are reversible, the spans remain identical. We now carry this out for each of the three types of Definition 6.8. Let $M \in \mathcal{M}_{m\times n}(K)$ be a matrix with rows $w_1, \ldots, w_m \in K^n$.

Type 3 ($E_i \leftrightarrow E_j$). Exchanging two rows of M merely reorders the list $w_1, \ldots, w_m$, and the span of a list does not depend on the order of its members. Hence $\mathrm{Rows}(M)$ is unchanged by such an operation.

Type 1 ($\lambda \cdot E_i \to E_i$, with $\lambda \in K \setminus \{0\}$). Replacing w_i by λw_i does not change the span of the rows either: the new row lies in $\mathrm{Sp}(w_1, \ldots, w_m)$, and conversely $w_i = \lambda^{-1}(\lambda w_i)$ lies in the span of the new list, which is where the hypothesis $\lambda \neq 0$ enters. So $\mathrm{Rows}(M)$ is left unchanged by such an operation as well.

Type 2 ($\lambda \cdot E_j + E_i \to E_i$, with $i \neq j$ and $\lambda \in K$). Here we claim that

$$\mathrm{Sp}(w_1, \ldots, w_i, \ldots, w_m) = \mathrm{Sp}(w_1, \ldots, w_i + \lambda w_j, \ldots, w_m),$$

where on the right-hand side the i-th member of the list has been replaced by $w_i + \lambda w_j$ and all other members are unchanged. Indeed, the inclusion

$$\mathrm{Sp}(w_1, \ldots, w_i + \lambda w_j, \ldots, w_m) \subseteq \mathrm{Sp}(w_1, \ldots, w_i, \ldots, w_m)$$

holds because every linear combination of the vectors on the left is obviously also a linear combination of $w_1, \dots, w_m$. But we also have the reverse inclusion

$$\mathrm{Sp}(w_1, \dots, w_i, \dots, w_m) \subseteq \mathrm{Sp}(w_1, \dots, w_i + \lambda w_j, \dots, w_m),$$

because $w_i = (w_i + \lambda w_j) - \lambda w_j$, and since $i \neq j$, the vector w_j itself is still one of the members of the list on the right. Consequently, whenever we write a linear combination of $w_1, \dots, w_m$, we may substitute this expression for w_i and obtain a linear combination of $w_1, \dots, w_i + \lambda w_j, \dots, w_m$. This proves the claim, and with it that an operation of Type 2 does not change $\mathrm{Rows}(M)$ either.

Finally, let A and B be row-equivalent. By Definition 6.9, there exists a finite sequence of matrices

$$A = A_0 \longrightarrow A_1 \longrightarrow \cdots \longrightarrow A_{r-1} \longrightarrow A_r = B$$

such that A_{i+1} is obtained from A_i by a single elementary row operation for every $0 \leq i \leq r-1$. By what we have just proved, each of these steps leaves the row space unchanged, and therefore

$$\mathrm{Rows}(A) = \mathrm{Rows}(A_0) = \mathrm{Rows}(A_1) = \cdots = \mathrm{Rows}(A_r) = \mathrm{Rows}(B).$$

Q.E.D.

Important remark 13.1: Row operations **do** change the column space, as seen by $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. However, they **preserve** the linear relations between columns: $\sum \alpha_i w_i = 0 \iff \sum \alpha_i w_i' = 0$.

This second assertion carries the entire weight of Theorem 13.5 below, which transfers a statement about the echelon form back to the original matrix. The notes do not prove it, so we record it here, together with the two consequences in which it is actually used.

AI-Lemma 13.1 (*Row Operations Preserve the Relations Between Columns*): Let $A, A' \in \mathcal{M}_{m \times n}(K)$ be row-equivalent matrices, and denote by $w_1, \dots, w_n$ the columns of A and by $w_1', \dots, w_n'$ the columns of A'. Then, for all scalars $\alpha_1, \dots, \alpha_n \in K$,

$$\alpha_1 w_1 + \cdots + \alpha_n w_n = 0 \quad \text{if and only if} \quad \alpha_1 w_1' + \cdots + \alpha_n w_n' = 0.$$

Consequently, for every subset of indices $S \subseteq \{1, \dots, n\}$:

(a) the columns $(w_i)_{i \in S}$ are linearly independent if and only if the columns $(w_i')_{i \in S}$ are linearly independent;

(b) for every index j, we have $w_j \in \mathrm{Sp}\left((w_i)_{i \in S}\right)$ if and only if $w_j' \in \mathrm{Sp}\left((w_i')_{i \in S}\right)$.

Proof:

Writing $x := (\alpha_1, \dots, \alpha_n)^{\mathsf{T}}$, the equation $\alpha_1 w_1 + \cdots + \alpha_n w_n = 0$ says precisely that $Ax = 0$, and likewise $\alpha_1 w_1' + \cdots + \alpha_n w_n' = 0$ says precisely that $A'x = 0$. Performing on $(A \mid 0)$ the very sequence of elementary row operations that leads from A to A' produces $(A' \mid 0)$, since a zero column stays a zero column under any row operation. So the two augmented matrices are row-equivalent, and Theorem 6.10 gives the two homogeneous systems the same solution set. This is the asserted equivalence.

For **(a)**, a linear relation among the columns indexed by S is the same thing as a linear relation among all n columns whose coefficients vanish outside S. The equivalence just proved matches such relations for A with such relations for A' coefficient by coefficient, so one family admits a non-trivial relation exactly when the other does.

For **(b)**, an expression $w_j = \sum_{i \in S} \gamma_i w_i$ is the relation $\sum_{i \in S} \gamma_i w_i - w_j = 0$, which again has vanishing coefficients outside $S \cup \{j\}$. Applying the equivalence to it, and reading the resulting relation

in the other direction, gives $w'_j = \sum_{i\in S} \gamma_i w'_i$, with the same coefficients γ_i; and symmetrically.
Q.E.D.

Both spaces now have a dimension, and these two numbers deserve names of their own.

Definition 13.4 (*Row Rank and Column Rank*): Let $A \in \mathcal{M}_{m\times n}(K)$. We define the "*row rank*" and the "*column rank*" of A as

$$\operatorname{rank}_{\text{row}}(A) := \dim \operatorname{Rows}(A), \qquad \operatorname{rank}_{\text{col}}(A) := \dim \operatorname{Cols}(A).$$

Remark 13.1: The notes flag right here that these two numbers always coincide, $\operatorname{rank}_{\text{row}}(A) = \operatorname{rank}_{\text{col}}(A)$, and postpone the proof. In this chapter the equality already falls out of the two basis theorems below, in Theorem 13.9; we return to it in Theorem 17.c.1, where it is proved once more, this time from the theory of linear maps and without any appeal to Gauss elimination.

Constructing Bases

Theorem 13.5 (*Basis for the Column Space*): The columns of A that correspond to the *"pivot columns"* of its row-reduced echelon form B constitute a basis for $\operatorname{Cols}(A)$.

Proof:

By the principle that row operations preserve the linear dependence relations between columns (AI-Lemma 13.1), we have:

$$\sum_{i=1}^{n} \alpha_i w_i = 0 \iff \sum_{i=1}^{n} \alpha_i w'_i = 0.$$

Thus, a set of columns in A is linearly independent (or spans the column space) if and only if the corresponding set of columns in B possesses the same property in $\operatorname{Cols}(B)$.

Consider the row-reduced echelon matrix $B \in \mathcal{M}_{m\times n}(K)$ with r pivots. We visualize its block structure and the specific indices of its pivot columns $j_1, j_2, \ldots, j_r$:

$$B = \left(\begin{array}{ccccccccc}
 & {\scriptstyle\text{col}}\, j_1 & & {\scriptstyle\text{col}}\, j_2 & & {\scriptstyle\text{col}}\, j_r & & & \\
 & \downarrow & & \downarrow & & \downarrow & & & \\
\ldots & \mathbf{1} & * & \mathbf{0} & * & \mathbf{0} & * & \ldots & \leftarrow 1 \\
\ldots & \mathbf{0} & \ldots & \mathbf{1} & \ldots & \mathbf{0} & * & \ldots & \leftarrow 2 \\
 & \vdots & & \vdots & \ddots & \vdots & \vdots & & \vdots \\
\ldots & \mathbf{0} & \ldots & \mathbf{0} & \ldots & \mathbf{1} & * & \ldots & \leftarrow r \\
\hline
 & 0 & & 0 & & 0 & 0 & \ldots & \\
 & \vdots & & \vdots & & \vdots & \vdots & \ddots & \\
 & 0 & & 0 & & 0 & 0 & \ldots &
\end{array}\right) \tag{13.2}$$

In this echelon form, the pivot columns are precisely the first r standard basis vectors of K^m:

$$e_1 = \begin{pmatrix}1\\0\\\vdots\\0\\\vdots\\0\end{pmatrix}, \quad e_2 = \begin{pmatrix}0\\1\\\vdots\\0\\\vdots\\0\end{pmatrix}, \quad \ldots, \quad e_r = \begin{pmatrix}0\\\vdots\\1\\0\\\vdots\\0\end{pmatrix}$$

where for each e_i, the entry 1 is located at the i-th row. Because B is in row-reduced echelon form, every entry in every column w'_j that lies below the r-th row must be zero. This implies that:

$$w'_j \in K^r \times \{0\}^{m-r} = \left\{(x_1, \ldots, x_r, 0, \ldots, 0)^\mathsf{T} \in K^m\right\}.$$

Since the standard basis vectors $e_1, \ldots, e_r$ are clearly linearly independent and they span the subspace $K^r \times \{0\}^{m-r}$, we establish the following chain of inclusions:

$$K^r \times \{0\}^{m-r} = \mathrm{Sp}(e_1, \ldots, e_r) \subseteq \mathrm{Cols}(B) \subseteq K^r \times \{0\}^{m-r}.$$

This double inclusion forces the equality $\mathrm{Cols}(B) = \mathrm{Sp}(e_1, \ldots, e_r) = K^r \times \{0\}^{m-r}$. Consequently, the pivot columns of B form a basis for its column space. By the preservation of linear relations, the corresponding columns in the original matrix A form a basis for $\mathrm{Cols}(A)$: they are linearly independent by part **(a)** of AI-Lemma 13.1, and every other column of A lies in their span by part **(b)**, since this is so for B. **Q.E.D.**

Theorem 13.6 (*Basis for the Row Space*): The non-zero rows of the row-reduced echelon matrix B form a basis for $\mathrm{Rows}(A)$.

Proof:

By Proposition 13.3, $\mathrm{Rows}(A) = \mathrm{Rows}(B)$. Let r be the number of pivots in B. The non-zero rows of B are $w'_1, \ldots, w'_r$. By definition, they span $\mathrm{Rows}(B)$.

To show linear independence, consider the indices of the pivot columns $j_1, \ldots, j_r$. For each row $i \in \{1, \ldots, r\}$, the entry in the j_i-th column is 1, while in all other rows $k \neq i$, the entry in the j_i-th column is 0. Suppose $\sum_{i=1}^r \alpha_i w'_i = 0$. Looking at the j_k-th component of this vector sum, we have:

$$\sum_{i=1}^r \alpha_i B_{i,j_k} = \alpha_k \cdot 1 + \sum_{i \neq k} \alpha_i \cdot 0 = \alpha_k = 0.$$

Since this holds for all k, all coefficients must be zero. Thus, the non-zero rows are linearly independent and form a basis. **Q.E.D.**

Corollary 13.7 (*Rank Invariance*): The number of pivots is independent of the sequence of row operations, as it always equals $\dim \mathrm{Rows}(A)$.

Proof:

By Theorem 13.6, the non-zero rows of any row-reduced echelon form B of A form a basis for $\mathrm{Rows}(A)$. The number of such rows is exactly the number of pivots. Since the dimension of a vector space is a fixed property, the number of pivots must be the same regardless of the specific row-reduction process used to reach an echelon form. **Q.E.D.**

The row basis theorem is not merely a structural statement; read in the right direction, it is a recipe. It answers the practical question of how to produce a basis for a subspace that has been handed to us as a span.

Corollary 13.8 (*Algorithm for Finding a Basis*): Let $U := \mathrm{Sp}(u_1, \ldots, u_m) \subseteq K^n$ be the subspace spanned by a list of vectors $u_1, \ldots, u_m \in K^n$, which we regard as row vectors. Form the matrix

$$A := \begin{pmatrix} — & u_1 & — \\ & \vdots & \\ — & u_m & — \end{pmatrix} \in \mathcal{M}_{m\times n}(K),$$

and apply Gauss elimination to A until a row-reduced echelon matrix M is reached. Then the non-zero rows of M form a basis for U.

Proof:

By construction, $U = \mathrm{Rows}(A)$. The matrix M is row-equivalent to A, so $\mathrm{Rows}(A) = \mathrm{Rows}(M)$ by Proposition 13.3, and the non-zero rows of M form a basis for that space by Theorem 13.6.

Q.E.D.

Theorem 13.9 (*The Rank Equality*): For any $A \in \mathcal{M}_{m\times n}(K)$, $\dim \mathrm{Rows}(A) = \dim \mathrm{Cols}(A)$, or equivalently, in the terminology of Definition 13.4, $\mathrm{rank}_{\mathrm{row}}(A) = \mathrm{rank}_{\mathrm{col}}(A)$. This shared dimension is the "*rank*" of A, denoted $\mathrm{rank}(A)$.

Proof:

Let r be the number of pivots in the RREF of A. By Theorem 13.5, $\dim \mathrm{Cols}(A) = r$. By Theorem 13.6, $\dim \mathrm{Rows}(A) = r$. Thus, the dimensions are exactly equal. **Q.E.D.**

The Transpose of a Matrix

The row space and the column space of a matrix are built from two different readings of the same array of scalars. The operation that turns one reading into the other is the transpose, which we introduce here and will use constantly from now on.

Definition 13.10 (*Transpose of a Matrix*): Let $A = (a_{ij}) \in \mathcal{M}_{m\times n}(K)$. The "*transposed matrix*" of A is the matrix $A^{\mathsf{T}} := (b_{ij}) \in \mathcal{M}_{n\times m}(K)$ whose entries are given by

$$b_{ij} := a_{ji} \qquad \text{for all } 1 \le i \le n,\ 1 \le j \le m.$$

Notation 13.1: The transposed matrix is sometimes also denoted by A^t.

Example 13.2 (*Transposing a Rectangular Matrix*): Transposing turns the rows of a matrix into columns and its columns into rows:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}^{\mathsf{T}} = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}.$$

Proposition 13.11 (*Simple Properties of the Transpose*): For all matrices $A, B \in \mathcal{M}_{m\times n}(K)$ and every scalar $\lambda \in K$, the following hold:

(a) $(A + B)^{\mathsf{T}} = A^{\mathsf{T}} + B^{\mathsf{T}}$.
(b) $(\lambda A)^{\mathsf{T}} = \lambda \cdot A^{\mathsf{T}}$.
(c) $(A^{\mathsf{T}})^{\mathsf{T}} = A$.
(d) $\mathrm{Rows}(A^{\mathsf{T}}) = \mathrm{Cols}(A)$ and $\mathrm{Cols}(A^{\mathsf{T}}) = \mathrm{Rows}(A)$.

Proof:

(a) The entry of $(A + B)^{\mathsf{T}}$ in position (i, j) is the entry of $A + B$ in position (j, i), namely $a_{ji} + b_{ji}$, which is exactly the entry of $A^{\mathsf{T}} + B^{\mathsf{T}}$ in position (i, j).
(b) Likewise, the entry of $(\lambda A)^{\mathsf{T}}$ in position (i, j) is λa_{ji}, which is the entry of $\lambda \cdot A^{\mathsf{T}}$ in position (i, j).
(c) Transposing twice returns the entry in position (i, j) to a_{ij}.
(d) By Definition 13.10, the i-th row of A^{T} consists of the entries $a_{1i}, \ldots, a_{mi}$, that is, it is the i-th column of A written as a row vector. The rows of A^{T} are therefore precisely the columns of A, so the two spans agree: $\mathrm{Rows}(A^{\mathsf{T}}) = \mathrm{Cols}(A)$. The second identity follows by applying the first one to A^{T} and using **(c)**.

Q.E.D.

Remark 13.2: Statement **(d)** is the reason the transpose is the natural bookkeeping device for the two ranks. Taking dimensions in Proposition 13.11 **(d)** gives

$$\operatorname{rank}_{\text{row}}(A^{\mathsf{T}}) = \operatorname{rank}_{\text{col}}(A), \qquad \operatorname{rank}_{\text{col}}(A^{\mathsf{T}}) = \operatorname{rank}_{\text{row}}(A),$$

so every statement about row rank is a statement about the column rank of the transposed matrix, and conversely.

AI-Note: Four pieces of the notes were missing from this transcript and are restored above: the case distinction in the answer to Questions 1 and 2, where an A' without zero rows forces $\operatorname{Sp}(w_1, \dots, w_n) = K^m$; the definition of row rank and column rank (Definition 13.4), used from Theorem 17.c.1 onwards but never introduced; Corollary 13.8, which the notes state as an algorithm; and the whole closing section on the transpose, whose notation A^{T} the transcript already used in this chapter and in fourteen later ones without ever defining it. The proof of Proposition 13.3 had been compressed to a single sentence; the notes treat the three types of row operation separately and prove the Type 2 case by double inclusion, so that argument is restored around the sentence that was there. Theorem 13.1 is the computational counterpart of Theorem 12.2, which is how the notes present it; the description of the span by the vanishing of b'' is put back into its proof.

Two things go beyond the notes. AI-Lemma 13.1 is not in Biran's text: the transcript asserts the preservation of linear relations between columns in an important remark and then rests Theorem 13.5 on it, so it is stated and proved here as an AI-Lemma, on the strength of Theorem 6.10. And the notes defer $\operatorname{rank}_{\text{row}}(A) = \operatorname{rank}_{\text{col}}(A)$ to a later lecture, while this chapter already reaches it in Theorem 13.9; the argument is sound and is kept, with the relation to Theorem 17.c.1 recorded in a remark. In the worked span computation on Page 79 the elimination now carries Prof. Biran's arrow labels and includes the normalisation step $-1 \cdot E_2 \to E_2$, without which the displayed chain never reaches a row-reduced echelon matrix.

Sums of Vector Spaces (Chapter 14)

Definition 14.1 (*Sum of Subspaces*): Let V be a vector space over K, and let $U, W \subseteq V$ be linear subspaces. We define the "*sum*" of U and W, denoted by $U + W$, as the set:

$$U + W := \{u + w \mid u \in U, w \in W\}.$$

In a similar way, we can define sums of more than two subspaces. If $U_1, \ldots, U_r \subseteq V$ are subspaces, we define their total sum as:

$$U_1 + \cdots + U_r := \{u_1 + \cdots + u_r \mid u_1 \in U_1, \ldots, u_r \in U_r\}.$$

Proposition 14.2 (*Properties of Sums*): Let $U, W \subseteq V$ be subspaces. Then, the following properties hold:

(a) $U + W = \mathrm{Sp}(U \cup W)$. In particular, $U + W \subseteq V$ is a subspace, and U and W are themselves subspaces of $U + W$.

(b) If U and W are finite-dimensional, then so is $U + W$. Moreover, one can systematically find a basis for $U + W$ by following these sequential steps: first, choose a basis $(v_1, \ldots, v_k)$ for $U \cap W$ (where $k = \dim(U \cap W)$). Second, extend this to a basis $(v_1, \ldots, v_k, u_1, \ldots, u_{\ell-k})$ for U (where $\ell = \dim U$). Third, extend the initial intersection basis to a basis $(v_1, \ldots, v_k, w_1, \ldots, w_{m-k})$ for W (where $m = \dim W$). Finally, the combined elements

$$(v_1, \ldots, v_k, u_1, \ldots, u_{\ell-k}, w_1, \ldots, w_{m-k})$$

form a basis for $U + W$.

(c) If U and W are finite-dimensional, the dimensions satisfy the "*dimension formula*"

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Proof:

(a) Clearly $U + W \subseteq \mathrm{Sp}(U \cup W)$, by the characterisation of $\mathrm{Sp}(U \cup W)$ as the set of all linear combinations of elements from $U \cup W$. But we also have $\mathrm{Sp}(U \cup W) \subseteq U + W$. Indeed, if $v \in \mathrm{Sp}(U \cup W)$, then

$$v = \underbrace{\sum_{i=1}^{s} a_i u_i}_{\in U} + \underbrace{\sum_{j=1}^{r} b_j w_j}_{\in W},$$

with $a_i, b_j \in K$, $u_i \in U$ and $w_j \in W$, which implies that $v \in U + W$. This proves $\mathrm{Sp}(U \cup W) \subseteq U + W$. Putting all together, we have $U + W = \mathrm{Sp}(U \cup W)$. The rest of the statements in **(a)** follow immediately.

(b) By definition, the subspace $U + W$ is generated by the combined vectors $(v_1, \ldots, v_k, u_1, \ldots, u_{\ell-k}, w_1, \ldots, w_{m-k})$. It remains to show that these vectors are linearly independent.

Suppose there exists a linear combination equal to the zero vector:

$$\alpha_1 v_1 + \cdots + \alpha_k v_k + \beta_1 u_1 + \cdots + \beta_{\ell-k} u_{\ell-k} + \gamma_1 w_1 + \cdots + \gamma_{m-k} w_{m-k} = 0. \tag{14.1}$$

We can rearrange this equation to isolate the terms belonging to U and W:

$$\underbrace{\beta_1 u_1 + \cdots + \beta_{\ell-k} u_{\ell-k}}_{\in U} = \underbrace{-\alpha_1 v_1 - \cdots - \alpha_k v_k - \gamma_1 w_1 - \cdots - \gamma_{m-k} w_{m-k}}_{\in W}.$$

Let us denote this common vector by x. Since the left side is entirely in U and the right side is entirely in W, we have $x \in U$ and $x \in W$, which implies $x \in U \cap W$.

Because $(v_1, \ldots, v_k)$ is a basis for $U \cap W$, there must exist scalars $\delta_1, \ldots, \delta_k \in K$ such that:

$$x = \delta_1 v_1 + \cdots + \delta_k v_k.$$

Equating this with our original expression for x in U, we obtain:

$$\beta_1 u_1 + \cdots + \beta_{\ell-k} u_{\ell-k} = \delta_1 v_1 + \cdots + \delta_k v_k,$$

or, after moving everything to one side,

$$\sum_{i=1}^{\ell-k} \beta_i u_i - \sum_{j=1}^{k} \delta_j v_j = 0.$$

However, the vectors $(v_1, \dots, v_k, u_1, \dots, u_{\ell-k})$ form a basis for U, meaning they are strictly linearly independent. Consequently, all coefficients in this sum must be zero. In particular, $\beta_1 = \cdots = \beta_{\ell-k} = 0$.

Substituting these zero values back into our initial equation (14.1), the terms involving u_i vanish, leaving:

$$\alpha_1 v_1 + \cdots + \alpha_k v_k + \gamma_1 w_1 + \cdots + \gamma_{m-k} w_{m-k} = 0.$$

Since the vectors $(v_1, \dots, v_k, w_1, \dots, w_{m-k})$ form a basis for W, they are linearly independent. This immediately implies that $\alpha_1 = \cdots = \alpha_k = 0$ and $\gamma_1 = \cdots = \gamma_{m-k} = 0$.

Because all coefficients α_i, β_j, and γ_k have been forced to zero, the combined family of vectors is linearly independent, and therefore constitutes a valid basis for $U + W$.

Statement **(c)** now follows by simply counting the basis vectors produced in **(b)**. The basis consists of the k vectors v_i, the $\ell - k$ vectors u_i, and the $m - k$ vectors w_j, and therefore,

$$\begin{aligned} \dim(U+W) &= k + (\ell - k) + (m - k) \\ &= \ell + m - k \\ &= \dim U + \dim W - \dim(U \cap W). \end{aligned}$$

Q.E.D.

Exercise 14.1 (*The Combined Family Spans the Sum*): The proof above verifies that the family

$$(v_1, \dots, v_k, u_1, \dots, u_{\ell-k}, w_1, \dots, w_{m-k})$$

is linearly independent. Check that it also spans $U + W$, which the notes leave to the reader.

Corollary 14.3 (*Equivalent Conditions for Trivial Intersection*): Let V be a vector space, and let $U, W \subseteq V$ be two finite-dimensional linear subspaces. The following statements are equivalent:

- **(a)** $\dim(U + W) = \dim U + \dim W$.
- **(b)** $\dim(U \cap W) = 0$.
- **(c)** $U \cap W = \{0_V\}$.
- **(d)** For every $v \in U + W$, there exist unique vectors $u \in U$ and $w \in W$ such that $v = u + w$.
- **(e)** The equality $0 = u + w$ with $u \in U$ and $w \in W$ implies that $u = 0$ and $w = 0$.

Proof:

The equivalence **(a)** $\iff$ **(b)** follows from the previous Proposition 14.2 **(c)**, whose dimension formula reads $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$. And **(b)** $\iff$ **(c)** holds because there is only one linear subspace of dimension 0, namely the space $\{0_V\}$.

The remaining three statements are tied to **(c)** by the cycle of implications **(c)** $\implies$ **(d)** $\implies$ **(e)** $\implies$ **(c)**.

(c) $\implies$ **(d):** Suppose that $v \in U + W$ can be written in two ways: $v = u + w = u' + w'$, where $u, u' \in U$ and $w, w' \in W$. Rearranging this equation yields $u - u' = w' - w$. The left side is in U and the right side is in W, which means the vector $u - u'$ resides in the intersection $U \cap W$. Since we assume $U \cap W = \{0_V\}$, it follows that $u - u' = 0$ and $w' - w = 0$. Therefore, $u = u'$ and $w = w'$, proving uniqueness.

(d) $\implies$ (e): Consider the zero vector, which can obviously be written as $0 = 0_U + 0_W$. If we also have $0 = u + w$ for some $u \in U$ and $w \in W$, the assumed uniqueness in statement **(d)** immediately forces $u = 0$ and $w = 0$.

(e) $\implies$ (c): Let $x \in U \cap W$. We can express the zero vector as $0 = x + (-x)$. Since $x \in U \cap W$, we know $x \in U$ and $-x \in W$. If statement **(e)** holds, this equality strictly implies $x = 0$. Consequently, the only element in the intersection is the zero vector, so $U \cap W = \{0_V\}$.

Q.E.D.

Definition 14.4 (*Complement of a Subspace*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. A subspace $W \subseteq V$ is called a "*complement*" of U if it satisfies the following two conditions:

$$U + W = V \quad \text{and} \quad U \cap W = \{0_V\}.$$

Remark 14.1: Note that if W is a complement of U, then U and W satisfy all of the five equivalent statements derived from Corollary 14.3.

Proposition 14.5 (*Existence of a Complement*): Let V be a finite-dimensional vector space, and let $U \subseteq V$ be a subspace. Then, there exists a subspace $W \subseteq V$ which is a complement to U.

Proof:

Choose a basis $\mathcal{B}_U = (u_1, \ldots, u_\ell)$ for U, where $\ell = \dim U$. By the Steinitz Exchange Lemma, we can extend this to a full basis for V:

$$\mathcal{B}_V = (u_1, \ldots, u_\ell, w_1, \ldots, w_m),$$

where $\ell + m = \dim V$. We define $W := \mathrm{Sp}(w_1, \ldots, w_m)$.

We claim that W is a valid complement of U. Indeed, since the combined basis spans the entirety of V, it is clear that $U + W = V$. To see that $U \cap W = \{0\}$, let $x \in U \cap W$. Because $x \in U$, we can write $x = \sum_{i=1}^{\ell} a_i u_i$. Because $x \in W$, we can also write $x = \sum_{j=1}^{m} b_j w_j$. Equating these expressions yields:

$$\sum_{i=1}^{\ell} a_i u_i = \sum_{j=1}^{m} b_j w_j \implies \sum_{i=1}^{\ell} a_i u_i - \sum_{j=1}^{m} b_j w_j = 0.$$

Since $\mathcal{B}_V$ is a basis for V, all of its constituent vectors are strictly linearly independent. This forces all scalar coefficients to be zero: $a_i = 0$ for all i and $b_j = 0$ for all j. Thus, $x = 0$, proving that $U \cap W = \{0\}$. **Q.E.D.**

Important remark 14.1: If $U \subsetneq V$, then there are many different subspaces $W \subseteq V$ which are complements of U. In general, there is no canonical choice for a complement W.

Solutions to the Exercises

Proof that the Combined Family Spans $U + W$ (on Page 87):

Write $\mathcal{S} := (v_1, \ldots, v_k, u_1, \ldots, u_{\ell-k}, w_1, \ldots, w_{m-k})$ for the combined family.

Every member of $\mathcal{S}$ lies in $U + W$: the vectors v_i and u_i lie in U, the vectors w_j lie in W, and both U and W are contained in $U + W$ by Proposition 14.2 **(a)**. Since $U + W$ is a subspace, it follows that $\mathrm{Sp}(\mathcal{S}) \subseteq U + W$.

For the reverse inclusion, let $x \in U + W$ and write $x = u + w$ with $u \in U$ and $w \in W$. Because $(v_1, \ldots, v_k, u_1, \ldots, u_{\ell-k})$ is a basis for U, there are scalars with

$$u = \sum_{i=1}^{k} \alpha_i v_i + \sum_{i=1}^{\ell-k} \beta_i u_i,$$

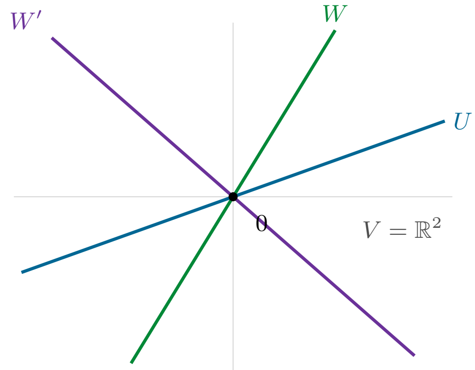


Figure 14.1: Two complements of the same subspace. Inside $V = \mathbb{R}^2$, take for U any line through the origin. Every other line through the origin meets U only in 0 and together with U spans the whole plane, so both W and W' are complements of U — and so is any of the infinitely many further lines through the origin.

and because $(v_1, \ldots, v_k, w_1, \ldots, w_{m-k})$ is a basis for W, there are scalars with

$$w = \sum_{i=1}^{k} \alpha_i' v_i + \sum_{j=1}^{m-k} \gamma_j w_j .$$

Adding the two expressions writes x as a linear combination of the members of $\mathcal{S}$, the coefficient of v_i being $\alpha_i + \alpha_i'$. Hence $U + W \subseteq \mathrm{Sp}(\mathcal{S})$, and the two inclusions give $\mathrm{Sp}(\mathcal{S}) = U + W$. **Q.E.D.**

AI-Note: Restored from the notes: the proof of statement **(a)** of Proposition 14.2, which the transcript skipped with the words *"We will explicitly prove statement (b)"*; the two dimension criteria in Corollary 14.3; and the closing remark on Page 88, with the sketch that accompanies it in the notes redrawn as Figure 14.1.

Corollary 14.3 deserves a word. The notes state **five** equivalent conditions, and the transcript kept only four of them: the two dimension criteria, $\dim(U+W) = \dim U + \dim W$ and $\dim(U \cap W) = 0$, were lost, and in their place stood a statement with no counterpart in the notes, namely that every $x \in U \cap W$ can be written uniquely as $x = u + w$. That statement is true, but it is a garbled duplicate of statement **(e)**, and it has been dropped in favour of Biran's list. That something was lost here is visible in the transcript itself: the remark following Definition 14.4 already spoke of *"the five equivalent statements"*. The hypothesis that U and W be finite-dimensional is the notes' own and is needed for the two dimension criteria to mean anything; the transcript had dropped it as well.

The notes prove the independence claim in Proposition 14.2 **(b)** by eliminating the W-coefficients first, whereas the transcript isolates the U-part first. The two routes are mirror images of one another and equally correct, so the transcript's version is kept. The notes also mark the spanning half of that statement as an exercise, *"exc. check this!"*; it is stated as an exercise on Page 87 and solved at the end of the chapter.

Linear Maps (Chapter 15)

Introduction to Linear Maps (Section 15.a)

Linear maps, also known as linear transformations, are homomorphisms of vector spaces. They are the primary objects of study in linear algebra, as they preserve the underlying algebraic structure of vector spaces.

Definition 15.a.1 (*Linear Map*): Let V and W be vector spaces over a field K. A map $T : V \to W$ is called a "*linear map*" if it satisfies the following two conditions:

(a) For all $v_1, v_2 \in V$, $T(v_1 + v_2) = T(v_1) + T(v_2)$.

(b) For all $\alpha \in K$ and $v \in V$, $T(\alpha \cdot v) = \alpha \cdot T(v)$.

Remark 15.1 (*Notation and Terminology*): In many contexts, we omit the parentheses and write Tv instead of $T(v)$. The conditions outlined in Definition 15.a.1 essentially state that $T : V \to W$ is a map that respects the operations defined on the vector spaces; specifically, the addition and scalar multiplication in V are mapped perfectly to the corresponding operations in W.

We denote the set of all linear maps from V to W by $\mathrm{Hom}_K(V, W)$, or simply $\mathrm{Hom}(V, W)$. While some authors use alternative notations such as $\mathrm{Lin}(V, W)$ or $\mathrm{hom}_K(V, W)$, $\mathrm{Hom}_K(V, W)$ remains the standard academic convention.

Exercise 15.1 (*Linearity Criterion*): Prove that a map $T : V \to W$ is linear if and only if for all $\alpha_1, \alpha_2 \in K$ and $v_1, v_2 \in V$:

$$T(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 T(v_1) + \alpha_2 T(v_2).$$

Example 15.1 (*Fundamental Examples of Linear Maps*):

(a) The identity map $\mathrm{id}_V : V \to V$, defined by $\mathrm{id}_V(v) := v$ for all $v \in V$.

(b) The zero map $0 : V \to W$, defined by $0(v) := 0_W$ for all $v \in V$.

Clearly, these initial examples are elementary, yet they serve as vital benchmarks.

(c) The derivative map $D : K[x] \to K[x]$. For any polynomial $p(x) = \sum_{k=0}^{n} a_k x^k$, we define its image as:

$$D(p(x)) := \sum_{k=1}^{n} k a_k x^{k-1}.$$

This map is linear, as the derivative of a linear combination of polynomials is simply the same linear combination of their derivatives.

(d) Consider the vector space $C([a, b])$ consisting of continuous functions $f : [a, b] \to \mathbb{R}$. The definite integral defines a linear map $T : C([a, b]) \to \mathbb{R}$, given by:

$$T(f) := \int_a^b f(x)\, dx.$$

(e) The shift map $S : K^\infty \to K^\infty$, defined on infinite sequences as:

$$S((a_1, a_2, a_3, \dots)) := (a_2, a_3, a_4, \dots).$$

Example 15.2 (*Matrix Transformation*): This is a central example in finite-dimensional linear algebra. Let $A \in \mathcal{M}_{m\times n}(K)$ be a matrix. We define a map $T_A : K^n \to K^m$ via matrix-vector multiplication:

$$T_A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} := A \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Recall from matrix arithmetic that $A \cdot (\alpha \cdot x) = \alpha \cdot (A \cdot x)$ and $A \cdot (x + y) = A \cdot x + A \cdot y$. Consequently, T_A is a linear map.

Proposition 15.a.2 (*Basic Properties of Linear Maps*)**:** Let V and W be vector spaces over a field K, and let $T : V \to W$ be a linear map. Then, the following properties hold:

(a) For all $n \in \mathbb{N}$, $\alpha_1, \ldots, \alpha_n \in K$, and $v_1, \ldots, v_n \in V$, we have:
$$T\left(\sum_{i=1}^{n} \alpha_i v_i\right) = \sum_{i=1}^{n} \alpha_i T(v_i).$$

(b) $T(0_V) = 0_W$.

Proof:

(a) The first statement follows directly from the definition of linearity, Definition 15.a.1, by applying induction on the number of vectors n.

(b) To prove the second statement, we evaluate the image of the zero vector:
$$0_W = T(0_V) - T(0_V) = T(0_V - 0_V) = T(0_V).$$
Therefore, any linear map must map the zero vector of the domain directly to the zero vector of the codomain.

Q.E.D.

Theorem 15.a.3 (*Existence and Uniqueness of Linear Maps on a Basis*)**:** Let V be a finite-dimensional vector space, and let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V. Given any arbitrary sequence of vectors $w_1, \ldots, w_n \in W$, there exists a unique linear map $T : V \to W$ such that:
$$T(v_i) = w_i, \quad \forall\, 1 \le i \le n.$$

Proof:

Existence: Let $v \in V$ be an arbitrary vector. Since $\mathcal{B}$ is a basis, there exist unique scalars $a_1, \ldots, a_n \in K$ such that $v = \sum_{i=1}^{n} a_i v_i$. We define the map T by setting:
$$T(v) := \sum_{i=1}^{n} a_i w_i. \tag{15.1}$$
Because the coordinates a_i are uniquely determined for each v, the map T is well-defined.

We claim that T is linear. Indeed, let $v, v' \in V$. There exist $a_1, \ldots, a_n \in K$ with $v = a_1 v_1 + \cdots + a_n v_n$ and $a'_1, \ldots, a'_n \in K$ with $v' = a'_1 v_1 + \cdots + a'_n v_n$, so that
$$v + v' = (a_1 + a'_1) v_1 + \cdots + (a_n + a'_n) v_n.$$
By our definition of T, this gives
$$T(v+v') = (a_1+a'_1)w_1+\cdots+(a_n+a'_n)w_n = (a_1 w_1+\cdots+a_n w_n)+(a'_1 w_1+\cdots+a'_n w_n) = T(v)+T(v').$$
Also, if $\alpha \in K$, then $\alpha v = \alpha a_1 v_1 + \cdots + \alpha a_n v_n$, and therefore,
$$T(\alpha v) = \alpha a_1 w_1 + \cdots + \alpha a_n w_n = \alpha(a_1 w_1 + \cdots + a_n w_n) = \alpha T(v).$$
This proves that $T : V \to W$ is a linear map.

We claim next that $T(v_i) = w_i$ for all $1 \le i \le n$. Indeed, writing $v_i = 0 \cdot v_1 + \cdots + 1 \cdot v_i + \cdots + 0 \cdot v_n$ exhibits the coordinates of v_i, whence
$$T(v_i) = 0 \cdot w_1 + \cdots + 1 \cdot w_i + \cdots + 0 \cdot w_n = w_i.$$
This concludes the proof of the existence of T.

Uniqueness: Suppose there exists another linear map $S : V \to W$ such that $S(v_i) = w_i$ for all $1 \le i \le n$. For any $v = \sum_{i=1}^{n} a_i v_i \in V$, the linearity of S dictates:
$$S(v) = S\left(\sum_{i=1}^{n} a_i v_i\right) = \sum_{i=1}^{n} a_i S(v_i) = \sum_{i=1}^{n} a_i w_i = T(v).$$

Since $S(v) = T(v)$ for all $v \in V$, the maps are functionally identical, establishing that $S = T$.
Q.E.D.

Example 15.3 (*Linear Maps on Coordinate Spaces*): Consider the coordinate space K^n equipped with the standard basis $\mathcal{E} = (e_1, \dots, e_n)$. Let $w_1, \dots, w_n \in K^m$ be any vectors. Theorem 15.a.3 guarantees a unique linear map $T : K^n \to K^m$ such that $T(e_i) = w_i$ for all $1 \le i \le n$.

Question 15.1: Can we provide an explicit matrix description for this map T?

Answer 15.1: Let $A \in \mathcal{M}_{m\times n}(K)$ be the matrix whose j-th column is precisely the vector w_j:

$$A = \begin{pmatrix} | & & | \\ w_1 & \dots & w_n \\ | & & | \end{pmatrix}.$$

The linear transformation $T_A : K^n \to K^m$ defined by $v \mapsto A \cdot v$ satisfies $T_A(e_j) = w_j$ for all j. Indeed, multiplying A by e_i picks out the i-th column of A:

$$T_A(e_i) = \begin{pmatrix} | & & | & & | \\ w_1 & \dots & w_i & \dots & w_n \\ | & & | & & | \end{pmatrix} \cdot \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} | \\ w_i \\ | \end{pmatrix},$$

where the 1 on the right sits in entry i. Due to the absolute uniqueness property established in Theorem 15.a.3, we immediately conclude that $T = T_A$.

Lemma 15.a.5 (*Matrix Representation of Linear Maps*): For every linear transformation $T : K^n \to K^m$, there exists a unique matrix $A \in \mathcal{M}_{m\times n}(K)$ such that $T = T_A$.

Proof:

Let $w_i := T(e_i)$ for each $i \in \{1, \dots, n\}$. As demonstrated in the preceding example, the matrix A formed by using these specific vectors as columns defines a linear map T_A that perfectly agrees with T on the standard basis $\mathcal{E}$. Because $\mathcal{E}$ is a basis for K^n, Theorem 15.a.3 strictly implies that T and T_A must be the exact same map.

The matrix is moreover unique, which the notes do not claim but which costs one line: if $B \in \mathcal{M}_{m\times n}(K)$ also satisfies $T = T_B$, then the i-th column of B equals $B \cdot e_i = T_B(e_i) = T(e_i) = w_i$ for every i, so B and A have the same columns and $B = A$. **Q.E.D.**

Kernel, Image, and the Rank-Nullity Theorem (Section 15.b)

In this section, we explore two fundamental subspaces associated with every linear map: the kernel and the image. These subspaces provide deep insight into the structure of the transformation and ultimately lead us to one of the most celebrated results in linear algebra, the Rank-Nullity Theorem.

Proposition 15.b.1 (*Kernel and Image*): Let V and W be vector spaces over a field K, and let $T : V \to W$ be a linear map.

- **(a)** The set $\ker(T) := \{v \in V \mid T(v) = 0_W\} \subseteq V$ is a linear subspace of V. It is called the "*kernel*" of T.
- **(b)** The set $\mathrm{Im}(T) := \{T(v) \mid v \in V\} \subseteq W$ is a linear subspace of W. It is called the "*image*" of T.

Remark 15.2 (*Set-Theoretic Perspective*): Using standard notation from set theory, we can concisely write the kernel as the fiber over the zero vector, $\ker(T) = T^{-1}(\{0_W\})$, and the image as the range of the entire domain, $\operatorname{Im}(T) = T(V)$. (Note: The image is sometimes denoted as $\operatorname{Image}(T)$).

Proof of Proposition 15.b.1:

(a) To show that $\ker(T) \subseteq V$ is a subspace, let $a, b \in K$, and let $v_1, v_2 \in \ker(T)$. By the linearity of T, we have:

$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) = a \cdot 0_W + b \cdot 0_W = 0_W.$$

This implies that $av_1 + bv_2 \in \ker(T)$. Furthermore, since $T(0_V) = 0_W$, the zero vector $0_V \in \ker(T)$. Thus, the kernel is a valid subspace.

(b) To show that $\operatorname{Im}(T) \subseteq W$ is a subspace, let $a, b \in K$, and let $w_1, w_2 \in \operatorname{Im}(T)$. By definition, there exist vectors $v_1, v_2 \in V$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$. Therefore, we can write:

$$aw_1 + bw_2 = aT(v_1) + bT(v_2) = T(av_1 + bv_2).$$

Since $av_1 + bv_2 \in V$, it follows immediately that $aw_1 + bw_2 \in \operatorname{Im}(T)$. Moreover, since $T(0_V) = 0_W$, we have $0_W \in \operatorname{Im}(T)$. Thus, the image is a subspace.

Q.E.D.

Example 15.4 (*The Kernel of a Matrix Transformation*): Let $A \in \mathcal{M}_{m\times n}(K)$ and consider the linear map $T_A : K^n \to K^m$, recalling that $T_A(x) = A \cdot x$ for all $x \in K^n$. Then

$$\ker(T_A) = \{x \in K^n \mid A \cdot x = 0\},$$

which is precisely the solution set of the homogeneous linear system with coefficient matrix A. The kernel is therefore not a new object at all: it is the solution set of a homogeneous system, seen from the point of view of linear maps.

Proposition 15.b.2 (*Injectivity, Surjectivity, Kernel and Image*): Let $T : V \to W$ be a linear map. Then:

(a) T is injective if and only if $\ker(T) = \{0_V\}$.

(b) T is surjective if and only if $\operatorname{Im}(T) = W$.

Recall that a map is called "*bijective*" if it is both injective and surjective.

Proof:

Statement **(b)** is a restatement of the definition of surjectivity, since $\operatorname{Im}(T) = T(V)$, so there is nothing to prove. We turn to **(a)**.

First, suppose T is injective. If $v \in \ker(T)$, then $T(v) = 0_W$. Since we also know that $T(0_V) = 0_W$, the injectivity of T forces $v = 0_V$. Thus, $\ker(T) = \{0_V\}$.

Conversely, suppose that $\ker(T) = \{0_V\}$. Let $v_1, v_2 \in V$ such that $T(v_1) = T(v_2)$. By linearity, we obtain:

$$T(v_1 - v_2) = T(v_1) - T(v_2) = 0_W.$$

This implies that $v_1 - v_2 \in \ker(T)$. By our assumption, the only element in the kernel is the zero vector, and therefore, $v_1 - v_2 = 0_V$, which yields $v_1 = v_2$. Consequently, T is injective.

Q.E.D.

Exercise 15.2 (*Images and Preimages of Subspaces*): Let $T : V \to W$ be a linear map. Show that:

(a) if $V' \subseteq V$ is a linear subspace, then $T(V') \subseteq W$ is a linear subspace;

(b) if $W' \subseteq W$ is a linear subspace, then $T^{-1}(W') \subseteq V$ is a linear subspace, where $T^{-1}(W') := \{v \in V \mid T(v) \in W'\}$;

(c) both statements are generalisations of Proposition 15.b.1. Explain why.

Isomorphisms

Recall from set theory that a map $f : X \to Y$ is called bijective if it is both injective and surjective, and that in this case there exists a unique inverse map $g : Y \to X$, denoted by f^{-1}, meaning that

$$g \circ f = \mathrm{id}_X \quad \text{and} \quad f \circ g = \mathrm{id}_Y \,.$$

Conversely, if $f : X \to Y$ is a map for which some $g : Y \to X$ satisfies $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$, then f is bijective. What is the right analogue of this for linear maps?

Definition 15.b.3 (*Isomorphism*): A linear map $T : V \to W$ is called an "*isomorphism*" if there exists a **linear** map $S : W \to V$ such that

$$S \circ T = \mathrm{id}_V \quad \text{and} \quad T \circ S = \mathrm{id}_W \,.$$

Two vector spaces V and W are defined to be *"isomorphic"*, denoted as $V \cong W$, if there exists an isomorphism between them.

$$V \underset{S}{\overset{T}{\rightleftarrows}} W$$

Remark 15.3: An isomorphism $T : V \to W$ is obviously bijective: the linear map S of Definition 15.b.3 is in particular an inverse map of T in the set-theoretic sense recalled above. The converse is the substantial question, and it is not a matter of definition, because the definition of an isomorphism demands that the inverse be **linear**: is every bijective linear map an isomorphism?

Lemma 15.b.4 (*Bijective Linear Maps are Isomorphisms*): Let $T : V \to W$ be a linear map which is bijective. Then the inverse map $T^{-1} : W \to V$ is linear. In other words, a bijective linear map is an isomorphism.

Proof:

Let $a, b \in K$ and $w_1, w_2 \in W$. Put $v_1 := T^{-1}(w_1)$ and $v_2 := T^{-1}(w_2)$, so that $T(v_1) = w_1$ and $T(v_2) = w_2$. Since T is linear, we have

$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) = aw_1 + bw_2.$$

Applying T^{-1} to both sides gives

$$T^{-1}(aw_1 + bw_2) = av_1 + bv_2 = a \cdot T^{-1}(w_1) + b \cdot T^{-1}(w_2),$$

so T^{-1} is a linear map. **Q.E.D.**

Summary 15.1: For linear maps, bijective = isomorphism.

Lemma 15.b.5 (*Compositions of Linear Maps are Linear*): Let $T : V \to W$ and $S : W \to U$ be linear maps. Then the composition $S \circ T : V \to U$ is also linear.

Proof:

Let $a, b \in K$ and $v_1, v_2 \in V$. Then

$$\begin{aligned}(S \circ T)(av_1 + bv_2) &= S\big(T(av_1 + bv_2)\big) = S\big(aT(v_1) + bT(v_2)\big) \\ &= a \cdot S(T(v_1)) + b \cdot S(T(v_2)) = a \cdot (S \circ T)(v_1) + b \cdot (S \circ T)(v_2).\end{aligned}$$

Q.E.D.

Exercise 15.3 (*Isomorphism is an Equivalence Relation*): Show that $\cong$, that is, being isomorphic, defines an equivalence relation on the set of vector spaces over K.

Definition 15.b.6 (*Endomorphisms and Automorphisms*): A linear map $T : V \to V$, whose domain and target are the same space V, is sometimes called an "*endomorphism*" of V. We denote the set of all endomorphisms of V by $\mathrm{End}(V)$, so that $\mathrm{End}(V) = \mathrm{Hom}(V, V)$. A linear map $T : V \to V$ which is an isomorphism is called an "*automorphism*" of V; in slogan form, endomorphism + isomorphism = automorphism.

Definition 15.b.7 (*Monomorphisms and Epimorphisms*): If a linear map T is injective, one says that T is a "*monomorphism*"; if T is surjective, one says that T is an "*epimorphism*".

Remark 15.4: We will not use the terminology of Definition 15.b.7 in this course, and will simply say injective and surjective instead.

The Rank Theorem

Lemma 15.b.8 (*The Image is Spanned by the Images of a Basis*): Let $T : V \to W$ be a linear map, and let $v_1, \ldots, v_n$ be a basis of V. Then

$$\mathrm{Im}(T) = \mathrm{Sp}\left(T(v_1), \ldots, T(v_n)\right).$$

Proof:

We first show that $\mathrm{Im}(T) \supseteq \mathrm{Sp}(T(v_1), \ldots, T(v_n))$. Let $a_1, \ldots, a_n \in K$. Then

$$\sum_{i=1}^{n} a_i T(v_i) = T\left(\sum_{i=1}^{n} a_i v_i\right) \in \mathrm{Im}(T),$$

which shows that $\mathrm{Sp}(T(v_1), \ldots, T(v_n)) \subseteq \mathrm{Im}(T)$.

We show now that $\mathrm{Im}(T) \subseteq \mathrm{Sp}(T(v_1), \ldots, T(v_n))$. Let $w \in \mathrm{Im}(T)$. By definition, there exists $v \in V$ such that $T(v) = w$. Since $v_1, \ldots, v_n$ is a basis for V, there exist $a_1, \ldots, a_n \in K$ such that $v = \sum_{i=1}^{n} a_i v_i$, and therefore,

$$w = T(v) = T\left(\sum_{i=1}^{n} a_i v_i\right) = \sum_{i=1}^{n} a_i T(v_i).$$

So $w \in \mathrm{Sp}(T(v_1), \ldots, T(v_n))$, which proves the second inclusion. **Q.E.D.**

The dimension of the image is the quantity the next theorem is about; it is given the name $\mathrm{rank}(T)$ at the end of this section, in Definition 15.b.13.

Theorem 15.b.9 (*Rank-Nullity Theorem*): Let V be a finite-dimensional vector space, and let $T : V \to W$ be a linear map. Then, the following dimension formula holds:

$$\dim V = \dim \ker(T) + \dim \mathrm{Im}(T).$$

Proof:

Let $k := \dim \ker(T)$ and $n := \dim V$. Let $(u_1, \ldots, u_k)$ be a basis for $\ker(T)$. By the Steinitz Exchange Lemma, we can extend this linearly independent set to a full basis for V, denoted by $\mathcal{B} = (u_1, \ldots, u_k, v_1, \ldots, v_{n-k})$. We claim that the family $\mathcal{C} = (T(v_1), \ldots, T(v_{n-k}))$ forms a basis for $\mathrm{Im}(T)$.

(a) **Spanning:** Let $w \in \mathrm{Im}(T)$. By definition, there exists some $v \in V$ such that $w = T(v)$. We can express v uniquely as a linear combination of the basis vectors in $\mathcal{B}$:

$$v = \sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{n-k} b_j v_j.$$

Applying the linear map T, and recalling that $T(u_i) = 0_W$ for all i, we find:

$$\begin{aligned} w &= T\left(\sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{n-k} b_j v_j\right) \\ &= \sum_{i=1}^{k} a_i T(u_i) + \sum_{j=1}^{n-k} b_j T(v_j) \\ &= 0_W + \sum_{j=1}^{n-k} b_j T(v_j). \end{aligned}$$

Thus, the vectors in $\mathcal{C}$ span $\operatorname{Im}(T)$.

(b) **Linear Independence:** Suppose there exist scalars $b_j \in K$ such that:

$$\sum_{j=1}^{n-k} b_j T(v_j) = 0_W.$$

By the linearity of T, this is equivalent to $T\left(\sum_{j=1}^{n-k} b_j v_j\right) = 0_W$, which implies that the vector $\sum_{j=1}^{n-k} b_j v_j$ lies in $\ker(T)$. Because $(u_1, \dots, u_k)$ is a basis for the kernel, there must exist scalars $a_i \in K$ such that:

$$\sum_{j=1}^{n-k} b_j v_j = \sum_{i=1}^{k} a_i u_i,$$

or, after moving everything to one side,

$$\sum_{j=1}^{n-k} b_j v_j - \sum_{i=1}^{k} a_i u_i = 0_V.$$

However, since $\mathcal{B}$ is a basis for V, all of its constituent vectors are linearly independent. This forces all coefficients to be zero, specifically $b_j = 0$ for all $1 \le j \le n-k$.

Therefore, $\mathcal{C}$ is linearly independent and spans the image, making it a basis for $\operatorname{Im}(T)$. Consequently, $\dim \operatorname{Im}(T) = n - k = \dim V - \dim \ker(T)$.

Q.E.D.

Remark 15.5: The spanning step above can be shortened: applying Lemma 15.b.8 to the basis $\mathcal{B}$ gives

$$\operatorname{Im}(T) = \operatorname{Sp}\left(\underbrace{T(u_1)}_{=0}, \dots, \underbrace{T(u_k)}_{=0}, T(v_1), \dots, T(v_{n-k})\right) = \operatorname{Sp}\left(T(v_1), \dots, T(v_{n-k})\right),$$

which is how the notes obtain it. Only the independence of $\mathcal{C}$ then remains to be checked.

The theorem also admits a second, more structural proof, which replaces the manipulation of coordinates by a single well-chosen decomposition of V. It uses the existence of a complement from Proposition 14.5 and the fact that isomorphisms carry bases to bases, Theorem 15.b.12 below; neither of these depends on the rank theorem, so no circularity arises.

Second proof of Theorem 15.b.9:

By Proposition 14.5, the subspace $\ker(T) \subseteq V$ admits a complement, that is, a subspace $U \subseteq V$ with

$$\ker(T) + U = V \quad \text{and} \quad \ker(T) \cap U = \{0_V\}.$$

We claim that the restriction $T|_U : U \to \operatorname{Im}(T)$ is an isomorphism.

It is injective: if $u \in U$ satisfies $T(u) = 0_W$, then $u \in \ker(T) \cap U = \{0_V\}$, so $u = 0_V$, and Proposition 15.b.2 **(a)** applies. It is surjective onto $\operatorname{Im}(T)$: given $w \in \operatorname{Im}(T)$, write $w = T(v)$ for some $v \in V$ and decompose $v = x + u$ with $x \in \ker(T)$ and $u \in U$; then

$$w = T(x+u) = T(x) + T(u) = 0_W + T(u) = T(u).$$

Being a bijective linear map, $T|_U$ is an isomorphism by Lemma 15.b.4. By Theorem 15.b.12, it carries a basis of U to a basis of $\operatorname{Im}(T)$, and therefore, $\dim U = \dim \operatorname{Im}(T)$.

Finally, the dimension formula of Proposition 14.2 **(c)**, applied to the subspaces $\ker(T)$ and U, gives

$$\dim V = \dim\big(\ker(T) + U\big) = \dim \ker(T) + \dim U - \underbrace{\dim\big(\ker(T) \cap U\big)}_{=0} = \dim \ker(T) + \dim \operatorname{Im}(T).$$

Q.E.D.

Corollary 15.b.10 (*Dimension Obstructions to Injectivity and Surjectivity*): Let $T : V \to W$ be a linear map, where V and W are finite-dimensional vector spaces. Then:

(a) If $\dim W < \dim V$, then T is **not** injective.

(b) If $\dim W > \dim V$, then T is **not** surjective.

(c) If $\dim W = \dim V$, then T is bijective if and only if T is injective, if and only if T is surjective.

Proof:

(a) Since $\operatorname{Im}(T)$ is a subspace of W, we have $\dim \operatorname{Im}(T) \le \dim W < \dim V$. By Theorem 15.b.9,

$$\dim \ker(T) = \dim V - \dim \operatorname{Im}(T) > 0,$$

so $\ker(T) \neq \{0_V\}$, and T is not injective by Proposition 15.b.2 **(a)**.

(b) By Theorem 15.b.9 and the assumption,

$$\dim \operatorname{Im}(T) = \dim V - \dim \ker(T) \le \dim V < \dim W,$$

so $\operatorname{Im}(T) \subsetneq W$, and T is not surjective by Proposition 15.b.2 **(b)**.

(c) Here the chain of equivalences reads

$$T \text{ injective} \iff \ker(T) = \{0_V\} \iff \dim \ker(T) = 0 \iff \dim V = \dim \operatorname{Im}(T) \iff \dim W = \dim \operatorname{Im}(T),$$

the third step by Theorem 15.b.9 and the last by the assumption $\dim V = \dim W$. Since $\operatorname{Im}(T)$ is a subspace of W, the final equality holds if and only if $\operatorname{Im}(T) = W$, that is, if and only if T is surjective. As each of the two properties implies the other, either one already makes T bijective.

Q.E.D.

Remark 15.6: By Lemma 15.b.4, the word *"bijective"* in statement **(c)** may equivalently be read as *"is an isomorphism"*. So for maps between spaces of equal finite dimension, injectivity alone already forces T to be an isomorphism.

Corollary 15.b.11 (*Isomorphism is Detected by Dimension*): Two finite-dimensional vector spaces V and W over K are isomorphic if and only if $\dim V = \dim W$.

Proof:

Suppose first that $T : V \to W$ is an isomorphism. By Definition 15.b.3, it admits a linear inverse and is in particular bijective, so $\ker(T) = \{0_V\}$ and $\operatorname{Im}(T) = W$ by Proposition 15.b.2. The rank theorem, Theorem 15.b.9, then gives

$$\dim W = \dim \operatorname{Im}(T) = \dim V - \dim \ker(T) = \dim V.$$

Conversely, assume that $\dim V = \dim W$, and denote this common dimension by n. Let $v_1, \dots, v_n$ be a basis for V and $w_1, \dots, w_n$ a basis for W. By Theorem 15.a.3, there is a linear map $T : V \to W$ with $T(v_i) = w_i$ for all $1 \le i \le n$. By Lemma 15.b.8,

$$\operatorname{Im}(T) = \operatorname{Sp}\left(T(v_1), \dots, T(v_n)\right) = \operatorname{Sp}(w_1, \dots, w_n) = W,$$

so T is surjective and $\dim \operatorname{Im}(T) = n$. By Theorem 15.b.9, $\dim \ker(T) = n - n = 0$, so T is injective as well. Being a bijective linear map, T is an isomorphism by Lemma 15.b.4, and hence $V \cong W$. **Q.E.D.**

Isomorphisms and Vector Space Equivalence

Isomorphisms allow us to classify vector spaces and identify when two ostensibly different spaces are essentially the *"same"* in terms of their linear algebraic structure.

Theorem 15.b.12 (*Isomorphisms Preserve Bases*): Let $T : V \to W$ be an isomorphism between two vector spaces V and W. Let $\mathcal{S} = \{v_i\}$ be a family of vectors in V. We define its image family as $T(\mathcal{S}) = \{T(v_i)\}_{v_i \in \mathcal{S}} \subseteq W$. Then, the following hold:

- **(a)** $\mathcal{S}$ is linearly independent in V if and only if $T(\mathcal{S})$ is linearly independent in W.
- **(b)** $\mathcal{S}$ spans V if and only if $T(\mathcal{S})$ spans W.
- **(c)** $\mathcal{S}$ is a basis of V if and only if $T(\mathcal{S})$ is a basis of W.

The notes state this theorem without proof. Since the second proof of Theorem 15.b.9 above appeals to it, we supply one; it is short, and it uses nothing beyond the definition of an isomorphism.

Proof:

Write $S := T^{-1} : W \to V$ for the inverse of T, which is linear because T is an isomorphism. Each statement is an equivalence, and in each case the implication from right to left follows by applying the already-proved implication from left to right to the isomorphism S and the family $T(\mathcal{S})$, since $S(T(\mathcal{S})) = \mathcal{S}$. It therefore suffices to prove the three implications from left to right.

- **(a)** Let $\mathcal{S}$ be linearly independent, and suppose that a finite subfamily satisfies $\sum_i a_i T(v_i) = 0_W$. By linearity, $T\left(\sum_i a_i v_i\right) = 0_W$, and applying S gives $\sum_i a_i v_i = 0_V$. The independence of $\mathcal{S}$ forces every a_i to vanish, so $T(\mathcal{S})$ is linearly independent.
- **(b)** Let $\mathcal{S}$ span V, and let $w \in W$. Then $S(w) \in V$ is a linear combination $S(w) = \sum_i a_i v_i$ of members of $\mathcal{S}$, and applying T gives
$$w = T(S(w)) = \sum_i a_i T(v_i),$$
so $T(\mathcal{S})$ spans W.
- **(c)** Immediate from the first two statements, since a basis is precisely a linearly independent spanning family.

Q.E.D.

Remark 15.7 (*The Significance of Isomorphism*): If two vector spaces are isomorphic ($V \cong W$), they possess identical linear algebraic properties, such as having the exact same dimension. We can conceptually think of V and W as the exact same space, just represented in two different ways.

However, it is crucial to note that identifying V with W requires the explicit choice of an isomorphism $T : V \to W$. Because there is usually no preferred (or *"canonical"*) choice for this map, it is generally better practice not to forcefully identify different vector spaces simply because they are isomorphic. We will explore the nuances of canonical choices later in the course.

Definition 15.b.13 (*Rank of a Linear Map*): Let $T \in \operatorname{Hom}(V, W)$ be a linear map such that $\dim \operatorname{Im}(T) < \infty$. We define the "*rank*" of T as the dimension of its image:

$$\operatorname{rank}(T) := \dim \operatorname{Im}(T).$$

Exercise 15.4 (*Rank of a Composition*): Let $T : V \to W$ and $S : W \to U$ be linear maps, where U, V and W are finite-dimensional. Show that:

(a) $\operatorname{rank}(S \circ T) \leq \min\{\operatorname{rank}(S), \operatorname{rank}(T)\}$;

(b) if S is injective, then $\operatorname{rank}(S \circ T) = \operatorname{rank}(T)$;

(c) if T is surjective, then $\operatorname{rank}(S \circ T) = \operatorname{rank}(S)$.

Solutions to the Exercises

Linearity Criterion (on Page 90):

Suppose first that T is linear. Then, for all $\alpha_1, \alpha_2 \in K$ and $v_1, v_2 \in V$,

$$T(\alpha_1 v_1 + \alpha_2 v_2) = T(\alpha_1 v_1) + T(\alpha_2 v_2) = \alpha_1 T(v_1) + \alpha_2 T(v_2),$$

using additivity for the first equality and homogeneity for the second.

Conversely, assume the displayed identity for all scalars and vectors. Taking $\alpha_1 = \alpha_2 = 1$ gives $T(v_1 + v_2) = T(v_1) + T(v_2)$, which is additivity. Taking $\alpha_2 = 0$ and $v_2 = 0_V$ gives

$$T(\alpha_1 v_1) = \alpha_1 T(v_1) + 0 \cdot T(0_V) = \alpha_1 T(v_1),$$

which is homogeneity. So T satisfies both conditions of Definition 15.a.1. **Q.E.D.**

Images and Preimages of Subspaces (on Page 93):

(a) Since $0_V \in V'$ and $T(0_V) = 0_W$, we have $0_W \in T(V')$. Let $a, b \in K$ and $w_1, w_2 \in T(V')$, say $w_1 = T(v_1)$ and $w_2 = T(v_2)$ with $v_1, v_2 \in V'$. Then $av_1 + bv_2 \in V'$, because V' is a subspace, and

$$aw_1 + bw_2 = aT(v_1) + bT(v_2) = T(av_1 + bv_2) \in T(V').$$

(b) Since $T(0_V) = 0_W \in W'$, we have $0_V \in T^{-1}(W')$. Let $a, b \in K$ and $v_1, v_2 \in T^{-1}(W')$, so that $T(v_1), T(v_2) \in W'$. Then

$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) \in W',$$

because W' is a subspace, and therefore $av_1 + bv_2 \in T^{-1}(W')$.

(c) Both parts of Proposition 15.b.1 are special cases. Taking $V' := V$ in **(a)** gives that $\operatorname{Im}(T) = T(V)$ is a subspace of W, and taking $W' := \{0_W\}$ in **(b)** gives that $\ker(T) = T^{-1}(\{0_W\})$ is a subspace of V — exactly the two set-theoretic descriptions recorded in the remark on Page 93.

Q.E.D.

Isomorphism is an Equivalence Relation (on Page 95):

Reflexivity. The identity $\mathrm{id}_V : V \to V$ is linear and satisfies $\mathrm{id}_V \circ \mathrm{id}_V = \mathrm{id}_V$, so it is an isomorphism, and $V \cong V$.

Symmetry. Let $V \cong W$, witnessed by an isomorphism $T : V \to W$ with linear inverse $S : W \to V$. Then S is itself linear with linear inverse T, since $T \circ S = \mathrm{id}_W$ and $S \circ T = \mathrm{id}_V$. So S is an isomorphism and $W \cong V$.

Transitivity. Let $T : V \to W$ and $T' : W \to U$ be isomorphisms with inverses S and S'. By Lemma 15.b.5, both $T' \circ T : V \to U$ and $S \circ S' : U \to V$ are linear, and

$$(S \circ S') \circ (T' \circ T) = S \circ (S' \circ T') \circ T = S \circ \mathrm{id}_W \circ T = S \circ T = \mathrm{id}_V,$$

and symmetrically $(T' \circ T) \circ (S \circ S') = \mathrm{id}_U$. So $T' \circ T$ is an isomorphism and $V \cong U$. **Q.E.D.**

Rank of a Composition (on Page 99):

Throughout, we use the identity

$$\mathrm{Im}(S \circ T) = \{S(T(v)) \mid v \in V\} = \{S(w) \mid w \in \mathrm{Im}(T)\} = S\big(\mathrm{Im}(T)\big),$$

so that $\mathrm{Im}(S \circ T)$ is the image of the restricted linear map $S|_{\mathrm{Im}(T)} : \mathrm{Im}(T) \to U$.

(a) Since $\mathrm{Im}(T) \subseteq W$, we have $S(\mathrm{Im}(T)) \subseteq S(W) = \mathrm{Im}(S)$, and both are subspaces of U by Page 93, whence $\mathrm{rank}(S \circ T) \leq \mathrm{rank}(S)$. Applying Theorem 15.b.9 to $S|_{\mathrm{Im}(T)}$ gives

$$\mathrm{rank}(S \circ T) = \dim \mathrm{Im}\left(S|_{\mathrm{Im}(T)}\right) = \dim \mathrm{Im}(T) - \dim \ker\left(S|_{\mathrm{Im}(T)}\right) \leq \mathrm{rank}(T),$$

and the two bounds together give $\mathrm{rank}(S \circ T) \leq \min\{\mathrm{rank}(S), \mathrm{rank}(T)\}$.

(b) If S is injective, then so is its restriction $S|_{\mathrm{Im}(T)}$, so $\ker(S|_{\mathrm{Im}(T)}) = \{0_W\}$ and the displayed computation becomes an equality: $\mathrm{rank}(S \circ T) = \dim \mathrm{Im}(T) = \mathrm{rank}(T)$.

(c) If T is surjective, then $\mathrm{Im}(T) = W$, and therefore $\mathrm{Im}(S \circ T) = S(W) = \mathrm{Im}(S)$, so that $\mathrm{rank}(S \circ T) = \mathrm{rank}(S)$.

Q.E.D.

AI-Note: This was the most heavily damaged chapter so far. Of the thirteen numbered statements in the notes for Section 15.b, the transcript kept five, and it hid the loss with a counter jump: a `\setcounter{theorem}{11}` placed just before Theorem 15.b.12, annotated as aligning the numbering with the notes. Restored above, in the notes' own order, so that the printed numbers now agree with the handwritten ones throughout: Definition 15.b.3, Lemma 15.b.4, Lemma 15.b.5, Definition 15.b.6, Definition 15.b.7, Lemma 15.b.8 and Corollary 15.b.11, along with the example of $\ker(T_A)$, the recollection on bijections that motivates Definition 15.b.3, and three of the notes' exercises. In Proposition 15.b.2 the surjectivity criterion was missing, and in Corollary 15.b.10 the two dimension obstructions — the statements that really do the work in practice — had been dropped in favour of the third part alone.

The gap that mattered most was Definition 15.b.3 itself. The transcript carried a definition of **isomorphic spaces** at the very end of the chapter, numbered 15.b.13, while the word *"isomorphism"* was already used in Corollary 15.b.10 and Theorem 15.b.12 above it. Worse, the old proof of Corollary 15.b.10 concluded from bijectivity that the map is an isomorphism, which is exactly Lemma 15.b.4 — the one result in the notes that was missing. That definition has been moved to the notes' position, 15.b.3, and merged with the notes' formulation.

Since Prof. Biran leaves Theorem 15.b.12 unproved, a proof is supplied here. It is also used for a second proof of Theorem 15.b.9, added above: rather than manipulating coordinates, one takes a complement U of $\ker(T)$ as in Proposition 14.5 and observes that $T|_U$ is an isomorphism onto $\mathrm{Im}(T)$, after which the dimension formula of Proposition 14.2 finishes the argument. Neither ingredient depends on the rank theorem.

In Section 15.a, the notes verify in full that the map constructed in Theorem 15.a.3 is linear and that it sends v_i to w_i; the transcript replaced both verifications with the phrase *"it is straightforward to show"*. They are restored, as is the computation showing that $A \cdot e_i$ picks out the i-th column of A. The uniqueness of the matrix in Lemma 15.a.5 is asserted by the transcript but not proved, and not claimed at all in the notes; a one-line proof is added. Two cross-references used `\eqref` on a definition and on a theorem, which typesets as a bare equation number; they are now `\cref`.

Linear Maps and Bases (Chapter 16)

Writing Linear Maps Using Bases and Coordinates

In this lecture, we will assume all vector spaces to be *"finite-dimensional"* unless otherwise stated. We will write bases of a vector space V as an ordered list (or tuple) $\mathcal{B} = (v_1, \dots, v_n)$.

Definition 16.1 (*Coordinate Vector*): Let V be a vector space over K, and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis of V. Let $v \in V$. We define the *"coordinate vector"* of v with respect to the basis $\mathcal{B}$ as

$$[v]_{\mathcal{B}} := \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in K^n,$$

where $a_1, \dots, a_n \in K$ are the unique scalars in K such that $v = a_1 v_1 + \dots + a_n v_n$.

Remark 16.1: Note that $[v]_{\mathcal{B}}$ is uniquely defined, since every $v \in V$ has a unique representation as a linear combination of the basis vectors $v_1, \dots, v_n$.

Define a map $\Phi_{\mathcal{B}} : V \to K^n$ given by:

$$\Phi_{\mathcal{B}}(v) := [v]_{\mathcal{B}} \in K^n.$$

Proposition 16.2 (*Isomorphism of the Coordinate Map*): The map $\Phi_{\mathcal{B}}$ is an isomorphism.

Proof:

We first show that $\Phi_{\mathcal{B}}$ is a linear map. Let $a, b \in K$, and let $v, v' \in V$. Write v and v' as linear combinations of the elements of $\mathcal{B}$:

$$v = a_1 v_1 + \dots + a_n v_n, \quad \text{and} \quad v' = b_1 v_1 + \dots + b_n v_n,$$

with $a_i, b_i \in K$. By definition, we have:

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \quad \text{and} \quad [v']_{\mathcal{B}} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

Now, $av + bv' = (aa_1 + bb_1)v_1 + \dots + (aa_n + bb_n)v_n$, and hence:

$$\Phi_{\mathcal{B}}(av + bv') = [av + bv']_{\mathcal{B}} = \begin{pmatrix} aa_1 + bb_1 \\ \vdots \\ aa_n + bb_n \end{pmatrix} = a \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + b \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = a\Phi_{\mathcal{B}}(v) + b\Phi_{\mathcal{B}}(v').$$

This proves that $\Phi_{\mathcal{B}}$ is linear.

We now claim that $\Phi_{\mathcal{B}}$ is an isomorphism. By previous results (Lemma 15.b.4), it is enough to show $\Phi_{\mathcal{B}}$ is bijective. Indeed, if $v \in \ker(\Phi_{\mathcal{B}})$, then $v = 0v_1 + \dots + 0v_n = 0$, which implies $\ker(\Phi_{\mathcal{B}}) = \{0\}$. This shows that $\Phi_{\mathcal{B}}$ is injective.

For the surjectivity of $\Phi_{\mathcal{B}}$, let $(a_1, \dots, a_n)^\mathsf{T} \in K^n$. Put $v := a_1 v_1 + \dots + a_n v_n \in V$. Then, $\Phi_{\mathcal{B}}(v) = (a_1, \dots, a_n)^\mathsf{T}$, which implies $\operatorname{Im}(\Phi_{\mathcal{B}}) = K^n$. **Q.E.D.**

Remark 16.2: The inverse map to $\Phi_{\mathcal{B}}$ is given by:

$$\Phi_{\mathcal{B}}^{-1} : K^n \to V, \quad \Phi_{\mathcal{B}}^{-1} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} := a_1 v_1 + \dots + a_n v_n.$$

Remark 16.3: The previous Proposition 16.2 gives yet another proof for the following Corollary.

Corollary 16.3 (*Classification of Finite-Dimensional Vector Spaces*): Let V be a vector space over K with $\dim V = n \in \mathbb{Z}_{\geq 0}$. Then, $V \cong K^n$.

Proof:

Choose a basis $\mathcal{B}$ for V. Then, $\Phi_{\mathcal{B}} : V \to K^n$ is an isomorphism. **Q.E.D.**

Remark 16.4:

- The Corollary does NOT hold for $\dim V = \infty$, at least not as stated.
- We have seen that $V \cong K^n$ where $n = \dim V < \infty$, but in general, there does not exist a *"canonical"* (or preferred) isomorphism $V \to K^n$. For example, $\Phi_{\mathcal{B}}$ depends on a choice of a basis $\mathcal{B}$.

Example 16.1 (*Coordinate Vector Calculation*): Consider $V = \mathbb{R}^2$, let $\mathcal{E} = \left(\binom{1}{0}, \binom{0}{1}\right)$ be the standard basis, and let $\mathcal{B} = \left(\binom{1}{1}, \binom{1}{-1}\right)$ be another basis (check that $\mathcal{B}$ is indeed a basis!). We have

$$\left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{E}} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \text{and we want to find} \quad \left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} a \\ b \end{pmatrix} =?$$

Let us first try to determine $\left[\binom{1}{0}\right]_{\mathcal{B}}$ and $\left[\binom{0}{1}\right]_{\mathcal{B}}$. We set $\left[\binom{1}{0}\right]_{\mathcal{B}} = \binom{a_1}{a_2}$, where

$$a_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + a_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

This leads to the system of linear equations:

$$\begin{cases} a_1 + a_2 = 1 \\ a_1 - a_2 = 0 \end{cases}$$

Solving this system yields $a_1 = a_2 = \frac{1}{2}$, and therefore, $\left[\binom{1}{0}\right]_{\mathcal{B}} = \binom{1/2}{1/2}$.
Similarly, for the second standard basis vector, we set

$$\left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad \text{where} \quad b_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + b_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The resulting system is:

$$\begin{cases} b_1 + b_2 = 0 \\ b_1 - b_2 = 1 \end{cases}$$

Solving this yields $b_1 = \frac{1}{2}$ and $b_2 = -\frac{1}{2}$, so $\left[\binom{0}{1}\right]_{\mathcal{B}} = \binom{1/2}{-1/2}$.
Finally, using the fact that $\Phi_{\mathcal{B}}$ is linear, we calculate:

$$\left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{B}} = x \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right]_{\mathcal{B}} + y \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} x/2 \\ x/2 \end{pmatrix} + \begin{pmatrix} y/2 \\ -y/2 \end{pmatrix} = \begin{pmatrix} \frac{x+y}{2} \\ \frac{x-y}{2} \end{pmatrix}.$$

Representation Matrices

Now we arrive at the main question. Let V and W be vector spaces over K, let $n := \dim V$ and $m := \dim W$, and let $T : V \to W$ be a linear map. Fix a basis $\mathcal{B}$ for V and a basis $\mathcal{C}$ for W. Consider the following diagram, in which the two paths from V to K^m agree, so that the diagram is commutative:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\Phi_{\mathcal{B}}"'] & W \arrow[d, "\Phi_{\mathcal{C}}"] \\ K^n \arrow[r, "\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}"'] & K^m \end{tikzcd}$$

Since $\Phi_{\mathcal{B}}^{-1}$, T, and $\Phi_{\mathcal{C}}$ are all linear maps, the map $\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} : K^n \to K^m$ is also linear. By previous results (Lemma 15.a.5), any linear map $K^n \to K^m$ can be written using a matrix $A \in \mathcal{M}_{m\times n}(K)$. In other words, there exists a unique matrix $A \in \mathcal{M}_{m\times n}(K)$ such that $\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} = T_A$.

The matrix A is called the "*representation matrix*" of T with respect to the bases $\mathcal{B}$ and $\mathcal{C}$. We also say that A represents T in the bases $\mathcal{B}$ and $\mathcal{C}$. We denote this matrix by $[T]_{\mathcal{C}}^{\mathcal{B}} \in \mathcal{M}_{m\times n}(K)$.

Question 16.1:

(a) How can we calculate the entries of A?

(b) How does A depend on different choices of the bases $\mathcal{B}$ and $\mathcal{C}$?

(c) If we have three vector spaces V, W, U with bases $\mathcal{B}, \mathcal{C}, \mathcal{D}$ respectively, and $T : V \to W$ and $S : W \to U$ are linear maps where T is represented by the matrix M and S by the matrix N, then what is the matrix representing $S \circ T : V \to U$ with respect to the bases $\mathcal{B}$ and $\mathcal{D}$?

Let $T : V \to W$ be a linear map, let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V, and let $\mathcal{C} = (w_1, \dots, w_m)$ be a basis for W. We wish to determine the entries of its representation matrix $A := [T]_{\mathcal{C}}^{\mathcal{B}} \in \mathcal{M}_{m\times n}(K)$. By definition, A is the unique matrix such that the linear map $T_A : K^n \to K^m$, given by left-multiplication by A, satisfies $T_A = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}$.

Recall that the j-th column of A is given by $T_A(e_j)$, where $e_j \in K^n$ is the j-th standard basis vector. We can evaluate $T_A(e_j)$ using the commutative diagram:

$$T_A(e_j) = \left(\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}\right)(e_j) = \Phi_{\mathcal{C}}\left(T\left(\Phi_{\mathcal{B}}^{-1}(e_j)\right)\right).$$

Recall that the coordinate vector of the basis vector $v_j \in V$ with respect to its own basis $\mathcal{B}$ is the j-th standard basis vector $e_j \in K^n$, meaning $\Phi_{\mathcal{B}}(v_j) = [v_j]_{\mathcal{B}} = e_j$. Applying the inverse coordinate map gives $\Phi_{\mathcal{B}}^{-1}(e_j) = v_j$. Substituting this into our expression, we have:

$$T_A(e_j) = \Phi_{\mathcal{C}}(T(v_j)) = [T(v_j)]_{\mathcal{C}}.$$

This means that the j-th column of the representation matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$ is precisely the coordinate vector of $T(v_j)$ with respect to the basis $\mathcal{C}$. Thus, we can write A in block-column form as:

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \dots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \in \mathcal{M}_{m\times n}(K).$$

Equivalently, if we write $A = (a_{ij}) \in \mathcal{M}_{m\times n}(K)$, the entries a_{ij} of the j-th column are determined by expressing $T(v_j) \in W$ as a linear combination of the basis vectors $\mathcal{C} = (w_1, \dots, w_m)$:

$$T(v_j) = a_{1j}w_1 + a_{2j}w_2 + \dots + a_{mj}w_m = \sum_{i=1}^{m} a_{ij}w_i.$$

Proposition 16.4 (*Evaluation via Representation Matrix*): Let $T : V \to W$ be a linear map between two finite-dimensional vector spaces V and W. Let $\mathcal{B}$ and $\mathcal{C}$ be bases for V and W, respectively. Then, for all $v \in V$:

$$[T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = [T(v)]_{\mathcal{C}}.$$

Proof:

First Proof (Computational): Let $v \in V$ be an arbitrary vector. We can express v uniquely as a linear combination of the basis vectors in $\mathcal{B} = (v_1, \dots, v_n)$:

$$v = \sum_{j=1}^{n} x_j v_j.$$

By definition, the coordinate vector of v with respect to $\mathcal{B}$ is $[v]_{\mathcal{B}} = (x_1, \dots, x_n)^{\mathsf{T}} \in K^n$. Since T is linear, we can apply it to v:

$$T(v) = T\left(\sum_{j=1}^{n} x_j v_j\right) = \sum_{j=1}^{n} x_j T(v_j).$$

Now, we take the coordinate vector of $T(v)$ with respect to the basis $\mathcal{C}$. Since the coordinate map $\Phi_{\mathcal{C}}$ is an isomorphism, it is linear, and we obtain:

$$[T(v)]_{\mathcal{C}} = \Phi_{\mathcal{C}}(T(v)) = \Phi_{\mathcal{C}}\left(\sum_{j=1}^{n} x_j T(v_j)\right) = \sum_{j=1}^{n} x_j [T(v_j)]_{\mathcal{C}}.$$

Observe that this last expression is exactly the definition of the matrix-vector product of the matrix $[T]^{\mathcal{B}}_{\mathcal{C}}$ (whose columns are $[T(v_j)]_{\mathcal{C}}$) and the column vector $[v]_{\mathcal{B}}$ (whose entries are x_j):

$$\sum_{j=1}^{n} x_j [T(v_j)]_{\mathcal{C}} = \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \dots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = [T]^{\mathcal{B}}_{\mathcal{C}} \cdot [v]_{\mathcal{B}}.$$

Therefore, we conclude that $[T]^{\mathcal{B}}_{\mathcal{C}} \cdot [v]_{\mathcal{B}} = [T(v)]_{\mathcal{C}}$.

Second Proof (Structural): Recall that the representation matrix $A := [T]^{\mathcal{B}}_{\mathcal{C}}$ is uniquely defined by the relation $T_A = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}$. Evaluating the linear transformation T_A at the coordinate vector $[v]_{\mathcal{B}} = \Phi_{\mathcal{B}}(v) \in K^n$ yields:

$$T_A([v]_{\mathcal{B}}) = \left(\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}\right)(\Phi_{\mathcal{B}}(v)) = \Phi_{\mathcal{C}}\left(T\left(\Phi_{\mathcal{B}}^{-1}(\Phi_{\mathcal{B}}(v))\right)\right) = \Phi_{\mathcal{C}}(T(v)) = [T(v)]_{\mathcal{C}}.$$

On the other hand, the map T_A acts on vectors in K^n by left-multiplication by the matrix A. Thus, we also have:

$$T_A([v]_{\mathcal{B}}) = A \cdot [v]_{\mathcal{B}} = [T]^{\mathcal{B}}_{\mathcal{C}} \cdot [v]_{\mathcal{B}}.$$

Equating both expressions gives $[T]^{\mathcal{B}}_{\mathcal{C}} \cdot [v]_{\mathcal{B}} = [T(v)]_{\mathcal{C}}$, which completes the proof. **Q.E.D.**

Example 16.2 (*Examples of Representation Matrices*):

(a) Let $V = K^n$ and $W = K^m$, and let $T : K^n \to K^m$ be given by $T = T_A$ for $A \in \mathcal{M}_{m \times n}(K)$, so $T(v) = A \cdot v$. Take $\mathcal{B} := (e_1, \dots, e_n)$ and $\mathcal{C} := (e_1, \dots, e_m)$ to be the standard bases. Then, $[T]^{\mathcal{B}}_{\mathcal{C}} = A$. *(Check the details!)*

(b) Let $V = K[x]_3$ and $W = K[x]_2$, and let $D : V \to W$ be the derivative map defined by $D(p(x)) = p'(x)$. Let $\mathcal{B} := (1, x, x^2, x^3)$ and $\mathcal{C} := (1, x, x^2)$. We calculate:

$$\begin{aligned} D(1) &= 0 = 0 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \\ D(x) &= 1 = 1 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \\ D(x^2) &= 2x = 0 \cdot 1 + 2 \cdot x + 0 \cdot x^2 \\ D(x^3) &= 3x^2 = 0 \cdot 1 + 0 \cdot x + 3 \cdot x^2 \end{aligned}$$

Therefore, the representation matrix is:

$$[D]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

(c) Revisiting the previous item, suppose we view D as a linear map $D : K[x]_3 \to K[x]_3$, so that the same map is now given the same basis on both sides. Then,

$$[D]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Representing Composition of Linear Maps by Matrices

Let $T : V \to W$ and $S : W \to U$ be linear maps. Let $\mathcal{A} = (v_1, \ldots, v_n)$ be a basis for V, let $\mathcal{B} = (w_1, \ldots, w_m)$ be a basis for W, and let $\mathcal{C} = (u_1, \ldots, u_p)$ be a basis for U. The dimensions are thus $n = \dim V$, $m = \dim W$, and $p = \dim U$.

Consider the composition $S \circ T : V \longrightarrow U$.

Question 16.2: What is the matrix $[S \circ T]_{\mathcal{C}}^{\mathcal{A}}$? How is it related to the matrices $[T]_{\mathcal{B}}^{\mathcal{A}}$ and $[S]_{\mathcal{C}}^{\mathcal{B}}$?

To answer this, let us write the representation matrices as follows:

$$[T]_{\mathcal{B}}^{\mathcal{A}} = (a_{ij}), \quad [S]_{\mathcal{C}}^{\mathcal{B}} = (b_{ij}), \quad [S \circ T]_{\mathcal{C}}^{\mathcal{A}} = (c_{ij}).$$

By definition, the images of the basis vectors are given by:

$$T(v_j) = \sum_{k=1}^{m} a_{kj} w_k, \quad \forall\, 1 \le j \le n, \tag{16.1}$$

$$S(w_k) = \sum_{i=1}^{p} b_{ik} u_i, \quad \forall\, 1 \le k \le m, \tag{16.2}$$

$$(S \circ T)(v_j) = \sum_{i=1}^{p} c_{ij} u_i, \quad \forall\, 1 \le j \le n. \tag{16.3}$$

But we can also calculate $(S \circ T)(v_j)$ using (16.1) and (16.2):

$$\begin{aligned}
(S \circ T)(v_j) &= S\left(\sum_{k=1}^{m} a_{kj} w_k\right) \\
&= \sum_{k=1}^{m} a_{kj} S(w_k) \\
&= \sum_{k=1}^{m} a_{kj} \left(\sum_{i=1}^{p} b_{ik} u_i\right) \\
&= \sum_{k=1}^{m} \sum_{i=1}^{p} a_{kj} b_{ik} u_i = \sum_{i=1}^{p} \left(\sum_{k=1}^{m} b_{ik} a_{kj}\right) \cdot u_i.
\end{aligned}$$

Comparing this with (16.3), we find that $c_{ij} = \sum_{k=1}^{m} b_{ik} a_{kj}$. In other words, row i of $[S]_{\mathcal{C}}^{\mathcal{B}}$ multiplied by column j of $[T]_{\mathcal{B}}^{\mathcal{A}}$ yields the entry c_{ij}:

$$\begin{pmatrix} b_{i1} & \ldots & b_{im} \end{pmatrix} \cdot \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} = c_{ij}.$$

This reveals that:

$$[S]_{\mathcal{C}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}} = [S \circ T]_{\mathcal{C}}^{\mathcal{A}},$$

where the matrix dimensions follow $(p \times m) \cdot (m \times n) = (p \times n)$.

In order to obtain $[S \circ T]_{\mathcal{C}}^{\mathcal{A}}$ from $[T]_{\mathcal{B}}^{\mathcal{A}}$ and $[S]_{\mathcal{C}}^{\mathcal{B}}$, we need to use a new operation on matrices. It is called *"matrix multiplication"*. But is this operation really new?

— Yes, but not entirely...

AI-Note: This chapter came through the transcription essentially intact: every statement of the notes is present, in the notes' order, and the printed numbers 16.1–16.4 already agreed with the handwritten ones. The abrupt final line is Prof. Biran's own; his notes end there, in the middle of the thought that chapter 17 picks up.

What needed repair was small and mostly typographic. The third example carried a `\setcounter{enumi}{1}`, annotated as matching the professor's numbering, which made the list print two items labelled **(b)**; the notes label that item $2'$ because it revisits the previous one, so the counter manipulation is dropped and the revisiting is said in words instead. The sentence introducing the commutative diagram ended in the word *"test"*, evidently a leftover. A `\usetikzlibrary{babel}` sat in the middle of the document, although the preamble already loads that library. Prof. Biran's aside *"check the details!"* on the first representation-matrix example is restored, and his closing line is now set as the reply it is rather than as a stray hyphen.

The proof that $\Phi_{\mathcal{B}}$ is an isomorphism cited the definition of isomorphic spaces for the step that bijectivity suffices. That step is Lemma 15.b.4, which the transcript of the previous chapter had dropped altogether and which is now restored; the citation points there.

One addition of the transcript is worth keeping and worth flagging: the notes prove Proposition 16.4 only structurally, by observing that the asserted identity is exactly the commutative diagram defining $[T]_{\mathcal{C}}^{\mathcal{B}}$. The computational proof that precedes it in the text above is not Prof. Biran's.

Matrices (Chapter 17)

Matrix Multiplication (Section 17.a)

Definition 17.a.1 (*Matrix Multiplication*): Let $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$. We define a new matrix, which is called the "*product*" (or "*multiplication*") of A and B, denoted by $A \cdot B \in \mathcal{M}_{m\times p}(K)$.

Write $A = (a_{ij})_{\substack{1\le i\le m\\ 1\le j\le n}}$ and $B = (b_{ij})_{\substack{1\le i\le n\\ 1\le j\le p}}$. The matrix $A \cdot B$ has entries $(c_{ij})_{\substack{1\le i\le m\\ 1\le j\le p}}$, where

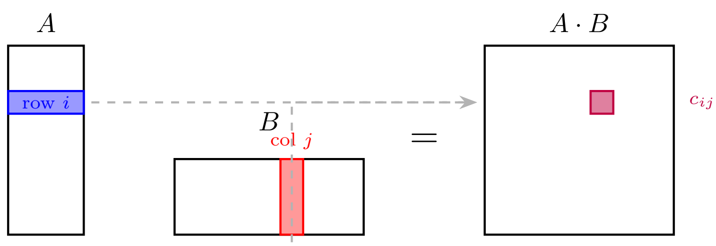
$$c_{ij} := a_{i1}b_{1j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}$$

We often write AB instead of $A \cdot B$.

Important remark 17.1: The product AB is defined only when the number of columns of A equals the number of rows of B.

An alternative description of $A \cdot B$ can be visualized by taking the dot product of the i-th row of A with the j-th column of B:

$$c_{ij} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B) = a_{i1}b_{1j} + \cdots + a_{ik}b_{kj} + \cdots + a_{in}b_{nj}$$



Another way to describe it is by considering the columns of B, denoted as $w_1, \ldots, w_p$. The matrix $A \cdot B$ is then formed by multiplying A with each column vector w_j:

$$A \cdot \begin{pmatrix} | & & | \\ w_1 & \ldots & w_p \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ Aw_1 & \ldots & Aw_p \\ | & & | \end{pmatrix}$$

Example 17.1 (*Matrix Multiplication Example*): We compute the product of a 2×3 matrix and a 3×2 matrix:

$$\begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 0 \end{pmatrix} = \begin{pmatrix} 3\cdot 1 + 2\cdot 0 + 1\cdot 4 & 3\cdot 2 + 2\cdot 1 + 1\cdot 0 \\ 1\cdot 1 + 0\cdot 0 + 2\cdot 4 & 1\cdot 2 + 0\cdot 1 + 2\cdot 0 \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 9 & 2 \end{pmatrix}$$

Conversely, multiplying the 3×2 matrix by the 2×3 matrix yields a 3×3 matrix:

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 2 & 5 \\ 1 & 0 & 2 \\ 12 & 8 & 4 \end{pmatrix}$$

If $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$, then $A \cdot B$ is defined, but $B \cdot A$ is not defined, unless $p = m$.

Some Special Cases

(a) If $A \in \mathcal{M}_{m\times n}(K)$ and $v \in K^n_{\text{col}}$, we have previously defined the matrix-vector product $A \cdot v \in K^m_{\text{col}}$. However, we can also identify $K^n_{\text{col}} = \mathcal{M}_{n\times 1}(K)$. If we view v as an $n \times 1$ matrix, then the matrix multiplication of A and v yields the exact same result as applying A to the vector v.

(b) If $A \in \mathcal{M}_{m\times n}(K)$ and $w \in K^m_{\text{row}} = \mathcal{M}_{1\times m}(K)$, then we can define the left-multiplication $w \cdot A \in \mathcal{M}_{1\times n}(K) = K^n_{\text{row}}$. If A is represented by its row vectors $v_1, \ldots, v_m$, this operation scales and sums the rows of A according to the entries of w. This gives the product a row-wise description matching the column-wise one above. If A has rows $v_1, \ldots, v_m$ and $B \in \mathcal{M}_{n\times p}(K)$ has columns $w_1, \ldots, w_p$, then

$$A \cdot B = \begin{pmatrix} | & & | \\ Aw_1 & \ldots & Aw_p \\ | & & | \end{pmatrix} = \begin{pmatrix} — & v_1 \cdot B & — \\ & \vdots & \\ — & v_m \cdot B & — \end{pmatrix}.$$

(c) Let $v = (a_1, \ldots, a_n) \in K^n_{\text{row}} = \mathcal{M}_{1\times n}(K)$, and let $w = (b_1, \ldots, b_n)^\mathsf{T} \in K^n_{\text{col}} = \mathcal{M}_{n\times 1}(K)$. Then the product $v \cdot w \in \mathcal{M}_{1\times 1}(K) = K$ is simply the standard scalar product:

$$v \cdot w = a_1 b_1 + \cdots + a_n b_n$$

Connections to Linear Maps

As a conclusion from our previous discussions on linear maps, let $T : V \to W$ and $S : W \to U$ be linear maps between finite-dimensional vector spaces. Let $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{E}$ be bases for V, W, and U, respectively. The matrix representation of the composition $S \circ T$ is given by the product of their respective representation matrices:

$$[S \circ T]^{\mathcal{B}}_{\mathcal{E}} = [S]^{\mathcal{C}}_{\mathcal{E}} \cdot [T]^{\mathcal{B}}_{\mathcal{C}}$$

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[rr, bend left, "S\circ T"] & W \arrow[r, "S"] & U \end{tikzcd}$$

Here $\mathcal{B}$ is a basis of the source V, $\mathcal{C}$ a basis of the space W in the middle, and $\mathcal{E}$ a basis of the target U; each matrix in the identity above carries the basis of its own source as a superscript and the basis of its own target as a subscript, and the two occurrences of $\mathcal{C}$ meet in the middle exactly where T and S meet.

Notation and The Identity Matrix

Let $\mathcal{I}$ be an index set. For $i, j \in \mathcal{I}$, we define the **Kronecker-delta** as:

$$\delta_{ij} := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Using this, we define the "*identity matrix*" $I_n \in \mathcal{M}_{n\times n}(K)$ as $I_n = (\delta_{ij})_{\substack{1\le i\le n \\ 1\le j\le n}}$. Explicitly, this is:

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

Remark 17.1: Let V be a finite-dimensional vector space, and let $n := \dim V$. Let $\mathrm{id}_V : V \to V$ be the identity map, and let $\mathcal{B}$ be any basis for V. Then, $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}} = I_n$. In other words, the matrix representing the identity map with respect to the basis $\mathcal{B}$ is precisely the identity matrix I_n.

Exercise 17.1 (*The Identity Map is Represented by I_n*): Prove Remark 17.1.

Properties of Matrix Multiplication

Proposition 17.a.2 (*Algebraic Properties of Matrix Multiplication*): Matrix multiplication satisfies the following algebraic properties:

(a) Associativity: For all $A \in \mathcal{M}_{m\times n}(K)$, $B \in \mathcal{M}_{n\times p}(K)$, and $C \in \mathcal{M}_{p\times q}(K)$, we have:

$$(AB)C = A(BC) \in \mathcal{M}_{m\times q}(K)$$

(b) Distributivity: For all $A, B \in \mathcal{M}_{m\times n}(K)$, $C \in \mathcal{M}_{n\times p}(K)$, and $C' \in \mathcal{M}_{q\times m}(K)$, we have:

$$(A+B)C = AC + BC \in \mathcal{M}_{m\times p}(K)$$

$$C'(A+B) = C'A + C'B \in \mathcal{M}_{q\times n}(K)$$

(c) Identity: For all $A \in \mathcal{M}_{m\times n}(K)$, we have $I_m A = A$ and $AI_n = A$.

(d) Scalar Multiplication: For all $\alpha \in K$, $A \in \mathcal{M}_{m\times n}(K)$, and $B \in \mathcal{M}_{n\times p}(K)$, we have:

$$\alpha(AB) = (\alpha A)B = A(\alpha B)$$

Proof:

Let $T_A : K^n \to K^m$, $T_B : K^p \to K^n$, and $T_C : K^q \to K^p$ be the linear maps defined by left-multiplication with the matrices A, B, and C, respectively (i.e., $T_A(v) = A\cdot v$ for all $v \in K^n$, $T_B(w) = B\cdot w$ for all $w \in K^p$, and $T_C(u) = C\cdot u$ for all $u \in K^q$). For any dimension r, we denote the standard basis of K^r by $\mathcal{E}_r$. We have already established that:

$$[T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} = A, \quad [T_B]_{\mathcal{E}_p}^{\mathcal{E}_n} = B, \quad \text{and} \quad [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p} = C$$

(a) We rely on the associativity of function composition, which states that $(T_A \circ T_B)\circ T_C = T_A \circ (T_B \circ T_C)$. These are both maps from K^q to K^m. We now pass to the matrix representation of these maps with respect to the standard bases $\mathcal{E}_q$ and $\mathcal{E}_m$:

$$[(T_A \circ T_B)\circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m} = [T_A \circ (T_B \circ T_C)]_{\mathcal{E}_q}^{\mathcal{E}_m}$$

By applying the rule for the matrix representation of composed maps, we get:

$$[T_A \circ T_B]_{\mathcal{E}_p}^{\mathcal{E}_m} \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p} = [T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot [T_B \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_n}$$

Expanding further, we obtain:

$$\left([T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot [T_B]_{\mathcal{E}_p}^{\mathcal{E}_n}\right) \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p} = [T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot \left([T_B]_{\mathcal{E}_p}^{\mathcal{E}_n} \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p}\right)$$

Substituting the matrices back in yields $(A\cdot B)\cdot C = A\cdot(B\cdot C)$.

(b) From the properties of linear maps, we know that $(T_A + T_B)\circ T_C = T_A \circ T_C + T_B \circ T_C$. Passing to the matrix representations, we have:

$$[(T_A + T_B)\circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m} = [T_A \circ T_C + T_B \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m}$$

This implies that:

$$[T_A + T_B]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_n} = [T_A \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m} + [T_B \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m}$$

Which directly gives us $(A+B)\cdot C = A\cdot C + B\cdot C$. The other distributive identity in **(b)** is proven similarly.

(c) The identity matrix I_m represents the identity map: $I_m = [\mathrm{id}_{K^m}]_{\mathcal{E}_m}^{\mathcal{E}_m}$. Since composing with the identity map does not change a function, we have $\mathrm{id}_{K^m} \circ T_A = T_A$. Therefore:

$$[\mathrm{id}_{K^m} \circ T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} = [T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} = A$$

This implies that $I_m \cdot A = A$. The proof for $AI_n = A$ follows the same logic.

(d) Let $T_A : K^n \to K^m$ and $T_B : K^p \to K^n$ be the linear maps associated with A and B, respectively. We want to show that $\alpha(AB) = (\alpha A)B = A(\alpha B)$. This is equivalent to showing that the corresponding linear maps are equal.

First, consider the map $Q : K^n \to K^n$ defined by $Q(v) := \alpha v$. This map is linear. If $\mathcal{E}_n$ is the standard basis for K^n, then for any $e_j \in \mathcal{E}_n$, $Q(e_j) = \alpha e_j$. Thus, the matrix representation of Q with respect to $\mathcal{E}_n$ is $[Q]_{\mathcal{E}_n}^{\mathcal{E}_n} = \alpha I_n$.

Now, let's look at the compositions:

- The linear map corresponding to $(\alpha A)B$ is $T_{\alpha A} \circ T_B$. We know that $T_{\alpha A}(v) = (\alpha A)v = \alpha(Av) = \alpha T_A(v)$, so $T_{\alpha A} = \alpha T_A$. Thus, $T_{(\alpha A)B} = (\alpha T_A) \circ T_B$.
- The linear map corresponding to $A(\alpha B)$ is $T_A \circ T_{\alpha B}$. Similarly, $T_{\alpha B} = \alpha T_B$. Thus, $T_{A(\alpha B)} = T_A \circ (\alpha T_B)$.

We need to show that $(\alpha T_A) \circ T_B = T_A \circ (\alpha T_B)$. For any $v \in K^p$:

$$\begin{aligned} ((\alpha T_A) \circ T_B)(v) &= (\alpha T_A)(T_B(v)) \\ &= \alpha(T_A(T_B(v))) \quad \text{(by definition of scalar multiplication for maps)} \end{aligned}$$

And for the other side:

$$\begin{aligned} (T_A \circ (\alpha T_B))(v) &= T_A((\alpha T_B)(v)) \\ &= T_A(\alpha(T_B(v))) \quad \text{(by definition of scalar multiplication for maps)} \\ &= \alpha(T_A(T_B(v))) \quad \text{(by linearity of } T_A\text{)} \end{aligned}$$

Since both compositions yield the same result, the corresponding matrices are equal: $(\alpha A)B = A(\alpha B)$.

Finally, to show $\alpha(AB) = (\alpha A)B$: The matrix $\alpha(AB)$ corresponds to the linear map $\alpha(T_A \circ T_B)$. The matrix $(\alpha A)B$ corresponds to the linear map $(\alpha T_A) \circ T_B$. We have already shown that $(\alpha T_A) \circ T_B = \alpha(T_A \circ T_B)$ (from the first part of the derivation above). Therefore, $\alpha(AB) = (\alpha A)B$.

Q.E.D.

The notes prepare statement **(d)** differently, through the following observation about the map that scales every vector by a fixed factor.

Claim 17.1 (*The Scaling Map is Represented by αI_n*): Let V be a finite-dimensional vector space over K, let $\alpha \in K$, and let $\mathcal{B}$ be a basis for V. Consider the map $Q : V \to V$ defined by $Q(v) := \alpha v$. Then Q is linear, and moreover

$$[Q]_{\mathcal{B}}^{\mathcal{B}} = \alpha \cdot I_n = \begin{pmatrix} \alpha & & 0 \\ & \ddots & \\ 0 & & \alpha \end{pmatrix}, \qquad \text{where } n = \dim V.$$

Exercise 17.2 (*Preparation for Statement (d)*): Prove Claim 17.1, and use it to prove statement **(d)** of Proposition 17.a.2.

Remark 17.2: One can prove Proposition 17.a.2 also by direct calculation with the entries. It is worth trying once, say for $(AB)C = A(BC)$, to see that it is not very pleasant — and to appreciate how much work the passage to linear maps saves.

Remark 17.3: In the space $\mathcal{M}_{n\times n}(K)$, we can add and subtract matrices, and we can also multiply matrices. There is additionally a multiplicative unit I_n. Because these operations satisfy associativity, distributivity, and other relevant axioms, it follows that $\mathcal{M}_{n\times n}(K)$ forms a (non-commutative) ring with a unity.

Important remark 17.2: Matrix multiplication is, in general, **not commutative**. There exist matrices A and B such that $A \cdot B \neq B \cdot A$. For example:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$
$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Invertible Matrices

Definition 17.a.3 (*Invertibility*): An $n \times n$ matrix $A \in \mathcal{M}_{n\times n}(K)$ is called "*invertible*" if there exists a matrix $B \in \mathcal{M}_{n\times n}(K)$ such that $A \cdot B = B \cdot A = I_n$.

Remark 17.4: If A is invertible, then there exists **only one** matrix B such that $A \cdot B = B \cdot A = I_n$.

Proof:

Assume there are two matrices B and C such that $AB = BA = I_n$ and $AC = CA = I_n$. Then, we can calculate:

$$B = B \cdot I_n = B \cdot (A \cdot C) = (B \cdot A) \cdot C = I_n \cdot C = C$$

This proves the uniqueness. **Q.E.D.**

Definition 17.a.4 (*Inverse Matrix and the General Linear Group*): If A is invertible, the unique matrix B such that $AB = BA = I_n$ is called the "*inverse*" of A and is denoted by A^{-1}. We have:

$$A \cdot A^{-1} = A^{-1} \cdot A = I_n$$

The set of all invertible $n \times n$ matrices over K is denoted by:

$$\mathrm{GL}_n(K) = \mathrm{GL}(n, K) := \{A \in \mathcal{M}_{n\times n}(K) \mid A \text{ is invertible}\}$$

Definition 17.a.5 (*Group*): Let G be a set and $\cdot : G \times G \to G$, $(a, b) \mapsto a \cdot b \in G$, be a binary operation. We say that $(G, \cdot)$ is a "*group*" if the following axioms hold:

- **(a) Associativity:** For all $g_1, g_2, g_3 \in G$, we have $g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3$.
- **(b) Identity Element:** There exists an element $e \in G$ such that $e \cdot g = g \cdot e = g$ for all $g \in G$. The element e is called the "*unit*" (or "*identity*") of G.
- **(c) Inverse Element:** For every $g \in G$, there exists an element $h \in G$ such that $g \cdot h = h \cdot g = e$. The element h is called the "*inverse*" of g.

Furthermore, if G satisfies $g_1 \cdot g_2 = g_2 \cdot g_1$ for all $g_1, g_2 \in G$, we say that G is a "*commutative group*" (or "*Abelian group*").

Proposition 17.a.6 (*Group Properties*): Let G be a group. Then:

- **(a)** The unit $e \in G$ is unique. (The unit e is sometimes denoted as $1 \in G$.)
- **(b)** For every $g \in G$, its inverse is uniquely defined. It is denoted by g^{-1} (so that $g \cdot g^{-1} = g^{-1} \cdot g = e$).
- **(c)** For every $g \in G$, we have $(g^{-1})^{-1} = g$.
- **(d)** For all $g_1, g_2 \in G$, we have $(g_1 \cdot g_2)^{-1} = g_2^{-1} \cdot g_1^{-1}$.

Proposition 17.a.7 (*The General Linear Group*): The set $\mathrm{GL}_n(K)$, endowed with matrix multiplication $(A, B) \mapsto A \cdot B$, is a group. The unity is the identity matrix I_n. The inverse of a matrix $A \in \mathrm{GL}_n(K)$ is A^{-1}. Moreover, for all $A, B \in \mathrm{GL}_n(K)$, we have:

$$(A^{-1})^{-1} = A \quad \text{and} \quad (AB)^{-1} = B^{-1}A^{-1}$$

Proof:

Let $A, B \in \mathrm{GL}_n(K)$. To establish that $\mathrm{GL}_n(K)$ forms a group, we first need to show closure, namely that $AB \in \mathrm{GL}_n(K)$ as well. Indeed, we can verify this by checking the inverse property:

$$(B^{-1} \cdot A^{-1})(AB) = B^{-1}A^{-1}AB = B^{-1}I_nB = B^{-1}B = I_n$$

and, multiplying from the right,

$$(AB)(B^{-1}A^{-1}) = ABB^{-1}A^{-1} = AI_nA^{-1} = AA^{-1} = I_n$$

This proves that AB is invertible with inverse $B^{-1}A^{-1}$, and therefore, $AB \in \mathrm{GL}_n(K)$. The other statements in the proposition follow immediately from what we have proved earlier regarding the uniqueness of inverses. **Q.E.D.**

Corollary 17.a.8 (*Unique Solution via Matrix Inverse*): Let $A \in \mathrm{GL}_n(K)$. Then, for all $b \in K^n$, the linear system of equations $A \cdot x = b$ has a unique solution, namely $x = A^{-1} \cdot b$.

Proof:

If $Ax = b$, then multiplying both sides from the left by A^{-1} yields:

$$x = I_nx = (A^{-1}A)x = A^{-1}(Ax) = A^{-1}b.$$

Conversely, if we substitute $x = A^{-1}b$ into the equation, we obtain:

$$A(A^{-1}b) = (AA^{-1})b = I_nb = b$$

This confirms both the existence and uniqueness of the solution. **Q.E.D.**

Proposition 17.a.9 (*Equivalence of Invertibility and Isomorphism*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then, $A \in \mathrm{GL}_n(K)$ (i.e., A is invertible) if and only if the associated linear map $T_A : K^n \to K^n$ is an isomorphism. Moreover, in this case, $(T_A)^{-1} = T_{A^{-1}}$.

Proof:

First, assume that A is invertible. We claim that $T_{A^{-1}} \circ T_A = \mathrm{id}_{K^n}$ and $T_A \circ T_{A^{-1}} = \mathrm{id}_{K^n}$, which will show that T_A is an isomorphism. Indeed, for all $v \in K^n$:

$$T_{A^{-1}} \circ T_A(v) = A^{-1} \cdot (T_A(v)) = A^{-1} \cdot A \cdot v = I_n \cdot v = v$$

Similarly, one shows that for all $v \in K^n$:

$$T_A \circ T_{A^{-1}}(v) = I_n \cdot v = v$$

This proves that T_A is an isomorphism.

Conversely, assume that $T_A : K^n \to K^n$ is an isomorphism. Let $S := (T_A)^{-1}$. By Lemma 15.a.5, we know that any linear map from K^n to K^n is given by left-multiplication with a matrix, so $S = T_B$ for some $B \in \mathcal{M}_{n\times n}(K)$. Now, for all $w \in K^n$:

$$w = S \circ T_A(w) = T_B(A \cdot w) = B \cdot A \cdot w = (BA) \cdot w$$

We apply this equality for the standard basis vectors $w = e_1, w = e_2, \dots, w = e_n$ and obtain that $BA = I_n$. Similarly, using $T_A \circ S = \mathrm{id}_{K^n}$, one shows that $AB = I_n$ too. Thus, A is invertible. **Q.E.D.**

Exercise 17.3 (*Properties of the Transpose Map*):

(a) Let $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$. Prove that $(AB)^\mathsf{T} = B^\mathsf{T} \cdot A^\mathsf{T}$. (*Hint: Proving this by direct calculation is not so bad.*)

(b) Let $A \in \mathrm{GL}_n(K)$. Prove that $A^\mathsf{T} \in \mathrm{GL}_n(K)$ and that $(A^\mathsf{T})^{-1} = (A^{-1})^\mathsf{T}$.

Definition 17.a.10 (*Triangular and Diagonal Matrices*): A matrix $A \in \mathcal{M}_{n\times n}(K)$, with entries $A = (a_{ij})_{\substack{1\le i\le n\\ 1\le j\le n}}$, is called:

- "*Upper triangular*" if $a_{ij} = 0$ for all $i > j$.
- "*Lower triangular*" if $a_{ij} = 0$ for all $i < j$.
- "*Diagonal*" if $a_{ij} = 0$ for all $i \neq j$. A diagonal matrix has the form:

$$\begin{pmatrix} * & & 0 \\ & \ddots & \\ 0 & & * \end{pmatrix}$$

Lemma 17.a.11 (*Closure of Triangular Matrices*): Let A and B be two diagonal matrices (respectively, both upper triangular, or both lower triangular). Then, their product $A \cdot B$ is of the same type.

The notes leave the proof as an exercise; it is carried out among the solutions at the end of this section, on Page 113.

Solutions to the Exercises

Proof of Remark 17.1:

Let $\mathcal{B} = (v_1, \ldots, v_n)$. The j-th column of $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}}$ is the coordinate vector of $\mathrm{id}_V(v_j) = v_j$ with respect to $\mathcal{B}$. Writing $v_j = 0 \cdot v_1 + \cdots + 1 \cdot v_j + \cdots + 0 \cdot v_n$ shows that this coordinate vector is e_j, whose i-th entry is δ_{ij}. So the (i,j)-entry of $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}}$ is δ_{ij}, which is exactly I_n. **Q.E.D.**

Proof of Claim 17.1 and of Proposition 17.a.2 (d):

The map $Q(v) := \alpha v$ is linear, since $Q(av + bv') = \alpha(av + bv') = a\alpha v + b\alpha v' = aQ(v) + bQ(v')$. For its matrix, the j-th column of $[Q]_{\mathcal{B}}^{\mathcal{B}}$ is $[Q(v_j)]_{\mathcal{B}} = [\alpha v_j]_{\mathcal{B}} = \alpha e_j$, and a matrix whose j-th column is αe_j for every j is precisely αI_n.

Now let $\alpha \in K$, $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$. Applying the claim in K^m with respect to the standard basis, multiplication by α on K^m is represented by αI_m, and therefore $\alpha C = (\alpha I_m)\cdot C$ for every $C \in \mathcal{M}_{m\times q}(K)$; likewise $\alpha A = (\alpha I_m)A$. Using associativity, Proposition 17.a.2 (a), twice:

$$\alpha(AB) = (\alpha I_m)(AB) = \big((\alpha I_m)A\big)B = (\alpha A)B.$$

For the remaining identity, note that $(\alpha I_m)A = A(\alpha I_n)$, since both sides have (i,j)-entry αa_{ij}. Hence

$$(\alpha A)B = \big(A(\alpha I_n)\big)B = A\big((\alpha I_n)B\big) = A(\alpha B),$$

again by associativity. Together these give $\alpha(AB) = (\alpha A)B = A(\alpha B)$. **Q.E.D.**

Proof of Exercise 17.3:

(a) Write $A = (a_{ij}) \in \mathcal{M}_{m\times n}(K)$ and $B = (b_{jk}) \in \mathcal{M}_{n\times p}(K)$, and put $C := AB$, so that $c_{ik} = \sum_{\ell=1}^{n} a_{i\ell} b_{\ell k}$. Both $(AB)^{\mathsf{T}}$ and $B^{\mathsf{T}}A^{\mathsf{T}}$ lie in $\mathcal{M}_{p\times m}(K)$, so it suffices to compare entries. The (k,i)-entry of $(AB)^{\mathsf{T}}$ is

$$c_{ik} = \sum_{\ell=1}^{n} a_{i\ell} b_{\ell k}.$$

On the other hand, the (k,i)-entry of $B^{\mathsf{T}}A^{\mathsf{T}}$ is the k-th row of B^{T} times the i-th column of A^{T}, that is,

$$\sum_{\ell=1}^{n} (B^{\mathsf{T}})_{k\ell} (A^{\mathsf{T}})_{\ell i} = \sum_{\ell=1}^{n} b_{\ell k} a_{i\ell},$$

which is the same sum, since the entries are scalars and K is commutative. Hence $(AB)^\intercal = B^\intercal A^\intercal$.

(b) Let $A \in \mathrm{GL}_n(K)$. Using **(a)** and $I_n^\intercal = I_n$,

$$A^\intercal \cdot (A^{-1})^\intercal = (A^{-1}A)^\intercal = I_n^\intercal = I_n, \qquad (A^{-1})^\intercal \cdot A^\intercal = (AA^{-1})^\intercal = I_n^\intercal = I_n.$$

So $A^\intercal$ has a two-sided inverse, which means $A^\intercal \in \mathrm{GL}_n(K)$ and $(A^\intercal)^{-1} = (A^{-1})^\intercal$. Note that no determinants are needed: part **(a)** alone produces the inverse explicitly.

Q.E.D.

Proof of Lemma 17.a.11:

We give the upper triangular case; the other two follow from it.

Let $A = (a_{ij})$ and $B = (b_{ij})$ be upper triangular, so that $a_{ik} = 0$ whenever $i > k$, and $b_{kj} = 0$ whenever $k > j$. Write $C := AB = (c_{ij})$, so that $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$.

Fix a pair of indices with $i > j$, and consider a single term $a_{ik}b_{kj}$ of that sum. Exactly one of the following two cases occurs:

- $k < i$. Then $a_{ik} = 0$, because A is upper triangular.
- $k \geq i$. Then $k \geq i > j$, so $k > j$, and hence $b_{kj} = 0$, because B is upper triangular.

In either case the term vanishes, so $c_{ij} = 0$ for all $i > j$, which says precisely that C is upper triangular.

For the lower triangular case, note that A is lower triangular if and only if $A^\intercal$ is upper triangular. So if A and B are lower triangular, then $A^\intercal$ and $B^\intercal$ are upper triangular, and by Exercise 17.3 **(a)** together with the case already proved,

$$(AB)^\intercal = B^\intercal A^\intercal \quad \text{is upper triangular,}$$

which makes AB lower triangular. Finally, a matrix is diagonal exactly when it is both upper and lower triangular, so the diagonal case follows from the two cases just proved. **Q.E.D.**

AI-Note: Every numbered statement of the notes for this section is present in the transcript, and 17.a.1–17.a.11 already matched the handwritten numbering. What was missing sits around them.

Restored: the preparation the notes give for statement **(d)** of Proposition 17.a.2, namely Claim 17.1, that scaling by α is represented by αI_n, which is how the notes ask for that statement to be proved; the remark that the proposition can also be proved by direct calculation, with Prof. Biran's warning that it is *"not very pleasant"*; and the row-wise description of a product in the second special case, which the notes give alongside the column-wise one.

Removed: a second, verbatim copy of the remark that $\mathcal{M}_{n\times n}(K)$ is a non-commutative ring. It appeared twice in the transcript, a few lines apart, and once in the notes.

Two things were wrong rather than missing. The diagram under *"Connections to Linear Maps"* placed $\mathrm{Sp}(\mathcal{B})$ below V joined by a rotated "$\in$", asserting $\mathrm{Sp}(\mathcal{B}) \in V$, which is not a relation between a span and the space containing it; the notes have no such diagram, and it is replaced by the composition diagram with the bookkeeping said in words. And the proof of Lemma 17.a.11 opened with the word *"Exercise"* and then argued, in the middle of the paragraph, that *"for a_{ik} to be non-zero, k must be less than or equal to i"* — the condition runs the other way. The conclusion was right and the argument was not.

The notes mark four things *"Exc."*: the identity matrix remark, the preparation claim and statement **(d)**, the two transpose identities, and Lemma 17.a.11. All four are now stated as exercises and solved on Page 113. The transcript had instead embedded *"proof sketches"* inside the exercise environment, and one of them derived $A^\intercal \in \mathrm{GL}_n(K)$ from $\det(A^\intercal) = \det(A)$ — determinants are three chapters away, and the transpose identity of part **(a)** produces the inverse directly.

Finally, terms being defined for the first time — the product, the Kronecker-delta, the identity matrix, invertible, the inverse, upper and lower triangular, diagonal — were marked with `\qt` rather than `\newterm`, which the house style reserves for exactly this.

Change of Basis (Section 17.b)

Question 17.1: How does the representation matrix $[T]$ of a linear map $T : V \to W$ depend on the choices of bases $\mathcal{B}$ and $\mathcal{C}$?

Corollary 17.b.1 (*Change of Basis Formula*): Let V and W be finite-dimensional vector spaces over K. Let $n := \dim V$ and $m := \dim W$. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for V, and let $\mathcal{C}$ and $\mathcal{C}'$ be two bases for W. Then, $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} \in \mathrm{GL}_n(K)$ and $[\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}'} \in \mathrm{GL}_m(K)$, with inverses given by:

$$[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} = ([\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1} \quad \text{and} \quad [\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}'} = ([\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}})^{-1}$$

Let $T : V \to W$ be a linear map. Then, its representation matrix changes according to the formula:

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

Proof:

We can write T as the composition $T = \mathrm{id}_W \circ T \circ \mathrm{id}_V = \mathrm{id}_W \circ (T \circ \mathrm{id}_V)$. Passing to the matrix representations, this yields:

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T \circ \mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}'} = [\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

Also, we have the identity $\mathrm{id}_V = \mathrm{id}_V \circ \mathrm{id}_V$. This implies that:

$$I_n = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} \cdot [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}}$$

and similarly:

$$I_n = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

This demonstrates that $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$ is invertible and that its inverse is indeed $([\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1}$. The proof for $[\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}'}$ follows the exact same logic. **Q.E.D.**

Definition 17.b.2 (*Change of Basis Matrix*): Let V be a finite-dimensional vector space over K, and let $n := \dim V$. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases of V. The matrix $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \in \mathrm{GL}_n(K)$ is called the "*change of basis matrix*" between $\mathcal{B}$ and $\mathcal{B}'$, or the "*transition matrix*" from basis $\mathcal{B}$ to basis $\mathcal{B}'$.

Remark 17.5: If $\mathcal{B} = (w_1, \ldots, w_n)$ and $\mathcal{B}' = (w_1', \ldots, w_n')$, then the transition matrix is explicitly constructed by placing the coordinate vectors of the old basis elements with respect to the new basis into the columns:

$$[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [w_1]_{\mathcal{B}'} & \ldots & [w_n]_{\mathcal{B}'} \\ | & & | \end{pmatrix}$$

Proposition 17.b.3 (*Representation of Isomorphisms*): Let $T : V \to W$ be an isomorphism between two finite-dimensional vector spaces. Let $\mathcal{B}$ and $\mathcal{C}$ be bases for V and W, respectively. Then, the matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$ is an invertible matrix, and furthermore:

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{C}}^{\mathcal{B}})^{-1}$$

Proof:

Let $n := \dim V = \dim W$. By the properties of composed linear maps and representation matrices, we have:

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I_n$$

and conversely:

$$[T]^{\mathcal{B}}_{\mathcal{C}} \cdot [T^{-1}]^{\mathcal{C}}_{\mathcal{B}} = [\mathrm{id}_W]^{\mathcal{C}}_{\mathcal{C}} = I_n$$

This proves the claim. **Q.E.D.**

Equivalence and Similarity of Matrices

Question 17.2: Suppose $T : V \to V$ is an endomorphism. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for V. What is the relation between $[T]^{\mathcal{B}'}_{\mathcal{B}'}$ and $[T]^{\mathcal{B}}_{\mathcal{B}}$?

Corollary 17.b.4 (*Similarity of Representation Matrices*): Let V be a finite-dimensional vector space, and let $T : V \to V$ be a linear map. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for V. Then:

$$[T]^{\mathcal{B}'}_{\mathcal{B}'} = [\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}'} \cdot [T]^{\mathcal{B}}_{\mathcal{B}} \cdot [\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}} = ([\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}})^{-1} \cdot [T]^{\mathcal{B}}_{\mathcal{B}} \cdot [\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}$$

Definition 17.b.5 (*Equivalence and Similarity*): We define two types of relations between matrices:

- **(a)** Let $A, B \in \mathcal{M}_{m\times n}(K)$. We say that A and B are "*equivalent*" if there exist matrices $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that $B = PAQ$.
- **(b)** Let $A, B \in \mathcal{M}_{n\times n}(K)$. We say that A and B are "*similar*" if there exists a matrix $P \in \mathrm{GL}_n(K)$ such that $B = P^{-1}AP$.

Exercise 17.4 (*Properties and Characterization of Matrix Equivalence*):

- **(a)** Show that *"equivalence"* between $m \times n$ matrices defines an equivalence relation on $\mathcal{M}_{m\times n}(K)$.
- **(b)** Show that two matrices $A, B \in \mathcal{M}_{m\times n}(K)$ are equivalent if and only if there exist two vector spaces V and W with $\dim V = n$ and $\dim W = m$, two bases $\mathcal{B}$ and $\mathcal{B}'$ of V, and two bases $\mathcal{C}$ and $\mathcal{C}'$ of W such that:

  $$A = [T]^{\mathcal{B}}_{\mathcal{C}} \quad \text{and} \quad B = [T]^{\mathcal{B}'}_{\mathcal{C}'}$$

  In other words, A is equivalent to B if and only if A and B are matrix representations of the exact same linear map T, but with respect to different choices of bases for V and W.
- **(c)** Show that *"similarity"* between $n \times n$ matrices defines an equivalence relation on $\mathcal{M}_{n\times n}(K)$.
- **(d)** Show that two square matrices $M, N \in \mathcal{M}_{n\times n}(K)$ are similar if and only if there exists an n-dimensional space V, an endomorphism $T : V \to V$, and two bases $\mathcal{B}$ and $\mathcal{B}'$ of V such that $M = [T]^{\mathcal{B}}_{\mathcal{B}}$ and $N = [T]^{\mathcal{B}'}_{\mathcal{B}'}$.

Proposition 17.b.6 (*Normal Form for Linear Maps*): Let V and W be finite-dimensional vector spaces over K, with dimensions $n = \dim V$ and $m = \dim W$. Let $T : V \to W$ be a linear map with $\mathrm{rank}(T) = r$. (Recall that $\mathrm{rank}(T) := \dim \mathrm{Im}(T)$). Then, there exists a basis $\mathcal{B} = (v_1, \ldots, v_n)$ of V and a basis $\mathcal{C} = (w_1, \ldots, w_m)$ of W such that:

$$[T]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} I_r & O_{r,n-r} \\ O_{m-r,r} & O_{m-r,n-r} \end{pmatrix}$$

where $O_{p,q}$ stands for the zero matrix in $\mathcal{M}_{p\times q}(K)$.

Proof:

Put $\ell := \dim \ker(T)$. By the rank theorem, we know that $r = n - \ell$. Let $v_1, \ldots, v_\ell$ be a basis of $\ker(T)$, and extend it to a full basis $(v_1, \ldots, v_\ell, v_{\ell+1}, \ldots, v_n)$ of V. It will be useful to work below with the rearranged basis:

$$\mathcal{B} = (v_{\ell+1}, \ldots, v_n, v_1, \ldots, v_\ell)$$

In the proof of the rank theorem, we have seen that $T(v_{\ell+1}), \dots, T(v_n)$ form a basis for $\mathrm{Im}(T)$. Define $w_i := T(v_{\ell+i})$ for $i = 1, \dots, r$ (i.e., $w_1 = T(v_{\ell+1}), \dots, w_r = T(v_{\ell+r})$), and extend this list to a basis of W:

$$\mathcal{C} := (w_1, \dots, w_r, w_{r+1}, \dots, w_m)$$

It now directly follows from these definitions that the representation matrix $[T]^{\mathcal{B}}_{\mathcal{C}}$ takes the exact block form described in the proposition. Indeed, the j-th column of $[T]^{\mathcal{B}}_{\mathcal{C}}$ is the coordinate vector of the image of the j-th member of $\mathcal{B}$. For $1 \leq j \leq r$ that member is $v_{\ell+j}$, whose image is w_j by construction, so the column is $[w_j]_{\mathcal{C}} = e_j \in K^m$. For $r < j \leq n$ the member is one of $v_1, \dots, v_\ell$, which lie in $\ker(T)$, so the column is $[0_W]_{\mathcal{C}} = 0$. A matrix whose first r columns are $e_1, \dots, e_r$ and whose remaining columns vanish is exactly the asserted block matrix. **Q.E.D.**

Corollary 17.b.7 (*Matrix Equivalence and Normal Form*): Let $A \in \mathcal{M}_{m \times n}(K)$. Then, there exist invertible matrices $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that:

$$PAQ = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$$

where $r = \mathrm{rank}_{\mathrm{col}}(A)$. In particular, $\mathrm{rank}_{\mathrm{col}}(A) \leq \min\{m, n\}$.

Proof:

Consider the linear map $T_A : K^n \to K^m$ defined by left-multiplication, $T_A(v) = A \cdot v$. Then, $r = \mathrm{rank}_{\mathrm{col}}(A) = \dim \mathrm{Im}(T_A)$, because we have previously seen that

$$\mathrm{Im}(T_A) = \mathrm{Sp}(T_A(e_1), \dots, T_A(e_n)) = \mathrm{Sp}(\text{columns of } A).$$

Now, we apply the previous proposition to T_A, yielding that:

$$[T_A]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$$

for some basis $\mathcal{B}$ of K^n and some basis $\mathcal{C}$ of K^m. Using the change of basis formula, we can rewrite this as:

$$\underbrace{[\mathrm{id}_{K^m}]^{\mathcal{C}}_{\mathcal{E}_m}}_{=:P} \cdot \underbrace{[T_A]^{\mathcal{E}_n}_{\mathcal{E}_m}}_{=A} \cdot \underbrace{[\mathrm{id}_{K^n}]^{\mathcal{B}}_{\mathcal{E}_n}}_{=:Q} = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$$

where $\mathcal{E}_n$ and $\mathcal{E}_m$ are the standard bases of K^n and K^m, respectively. Setting $P = [\mathrm{id}_{K^m}]^{\mathcal{C}}_{\mathcal{E}_m}$ and $Q = [\mathrm{id}_{K^n}]^{\mathcal{B}}_{\mathcal{E}_n}$ concludes the proof. **Q.E.D.**

Remark 17.6: The fact that $\mathrm{rank}_{\mathrm{col}}(A) \leq \min\{m, n\}$ also follows from observing the linear map $T_A : K^n \to K^m$. We have:

$$\mathrm{rank}_{\mathrm{col}}(A) = \dim \mathrm{Im}(T_A) = n - \dim \ker(T_A) \leq n$$

and simultaneously,

$$\mathrm{rank}_{\mathrm{col}}(A) = \dim \mathrm{Im}(T_A) \leq \dim K^m = m$$

Question 17.3: Recall that we have an equivalence relation on $\mathcal{M}_{m \times n}(K)$: $A \sim B$ if there exist $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that $B = PAQ$. Can we describe the equivalence classes of this equivalence relation?

Corollary 17.b.8 (*Classification of Matrices by Rank*): Let $A, B \in \mathcal{M}_{m \times n}(K)$. Then, $A \sim B$ if and only if $\mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{col}}(B)$. Moreover, for every $0 \leq r \leq \min\{m, n\}$, there is precisely one equivalence class of matrices A with $\mathrm{rank}_{\mathrm{col}} = r$.

Proof:

Let $A, B \in \mathcal{M}_{m\times n}(K)$. First, assume that $\operatorname{rank}_{\text{col}}(A) = \operatorname{rank}_{\text{col}}(B)$, and denote this common column rank by r. By Corollary 17.b.7, we know that $A \sim \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ and $B \sim \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$. Since $\sim$ is an equivalence relation, it naturally follows that $A \sim B$.

Conversely, assume that $A \sim B$. This implies that there exist matrices $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that $B = PAQ$. Transitioning to the associated linear maps, this means $T_B = T_P \circ T_A \circ T_Q$. We analyze the image of T_B:

$$\operatorname{Im}(T_B) = T_B(K^n) = T_P \circ T_A(T_Q(K^n)) = T_P \circ T_A(K^n) = T_P(\operatorname{Im}(T_A))$$

Note that $T_Q(K^n) = K^n$ because T_Q is an isomorphism. Consequently, the dimension of the image is:

$$\operatorname{rank}_{\text{col}}(B) = \dim \operatorname{Im}(T_B) = \dim T_P(\operatorname{Im}(T_A)) = \dim \operatorname{Im}(T_A) = \operatorname{rank}_{\text{col}}(A)$$

where the second to last equality holds because T_P is an isomorphism, preserving the dimension of subspaces. **Q.E.D.**

Solutions to the Exercises

Proof of Exercise 17.4:

Throughout, write $A \sim B$ for equivalence and $A \approx B$ for similarity.

(a) Equivalence is an equivalence relation. It is reflexive, since $A = I_m A I_n$ with $I_m \in \mathrm{GL}_m(K)$ and $I_n \in \mathrm{GL}_n(K)$. It is symmetric: if $B = PAQ$ with $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$, then multiplying by P^{-1} on the left and Q^{-1} on the right gives $A = P^{-1}BQ^{-1}$, and P^{-1}, Q^{-1} are again invertible by Proposition 17.a.7. It is transitive: if $B = PAQ$ and $C = P'BQ'$, then

$$C = P'(PAQ)Q' = (P'P)A(QQ'),$$

and $P'P \in \mathrm{GL}_m(K)$, $QQ' \in \mathrm{GL}_n(K)$, again by Proposition 17.a.7.

(b) Equivalent matrices represent one map in two pairs of bases. Suppose first that $A = [T]^{\mathcal{B}}_{\mathcal{C}}$ and $B = [T]^{\mathcal{B}'}_{\mathcal{C}'}$ for a single linear map $T : V \to W$. By Corollary 17.b.1,

$$B = \underbrace{[\mathrm{id}_W]^{\mathcal{C}}_{\mathcal{C}'}}_{\in \mathrm{GL}_m(K)} \cdot A \cdot \underbrace{[\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}}_{\in \mathrm{GL}_n(K)},$$

so $A \sim B$.

Conversely, suppose $B = PAQ$ with $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$. Take $V := K^n$, $W := K^m$ and $T := T_A$, and let $\mathcal{E}_n$, $\mathcal{E}_m$ be the standard bases, so that $A = [T]^{\mathcal{E}_n}_{\mathcal{E}_m}$. Let $\mathcal{B}'$ be the list of columns of Q and let $\mathcal{C}'$ be the list of columns of P^{-1}; both are bases, since a square matrix is invertible precisely when its columns form a basis. The matrix $[\mathrm{id}]^{\mathcal{B}'}_{\mathcal{E}_n}$ has as its j-th column the coordinate vector of the j-th member of $\mathcal{B}'$ with respect to the standard basis, that is, the j-th column of Q; hence $[\mathrm{id}_{K^n}]^{\mathcal{B}'}_{\mathcal{E}_n} = Q$, and likewise $[\mathrm{id}_{K^m}]^{\mathcal{C}'}_{\mathcal{E}_m} = P^{-1}$, so that $[\mathrm{id}_{K^m}]^{\mathcal{E}_m}_{\mathcal{C}'} = P$ by Corollary 17.b.1. That same corollary now gives

$$[T]^{\mathcal{B}'}_{\mathcal{C}'} = [\mathrm{id}_{K^m}]^{\mathcal{E}_m}_{\mathcal{C}'} \cdot [T]^{\mathcal{E}_n}_{\mathcal{E}_m} \cdot [\mathrm{id}_{K^n}]^{\mathcal{B}'}_{\mathcal{E}_n} = PAQ = B,$$

so A and B do represent the same map in two pairs of bases.

(c) Similarity is an equivalence relation. Reflexivity: $A = I_n^{-1} A I_n$. Symmetry: if $B = P^{-1}AP$, then $A = PBP^{-1} = (P^{-1})^{-1}B(P^{-1})$. Transitivity: if $B = P^{-1}AP$ and $C = R^{-1}BR$, then

$$C = R^{-1}P^{-1}APR = (PR)^{-1}A(PR),$$

using $(PR)^{-1} = R^{-1}P^{-1}$ from Proposition 17.a.7.

(d) Similar matrices represent one endomorphism in two bases. If $M = [T]^{\mathcal{B}}_{\mathcal{B}}$ and $N = [T]^{\mathcal{B}'}_{\mathcal{B}'}$ for an endomorphism $T : V \to V$, then Corollary 17.b.4 gives $N = P^{-1}MP$ with $P := [\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}} \in \mathrm{GL}_n(K)$, so $M \approx N$.
Conversely, let $N = P^{-1}MP$ with $P \in \mathrm{GL}_n(K)$. Take $V := K^n$, $T := T_M$ and $\mathcal{B} := \mathcal{E}_n$, so that $M = [T]^{\mathcal{B}}_{\mathcal{B}}$. Let $\mathcal{B}'$ be the list of columns of P, a basis of K^n, so that $[\mathrm{id}_{K^n}]^{\mathcal{B}'}_{\mathcal{E}_n} = P$ as in **(b)**. Then Corollary 17.b.4 gives

$$[T]^{\mathcal{B}'}_{\mathcal{B}'} = \left([\mathrm{id}_{K^n}]^{\mathcal{B}'}_{\mathcal{E}_n}\right)^{-1} \cdot [T]^{\mathcal{E}_n}_{\mathcal{E}_n} \cdot [\mathrm{id}_{K^n}]^{\mathcal{B}'}_{\mathcal{E}_n} = P^{-1}MP = N.$$

Q.E.D.

AI-Note: Every numbered statement of the notes for this section is present in the transcript, in the notes' order, and 17.b.1–17.b.8 already matched the handwritten numbering; the proofs follow Prof. Biran's.

The four parts of Exercise 17.4 are his, marked *"Exc."* in the notes and left unanswered by the transcript; they are solved above. Two of them are the substantial ones: equivalence of matrices means representing the same linear map in two pairs of bases, and similarity means representing the same endomorphism in two bases. Both directions need the observation that for an invertible P, the columns of P form a basis whose transition matrix from the standard basis is P itself.

Also here: the block form asserted at the end of the proof of Proposition 17.b.6 is now verified column by column rather than declared to follow from the definitions, and the notation was brought to the house macros in three places, where $\mathcal{M}$ and GL had been typeset as plain M and GL.

Column Rank and Row Rank (Section 17.c)

Theorem 17.c.1 (*Equality of Column and Row Rank*): Let $A \in \mathcal{M}_{m\times n}(K)$. Then, $\mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{row}}(A)$.

Remark 17.7 (*Notation and Properties of Rank*):

(a) Because of this theorem, we will from now on denote $\mathrm{rank}(A) := \mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{row}}(A)$.

(b) Suppose A' is a row-reduced echelon matrix that is row-equivalent to A. Then, we have:

$$\mathrm{rank}(A) = \mathrm{rank}_{\mathrm{row}}(A) = \mathrm{rank}_{\mathrm{row}}(A') = \text{number of pivots in } A'$$

(c) Recall that for a linear map $T : V \to W$, we defined $\mathrm{rank}(T) := \dim \mathrm{Im}(T)$. Moreover, if $A \in \mathcal{M}_{m\times n}(K)$, then $\mathrm{rank}(T_A) = \dim \mathrm{Im}(T_A) = \mathrm{rank}_{\mathrm{col}}(A)$.

Preparation for the Proof

Lemma 17.c.2 (*Rank Invariance under Composition with Isomorphisms*): Let $T : V \to W$ be a linear map, and let $S : W \to W'$ and $L : P \to V$ be isomorphisms. Then:

(a) $\mathrm{rank}(S \circ T) = \mathrm{rank}(T)$.

(b) $\mathrm{rank}(T \circ L) = \mathrm{rank}(T)$.

Proof:

(a) By definition, $\mathrm{rank}(S\circ T) = \dim(S(T(V)))$. Since S is an isomorphism, its restriction to the subspace $T(V) \subseteq W$ is injective. An injective linear map preserves the dimension of a vector space, and therefore:

$$\dim(S(T(V))) = \dim(T(V)) = \mathrm{rank}(T)$$

(b) By definition, $\mathrm{rank}(T \circ L) = \dim(T(L(P)))$. Since L is an isomorphism, it is surjective, which implies that $L(P) = V$. Consequently:

$$\dim(T(L(P))) = \dim(T(V)) = \mathrm{rank}(T)$$

Q.E.D.

Lemma 17.c.3 (*Invertibility and Isomorphism*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then A is invertible if and only if $T_A : K^n \to K^n$ is an isomorphism.

Proof:

Suppose first that A is invertible. Then $T_{A^{-1}} = (T_A)^{-1}$, so T_A is an isomorphism.

Conversely, suppose that T_A is an isomorphism. We know that $A = [T_A]_{\mathcal{E}_n}^{\mathcal{E}_n}$, and by a previous result the representation matrix of an isomorphism is invertible, so A is invertible. **Q.E.D.**

Remark 17.8 (*Invertibility of the Transpose*): It is worth recording alongside this that $A \in \mathrm{GL}_n(K)$ if and only if $A^\top \in \mathrm{GL}_n(K)$, with $(A^\top)^{-1} = (A^{-1})^\top$. This is Exercise 17.3 **(b)** of Section 17.a, and it needs nothing from the present section: applying $(XY)^\top = Y^\top X^\top$ to $A^{-1}A = AA^{-1} = I_n$ exhibits $(A^{-1})^\top$ as a two-sided inverse of $A^\top$, and the converse follows by applying the same argument to $A^\top$ and using $(A^\top)^\top = A$.

Lemma 17.c.4 (*Rank Invariance under Matrix Equivalence*): Let $A, B \in \mathcal{M}_{m\times n}(K)$ such that $B = PAQ$, where $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$. Then:

(a) $\mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{col}}(B)$.

(b) $\mathrm{rank}_{\mathrm{row}}(A) = \mathrm{rank}_{\mathrm{row}}(B)$.

Proof:

(a) Transitioning to the associated linear maps, we have $T_B = T_P \circ T_A \circ T_Q$. Since P and Q are invertible matrices, both T_P and T_Q are isomorphisms. Applying Lemma 17.c.2, we deduce that $\mathrm{rank}(T_B) = \mathrm{rank}(T_A)$. Recalling that the rank of a linear map given by a matrix matches the column rank of that matrix, we obtain $\mathrm{rank}_{\mathrm{col}}(B) = \mathrm{rank}_{\mathrm{col}}(A)$.

(b) Taking the transpose of the equation $B = PAQ$ yields $B^\top = Q^\top A^\top P^\top$. Since P and Q are invertible matrices, their transposes $P^\top$ and $Q^\top$ are also invertible (we have previously seen that $(P^\top)^{-1} = (P^{-1})^\top$). We can now apply statement **(a)** of this Lemma 17.c.4, which we have already proven, to the matrices $A^\top$ and $B^\top$. This gives us $\mathrm{rank}_{\mathrm{col}}(B^\top) = \mathrm{rank}_{\mathrm{col}}(A^\top)$. By the definition of the transpose, this immediately translates to $\mathrm{rank}_{\mathrm{row}}(B) = \mathrm{rank}_{\mathrm{row}}(A)$.

Q.E.D.

Proof of the Main Theorem

We are now ready for the proof of Theorem 17.c.1.

Proof of Theorem 17.c.1:

Let $r := \mathrm{rank}_{\mathrm{col}}(A)$. By Corollary 17.b.7, we know that there exist invertible matrices $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that:

$$PAQ = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix} =: B$$

By Lemma 17.c.4, we know that $\mathrm{rank}_{\mathrm{row}}(A) = \mathrm{rank}_{\mathrm{row}}(B)$. Thus, it suffices to determine the row rank of B.

The matrix B is of the block form $\left(\begin{smallmatrix} I_r & O \\ O & O \end{smallmatrix}\right)$. The non-zero rows of B are precisely the first r standard basis vectors $e_1^\top, e_2^\top, \ldots, e_r^\top \in K^n_{\mathrm{row}}$. Since these vectors form a linearly independent set, the dimension of the row space of B is exactly r. Consequently, $\mathrm{rank}_{\mathrm{row}}(B) = r$.

Therefore, we conclude that $\mathrm{rank}_{\mathrm{row}}(A) = r = \mathrm{rank}_{\mathrm{col}}(A)$, completing the proof. **Q.E.D.**

Important remark 17.3 (*Column Space versus Row Space*): Although $\text{rank}_{\text{col}}(A) = \text{rank}_{\text{row}}(A)$, the column space and the row space are, in general, different vector spaces. First of all, if $A \in \mathcal{M}_{m\times n}(K)$, then the column space is a subspace of K^m, whereas the row space is a subspace of K^n_{row} (which is isomorphic to K^n). But even in the case where $m = n$ (i.e., $A \in \mathcal{M}_{n\times n}(K)$), these two spaces are generally distinct! They share the exact same dimension, but they are not the same space.

AI-Note: The transcript has all four numbered statements of this section, but 17.c.3 is not the one in the notes. Prof. Biran's Lemma 2 records that A is invertible if and only if T_A is an isomorphism, in two lines; the transcript put in its place a lemma on the invertibility of A^T, with a proof that runs through row-reduced echelon forms and pivots. That lemma is true, and its first sentence is precisely Prof. Biran's Lemma 2 used as a step, but the statement is not his and nothing in this section uses it. The notes' lemma is restored as Lemma 17.c.3, and the transpose statement is kept as Remark 17.8, where it belongs: it is Exercise 17.3 **(b)** of Section 17.a, needs nothing from this section, and comes with the inverse written out.

Two citations were vague — *"a previous proposition regarding the equivalence of matrices"* is Corollary 17.b.7 — and GL was typeset as plain GL in the main proof.

More on Matrices and Linear Equations (Section 17.d)

Let $A \in \mathcal{M}_{m\times n}(K)$, and consider the homogeneous system of linear equations $A\cdot x = 0$. We denote the space of solutions of this system by:

$$\text{Sol}(A, 0) := \{x \in K^n \mid Ax = 0\} = \ker(T_A)$$

Note that by the Rank-Nullity Theorem, we have

$$\dim \text{Sol}(A, 0) = n - \text{rank}(A),$$

because $\dim \ker(T_A) = n - \dim \text{Im}(T_A)$.

Let A' be a row-reduced echelon matrix that is row-equivalent to A (for example, A' is obtained from A by Gauss elimination). We then know that the number of pivots of A' equals $\text{rank}(A)$, and the number of free variables is $n - \text{rank}(A)$.

We can visualize the structure of A' and its corresponding variables $x_1, \ldots, x_n$ as in Figure 17.1. The columns containing the leading 1's (pivots) correspond to the pivot variables, while the remaining columns correspond to the free variables. Clearly, the solution spaces are identical:

$$\text{Sol}(A, 0) = \text{Sol}(A', 0)$$

Denote by $x_{i_1}, \ldots, x_{i_\ell}$ the free variables, where $1 \le i_1 < \cdots < i_\ell \le n$ and $\ell = n - \text{rank}(A)$. Similarly, denote by $x_{j_1}, \ldots, x_{j_r}$ the pivot variables, where $1 \le j_1 < \cdots < j_r \le n$ and $r = \text{rank}(A)$.

For each choice of values for the free variables $x_{i_1}, \ldots, x_{i_\ell} \in K$, the pivot variables $x_{j_1}, \ldots, x_{j_r}$ are uniquely determined. So, we obtain a map $\Phi : K^\ell \to \text{Sol}(A, 0)$, where $\ell = n - \text{rank}(A)$, defined by:

$$\Phi(\lambda_1, \ldots, \lambda_\ell) = \text{the unique solution of } Ax = 0 \text{ (which is the solution of } A'x = 0)$$

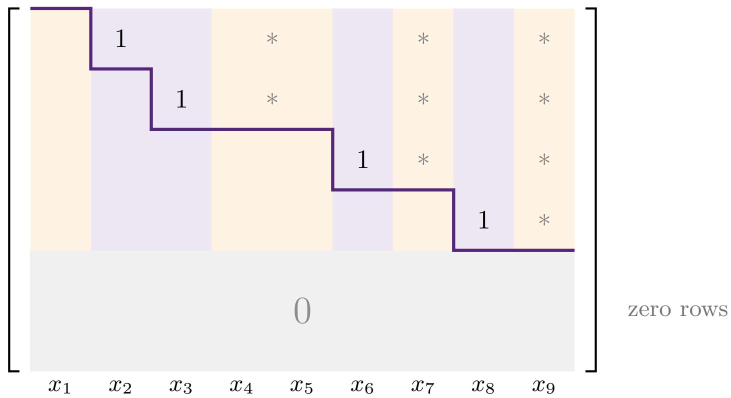
with $x_{i_1} = \lambda_1, \ldots, x_{i_k} = \lambda_k, \ldots, x_{i_\ell} = \lambda_\ell$.

Proposition 17.d.1 (*Linearity and Isomorphism of the Solution Map*): The map Φ is linear. Moreover, it is an isomorphism.

Proof:

Linearity of Φ: The k-th non-zero row of A' gives the following equation:

$$x_{j_k} + (\text{linear combination of the free variables that come after } j_k) = 0$$



pivot columns: x_2, x_3, x_6, x_8 **(**$r = \mathrm{rank}(A) = 4$ **of them)**
free columns: x_1, x_4, x_5, x_7, x_9 **(**$\ell = n - \mathrm{rank}(A) = 5$ **of them)**

Figure 17.1: The staircase shape of a row-reduced echelon matrix A', here with $n = 9$ columns and $\mathrm{rank}(A) = 4$. Every entry below the purple staircase vanishes. Each of the r non-zero rows contributes exactly one leading 1, whose column is a **pivot** column; the remaining $\ell = n - r$ columns are the **free** columns. Choosing the ℓ free variables arbitrarily forces the r pivot variables, which is precisely the map Φ below.

This allows us to isolate the pivot variable x_{j_k}:

$$x_{j_k} = -\sum_{\{q \mid 1 \le q \le \ell,\, i_q > j_k\}} a'_{k,i_q} \cdot x_{i_q}$$

This explicitly shows that the pivot variables $x_{j_1}, \ldots, x_{j_r}$ depend linearly on the free variables $x_{i_1}, \ldots, x_{i_\ell}$. This proves the linearity of Φ.

Isomorphism: To prove that Φ is an isomorphism, it is enough to show that $\ker(\Phi) = \{0\}$, because $\Phi : K^\ell \to \mathrm{Sol}(A, 0)$ is a linear map between two spaces of the exact same dimension ℓ.

But this clearly holds: if $\lambda = (\lambda_1, \ldots, \lambda_\ell) \in \ker(\Phi)$, which means $\Phi(\lambda_1, \ldots, \lambda_\ell) = 0$, then by the definition of Φ, we must have $\lambda_1 = \cdots = \lambda_\ell = 0$. This is simply because each λ_k is an actual entry in the output vector $\Phi(\lambda_1, \ldots, \lambda_\ell)$.

Furthermore, we can explicitly state the inverse map Φ^{-1}. For any $x \in \mathrm{Sol}(A, 0)$, we extract the entries corresponding to the free variables:

$$\Phi^{-1}(x) = \begin{pmatrix} x_{i_1} \\ \vdots \\ x_{i_\ell} \end{pmatrix}$$

This completes the proof. **Q.E.D.**

Corollary 17.d.2 (*Structure of Non-Homogeneous Solutions*): Let $b \in K^m$, and $A \in \mathcal{M}_{m \times n}(K)$. Then:

(a) $\mathrm{Sol}(A, b) := \{x \in K^n \mid A \cdot x = b\} \neq \emptyset$ if and only if $b \in \mathrm{Cols}(A)$.

(b) If $\mathrm{Sol}(A, b) \neq \emptyset$ and $y \in \mathrm{Sol}(A, b)$, then:

$$\mathrm{Sol}(A, b) = y + \mathrm{Sol}(A, 0) := \{y + x \mid x \in \mathrm{Sol}(A, 0)\}$$

Exercise 17.5 (*Proof of Corollary 17.d.2*): The notes leave this proof as an exercise, with the following hints.

(a) Write the matrix A in terms of its column vectors $v_1, \dots, v_n$:

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$$

For any vector $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in K^n$, the matrix-vector product is a linear combination of the columns:

$$A \cdot x = x_1 \begin{pmatrix} | \\ v_1 \\ | \end{pmatrix} + \dots + x_n \begin{pmatrix} | \\ v_n \\ | \end{pmatrix}$$

(b) Let $y \in \mathrm{Sol}(A, b)$. First, assume $x \in \mathrm{Sol}(A, 0)$. Then $A \cdot x = 0$ and $A \cdot y = b$. Adding these together yields:

$$A(y + x) = A \cdot y + A \cdot x = b + 0 = b$$

Conversely, if $z \in \mathrm{Sol}(A, b)$, meaning $A \cdot z = b$, then define $x := z - y$. We obviously have $z = y + x$, and checking the multiplication gives:

$$A \cdot x = A \cdot (z - y) = A \cdot z - A \cdot y = b - b = 0$$

This shows that any solution z can be written as $y + x$ for some $x \in \mathrm{Sol}(A, 0)$.

Proposition 17.d.3 (*Rank Criterion for Solvability*): Let $A \in \mathcal{M}_{m \times n}(K)$ and $b \in K^m$. The following statements are equivalent:

(a) $\mathrm{rank}(A|b) = \mathrm{rank}(A)$.

(b) The system of equations $Ax = b$ has a solution.

Proof:

Write A in terms of its columns,

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}.$$

We have the following chain of equivalences:

$$\begin{aligned} \mathrm{rank}(A|b) = \mathrm{rank}(A) &\iff \mathrm{rank}_{\mathrm{col}}(A|b) = \mathrm{rank}_{\mathrm{col}}(A) \\ &\iff \dim \mathrm{Sp}(v_1, \dots, v_n, b) = \dim \mathrm{Sp}(v_1, \dots, v_n) \\ &\iff b \in \mathrm{Sp}(v_1, \dots, v_n) \\ &\iff b \in \mathrm{Cols}(A) \\ &\iff \mathrm{Sol}(A, b) \neq \emptyset \end{aligned}$$

The last step holds directly by Corollary 17.d.2, thus finishing the proof. **Q.E.D.**

Proposition 17.d.4 (*Criterion for a Unique Solution*): Let $A \in \mathcal{M}_{m \times n}(K)$ and $b \in K^m$. The following statements are equivalent:

(a) $\mathrm{rank}(A) = \mathrm{rank}(A|b) = n$.

(b) The system $Ax = b$ has a unique solution.

Exercise 17.6 (*Proof of Proposition 17.d.4*): The notes leave this proof as an exercise, with the following hint. Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}.$$

Note that:

$$\operatorname{rank}(A) = n \iff \dim \operatorname{Sp}(v_1, \dots, v_n) = n$$

Since there are n column vectors, this means that $v_1, \dots, v_n$ are linearly independent, and therefore, they form a basis for $\operatorname{Cols}(A)$.

Solutions to the Exercises

Proof of Corollary 17.d.2:

(a) As the hint records, for $x = (x_1, \dots, x_n)^{\mathsf{T}} \in K^n$ the product $A \cdot x$ is the linear combination $x_1 v_1 + \dots + x_n v_n$ of the columns of A. Hence

$$\operatorname{Sol}(A, b) \neq \emptyset \iff \exists x \in K^n : x_1 v_1 + \dots + x_n v_n = b \iff b \in \operatorname{Sp}(v_1, \dots, v_n) = \operatorname{Cols}(A),$$

the last equality being the definition of the column space.

(b) Let $y \in \operatorname{Sol}(A, b)$. If $x \in \operatorname{Sol}(A, 0)$, then $A(y+x) = Ay + Ax = b + 0 = b$, so $y + x \in \operatorname{Sol}(A, b)$, which gives the inclusion $y + \operatorname{Sol}(A, 0) \subseteq \operatorname{Sol}(A, b)$. Conversely, if $z \in \operatorname{Sol}(A, b)$, put $x := z - y$; then $z = y + x$ and

$$Ax = A(z - y) = Az - Ay = b - b = 0,$$

so $x \in \operatorname{Sol}(A, 0)$ and $z \in y + \operatorname{Sol}(A, 0)$. The two inclusions give the asserted equality.

Q.E.D.

Proof of Proposition 17.d.4:

Suppose first that $\operatorname{rank}(A) = \operatorname{rank}(A \mid b) = n$. Since the two ranks agree, Proposition 17.d.3 gives a solution $y \in \operatorname{Sol}(A, b)$. Since $\operatorname{rank}(A) = n$, the computation at the start of this section gives

$$\dim \operatorname{Sol}(A, 0) = n - \operatorname{rank}(A) = 0,$$

so $\operatorname{Sol}(A, 0) = \{0\}$. By Corollary 17.d.2 **(b)**, $\operatorname{Sol}(A, b) = y + \{0\} = \{y\}$, so the solution is unique.

Conversely, suppose $Ax = b$ has a unique solution y. In particular a solution exists, so $\operatorname{rank}(A \mid b) = \operatorname{rank}(A)$ by Proposition 17.d.3. If now $x \in \operatorname{Sol}(A, 0)$, then $y + x$ is again a solution of $Ax = b$ by Corollary 17.d.2 **(b)**, and uniqueness forces $y + x = y$, that is, $x = 0$. So $\operatorname{Sol}(A, 0) = \{0\}$, and therefore

$$0 = \dim \operatorname{Sol}(A, 0) = n - \operatorname{rank}(A),$$

which gives $\operatorname{rank}(A) = n$. In the language of the hint: the columns $v_1, \dots, v_n$ are then linearly independent and form a basis of $\operatorname{Cols}(A)$, which is exactly the statement that the coefficients in $b = x_1 v_1 + \dots + x_n v_n$ are uniquely determined. **Q.E.D.**

AI-Note: This section is complete in the transcript: all four numbered statements, in the notes' order, with the notes' proofs and with Prof. Biran's hints for the two he leaves as exercises. The staircase picture that the notes draw by hand is rendered as Figure 17.1.

The two proofs the notes mark *"Exc."* stood in the text as `\begin{proof}` blocks reading *"This is left as an exercise"*, followed by the hints. They are now exercises carrying those hints, and both are solved above. The second one needs a little more than the hint gives: the uniqueness criterion follows by combining Proposition 17.d.3 with the dimension count $\dim \operatorname{Sol}(A, 0) = n - \operatorname{rank}(A)$ from the opening of the section.

Elementary Row Operations Revisited (Section 17.e)

Let $A \in \mathcal{M}_{n\times p}(K)$. We have previously learned about elementary row operations, which are utilized as an integral part of the Gaussian elimination algorithm. The main point of this section is to demonstrate that all of these operations can be expressed through matrix multiplications.

Notation 17.1: Let $n \in \mathbb{Z}_{\geq 1}$, and let $1 \leq i \leq n$, $1 \leq j \leq n$. We denote by $E_{ij} \in \mathcal{M}_{n\times n}(K)$ the matrix whose entries are all zero, except for the entry in the i-th row and j-th column, which is equal to 1.

We will now define three types of elementary matrices in $\mathcal{M}_{n\times n}(K)$. The notes introduce them without a number of their own, so that the numbering of this section starts at Lemma 17.e.1 below.

Definition (*Elementary Matrices*): We define the following three types of elementary matrices:

(a) **Type I:** Let $1 \leq i \leq n$, $1 \leq j \leq n$ with $i \neq j$, and let $\alpha \in K$. We define $Q_{ij}(\alpha)$ to be the matrix obtained from I_n by applying the row operation $R_i + \alpha R_j \longrightarrow R_i$ (i.e., replacing the i-th row by the sum of the i-th row and α times the j-th row). Explicitly, we have:

$$Q_{ij}(\alpha) = I_n + \alpha \cdot E_{ij}$$

(b) **Type II:** Let $1 \leq i \leq n$, $1 \leq j \leq n$ with $i \neq j$. We define P_{ij} to be the matrix obtained from I_n by permuting (exchanging) the i-th row and the j-th row, denoted by $R_i \leftrightarrow R_j$.

(c) **Type III:** Let $1 \leq i \leq n$, and let $0 \neq \alpha \in K$. We denote by $S_i(\alpha)$ the matrix obtained from I_n by applying the row operation $\alpha R_i \longrightarrow R_i$ (i.e., multiplying the i-th row by the non-zero scalar α).

Exercise 17.7 (*Explicit Form of the Permutation Matrix*): Show that the Type II elementary matrix can be written explicitly as $P_{ij} = I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$.

Example 17.2 (*Examples of Elementary Matrices*): Consider the space $\mathcal{M}_{4\times 4}(K)$. Examples of the three types of elementary matrices are:

$$Q_{1,3}(\alpha) = \begin{pmatrix} 1 & 0 & \alpha & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$P_{2,4} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$S_3(\alpha) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \alpha & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Lemma 17.e.1 (*Matrix Multiplication and Row Operations*): Let $A \in \mathcal{M}_{n\times p}(K)$. Multiplying A from the left by an elementary matrix performs the corresponding elementary row operation on A:

(a) Multiplying A from the left with $Q_{ij}(\alpha)$ results in the row operation $R_i + \alpha R_j \longrightarrow R_i$:

$$A \xrightarrow{R_i + \alpha R_j \longrightarrow R_i} Q_{ij}(\alpha) \cdot A$$

(b) Multiplying A from the left with P_{ij} results in the row operation $R_i \leftrightarrow R_j$:

$$A \xrightarrow{R_i \leftrightarrow R_j} P_{ij} \cdot A$$

(c) Multiplying A from the left with $S_i(\alpha)$ results in the row operation $\alpha R_i \longrightarrow R_i$:

$$A \xrightarrow{\alpha R_i \longrightarrow R_i} S_i(\alpha) \cdot A$$

Important remark 17.4 (*Column Operations*): The same principle holds true for elementary column operations, but via right-multiplication:

(a') Multiplying A from the right with $Q_{ij}(\alpha)$ results in the column operation $C_j + \alpha C_i \longrightarrow C_j$:

$$A \xrightarrow{C_j + \alpha C_i \longrightarrow C_j} A \cdot Q_{ij}(\alpha)$$

(b') Multiplying A from the right with P_{ij} results in the column operation $C_i \leftrightarrow C_j$:

$$A \xrightarrow{C_i \leftrightarrow C_j} A \cdot P_{ij}$$

(c') Multiplying A from the right with $S_i(\alpha)$ results in the column operation $\alpha C_i \longrightarrow C_i$:

$$A \xrightarrow{\alpha C_i \longrightarrow C_i} A \cdot S_i(\alpha)$$

Proof:

Statements **(a)**, **(b)**, and **(c)** can be proven directly by calculating the matrix multiplications; the notes leave this calculation to the reader, and it is carried out for **(a)** among the solutions at the end of the section.

Statements **(a')**, **(b')**, and **(c')** can be deduced from the row operation versions by passing to the transposed matrix. For example, to prove **(a')**:

$$(A \cdot Q_{ij}(\alpha))^{\mathsf{T}} = Q_{ij}(\alpha)^{\mathsf{T}} \cdot A^{\mathsf{T}} = Q_{ji}(\alpha) \cdot A^{\mathsf{T}}$$

By part **(a)**, this is exactly the matrix obtained from A^{T} after applying to it the row operation $R_j + \alpha R_i \longrightarrow R_j$. But taking the transpose again, this is equal to:

$$(\text{matrix obtained from } A \text{ after applying } C_j + \alpha C_i \longrightarrow C_j)^{\mathsf{T}}$$

Taking the transpose of both sides, it follows that $A \cdot Q_{ij}(\alpha)$ is the matrix obtained from A after the column operation $C_j + \alpha C_i \longrightarrow C_j$. The proofs for (b') and (c') follow a similar logic. **Q.E.D.**

Lemma 17.e.2 (*Inverses of Elementary Matrices*): Every elementary matrix is invertible, and the inverse of each is an elementary matrix of the exact same type:

$$Q_{ij}(\alpha)^{-1} = Q_{ij}(-\alpha), \quad P_{ij}^{-1} = P_{ij}, \quad \text{and} \quad S_i(\alpha)^{-1} = S_i(\alpha^{-1})$$

Exercise 17.8 (*Proof of Lemma 17.e.2*): The notes leave this proof as an exercise. *(Hint:* Note that $P_{ij} \cdot P_{ij}$ is the matrix obtained from P_{ij} by exchanging rows i and j, which brings us immediately back to I_n. Apply similar logical steps for the other types.*)*

Theorem 17.e.3 (*Decomposition into Elementary Matrices*): For every invertible matrix $A \in \mathrm{GL}_n(K)$, there exists an integer $k \geq 1$ and a sequence of elementary matrices $T_1, \ldots, T_k$ such that:

$$T_k \cdot T_{k-1} \ldots T_1 \cdot A = I_n$$

Furthermore, we have $A = T_1^{-1} \ldots T_k^{-1}$ and $A^{-1} = T_k \ldots T_1$. In particular, any invertible matrix A can be written as a finite product of elementary matrices.

Proof:

We already know that every matrix $A \in \mathcal{M}_{n \times n}(K)$ can be brought to a row-reduced echelon matrix, which we will call A', using the Gaussian elimination algorithm.

In our case, A is invertible, and hence $\mathrm{rank}(A) = n$. Because row operations do not change the rank, we have $\mathrm{rank}(A') = \mathrm{rank}(A)$, which implies that $\mathrm{rank}(A') = n$. The only $n \times n$ row-reduced echelon matrix with rank n is the identity matrix, meaning $A' = I_n$.

Since each step in the Gaussian elimination algorithm corresponds to an elementary operation, we can deduce from Lemma 17.e.1 that there exist elementary matrices $T_1, \ldots, T_k$ (of types I, II, and III) such that:

$$T_k \cdot T_{k-1} \ldots T_1 \cdot A = I_n$$

By left-multiplying successively with the inverses of these elementary matrices, we isolate A and deduce that $A = T_1^{-1} \ldots T_k^{-1}$. Taking the inverse of A yields $A^{-1} = T_k \ldots T_1$. **Q.E.D.**

Theorem 17.e.4 (*Matrix Inversion Algorithm*): Let $A \in \mathcal{M}_{n\times n}(K)$. Construct the augmented matrix by placing A alongside I_n, denoted as $(A \mid I_n)$. Then, A is invertible if and only if $(A \mid I_n)$ can be brought, via a sequence of elementary row operations, to the form $(I_n \mid B)$ for some matrix $B \in \mathcal{M}_{n\times n}(K)$. Furthermore, if this is the case, we have $A^{-1} = B$.

Before proving this theorem, we require an intermediate technical result.

Lemma 17.e.5 (*Block Matrix Multiplication*): Let $B, C, D \in \mathcal{M}_{n\times n}(K)$. Then:

$$B \cdot (C \mid D) = (B \cdot C \mid B \cdot D)$$

Exercise 17.9 (*Proof of Lemma 17.e.5*): The notes leave this proof as an exercise: it follows directly from the definition of matrix multiplication. *(Hint:* Consider the (k, ℓ)-th entry of $B \cdot (C \mid D)$ and show that it matches the (k, ℓ)-th entry of $(B \cdot C \mid B \cdot D)$.*)*

Proof of Theorem 17.e.4:

Assume that A is invertible. By Theorem 17.e.3, there exist elementary matrices $T_1, \ldots, T_k$ such that $T_k \ldots T_1 \cdot A = I_n$. Using Lemma 17.e.5, we can multiply the augmented matrix $(A \mid I_n)$ from the left by this chain of elementary matrices:

$$T_k \ldots T_1 \cdot (A \mid I_n) = (T_k \ldots T_1 \cdot A \mid T_k \ldots T_1 \cdot I_n) = (I_n \mid T_k \ldots T_1)$$

Let us define $B := T_k \ldots T_1$. We can clearly see that:

$$B \cdot A = (T_k \ldots T_1) \cdot A = I_n$$

and using the decomposition of A provided by Theorem 17.e.3,

$$A \cdot B = (T_1^{-1} \ldots T_k^{-1}) \cdot (T_k \ldots T_1) = I_n$$

This explicitly implies that $A^{-1} = B$.

Conversely, assume that $(A \mid I_n)$ can be brought to $(I_n \mid B)$ by a sequence of elementary row operations, for some matrix $B \in \mathcal{M}_{n\times n}(K)$. This directly implies that the matrix A itself is row-equivalent to I_n. Consequently, A must have rank n, which inherently means that A is invertible. This concludes the proof. **Q.E.D.**

Example 17.3 (*Computing the Inverse of a Two-by-Two Matrix*): Let us find the inverse of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathcal{M}_{2\times 2}(\mathbb{Q})$. We place A and I_2 together in an augmented matrix and apply Gaussian elimination:

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right) \xrightarrow{R_2 - 3R_1 \longrightarrow R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right)$$

Next, we scale the second row to create a pivot of 1:

$$\xrightarrow{-\frac{1}{2}R_2 \longrightarrow R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right)$$

Finally, we eliminate the entry above the second pivot:

$$\xrightarrow{R_1-2R_2\longrightarrow R_1}\left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right)$$

The left side is now the identity matrix I_2, which means the right side is our inverse matrix. We conclude that:

$$A^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

Solutions to the Exercises

Proof of Exercise 17.7:

Both sides are $n\times n$ matrices, so it suffices to compare entries. Write $M := I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$, and recall that the (k,ℓ)-entry of E_{pq} is 1 if $(k,\ell)=(p,q)$ and 0 otherwise, while the (k,ℓ)-entry of I_n is $\delta_{k\ell}$. Since $i\neq j$, the five matrix units involved sit in five distinct positions, and the entries of M are:

- $M_{kk}=1$ for $k\notin\{i,j\}$, from I_n alone;
- $M_{ii}=1-1=0$ and $M_{jj}=1-1=0$;
- $M_{ij}=0+1=1$ and $M_{ji}=0+1=1$;
- $M_{k\ell}=0$ in every other position.

That is exactly the matrix obtained from I_n by exchanging rows i and j: row i of M has its single 1 in column j, row j has its single 1 in column i, and every other row is unchanged. Hence $M=P_{ij}$. **Q.E.D.**

Proof of Lemma 17.e.1 (a):

Let $A=(a_{k\ell})\in\mathcal{M}_{n\times p}(K)$. Using $Q_{ij}(\alpha)=I_n+\alpha E_{ij}$ and distributivity,

$$Q_{ij}(\alpha)\cdot A=(I_n+\alpha E_{ij})A=A+\alpha\,E_{ij}A.$$

Now $E_{ij}A$ has (k,ℓ)-entry $\sum_q (E_{ij})_{kq}a_{q\ell}$, and $(E_{ij})_{kq}$ vanishes unless $k=i$ and $q=j$. So $E_{ij}A$ has the j-th row of A sitting in its i-th row and zeros everywhere else. Adding α times this to A therefore leaves every row of A untouched except the i-th, which becomes

$$(\text{row } i \text{ of } A)+\alpha\cdot(\text{row } j \text{ of } A),$$

which is precisely the operation $R_i+\alpha R_j\longrightarrow R_i$. The calculations for P_{ij} and $S_i(\alpha)$ are of the same kind. **Q.E.D.**

Proof of Lemma 17.e.2:

For Type II, the hint does it: $P_{ij}\cdot P_{ij}$ is, by Lemma 17.e.1 **(b)**, the matrix obtained from P_{ij} by exchanging its rows i and j, and exchanging them a second time restores I_n. So $P_{ij}^{-1}=P_{ij}$.

For Type I, apply Lemma 17.e.1 **(a)** twice, or compute directly using $i\neq j$, which gives $E_{ij}E_{ij}=0$:

$$Q_{ij}(\alpha)\,Q_{ij}(-\alpha)=(I_n+\alpha E_{ij})(I_n-\alpha E_{ij})=I_n-\alpha E_{ij}+\alpha E_{ij}-\alpha^2E_{ij}E_{ij}=I_n,$$

and the same computation with α and $-\alpha$ interchanged gives $Q_{ij}(-\alpha)Q_{ij}(\alpha)=I_n$. Hence $Q_{ij}(\alpha)^{-1}=Q_{ij}(-\alpha)$.

For Type III, $S_i(\alpha)S_i(\alpha^{-1})$ scales the i-th row of $S_i(\alpha^{-1})$ by α, turning its (i,i)-entry α^{-1} into 1 and leaving the other rows alone, so the product is I_n; symmetrically for the other order. This uses $\alpha\neq 0$, which is part of the definition of $S_i(\alpha)$. In every case the inverse is an elementary matrix of the same type. **Q.E.D.**

Proof of Lemma 17.e.5:

Write $(C \mid D) \in \mathcal{M}_{n\times 2n}(K)$ for the matrix whose first n columns are those of C and whose last n columns are those of D. By the column-wise description of a product, multiplying on the left by B acts on each column separately:

$$B \cdot \begin{pmatrix} | & & | \\ u_1 & \dots & u_{2n} \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ Bu_1 & \dots & Bu_{2n} \\ | & & | \end{pmatrix}.$$

The first n columns of $(C \mid D)$ are the columns $c_1, \dots, c_n$ of C, and the last n are the columns $d_1, \dots, d_n$ of D. So the first n columns of $B \cdot (C \mid D)$ are $Bc_1, \dots, Bc_n$, that is, the columns of $B \cdot C$, and the last n are $Bd_1, \dots, Bd_n$, the columns of $B \cdot D$. Hence $B \cdot (C \mid D) = (B \cdot C \mid B \cdot D)$.

Q.E.D.

AI-Note: All five numbered statements of the notes are present in the transcript, in the notes' order and with the notes' proofs, but the printed numbers were each one too large: the three types of elementary matrix are introduced in the notes under *"Notation"*, without a number, while the transcript wrapped them in a numbered `definition`, which consumed 17.e.1 and pushed everything after it. That definition is now unnumbered, and 17.e.1 through 17.e.5 line up with the handwritten ones again.

The notes call the matrix units E_{ij}; the transcript renamed them R_{ij}, which collides with the R_i that this very section uses for rows — R_{ij} and $R_i + \alpha R_j \to R_i$ appear within two lines of each other. The notes' E_{ij} is restored. Note that the notes switch from the E_i of the chapter on linear systems to R_i here for rows; that switch is Prof. Biran's own and is left alone.

The notes mark four things for the reader: the explicit form of P_{ij}, the direct calculation behind Lemma 17.e.1, Lemma 17.e.2, and Lemma 17.e.5. All four are now exercises with solutions above. The proof of Lemma 17.e.5 in the transcript also carried its opening sentence twice, once with an exercise prompt and once with a hint.

Finally, the transcript's example of a Type II matrix was $P_{3,4}$, correctly computed but not the notes' example; it is back to Prof. Biran's $P_{2,4}$.

Constructing New Vector Spaces Out of Old Ones (Chapter 18)

The Homomorphism Space, The Dual Space And The Dual Map (Section 18.a)

The Homomorphism Space

Proposition 18.a.1 (*The Vector Space Structure of the Homomorphism Space*): Let V and W be two vector spaces over K. Then, $\operatorname{Hom}(V,W)$ has the structure of a vector space over K, if we endow it with the following operations:

- For all $T_1, T_2 \in \operatorname{Hom}(V,W)$, $(T_1+T_2)(v) := T_1(v) + T_2(v)$ for all $v \in V$.
- For all $T \in \operatorname{Hom}(V,W)$ and $\alpha \in K$, $(\alpha \cdot T)(v) := \alpha \cdot T(v)$ for all $v \in V$.

Proof:

We claim that $T_1 + T_2 \in \operatorname{Hom}(V,W)$, meaning $T_1 + T_2$ is a linear map. Indeed, for a scalar a and a vector v, we have

$$\begin{aligned}(T_1+T_2)(a\cdot v) &= T_1(a\cdot v) + T_2(a\cdot v) = a\cdot T_1(v) + a\cdot T_2(v)\\ &= a\cdot (T_1(v)+T_2(v)) = a\cdot (T_1+T_2)(v),\end{aligned}$$

and for vectors v_1, v_2, we have

$$\begin{aligned}(T_1+T_2)(v_1+v_2) &= T_1(v_1+v_2) + T_2(v_1+v_2) = T_1(v_1) + T_1(v_2) + T_2(v_1) + T_2(v_2)\\ &= (T_1+T_2)(v_1) + (T_1+T_2)(v_2).\end{aligned}$$

Consequently, $T_1 + T_2$ is a linear map. The proof that $\alpha \cdot T$ is linear is analogous.

So far, we have defined $+\colon \operatorname{Hom}(V,W)\times\operatorname{Hom}(V,W)\to\operatorname{Hom}(V,W)$ and $\cdot\colon K\times\operatorname{Hom}(V,W)\to\operatorname{Hom}(V,W)$.

Regarding the neutral element, the map $0\colon V\to W$ defined by $v\mapsto 0$ for all $v\in V$ is linear. We claim that $0+T=T$. Indeed,

$$(0+T)(v) = 0(v) + T(v) = 0 + T(v) = T(v)$$

for all $v \in V$. Similarly, one shows that $T + 0 = T$. **Q.E.D.**

Exercise 18.1 (*The Axioms for* $\operatorname{Hom}(V,W)$): Complete the proof that $\operatorname{Hom}(V,W)$ is a vector space by checking that it satisfies all the axioms of a vector space, using the zero map $0\colon V\to W$ as the additive identity. Show also that $\alpha\cdot T$ is linear for all $\alpha\in K$ and $T\in\operatorname{Hom}(V,W)$, and that $\Psi^{\mathcal{B}}_{\mathcal{C}}(T_1+T_2) = \Psi^{\mathcal{B}}_{\mathcal{C}}(T_1) + \Psi^{\mathcal{B}}_{\mathcal{C}}(T_2)$ in Theorem 18.a.2.

For example, consider the distributivity axiom. For all $\alpha\in K$ and $T_1,T_2\in\operatorname{Hom}(V,W)$, we have $\alpha\cdot(T_1+T_2) = \alpha\cdot T_1 + \alpha\cdot T_2$. Indeed, for $v\in V$,

$$\begin{aligned}(\alpha\cdot(T_1+T_2))(v) &= \alpha\cdot(T_1+T_2)(v) = \alpha(T_1(v)+T_2(v))\\ &= \alpha\cdot T_1(v) + \alpha\cdot T_2(v) = (\alpha\cdot T_1)(v) + (\alpha\cdot T_2)(v)\\ &= (\alpha\cdot T_1 + \alpha\cdot T_2)(v).\end{aligned}$$

Therefore, $\alpha\cdot(T_1+T_2) = \alpha\cdot T_1 + \alpha\cdot T_2$.

Theorem 18.a.2 (*The Canonical Isomorphism from the Homomorphism Space* $\operatorname{Hom}(V,W)$ *to the Matrix Space* $\mathcal{M}_{m\times n}(K)$): Let V and W be finite-dimensional vector spaces over K, let $\mathcal{B} =$

$(v_1, \dots, v_n)$ be a basis for V, and let $\mathcal{C} = (w_1, \dots, w_m)$ be a basis for W. Then, $\Psi_{\mathcal{C}}^{\mathcal{B}} \colon \operatorname{Hom}(V, W) \to \mathcal{M}_{m\times n}(K)$, defined by

$$\Psi_{\mathcal{C}}^{\mathcal{B}}(T) := [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \dots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix},$$

is a linear map, and moreover, an isomorphism.

Proof:

Recall that $T_{[T]_{\mathcal{C}}^{\mathcal{B}}} = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}$. Recall also that $\Phi_{\mathcal{B}}(v) = [v]_{\mathcal{B}} \in K^n$ for all $v \in V$. Furthermore, $\Phi_{\mathcal{C}}(w) = [w]_{\mathcal{C}} \in K^m$ for all $w \in W$. We have

$$\Phi_{\mathcal{B}}^{-1}\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1 v_1 + \dots + a_n v_n, \quad \Phi_{\mathcal{C}}^{-1}\begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} = b_1 w_1 + \dots + b_m w_m.$$

We claim that the map $\Psi_{\mathcal{C}}^{\mathcal{B}}$ is bijective. Indeed, define $\Theta \colon \mathcal{M}_{m\times n}(K) \to \operatorname{Hom}(V, W)$ by $\Theta(A) := \Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}} \in \operatorname{Hom}(V, W)$ for $A \in \mathcal{M}_{m\times n}(K)$. We have:

$$\begin{aligned} \Psi_{\mathcal{C}}^{\mathcal{B}} \circ \Theta(A) &= \Psi_{\mathcal{C}}^{\mathcal{B}}(\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}) \\ &= \big([\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}(v_1)]_{\mathcal{C}} \dots [\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}(v_n)]_{\mathcal{C}}\big) \\ &= \begin{pmatrix} | & & | \\ T_A\Phi_{\mathcal{B}}(v_1) & \dots & T_A\Phi_{\mathcal{B}}(v_n) \\ | & & | \end{pmatrix} \\ &= \begin{pmatrix} | & & | \\ T_A(e_1) & \dots & T_A(e_n) \\ | & & | \end{pmatrix} = A. \end{aligned}$$

This uses the facts that $[\Phi_{\mathcal{C}}^{-1}(b)]_{\mathcal{C}} = b$ for all $b \in K^m$, and $\Phi_{\mathcal{B}}(v_k) = e_k$. This proves that $\Psi_{\mathcal{C}}^{\mathcal{B}} \circ \Theta = \mathrm{id}_{\mathcal{M}_{m\times n}(K)}$.

Now, by definition, $\Theta \circ \Psi_{\mathcal{C}}^{\mathcal{B}}(T) = \Phi_{\mathcal{C}}^{-1} \circ T_{\Psi_{\mathcal{C}}^{\mathcal{B}}(T)} \circ \Phi_{\mathcal{B}} = \Phi_{\mathcal{C}}^{-1} \circ T_{[T]_{\mathcal{C}}^{\mathcal{B}}} \circ \Phi_{\mathcal{B}} = T$. This shows that $\Theta \circ \Psi_{\mathcal{C}}^{\mathcal{B}} = \mathrm{id}_{\operatorname{Hom}(V,W)}$. Therefore, $\Psi_{\mathcal{C}}^{\mathcal{B}}$ is bijective.

It remains to prove that $\Psi_{\mathcal{C}}^{\mathcal{B}}$ is linear. Indeed, let $\alpha \in K$, and let $T \in \operatorname{Hom}(V, W)$. Then,

$$\begin{aligned} \Psi_{\mathcal{C}}^{\mathcal{B}}(\alpha \cdot T) &= ([(\alpha T)(v_1)]_{\mathcal{C}} \dots [(\alpha T)(v_n)]_{\mathcal{C}}) \\ &= ([\alpha T(v_1)]_{\mathcal{C}} \dots [\alpha T(v_n)]_{\mathcal{C}}) \\ &= (\alpha[T(v_1)]_{\mathcal{C}} \dots \alpha[T(v_n)]_{\mathcal{C}}) \\ &= \alpha \cdot \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \dots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \\ &= \alpha \cdot \Psi_{\mathcal{C}}^{\mathcal{B}}(T). \end{aligned}$$

Similarly, one shows that $\Psi_{\mathcal{C}}^{\mathcal{B}}(T_1 + T_2) = \Psi_{\mathcal{C}}^{\mathcal{B}}(T_1) + \Psi_{\mathcal{C}}^{\mathcal{B}}(T_2)$. **Q.E.D.**

Corollary 18.a.3 (*Dimensionality of the Homomorphism Space*): If V and W are finite-dimensional vector spaces, then the homomorphism space $\operatorname{Hom}(V, W)$ is also finite-dimensional, and $\dim \operatorname{Hom}(V, W) = \dim V \cdot \dim W$.

Proof:

Let $n := \dim V$ and $m := \dim W$. By Theorem 18.a.2, we have an isomorphism

$$\operatorname{Hom}(V, W) \cong \mathcal{M}_{m \times n}(K).$$

This means $\operatorname{Hom}(V, W)$ is a finite-dimensional space. Furthermore,

$$\dim \operatorname{Hom}(V, W) = \dim \mathcal{M}_{m \times n}(K) = m \cdot n.$$

Q.E.D.

Rings of Endomorphisms

The notes say at this point only *"explain briefly the structure of a ring"* and *"explain what is a homomorphism of rings"*, since both were given orally. The book uses both notions from here on — and already used the word *"ring"* in Page 110 — so we record them.

AI-Definition 18.1 (*Ring*): A "*ring*" is a set R together with two operations, an addition $+: R \times R \to R$ and a multiplication $\cdot: R \times R \to R$, such that:

- **(a)** $(R, +)$ is a commutative group, whose neutral element is denoted by 0;
- **(b)** the multiplication is associative: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$;
- **(c)** the two operations are connected by the distributive laws

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (b + c) \cdot a = b \cdot a + c \cdot a \qquad \text{for all } a, b, c \in R.$$

If there is an element $1 \in R$ with $1 \cdot a = a \cdot 1 = a$ for all $a \in R$, we call R a "*ring with a unity*", and 1 its "*unity*". If moreover $a \cdot b = b \cdot a$ for all $a, b \in R$, the ring is called "*commutative*".

AI-Definition 18.2 (*Homomorphism of Rings*): Let R and S be rings with unity. A map $\varphi: R \to S$ is a "*homomorphism of rings*" if for all $a, b \in R$,

$$\varphi(a + b) = \varphi(a) + \varphi(b), \qquad \varphi(a \cdot b) = \varphi(a) \cdot \varphi(b), \qquad \varphi(1_R) = 1_S.$$

A bijective homomorphism of rings is called an "*isomorphism of rings*"; its inverse is then automatically a homomorphism of rings as well.

Example 18.1 (*Examples of Rings*):

- **(a)** Every field K is also a commutative ring.
- **(b)** $\mathbb{Z}$ is a commutative ring.
- **(c)** $\mathcal{M}_{n \times n}(K)$ is a ring, where $A \cdot B$ is matrix multiplication, and the unity is I_n. Clearly, it is not commutative.

Corollary 18.a.4 (*The Ring of Endomorphisms*): Let V be a vector space over K. Then, $\operatorname{End}(V) := \operatorname{Hom}(V, V)$ is a ring with a unity. This ring is, in general, not commutative. The multiplication in $\operatorname{End}(V)$ is given by composition $T_1 \cdot T_2 := T_1 \circ T_2$ for all $T_1, T_2 \in \operatorname{Hom}(V, V)$, and the unity is id_V. Moreover, if V is finite-dimensional with $\dim V = n$, and $\mathcal{B} = (w_1, \ldots, w_n)$ is a basis for V, then $\Psi^{\mathcal{B}}_{\mathcal{B}}: \operatorname{Hom}(V, V) \to \mathcal{M}_{n \times n}(K)$ is an isomorphism of rings.

Proof:

First, we verify that $(\operatorname{End}(V), +, \circ)$ satisfies the ring axioms. We already know that $(\operatorname{End}(V), +)$ is an abelian group from the proposition regarding the vector space structure of $\operatorname{Hom}(V, W)$. We now verify the properties related to multiplication (composition):

- **(a)** **Associativity of Multiplication**: For any $T_1, T_2, T_3 \in \operatorname{End}(V)$, function composition is inherently associative. That is, $(T_1 \circ T_2) \circ T_3 = T_1 \circ (T_2 \circ T_3)$.

(b) **Existence of Multiplicative Identity**: The identity map $\mathrm{id}_V : V \to V$ is an endomorphism. For any $T \in \mathrm{End}(V)$, we have $T \circ \mathrm{id}_V = T$ and $\mathrm{id}_V \circ T = T$. Thus, id_V is the multiplicative identity.

(c) **Distributivity**: For any $T, T_1, T_2 \in \mathrm{End}(V)$, we must show both left and right distributivity.

- **Left Distributivity**: For any $v \in V$,
$$(T \circ (T_1 + T_2))(v) = T((T_1 + T_2)(v)) = T(T_1(v) + T_2(v)).$$
By the linearity of T, this equals $T(T_1(v)) + T(T_2(v)) = (T \circ T_1)(v) + (T \circ T_2)(v) = (T \circ T_1 + T \circ T_2)(v)$. Hence, $T \circ (T_1 + T_2) = T \circ T_1 + T \circ T_2$.
- **Right Distributivity**: For any $v \in V$,
$$\begin{aligned}((T_1 + T_2) \circ T)(v) &= (T_1 + T_2)(T(v)) \\ &= T_1(T(v)) + T_2(T(v)) \\ &= (T_1 \circ T)(v) + (T_2 \circ T)(v) \\ &= (T_1 \circ T + T_2 \circ T)(v).\end{aligned}$$
Hence, $(T_1 + T_2) \circ T = T_1 \circ T + T_2 \circ T$.

Thus, $\mathrm{End}(V)$ is a ring with unity id_V. It is generally not commutative, as demonstrated by matrix multiplication.

Now, assume V is finite-dimensional with $\dim V = n$, and let $\mathcal{B}$ be a basis for V. We have already established that $\Psi^{\mathcal{B}}_{\mathcal{B}} \colon \mathrm{End}(V) \to \mathcal{M}_{n\times n}(K)$ is a linear isomorphism. To prove it is a ring isomorphism, we must show it preserves the multiplicative structure and the multiplicative identity.

(a) **Preservation of Multiplication**: For any $T_1, T_2 \in \mathrm{End}(V)$, a fundamental result from Linear Algebra I states that the matrix representation of a composition of linear maps is the product of their matrix representations. That is,
$$\Psi^{\mathcal{B}}_{\mathcal{B}}(T_1 \circ T_2) = [T_1 \circ T_2]^{\mathcal{B}}_{\mathcal{B}} = [T_1]^{\mathcal{B}}_{\mathcal{B}} \cdot [T_2]^{\mathcal{B}}_{\mathcal{B}} = \Psi^{\mathcal{B}}_{\mathcal{B}}(T_1) \cdot \Psi^{\mathcal{B}}_{\mathcal{B}}(T_2).$$

(b) **Preservation of Multiplicative Identity**: The matrix representation of the identity endomorphism with respect to any basis $\mathcal{B}$ is the identity matrix I_n.
$$\Psi^{\mathcal{B}}_{\mathcal{B}}(\mathrm{id}_V) = [\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}} = I_n.$$

Since $\Psi^{\mathcal{B}}_{\mathcal{B}}$ is a bijective linear map that preserves both multiplication and the multiplicative identity, it is an isomorphism of rings. **Q.E.D.**

The Dual Space

An important special case of $\mathrm{Hom}(V, W)$ is when $W = K$.

Definition 18.a.5 (*The Dual Space and Linear Functionals*): Let V be a vector space over K. The "*dual space*" of V is defined to be the vector space $V^* := \mathrm{Hom}(V, K)$. The elements of V^* are linear maps $\ell \colon V \to K$, and are sometimes called "*functionals*".

Corollary 18.a.6 (*Dimensionality of the Dual Space*): Let V be a finite-dimensional vector space over K. Then, $\dim V^* = \dim V$.

Proof:

We have $\dim V^* = \dim \mathrm{Hom}(V, K) = \dim V \cdot \dim K = \dim V \cdot 1 = \dim V$. **Q.E.D.**

Let $V = K^n_{\mathrm{col}}$. Then, we can identify V^* with K^n_{row} as follows: If $\ell \in K^n_{\mathrm{row}}$, then ℓ defines a functional $f_\ell \colon K^n_{\mathrm{col}} \to K$ by $f_\ell(v) := \ell \cdot v \in \mathcal{M}_{1\times 1}(K) \cong K$. The map $D \colon K^n_{\mathrm{row}} \to (K^n_{\mathrm{col}})^*$ defined by $D(\ell) := f_\ell$ is a linear map, and it is in fact an isomorphism; the notes leave the verification to the reader, and it is carried out among the solutions at the end of this section.

Exercise 18.2 (*The Row Space as the Dual of the Column Space*): Show that the map $D\colon K^n_{\text{row}} \to (K^n_{\text{col}})^*$, $D(\ell) := f_\ell$, is a linear map and an isomorphism.

Let V be a finite-dimensional vector space over K, and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V. For $1 \le i \le n$, define a functional $v_i^* \in V^*$ as follows: $v_i^*(v_j) := \delta_{ij}$. This means that $v_i^*(v_i) = 1$, and $v_i^*(v_j) = 0$ for all $j \neq i$. Since $v_1, \dots, v_n$ form a basis for V, our definition determines uniquely a well-defined functional $v_i^*\colon V \to K$.

Lemma 18.a.7 (*Construction of the Dual Basis*): The elements $v_1^*, \dots, v_n^* \in V^*$ form a basis for V^*. This basis is called the "*dual basis*" to $\mathcal{B}$. We denote this basis by $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$. Moreover, for all $\ell \in V^*$, $\ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*$.

Proof:

Let $\ell \in V^*$. We claim that $\ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*$, or in other words, we claim that $\ell(v) = \sum_{i=1}^n \ell(v_i) \cdot v_i^*(v)$. Indeed, both sides of the equation are linear maps $V \to K$, and therefore, it is enough to check that it holds for $v = v_1, v = v_2, \dots, v = v_n$, because $v_1, \dots, v_n$ form a basis for V. And indeed, for $v = v_k$, we have:

$$\sum_{i=1}^n \ell(v_i) \cdot v_i^*(v_k) = \sum_{i=1}^n \ell(v_i) \cdot \delta_{ik} = \ell(v_k).$$

The claim just proven shows that $v_1^*, \dots, v_n^*$ span V^*. Since $\dim V^* = \dim V = n$, it follows that $v_1^*, \dots, v_n^*$ form a basis for V^*. **Q.E.D.**

Question 18.1: Let V be a finite-dimensional vector space over K, and let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V. $\mathcal{B}^*$ and $\mathcal{C}^*$ are two bases for V^*. What is the relation between $[\mathrm{id}_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*}$ and $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{C}}$?

Proposition 18.a.8 (*Dual Change of Basis and Transposition*): The relationship between transition matrices in the dual space and the primal space is characterized by transposition and inversion. Specifically, the transition matrix between dual bases $\mathcal{C}^*$ and $\mathcal{B}^*$ is the transpose of the inverse transition matrix between the original bases $\mathcal{C}$ and $\mathcal{B}$:

$$[\mathrm{id}_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*} = \left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}}\right)^{\mathsf{T}} = \left(\left([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{C}}\right)^{-1}\right)^{\mathsf{T}}.$$

Thus, the coordinate transformation in the dual space corresponds to the transpose of the inverse transformation in the primal space.

Proof:

Let $\mathcal{B} = (v_1, \dots, v_n)$, and let $\mathcal{C} = (w_1, \dots, w_n)$. We have

$$[\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{C}} & \dots & [v_n]_{\mathcal{C}} \\ | & & | \end{pmatrix}.$$

Let $A := [\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}}$ and write $A = (a_{ij})$. By definition, $v_i = \sum_{k=1}^n a_{ki} w_k$. Let $\mathcal{C}^* = (w_1^*, \dots, w_n^*)$ be the dual basis of $\mathcal{C}$. We apply the previous lemma to $\ell = w_j^*$ and the basis $\mathcal{B}$:

$$w_j^* = \sum_{i=1}^n w_j^*(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^* \left(\sum_{k=1}^n a_{ki} w_k\right) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*.$$

Here, only $k = j$ remains from the inner sum. This implies that $[\mathrm{id}_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*} = A^{\mathsf{T}} = ([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}})^{\mathsf{T}}$. **Q.E.D.**

The Dual Map

Lemma 18.a.9 (*Definition of the Dual Map*): Let V and W be vector spaces over K, and let $T\colon V \to W$ be a linear map. Define a new map $T^*\colon W^* \to V^*$ by $T^*(\ell) := \ell \circ T$ for all $\ell \in W^*$. Then, T^* is a linear map. It is called the "*dual map*" to T.

Proof:

Let $\ell \in W^*$, meaning $\ell\colon W \to K$ is linear. Clearly, $\ell \circ T\colon V \to K$ is a linear map because both T and ℓ are linear, so $T^*(\ell) := \ell \circ T \in V^*$. This shows that $T^*\colon W^* \to V^*$ has its target indeed in V^*, as claimed.

We will show now that T^* is a linear map. Let $\ell \in W^*$, and let $\alpha \in K$. We have that $T^*(\alpha\ell) \in V^*$ is the map:

$$T^*(\alpha\ell)(v) = ((\alpha\ell) \circ T)(v) = (\alpha\ell)(T(v)) = \alpha\ell(T(v)) = \alpha \cdot (\ell \circ T)(v) = \alpha(T^*\ell)(v).$$

So, $T^*(\alpha\ell) = \alpha \cdot T^*(\ell)$. Similarly, one shows that $T^*(\ell_1 + \ell_2) = T^*(\ell_1) + T^*(\ell_2)$. **Q.E.D.**

Proposition 18.a.10 (*Functoriality and Composition of Duals*): Let U, V, W be vector spaces over K, and let $T\colon V \to W$ and $S\colon W \to U$ be linear maps. Consider the dual maps $T^*\colon W^* \to V^*$ and $S^*\colon U^* \to W^*$. Then, $(S \circ T)^* = T^* \circ S^*$.

The notes mark this proof as an exercise; it is carried out among the solutions at the end of the section, together with the following one.

Exercise 18.3 (*Duals of Compositions, of the Identity and of the Zero Map*): Let V be a vector space over K. Prove Proposition 18.a.10, and show that:

- **(a)** $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.
- **(b)** $(0\colon V \to V)^* = (0\colon V^* \to V^*)$.

Summary 18.1: In summary, applying $(-)^*$, or $\mathrm{Hom}(-, K)$, takes every vector space V and gives us a new vector space V^*. It also takes any linear map $T\colon V \to W$ and gives us a new map T^* between the duals of V and W, but the arrow is reversed: $T^*\colon W^* \to V^*$. This is an example of a "*contravariant functor*".

Lemma 18.a.11 (*Transposition of the Representation Matrix*): Let $S\colon V \to W$ be a linear map between two finite-dimensional vector spaces V and W over K. Let $\mathcal{B}$ be a basis for V, and let $\mathcal{C}$ be a basis for W. Consider $S^*\colon W^* \to V^*$ and the bases $\mathcal{C}^*, \mathcal{B}^*$ for W^*, V^*, respectively. Then, $[S^*]_{\mathcal{B}^*}^{\mathcal{C}^*} = ([S]_{\mathcal{C}}^{\mathcal{B}})^\mathsf{T}$.

Proof:

Write $\mathcal{B} = (v_1, \ldots, v_n)$, $\mathcal{C} = (w_1, \ldots, w_m)$, and $A := [S]_{\mathcal{C}}^{\mathcal{B}} = (a_{ij})$. The matrix A satisfies $S(v_j) = \sum_{i=1}^m a_{ij} w_i$. Now, for $w_j^* \in \mathcal{C}^*$, we have

$$S^*(w_j^*) = w_j^* \circ S = \sum_{i=1}^n (w_j^* \circ S)(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^*(S(v_i)) \cdot v_i^*,$$

where the second equality follows from the Lemma on the Dual Basis. Substituting $S(v_i)$, we get

$$\sum_{i=1}^n w_j^* \left(\sum_{k=1}^m a_{ki} w_k \right) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*.$$

The coordinates of $S^*(w_j^*)$ in the basis $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$ are $\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix}$, which corresponds to row j in A, represented as a column (i.e., column j in A^T). **Q.E.D.**

Reflexivity

Let V be a vector space over K. We have V^*, but also $V^{**} := (V^*)^*$, obtained by applying the dual operation one more time to V^*. V^{**} is sometimes called the "*bidual space*" of V. Is there any relation between V^{**} and V?

Theorem 18.a.12 (*The Canonical Isomorphism to the Bidual Space*): Let V be a finite-dimensional vector space over K. Then, there is a canonical isomorphism $\tau\colon V \to (V^*)^*$ given by: for $v \in V$, $\tau(v)\colon V^* \to K$ is the map $\tau(v)(\ell) := \ell(v)$ for all $\ell \in V^*$.

Remark 18.1: The map $\tau\colon V \to (V^*)^*$ exists and is defined in the same way as above regardless of V being finite- or infinite-dimensional. Moreover, τ is always injective. However, when V is infinite-dimensional, τ fails to be surjective.

Proof:

We claim that τ is a linear map. Indeed, if $v \in V$ and $\alpha \in K$, we have

$$\begin{aligned}\tau(\alpha v) &= (V^* \ni \ell \mapsto \ell(\alpha v) \in K)\\ &= (V^* \ni \ell \mapsto \alpha\ell(v) \in K)\\ &= \alpha \cdot (V^* \ni \ell \mapsto \ell(v) \in K) = \alpha\tau(v).\end{aligned}$$

Similarly, one shows that $\tau(v' + v'') = \tau(v') + \tau(v'')$.

We now claim that τ is injective. To see this, we will show that $\ker(\tau) = 0$. Indeed, let $v \in \ker(\tau)$, meaning $\tau(v) = 0$. Then, $\ell(v) = 0$ for all $\ell \in V^*$. Choose a basis $\mathcal{B} = (v_1, \dots, v_n)$ for V. Write $v = c_1v_1 + \cdots + c_nv_n$. Apply v_k^* to v, and we obtain $v_k^*(c_1v_1 + \cdots + c_nv_n) = 0$. But $v_k^*(c_1v_1 + \cdots + c_nv_n) = c_k$. Therefore, $c_k = 0$, and this holds for $1 \le k \le n$, which implies that $v = 0$. This proves that τ is injective.

To finish the proof, note that $\dim(V^*)^* = \dim V^* = \dim V$. Therefore, since $\tau\colon V \to (V^*)^*$ is injective, it must be an isomorphism. **Q.E.D.**

Remark 18.2: Theorem 18.a.12 is not true for infinite-dimensional vector spaces. More specifically, $\tau\colon V \to (V^*)^*$ is injective, but not surjective. In fact, when V is infinite-dimensional, $(V^*)^*$ is not isomorphic to V.

Naturality

The map $\tau\colon V \to (V^*)^*$ exists for every space V, and it is defined in the same way. To distinguish τ's for different spaces, we denote it by $\tau_V\colon V \to (V^*)^*$. Let V and W be two vector spaces over K. Then, for any linear map $T\colon V \to W$, we have the following naturality condition, represented by a commutative diagram:

$$\begin{tikzcd} V \arrow[r, "\tau_V"] \arrow[d, "T"'] & (V^*)^* \arrow[d, "(T^*)^*"] \\ W \arrow[r, "\tau_W"'] & (W^*)^* \end{tikzcd}$$

Exercise 18.4 (*Canonical Evaluation on Dual Bases*): Let V be a finite-dimensional vector space over K, and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V. Let $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ be the dual basis, and let $(\mathcal{B}^*)^* = ((v_1^*)^*, \dots, (v_n^*)^*)$ be the basis for $(V^*)^*$ which is dual to $\mathcal{B}^*$. Show that $\tau(\mathcal{B}) = (\mathcal{B}^*)^*$, meaning $\tau(v_i) = (v_i^*)^*$ for all $1 \le i \le n$.

Solutions to the Exercises

Proof of Exercise 18.1:

For the linearity of $\alpha \cdot T$: for $a \in K$ and $v, v' \in V$,

$$(\alpha T)(av + v') = \alpha T(av + v') = \alpha\big(aT(v) + T(v')\big) = a \cdot \alpha T(v) + \alpha T(v') = a\,(\alpha T)(v) + (\alpha T)(v').$$

The vector space axioms for $\mathrm{Hom}(V, W)$ all reduce to the corresponding axioms in W, because both operations are defined pointwise. For instance, associativity of addition: for all $v \in V$,

$$\big((T_1 + T_2) + T_3\big)(v) = \big(T_1(v) + T_2(v)\big) + T_3(v) = T_1(v) + \big(T_2(v) + T_3(v)\big) = \big(T_1 + (T_2 + T_3)\big)(v),$$

using associativity in W in the middle; since two maps agreeing at every v are equal, $(T_1+T_2)+T_3 = T_1 + (T_2 + T_3)$. The same pattern gives commutativity, $1 \cdot T = T$, $(\alpha\beta) \cdot T = \alpha \cdot (\beta \cdot T)$, and the two distributive laws — the one for $\alpha \cdot (T_1 + T_2)$ is written out in the text above. The additive inverse of T is $(-1) \cdot T$, since $(T + (-1)T)(v) = T(v) - T(v) = 0$ for all v.

For the additivity of $\Psi^{\mathcal{B}}_{\mathcal{C}}$: its j-th column at $T_1 + T_2$ is

$$[(T_1 + T_2)(v_j)]_{\mathcal{C}} = [T_1(v_j) + T_2(v_j)]_{\mathcal{C}} = [T_1(v_j)]_{\mathcal{C}} + [T_2(v_j)]_{\mathcal{C}},$$

because the coordinate map $\Phi_{\mathcal{C}}$ is linear. Columnwise addition of matrices is addition of matrices, so $\Psi^{\mathcal{B}}_{\mathcal{C}}(T_1 + T_2) = \Psi^{\mathcal{B}}_{\mathcal{C}}(T_1) + \Psi^{\mathcal{B}}_{\mathcal{C}}(T_2)$. **Q.E.D.**

Proof of Exercise 18.2:

Linearity of D: for $\ell, \ell' \in K^n_{\mathrm{row}}$, $\alpha \in K$ and every $v \in K^n_{\mathrm{col}}$,

$$f_{\alpha\ell+\ell'}(v) = (\alpha\ell + \ell') \cdot v = \alpha(\ell \cdot v) + \ell' \cdot v = \alpha f_\ell(v) + f_{\ell'}(v),$$

by the distributivity of matrix multiplication, so $D(\alpha\ell + \ell') = \alpha D(\ell) + D(\ell')$.

Injectivity: if $D(\ell) = 0$, then $\ell \cdot v = 0$ for every $v \in K^n_{\mathrm{col}}$; taking $v = e_i$ picks out the i-th entry of ℓ, so every entry of ℓ vanishes and $\ell = 0$. Since $\dim K^n_{\mathrm{row}} = n = \dim K^n_{\mathrm{col}} = \dim(K^n_{\mathrm{col}})^*$ by Corollary 18.a.6, an injective linear map between these two spaces is an isomorphism by Corollary 15.b.10. **Q.E.D.**

Proof of Proposition 18.a.10 and of Exercise 18.3:

For the composition rule, both $(S \circ T)^*$ and $T^* \circ S^*$ are maps $U^* \to V^*$, so it suffices to evaluate them at an arbitrary $\ell \in U^*$:

$$(S \circ T)^*(\ell) = \ell \circ (S \circ T) = (\ell \circ S) \circ T = T^*(\ell \circ S) = T^*\big(S^*(\ell)\big) = (T^* \circ S^*)(\ell),$$

using only the associativity of composition. Hence $(S \circ T)^* = T^* \circ S^*$.

(a) For $\ell \in V^*$ we have $(\mathrm{id}_V)^*(\ell) = \ell \circ \mathrm{id}_V = \ell$, so $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.

(b) For $\ell \in V^*$ and $v \in V$, $\big(0^*(\ell)\big)(v) = \ell(0(v)) = \ell(0_V) = 0$, so $0^*(\ell)$ is the zero functional for every ℓ, that is, 0^* is the zero map $V^* \to V^*$.

Q.E.D.

Proof of Exercise 18.4:

Both $\tau(v_i)$ and $(v_i^*)^*$ are elements of $(V^*)^*$, that is, functionals on V^*. Since $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$ is a basis of V^*, it suffices to check that they agree on each v_j^*. By the definition of τ,

$$\tau(v_i)(v_j^*) = v_j^*(v_i) = \delta_{ji} = \delta_{ij},$$

while by the definition of the basis dual to $\mathcal{B}^*$,

$$(v_i^*)^*(v_j^*) = \delta_{ij}.$$

The two agree on every member of a basis of V^*, hence $\tau(v_i) = (v_i^*)^*$ for all i, that is, $\tau(\mathcal{B}) = (\mathcal{B}^*)^*$. **Q.E.D.**

Direct Sums (Section 18.b)

The External Direct Sum

Proposition 18.b.1 (*Vector Space Structure of the Cartesian Product*): Let V and W be two vector spaces over K. Then, the Cartesian product $V \times W$ becomes a vector space over K, provided we endow it with the following operations:

- $(v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2)$ for all $(v_1, w_1), (v_2, w_2) \in V \times W$.
- $\alpha \cdot (v, w) := (\alpha v, \alpha w)$ for all $\alpha \in K$ and $(v, w) \in V \times W$.

The zero element of this space is given by $0_{V \times W} := (0_V, 0_W)$. This new vector space $V \times W$ is usually denoted by $V \oplus W$ and is called the direct sum of V and W.

Exercise 18.5 (*The Axioms for $V \oplus W$, and the Canonical Maps*): Verify that all vector space axioms are satisfied by $V \times W$ under the componentwise operations above, so that Proposition 18.b.1 holds. Check also that the four canonical maps of the following remark are linear, that p_V and p_W are surjective, and that i_V and i_W are injective.

Remark 18.3 (*Canonical Maps associated with Direct Sums*): The direct sum $V \oplus W$ is naturally equipped with several canonical linear maps:

(a) **Canonical Embeddings**:
- $i_V : V \to V \oplus W$ defined by $i_V(v) := (v, 0)$.
- $i_W : W \to V \oplus W$ defined by $i_W(w) := (0, w)$.

(b) **Canonical Projections**:
- $p_V : V \oplus W \to V$ defined by $p_V(v, w) := v$.
- $p_W : V \oplus W \to W$ defined by $p_W(v, w) := w$.

Clearly, all these maps are linear. Furthermore, the projections p_V and p_W are surjective, while the embeddings i_V and i_W are injective. By virtue of these embeddings, we can often identify V and W as subspaces of $V \oplus W$ through the identifications $v \leftrightarrow (v, 0)$ and $w \leftrightarrow (0, w)$.

Internal Direct Sums and Complements

Proposition 18.b.2 (*Isomorphism to a Complementary Subspace*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. Suppose $U' \subseteq V$ is another subspace that acts as a complement of U in V, which implies that $U + U' = V$ and $U \cap U' = \{0\}$. Then, the map

$$\varphi \colon U \oplus U' \to V, \quad \varphi(u, u') := u + u'$$

is a linear isomorphism.

Proof:

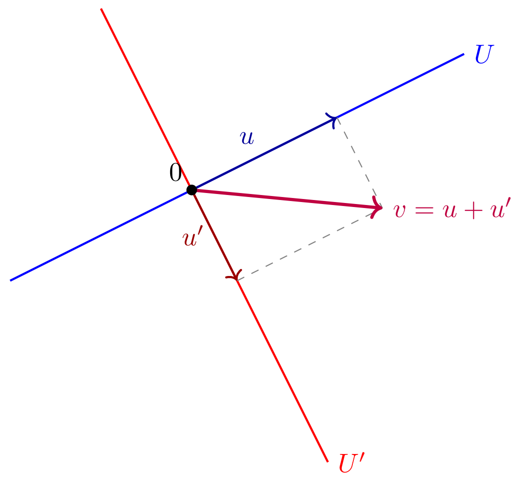
The linearity of φ follows immediately from the definitions of the operations in $U \oplus U'$ and the linearity of addition in V.

To establish injectivity, suppose $\varphi(u, u') = 0$, which implies $u + u' = 0$, and therefore $u = -u'$. Since $u \in U$ and $u' \in U'$, it follows that $u = -u' \in U \cap U'$. However, as U' is a complement of U, we have $U \cap U' = \{0\}$ by definition. Consequently, $u = u' = 0$, which shows that $\ker(\varphi) = \{0_{U \oplus U'}\}$. Thus, φ is injective.

Regarding surjectivity, we observe that since U' is a complement of U in V, we have $U + U' = V$ by definition. Thus, for any $v \in V$, there exist $u \in U$ and $u' \in U'$ such that $u + u' = v$. This implies $\varphi(u, u') = v$, and therefore $\mathrm{Im}(\varphi) = V$. This proves that φ is surjective, and consequently, an isomorphism. **Q.E.D.**

Remark 18.4: While $U \oplus U'$ is formally a different vector space from V, the existence of the canonical isomorphism φ allows us to treat V as the direct sum of its internal subspaces whenever a complement is specified.

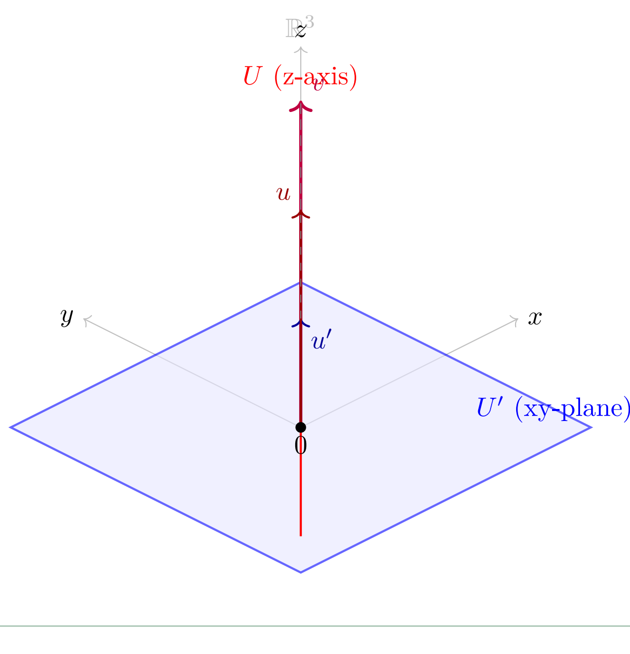
Geometrically, in $\mathbb{R}^2$, this represents the decomposition of any vector v uniquely into a sum of a vector $u \in U$ and a vector $u' \in U'$, constructed via a parallelogram parallel to the complementary subspaces.



Similarly, we can extend this intuition to $\mathbb{R}^3$. Consider U to be the z-axis (a line through the origin) and U' to be the xy-plane (a plane through the origin). Then U and U' are subspaces of $\mathbb{R}^3$.

- $U = \mathrm{Sp}(e_3) = \{(0,0,z) \mid z \in \mathbb{R}\}$
- $U' = \mathrm{Sp}(e_1, e_2) = \{(x,y,0) \mid x, y \in \mathbb{R}\}$

Any vector $v = (x,y,z) \in \mathbb{R}^3$ can be uniquely written as a sum of a vector $u \in U$ and a vector $u' \in U'$: $v = (0,0,z) + (x,y,0)$. Here, $u = (0,0,z) \in U$ and $u' = (x,y,0) \in U'$. Furthermore, $U \cap U' = \{(0,0,0)\}$, which is the zero vector. Thus, $\mathbb{R}^3 = U \oplus U'$. Geometrically, this means any point in 3D space can be reached by first moving along the z-axis and then moving in the xy-plane.



Generalization to Multiple Summands

The concept of a direct sum can be extended to any finite collection of vector spaces $U_1, \dots, U_m$ over K. We define their direct sum as:

$$U_1 \oplus \cdots \oplus U_m := U_1 \times \cdots \times U_m,$$

endowed with coordinatewise operations.

For an infinite family of vector spaces $\{U_i\}_{i\in I}$, we define the direct sum $\bigoplus_{i\in I} U_i$ as the set of functions $t\colon I \to \bigcup_{i\in I} U_i$ such that $t(i) \in U_i$ for all $i \in I$, and $t(i) \neq 0$ for at most finitely many indices $i \in I$. This ensures that any element of the direct sum can be represented as a finite formal sum of components.

Solutions to the Exercises

Proof of Exercise 18.5:

Every axiom for $V \times W$ is checked coordinatewise and reduces to the same axiom in V and in W. For instance, for $(v_1, w_1), (v_2, w_2), (v_3, w_3) \in V \times W$,

$$\big((v_1, w_1) + (v_2, w_2)\big) + (v_3, w_3) = (v_1 + v_2 + v_3,\ w_1 + w_2 + w_3) = (v_1, w_1) + \big((v_2, w_2) + (v_3, w_3)\big),$$

using associativity in V in the first coordinate and in W in the second. The same pattern gives commutativity, the neutral element $(0_V, 0_W)$, the additive inverse $-(v, w) = (-v, -w)$, and the four axioms involving scalars, for instance $\alpha \cdot \big((v_1, w_1) + (v_2, w_2)\big) = (\alpha v_1 + \alpha v_2, \alpha w_1 + \alpha w_2) = \alpha(v_1, w_1) + \alpha(v_2, w_2)$.

The canonical maps are linear for the same reason: $i_V(\alpha v + v') = (\alpha v + v', 0) = \alpha(v, 0) + (v', 0) = \alpha\, i_V(v) + i_V(v')$, and

$$p_V\big(\alpha(v, w) + (v', w')\big) = p_V(\alpha v + v',\ \alpha w + w') = \alpha v + v' = \alpha\, p_V(v, w) + p_V(v', w'),$$

and symmetrically for i_W and p_W.

The projections are surjective, since $p_V(v, 0) = v$ for every $v \in V$ and $p_W(0, w) = w$ for every $w \in W$. The embeddings are injective, since $i_V(v) = (v, 0) = (0, 0)$ forces $v = 0$, so $\ker(i_V) = \{0\}$, and likewise for i_W. **Q.E.D.**

Quotient Spaces (Section 18.c)

Equivalence Relation and the Quotient Space

Let V be a vector space over K, and let $U \subseteq V$ be a subspace. We will define a new vector space over K, denoted by V/U, together with a linear map $\pi\colon V \to V/U$ such that π is surjective and $\ker(\pi) = U$. The space V/U will have some other good properties. The space V/U will be called the "*quotient*" of V by U, or V modulo U, and the map π the "*quotient map*", or the "*projection onto the quotient space*".

We begin by defining a relation on V. For $v_1, v_2 \in V$, we define $v_1 \sim v_2$ if and only if $v_1 - v_2 \in U$.

Lemma (*Equivalence Relation on V*): The relation $\sim$ defines an equivalence relation on V.

Proof:

Reflexivity: For all $v \in V$, $v \sim v$ because $v - v = 0 \in U$.

Symmetry: If $v_1 \sim v_2$, then $v_1 - v_2 \in U$. Since U is a subspace, $-(v_1 - v_2) = v_2 - v_1 \in U$, which implies $v_2 \sim v_1$.

Transitivity: If $v_1 \sim v_2$ and $v_2 \sim v_3$, then $v_1 - v_2 \in U$ and $v_2 - v_3 \in U$. Because U is a subspace, their sum is in U:

$$(v_1 - v_2) + (v_2 - v_3) = v_1 - v_3 \in U,$$

which means $v_1 \sim v_3$. **Q.E.D.**

We denote the equivalence class of $v \in V$ by $[v]$. Thus,

$$[v] = v + U = \{v + u \mid u \in U\}.$$

Definition 18.c.1 (*The Quotient Space*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. We define $V/U := V/\sim$ to be the set of equivalence classes of the equivalence relation $\sim$. So, the elements of V/U are $[v]$, where $v \in V$.

Alternatively, each element in V/U is a set of the form $v + U$, where $v \in V$. The space V/U is called the quotient of V by U (or modulo U).

Geometrically, the nature of the quotient space $\mathbb{R}^2/U$ depends on the dimension of U:

- If $U = \{0\}$ is the trivial subspace, then each equivalence class $[v] = v + \{0\} = \{v\}$ is simply a single **point**. In this case, no points are *"collapsed"* together, and therefore $\mathbb{R}^2/\{0\} \cong \mathbb{R}^2$.
- If U is a **line** through the origin, the equivalence classes $[v]$ forming the space $\mathbb{R}^2/U$ are the lines parallel to U, as seen in Figure 18.1.

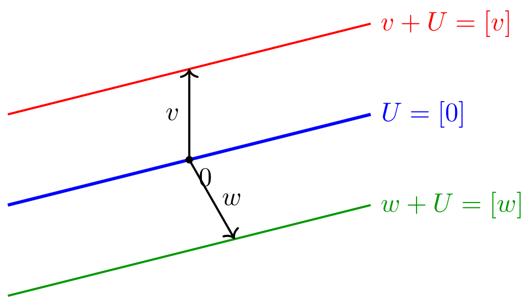


Figure 18.1: Geometric interpretation of the quotient space $\mathbb{R}^2/U$.

Similarly, we can extend this intuition to $\mathbb{R}^3$, as illustrated in Figure 18.2:

- If $U = \{0\}$ is the trivial subspace, then each equivalence class $[v] = \{v\}$ is simply a **point**. Since no two distinct vectors are identified, the quotient space $\mathbb{R}^3/\{0\}$ is isomorphic to $\mathbb{R}^3$ itself.
- If U is a **plane** through the origin, then $\mathbb{R}^3/U$ is 1-dimensional (by Proposition 18.c.4) and therefore isomorphic to a line. Each element $[v]$ is a plane parallel to U. The quotient space can be viewed as the set of all such parallel planes *"stacked"* along a transversal direction (see Figure 18.2a and Proposition 18.c.7).
- If U is a **line** through the origin, then $\mathbb{R}^3/U$ is 2-dimensional (by Proposition 18.c.4) and therefore isomorphic to a plane. Each element $[v]$ is a line parallel to U (see Figure 18.2b and Proposition 18.c.7).
- If $U = \mathbb{R}^3$ is the entire space, then there is only one equivalence class, $[v] = \mathbb{R}^3$. The quotient space $\mathbb{R}^3/\mathbb{R}^3$ is the zero vector space, with dimension 0.

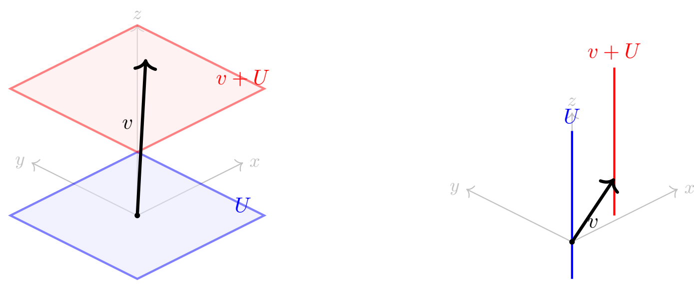
Remark 18.5: The crucial insight is that an equivalence class $[v] = v + U = \{v + u \mid u \in U\}$ is not merely a single vector, but a translate of the subspace U. Consequently, the "shape" or dimension of $[v]$ is identical to the dimension of U. Specifically:

- If U is a line, then every $[v]$ is a line parallel to U.

- If U is a plane, then every $[v]$ is a plane parallel to U.

The dimension of the quotient space V/U represents the number of independent directions available to shift these objects. For instance, in $\mathbb{R}^3$, if U is a line (1-dimensional), there are 2 independent directions in which one can move this line to obtain a distinct parallel line (effectively moving the line across a 2-dimensional plane).

Thus, the space of all such lines is 2-dimensional. This geometric intuition aligns perfectly with the dimension formula established in Proposition 18.c.4. In the case where $U = \{0\}$, the equivalence class is a point (0-dimensional), and we retain all 3 degrees of freedom of the ambient space.



(a) Plane case ($1D$-quotient) (b) Line case ($2D$-quotient)

Figure 18.2: Geometric interpretation of the quotient space $\mathbb{R}^3/U$.

The Vector Space Structure of V/U

We want to turn V/U into a vector space over K. We define addition as $[v_1] + [v_2] := [v_1 + v_2]$ for all $[v_1], [v_2] \in V/U$. Alternatively, this can be written as $(v_1 + U) + (v_2 + U) := (v_1 + v_2) + U$. For scalar multiplication, let $\alpha \in K$ and $[v] \in V/U$. We define $\alpha \cdot [v] := [\alpha v]$. The zero element is defined as $0_{V/U} := [0_V] = U$.

Proposition 18.c.2 (*Well-definedness of Operations in V/U*): The operations defined above are well-defined. Moreover, they give V/U the structure of a vector space over K.

Proof:

To establish well-definedness, we must show that the operations do not depend on the choice of representatives. For scalar multiplication, suppose $[v] = [v']$, and let $\alpha \in K$. We need to show that $[\alpha v] = [\alpha v']$. Indeed, since $[v] = [v']$, we have $v - v' \in U$. Because U is a subspace, $\alpha(v - v') = \alpha v - \alpha v' \in U$. Consequently, $[\alpha v] = [\alpha v']$.

Similarly, for addition, suppose $[v_1] = [v_1']$ and $[v_2] = [v_2']$. We need to show that $[v_1 + v_2] = [v_1' + v_2']$. By assumption, $v_1 - v_1' \in U$ and $v_2 - v_2' \in U$. Adding these yields:

$$(v_1 + v_2) - (v_1' + v_2') = (v_1 - v_1') + (v_2 - v_2') \in U.$$

Therefore, $v_1 + v_2 \sim v_1' + v_2'$, which implies $[v_1 + v_2] = [v_1' + v_2']$. This proves that the operations are well-defined.

It remains to show that they satisfy the axioms of a vector space. These follow directly from the corresponding axioms in V; the notes leave that verification to the reader, and it is carried out among the solutions at the end of this section. **Q.E.D.**

Exercise 18.6 (*The Axioms for V/U*): Complete the proof of Proposition 18.c.2 by checking that the operations on V/U satisfy the axioms of a vector space.

We now define a map $\pi\colon V \to V/U$ by $\pi(v) := [v] \in V/U$ for all $v \in V$.

Proposition 18.c.3 (*The Quotient Map*): The map $\pi\colon V \to V/U$ is a linear map. Moreover, it is surjective (i.e., $\operatorname{Im}(\pi) = V/U$), and $\ker(\pi) = U$.

Proof:

Linearity: For $v_1, v_2 \in V$, $\pi(v_1 + v_2) = [v_1 + v_2] = [v_1] + [v_2] = \pi(v_1) + \pi(v_2)$. For $\alpha \in K$ and $v \in V$, $\pi(\alpha v) = [\alpha v] = \alpha[v] = \alpha\pi(v)$.

Surjectivity: Let $x \in V/U$. By definition, x is an equivalence class $[v]$ of some element $v \in V$. Thus, $x = \pi(v)$.

Kernel: We claim that $\ker(\pi) = U$. First, let $v \in \ker(\pi)$, meaning $\pi(v) = 0$. Then, $[v] = [0]$, which implies $v = v - 0 \in U$. Hence, $\ker(\pi) \subseteq U$. Conversely, let $u \in U$. Then, $u \sim 0$ because $u - 0 \in U$. Therefore, $\pi(u) = [u] = [0] = 0$, so $u \in \ker(\pi)$. This shows that $U \subseteq \ker(\pi)$, completing the proof. **Q.E.D.**

Proposition 18.c.4 (*Dimension of the Quotient Space*): Suppose that V is a finite-dimensional vector space. Then, V/U is also finite-dimensional, and $\dim V/U = \dim V - \dim U$.

Proof:

We have seen that $\pi\colon V \to V/U$ is a surjective linear map. This guarantees that V/U is finite-dimensional. (Generally, if $T\colon W' \to W''$ is a surjective linear map and W' is finite-dimensional, then W'' is also finite-dimensional, because the images of a basis of W' span W'').

We now apply the Rank-Nullity Theorem to π:

$$\dim V = \dim\ker(\pi) + \dim\operatorname{Im}(\pi).$$

Since $\ker(\pi) = U$ and $\operatorname{Im}(\pi) = V/U$, we obtain $\dim V/U = \dim V - \dim U$. **Q.E.D.**

The Isomorphism Theorem and Universal Property

Theorem 18.c.5 (*The First Isomorphism Theorem*): Let V and W be vector spaces over K, and let $T\colon V \to W$ be a linear map. We define a new map $\overline{T}\colon V/\ker(T) \to \operatorname{Im}(T)$ as follows: For $x \in V/\ker(T)$, choose a representative $v \in V$ such that $x = [v]$. Define $\overline{T}(x) := T(v)$. Then:

(a) $\overline{T}$ is well-defined and is a linear map.

(b) $\overline{T}\colon V/\ker(T) \to \operatorname{Im}(T)$ is an isomorphism.

(c) The following diagram commutes: $T = i \circ \overline{T} \circ \pi$, where $i\colon \operatorname{Im}(T) \to W$ is the inclusion map.

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[d, "\pi"'] & W \\
V/\ker(T) \arrow[r, "\overline{T}\ (\cong)"] & \operatorname{Im}(T) \arrow[u, "i"']
\end{tikzcd}$$

Proof:

(a) To show that $\overline{T}$ is well-defined, suppose $x = [v] = [v']$. We must show that $T(v) = T(v')$. Indeed, $[v] = [v']$ implies that $v - v' \in \ker(T)$, meaning $T(v - v') = 0$. By linearity, $T(v) = T(v')$.

The linearity of $\overline{T}$ is straightforward: $\overline{T}(\alpha[v]) = \overline{T}([\alpha v]) = T(\alpha v) = \alpha T(v) = \alpha\overline{T}([v])$. Also, $\overline{T}([v_1]+[v_2]) = \overline{T}([v_1+v_2]) = T(v_1+v_2) = T(v_1)+T(v_2) = \overline{T}([v_1]) + \overline{T}([v_2])$.

(b) To show that $\overline{T}$ is injective, let $x \in \ker(\overline{T})$, and write $x = [v]$ for some $v \in V$. Then, $\overline{T}(x) = T(v) = 0$, meaning $v \in \ker(T)$. This implies that $x = [v] = 0_{V/\ker(T)}$.
To show that $\overline{T}$ is surjective, let $w \in \mathrm{Im}(T)$. Then, there exists $v \in V$ such that $T(v) = w$. Consequently, $\overline{T}([v]) = T(v) = w$, proving surjectivity.

(c) The commutativity of the diagram follows immediately from the definitions; the notes leave this to the reader, and it is checked among the solutions at the end of this section.

Q.E.D.

Exercise 18.7 (*Commutativity of the Diagram in Theorem 18.c.5*): Check that $T = i \circ \overline{T} \circ \pi$, where $i\colon \mathrm{Im}(T) \to W$ is the inclusion.

Theorem 18.c.6 (*Universal Property of Quotient Spaces*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. The quotient space V/U has the following universal property: For every vector space W and every linear map $T\colon V \to W$ for which $T(U) = 0$ (i.e., for which $\ker(T) \supseteq U$), there exists a unique linear map $T'\colon V/U \to W$ such that $T = T' \circ \pi$. Moreover, $\ker(T') = \ker(T)/U$.

Remark 18.6: In algebraic jargon, every linear map $T\colon V \to W$ that maps U to 0 factors through V/U via $T = T' \circ \pi$, and moreover, in a unique way. We say that T induces the map T', or that T descends to a map $T'\colon V/U \to W$.

Proof:

We define $T'\colon V/U \to W$ as follows: let $x \in V/U$. Choose $v \in V$ such that $x = [v]$. Define $T'(x) := T(v)$.

Exercise 18.8 (*Well-definedness and Linearity of the Induced Map*):

(a) Show that T' is well-defined (i.e., if $[v] = [v']$, then $T(v) = T(v')$).

(b) Show that T' is linear.

The diagram commutes by construction, since for all $v \in V$, $T' \circ \pi(v) = T'([v]) = T(v)$.

For uniqueness, suppose $T = T'' \circ \pi$ is another factorization. We will show that $T'' = T'$. Let $x \in V/U$. Since π is surjective, there exists $v \in V$ such that $x = \pi(v)$. We then have $T(v) = T'(\pi(v))$ and $T(v) = T''(\pi(v))$, which immediately yields $T'(x) = T''(x)$. **Q.E.D.**

Exercise 18.9 (*Kernel of the Induced Quotient Map*): Prove that $\ker(T') = \ker(T)/U$.

Quotients, Complements, and Dual Spaces

Let V be a vector space over K, and let $U \subseteq V$ be a subspace. Let $W \subseteq V$ be a complement of U (i.e., W is a subspace such that $U \cap W = \{0\}$ and $U + W = V$).

Proposition 18.c.7 (*Canonical Isomorphism from a Complement*): Under the assumptions above, there is a canonical isomorphism $Q\colon W \to V/U$, defined by $Q(w) := [w] \in V/U$ for all $w \in W$.

Remark 18.7: The isomorphism Q is canonical only once the complement W is specified. In general, there are many choices of complements to U, and there is no preferred choice.

Proof:

Consider the inclusion map $i\colon W \to V$ and the projection map $\pi\colon V \to V/U$. Since $Q = \pi \circ i$, Q is a linear map.

We claim that $\ker(Q) = \{0\}$. Indeed, if $w \in \ker(Q)$, then $[w] = 0$, implying that $w \in U$. However, since $w \in W$, we must have $w \in U \cap W = \{0\}$, and therefore $w = 0$. This shows that $\ker(Q) = \{0\}$.

We also claim that Q is surjective. Let $x \in V/U$ and choose $v \in V$ such that $x = [v]$. Since $V = U + W$, there exist $u \in U$ and $w \in W$ such that $v = w + u$. It follows that $[v] = [w+u] = [w]$, so $Q(w) = x$. This proves that Q is surjective, and consequently, an isomorphism. **Q.E.D.**

Exercise 18.10 (*Inverse Isomorphism from a Complement*): Describe the inverse of Q, $Q^{-1} \colon V/U \xrightarrow{\cong} W$.

Quotients, Complements, and Annihilators

Proposition 18.c.8 (*Annihilators and Dual Quotients*): Let $U \subseteq V$ be a subspace, and define the "*annihilator*" $U^{\perp} := \{\ell \in V^* \mid \ell|_U = 0\}$. Then:

(a) $U^{\perp} \subseteq V^*$ is a linear subspace.

(b) There is a canonical isomorphism $(V/U)^* \cong U^{\perp}$.

(c) If V is finite-dimensional, then $U^* \cong V^*/U^{\perp}$.

Proof:

(a) Clearly, the zero functional vanishes on U. For $\ell_1, \ell_2 \in U^{\perp}$ and $\alpha, \beta \in K$, we have $(\alpha\ell_1 + \beta\ell_2)(u) = \alpha\ell_1(u) + \beta\ell_2(u) = 0$ for all $u \in U$, establishing that $U^{\perp}$ is a subspace.

(b) We define a map $L \colon (V/U)^* \to U^{\perp}$ as follows: Let $S \in (V/U)^*$, i.e., $S \colon V/U \to K$. Define $L(S) \in V^*$ to be $L(S) := S \circ \pi$.

Exercise 18.11 (*The Map L Lands in $U^{\perp}$ and is Linear*):

(a) Show that for all $S \in (V/U)^*$, $S \circ \pi \in U^{\perp}$.

(b) Show that L is a linear map.

We claim that L is an isomorphism. To see this, we define a map $P \colon U^{\perp} \to (V/U)^*$ as follows. Let $T \in U^{\perp}$, i.e., $T \colon V \to K$ such that $T|_U = 0$. By the Universal Property of Quotients, there exists a unique $T' \colon V/U \to K$ such that $T = T' \circ \pi$. Define $P(T) := T'$.

We claim that $L \circ P = \mathrm{id}_{U^{\perp}}$ and $P \circ L = \mathrm{id}_{(V/U)^*}$. Indeed, for all $S \colon V/U \to K$, we have $P \circ L(S) = P(S \circ \pi) = S$, because of the uniqueness statement in the theorem about the universal property of quotient spaces. Also, for all $T \in U^{\perp}$, $L \circ P(T) = L(T') = T' \circ \pi = T$. This proves that L is bijective, and since we know L is linear, it is an isomorphism. It follows that P is also a linear isomorphism.

(c) Assume V is finite-dimensional. Consider the restriction map $R \colon V^* \to U^*$ defined by $R(\varphi) := \varphi|_U$ for all $\varphi \in V^*$.

Exercise 18.12 (*Linearity of the Restriction Map*): Show that R is a linear map.

We claim that R is surjective. Indeed, let $\psi \in U^*$. We need to show that there exists $\varphi \in V^*$ such that $\varphi|_U = \psi$. The finite-dimensionality of V is important here. Pick a basis $u_1, \ldots, u_k$ for U, and extend it to a basis $u_1, \ldots, u_k, v_{k+1}, \ldots, v_n$ of V. Define a linear map $\varphi \colon V \to K$ by $\varphi(u_i) := \psi(u_i)$ for all $1 \leq i \leq k$, and $\varphi(v_j) := 0$ for all $k < j \leq n$. Clearly, $\varphi|_U = \psi$ because φ and ψ coincide on the elements of a basis of U. This proves $\mathrm{Im}(R) = U^*$.

We claim that $\ker(R) = U^{\perp}$. Indeed, this follows directly from definitions: $\ker(R) = \{\varphi \in V^* \mid \varphi|_U = 0\} = U^{\perp}$. By the First Isomorphism Theorem, R induces an isomorphism $\overline{R} \colon V^*/U^{\perp} \xrightarrow{\cong} U^*$.

Q.E.D.

Exercise 18.13 (*Relationship between Annihilator of Image and Kernel of Dual Map*): Let $T\colon V \to W$ be a linear map between vector spaces. Prove that the annihilator of the image of T is equal to the kernel of its dual map:

$$(\operatorname{Im} T)^{\perp} = \ker(T^*).$$

Solutions to the Exercises

Proof of Exercise 18.6:

Every axiom is inherited from V by passing to classes. For instance, associativity: for $v_1, v_2, v_3 \in V$,

$$\big([v_1] + [v_2]\big) + [v_3] = [v_1 + v_2] + [v_3] = [(v_1 + v_2) + v_3] = [v_1 + (v_2 + v_3)] = [v_1] + \big([v_2] + [v_3]\big),$$

where the middle step is associativity in V. Commutativity is identical. The class $[0_V] = U$ is neutral, since $[v] + [0_V] = [v + 0_V] = [v]$, and $[-v]$ is an additive inverse of $[v]$, since $[v] + [-v] = [0_V]$. For scalars, $\alpha([v_1] + [v_2]) = [\alpha(v_1 + v_2)] = [\alpha v_1 + \alpha v_2] = \alpha[v_1] + \alpha[v_2]$, and similarly $(\alpha + \beta)[v] = \alpha[v] + \beta[v]$, $(\alpha\beta)[v] = \alpha(\beta[v])$ and $1 \cdot [v] = [v]$. Each of these is the corresponding axiom in V with brackets placed around it, which is legitimate precisely because the operations were shown to be well defined. **Q.E.D.**

Proof of Exercise 18.7:

Both T and $i \circ \overline{T} \circ \pi$ are maps $V \to W$, so we evaluate at an arbitrary $v \in V$:

$$\big(i \circ \overline{T} \circ \pi\big)(v) = i\big(\overline{T}([v])\big) = i\big(T(v)\big) = T(v),$$

using the definition of π, then the definition of $\overline{T}$ on the class $[v]$ with representative v, and finally that i is the inclusion of $\operatorname{Im}(T)$ into W. **Q.E.D.**

Proof of Exercise 18.8:

(a) Suppose $[v] = [v']$. Then $v - v' \in U \subseteq \ker(T)$ by the hypothesis $T(U) = 0$, so $T(v - v') = 0$ and hence $T(v) = T(v')$. So the value $T'([v]) := T(v)$ does not depend on the representative chosen.

(b) For $\alpha \in K$ and $[v], [v'] \in V/U$,

$$T'\big(\alpha[v] + [v']\big) = T'\big([\alpha v + v']\big) = T(\alpha v + v') = \alpha T(v) + T(v') = \alpha\, T'([v]) + T'([v']),$$

using the definition of the operations on V/U, then the definition of T', then the linearity of T.

Q.E.D.

Proof of Exercise 18.9:

First note that $\ker(T) \supseteq U$, so the quotient $\ker(T)/U$ makes sense and is a subspace of V/U: its elements are the classes $[v]$ with $v \in \ker(T)$.

Let $x \in V/U$ and write $x = [v]$. Then

$$x \in \ker(T') \iff T'([v]) = 0 \iff T(v) = 0 \iff v \in \ker(T) \iff x \in \ker(T)/U,$$

where the second step is the definition of T'. The middle equivalence is legitimate for either representative of x, since T' is well defined. Hence $\ker(T') = \ker(T)/U$. **Q.E.D.**

Proof of Exercise 18.10:

Let $x \in V/U$. Since $V = U + W$, any representative v of x can be written as $v = u + w$ with $u \in U$ and $w \in W$, and then $x = [v] = [w]$. The map

$$Q^{-1}\colon V/U \to W, \qquad Q^{-1}(x) := w, \quad \text{where } x = [u+w] \text{ with } u \in U,\, w \in W,$$

is the inverse of Q. It is well defined: if $u + w = u' + w'$ with $u, u' \in U$ and $w, w' \in W$, then $w - w' = u' - u \in U \cap W = \{0\}$, so $w = w'$. And indeed $Q^{-1}(Q(w)) = Q^{-1}([w]) = w$ for $w \in W$, while $Q(Q^{-1}([v])) = Q(w) = [w] = [v]$.

Equivalently, and more compactly: $Q^{-1} = p_W \circ (\text{choice of representative})$, where p_W is the projection of $V = U \oplus W$ onto W furnished by Proposition 18.b.2; the content of the verification above is exactly that this projection is well defined on classes. **Q.E.D.**

Proof of Exercise 18.11:

(a) Let $S \in (V/U)^*$ and $u \in U$. Then $\pi(u) = [u] = 0_{V/U}$, because $u \in U = \ker(\pi)$, and therefore

$$(S \circ \pi)(u) = S(0_{V/U}) = 0.$$

So $S \circ \pi$ vanishes on U, that is, $S \circ \pi \in U^\perp$.

(b) For $S, S' \in (V/U)^*$ and $\alpha \in K$, and every $v \in V$,

$$L(\alpha S{+}S')(v) = \big((\alpha S{+}S')\circ\pi\big)(v) = (\alpha S{+}S')\big(\pi(v)\big) = \alpha\, S(\pi(v)){+}S'(\pi(v)) = \big(\alpha L(S){+}L(S')\big)(v),$$

so $L(\alpha S + S') = \alpha L(S) + L(S')$.

Q.E.D.

Proof of Exercise 18.12:

For $\varphi, \varphi' \in V^*$, $\alpha \in K$ and every $u \in U$,

$$R(\alpha\varphi + \varphi')(u) = (\alpha\varphi + \varphi')|_U(u) = \alpha\varphi(u) + \varphi'(u) = \big(\alpha R(\varphi) + R(\varphi')\big)(u),$$

since the operations on V^* are defined pointwise. Two functionals on U agreeing at every $u \in U$ are equal, so $R(\alpha\varphi + \varphi') = \alpha R(\varphi) + R(\varphi')$. **Q.E.D.**

Proof of Exercise 18.13:

Both sides are subsets of W^*. For $\ell \in W^*$, the notes' chain of equivalences reads

$$\ell \in (\operatorname{Im} T)^\perp \iff \ell|_{\operatorname{Im}(T)} = 0 \iff \ell \circ T = 0 \iff T^*(\ell) = 0 \iff \ell \in \ker(T^*).$$

The middle equivalence is the only one with content: $\ell \circ T$ is the zero map on V exactly when $\ell(T(v)) = 0$ for every $v \in V$, and the values $T(v)$ are precisely the elements of $\operatorname{Im}(T)$, so this says exactly that ℓ vanishes on $\operatorname{Im}(T)$. The remaining steps are the definitions of $U^\perp$, of T^*, and of the kernel. **Q.E.D.**

AI-Note: The three lectures behind this chapter — 18.a on Hom, duals and biduals, 18.b on direct sums, 18.c on quotients — are all present in the transcript, statement for statement, and the printed numbers 18.a.1–18.a.12, 18.b.1–18.b.2 and 18.c.1–18.c.8 already agreed with the handwritten ones. What the audit found here was damage of a different kind: repetitions, one broken inference, and a great many exercises left unanswered.

Repetitions and misplacements. The paragraph introducing the bidual appeared twice, once inside a `summary` environment that had swallowed a `\subsection` heading along with it, and once again under *“Reflexivity”*; the first copy is removed and the summary now ends where the notes end it. Corollary 18.a.4 carried two proofs in a row, a sketch and a full verification; the sketch was contained in the verification and is dropped. A `definition*` naming the quotient and the quotient map stood before either had been constructed, duplicating the naming that comes with Definition 18.c.1; it is folded into the sentence that announces the construction. Four cross-references read *"Figure Figure 18.1"*, which typesets the word *"Figure"* twice.

One broken inference. Of the map $D\colon K^n_{\text{row}} \to (K^n_{\text{col}})^*$ the transcript said it *"is a linear map, and therefore, it is an isomorphism"* — linearity does not give that, and the notes do not claim it does: they say the map is an iso and mark the verification *"exc. Prove this!"*. It is now Exercise 18.2, solved on Page 137. Also repaired: the proof of Proposition 18.a.8 set $A := [\mathrm{id}_V]$ with no bases on it, and the matrix it means is $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{C}}$.

Rings. At the point where the notes say only *"explain briefly the structure of a ring"* and *"explain what is a homomorphism of rings"* — both given orally — the book had nothing, although it uses the word from Page 110 onwards and states Corollary 18.a.4 as an isomorphism of rings. AI-Definition 18.1 and AI-Definition 18.2 fill that gap; they are not Prof. Biran's text and are marked as AI-Definitions.

Exercises. The notes mark fourteen things *"Exc."* across the three lectures, from the vector space axioms for $\mathrm{Hom}(V, W)$ to $(\operatorname{Im} T)^\perp = \ker(T^*)$. Several stood in the transcript as `\begin{proof}` blocks reading *"Exercise"*, which leaves a stated result with no proof anywhere. All of them are now exercises, and each of the three sections ends with the solutions.

Miscellaneous Topics and Motivation (Chapter 19)

Geometric Motivation: Why Abstraction Matters (Section 19.a)

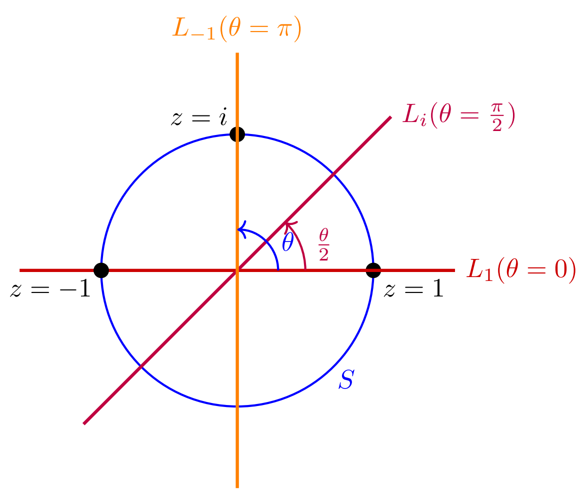
It is a common temptation to simply identify every n-dimensional vector space over K with the standard coordinate space K^n. However, while they are algebraically isomorphic for any fixed parameter, they may not be *"continuously"* isomorphic when the vector spaces depend dynamically on parameters.

The Möbius Vector Bundle

Definition 19.1 (*A Continuous Family of Subspaces*): Let $S \subseteq \mathbb{C}$ be the unit circle defined by $S = \{e^{i\theta} \mid \theta \in [0, 2\pi]\}$. For each $z = e^{i\theta} \in S$, we define a 1-dimensional subspace $L_z \subseteq \mathbb{R}^2$ as follows:

$$L_z := \operatorname{Sp}\left(\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\right).$$

Notice a topological anomaly: $z = 1$ corresponds to both $\theta = 0$ and $\theta = 2\pi$. In the first case ($\theta = 0$), the spanning vector is $(1, 0)$, so $L_1 = \operatorname{Sp}(1, 0)$. In the second case ($\theta = 2\pi$), the spanning vector is $(-1, 0)$, which gives the exact same subspace $L_1 = \operatorname{Sp}(-1, 0)$. The line L_z rotates at exactly half the speed of the point z.



Proposition 19.2 (*Non-triviality of the Möbius Bundle*): There does not exist a family of linear isomorphisms $T_z \colon L_z \to \mathbb{R}$ that depends continuously on the parameter $z \in S$.

Proof Sketch:

Consider the disjoint union of all these lines parameterized by z, which forms a space $M = \{(z, v) \mid z \in S, v \in L_z\}$. Topologically, this forms a vector bundle over S^1 known as the Möbius strip.

If such a continuous family of isomorphisms T_z existed, it would allow us to construct a global homeomorphism $\Phi \colon M \to S \times \mathbb{R}$. Removing the zero-section (the center circle where $v = 0$) from both spaces would imply that $M \setminus \{0\} \cong S \times (\mathbb{R} \setminus \{0\})$.

However, $S \times (\mathbb{R} \setminus \{0\})$ consists of two disconnected cylinders, an *"upper"* and a *"lower"* half. In contrast, if you cut a Möbius strip down its centre line, it remains a single connected piece. This topological contradiction demonstrates that we cannot consistently and continuously identify all L_z

with $\mathbb{R}$. Abstraction is therefore necessary to handle *"twisted"* families of spaces. This belongs to a branch of mathematics called topology. The conclusion for us is the one the notes draw: each L_z is a one-dimensional vector space over $\mathbb{R}$, so $L_z \cong \mathbb{R}$ for every single z, and yet the whole family cannot be identified with $\mathbb{R}$ in a continuous way. **Q.E.D.**

Affine Geometry (Section 19.b)

In our strict definition of a vector space, there is always a distinguished origin 0. Affine geometry studies spaces that *"look like"* vector spaces locally, but have forgotten their origin.

Definition 19.3 (*Affine Space*): A subset $X \subseteq V$ is called an "*affine space*" if there exists a linear subspace $U \subseteq V$ and a translation vector $x_0 \in V$ such that $X = x_0 + U$.

We say that X is modeled on the subspace U, and we write the "*direction space*" as $\vec{X} := U$. The elements of X are called "*points*", while the elements of $\vec{X}$ are called "*vectors*". The "*dimension*" of X is defined as $\dim X := \dim U$.

Remark 19.1: The point x_0 in Definition 19.3 is not uniquely determined by X, unless $\dim X = 0$: any point of X will serve. The subspace U, on the other hand, **is** uniquely determined by X, which is what makes the notation $\vec{X}$ legitimate. This is left as an exercise.

Proposition 19.4 (*Algebra of Affine Spaces*): Let X be an affine space modeled on $\vec{X}$. Then:

(a) We cannot inherently add two points $p, q \in X$, as $p + q \notin X$ in general.

(b) We can translate a point $p \in X$ by a vector $v \in \vec{X}$ to obtain a new point $p + v \in X$.

(c) For any two points $p, q \in X$, their difference defines a unique vector $\vec{pq} := q - p \in \vec{X}$.

Proof:

For **(b)**, write $p = x_0 + u$ with $u \in \vec{X}$. Then $p + v = x_0 + (u + v) \in x_0 + \vec{X} = X$, since $u + v \in \vec{X}$. For **(c)**, write $p = x_0 + u$ and $q = x_0 + v$ with $u, v \in \vec{X}$; then $q - p = v - u \in \vec{X}$. Statement **(a)** is the observation that on X there is no longer a zero: an affine space has forgotten its origin. **Q.E.D.**

Note also that the two operations fit together as one expects: $p + \vec{pq} = q$ for all $p, q \in X$.

Example 19.1 (*Affine Spaces*):

(a) An "*affine point*" is an affine space $X = \{x\} \subseteq V$ with $\vec{X} = \{0\}$, so $\dim X = 0$.

(b) An "*affine line*": let $x, y \in V$ be distinct and put $v := \vec{xy}$. Then

$$\ell_{xy} := \{x + tv \mid t \in K\}$$

is an affine line, modeled on the one-dimensional subspace $\mathrm{Sp}(v)$.

(c) In the same way one considers affine 2-dimensional planes, and generally affine k-dimensional planes, modeled on k-dimensional subspaces.

(d) Let $A \in \mathcal{M}_{m\times n}(K)$ and $b \in K^m$, and let $x_0 \in K^n$ be a solution of $Ax = b$. Then, by Corollary 17.d.2,

$$\underbrace{\mathrm{Sol}(A, b)}_{\text{affine space}} = x_0 + \underbrace{\mathrm{Sol}(A, 0)}_{\text{vector space}},$$

so the solution set of an inhomogeneous linear system is exactly an affine space modeled on the solution space of the associated homogeneous system.

Remark 19.2: One can go further and define "*affine maps*" $f : X \to Y$ between affine spaces, and, given an affine space X and a subspace $U \subseteq \vec{X}$, a new affine space X/U modeled on $\vec{X}/U$.

Definition 19.5 (*Parallel Affine Spaces*): Let $X, Y \subseteq V$ be affine spaces. We say that X and Y are "*parallel*", written $X \parallel Y$, if $\vec{X} = \vec{Y}$; equivalently, if $X = p + U$ and $Y = q + U$ for one and the same subspace U. We say that X is "*weakly parallel*" to Y if $\vec{X} \subseteq \vec{Y}$.

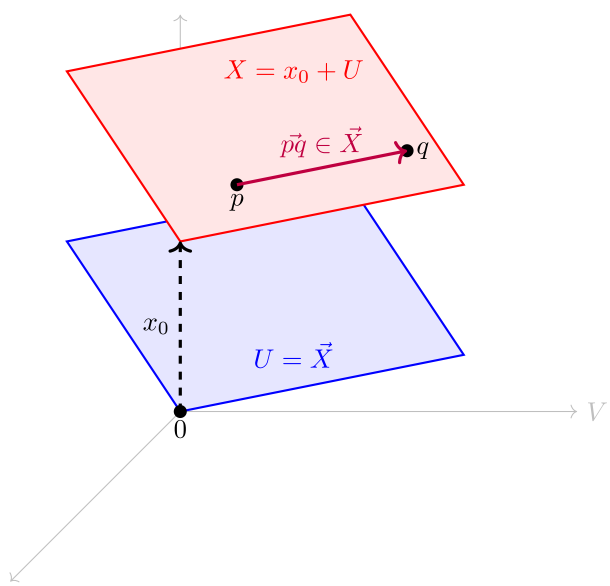
Important remark 19.1 (*Ratios of Vectors on a Line*): If L is a **one-dimensional** vector space over K, then vectors in L can be divided by one another. Indeed, let $u, v \in L$ with $u \neq 0$. Since $\dim L = 1$, the vector u alone is a basis of L, so there is a unique scalar $\lambda \in K$ with $v = \lambda \cdot u$, and we define

$$\frac{v}{u} := \lambda \in K.$$

Consequently, if X is an affine line and $a, b, c \in X$ with $a \neq b$, then $\vec{ab}$ and $\vec{ac}$ both lie in the one-dimensional space $\vec{X}$ and $\vec{ab} \neq 0$, so the ratio

$$\frac{\vec{ac}}{\vec{ab}} \in K$$

is defined. These are proportions, not distances: no notion of length is involved, and none is available, since K need not be $\mathbb{R}$.



Theorem 19.6 (*Thales' Theorem in Affine Geometry*): Let V be a 3-dimensional vector space over K, and let H, H', H'' be three distinct parallel 2-dimensional affine planes in V. Let ℓ_1, ℓ_2 be two affine lines, neither of them weakly parallel to H, and put

$$p_1 := H \cap \ell_1, \quad p_1' := H' \cap \ell_1, \quad p_1'' := H'' \cap \ell_1, \qquad p_2 := H \cap \ell_2, \quad p_2' := H' \cap \ell_2, \quad p_2'' := H'' \cap \ell_2.$$

Then

$$\frac{\vec{p_1p_1''}}{\vec{p_1p_1'}} = \frac{\vec{p_2p_2''}}{\vec{p_2p_2'}}.$$

In other words, this ratio does not depend on the line ℓ chosen, but only on the three planes H, H', H''.

Proof:

Put $U := \vec{H} = \vec{H'} = \vec{H''}$, which is a 2-dimensional subspace of V, the three planes being parallel. Consider the quotient V/U and the projection $\pi \colon V \to V/U$. By Proposition 18.c.4,

$$\dim V/U = \dim V - \dim U = 3 - 2 = 1,$$

so V/U is a line, and ratios of its non-zero vectors are defined as in Page 151.

Both ratios in the statement are defined: p_1, p_1', p_1'' lie on the affine line ℓ_1, whose direction space is one-dimensional, and $\vec{p_1p_1'} \neq 0$ because $p_1 \in H$ and $p_1' \in H'$ are points of distinct parallel planes. Put

$$\lambda_1 := \frac{\vec{p_1p_1''}}{\vec{p_1p_1'}}, \qquad \lambda_2 := \frac{\vec{p_2p_2''}}{\vec{p_2p_2'}},$$

so that $\vec{p_1p_1''} = \lambda_1\vec{p_1p_1'}$ and $\vec{p_2p_2''} = \lambda_2\vec{p_2p_2'}$. Our task is to show $\lambda_1 = \lambda_2$.

Now $\pi(\vec{p_1p_1'}) = \pi(\vec{p_2p_2'})$. Indeed,

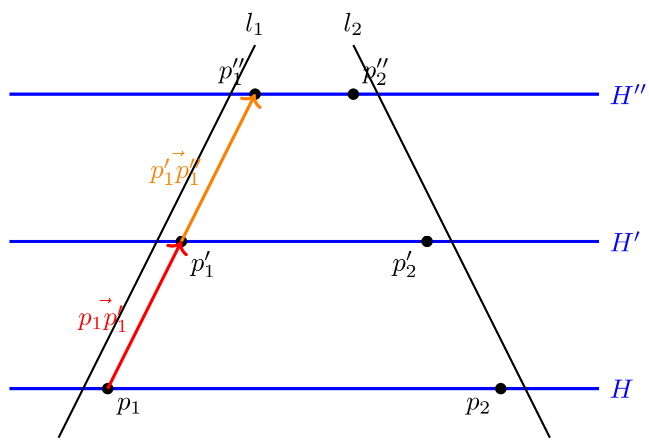
$$\vec{p_1p_1'} - \vec{p_2p_2'} = \vec{p_1p_2} + \vec{p_2'p_1'} \in U,$$

because $p_1, p_2 \in H$ gives $\vec{p_1p_2} \in \vec{H} = U$, and $p_2', p_1' \in H'$ gives $\vec{p_2'p_1'} \in \vec{H'} = U$; and $\ker(\pi) = U$ by Proposition 18.c.3. The same argument with H'' in place of H' gives $\pi(\vec{p_1p_1''}) = \pi(\vec{p_2p_2''})$. Combining, and using the linearity of π,

$$\lambda_1\,\pi(\vec{p_1p_1'}) = \pi(\lambda_1\vec{p_1p_1'}) = \pi(\vec{p_1p_1''}) = \pi(\vec{p_2p_2''}) = \pi(\lambda_2\vec{p_2p_2'}) = \lambda_2\,\pi(\vec{p_2p_2'}) = \lambda_2\,\pi(\vec{p_1p_1'}).$$

Finally $\pi(\vec{p_1p_1'}) \neq 0$: otherwise $\vec{p_1p_1'} \in \ker(\pi) = U$, and since $\vec{p_1p_1'}$ spans the direction space of ℓ_1, this would make ℓ_1 weakly parallel to H, which we excluded. Cancelling the non-zero vector $\pi(\vec{p_1p_1'})$ in the one-dimensional space V/U yields $\lambda_1 = \lambda_2$. **Q.E.D.**

Remark 19.3: Observe what this proof did **not** use: no coordinates, no lengths, no angles, and no assumption on K beyond its being a field. The whole argument is the observation that collapsing the direction U of the three planes turns them into three points of a line, where the ratio is visible at once. The four points p_1, p_2, p_1', p_2' need not lie in a common plane, and the same goes for p_1, p_2, p_1'', p_2''.



Introduction to Eigenvalues: Fibonacci Sequences (Section 19.c)

The Vector Space of Fibonacci Sequences

Consider the set $\mathrm{Fib} \subseteq \mathbb{R}^\infty$ of all infinite real sequences $(a_0, a_1, a_2, \dots)$ that satisfy the linear recurrence relation $a_n = a_{n-1} + a_{n-2}$ for all $n \geq 2$.

Proposition 19.7 (*Structure of* Fib): The set Fib is a 2-dimensional subspace of $\mathbb{R}^\infty$. The evaluation map $F\colon \mathrm{Fib} \to \mathbb{R}^2$ sending a sequence to its first two terms, $(a_n) \mapsto (a_0, a_1)$, is a linear isomorphism.

To find a closed-form formula for the Fibonacci numbers (like Binet's formula), we study the **shift map** $S\colon \mathbb{R}^\infty \to \mathbb{R}^\infty$ defined by shifting the sequence one position to the left:

$$S(a_0, a_1, a_2, \dots) := (a_1, a_2, a_3, \dots).$$

Since the shifted sequence also satisfies the Fibonacci recurrence, this map restricts to a linear endomorphism $S'\colon \mathrm{Fib} \to \mathrm{Fib}$.

By utilizing the isomorphism F, we can translate the abstract shift operator S' into concrete matrix multiplication on $\mathbb{R}^2$. The sequence (a_0, a_1) shifts to $(a_1, a_2) = (a_1, a_0 + a_1)$, which corresponds to multiplication by the matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$.

$$\begin{tikzcd}
\mathrm{Fib} \arrow[r, "S'"] \arrow[d, "F"', "\cong"] & \mathrm{Fib} \arrow[d, "\cong"', "F"] \\
\mathbb{R}^2 \arrow[r, "T_A"'] & \mathbb{R}^2
\end{tikzcd}$$

Eigenvalues and Eigenvectors

The problem of finding a geometric sequence $(1, \lambda, \lambda^2, \dots)$ inside Fib is equivalent to finding sequences that only scale by λ when shifted. That is, we seek vectors $\mathcal{F}$ such that $S'(\mathcal{F}) = \lambda\mathcal{F}$.

Definition 19.8 (*Eigenvalues and Eigenvectors*): Let V be a vector space and $T \in \mathrm{End}(V)$. A scalar $\lambda \in K$ is called an "*eigenvalue*" of T if there exists a non-zero vector $v \in V$ such that:

$$T(v) = \lambda v.$$

The vector v is called an "*eigenvector*" of T associated with the eigenvalue λ.

Remark 19.4 (*Reformulations of the Eigenvalue Condition*): Unwinding the definition, λ is an eigenvalue of T if and only if the linear map $T - \lambda \cdot \mathrm{id}_V \colon V \to V$ has a non-trivial kernel, since $T(v) = \lambda v$ says exactly that $(T - \lambda\,\mathrm{id}_V)(v) = 0$.

If we assume in addition that V is finite-dimensional, then by Corollary 15.b.10 **(c)** the last condition is equivalent to

$$T - \lambda \cdot \mathrm{id}_V \colon V \to V \quad \text{is } \textbf{not} \text{ an isomorphism},$$

and, passing to a basis $\mathcal{B}$ of V and writing $n := \dim V$, to

$$[T]_{\mathcal{B}}^{\mathcal{B}} - \lambda \cdot I_n \in \mathcal{M}_{n\times n}(K) \quad \text{is } \textbf{not} \text{ invertible}.$$

The last equivalence uses Proposition 17.b.3 together with $[\lambda \cdot \mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = \lambda I_n$ from Claim 17.1. This is the form that will let us actually compute eigenvalues, once we have determinants: it turns the question into a question about a single matrix.

Proposition 19.9 (*The Golden Ratio Diagonalises the Shift*): Put $\varphi := \frac{1+\sqrt{5}}{2}$ and $\psi := \frac{1-\sqrt{5}}{2}$. Then the two geometric sequences

$$x_{1,\varphi} := (1, \varphi, \varphi^2, \dots), \qquad x_{1,\psi} := (1, \psi, \psi^2, \dots)$$

belong to Fib, and they form a basis of Fib. Moreover φ and ψ are eigenvalues of the shift $S'\colon \mathrm{Fib} \to \mathrm{Fib}$, with $x_{1,\varphi}$ and $x_{1,\psi}$ as associated eigenvectors.

Proof:

A geometric sequence $(1, \lambda, \lambda^2, \dots)$ lies in Fib precisely when $\lambda^n = \lambda^{n-1} + \lambda^{n-2}$ for all $n \geq 2$, that is, when $\lambda^2 = \lambda + 1$. The roots of $\lambda^2 - \lambda - 1$ are exactly φ and ψ, so both sequences lie in Fib.

They are linearly independent: applying the isomorphism F of Proposition 19.7 sends them to $(1,\varphi)$ and $(1,\psi)$ in $\mathbb{R}^2$, which are independent because $\varphi \neq \psi$. Since $\dim \mathrm{Fib} = 2$, they form a basis.

Finally, shifting a geometric sequence multiplies it by its ratio:

$$S'(1,\lambda,\lambda^2,\dots) = (\lambda,\lambda^2,\lambda^3,\dots) = \lambda\cdot(1,\lambda,\lambda^2,\dots),$$

so $x_{1,\varphi}$ and $x_{1,\psi}$ are eigenvectors of S' with eigenvalues φ and ψ. **Q.E.D.**

All that is left in order to recover the closed formula is to express the Fibonacci sequence itself in this basis. The Fibonacci sequence is $x_{0,1} = (0,1,1,2,3,5,\dots)$, and we seek $\alpha,\beta\in\mathbb{R}$ with $\alpha\, x_{1,\varphi} + \beta\, x_{1,\psi} = x_{0,1}$. Comparing the first two entries gives the system

$$\begin{cases} \alpha\cdot 1 + \beta\cdot 1 = 0, \\ \alpha\cdot\varphi + \beta\cdot\psi = 1, \end{cases}$$

whose solution is $\alpha = \frac{1}{\sqrt5}$ and $\beta = -\frac{1}{\sqrt5}$. Reading off the n-th entry recovers the formula from the first chapter,

$$f_n = \frac{1}{\sqrt5}\left(\left(\frac{1+\sqrt5}{2}\right)^n - \left(\frac{1-\sqrt5}{2}\right)^n\right) \qquad \text{for all } n\geq 0.$$

This is the whole method in miniature, and it is the motivation for the chapters to come: find the eigenvalues of an endomorphism, use the associated eigenvectors as a basis, and in that basis the map becomes multiplication by scalars, one coordinate at a time. What is still missing is a way to find the eigenvalues, and that is what determinants will provide.

AI-Note: This chapter is the one place in the book where the notes are mostly pictures and asides, and the transcript kept the pictures but lost several of the asides. Restored above:

In the affine section, the dimension of an affine space; the observation that the base point x_0 is not determined by X while the direction space is (Page 150); the identity $p+\vec{pq} = q$; Prof. Biran's four examples, of which the fourth is the pleasant one — the solution set of $Ax = b$ **is** an affine space, modeled on the solution space of $Ax = 0$; the mention of affine maps and affine quotients; the definition of parallel and weakly parallel; and, above all, Page 151: in a one-dimensional space one vector may be divided by another. Without that observation the fraction in the statement of Theorem 19.6 has no meaning at all, and the transcript stated Thales without it.

Theorem 19.6 itself stood without a proof and with a hypothesis too loose to be true as stated — the ambient space must be three-dimensional, the planes two-dimensional, and the two lines must not be weakly parallel to H, or the intersections need not be single points. Prof. Biran's proof is restored: collapse the common direction U of the three planes, so that V/U becomes a line, and the equality of ratios falls out. He annotates it *"a truly coordinate-free proof"*, and it is a striking use of the quotient construction from the previous chapter.

In the closing section, the reformulations of the eigenvalue condition — non-trivial kernel, failure to be an isomorphism, non-invertibility of $[T]^{\mathcal{B}}_{\mathcal{B}} - \lambda I_n$ — which are what makes the notion computable later. And the payoff the chapter was building towards: that φ and ψ are the eigenvalues of the shift on Fib, that the two geometric sequences form a basis, and that expanding the Fibonacci sequence in that basis returns the formula from chapter 1. In the transcript this last part existed only as two commented-out lines at the end of the file, one of them a corrected copy of the other.

Also: the proposition on Fib had `$Fib$` in its title, typeset as a product of three italic letters rather than with the `\Fib` macro, and five raw quotation marks became `\qt`.

Determinants (Chapter 20)

Introduction and Motivation (Section 20.a)

The special case of 2×2 matrices

In linear algebra, determining whether a matrix is invertible is a fundamental problem. Let us begin by analyzing the familiar 2×2 case. Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathcal{M}_{2\times 2}(K)$. There is a very simple, well-known algebraic criterion to check whether A is invertible.

Proposition 20.a.1 (*Invertibility of 2×2 Matrices*): The matrix A is invertible if and only if $ad - bc \neq 0$. Moreover, if A is invertible, then its inverse is given by the explicit formula:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The scalar quantity $ad - bc$ is called the "*determinant*" of A, denoted $\det(A)$.

Proof:

First, we show that A is not invertible if and only if its rank is strictly less than 2. Because A is a 2×2 matrix, having a rank less than 2 implies that its column vectors are linearly dependent.

Exercise 20.1 (*Dependence of Two Columns*): Check that the columns of A being linearly dependent means that either the first column is a scalar multiple of the second, $\binom{a}{c} = \lambda \binom{b}{d}$ for some $\lambda \in K$, or the second is a scalar multiple of the first, $\binom{b}{d} = \tau \binom{a}{c}$ for some $\tau \in K$.

If $\binom{a}{c} = \lambda \binom{b}{d}$, then $a = \lambda b$ and $c = \lambda d$. Consequently, $ad - bc = (\lambda b)d - b(\lambda d) = 0$. If, on the other hand, $\binom{b}{d} = \tau \binom{a}{c}$, then $b = \tau a$ and $d = \tau c$. Consequently, $ad - bc = a(\tau c) - (\tau a)c = 0$. Thus, we have proved that if A is not invertible, then $ad - bc = 0$.

Conversely, assume $ad - bc = 0$. If $a = 0$, then $bc = 0$, so either $b = 0$ or $c = 0$. This means either $A = \left(\begin{smallmatrix} 0 & 0 \\ c & d \end{smallmatrix}\right)$ or $A = \left(\begin{smallmatrix} 0 & b \\ 0 & d \end{smallmatrix}\right)$. In both of these cases, A, clearly, has linearly dependent columns and is therefore not invertible.

Now, assume $a \neq 0$. This implies $d = \frac{bc}{a}$, and thus we can write $A = \begin{pmatrix} a & b \\ c & \frac{bc}{a} \end{pmatrix}$. We can observe that the second column of A is exactly a multiple of the first column because $\binom{b}{\frac{bc}{a}} = \frac{b}{a} \binom{a}{c}$. This implies that the columns of A are linearly dependent. Hence, $\operatorname{rank}(A) < 2$, which implies A is not invertible. **Q.E.D.**

To confirm the sufficiency of this condition and to find the explicit inverse, the formula for A^{-1} can be verified by direct matrix multiplication:

$$\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} ad - bc & bd - bd \\ -ac + ac & -bc + ad \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Goals 20.1: While the 2×2 formula is easy to memorize, it does not immediately suggest how to handle larger matrices. Our main goals now are to make the determinant conceptually rigorous and to generalize the determinant function to arbitrary $n \times n$ matrices.

Multilinear and Alternating Functions

We wish to generalize the determinant to arbitrary dimensions. To this end, we should view an $n \times n$ matrix not just as a grid of numbers, but as a collection of n row vectors.

Definition 20.a.2 (n-Linearity): Let $D : \mathcal{M}_{n\times n}(K) \to K$ be a function. We say that D is "n-*linear*" if for every $1 \le i \le n$, D is linear as a function of the i-th row when all the other $n-1$ rows are held fixed.

We can think of the matrix space $\mathcal{M}_{n\times n}(K)$ as a Cartesian product of row vectors: $K^n_{\text{row}} \times \cdots \times K^n_{\text{row}}$. If A has rows $\alpha_1, \ldots, \alpha_n$, we write $D(A) = D(\alpha_1, \ldots, \alpha_n)$. Saying that D is n-linear means that for any chosen row index i, the following two linearity conditions hold:

(a) $D(\alpha_1, \ldots, \alpha_i + \alpha_i', \ldots, \alpha_n) = D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + D(\alpha_1, \ldots, \alpha_i', \ldots, \alpha_n)$.

(b) $D(\alpha_1, \ldots, c\cdot\alpha_i, \ldots, \alpha_n) = c \cdot D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n)$ for all scalars $c \in K$.

Example 20.1 (*Product of Row Entries*): For a matrix $A \in \mathcal{M}_{n\times n}(K)$, write $A(i,j)$ for its entry at place i,j. Let $k_1, \ldots, k_n$ be a fixed sequence of integers such that $1 \le k_i \le n$ for all i. Fix also a scalar $c \in K$. Consider $D : \mathcal{M}_{n\times n}(K) \to K$ defined by the product of one entry from each row:

$$D(A) := c \cdot A(1,k_1) \cdot A(2,k_2) \cdots A(n,k_n).$$

We claim that D is an n-linear function.

To see this, consider $D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) = b \cdot A(i,k_i)$, where $b \in K$ is the product of all the fixed terms. Notice that b depends only on the rows $\alpha_1, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_n$ and does not depend on row i. Let $\alpha_i' = (\alpha_{i,1}', \ldots, \alpha_{i,n}')$ be another possible i-th row. Then, by standard distribution:

$$D(\ldots, \alpha_i + \alpha_i', \ldots) = b \cdot (A(i,k_i) + \alpha_{i,k_i}') = D(\ldots, \alpha_i, \ldots) + D(\ldots, \alpha_i', \ldots).$$

Similarly, for any scalar $\lambda \in K$:

$$D(\ldots, \lambda \cdot \alpha_i, \ldots) = b \cdot (\lambda A(i,k_i)) = \lambda \cdot D(\ldots, \alpha_i, \ldots).$$

Therefore, D is indeed n-linear.

Example 20.2 (*Diagonal Product*): A more concrete example: $D(A) := A_{11} \cdot A_{22} \cdot \cdots \cdot A_{nn}$ (i.e., the product of the terms on the main diagonal) is an n-linear function.

Now, let us try to find all 2-linear functions $D : \mathcal{M}_{2\times 2}(K) \to K$. Write the identity matrix as $I = \begin{pmatrix} -\epsilon_1- \\ -\epsilon_2- \end{pmatrix}$, where $\epsilon_1 = (1,0)$ and $\epsilon_2 = (0,1)$.

If D is 2-linear, then we have:

$$\begin{aligned} D(A) &= D(A_{11}\epsilon_1 + A_{12}\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}D(\epsilon_1, A_{21}\epsilon_1 + A_{22}\epsilon_2) + A_{12}D(\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}A_{21}D(\epsilon_1,\epsilon_1) + A_{11}A_{22}D(\epsilon_1,\epsilon_2) + A_{12}A_{21}D(\epsilon_2,\epsilon_1) + A_{12}A_{22}D(\epsilon_2,\epsilon_2). \end{aligned}$$

Therefore, D is completely determined by its values on the following four pairings: $D(\epsilon_1,\epsilon_1)$, $D(\epsilon_1,\epsilon_2)$, $D(\epsilon_2,\epsilon_1)$, and $D(\epsilon_2,\epsilon_2)$.

Lemma 20.a.3 (*Closure of n-Linear Functions*): Any linear combination of n-linear functions is again an n-linear function.

Proof:

It is enough to prove this for a linear combination of two n-linear functions. Let $D, E : \mathcal{M}_{n\times n}(K) \to K$ be n-linear functions and $a, b \in K$. We define the combined function as $(aD + bE)(A) := aD(A) + bE(A)$. By expanding $(aD+bE)(\ldots, \alpha_i + \alpha_i', \ldots)$ and $(aD+bE)(\ldots, \lambda\alpha_i, \ldots)$ using the individual linearities of D and E, we see the result naturally holds. **Q.E.D.**

Example 20.3 (*Two-by-Two Determinant Function*): Consider $D : \mathcal{M}_{2\times 2}(K) \to K$ defined by $D(A) := A_{11}A_{22} - A_{12}A_{21}$. The function D is 2-linear because it can be written as $D = D_1 + D_2$, where $D_1(A) = A_{11}A_{22}$ and $D_2(A) = -A_{12}A_{21}$, both of which are 2-linear by our previous example. Furthermore, $D(I) = 1$, and if A' is obtained from A by interchanging the rows, then $D(A') = -D(A)$.

Indeed, $D(A') = D\left(\begin{smallmatrix} A_{21} & A_{22} \\ A_{11} & A_{12} \end{smallmatrix}\right) = A_{21} \cdot A_{12} - A_{22}A_{11} = -(A_{11}A_{22} - A_{12}A_{21}) = -D(A)$.

Definition 20.a.4 (*Alternating Function*): Let $D : \mathcal{M}_{n\times n}(K) \to K$ be an n-linear function. We say that D is "*alternating*" if for every matrix A in which some two rows are equal, we have $D(A) = 0$.

Proposition 20.a.5 (*Adjacent Swap Property*): Let D be a "*alternating*" function. Assume that D has the property that whenever A has two equal adjacent rows, then $D(A) = 0$. Then the following are equivalent:

(a) D is alternating.

(b) For any matrix B, if B' is obtained from B by interchanging two of its rows, then $D(B') = -D(B)$.

Proof:

We begin with the proof of **(b)**. Assume first that B' is obtained from B by interchanging two adjacent rows (say, row k and row $k+1$). Consider evaluating D on a matrix where both the k-th and $(k+1)$-th rows are equal to $\beta_k + \beta_{k+1}$. By our assumption, this yields 0. Expanding this via n-linearity gives:

$$\begin{aligned} D(\dots, \beta_k + \beta_{k+1}, \beta_k + \beta_{k+1}, \dots) &= D(\dots, \beta_k, \beta_k, \dots) + D(\dots, \beta_k, \beta_{k+1}, \dots) \\ &\quad + D(\dots, \beta_{k+1}, \beta_k, \dots) + D(\dots, \beta_{k+1}, \beta_{k+1}, \dots) = 0. \end{aligned}$$

By assumption, the terms with adjacent equal rows are zero, leaving $D(B) + D(B') = 0$, which naturally implies $D(B') = -D(B)$.

Now suppose B' is obtained from B by interchanging rows k and ℓ where $k < \ell$. We can obtain B' by doing a sequence of adjacent row interchanges. Moving row ℓ to position k requires $r = \ell - k$ adjacent interchanges. Next, moving the original row k down to position ℓ requires $r - 1$ adjacent interchanges. The total number of interchanges is $2r - 1$. Since this is always an odd number, we flip the sign an odd number of times. Thus, $D(B') = (-1)^{2r-1}D(B) = -D(B)$.

We now prove **(a)**. Let A be a matrix in which row i equals row j where $i < j$. If $j = i+1$, then by assumption $D(A) = 0$. If $j > i+1$, we interchange row j with row $i+1$ to obtain a matrix A' with two equal adjacent rows. By assumption $D(A') = 0$, but by **(b)**, $D(A') = -D(A)$ (since we performed an odd number of adjacent swaps), which implies $D(A) = 0$. **Q.E.D.**

The Determinant Function

Definition 20.a.6 (*Determinant Function*): A function $D : \mathcal{M}_{n\times n}(K) \to K$ is called a "*determinant function*" if it is n-linear, it is alternating, and it normalizes the identity matrix such that $D(I) = 1$.

Example 20.4 (*Warmup*): Let us find all determinant functions on $M_{2\times 2}(K)$. Write $I = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} -\epsilon_1- \\ -\epsilon_2- \end{smallmatrix}\right)$, meaning $\epsilon_1 = (1,0)$ and $\epsilon_2 = (0,1)$. If D is a 2-linear function, we can expand it algebraically:

$$\begin{aligned} D(A) &= D(A_{11}\epsilon_1 + A_{12}\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}D(\epsilon_1, A_{21}\epsilon_1 + A_{22}\epsilon_2) + A_{12}D(\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}A_{21}D(\epsilon_1, \epsilon_1) + A_{11}A_{22}D(\epsilon_1, \epsilon_2) \\ &\quad + A_{12}A_{21}D(\epsilon_2, \epsilon_1) + A_{12}A_{22}D(\epsilon_2, \epsilon_2). \end{aligned}$$

Consequently, D is completely determined by its values on these standard basis row pairings.

If D is alternating, then evaluating equal rows gives zero, meaning $D(\epsilon_1, \epsilon_1) = D(\epsilon_2, \epsilon_2) = 0$. Additionally, swapping rows flips the sign, so $D(\epsilon_2, \epsilon_1) = -D(\epsilon_1, \epsilon_2) = -D(I)$. Finally, if D is a determinant function, $D(I) = 1$. Substituting these values back into our expansion, we perfectly recover the familiar formula:

$$D(A) = A_{11}A_{22}(1) + A_{12}A_{21}(-1) = A_{11}A_{22} - A_{12}A_{21}.$$

Therefore, the 2×2 determinant function is completely unique!

Definition 20.a.7 (*Submatrix*): Let $A \in \mathcal{M}_{n\times n}(K)$, $n \geq 2$, and $1 \leq i \leq n$, $1 \leq j \leq n$. We denote by $A(i|j) \in \mathcal{M}_{(n-1)\times(n-1)}(K)$ the *"submatrix"* obtained from A by deleting row i and column j. If D is an $(n-1)$-linear function, we intuitively define $D_{ij}(A) := D(A(i|j))$.

Theorem 20.a.8 (*Recursive Construction of Determinants*): Let $n \geq 2$, and let D be an $(n-1)$-linear function which is alternating. Let $1 \leq j \leq n$. Define $E_j : \mathcal{M}_{n\times n}(K) \to K$ by

$$E_j(A) := \sum_{i=1}^{n} (-1)^{i+j} A_{ij} D_{ij}(A).$$

Then, E_j is an n-linear function and it is alternating. Furthermore, if D is a determinant function, then so is E_j.

Proof:

Clearly, $D_{ij}(A)$ is independent of the i-th row of A because that row is deleted in the construction of $A(i|j)$. Since D is $(n-1)$-linear, the map $A \mapsto D_{ij}(A)$ is linear when viewed as a function of each of its remaining rows. The function $A \mapsto A_{ij}D_{ij}(A)$ is therefore n-linear. Since linear combinations of n-linear functions are n-linear, $E_j(A)$ is n-linear too.

We show that E_j is alternating. By Proposition 20.a.5, it is enough to show $E_j(A) = 0$ whenever A has two equal adjacent rows. Assume $\alpha_k = \alpha_{k+1}$ for some index k. For any $i \neq k, k+1$, the submatrix $A(i|j)$ still has two equal rows, hence $D_{ij}(A) = 0$. This leaves only two non-zero terms in the sum:

$$E_j(A) = (-1)^{k+j} A_{kj} D_{kj}(A) + (-1)^{k+1+j} A_{(k+1)j} D_{(k+1)j}(A).$$

But, since $\alpha_k = \alpha_{k+1}$, we have $A_{kj} = A_{(k+1)j}$ and $A(k|j) = A(k+1|j)$, meaning $D_{kj}(A) = D_{(k+1)j}(A)$. Thus, the two terms perfectly cancel out, yielding $E_j(A) = 0$. This proves E_j is alternating.

Finally, assume D is a determinant function, meaning $D(I_{n-1}) = 1$. Note that $I_n(j|j) = I_{n-1}$ for all j. When computing $E_j(I_n)$, the term $(I_n)_{ij}$ is 1 if $i = j$, and 0 if $i \neq j$. Therefore, the sum collapses to a single term:

$$E_j(I_n) = (-1)^{j+j} D_{jj}(I_n) = D(I_n(j|j)) = 1.$$

This shows that E_j is a valid determinant function. **Q.E.D.**

Corollary 20.a.9 (*Existence of Determinants*): For all $n \geq 1$, at least one determinant function exists.

Example 20.5 (*Three-by-Three Determinant Expansion*): Consider $A \in \mathcal{M}_{3\times 3}(K)$. Using the uniquely defined 2×2 determinant function $|B| := B_{11}B_{22} - B_{12}B_{21}$, we can explicitly write out $E_1(A)$ (expanding along the first column) as:

$$E_1(A) = A_{11} \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{21} \begin{vmatrix} A_{12} & A_{13} \\ A_{32} & A_{33} \end{vmatrix} + A_{31} \begin{vmatrix} A_{12} & A_{13} \\ A_{22} & A_{23} \end{vmatrix}.$$

We can similarly construct $E_2(A)$ and $E_3(A)$ along the other columns, and all of these form valid determinant functions on $M_{3\times 3}(K)$.

Uniqueness of the Determinant (Section 20.b)

We have shown that a determinant function exists. Now, we will prove that it is absolutely unique. Let $D : \mathcal{M}_{n\times n}(K) \to K$ be an n-linear alternating function. Let $A \in \mathcal{M}_{n\times n}(K)$ be a matrix with rows $\alpha_1, \ldots, \alpha_n$. We denote the identity matrix as I, whose rows are the standard basis vectors $\epsilon_1, \ldots, \epsilon_n$ of K^n_{row}.

Clearly, we can express each row of A in terms of the standard basis: $\alpha_i = \sum_{j=1}^n A(i,j)\epsilon_j$ for $1 \le i \le n$. Using the n-linearity of D, we can expand $D(A)$ as follows:

$$\begin{aligned}
D(A) &= D(\alpha_1, \ldots, \alpha_n) \\
&= D\left(\sum_{j=1}^n A(1,j)\epsilon_j, \alpha_2, \ldots, \alpha_n\right) \\
&= \sum_{j=1}^n A(1,j) \cdot D(\epsilon_j, \alpha_2, \ldots, \alpha_n) \\
&= \sum_{j=1}^n A(1,j) \cdot D\left(\epsilon_j, \sum_{k=1}^n A(2,k)\epsilon_k, \ldots, \alpha_n\right) \\
&= \sum_{j,k} A(1,j) \cdot A(2,k) \cdot D(\epsilon_j, \epsilon_k, \ldots, \alpha_n)
\end{aligned}$$

where both j and k run between 1 and n. Continuing this expansion for all n rows yields an enormous sum:

$$D(A) = \sum_{k_1,\ldots,k_n} A(1,k_1) \cdot A(2,k_2) \cdots A(n,k_n) \cdot D(\epsilon_{k_1}, \epsilon_{k_2}, \ldots, \epsilon_{k_n})$$

where each of the indices $k_1, \ldots, k_n$ runs between 1 and n.

Thus far, we have not used the fact that D is alternating. Because D is alternating, the term $D(\epsilon_{k_1}, \ldots, \epsilon_{k_n}) = 0$ whenever two of the indices $k_1, \ldots, k_n$ are equal. Thus, we are interested only in ordered sequences $(k_1, \ldots, k_n)$ in which $k_i \in \{1, \ldots, n\}$ for all i, and $k_i \neq k_j$ for all $i \neq j$.

Such a sequence is called a "*permutation*" of $\{1, \ldots, n\}$, or a permutation of degree n. Alternatively, we can think of a permutation of degree n as a bijective function $\sigma : \{1, \ldots, n\} \to \{1, \ldots, n\}$. Then, the sequence $(k_1, \ldots, k_n) = (\sigma(1), \sigma(2), \ldots, \sigma(n))$ is a permutation as defined earlier. In other words, a permutation of degree n is simply a re-ordering of a list of n elements: $(1, \ldots, n) \mapsto (\sigma(1), \ldots, \sigma(n))$.

We can nicely rewrite our expansion of $D(A)$ as:

$$D(A) = \sum_{\sigma} A(1,\sigma(1)) \cdot A(2,\sigma(2)) \cdots A(n,\sigma(n)) \cdot D(\epsilon_{\sigma(1)}, \epsilon_{\sigma(2)}, \ldots, \epsilon_{\sigma(n)})$$

where σ runs over all possible permutations of degree n.

Properties of Permutations

An important property about permutations is that if σ is a permutation of degree n, one can pass from the sequence $(1, \ldots, n)$ to $(\sigma(1), \ldots, \sigma(n))$ by performing a finite sequence of interchanges of pairs.

More precisely, a "*transposition*" is a permutation $\theta : \{1, \ldots, n\} \to \{1, \ldots, n\}$ for which there exist $i, j \in \{1, \ldots, n\}$ with $i \neq j$ such that $\theta(i) = j$, $\theta(j) = i$, and $\theta(k) = k$ for all $k \neq i, j$.

Every permutation σ can be written as a composition of transpositions:

$$\sigma = \theta_1 \circ \cdots \circ \theta_m \tag{20.1}$$

where $\theta_1, \ldots, \theta_m$ are transpositions. However, these θ_i's and the total number m are not unique. The same permutation σ can be written as $\sigma = \theta_1 \circ \cdots \circ \theta_m = \theta'_1 \circ \cdots \circ \theta'_{m'}$.

Remark 20.1: Why is the decomposition (20.1) always possible? Start with the sorted list $(1, \ldots, n)$. If $\sigma(1) \neq 1$, then interchange 1 and $\sigma(1)$. We now have $(\sigma(1), \ldots, 1, \ldots, n)$. Next, look at the second position. If it is not $\sigma(2)$, interchange it with $\sigma(2)$, and proceed similarly down the line. Of course, there are other ways to do it. For example, the permutation $(3, 2, 1)$ can be obtained in 3 steps:

$$(1, 2, 3) \to (2, 1, 3) \to (2, 3, 1) \to (3, 2, 1)$$

But, it can also be obtained in just one step: $(1, 2, 3) \to (3, 2, 1)$.

Lemma 20.b.1 (*Parity of Transpositions*): If $(\sigma(1), \ldots, \sigma(n))$ is obtained from $(1, \ldots, n)$ in two different ways—one time by a sequence of m transpositions, and another time by a sequence of m' transpositions—then m and m' must have the same parity. That is, m' is even if and only if m is even, and m' is odd if and only if m is odd.

In other words, $(-1)^{m'} = (-1)^m$. The parity depends only on the permutation σ, and not on the particular sequence of transpositions used.

A permutation σ is called "*even*" if m is even, and "*odd*" if m is odd. The number $\operatorname{sgn}(\sigma) := (-1)^m \in \{-1, 1\}$ is called the "*sign*" of σ.

Proof of Lemma:

We know from previous results (Corollary 20.a.9) that determinant functions $E : \mathcal{M}_{n\times n}(K) \to K$ exist for $n \geq 1$. Let us fix one such determinant function E (we denote it by E to distinguish it from D in the previous discussion).

Consider the evaluation $E(\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)})$. If $(\sigma(1), \ldots, \sigma(n))$ can be obtained from $(1, \ldots, n)$ by a sequence of m transpositions, then the matrix with rows $\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)}$ can be obtained from the identity matrix I by a sequence of m interchanges of pairs of rows.

Because E is alternating, swapping two rows flips the sign of the output. Therefore:

$$E(\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)}) = (-1)^m E(\epsilon_1, \ldots, \epsilon_n) = (-1)^m E(I) = (-1)^m.$$

But, this exact same logic applies to the sequence of m' transpositions. Thus:

$$E(\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)}) = (-1)^{m'} E(\epsilon_1, \ldots, \epsilon_n) = (-1)^{m'} E(I) = (-1)^{m'}.$$

This identically implies that $(-1)^m = (-1)^{m'}$, proving the lemma. **Q.E.D.**

Returning to our original function $D(A)$, we have seen that:

$$D(A) = \sum_{\sigma} A(1, \sigma(1)) \cdots A(n, \sigma(n)) \cdot D(\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)}).$$

By the alternating property of D, performing the transpositions necessary to sort $(\sigma(1), \ldots, \sigma(n))$ back to $(1, \ldots, n)$ yields:

$$D(\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)}) = \operatorname{sgn}(\sigma) \cdot D(\epsilon_1, \ldots, \epsilon_n) = \operatorname{sgn}(\sigma) \cdot D(I). \tag{20.2}$$

Theorem 20.b.2 (*Leibniz Formula & Uniqueness*): For all $n \geq 1$, there exists a unique determinant function on $\mathcal{M}_{n\times n}(K)$. We will denote it by $\det$. It is mathematically given by the formula:

$$\det(A) = \sum_{\sigma} \operatorname{sgn}(\sigma) \cdot A(1, \sigma(1)) \cdot A(2, \sigma(2)) \cdots A(n, \sigma(n)).$$

Proof:

This follows immediately from (20.2). Indeed, if D is a determinant function, then $D(I) = 1$. Substituting this into (20.2) yields the exact formula for $D(A)$ stated in the theorem, proving that the function is entirely unique. **Q.E.D.**

Corollary 20.b.3 (*Scaling of Alternating Functions*): For any n-linear alternating function $D : \mathcal{M}_{n\times n}(K) \to K$, we have $D(A) = \det(A) \cdot D(I)$ for all $A \in \mathcal{M}_{n\times n}(K)$.

The Symmetric Group

Denote by S_n the set of permutations $\sigma : \{1,\ldots,n\} \to \{1,\ldots,n\}$. The set S_n has the algebraic structure of a group under composition: if $\sigma,\tau \in S_n$, their composition is $(\sigma\circ\tau)(x) = \sigma(\tau(x))$. The identity function id is the neutral element. The inverse σ^{-1} exists because σ is bijective. Furthermore, the size of the symmetric group is $|S_n| = n!$.

Exercise 20.2 (*The Order of the Symmetric Group*): Show that $|S_n| = n!$.

Lemma 20.b.4 (*Multiplicativity of Signum*): For all $\sigma,\tau \in S_n$, we have $\operatorname{sgn}(\sigma\circ\tau) = \operatorname{sgn}(\sigma)\cdot\operatorname{sgn}(\tau)$.

Proof:

Suppose $\sigma = \theta_1 \circ \cdots \circ \theta_m$ and $\tau = \lambda_1 \circ \cdots \circ \lambda_\ell$, where the θ_i's and λ_j's are all transpositions. Then, their composition is simply $\sigma\circ\tau = \theta_1\circ\cdots\circ\theta_m\circ\lambda_1\circ\cdots\circ\lambda_\ell$, which explicitly consists of $m+\ell$ transpositions. Therefore:

$$\operatorname{sgn}(\sigma\circ\tau) = (-1)^{m+\ell} = (-1)^m \cdot (-1)^\ell = \operatorname{sgn}(\sigma)\cdot\operatorname{sgn}(\tau).$$

Q.E.D.

Multiplicativity of the Determinant

Theorem 20.b.5 (*Multiplicativity of the Determinant*): For any matrices $A, B \in M_{n\times n}(K)$, we have $\det(A\cdot B) = \det(A)\cdot\det(B)$.

Proof:

Fix a matrix $B \in \mathcal{M}_{n\times n}(K)$, and consider a function $D : \mathcal{M}_{n\times n}(K) \to K$ defined by $D(A) := \det(A\cdot B)$.

Claim: D is n-linear and alternating.

Proof of claim: Let the rows of A be $\alpha_1,\ldots,\alpha_n$. If we view α_i as a $1\times n$ matrix, then the matrix product $\alpha_i\cdot B$ is also $1\times n$. In fact, the rows of the product matrix $A\cdot B$ are exactly $\alpha_1\cdot B,\ldots,\alpha_n\cdot B$. So, $D(A) = \det(\alpha_1\cdot B,\ldots,\alpha_n\cdot B)$.

Clearly, for two row vectors α_i and α_i', we have $(\alpha_i+\alpha_i')\cdot B = \alpha_i\cdot B + \alpha_i'\cdot B$. Also, for any scalar $c \in K$, $(c\alpha_i)\cdot B = c(\alpha_i\cdot B)$. Since the determinant function det is n-linear with respect to its rows, it heavily follows that D is also n-linear.

Furthermore, if two rows of A are equal (i.e., $\alpha_i = \alpha_j$), then the corresponding rows of $A\cdot B$ are also equal ($\alpha_i\cdot B = \alpha_j\cdot B$). Because det is alternating, $D(A) = \det(\alpha_1\cdot B,\ldots,\alpha_n\cdot B) = 0$. This rigorously proves the claim that D is alternating.

We continue with the proof of Theorem 20.b.5. By Corollary 20.b.3, any n-linear alternating function satisfies $D(A) = \det(A)\cdot D(I)$. We compute $D(I)$ using our definition: $D(I) = \det(I\cdot B) = \det(B)$. Substituting this back yields $D(A) = \det(A)\cdot\det(B)$, completing the proof. Q.E.D.

Corollary 20.b.6 (*Determinant of an Inverse*): If A is invertible, then $\det(A) \neq 0$. Moreover, $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof:

Assume A is invertible, and denote its inverse by A^{-1}. Since $A \cdot A^{-1} = I$, taking the determinant of both sides and applying Theorem 20.b.5 yields $\det(A) \cdot \det(A^{-1}) = \det(I) = 1$. This automatically implies that $\det(A)$ cannot be zero, and dividing by $\det(A)$ yields the expected formula.

Q.E.D.

More on Determinants: Properties of Determinants and the Transpose (Section 20.c)

Recall that for any matrix $M \in M_{m\times n}(K)$, we have the transposed matrix $M^\mathsf{T} \in M_{n\times m}(K)$, where the entries are given by $M^\mathsf{T}(i,j) = M(j,i)$.

Theorem 20.c.1 (*Determinant of the Transpose*): For all $A \in \mathcal{M}_{n\times n}(K)$, we have $\det(A^\mathsf{T}) = \det(A)$.

Proof:

If σ is a permutation of degree n, then by definition of the transpose, $A^\mathsf{T}(i,\sigma(i)) = A(\sigma(i),i)$. Therefore, expanding the determinant definition gives:

$$\det(A^\mathsf{T}) = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \cdot A(\sigma(1),1) \cdots A(\sigma(n),n).$$

If we let $j = \sigma(i)$, then $i = \sigma^{-1}(j)$, and consequently $A(\sigma(i),i) = A(j,\sigma^{-1}(j))$. Hence, we can rearrange the product completely:

$$A(\sigma(1),1) \cdots A(\sigma(n),n) = A(1,\sigma^{-1}(1)) \cdots A(n,\sigma^{-1}(n)) \tag{20.3}$$

This holds because every factor on the left-hand side of (20.3) will appear exactly once on the right-hand side, and vice-versa.

Now, since $\sigma\circ\sigma^{-1} = \mathrm{id}$, we have $\operatorname{sgn}(\sigma)\cdot\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\mathrm{id}) = 1$, which implies $\operatorname{sgn}(\sigma) = \operatorname{sgn}(\sigma^{-1})$. When σ runs over S_n (all possible permutations of degree n), its inverse σ^{-1} also naturally runs over the entirety of S_n.

This implies:

$$\begin{aligned}
\det(A^\mathsf{T}) &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) A(\sigma(1),1) \cdots A(\sigma(n),n) \\
&= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma^{-1}) A(1,\sigma^{-1}(1)) \cdots A(n,\sigma^{-1}(n)) \\
&= \sum_{\tau\in S_n} \operatorname{sgn}(\tau) A(1,\tau(1)) \cdots A(n,\tau(n)) \\
&= \det(A).
\end{aligned}$$

Q.E.D.

Example 20.6 (*Determinant of a Transposed Matrix*): Let $A = \left(\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right)$. Its transpose is $A^\mathsf{T} = \left(\begin{smallmatrix}1&3\\2&4\end{smallmatrix}\right)$. We have $\det(A) = (1)(4) - (2)(3) = -2$, and $\det(A^\mathsf{T}) = (1)(4) - (3)(2) = -2$. Thus, the theorem holds.

Corollary 20.c.2 (*Column Alternating Property*): If $D : \mathcal{M}_{n\times n}(K) \to K$ is an alternating n-linear function of the rows of the matrix, then D is also an alternating n-linear function of the columns of the matrix. In fact, $D(A^\mathsf{T}) = D(A)$ for all $A \in \mathcal{M}_{n\times n}(K)$. In particular, this holds for $D = \det$.

Proof:

For any $A \in M_{n\times n}(K)$, we have previously established $D(A) = \det(A)\cdot D(I)$. Applying this to A^T gives $D(A^\mathsf{T}) = \det(A^\mathsf{T})\cdot D(I)$. Since $\det(A^\mathsf{T}) = \det(A)$, this evaluates exactly to $\det(A)\cdot D(I) = D(A)$. **Q.E.D.**

Because the determinant acts symmetrically on rows and columns, we can perform column operations just as we do row operations.

Theorem 20.c.3 (*Invariance Under Row Operations*): Let $A \in \mathcal{M}_{n\times n}(K)$.

(a) If B is obtained from A by adding a multiple of one row of A to another row (i.e., $c\cdot E_j + E_i \to E_i$ for $i \neq j$), or by doing the analogous operation on columns, then $\det(B) = \det(A)$.

(b) If B is obtained from A by interchanging two rows (i.e., $E_i \leftrightarrow E_j$ for $i \neq j$), or columns, then $\det(B) = -\det(A)$.

(c) If B is obtained from A by multiplying a row by a scalar (i.e., $c\cdot E_i \to E_i$), then $\det(B) = c\det(A)$.

Proof:

Statement **(b)** follows directly from the fact that the determinant is an alternating function. Statement **(c)** follows from the n-linearity of the determinant.

To prove **(a)**, let A have rows $\alpha_1,\dots,\alpha_n$. Assume $j<i$. Expanding by linearity on the i-th row gives:

$$\begin{aligned}\det(B) &= \det(\alpha_1,\dots,\alpha_i + c\alpha_j,\dots,\alpha_j,\dots,\alpha_n)\\ &= \det(\alpha_1,\dots,\alpha_i,\dots,\alpha_n) + c\det(\alpha_1,\dots,\alpha_j,\dots,\alpha_j,\dots,\alpha_n)\\ &= \det(A) + 0 = \det(A).\end{aligned}$$

The second term identically vanishes because two rows are equal (α_j appears twice). If $j>i$, the proof is similar. **Q.E.D.**

Determinants of Block Matrices

Calculating the determinant of a large matrix by expanding all permutations is computationally expensive. Block matrices provide a powerful shortcut. Consider a block matrix M of the form $M = \left(\begin{smallmatrix} A & B\\ 0 & C\end{smallmatrix}\right)$, where $A \in \mathcal{M}_{r\times r}(K)$, $C \in \mathcal{M}_{s\times s}(K)$, and $B \in \mathcal{M}_{r\times s}(K)$.

Proposition 20.c.4 (*Determinant of Block Matrices*): For a block matrix M defined as above, $\det(M) = \det(A)\cdot\det(C)$, where $A \in \mathcal{M}_{r\times r}(K)$, $C \in \mathcal{M}_{s\times s}(K)$, and $B \in \mathcal{M}_{r\times s}(K)$.

Proof:

Define a helper function $D(A,B,C) := \det\left(\begin{smallmatrix} A & B\\ 0 & C\end{smallmatrix}\right)$ as a function $D : \mathcal{M}_{r\times r}(K)\times\mathcal{M}_{r\times s}(K)\times\mathcal{M}_{s\times s}(K)\to K$.

If we fix the first two matrices A and B, then the function mapping $C \mapsto D(A,B,C)$ is s-linear and alternating. By Corollary 20.b.3, any such function satisfies:

$$D(A,B,C) = \det(C)\cdot D(A,B,I_s). \tag{20.4}$$

Let us calculate $D(A,B,I_s) = \det\left(\begin{smallmatrix} A & B\\ 0 & I_s\end{smallmatrix}\right)$. By doing row operations of the type $R_i + cR_j \to R_i$ (where $j\neq i$), we can use the last s rows of the identity matrix I_s to eliminate the entries in B entirely. This transforms the matrix into $\left(\begin{smallmatrix} A & 0\\ 0 & I_s\end{smallmatrix}\right)$.

Since these operations do not change the determinant, we have:

$$\det\begin{pmatrix} A & B\\ 0 & I_s\end{pmatrix} = \det\begin{pmatrix} A & 0\\ 0 & I_s\end{pmatrix}. \tag{20.5}$$

The function $A \mapsto \det\left(\begin{smallmatrix} A & 0 \\ 0 & I_s \end{smallmatrix}\right)$ is r-linear and alternating. By Corollary 20.b.3 again, it evaluates to $\det(A) \cdot \det\left(\begin{smallmatrix} I_r & 0 \\ 0 & I_s \end{smallmatrix}\right) = \det(A) \cdot \det(I_{r+s}) = \det(A) \cdot 1 = \det(A)$.

Going back through (20.5) and then to (20.4), we finally get that $D(A, B, C) = \det(A) \cdot \det(C)$.

Q.E.D.

Exercise 20.3 (*The Other Block Shape*): Show that $\det\left(\begin{smallmatrix} A & 0 \\ B & C \end{smallmatrix}\right) = \det(A) \cdot \det(C)$, where $A \in \mathcal{M}_{r\times r}(K)$, $C \in \mathcal{M}_{s\times s}(K)$, and $B \in \mathcal{M}_{s\times r}(K)$. (Hint: Use column operations to reduce B to zero, or consider the transpose of the matrix and apply Proposition 20.c.4.)

Example 20.7 (*Calculation of a 4×4 Determinant*): Let $A = \left(\begin{smallmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 1 & -1 & -1 \\ 1 & 2 & 3 & 0 \end{smallmatrix}\right) \in \mathcal{M}_{4\times 4}(\mathbb{R})$. We begin by performing row operations $R_2 - 2R_1 \to R_2$, $R_3 - 4R_1 \to R_3$, and $R_4 - R_1 \to R_4$. This transforms A into:

$$\begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix}$$

Now, we clean the second column below the diagonal using $R_3 - \frac{5}{4}R_2 \to R_3$ and $R_4 - \frac{3}{4}R_2 \to R_4$. This reveals a beautiful block structure:

$$\left(\begin{array}{cc|cc} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ \hline 0 & 0 & -4 & -8 \\ 0 & 0 & 4 & 0 \end{array}\right)$$

By applying Proposition 20.c.4 to this partitioned form, we can see that the determinant of the 4×4 matrix is simply the product of the determinants of the 2×2 diagonal blocks:

$$\det(A) = \det\begin{pmatrix} 1 & -1 \\ 0 & 4 \end{pmatrix} \cdot \det\begin{pmatrix} -4 & -8 \\ 4 & 0 \end{pmatrix}$$

Consequently, we evaluate the individual pieces:

$$\det(A) = (4) \cdot (32) = 128.$$

Cofactor Expansion and the Classical Adjoint

Recall that if D is a determinant function on $\mathcal{M}_{(n-1)\times(n-1)}(K)$, then for any fixed $j \in \{1, \dots, n\}$, the function $E_j(A) := \sum_{i=1}^{n} (-1)^{i+j} A_{ij} D(A(i|j))$ is a determinant function as well. But, we have seen that on $\mathcal{M}_{n\times n}(K)$ there is a strictly unique determinant function, $\det$.

Corollary 20.c.5 (*Laplace Expansion*): For all $A \in \mathcal{M}_{n\times n}(K)$, we have the following n different formulas for $\det(A)$ (expanding along column j):

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} A_{ij} \det A(i|j), \quad \text{for any } j = 1, \dots, n.$$

The scalars $C_{ij} := (-1)^{i+j} \det A(i|j)$ are called the "*i, j cofactor*" of A. We can succinctly write the expansion of $\det(A)$ by the cofactors of the j-th column as $\det(A) = \sum_{i=1}^{n} A_{ij} \cdot C_{ij}$.

Lemma (*Orthogonality of Cofactors*): If we replace A_{ij} in the formula by A_{ik} (where k is fixed), we always get 0. In other words, for all $A \in \mathcal{M}_{n\times n}(K)$ and for $j \neq k$, we have $\sum_{i=1}^{n} A_{ik} \cdot C_{ij} = 0$.

Proof:

Let $B \in \mathcal{M}_{n\times n}(K)$ be the matrix obtained from A by replacing (not exchanging!) column j of A by column k of A. So, in B, column j equals column k. Clearly, $\det(B) = 0$ since B has two equal columns.

Expanding $\det(B)$ along the j-th column gives:

$$0 = \det(B) = \sum_{i=1}^{n} (-1)^{i+j} B_{ij} \det B(i|j).$$

By construction, $B_{ij} = A_{ik}$ for all i. Furthermore, since we only modified column j, deleting column j from both matrices leaves identical submatrices: $B(i|j) = A(i|j)$ for all i. Thus:

$$0 = \sum_{i=1}^{n} (-1)^{i+j} A_{ik} \det A(i|j) = \sum_{i=1}^{n} A_{ik} \cdot C_{ij}.$$

Q.E.D.

Summary 20.1: For $A \in \mathcal{M}_{n\times n}(K)$, we have $\sum_{i=1}^{n} A_{ik} \cdot C_{ij} = \delta_{jk} \cdot \det(A)$, where δ_{jk} is the Kronecker delta (which is 1 if $j = k$, and 0 otherwise).

Definition (*Classical Adjoint*): The transpose of the matrix of cofactors, $(C_{ij})^{\mathsf{T}}$, is called the "*classical adjoint*" (or "*adjugate*") of A, denoted $\operatorname{adj} A \in \mathcal{M}_{n\times n}(K)$. Explicitly, $(\operatorname{adj} A)_{ij} = C_{ji} = (-1)^{i+j} \det A(j|i)$.

Our summary formula above immediately implies the matrix equation:

$$(\operatorname{adj} A) \cdot A = (\det A) \cdot I \tag{20.6}$$

Lemma (*Left Adjugate Identity*): It is also true that $A \cdot (\operatorname{adj} A) = (\det A) \cdot I$.

Proof:

Since $A^{\mathsf{T}}(i|j) = (A(j|i))^{\mathsf{T}}$, we have

$$(\operatorname{adj} A^{\mathsf{T}})_{ij} = (-1)^{i+j} \det A^{\mathsf{T}}(j|i) = (-1)^{i+j} \det(A(i|j)^{\mathsf{T}}) = (-1)^{i+j} \det A(i|j).$$

This perfectly matches $(\operatorname{adj} A)_{ji}$, which is exactly $((\operatorname{adj} A)^{\mathsf{T}})_{ij}$. Thus, $\operatorname{adj} A^{\mathsf{T}} = (\operatorname{adj} A)^{\mathsf{T}}$. Applying (20.6) to the matrix A^{T} gives $(\operatorname{adj} A^{\mathsf{T}}) \cdot A^{\mathsf{T}} = (\det A^{\mathsf{T}}) \cdot I$. Using our transpose identities, this becomes $(\operatorname{adj} A)^{\mathsf{T}} \cdot A^{\mathsf{T}} = (\det A) \cdot I$. Taking the transpose of both sides finally yields $A \cdot (\operatorname{adj} A) = (\det A) \cdot I$. **Q.E.D.**

Theorem 20.c.6 (*Inverse via Adjugate*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then, A is invertible if and only if $\det A \neq 0$. When A is invertible, its inverse is given by $A^{-1} = \frac{1}{\det A} \cdot (\operatorname{adj} A)$.

Proof:

One direction is Corollary 20.b.6: if A is invertible then $\det A \neq 0$.

Conversely, assume $\det A \neq 0$. Then the scalar $\frac{1}{\det A} \in K$ exists, and the two adjugate identities (20.6) and Page 165 give

$$A \cdot \left(\frac{1}{\det A} \operatorname{adj} A\right) = \frac{1}{\det A}(A \cdot \operatorname{adj} A) = \frac{1}{\det A}(\det A) \cdot I = I,$$

and in the same way $\left(\frac{1}{\det A} \operatorname{adj} A\right) \cdot A = I$. So $\frac{1}{\det A} \operatorname{adj} A$ is a two-sided inverse of A, which makes A invertible with $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$. **Q.E.D.**

Cramer's Rule

Let $A \in \mathcal{M}_{n\times n}(K)$. We would like an explicit formula to solve the system of linear equations $A \cdot x = b$, where $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ are the unknowns and $b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in K^n$ is a given vector.

Assume $A \cdot x = b$. Multiplying both sides by the adjoint gives $(\operatorname{adj} A) \cdot A \cdot x = (\operatorname{adj} A) \cdot b$. This directly implies $(\det A) \cdot x = (\operatorname{adj} A) \cdot b$. Looking at the j-th row of this vector equation, we get:

$$\begin{aligned} (\det A) \cdot x_j &= \sum_{i=1}^{n} (\operatorname{adj} A)_{ji} \cdot b_i \\ &= \sum_{i=1}^{n} (-1)^{i+j} b_i \det A(i|j) \\ &= \det(A(j,b)) \end{aligned}$$

where $A(j,b)$ is the matrix obtained from A by replacing column j with the vector b.

Theorem 20.c.7 (*Cramer's Rule*): Let $A \in \mathcal{M}_{n\times n}(K)$ with $\det A \neq 0$. Let $b = (b_1, \ldots, b_n)^\mathsf{T} \in K^n$. Then, the system $Ax = b$, where $x = (x_1, \ldots, x_n)^\mathsf{T}$, has a unique solution, and it is given by the formula:

$$x_j = \frac{\det A(j,b)}{\det A} \quad \text{for all } 1 \leq j \leq n,$$

where $A(j,b)$ is the matrix obtained from A by replacing column j with b.

Proof of Theorem 20.c.7:

We have seen previously in Linear Algebra I that if A is invertible, then $Ax = b$ has a unique solution. We have just shown above that $(\det A) \cdot x_j = \det A(j,b)$. Since $\det A \neq 0$, we can safely divide to obtain $x_j = \frac{\det A(j,b)}{\det A}$. **Q.E.D.**

Example 20.8 (*Solving a System using Cramer's Rule*): Consider the system $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right)\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 5 \\ 5 \end{smallmatrix}\right)$. We have $\det(A) = 2(3) - 1(1) = 5$. Using Cramer's rule:

$$x_1 = \frac{\det \begin{pmatrix} 5 & 1 \\ 5 & 3 \end{pmatrix}}{5} = \frac{15-5}{5} = 2, \quad x_2 = \frac{\det \begin{pmatrix} 2 & 5 \\ 1 & 5 \end{pmatrix}}{5} = \frac{10-5}{5} = 1.$$

Thus, the solution is $x = \binom{2}{1}$.

Determinants of Endomorphisms

Warm up 20.1: Let $A \in \mathcal{M}_{n\times n}(K)$, and let $P \in \mathrm{GL}(n, K)$ (meaning P is an invertible $n \times n$ matrix). Then, by multiplicativity (Theorem 20.b.5):

$$\det(P^{-1} \cdot A \cdot P) = (\det P^{-1}) \cdot (\det A) \cdot (\det P) = \frac{1}{\det P} \cdot \det A \cdot \det P = \det A.$$

In other words, similar matrices mathematically have the exact same determinant.

Let V be a finite-dimensional vector space over K, and put $n := \dim V$. Let $T : V \to V$ be a linear map (an endomorphism). Pick a basis $\mathcal{B}$ for V. We have a matrix representation $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathcal{M}_{n\times n}(K)$, and we can consider its determinant $\det[T]_{\mathcal{B}}^{\mathcal{B}} \in K$.

Question 20.1: Does $\det[T]_{\mathcal{B}}^{\mathcal{B}}$ depend on the specific choice of the basis $\mathcal{B}$?

Answer 20.1: Let $\mathcal{B}'$ be another arbitrary basis for V. Denote by $P := [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$ the transition matrix between the basis $\mathcal{B}'$ and $\mathcal{B}$. Recall that $[T]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$. We also formally have $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = ([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1} = P^{-1}$. So, $[T]_{\mathcal{B}'}^{\mathcal{B}'} = P^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot P$. This implies $\det[T]_{\mathcal{B}'}^{\mathcal{B}'} = \det[T]_{\mathcal{B}}^{\mathcal{B}}$. Recall also that if $T, S : V \to V$ are two linear maps, then $[S \circ T]_{\mathcal{B}}^{\mathcal{B}} = [S]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}}$. If T is an isomorphism, then $[T^{-1}]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^{-1}$. Finally, $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I$.

Corollary 20.c.8 (*Determinant of an Endomorphism*): Let V be a finite-dimensional vector space over K, and let $T : V \to V$ be a linear map. Then, $\det([T]_{\mathcal{B}}^{\mathcal{B}})$ does NOT depend on the choice of the basis $\mathcal{B}$ of V. We denote this intrinsic scalar simply by $\det(T) \in K$.

The function $\det : \mathrm{End}(V) \to K, T \mapsto \det(T)$ has the following elegant properties:

(a) $\det(S \circ T) = \det(S) \cdot \det(T)$ for all $S, T \in \mathrm{End}(V)$.

(b) T is an isomorphism if and only if $\det T \neq 0$. If T is an isomorphism, then $\det(T^{-1}) = \frac{1}{\det T}$.

(c) $\det(\mathrm{id}_V) = 1$.

(d) $\det(0) = 0 \in K$ (where 0 is the zero linear map).

Important remark 20.1: In contrast to an endomorphism $T : V \to V$, if we take a linear map $T : V \to W$ between two vector spaces of the same dimension and bases $\mathcal{B}, \mathcal{C}$ of V, W, then $\det[T]_{\mathcal{C}}^{\mathcal{B}}$ generally depends on the choice of $\mathcal{B}$ and $\mathcal{C}$.

Solutions to the Exercises

Proof of Exercise 20.1:

Write $u := \binom{a}{c}$ and $v := \binom{b}{d}$ for the two columns. By Proposition 9.7, the list (u, v) is linearly dependent exactly when one of the two vectors is a linear combination of the other. Explicitly: dependence gives scalars $\mu, \nu \in K$, not both zero, with $\mu u + \nu v = 0$. If $\mu \neq 0$ then $u = -\frac{\nu}{\mu} v$, which is the first alternative with $\lambda := -\frac{\nu}{\mu}$; and if $\mu = 0$ then $\nu \neq 0$, so $v = 0 = 0 \cdot u$, which is the second alternative with $\tau := 0$. Conversely, either alternative exhibits a non-trivial vanishing combination, so the columns are dependent. Note that the case $u = 0$ is contained in the first alternative, taking $\lambda := 0$. **Q.E.D.**

Proof of Exercise 20.2:

A permutation $\sigma \in S_n$ is determined by the list $(\sigma(1), \dots, \sigma(n))$, which is an arrangement of $1, \dots, n$ with no repetitions. Building such a list, there are n choices for $\sigma(1)$; once it is fixed, $\sigma(2)$ may be any of the remaining $n-1$ values; then $n-2$ for $\sigma(3)$, and so on, with a single choice left for $\sigma(n)$. Distinct sequences of choices give distinct permutations and every permutation arises exactly once, so

$$|S_n| = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1 = n!.$$

Q.E.D.

Proof of Exercise 20.3:

Take the transpose. By Theorem 20.c.1, $\det\left(\begin{smallmatrix} A & 0 \\ B & C \end{smallmatrix}\right) = \det\left(\left(\begin{smallmatrix} A & 0 \\ B & C \end{smallmatrix}\right)^{\mathsf{T}}\right)$, and transposing a block matrix transposes each block and reflects the arrangement, giving

$$\left(\begin{smallmatrix} A & 0 \\ B & C \end{smallmatrix}\right)^{\mathsf{T}} = \begin{pmatrix} A^{\mathsf{T}} & B^{\mathsf{T}} \\ 0 & C^{\mathsf{T}} \end{pmatrix},$$

where now $A^{\mathsf{T}} \in \mathcal{M}_{r \times r}(K)$, $C^{\mathsf{T}} \in \mathcal{M}_{s \times s}(K)$ and $B^{\mathsf{T}} \in \mathcal{M}_{r \times s}(K)$. This is exactly the shape of Proposition 20.c.4, so

$$\det \begin{pmatrix} A & 0 \\ B & C \end{pmatrix} = \det \begin{pmatrix} A^{\mathsf{T}} & B^{\mathsf{T}} \\ 0 & C^{\mathsf{T}} \end{pmatrix} = \det(A^{\mathsf{T}}) \cdot \det(C^{\mathsf{T}}) = \det(A) \cdot \det(C),$$

using Theorem 20.c.1 once more on each factor. **Q.E.D.**

AI-Note: This chapter came through in good order. All of 20.a.1–20.a.9, 20.b.1–20.b.6 and 20.c.1–20.c.8 are present, in the notes' order, with the notes' proofs, and the printed numbers already matched — including the deliberate use of unnumbered `lemma*` and `definition*` around the cofactor identity and the classical adjoint, which is what keeps 20.c.6 landing on the theorem about the inverse.

Three repairs. Theorem 20.c.6 was stated without any proof; it is the point the whole cofactor discussion has been building to, and the two lines it needs are supplied. The exercise inside the proof of Proposition 20.a.1 stated the dichotomy as *"either the first column is the zero vector or the second column is a scalar multiple of the first"*, while the proof two lines below it uses the notes' version — either column is a multiple of the other — and the zero case is subsumed there; the statement now matches what is used. And a sentence in the proof of Page 165 was left behind a second time as a commented-out copy.

Prof. Biran marks three things *"exc."*: the column dichotomy just mentioned, $|S_n| = n!$, and the lower-triangular block shape. Two of them stood in the text as `remark[Exercise]` and one as a parenthesis; all three are now exercises, solved above.

Eigenvalues & Eigenvectors (Chapter 21)

Motivation (Section 21.a)

Fibonacci revisited

Consider the sequence $f_0, f_1, f_2, f_3, \dots, f_n, \dots$ defined by $f_0 = 0$, $f_1 = 1$, and $f_n := f_{n-1} + f_{n-2}$ for $n \geq 2$. This yields the sequence: $0, 1, 1, 2, 3, 5, 8, 13, 21, \dots$

We found a formula for f_n:

$$f_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right) \quad \forall n \geq 0.$$

We defined $\mathrm{Fib} :=$ the set of all sequences $(a_0, a_1, a_2, \dots, a_n, \dots)$ that satisfy $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$.

Claim: $\mathrm{Fib} \subseteq \mathbb{R}^\infty$ is a 2-dimensional vector subspace of the space of all infinite sequences

$$(x_0, x_1, x_2, \dots, x_n, \dots).$$

In fact, the map $F : \mathrm{Fib} \longrightarrow \mathbb{R}^2$ given by $F(a_0, a_1, \dots, a_n, \dots) := (a_0, a_1) \in \mathbb{R}^2$ is an isomorphism.

Claim: Put $\varphi := \frac{1+\sqrt{5}}{2}$ and $\psi := \frac{1-\sqrt{5}}{2}$. Then the two geometric sequences $(1, \varphi, \varphi^2, \dots, \varphi^n, \dots)$ and $(1, \psi, \psi^2, \dots, \psi^n, \dots)$ belong to Fib and, moreover, they form a basis for Fib.

We also saw that using these two sequences, we could easily find an explicit formula for all Fibonacci sequences. But how did we find these two special sequences?

The shift map $S : \mathrm{Fib} \longrightarrow \mathrm{Fib}$ defined by $S(a_0, a_1, \dots, a_n, \dots) := (a_1, a_2, \dots, a_{n+1}, \dots)$ is a linear map. Let us try to find a Fibonacci sequence $\mathcal{F} \neq 0$ such that $S(\mathcal{F}) = \lambda \cdot \mathcal{F}$ for some λ.

$$S(\mathcal{F}) = \lambda \cdot \mathcal{F} \iff \lambda = \frac{1 \pm \sqrt{5}}{2} \text{ and } \mathcal{F} = (a, a\lambda, a\lambda^2, \dots, a\lambda^n, \dots)$$

Definition 21.a.1 (*Eigenvalue and Eigenvector*): Let V be a vector space over K and $T : V \to V$ a linear map (an endomorphism of V). We say that $\lambda \in K$ is an "*eigenvalue*" (a.k.a. "*characteristic value*") of T if there exists a vector $v \neq 0$ such that $Tv = \lambda v$. Any vector $v \neq 0$ with the property that $Tv = \lambda v$ is called an "*eigenvector*" (of T) for the eigenvalue λ.

Definition 21.a.2 (*Diagonalizable Endomorphism*): Let V be a finite-dimensional vector space over K, put $n := \dim V$, and let $T \in \mathrm{End}(V)$. We say that T is "*diagonalizable*" if there exists a basis $\mathcal{B} := (v_1, \dots, v_n)$ for V such that the representation matrix of T with respect to $\mathcal{B}$ is diagonal, that is,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix} \tag{21.1}$$

for some $\lambda_1, \dots, \lambda_n \in K$.

Lemma 21.a.3 (*Diagonalizable Matrix Criterion*): T is diagonalizable if and only if there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ consisting of eigenvectors of T (i.e., $\forall i$, $Tv_i = \lambda_i \cdot v_i$ for some $\lambda_i \in K$). Moreover, if for some basis $\mathcal{B} = (v_1, \dots, v_n)$ the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the shape (21.1), then for all i, λ_i is an eigenvalue of T and v_i is an eigenvector for λ_i.

Proof:

Recall that $[T]_{\mathcal{B}}^{\mathcal{B}} = ([Tv_1]_{\mathcal{B}} \dots [Tv_n]_{\mathcal{B}})$ and $[Tv_j]_{\mathcal{B}} = \begin{pmatrix} \alpha_{1j} \\ \vdots \\ \alpha_{nj} \end{pmatrix}$, where $Tv_j = \alpha_{1j}v_1 + \cdots + \alpha_{nj}v_n$.

The matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the shape (21.1) if and only if $\alpha_{ij} = 0$ for all $i \neq j$, and $\alpha_{kk} = \lambda_k \iff Tv_k = \lambda_k \cdot v_k$. **Q.E.D.**

Remark 21.1: We saw that for all $A \in \mathcal{M}_{n\times n}(K)$, we have a linear map $T_A : K^n \to K^n$, $T_A(v) := A \cdot v$. We say that a matrix $A \in \mathcal{M}_{n\times n}(K)$ is diagonalizable if the linear map T_A is diagonalizable.

Example 21.1 (*Stretching Transformation in $\mathbb{R}^2$*): Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $K = \mathbb{R}$. $T_A\binom{1}{1} = \binom{3}{3} = 3\binom{1}{1}$. Thus, $\lambda = 3$ is an eigenvalue and $v = \binom{1}{1}$ is a corresponding eigenvector. Furthermore, $T_A\binom{1}{-1} = \binom{1}{-1}$. This implies that $\lambda = 1$ is an eigenvalue and $v = \binom{1}{-1}$ is a corresponding eigenvector. Geometrically, this represents a stretching by a factor of 3 in $\mathbb{R}^2$ along one axis and the identity map along the other axis.

Example 21.2 (*Finding Eigenvalues via Quadratic Equations*): Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Let us find all possible eigenvalues and eigenvectors. We are looking for $A \cdot v = \lambda v$ with $\lambda \in \mathbb{R}$ and $0 \neq v \in \mathbb{R}^2$. This gives $(A - \lambda I)v = 0$. This implies that $A - \lambda I$ is not invertible, which in turn means $\det(A - \lambda I) = 0$ (which leads to a quadratic equation).

Proposition 21.a.4 (*Linear Independence of Eigenvectors*): Let V be a vector space over K and $T : V \to V$ a linear map. Suppose that $\lambda_1, \dots, \lambda_n \in K$ are eigenvalues of T which are pairwise distinct (i.e., $\lambda_i \neq \lambda_j$ for all $i \neq j$). Let $v_1, \dots, v_n$ be eigenvectors for $\lambda_1, \dots, \lambda_n$ respectively. Then $v_1, \dots, v_n$ are linearly independent. (Put shortly: eigenvectors corresponding to a list of distinct eigenvalues are always linearly independent).

Proof:

We proceed by induction on n. Base case $n = 1$: v_1 is an eigenvector for λ_1. By Definition 21.a.1, $v_1 \neq 0$, which implies that v_1 is linearly independent.

Inductive step for $n \geq 1$: Suppose the proposition holds for any list of n eigenvalues and eigenvectors. We will prove now that the proposition holds also for a list of $n+1$ eigenvalues and eigenvectors. Let $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$ be pairwise distinct eigenvalues of T, and $v_1, \dots, v_n, v_{n+1}$ a given list of eigenvectors of $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$ respectively. We need to prove that $v_1, \dots, v_n, v_{n+1}$ are linearly independent. Assume that

$$a_1v_1 + \cdots + a_nv_n + a_{n+1}v_{n+1} = 0. \tag{21.2}$$

We need to show that $a_1 = \cdots = a_n = a_{n+1} = 0$. Apply the transformation T to (21.2):

$$a_1T(v_1) + \cdots + a_nT(v_n) + a_{n+1}T(v_{n+1}) = 0.$$

But since $T(v_i) = \lambda_i v_i$, we have:

$$a_1\lambda_1v_1 + \cdots + a_n\lambda_nv_n + a_{n+1}\lambda_{n+1}v_{n+1} = 0. \tag{21.3}$$

Consider now the combination $(21.3) - \lambda_{n+1} \cdot (21.2)$:

$$a_1(\lambda_1 - \lambda_{n+1})v_1 + \cdots + a_n(\lambda_n - \lambda_{n+1})v_n + a_{n+1}(\lambda_{n+1} - \lambda_{n+1})v_{n+1} = 0.$$

By the induction hypothesis, $v_1, \dots, v_n$ are linearly independent. Therefore, $a_1(\lambda_1 - \lambda_{n+1}) = \cdots = a_n(\lambda_n - \lambda_{n+1}) = 0$. By assumption, the eigenvalues are distinct, so $\lambda_i \neq \lambda_{n+1}$. Consequently, $a_1 = \cdots = a_n = 0$. Going back to (21.2), we get $a_{n+1}v_{n+1} = 0$. But since $v_{n+1} \neq 0$ (because v_{n+1} is an eigenvector), this implies that $a_{n+1} = 0$. **Q.E.D.**

The Characteristic Polynomial

Proposition 21.a.5 (*Eigenvalue Injective Criterion*): Let V be a vector space over K and $T \in \operatorname{End}(V)$. Then λ is an eigenvalue of T if and only if the linear map $T - \lambda \cdot \mathrm{id}_V$ is not injective.

Proof:

The scalar λ is an eigenvalue if and only if there exists a non-zero vector $v \in V$ such that $Tv - \lambda v = 0$. This is equivalent to stating that $(T - \lambda \cdot \mathrm{id}_V)v = 0$, which implies that $\ker(T - \lambda \cdot \mathrm{id}_V) \neq \{0\}$. This implies that $T - \lambda \cdot \mathrm{id}_V$ is not injective. **Q.E.D.**

Definition 21.a.6 (*Characteristic Polynomial*): Let V be a finite-dimensional vector space over K and $T \in \operatorname{End}(V)$. We define the "*characteristic polynomial*" of T as $P_T(x) := \det(T - x \cdot \mathrm{id}_V)$, where x is a formal variable. We define the characteristic polynomial of a matrix $A \in \mathcal{M}_{n\times n}(K)$ in the same way: $P_A(x) := \det(A - x \cdot I_n)$.

Definition 21.a.7 (*Eigenspace*): Let V be a vector space over K and $T \in \operatorname{End}(V)$. Let λ be an eigenvalue of T. The "*eigenspace*" corresponding to λ is defined as:

$$\operatorname{Eig}_T(\lambda) := \{v \in V \mid Tv = \lambda v\} = \ker(T - \lambda \cdot \mathrm{id}_V) = \{v \in V \mid v \text{ is an eigenvector of } \lambda\} \cup \{0\}.$$

Clearly (by this definition), $\operatorname{Eig}_T(\lambda) \subseteq V$ is a linear subspace.

Lemma 21.a.8 (*Properties of the Characteristic Polynomial*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then:

(a) $P_A(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$ (i.e., $P_A(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + \cdots + c_0$, with $c_i \in K$ for all i, and $c_n = (-1)^n$).

(b) $P_{A^\intercal}(x) = P_A(x)$.

(c) If $A = \begin{pmatrix} \lambda_1 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}$ has upper triangular form, then $P_A(x) = (\lambda_1 - x) \cdot \dots \cdot (\lambda_n - x)$.

(d) If $A = \left(\begin{array}{c|c} B_1 & * \\ \hline 0 & B_2 \end{array}\right)$ is a block matrix, then $P_A(x) = P_{B_1}(x) \cdot P_{B_2}(x)$.

(e) If A and B are similar matrices (i.e., $B = P^{-1}AP$ for some invertible matrix P), then $P_A(x) = P_B(x)$.

Proof:

(a) Let

$$P_A(x) = \det(A - x \cdot I_n) = \det \begin{pmatrix} A_{11} - x & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} - x & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} - x \end{pmatrix}.$$

Denote this matrix (inside the determinant) by B. This determinant equals

$$\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot B_{1,\sigma 1} \dots B_{n,\sigma n} = (A_{11} - x) \cdot (A_{22} - x) \dots (A_{nn} - x)$$
$$+ (\text{terms from the permutations } \sigma \neq \mathrm{id}).$$

If $\sigma \neq \mathrm{id} = \left(\begin{smallmatrix} 1 & 2 & \cdots & n \\ 1 & 2 & \cdots & n \end{smallmatrix}\right)$, then at least one factor in the product $B_{1,\sigma 1} \dots B_{n,\sigma n}$ does not contain x, so each such term has degree $\leq n-1$. Consequently, the whole determinant equals

$$(-1)^n x^n + (\text{terms of degree} \leq n-1).$$

(b) $P_{A^\intercal}(x) = \det(A^\intercal - x \cdot I_n) = \det((A - x \cdot I_n)^\intercal) = \det(A - x \cdot I_n) = P_A(x)$.

(c)+(d) These properties follow directly from the definition of the determinant and its behavior with block matrices, as established in Chapter 20 (Determinants). For example, for (c), the determinant of an upper triangular matrix is the product of its diagonal entries. For (d), the determinant of a block triangular matrix is the product of the determinants of its diagonal blocks.

(e) $P_B(x) = \det(B - x \cdot I_n) = \det(P^{-1}AP - x \cdot I_n) = \det(P^{-1}(A - x \cdot I_n)P) = \det(A - x \cdot I_n) = P_A(x)$.

Q.E.D.

Remark 21.2: If V is an n-dimensional vector space over K and $T \in \mathrm{End}(V)$, then $P_T(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$.

Proof:

Let $\mathcal{B}$ be a basis for V. Then $[T - x \cdot \mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}} = [T]^{\mathcal{B}}_{\mathcal{B}} - x \cdot I_n$. This implies that $P_T(x) = \det([T - x \cdot \mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}}) = \det([T]^{\mathcal{B}}_{\mathcal{B}} - x \cdot I_n) = P_{[T]^{\mathcal{B}}_{\mathcal{B}}}(x)$. The claim of the remark now follows from **(a)** of Lemma 21.a.8.

Q.E.D.

Definition 21.a.9 (*Trace*): Let $A \in \mathcal{M}_{n \times n}(K)$. Define the "*trace*" (in German: „*Spur*") of A, denoted $\mathrm{Tr}(A) := \sum_{i=1}^{n} A_{ii} \in K$ (which is the sum of the entries along the diagonal of A). If V is a finite-dimensional vector space over K and $T \in \mathrm{End}(V)$, define $\mathrm{Tr}(T) := \mathrm{Tr}([T]^{\mathcal{B}}_{\mathcal{B}})$, where $\mathcal{B}$ is some basis for V. We will see soon that $\mathrm{Tr}(T)$ is well defined (i.e., it does not depend on the choice of $\mathcal{B}$).

Lemma 21.a.10 (*Properties of Trace*): For all $A, B \in \mathcal{M}_{n \times n}(K)$, we have:

- **(a)** $\mathrm{Tr}(A \cdot B) = \mathrm{Tr}(B \cdot A)$.
- **(b)** If A and B are similar matrices, then $\mathrm{Tr}(A) = \mathrm{Tr}(B)$.

In particular, if $\mathcal{B}$ and $\mathcal{B}'$ are two bases for V, then $\mathrm{Tr}([T]^{\mathcal{B}}_{\mathcal{B}}) = \mathrm{Tr}([T]^{\mathcal{B}'}_{\mathcal{B}'})$. Thus, $\mathrm{Tr}(T)$ is well defined.

Proof:

(a) $\mathrm{Tr}(A \cdot B) = \sum_{i=1}^{n} (A \cdot B)_{ii} = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} \cdot B_{ji} = \sum_{i,j} A_{ij} B_{ji} = \sum_{i,j} A_{ji} B_{ij} = \sum_{i=1}^{n} \sum_{j=1}^{n} B_{ij} A_{ji} = \sum_{i=1}^{n} (B \cdot A)_{ii} = \mathrm{Tr}(B \cdot A)$.

(b) If $B = P^{-1}AP$, then $\mathrm{Tr}(B) = \mathrm{Tr}(P^{-1}AP) = \mathrm{Tr}(P^{-1} \cdot (AP)) = \mathrm{Tr}(AP \cdot P^{-1}) = \mathrm{Tr}(A)$.

Regarding the claim about $\mathrm{Tr}(T)$, recall that $[T]^{\mathcal{B}'}_{\mathcal{B}'} = [\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}'} \cdot [T]^{\mathcal{B}}_{\mathcal{B}} \cdot [\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}$, and $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}'} = ([\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}})^{-1}$. Use **(b)** to conclude the proof.

Q.E.D.

Why is the trace interesting? Let $A \in \mathcal{M}_{n \times n}(K)$ and $P_A(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0$ its characteristic polynomial. We have seen that $c_n = (-1)^n$.

Lemma 21.a.11 (*Coefficients of the Characteristic Polynomial*): $c_{n-1} = (-1)^{n-1} \cdot \mathrm{Tr}(A)$ and $c_0 = \det(A)$. In other words, $P_A(x) = (-1)^n x^n + (-1)^{n-1} \mathrm{Tr}(A) \cdot x^{n-1} + c_{n-2} x^{n-2} + \cdots + c_1 x + \det(A)$.

Proof:

$$P_A(x) = \det \begin{pmatrix} A_{11} - x & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} - x & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} - x \end{pmatrix}.$$ Denote this matrix by B.

$$P_A(x) = (A_{11} - x) \cdot (A_{22} - x) \dots (A_{nn} - x) + \sum_{\sigma \in S_n, \sigma \neq \mathrm{id}} \mathrm{sgn}(\sigma) B_{1,\sigma 1} \dots B_{n,\sigma n}. \tag{21.4}$$

If $\sigma \neq \text{id}$, then there exist at least two indices $i \neq j$ such that $i \neq \sigma i$ and $j \neq \sigma j$. Thus, the product contains at least two factors ($B_{i,\sigma i}$ and $B_{j,\sigma j}$) that are not from the diagonal of B, and so do not contain x. Therefore, the degree of the sum in (21.4) is $\leq n-2$. $P_A(x) = (A_{11}-x)(A_{22}-x)\dots(A_{nn}-x) + (\text{another polynomial of degree} \leq n-2)$. $= (-1)^n x^n + \left[A_{11}(-x)^{n-1} + \dots + A_{nn}(-x)^{n-1}\right] + \dots$ Expanding the product $(A_{11}-x)(A_{22}-x)\dots(A_{nn}-x)$ yields $(-1)^n x^n + (-1)^{n-1}(A_{11}+\dots+A_{nn})x^{n-1} + \dots$. $= (-1)^n x^n + (-1)^{n-1}\cdot \operatorname{Tr}(A)x^{n-1} + \dots + c_1 x + c_0$.

Finally, $c_0 = P_A(0) = \det(A - 0 \cdot I_n) = \det(A)$ (obtained by substituting $x=0$). **Q.E.D.**

Definition 21.a.12 (*Direct Sum*): Let V be a vector space over K and $U_1,\dots,U_r \subseteq V$ subspaces. Denote $W := U_1 + \dots + U_r \subseteq V$. We say that W is a "*direct sum*" of $U_1,\dots,U_r$ (or that $U_1+\dots+U_r$ is a direct sum) if every $w \in W$ has a unique representation as $w = u_1+\dots+u_r$, with $u_i \in U_i$ for all i.

Exercise 21.1 (*Equivalent Conditions for Direct Sums*): $U_1+\dots+U_r$ is a direct sum if and only if any of the following conditions hold:

(a) The only way to write $0 = u_1+\dots+u_r$, with $u_i \in U_i$ for all i, is with $u_1 = \dots = u_r = 0$.

(b) For all $2 \leq j \leq r$, we have $U_j \cap (U_1+\dots+U_{j-1}) = \{0\}$.

Under the additional assumption that $U_1,\dots,U_r$ are all finite-dimensional, $U_1+\dots+U_r$ is a direct sum if and only if whenever $\mathcal{B}_1,\dots,\mathcal{B}_r$ are bases for $U_1,\dots,U_r$ respectively, then $(\mathcal{B}_1,\dots,\mathcal{B}_r)$ forms a basis for $U_1+\dots+U_r$.

Exercise 21.2 (*Isomorphism between External and Internal Direct Sums*): If $U_1+\dots+U_r$ is a direct sum, then there exists a canonical isomorphism $U_1 \oplus \dots \oplus U_r \xrightarrow{\sim} U_1+\dots+U_r$, given by $(u_1,\dots,u_r) \mapsto u_1+\dots+u_r$. So when $U_1+\dots+U_r$ is a direct sum, we can identify $U_1+\dots+U_r$ with $U_1\oplus\dots\oplus U_r$.

Proposition 21.a.13 (*Direct Sum of Eigenspaces*): Let V be a vector space over K and $T \in \operatorname{End}(V)$. Let $\lambda_1,\dots,\lambda_r \in K$ be pairwise distinct eigenvalues of T. Then the sum $\operatorname{Eig}_T(\lambda_1)+\dots+\operatorname{Eig}_T(\lambda_r)$ is a direct sum.

Proof:

Assume $u_1+\dots+u_r = u_1'+\dots+u_r'$, where $u_i, u_i' \in \operatorname{Eig}_T(\lambda_i)$ for all $1 \leq i \leq r$. We need to show that $u_i = u_i'$ for all $i \iff (u_i - u_i' = 0 \text{ for all } i)$. We have:

$$(u_1 - u_1') + \dots + (u_r - u_r') = 0. \tag{21.5}$$

Assume by contradiction that there exists an i such that $u_i - u_i' \neq 0$. Remove from the list $u_1-u_1',\dots,u_r-u_r'$ all those vectors that are 0. We get a non-empty list $u_{i_1}-u'_{i_1},\dots,u_{i_\ell}-u'_{i_\ell}$ of non-zero vectors, and by (21.5), we still have

$$(u_{i_1}-u'_{i_1}) + \dots + (u_{i_\ell}-u'_{i_\ell}) = 0. \tag{21.6}$$

where $u_{i_k}-u'_{i_k} \in \operatorname{Eig}_T(\lambda_{i_k})$ Now for all k, $u_{i_k}-u'_{i_k}$ is either an eigenvector of λ_{i_k} or 0, but by our assumptions, $u_{i_k}-u'_{i_k} \neq 0$. Recall from Proposition 21.a.4 that eigenvectors corresponding to pairwise distinct eigenvalues are linearly independent. $u_{i_1}-u'_{i_1},\dots,u_{i_\ell}-u'_{i_\ell}$ are linearly independent vectors, and by (21.6), their sum is 0. This is a contradiction. **Q.E.D.**

Corollary 21.a.14 (*Diagonalizability via Eigenspaces*): Let V be finite-dimensional (over K) and $T \in \operatorname{End}(V)$ with distinct eigenvalues $\lambda_1,\dots,\lambda_r \in K$. Then, T is diagonalizable if and only if the dimensions of its eigenspaces sum to $\dim V$:

$$\sum_{i=1}^{r} \dim \operatorname{Eig}_T(\lambda_i) = \dim V.$$

Proof:

Choose a basis $\mathcal{B}_i$ for $\operatorname{Eig}_T(\lambda_i)$. Recall that $W := \operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_r)$ is a direct sum. This implies that $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ is a basis for W. Now $W \subseteq V$ is a linear subspace, and $\dim W = \sum_{i=1}^r |\mathcal{B}_i| = \sum_{i=1}^r \dim \operatorname{Eig}_T(\lambda_i) \leq \dim V$. Since $W \subseteq V$ is a subspace of the finite-dimensional space V, we have $W = V$ if and only if $\dim W = \dim V$, that is, if and only if $\sum_{i=1}^r \dim \operatorname{Eig}_T(\lambda_i) = \dim V$.

It remains to connect this to diagonalizability. If the dimension equality holds, then $W = V$, so $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ is a basis for V consisting of eigenvectors of T, and therefore, T is diagonalizable by Lemma 21.a.3. Conversely, if T is diagonalizable, pick a basis $\mathcal{C} = (w_1, \ldots, w_n)$ of V consisting of eigenvectors. Every w_j lies in $\operatorname{Eig}_T(\lambda_i)$ for some i (the eigenvalues of T are exactly $\lambda_1, \ldots, \lambda_r$), hence $w_j \in W$ for all j. Consequently, $V = \operatorname{Sp}(w_1, \ldots, w_n) \subseteq W$, so $W = V$ and the dimension equality holds. **Q.E.D.**

Exercise 21.3: If V is finite-dimensional, then T can have only finitely many eigenvalues. (Of course, it is implicitly assumed $\lambda_j \neq \lambda_k$ for all $j \neq k$).

Solutions to the Exercises

Proof of Exercise 21.1:

Throughout, write $W := U_1 + \cdots + U_r$.

Directness implies (a). The zero vector always admits the representation $0 = 0 + \cdots + 0$ with $0 \in U_i$ for every i. If the sum is direct, this representation is the only one, which is precisely **(a)**.

(a) implies directness. Let $w \in W$ and suppose $w = u_1 + \cdots + u_r = u_1' + \cdots + u_r'$ with $u_i, u_i' \in U_i$. Subtracting the two representations gives $0 = (u_1 - u_1') + \cdots + (u_r - u_r')$, and $u_i - u_i' \in U_i$ because U_i is a subspace. By **(a)**, every summand vanishes, so $u_i = u_i'$ for all i; the representation of w is therefore unique.

(a) implies (b). Fix $2 \leq j \leq r$ and let $u \in U_j \cap (U_1 + \cdots + U_{j-1})$. Write $u = u_1 + \cdots + u_{j-1}$ with $u_i \in U_i$. Then

$$u_1 + \cdots + u_{j-1} + (-u) + 0 + \cdots + 0 = 0$$

is a representation of 0 whose i-th entry lies in U_i for every i (the j-th entry being $-u \in U_j$). By **(a)** all entries vanish, and in particular $u = 0$.

(b) implies (a). Suppose $0 = u_1 + \cdots + u_r$ with $u_i \in U_i$ and not all u_i equal to 0. Let j be the largest index with $u_j \neq 0$. If $j = 1$, then $0 = u_1$ contradicts $u_1 \neq 0$. If $j \geq 2$, then

$$u_j = -(u_1 + \cdots + u_{j-1}) \in U_j \cap (U_1 + \cdots + U_{j-1}) = \{0\},$$

again contradicting $u_j \neq 0$. Hence all the u_i vanish.

The basis criterion. Assume in addition that every U_i is finite-dimensional, and let $\mathcal{B}_i := (b_1^{(i)}, \ldots, b_{d_i}^{(i)})$ be a basis for U_i. The concatenated list $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ always spans W, since every $u_i \in U_i$ is a combination of $\mathcal{B}_i$. So the only question is linear independence.

If the sum is direct, consider a vanishing combination of the concatenated list and collect, for each i, the terms coming from $\mathcal{B}_i$ into a single vector $u_i \in U_i$. Then $u_1 + \cdots + u_r = 0$, so $u_i = 0$ for every i by **(a)**, and the linear independence of $\mathcal{B}_i$ forces all the coefficients in the i-th group to vanish. Thus $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ is a basis for W, for every choice of the bases $\mathcal{B}_i$.

Conversely, suppose $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ is a basis for W for some choice of bases $\mathcal{B}_i$, and let $0 = u_1 + \cdots + u_r$ with $u_i \in U_i$. Expanding each u_i in the basis $\mathcal{B}_i$ exhibits a vanishing combination of the concatenated list; by its linear independence all coefficients are 0, hence $u_i = 0$ for every i. This is **(a)**, so the sum is direct. **Q.E.D.**

Proof of Exercise 21.2:

Consider the map

$$\Phi : U_1 \oplus \cdots \oplus U_r \longrightarrow U_1 + \cdots + U_r, \qquad \Phi(u_1, \ldots, u_r) := u_1 + \cdots + u_r.$$

Addition and scalar multiplication in the external direct sum $U_1 \oplus \cdots \oplus U_r$ are performed componentwise, so

$$\Phi\big((u_1, \ldots, u_r) + \mu(u_1', \ldots, u_r')\big) = \sum_{i=1}^{r}(u_i + \mu u_i') = \Phi(u_1, \ldots, u_r) + \mu\Phi(u_1', \ldots, u_r'),$$

and Φ is linear. It is surjective, because by the very definition of $U_1 + \cdots + U_r$ every element of the target is of the form $u_1 + \cdots + u_r$ with $u_i \in U_i$.

For injectivity, let $(u_1, \ldots, u_r) \in \ker\Phi$, that is, $u_1 + \cdots + u_r = 0$ with $u_i \in U_i$. Since the sum is direct, part **(a)** of Exercise 21.1 gives $u_1 = \cdots = u_r = 0$, so $\ker\Phi = \{0\}$. Hence Φ is an isomorphism.

The isomorphism is canonical in the sense that its definition involves no choice whatsoever — no bases, no complements — only the addition of V. This is what licenses the identification of $U_1 + \cdots + U_r$ with $U_1 \oplus \cdots \oplus U_r$ whenever the sum is direct. **Q.E.D.**

Proof of Exercise 21.3:

Put $n := \dim V$ and let $\lambda_1, \ldots, \lambda_m \in K$ be pairwise distinct eigenvalues of T. Choose, for each i, an eigenvector $v_i \neq 0$ for λ_i. By Proposition 21.a.4, the vectors $v_1, \ldots, v_m$ are linearly independent, and a linearly independent list in an n-dimensional space has at most n members. Hence $m \leq n$.

Since m was the number of **any** list of pairwise distinct eigenvalues, T has at most $n = \dim V$ distinct eigenvalues; in particular, only finitely many. **Q.E.D.**

Multiplicities of Eigenvalues (Section 21.b)

Example 21.3 (*Identical Characteristic Polynomials*): Consider the matrices $A := \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $B := \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. We have that their characteristic polynomials are identical: $P_A(x) = P_B(x) = (1-x)^2 = (x-1)^2$.

Example 21.4 (*Jordan Block*): Consider the matrix $J_{\lambda,n} \in \mathcal{M}_{n\times n}(K)$, for $\lambda \in K$.

$$J := J_{\lambda,n} := \begin{pmatrix} \lambda & 1 & 0 & \ldots & 0 \\ 0 & \lambda & 1 & \ldots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & 0 & \lambda \end{pmatrix} \tag{21.7}$$

This is called a "*Jordan block*". It features λ's along the main diagonal, 1's along the *"diagonal"* just above the main diagonal, and 0's everywhere else. The characteristic polynomial is $P_{J_{\lambda,n}}(x) = (\lambda - x)^n$. This implies that J has only one eigenvalue, namely λ.

Let us find the eigenvectors for the eigenvalue λ:

$$J - \lambda \cdot I_n = \begin{pmatrix} 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

This matrix clearly has $n-1$ linearly independent columns. Therefore, the rank is $\operatorname{rank}(J-\lambda\cdot I_n) = n-1$. By the rank-nullity theorem, $\dim\ker(J-\lambda\cdot I_n)=1$. Consequently, the eigenspace $\operatorname{Eig}_J(\lambda)$ is 1-dimensional. Clearly, $e_1=(1,0,\dots,0)^\intercal$ is an eigenvector for λ. Thus, $\operatorname{Eig}_J(\lambda)=\operatorname{Sp}(e_1)$.

Important remark 21.1: There is an important difference between polynomials $p(x)\in K[x]$ and the functions they define $K\to K$. For example, if $K=\mathbb{Z}_2$, then $p(x)=x$ and $q(x)=x^2$ are different polynomials, but they define the same function $\mathbb{Z}_2\to\mathbb{Z}_2$. (What happens for $K=\mathbb{Q},\mathbb{R},\mathbb{C},\dots$?)

Exercise 21.4 (*Polynomials versus Polynomial Functions*): Settle the question raised at the end of Important remark 21.1: for which fields K do two distinct polynomials $p(x),q(x)\in K[x]$ always define distinct functions $K\to K$? In particular, what happens for $K=\mathbb{Q},\mathbb{R},\mathbb{C}$?

Lemma 21.b.1 (*Decomposition of Diagonalizable Matrices*): Let $A\in\mathcal{M}_{n\times n}(K)$ (or $T\in\operatorname{End}(V)$) be diagonalizable. Then $P_A(x)$ decomposes as a product of linear (i.e., degree 1) factors:

$$P_A(x)=(\lambda_1-x)\cdot\dots\cdot(\lambda_n-x),$$

where $\lambda_1,\dots,\lambda_n$ are the eigenvalues of A (but there might be repetitions among the λ_i's).

Proof:

By assumption, there exists an invertible matrix P such that:

$$P^{-1}AP=\begin{pmatrix}\lambda_1 & \dots & 0\\ \vdots & \ddots & \vdots\\ 0 & \dots & \lambda_n\end{pmatrix}=:D. \tag{21.8}$$

(i.e., A is similar to D). Since similar matrices have the same characteristic polynomial, we have:

$$P_A(x)=P_D(x)=\det\begin{pmatrix}\lambda_1-x & \dots & 0\\ \vdots & \ddots & \vdots\\ 0 & \dots & \lambda_n-x\end{pmatrix}=(\lambda_1-x)\dots(\lambda_n-x).$$

Q.E.D.

Example 21.5 (*Rotation Matrix without Real Eigenvalues*): Let $A=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$ over $K=\mathbb{R}$. The associated linear map $T_A:\mathbb{R}^2\to\mathbb{R}^2$ represents a rotation by 90° in the counter-clockwise direction. The characteristic polynomial is $P_A(x)=\det\left(\begin{smallmatrix}-x&-1\\1&-x\end{smallmatrix}\right)=x^2+1$. This polynomial has no roots in $\mathbb{R}$. And, of course, A has no real eigenvalues.

A quick digression about polynomials...

Let $0\neq p(x)\in K[x]$ and $\lambda\in K$. By dividing $p(x)$ by $(x-\lambda)$, we can write:

$$p(x)=(x-\lambda)\cdot q(x)+R, \tag{21.9}$$

where $\deg q(x)=\deg p(x)-1$, and $R\in K$ is the remainder. If we substitute $x=\lambda$ into (21.9), we obtain $p(\lambda)=0\cdot q(\lambda)+R=R$. Thus, $p(x)=(x-\lambda)\cdot q(x)+p(\lambda)$. Conclusion: $p(\lambda)=0 \iff p(x)=(x-\lambda)\cdot q(x)$ for some $q(x)\in K[x]$. We say that λ is a “*zero*” of $p(x)$ (in German: „*Nullstelle*“), or a “*root*” of $p(x)$. Sometimes, λ is also a zero of $q(x)$, so $q(x)=(x-\lambda)\cdot h(x)$, which implies $p(x)=(x-\lambda)^2\cdot h(x)$, etc.

Definition 21.b.2 (*Multiplicity of a Zero*): Let $0\neq p(x)\in K[x]$ and $\lambda\in K$. We say that λ is a zero of “*multiplicity*” $m\in\mathbb{N}_{>0}$ if $p(x)=(x-\lambda)^m\cdot g(x)$, where $g(x)\in K[x]$ and $g(\lambda)\neq 0$ (i.e., λ is not a zero of $g(x)$). If $m=1$, we say λ is a “*simple zero*” of $p(x)$. If $m\geq 2$, we say λ is a “*multiple zero*” of $p(x)$.

Geometric & Algebraic Multiplicity of Eigenvalues

Definition 21.b.3 (*Geometric and Algebraic Multiplicity*): Let V be a vector space over K and $T \in \operatorname{End}(V)$. Let λ be an eigenvalue of T. We define the "*geometric multiplicity*" of λ to be:

$$m_g(T;\lambda) := \dim \operatorname{Eig}_T(\lambda).$$

Assume now, in addition, that V is finite-dimensional. We define the "*algebraic multiplicity*" of λ to be the multiplicity of the zero λ in the characteristic polynomial $P_T(x)$ of T. We denote it by $m_a(T;\lambda)$.

Proposition 21.b.4 (*Multiplicity Inequality*): For any eigenvalue λ, the geometric multiplicity is less than or equal to the algebraic multiplicity:

$$m_g(T;\lambda) \leq m_a(T;\lambda).$$

Proof:

Let $v_1, \dots, v_k$ be a basis for the eigenspace $\operatorname{Eig}_T(\lambda)$. We extend this basis to a basis $\mathcal{B} := (v_1, \dots, v_k, w_{k+1}, \dots, w_n)$ of V. Then the matrix of T with respect to the basis $\mathcal{B}$ has the block form:

$$[T]^{\mathcal{B}}_{\mathcal{B}} = ([Tv_1]_{\mathcal{B}} \dots [Tv_k]_{\mathcal{B}} \mid [Tw_{k+1}]_{\mathcal{B}} \dots [Tw_n]_{\mathcal{B}})$$

$$[T]^{\mathcal{B}}_{\mathcal{B}} = \left(\begin{array}{c|c} \lambda I_k & C_1 \\ \hline 0 & C \end{array}\right) \tag{21.10}$$

where $C \in \mathcal{M}_{(n-k)\times(n-k)}(K)$. We compute the characteristic polynomial by utilizing the determinant property for block matrices (Proposition 20.c.4):

$$P_T(x) = \det([T]^{\mathcal{B}}_{\mathcal{B}} - x \cdot I_n) = (\lambda - x)^k \cdot \det(C - xI_{n-k})$$
$$P_T(x) = (\lambda - x)^k \cdot P_C(x) = (-1)^k (x-\lambda)^k \cdot P_C(x).$$

This factorization directly shows that the multiplicity of λ as a zero of $P_T(x)$ is at least k. Therefore, we conclude that $m_a(T;\lambda) \geq k = m_g(T;\lambda)$. **Q.E.D.**

Example 21.6 (*Multiplicities of a Jordan Block*): Revisiting the Jordan block $J := J_{\lambda,n}$ defined in (21.7): We previously saw that $P_J(x) = (\lambda - x)^n$, which directly implies that $m_a(J;\lambda) = n$. We also found that $\operatorname{Eig}_J(\lambda)$ is a 1-dimensional space, which means $m_g(J;\lambda) = 1$.

Solutions to the Exercises

Solution of Exercise 21.4:

The assignment $p(x) \mapsto \hat{p}$, where $\hat{p} : K \to K$ denotes the function $\hat{p}(a) := p(a)$, is K-linear, so two polynomials $p(x), q(x) \in K[x]$ define the same function if and only if their difference $r(x) := p(x) - q(x)$ vanishes at **every** point of K. The question is therefore: which non-zero polynomials vanish identically as functions?

A non-zero polynomial has at most deg **many zeros.** We claim that a polynomial $0 \neq r(x) \in K[x]$ of degree d has at most d distinct zeros in K. We induct on d. If $d = 0$, then r is a non-zero constant and has no zeros at all. Let $d \geq 1$. If r has no zero, there is nothing to prove; otherwise pick a zero $\lambda \in K$. By the Conclusion of the digression above, $r(x) = (x - \lambda) \cdot q(x)$ with $\deg q(x) = d - 1$. If $\mu \neq \lambda$ is another zero of r, then $(\mu - \lambda) \cdot q(\mu) = 0$ with $\mu - \lambda \neq 0$, and since a field has no zero divisors, $q(\mu) = 0$. So every zero of r other than λ is a zero of q, of which there are at most $d - 1$ by the inductive hypothesis. Hence r has at most d zeros.

Infinite fields. If K is infinite, then no non-zero polynomial can vanish at every point of K, since it has only finitely many zeros. Consequently $r(x) = 0$, that is, $p(x) = q(x)$: over an infinite field, distinct polynomials always define distinct functions. This settles $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}$ — for these fields the phenomenon of Important remark 21.1 cannot occur, and one may safely identify a polynomial with the function it defines.

Finite fields. If K is finite, the phenomenon always occurs. Indeed, put

$$r(x) := \prod_{a\in K}(x-a) \in K[x].$$

This polynomial is non-zero (it is monic of degree $|K|$), yet $\hat{r}(a) = 0$ for every $a \in K$. Thus $p(x)$ and $p(x)+r(x)$ are distinct polynomials defining the same function, for any $p(x)$ whatsoever. For $K = \mathbb{Z}_2$ this gives $r(x) = x\cdot(x-1) = x^2 - x$, and adding it to $p(x) = x$ returns exactly Prof. Biran's $q(x) = x^2$.

There is also a proof by counting, which shows that a collision must occur without exhibiting one: if $|K| = m < \infty$, then there are exactly m^m functions $K \to K$, whereas $K[x]$ is infinite, so the map $p(x) \mapsto \hat{p}$ cannot possibly be injective. **Q.E.D.**

Diagonalizability (Section 21.c)

Theorem 21.c.1 (*Equivalent Conditions for Diagonalizability*): Let V be an n-dimensional vector space over K and $T \in \mathrm{End}(V)$. The following statements are equivalent:

- **(a)** T is diagonalizable.
- **(b)** There exists a basis for V whose elements are eigenvectors of T.
- **(c)** The characteristic polynomial $P_T(x)$ splits into a product of linear factors and, for every eigenvalue of T, the geometric and algebraic multiplicities are equal.
- **(d)** $\sum_{i=1}^k \dim \mathrm{Eig}_T(\lambda_i) = \dim V$, where $\lambda_1,\dots,\lambda_k \in K$ are all the distinct eigenvalues of T.
- **(e)** $V \cong \bigoplus_{i=1}^k \mathrm{Eig}_T(\lambda_i)$.

Proof:

(a) $\iff$ **(b)** has already been proved.

(a) $\implies$ **(c)**: Let $\mathcal{B}$ be a basis in which $[T]_{\mathcal{B}}^{\mathcal{B}}$ has diagonal form. By changing the order of the elements in $\mathcal{B}$, we can arrange that $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the following block-diagonal shape:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|c|c} \begin{matrix}\lambda_1 & \dots & 0\\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_1\end{matrix} & 0 & 0 \\ \hline 0 & \ddots & 0 \\ \hline 0 & 0 & \begin{matrix}\lambda_k & \dots & 0\\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_k\end{matrix} \end{array}\right) \tag{21.11}$$

where $\mathcal{B} = (v_1^{(1)},\dots,v_{\ell_1}^{(1)},v_1^{(2)},\dots,v_{\ell_2}^{(2)},\dots,v_1^{(k)},\dots,v_{\ell_k}^{(k)})$. Here, $\lambda_1,\dots,\lambda_k$ are the eigenvalues of T (with $\lambda_i \neq \lambda_j$ for all $i \neq j$) and $v_1^{(i)},\dots,v_{\ell_i}^{(i)}$ are eigenvectors of λ_i for all i. The characteristic polynomial is:

$$P_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x\cdot I_n) = (\lambda_1 - x)^{\ell_1}\cdot\dots\cdot(\lambda_k - x)^{\ell_k}. \tag{21.12}$$

This shows that $P_T(x)$ splits as a product of linear factors. It also shows that $\lambda_1,\dots,\lambda_k$ are all the eigenvalues of T. It remains to show that $m_g(T;\lambda_i) = m_a(T;\lambda_i)$ for all i. Clearly, $v_1^{(i)},\dots,v_{\ell_i}^{(i)} \in \mathrm{Eig}_T(\lambda_i)$ are linearly independent; hence $\ell_i \le \dim \mathrm{Eig}_T(\lambda_i) = m_g(T;\lambda_i)$ by definition. But from (21.12), we also have $\ell_i = m_a(T;\lambda_i)$. So $m_a(T;\lambda_i) \le m_g(T;\lambda_i)$ for all i. At the same time, we always have the inequality $m_g(T;\lambda_i) \le m_a(T;\lambda_i)$. Therefore, $m_g(T;\lambda_i) = m_a(T;\lambda_i)$ for all i. (The proof also shows that $\ell_i = m_g(T;\lambda_i) = m_a(T;\lambda_i)$).

(c) $\implies$ **(d)**: We assume that $P_T(x) = (\lambda_1 - x)^{\ell_1} \dots (\lambda_k - x)^{\ell_k}$ and $m_g(T;\lambda_i) = m_a(T;\lambda_i)$. Then $\lambda_1, \dots, \lambda_k$ are exactly the eigenvalues of T. By definition, $\ell_i = m_a(T;\lambda_i)$. We have $n = \dim V = \sum_{i=1}^k m_a(T;\lambda_i)$ (because $\deg P_T(x) = n$). By assumption, this equals

$$\sum_{i=1}^k m_g(T;\lambda_i) = \sum_{i=1}^k \dim \operatorname{Eig}_T(\lambda_i).$$

(d) $\implies$ **(e)**: Recall from Proposition 21.a.13 that $\operatorname{Eig}_T(\lambda_1) + \dots + \operatorname{Eig}_T(\lambda_k)$ is a direct sum. Consequently,

$$\dim(\operatorname{Eig}_T(\lambda_1) + \dots + \operatorname{Eig}_T(\lambda_k)) = \dim \operatorname{Eig}_T(\lambda_1) + \dots + \dim \operatorname{Eig}_T(\lambda_k) = \dim V$$

by assumption. Since $\operatorname{Eig}_T(\lambda_1) + \dots + \operatorname{Eig}_T(\lambda_k) \subseteq V$ is a linear subspace, we have $\operatorname{Eig}_T(\lambda_1) + \dots + \operatorname{Eig}_T(\lambda_k) = V$. This implies that $\operatorname{Eig}_T(\lambda_1) + \dots + \operatorname{Eig}_T(\lambda_k) \cong \operatorname{Eig}_T(\lambda_1) \oplus \dots \oplus \operatorname{Eig}_T(\lambda_k)$.

(e) $\implies$ **(a)**: If $V \cong \operatorname{Eig}_T(\lambda_1) \oplus \dots \oplus \operatorname{Eig}_T(\lambda_k)$, then since $\operatorname{Eig}_T(\lambda_1) + \dots + \operatorname{Eig}_T(\lambda_k)$ is a direct sum and also a linear subspace of V, we must have $\operatorname{Eig}_T(\lambda_1) + \dots + \operatorname{Eig}_T(\lambda_k) = V$. If we choose a basis $v_1^{(i)}, \dots, v_{\ell_i}^{(i)}$ for $\operatorname{Eig}_T(\lambda_i)$ for all i (where $\ell_i = m_g(T;\lambda_i)$), then the union $(v_1^{(1)}, \dots, v_{\ell_1}^{(1)}, \dots, v_1^{(k)}, \dots, v_{\ell_k}^{(k)})$ is a basis of V and all the vectors in it are eigenvectors of T.

Q.E.D.

Corollary 21.c.2 (*Sufficient Condition for Diagonalizability*): If $P_T(x)$ splits as a product of linear factors and each eigenvalue λ of T has algebraic multiplicity 1, then T is diagonalizable.

Proof:

If $m_a(T;\lambda) = 1$, this implies that $m_g(T;\lambda) = 1$. Now use point **(c)** of Theorem 21.c.1. **Q.E.D.**

Trigonalization

(= bringing a matrix/endomorphism to upper triangular form)

Question 21.1: Let $T \in \operatorname{End}(V)$. We want to find a basis $\mathcal{B}$ for T such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ has as simple a form as possible.

- If T is diagonalizable, we are okay. But we have seen that this is not always the case. For example, $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is not diagonalizable (we saw: $m_a(A;1) = 2, m_g(A;1) = 1$).
- A weaker (hence more general) condition is trigonalizability.

Definition 21.c.3 (*Trigonalizable Endomorphism*): We say that $T \in \operatorname{End}(V)$ is "*trigonalizable*" if there exists a basis $\mathcal{B}$ for V such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular, i.e.,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix} \quad (n = \dim V). \tag{21.13}$$

Theorem 21.c.4 (*Trigonalizability Condition*): Let V be a finite-dimensional vector space over K and $T \in \operatorname{End}(V)$. Then T is trigonalizable if and only if $P_T(x)$ splits as a product of linear factors, i.e., $P_T(x) = (\lambda_1 - x)^{\ell_1} \dots (\lambda_k - x)^{\ell_k}$.

Short Digression

Theorem 21.c.5 (*Fundamental Theorem of Algebra*): Every polynomial $p(x) \in \mathbb{C}[x]$ with complex coefficients splits as a product of linear factors and a constant, i.e., $p(x) = C \cdot (x - a_1)^{n_1} \dots (x - a_r)^{n_r}$, with $C \in \mathbb{C}$, $r \geq 0$, $a_i \in \mathbb{C}$, and $n_j \in \mathbb{N}_{>0}$.

Remark 21.3: The proof is typically covered in an analysis course or a complex analysis course in the third semester.

Corollary 21.c.6 (*Trigonalizability over $\mathbb{C}$*): Let V be a finite-dimensional vector space over $\mathbb{C}$. Then every $T \in \mathrm{End}(V)$ is trigonalizable.

Proof:

The characteristic polynomial $P_T(x)$ lies in $\mathbb{C}[x]$, so by Theorem 21.c.5 it splits as a product of linear factors. Now apply Theorem 21.c.4. **Q.E.D.**

Exercise 21.5 (*Splitting Passes to the Cofactor*): Let $\lambda \in K$ and let $p(x), q(x) \in K[x]$ be non-zero polynomials with $p(x) = (\lambda - x) \cdot q(x)$. Show that if $p(x)$ splits into a product of linear factors over K, then so does $q(x)$. (This is the step Prof. Biran leaves as an exercise in both proofs of Theorem 21.c.4, applied to $P_T(x) = (\lambda_1 - x) \cdot P_A(x)$.)

Proof of the Theorem on Trigonalizability

Proof of Theorem 21.c.4:

$\Longrightarrow$: If there exists a basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular as in (21.13), then:

$$P_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n) = \det \begin{pmatrix} \lambda_1 - x & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n - x \end{pmatrix} = (\lambda_1 - x) \dots (\lambda_n - x).$$

$\Longleftarrow$: We will use induction on $n = \dim V$.

Base case $n = 1$: In this case, V is 1-dimensional and $[T]_{\mathcal{B}}^{\mathcal{B}} = (\lambda)$, so T is even diagonalizable.

Inductive step: Let $n \geq 1$. Assume that the theorem holds for all vector spaces of dimension $\leq n$ and every endomorphism of such a space. Let V be an $(n+1)$-dimensional vector space and $T \in \mathrm{End}(V)$ such that $P_T(x)$ splits as a product of linear factors.

Let λ_1 be a zero of $P_T(x)$. This implies that there exists an eigenvector $u_1 \in \ker(T - \lambda_1 \cdot \mathrm{id}_V)$. Extend u_1 to a basis $\mathcal{B}' = (u_1, u_2, \dots, u_{n+1})$ of V. Then the matrix of T in basis $\mathcal{B}'$ has the block form:

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = \left(\begin{array}{c|c} \lambda_1 & * \cdots * \\ \hline 0 & \\ \vdots & A \\ 0 & \end{array}\right) =: M \tag{21.14}$$

where A is an $n \times n$ matrix in $\mathcal{M}_{n \times n}(K)$. By block determinant properties (Proposition 20.c.4), we have $P_T(x) = (\lambda_1 - x) \cdot P_A(x)$. Since $P_T(x)$ splits as a product of linear factors, so does $P_A(x)$ (Exercise 21.5).

Consider the linear map $T_A : K^n \to K^n$ defined by $T_A(u) = A \cdot u$. By the induction hypothesis, there exists a basis $\mathcal{C}$ of K^n such that:

$$[T_A]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} \lambda_2 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_{n+1} \end{pmatrix}. \tag{21.15}$$

Denote by $\mathcal{E}$ the standard basis of K^n. We know that $[T_A]_{\mathcal{C}}^{\mathcal{C}} = [\mathrm{id}]_{\mathcal{C}}^{\mathcal{E}} \cdot [T_A]_{\mathcal{E}}^{\mathcal{E}} \cdot [\mathrm{id}]_{\mathcal{E}}^{\mathcal{C}} = Q^{-1}AQ$, where $Q = [\mathrm{id}]_{\mathcal{C}}^{\mathcal{E}}$. Consequently, $Q^{-1}AQ = \begin{pmatrix} \lambda_2 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_{n+1} \end{pmatrix}$.

Now, define the $(n+1) \times (n+1)$ block matrix:

$$\underline{P} := \left(\begin{array}{c|c} 1 & 0 \dots 0 \\ \hline 0 & \\ \vdots & Q \\ 0 & \end{array}\right) \in \mathcal{M}_{(n+1)\times(n+1)}(K). \tag{21.16}$$

Claim: $\underline{P}$ is invertible with $\underline{P}^{-1} = \left(\begin{array}{c|c} 1 & 0 \dots 0 \\ \hline 0 & \\ \vdots & Q^{-1} \\ 0 & \end{array}\right)$, and $\underline{P}^{-1}M\underline{P} = \left(\begin{array}{c|c} \lambda_1 & * \cdots * \\ \hline 0 & \\ \vdots & Q^{-1}AQ \\ 0 & \end{array}\right)$.

Proof of Claim:

This follows from direct block matrix multiplication. First, we verify the product $\underline{P} \cdot \underline{P}^{-1}$ to confirm the inverse:

$$\underline{P} \cdot \underline{P}^{-1} = \left(\begin{array}{c|c} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot Q^{-1} \\ \hline 0 \cdot 1 + Q \cdot 0 & 0 \cdot 0 + Q \cdot Q^{-1} \end{array}\right) = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & I_n \end{array}\right).$$

Next, we compute the similarity transformation $\underline{P}^{-1}M\underline{P}$ in two steps. Let us first multiply $\underline{P}^{-1}$ and M:

$$\begin{aligned} \underline{P}^{-1}M &= \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q^{-1} \end{array}\right) \left(\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & A \end{array}\right) \\ &= \left(\begin{array}{c|c} 1 \cdot \lambda_1 + 0 \cdot 0 & 1 \cdot * + 0 \cdot A \\ \hline 0 \cdot \lambda_1 + Q^{-1} \cdot 0 & 0 \cdot * + Q^{-1} \cdot A \end{array}\right) = \left(\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & Q^{-1}A \end{array}\right). \end{aligned}$$

Finally, we multiply the resulting matrix by $\underline{P}$ from the right:

$$\begin{aligned} (\underline{P}^{-1}M)\underline{P} &= \left(\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & Q^{-1}A \end{array}\right) \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q \end{array}\right) \\ &= \left(\begin{array}{c|c} \lambda_1 \cdot 1 + * \cdot 0 & \lambda_1 \cdot 0 + * \cdot Q \\ \hline 0 \cdot 1 + Q^{-1}A \cdot 0 & 0 \cdot 0 + Q^{-1}A \cdot Q \end{array}\right) \\ &= \left(\begin{array}{c|c} \lambda_1 & *' \\ \hline 0 & Q^{-1}AQ \end{array}\right), \end{aligned}$$

where $*'$ denotes the new row vector $*Q$. Since $Q^{-1}AQ$ is upper triangular with diagonal entries $\lambda_2, \dots, \lambda_{n+1}$ (as guaranteed by the induction hypothesis), the block structure ensures the entire matrix $\underline{P}^{-1}M\underline{P}$ is upper triangular. This proves the claim. **Q.E.D.**

We continue with the proof of the Theorem. Recall we have the basis $\mathcal{B}' = (u_1, u_2, \dots, u_{n+1})$, where $[T]_{\mathcal{B}'}^{\mathcal{B}'} = M$.

Claim: There exists a basis $\mathcal{B}$ for V such that $[\mathrm{id}]_{\mathcal{B}'}^{\mathcal{B}} = \underline{P}$.

Proof of Claim:

Write the entries of $\underline{P}$ as P_{ij}. Define $m := n+1$ and set vectors $v_1, \dots, v_m$ as linear combinations of $\mathcal{B}'$ using the columns of $\underline{P}$:

$$v_k := \sum_{j=1}^{m} P_{jk} u_j \quad \text{for } k = 1, \dots, m.$$

The set $\mathcal{B} := (v_1, \dots, v_m)$ is a basis for V. To see this, assume $c_1 v_1 + \cdots + c_m v_m = 0$. Substituting the definitions of v_k and using the linear independence of $(u_1, \dots, u_m)$ yields the matrix equation $\underline{P} \cdot (c_1, \dots, c_m)^{\mathsf{T}} = 0$. Because $\underline{P}$ is invertible, $(c_1, \dots, c_m)^{\mathsf{T}} = 0$, proving linear independence.

Q.E.D.

We now have the basis $\mathcal{B}$. Computing the matrix of T in this new basis yields:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}]_{\mathcal{B}}^{\mathcal{B}'} \cdot [T]_{\mathcal{B}'}^{\mathcal{B}'} \cdot [\mathrm{id}]_{\mathcal{B}'}^{\mathcal{B}} = \underline{P}^{-1} \cdot M \cdot \underline{P}. \tag{21.17}$$

As proven above, $\underline{P}^{-1} M \underline{P}$ is upper triangular, with $\lambda_1, \lambda_2, \dots, \lambda_{n+1}$ along its diagonal. Therefore, T is trigonalizable. The induction is now complete. **Q.E.D.**

An Alternative (And More Elegant) Proof

Exercise 21.6 (*Bases of a Quotient Space*): Let V be a finite-dimensional vector space over K, let $U \subseteq V$ be a linear subspace, and let $(u_1, \dots, u_r)$ be a basis for U, extended to a basis $(u_1, \dots, u_r, w_{r+1}, \dots, w_n)$ of V. Show that the equivalence classes $([w_{r+1}], \dots, [w_n])$ form a basis for the quotient space V/U. (In particular, $\dim V/U = \dim V - \dim U$.)

Alternative proof of the direction $\Longleftarrow$ of Theorem 21.c.4:

$\Longleftarrow$: We will use induction on $n = \dim V$, and also quotient spaces.

Let us first recall the latter. If $U \subseteq V$ is a linear subspace, we have the quotient space V/U, whose elements are the equivalence classes $[v]$ of the vectors $v \in V$, where $[v] = [v']$ if and only if $v - v' \in U$. Recall also that if V is finite-dimensional, then so is V/U, and

$$\dim V/U = \dim V - \dim U.$$

Now to the proof.

Base case $n = 1$: In this case, V is 1-dimensional and $[T]_{\mathcal{B}}^{\mathcal{B}} = (\,\lambda\,)$, so T is even diagonalizable.

Inductive step: Let $n \geq 1$. Assume that the direction $\Longleftarrow$ of the theorem holds for all vector spaces of dimension $\leq n$ and every endomorphism of such a space. Let V be an $(n+1)$-dimensional vector space and $T \in \mathrm{End}(V)$ such that $P_T(x)$ splits as a product of linear factors.

Let $\lambda_1 \in K$ be a zero of $P_T(x)$. This implies that there exists an eigenvector $v_1 \in \ker(T - \lambda_1 \cdot \mathrm{id}_V)$.

Define the 1-dimensional subspace $U := \mathrm{Sp}(v_1)$ and consider the quotient space V/U. Since v_1 is an eigenvector, $T(U) \subseteq U$. This invariance property allows us to obtain a well-defined induced endomorphism on the quotient space, $\overline{T} : V/U \to V/U$, defined by $\overline{T}([v]) := [Tv]$.

Extend v_1 to a basis $\mathcal{B}_1 = (v_1, w_2, \dots, w_{n+1})$ of V. The equivalence classes $[w_2], \dots, [w_{n+1}]$ form a basis for V/U (Exercise 21.6, applied with $r = 1$ and the basis (v_1) of U). In this basis, $\overline{T}$ is represented by an $n \times n$ matrix A, and the full transformation T in basis $\mathcal{B}_1$ takes the block form:

$$[T]_{\mathcal{B}_1}^{\mathcal{B}_1} = \left(\begin{array}{c|ccc} \lambda_1 & * & \dots & * \\ \hline 0 & & & \\ \vdots & & A & \\ 0 & & & \end{array}\right). \tag{21.18}$$

By the properties of block-triangular determinants (Proposition 20.c.4), we have $P_T(x) = (\lambda_1 - x) \cdot P_A(x)$. Because $\overline{T}$ has the matrix representation A, we know $P_{\overline{T}}(x) = P_A(x)$, which gives us:

$$P_T(x) = (\lambda_1 - x) \cdot P_{\overline{T}}(x).$$

Since $P_T(x)$ splits into a product of linear factors by our initial assumption, $P_{\overline{T}}(x)$ must also split into linear factors (Exercise 21.5).

Because $\dim(V/U) = n$, we can apply the induction hypothesis to $\overline{T}$. There exists a basis $\mathcal{C} := (z_2, \ldots, z_{n+1})$ for V/U such that its matrix is upper triangular:

$$[\overline{T}]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} \lambda_2 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_{n+1} \end{pmatrix}. \tag{21.19}$$

Now, each element in $\mathcal{C}$ is represented by some vector $v_i \in V$, meaning $z_i = [v_i]$. (Such a v_i is not unique, but we pick one specific representative for every i). Define the list of vectors $\mathcal{B} := (v_1, v_2, \ldots, v_{n+1})$. We claim that $\mathcal{B}$ is a basis for V.

To show this, we check for linear independence. Assume:

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_{n+1} v_{n+1} = 0.$$

Applying the quotient map to this equation (where $[v_1] = 0$), we get:

$$\alpha_2 [v_2] + \cdots + \alpha_{n+1} [v_{n+1}] = 0.$$

This implies that $\alpha_2 z_2 + \cdots + \alpha_{n+1} z_{n+1} = 0$. Since the z_i vectors form a basis for V/U, they are linearly independent, which forces $\alpha_2 = \cdots = \alpha_{n+1} = 0$. Plugging this back into our original sum leaves $\alpha_1 v_1 = 0$. Since $v_1 \neq 0$, we must have $\alpha_1 = 0$. Because the $n+1$ vectors are linearly independent and $\dim V = n+1$, $\mathcal{B}$ is indeed a basis for V.

Finally, we claim that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular. For the first column, we already know $T(v_1) = \lambda_1 v_1$. For the remaining columns ($2 \leq i \leq n+1$), we read from the upper-triangular matrix (21.19) that:

$$[\overline{T} z_i]_{\mathcal{C}} = \alpha_{2i} z_2 + \cdots + \alpha_{i-1,i} z_{i-1} + \lambda_i z_i$$

for some scalars $\alpha \in K$. Translating this back to the equivalence classes:

$$[\overline{T} v_i] = \alpha_{2i} [v_2] + \cdots + \alpha_{i-1,i} [v_{i-1}] + \lambda_i [v_i].$$

This implies that the difference between $T v_i$ and the linear combination of the v's must belong to the kernel of the quotient map, which is U:

$$T v_i = \alpha_{2i} v_2 + \cdots + \alpha_{i-1,i} v_{i-1} + \lambda_i v_i + u \quad \text{for some } u \in U.$$

Since $U = \operatorname{Sp}(v_1)$, we can write $u = \beta_i v_1$ for some scalar β_i. Thus,

$$T v_i = \beta_i v_1 + \alpha_{2i} v_2 + \cdots + \alpha_{i-1,i} v_{i-1} + \lambda_i v_i.$$

Because $T v_i$ can be expressed as a linear combination of $v_1, \ldots, v_i$ (and not $v_{i+1}, \ldots, v_{n+1}$), $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \beta_2 & \dots & \beta_{n+1} \\ 0 & \lambda_2 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda_{n+1} \end{pmatrix}.$$

This completes the inductive step, showing that T is trigonalizable. **Q.E.D.**

Solutions to the Exercises

Proof of Exercise 21.5:

If $q(x)$ is a non-zero constant there is nothing to prove, so assume $\deg q(x) \geq 1$ and hence $\deg p(x) \geq 2$. By hypothesis

$$p(x) = c \cdot (\mu_1 - x) \cdot \dots \cdot (\mu_m - x)$$

for some $c \in K \setminus \{0\}$, some $\mu_1, \dots, \mu_m \in K$ and $m := \deg p(x)$.

First, λ occurs among the μ_i. Indeed, substituting $x = \lambda$ into $p(x) = (\lambda - x)q(x)$ gives $p(\lambda) = 0$, so

$$c \cdot (\mu_1 - \lambda) \cdot \dots \cdot (\mu_m - \lambda) = 0.$$

A field has no zero divisors and $c \neq 0$, so $\mu_j = \lambda$ for at least one index j; after renumbering we may assume $\mu_1 = \lambda$.

Consequently,

$$(\lambda - x) \cdot q(x) = p(x) = (\lambda - x) \cdot \big(c \cdot (\mu_2 - x) \cdot \dots \cdot (\mu_m - x)\big).$$

The polynomial ring $K[x]$ over a field has no zero divisors — the degree of a product is the sum of the degrees — so we may cancel the non-zero factor $(\lambda - x)$ and obtain

$$q(x) = c \cdot (\mu_2 - x) \cdot \dots \cdot (\mu_m - x),$$

which is a product of linear factors and a constant, as claimed. **Q.E.D.**

Proof of Exercise 21.6:

Recall that the quotient map $\pi : V \to V/U$, $\pi(v) := [v]$, is linear and satisfies $[u] = 0$ exactly for $u \in U$.

Spanning. Let $[v] \in V/U$ be arbitrary and expand v in the basis of V:

$$v = a_1 u_1 + \dots + a_r u_r + b_{r+1} w_{r+1} + \dots + b_n w_n.$$

Applying π and using $[u_i] = 0$ for $i \leq r$ leaves $[v] = b_{r+1}[w_{r+1}] + \dots + b_n[w_n]$. Hence $([w_{r+1}], \dots, [w_n])$ spans V/U.

Linear independence. Suppose $b_{r+1}[w_{r+1}] + \dots + b_n[w_n] = 0$, that is, $b_{r+1}w_{r+1} + \dots + b_n w_n \in U$. Since $(u_1, \dots, u_r)$ is a basis for U, there are scalars $a_1, \dots, a_r$ with

$$b_{r+1}w_{r+1} + \dots + b_n w_n = a_1 u_1 + \dots + a_r u_r,$$

and therefore

$$a_1 u_1 + \dots + a_r u_r - b_{r+1} w_{r+1} - \dots - b_n w_n = 0.$$

This is a vanishing linear combination of the basis $(u_1, \dots, u_r, w_{r+1}, \dots, w_n)$ of V, so all its coefficients vanish; in particular $b_{r+1} = \dots = b_n = 0$.

Thus $([w_{r+1}], \dots, [w_n])$ is a basis for V/U, and counting its members gives $\dim V/U = n - r = \dim V - \dim U$. **Q.E.D.**

The Minimal Polynomial & The Cayley-Hamilton Theorem (Section 21.d)

A powerful tool in linear algebra is the ability to evaluate polynomials not just at scalar numbers, but at matrices and endomorphisms. Let $A \in \mathcal{M}_{n\times n}(K)$. For any integer $k \geq 1$, the matrix power $A^k = \underbrace{A \cdot \dots \cdot A}_{k}$ is well-defined. Therefore, for any formal polynomial $g(x) = a_d x^d + a_{d-1}x^{d-1} + \dots + a_1 x + a_0 \in K[x]$, we can define the matrix evaluation $g(A) \in \mathcal{M}_{n\times n}(K)$ naturally as:

$$g(A) := a_d A^d + a_{d-1}A^{d-1} + \dots + a_1 A + a_0 I_n.$$

A natural question arises: can we find a polynomial that completely *"kills"* or annihilates a given matrix?

Claim (*Existence of an Annihilating Polynomial*): For every matrix $A \in \mathcal{M}_{n\times n}(K)$, there exists a non-zero polynomial $g \in K[x]$ such that $g(A) = 0$.

Proof:

Consider the sequence of successive matrix powers. Since the vector space $\mathcal{M}_{n\times n}(K)$ has dimension n^2, we can choose an integer $r \geq n^2$ to construct a sequence of length $r+1$:

$$\underbrace{I_n,\ A,\ A^2,\ A^3,\ \ldots,\ A^r}_{r+1 \text{ elements in a space of dimension } n^2} \tag{21.20}$$

Because the number of elements $(r+1)$ strictly exceeds the dimension of the ambient space (n^2), the matrices in (21.20) cannot be linearly independent. This implies there exist scalars $a_0, a_1, \ldots, a_r \in K$, not all zero, such that:

$$a_0 I_n + a_1 A + a_2 A^2 + \cdots + a_r A^r = 0.$$

By setting $g(x) := a_r x^r + \cdots + a_1 x + a_0$, we obtain the desired non-zero annihilating polynomial.

Q.E.D.

A similar logic applies seamlessly to any abstract operator $T \in \operatorname{End}(V)$, where V is a finite-dimensional vector space over K, writing $T^k := \underbrace{T \circ \cdots \circ T}_{k}$ and defining $g(T) := a_d T^d + \cdots + a_1 T + a_0 \operatorname{id}_V$. Since annihilating polynomials exist, it is mathematically fruitful to look for the *"simplest"* one, which acts as a sort of algebraic DNA for the operator.

Definition 21.d.1 (*The Minimal Polynomial*): Let V be a vector space over K and $T \in \operatorname{End}(V)$. A "*minimal polynomial*" for T is a non-zero, monic polynomial $m_T(x) \in K[x]$ of the minimal possible degree such that $m_T(T) = 0$.

While the minimal polynomial guarantees that an operator satisfies *some* polynomial equation, one of the most celebrated theorems in mathematics provides a canonical polynomial that always works: the characteristic polynomial.

Theorem 21.d.2 (*Cayley-Hamilton*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then

(a) $P_A(A) = 0$.

(b) Let V be a finite-dimensional vector space over K, and $T \in \operatorname{End}(V)$. Then $P_T(T) = 0$.

Exercise 21.7 (*Cayley-Hamilton for 2×2 Matrices*): Let $A := \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Its characteristic polynomial is

$$P_A(x) = \det\begin{pmatrix} a-x & b \\ c & d-x \end{pmatrix} = (a-x)(d-x) - bc = x^2 - \underbrace{(a+d)}_{\operatorname{Tr}(A)}\, x + \underbrace{(ad-bc)}_{\det(A)}.$$

Verify Theorem 21.d.2 in this case by calculating

$$P_A(A) = A^2 - (a+d)\cdot A + (ad-bc)\cdot I_2$$

directly, and obtain $\left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$.

Exercise 21.8 (*Expressing Matrix Inverses via Cayley-Hamilton*): Let $A \in \mathcal{M}_{n\times n}(K)$ be an invertible matrix. Use Theorem 21.d.2 to prove that the inverse matrix A^{-1} can be expressed as a polynomial in A of degree at most $n-1$.

Proof of Cayley-Hamilton Theorem (Analytic Approach)

Before we dive into a purely algebraic mechanism, it is highly instructive to view this theorem through the lens of topology. The core idea is elegant: the theorem is trivially true for diagonal matrices. If we can show that diagonalizable matrices are *"dense"* (meaning almost all matrices are diagonalizable), we can use the continuity of polynomials to force the theorem to hold for *every* matrix.

First, we note that it is enough to prove part **(a)** (for matrices). Suppose we know that **(a)** is true, and let $T \in \mathrm{End}(V)$. Pick a basis $\mathcal{B}$ for V, and assign $A := [T]_{\mathcal{B}}^{\mathcal{B}} \in \mathcal{M}_{n\times n}(K)$.

Claim (*Matrix Representation of Polynomials*): $[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} = P_A(A)$.

Proof of Claim:

The map assigning an endomorphism to its matrix representation is an algebra isomorphism. Thus, $[R+S]_{\mathcal{B}}^{\mathcal{B}} = [R]_{\mathcal{B}}^{\mathcal{B}} + [S]_{\mathcal{B}}^{\mathcal{B}}$, and $[R \circ S]_{\mathcal{B}}^{\mathcal{B}} = [R]_{\mathcal{B}}^{\mathcal{B}} \cdot [S]_{\mathcal{B}}^{\mathcal{B}}$. Since $P_T(x) = P_A(x)$ by definition, we write $P_A(x) = c_n x^n + \cdots + c_0$. Then:

$$\begin{aligned}[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} &= [c_n T^n + \cdots + c_1 T + c_0 \,\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} \\ &= c_n [T^n]_{\mathcal{B}}^{\mathcal{B}} + \cdots + c_1 [T]_{\mathcal{B}}^{\mathcal{B}} + c_0 I_n \\ &= c_n A^n + \cdots + c_1 A + c_0 I_n = P_A(A).\end{aligned}$$

Q.E.D.

If Cayley-Hamilton holds for matrices, then $P_A(A) = 0$, meaning $[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} = 0$. Since the zero matrix uniquely represents the zero endomorphism, we conclude $P_T(T) = 0$. We now concentrate entirely on proving the theorem for matrices, beginning over the complex field $\mathbb{C}$.

Lemma 21.d.3 (*Cayley-Hamilton for Diagonalizable Matrices*): Suppose $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is diagonalizable. Then $P_A(A) = 0$.

Proof:

By assumption, there exists an invertible matrix Q such that

$$Q^{-1}AQ = \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

Since the roots of the characteristic polynomial are exactly the eigenvalues, we know $P_A(\lambda_i) = 0$ for all i. Writing $P_A(x) = \sum c_j x^j$, we substitute A:

$$Q^{-1}P_A(A)Q = \sum c_j Q^{-1}A^j Q = \sum c_j (Q^{-1}AQ)^j = \sum c_j \Lambda^j = P_A(\Lambda).$$

Because Λ is diagonal, $P_A(\Lambda)$ is simply the diagonal matrix with entries $P_A(\lambda_1), \ldots, P_A(\lambda_n)$. Since every entry is zero, $Q^{-1}P_A(A)Q = 0$, which immediately implies $P_A(A) = 0$. **Q.E.D.**

Lemma 21.d.4 (*Density of Diagonalizable Matrices*): Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be any arbitrary matrix. Then there exists a sequence of matrices $(A_k)_{k=1}^{\infty}$ such that:

(a) Each A_k is diagonalizable.

(b) $\lim_{k\to\infty} A_k = A$, in the entrywise sense that $\lim_{k\to\infty} A_k(i,j) = A(i,j)$ for all $1 \le i \le n$ and $1 \le j \le n$.

Proof:

Because $\mathbb{C}$ is algebraically closed, every matrix is trigonalizable (Corollary 21.c.6). Thus, there exists an invertible matrix $P \in \mathrm{GL}_n(\mathbb{C})$ such that

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & & b_{ij} \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

where $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ are *"the"* eigenvalues of A — or more precisely the zeros of $P_A(x)$, each of them appearing on the diagonal as many times as its algebraic multiplicity — and $b_{ij} \in \mathbb{C}$ for $1 \leq i < j \leq n$. Note that, equivalently,

$$A = P \begin{pmatrix} \lambda_1 & & b_{ij} \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} P^{-1}. \tag{21.21}$$

To ensure diagonalizability, we must prevent repeated eigenvalues, and we fix this by a tiny perturbation of the diagonal — one chosen afresh for each k. Let $k \in \mathbb{Z}_{\geq 1}$ and pick $\lambda_1(k), \ldots, \lambda_n(k) \in \mathbb{C}$ such that

(a) $|\lambda_i(k) - \lambda_i| < \frac{1}{2023 \cdot k}$ for every i, and

(b) $\lambda_1(k), \ldots, \lambda_n(k)$ are pairwise distinct.

This is always possible: distinct points may be chosen inside any collection of discs of positive radius, however small. Now define

$$A_k := P \begin{pmatrix} \lambda_1(k) & & b_{ij} \\ & \ddots & \\ 0 & & \lambda_n(k) \end{pmatrix} P^{-1},$$

keeping the very same P and the very same entries b_{ij} above the diagonal.

Clearly A_k has n distinct eigenvalues, namely $\lambda_1(k), \ldots, \lambda_n(k)$, and is therefore diagonalizable. (Alternatively: $P_{A_k}(x)$ splits into the n distinct linear factors $(\lambda_1(k) - x) \cdot \cdots \cdot (\lambda_n(k) - x)$, so $m_a(A_k; \lambda_i(k)) = 1$ for every i, and Corollary 21.c.2 applies.)

Finally, by condition **(a)** the middle matrix converges entrywise to the middle matrix of (21.21) as $k \to \infty$, and the map $X \mapsto PXP^{-1}$ is continuous (Exercise 21.9). Consequently,

$$\lim_{k\to\infty} A_k = P \left(\lim_{k\to\infty} \begin{pmatrix} \lambda_1(k) & & b_{ij} \\ & \ddots & \\ 0 & & \lambda_n(k) \end{pmatrix} \right) P^{-1} = P \begin{pmatrix} \lambda_1 & & b_{ij} \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} P^{-1} = A.$$

Q.E.D.

Exercise 21.9 (*Continuity of Two-Sided Multiplication*): Let $P, Q \in \mathcal{M}_{n\times n}(\mathbb{C})$ be given matrices. Show that the map

$$\mathcal{M}_{n\times n}(\mathbb{C}) \longrightarrow \mathcal{M}_{n\times n}(\mathbb{C}), \qquad X \longmapsto PXQ,$$

is continuous. (This is what licenses pulling the limit through P and P^{-1} at the end of the proof above.)

Lemma 21.d.5 (*Cayley-Hamilton over* $\mathbb{C}$): Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be any complex matrix. The Cayley-Hamilton theorem holds for A.

Proof:

By our construction in Lemma 21.d.4, we can find a sequence of diagonalizable matrices $A_k \to A$. By Lemma 21.d.3, the theorem holds for all of them: $P_{A_k}(A_k) = 0$.

Notice that the mapping $\Phi : \mathcal{M}_{n\times n}(\mathbb{C}) \to \mathcal{M}_{n\times n}(\mathbb{C})$ defined by $\Phi(M) = P_M(M)$ is built entirely from taking determinants, adding matrices, and multiplying matrices. Therefore, Φ is a continuous polynomial function of the matrix entries. Because continuous functions commute with limits, we have:

$$P_A(A) = \Phi(A) = \Phi\left(\lim_{k\to\infty} A_k\right) = \lim_{k\to\infty} \Phi(A_k) = \lim_{k\to\infty} 0 = 0.$$

Q.E.D.

To generalize this result from $\mathbb{C}$ to any arbitrary field K, we rely on a formal algebraic identity regarding multi-variable polynomials.

Here $\mathbb{R}[x_1, \dots, x_r]$ denotes the set of all polynomials in the variables $x_1, \dots, x_r$ with coefficients in $\mathbb{R}$, that is, the finite sums

$$f(x_1, \dots, x_r) = \sum_{\underline{i}\in\mathcal{I}} c_{\underline{i}}\, x_1^{i_1} \cdot \dots \cdot x_r^{i_r}, \qquad c_{\underline{i}} \in \mathbb{R},$$

taken over a finite set $\mathcal{I} \subseteq (\mathbb{Z}_{\geq 0})^r$ of multi-indices $\underline{i} = (i_1, \dots, i_r)$.

Lemma 21.d.6 (*Formal Zero of Multi-Variable Polynomials*)**:** Suppose $f(x_1, \dots, x_r) \in \mathbb{R}[x_1, \dots, x_r]$ satisfies $f(a_1, \dots, a_r) = 0$ for all $(a_1, \dots, a_r) \in \mathbb{R}^r$. Then $f \equiv 0$ as a formal polynomial (all coefficients are 0).

Proof:

Write $f(x_1, \dots, x_r) = \sum_{\underline{i}\in\mathcal{I}} c_{\underline{i}} x_1^{i_1} \dots x_r^{i_r}$, fix a multi-index $\underline{i} = (i_1, \dots, i_r) \in \mathcal{I}$, and put $m := i_1 + \dots + i_r$. We show that $c_{\underline{i}} = 0$.

Since $f(a_1, \dots, a_r) = 0$ for every $(a_1, \dots, a_r) \in \mathbb{R}^r$, the function f is identically zero on $\mathbb{R}^r$, and hence so are all of its partial derivatives, of any order, all over $\mathbb{R}^r$. In particular,

$$\left.\frac{\partial^m f}{\partial x_1^{i_1} \dots \partial x_r^{i_r}}\right|_{x_1=0,\dots,x_r=0} = 0. \tag{21.22}$$

On the other hand, we can compute that same derivative monomial by monomial. For a multi-index $\underline{j} = (j_1, \dots, j_r)$,

$$\frac{\partial^m}{\partial x_1^{i_1} \dots \partial x_r^{i_r}} \left(x_1^{j_1} \cdot \dots \cdot x_r^{j_r}\right) = \begin{cases} 0, & \text{if } j_k < i_k \text{ for some } k, \\ \displaystyle\prod_{k=1}^{r} j_k(j_k-1)\dots(j_k-i_k+1)\, x_k^{j_k-i_k}, & \text{if } j_k \geq i_k \text{ for all } k. \end{cases}$$

Now evaluate at the origin. In the first case the result is 0. In the second case, if $\underline{j} \neq \underline{i}$, then $j_k > i_k$ for at least one k, so a positive power of x_k survives and the value at the origin is again 0. Only $\underline{j} = \underline{i}$ contributes, and it contributes

$$\prod_{k=1}^{r} i_k(i_k-1)\dots 1 = i_1! \cdot \dots \cdot i_r!.$$

Summing over the monomials of f therefore gives

$$\left.\frac{\partial^m f}{\partial x_1^{i_1} \dots \partial x_r^{i_r}}\right|_{x_1=0,\dots,x_r=0} = c_{\underline{i}} \cdot i_1! \cdot \dots \cdot i_r!.$$

Comparing with (21.22) and dividing by the non-zero real number $i_1! \dots i_r!$ yields $c_{\underline{i}} = 0$, as claimed. **Q.E.D.**

Exercise 21.10 (*Where the Argument Breaks over Other Fields*)**:** What fails in the proof of Lemma 21.d.6 if we replace $\mathbb{R}[x_1, \dots, x_r]$ by $K[x_1, \dots, x_r]$ — even for $r = 1$ — with, for example, $K = \mathbb{Z}_2$, or some other field, finite or infinite?

Analytic Proof of Cayley-Hamilton over every field K:

The point of departure is that the determinant is given by one and the same formula over every field: for every $n \times n$ matrix B over every field K,

$$\det B = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot B_{1,\sigma 1} \cdot \dots \cdot B_{n,\sigma n}. \tag{21.23}$$

Consequently $P_A(x) = \det(A - x \cdot I_n)$ may be regarded as an element of $K[x, A_{11}, A_{12}, \dots, A_{nn}]$, a polynomial in the n^2+1 variables $x, A_{11}, \dots, A_{nn}$. But (21.23) shows more: the coefficients produced by that formula are sums of products of ± 1, so they lie in $\mathbb{Z}$, and hence in $\mathbb{R}$. Thus, $P_A(x) \in \mathbb{R}[x, A_{11}, \dots, A_{nn}]$, and the polynomial does not depend on K at all.

For an $n \times n$ matrix A, the entries of $P_A(A)$ are given by multi-variable polynomials in the n^2 variables $A_{11}, A_{12}, \dots, A_{nn}$. These polynomials have integer (and hence real) coefficients. Let the polynomial representing the (i,j)-th entry be $q_{ij}(A_{11}, \dots, A_{nn})$.

Since Cayley-Hamilton holds for $\mathbb{C}$ (and thus for any matrix in $\mathbb{R}$), we know $q_{ij}(A_{11}, \dots) = 0$ for all evaluations in $\mathbb{R}^{n^2}$. By Lemma 21.d.6, $q_{ij} \equiv 0$ as formal polynomials. Because they are formally the zero polynomial, evaluating them over *any* field K will always yield 0. Thus, $P_A(A) = 0$ for any matrix over any field. **Q.E.D.**

Example 21.7 (*The Polynomials q_{ij} for $n = 2$*): Let $n = 2$ and $A := \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, so that $P_A(x) = (a-x)(d-x) - bc = x^2 - (a+d)x + (ad-bc)$ as in Exercise 21.7. Then

$$\begin{aligned} P_A(A) &= \begin{pmatrix} a & b \\ c & d \end{pmatrix}^2 - (a+d) \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} \\ &= \begin{pmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{pmatrix} - (a+d) \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} \\ &= \begin{pmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{pmatrix}, \qquad q_{ij} \in \mathbb{R}[a,b,c,d]. \end{aligned}$$

This displays concretely what the proof above uses: the four entries of $P_A(A)$ are polynomial expressions in the entries of A, with coefficients that are plain integers and owe nothing to the field the entries are drawn from.

A Purely Algebraic Proof of Cayley-Hamilton Thm

While the analytic proof is mathematically beautiful, it relies on topological machinery (limits and continuity) that feels slightly alien in pure algebra. Let us now construct a purely structural, algebraic proof using invariant subspaces.

Definition 21.d.7 (*T-Cyclic Subspace*): Let V be a vector space over K and $T \in \operatorname{End}(V)$. Let $v \in V$. We define the "*T-cyclic subspace*" generated by v as the span of the orbit of v under T:

$$\langle v \rangle_T := \operatorname{Sp}\{T^k v \mid k \in \mathbb{Z}_{\geq 0}\} = \operatorname{Sp}(v, Tv, T^2 v, T^3 v, \dots) \subseteq V,$$

where, as usual, $T^0 := \mathrm{id}_V$.

Exercise 21.11 (*Minimality of the Cyclic Subspace*): Let V be a vector space over K, let $T \in \operatorname{End}(V)$ and let $v \in V$. Prove:

(a) $\langle v \rangle_T$ is invariant under T, that is, $T(\langle v \rangle_T) \subseteq \langle v \rangle_T$.

(b) $\langle v \rangle_T$ is the smallest linear subspace of V that contains v and is T-invariant. That is, if $U \subseteq V$ is a linear subspace with $v \in U$ and $T(U) \subseteq U$, then $\langle v \rangle_T \subseteq U$.

Lemma 21.d.8 (*Basis and Companion Matrix*): Suppose that $v, Tv, \dots, T^{d-1}v$ $(d \geq 1)$ are linearly independent, and $T^d v = c_0 v + c_1 T v + \dots + c_{d-1} T^{d-1} v$. Then:

(a) $\mathcal{B} = (v, Tv, \ldots, T^{d-1}v)$ forms a basis for $\langle v \rangle_T$.

(b) Let $S = T|_{\langle v \rangle_T}$. Then $[S]_{\mathcal{B}}^{\mathcal{B}}$ is a companion matrix:

$$[S]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & 0 & \ldots & 0 & c_0 \\ 1 & 0 & \ldots & 0 & c_1 \\ 0 & 1 & \ldots & 0 & c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 1 & c_{d-1} \end{pmatrix}.$$

Proof:

(a) Let $U := \mathrm{Sp}(\mathcal{B})$. Applying T to any basis vector $T^k v$ yields either the next basis element $T^{k+1}v \in U$, or (in the case of $k = d-1$) it yields $T^d v \in U$ due to the given linear combination. Thus $T(U) \subseteq U$, so U is a T-invariant subspace of V containing v. Clearly $U \subseteq \langle v \rangle_T$; and by part **(b)** of Exercise 21.11 — which identifies $\langle v \rangle_T$ as the smallest T-invariant subspace containing v — we also have $\langle v \rangle_T \subseteq U$. Hence $U = \langle v \rangle_T$. Since the elements of $\mathcal{B}$ are linearly independent by assumption and span this space, they form a basis for it.

(b) This follows directly from the definition of the matrix representation; each vector maps to the next standard basis vector, creating the 1s on the subdiagonal, and the last vector maps to the prescribed linear combination, populating the final column.

Q.E.D.

Lemma 21.d.9 (*Characteristic Polynomial of a Companion Matrix*): Under the assumptions of Lemma 21.d.8, the characteristic polynomial of the restriction S is given by:

$$P_S(x) = (-1)^d (x^d - c_{d-1}x^{d-1} - \cdots - c_1 x - c_0).$$

For $d = 1$ this reads $P_S(x) = -(x - c_0)$.

Proof:

We compute $\det([S]_{\mathcal{B}}^{\mathcal{B}} - xI_d)$ using cofactor expansion along the first row:

$$P_S(x) = \begin{vmatrix} -x & 0 & \ldots & 0 & c_0 \\ 1 & -x & \ldots & 0 & c_1 \\ 0 & 1 & \ldots & 0 & c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 1 & c_{d-1} - x \end{vmatrix}$$

Expanding across the first row, we get two non-zero terms:

$$P_S(x) = (-x)\det(M_{11}) + (-1)^{1+d} c_0 \det(M_{1d})$$

where M_{11} is the submatrix obtained by deleting the first row and first column, and M_{1d} is the submatrix obtained by deleting the first row and d-th column. Notice that M_{1d} is an upper triangular matrix with entirely 1s on its main diagonal. Thus, $\det(M_{1d}) = 1$. The matrix M_{11} is exactly the $(d-1) \times (d-1)$ matrix of the same shape, built from the constants $c_1, \ldots, c_{d-1}$ in place of $c_0, \ldots, c_{d-1}$.

We can now run the induction on d explicitly. For $d = 1$ the matrix is the 1×1 matrix $(c_0 - x)$, whose determinant is $c_0 - x = (-1)^1(x - c_0)$, which is the asserted formula. Let $d \geq 2$ and assume the formula for size $d - 1$. Applied to M_{11}, whose constants are $c_1, \ldots, c_{d-1}$, it gives

$$\det(M_{11}) = (-1)^{d-1} \left(x^{d-1} - c_{d-1}x^{d-2} - \cdots - c_2 x - c_1\right).$$

Substituting this and $\det(M_{1d}) = 1$ into the cofactor expansion above yields

$$\begin{aligned} P_S(x) &= (-x)\cdot(-1)^{d-1}\left(x^{d-1} - c_{d-1}x^{d-2} - \cdots - c_1\right) + (-1)^{d+1}c_0 \\ &= (-1)^d\left(x^d - c_{d-1}x^{d-1} - \cdots - c_1 x\right) + (-1)^d(-c_0) \\ &= (-1)^d\left(x^d - c_{d-1}x^{d-1} - \cdots - c_1 x - c_0\right), \end{aligned}$$

which is exactly the formula claimed. Here we used $(-1)^{d+1} = (-1)^{d-1} = -(-1)^d$ in both of the last two steps. **Q.E.D.**

Algebraic Proof of Cayley-Hamilton:

Let V be finite-dimensional, and $T \in \mathrm{End}(V)$. We want to show $P_T(T) = 0$, meaning $P_T(T)v = 0$ for all $v \in V$.

Let $v \in V$, $v \neq 0$. Consider the cyclic subspace $\langle v\rangle_T$. Take the maximal integer $d \geq 1$ such that $v, Tv, \ldots, T^{d-1}v$ are linearly independent. Then $T^d v = c_0 v + \cdots + c_{d-1}T^{d-1}v$ for some $c_0, \ldots, c_{d-1} \in K$, and $\dim\langle v\rangle_T = d$ by Lemma 21.d.8. Put $n := \dim V$. If $d < n$, extend this basis $\mathcal{B}$ to a full basis $\mathcal{C} := (v, \ldots, T^{d-1}v, w_1, \ldots, w_{n-d})$ for V. (If $d = n$, take $\mathcal{C} := \mathcal{B}$; the block C below is then empty and $P_C(x) = 1$, so the argument goes through unchanged.) The matrix representation of T with respect to $\mathcal{C}$ is block upper-triangular:

$$[T]^{\mathcal{C}}_{\mathcal{C}} = \begin{pmatrix} [S]^{\mathcal{B}}_{\mathcal{B}} & * \\ 0 & [C] \end{pmatrix}$$

Because the determinant of a block-triangular matrix is the product of the determinants of its diagonal blocks (Proposition 20.c.4), we have $P_T(x) = P_S(x)\cdot P_C(x)$. Consider the endomorphism $R := P_T(T) \in \mathrm{End}(V)$. Since polynomials evaluated at T commute with one another, we can factor R:

$$R = P_C(T) \circ P_S(T).$$

Therefore, $Rv = P_C(T)\Big(P_S(T)v\Big)$. However, applying Lemma 21.d.9 directly to v:

$$P_S(T)v = (-1)^d(T^d v - c_{d-1}T^{d-1}v - \cdots - c_0 v).$$

By the very definition of our constants c_i, this evaluates exactly to 0. Hence $Rv = P_C(T)(0) = 0$. Since $v \neq 0$ was chosen completely arbitrarily, $P_T(T)$ evaluates to zero on every vector in V, meaning $P_T(T) = 0$. **Q.E.D.**

Remark 21.4 (*The Lazy Proof*): A completely incorrect *"proof"* often presented by students is the following: $P_A(A) = \det(A - A\cdot I) = \det(0) = 0$. This is fundamentally invalid. The expression $P_A(x) = \det(A - xI)$ evaluates to a polynomial with scalar coefficients. Substituting matrices into a scalar expression *before* expanding the determinant leads to a severe category error. One must compute the scalar polynomial first, and only then evaluate it at the matrix A. Note also that the two sides do not even live in the same place: $\det(A - xI)$ is a **scalar**, whereas $P_A(A)$ is an $n \times n$ **matrix**, so the very first equality of the *"proof"* equates a scalar with a matrix.

Solutions to the Exercises

Solution of Exercise 21.7:

Write $A := \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ and compute the three summands separately:

$$A^2 = \begin{pmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{pmatrix}, \qquad -(a+d)\cdot A = \begin{pmatrix} -a^2-ad & -ab-bd \\ -ac-cd & -ad-d^2 \end{pmatrix},$$

$$(ad-bc)\cdot I_2 = \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix}.$$

Adding them entry by entry:

$$\begin{aligned} (1,1):&\quad (a^2+bc)+(-a^2-ad)+(ad-bc)=0,\\ (1,2):&\quad (ab+bd)+(-ab-bd)+0=0,\\ (2,1):&\quad (ac+cd)+(-ac-cd)+0=0,\\ (2,2):&\quad (bc+d^2)+(-ad-d^2)+(ad-bc)=0. \end{aligned}$$

Hence $P_A(A) = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, confirming Theorem 21.d.2 for $n=2$. Note that every cancellation above is an identity in the four symbols a,b,c,d with integer coefficients — it uses nothing about the field K, which is precisely the observation exploited in Example 21.7. **Q.E.D.**

Proof of Exercise 21.8:

Write $P_A(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$. Its constant coefficient is

$$c_0 = P_A(0) = \det(A - 0\cdot I_n) = \det(A),$$

which is non-zero precisely because A is invertible. By Theorem 21.d.2,

$$c_n A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0,$$

so, moving the constant term to the other side and factoring out one copy of A (which is legitimate, since A commutes with every power of itself),

$$c_0 I_n = -\left(c_n A^n + \cdots + c_1 A\right) = A\cdot\left(-c_n A^{n-1} - c_{n-1}A^{n-2} - \cdots - c_1 I_n\right).$$

Dividing by the non-zero scalar $c_0 = \det(A)$ and multiplying by A^{-1} gives

$$A^{-1} = -\frac{1}{\det(A)}\left(c_n A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_2 A + c_1 I_n\right),$$

which exhibits A^{-1} as a polynomial in A of degree at most $n-1$, as required.

For $n=2$ this recovers the familiar formula: there $c_2 = 1$ and $c_1 = -(a+d) = -\operatorname{Tr}(A)$, so

$$A^{-1} = \frac{1}{\det(A)}\left(\operatorname{Tr}(A)\cdot I_2 - A\right) = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Q.E.D.

Proof of Exercise 21.9:

Identify $\mathcal{M}_{n\times n}(\mathbb{C})$ with $\mathbb{C}^{n^2}$ by listing the entries; under this identification, convergence of matrices is exactly entrywise convergence, which is what the statement of Lemma 21.d.4 asks for. A map into $\mathbb{C}^{n^2}$ is continuous if and only if each of its n^2 coordinate functions is continuous, so it suffices to look at one entry of PXQ at a time. By the definition of matrix multiplication,

$$(PXQ)_{ij} = \sum_{k=1}^{n}\sum_{\ell=1}^{n} P_{ik}\, X_{k\ell}\, Q_{\ell j}.$$

Here P_{ik} and $Q_{\ell j}$ are fixed complex numbers, so the right-hand side is a linear — in particular polynomial — expression in the n^2 coordinates $X_{k\ell}$ of X. Polynomial functions $\mathbb{C}^{n^2} \to \mathbb{C}$ are continuous, being built from the (continuous) coordinate projections by finitely many additions and multiplications. Hence every entry of PXQ depends continuously on X, and so does PXQ itself. Taking $Q := P^{-1}$ gives the case used in Lemma 21.d.4: we have $\lim_{k\to\infty} PB_kP^{-1} = P\,(\lim_{k\to\infty} B_k)\,P^{-1}$ whenever the limit on the right exists. **Q.E.D.**

Solution of Exercise 21.10:

Two separate things go wrong, and it is worth keeping them apart.

The proof breaks immediately. The argument is analytic: it differentiates f as a **function** on $\mathbb{R}^r$, and differentiation rests on limits, which a bare field K does not provide. One can of course define the formal derivative of a polynomial without any analysis, but the step that made the proof work — *"f vanishes identically, therefore all its derivatives vanish identically"* — is a statement about functions and is no longer available. Even granting formal derivatives, the last step divides by $i_1! \cdot \dots \cdot i_r!$, and in a field of characteristic p those factorials can be 0: for $K = \mathbb{Z}_2$ and $i = 2$ already $2! = 0$ in K.

For finite K the statement itself is false, so no repair of the proof is possible. Take $K = \mathbb{Z}_2$ and $r = 1$. Then $f(x) := x^2 + x$ satisfies $f(0) = 0$ and $f(1) = 1 + 1 = 0$, so f vanishes at every point of K, yet f is not the zero polynomial. This is exactly the phenomenon of Important remark 21.1, and Exercise 21.4 shows that it occurs over every finite field.

For infinite K the statement survives, but it must be proved algebraically rather than analytically. Induct on r. The case $r = 1$ is the content of Exercise 21.4: a non-zero polynomial in one variable has at most deg many zeros, so over an infinite field it cannot vanish everywhere. For $r \geq 2$, collect f by powers of the last variable,

$$f(x_1, \dots, x_r) = \sum_j g_j(x_1, \dots, x_{r-1}) \cdot x_r^j,$$

and fix any $(a_1, \dots, a_{r-1}) \in K^{r-1}$. The one-variable polynomial $\sum_j g_j(a_1, \dots, a_{r-1})\, x_r^j$ then vanishes at every point of K, so by the case $r = 1$ all of its coefficients $g_j(a_1, \dots, a_{r-1})$ are 0. As $(a_1, \dots, a_{r-1})$ was arbitrary, each g_j vanishes identically on K^{r-1}, hence $g_j \equiv 0$ by the inductive hypothesis, and therefore $f \equiv 0$.

This is why the proof of Cayley-Hamilton over an arbitrary field routes through $\mathbb{R}$: Lemma 21.d.6 is applied only over $\mathbb{R}$, where it is both true and easy, and its conclusion — that the q_{ij} are the zero polynomial **formally** — is then field-independent. **Q.E.D.**

Proof of Exercise 21.11:

(a) Let $u \in \langle v \rangle_T$. By the definition of the span, $u = a_0 v + a_1 Tv + \dots + a_m T^m v$ for some $m \in \mathbb{Z}_{\geq 0}$ and $a_0, \dots, a_m \in K$. Applying the linear map T gives

$$Tu = a_0 Tv + a_1 T^2 v + \dots + a_m T^{m+1} v,$$

which is again a linear combination of vectors of the form $T^k v$, and hence lies in $\langle v \rangle_T$. Therefore $T(\langle v \rangle_T) \subseteq \langle v \rangle_T$.

(b) Let $U \subseteq V$ be a linear subspace with $v \in U$ and $T(U) \subseteq U$. We show by induction on k that $T^k v \in U$ for every $k \in \mathbb{Z}_{\geq 0}$. For $k = 0$ this says $v \in U$, which holds by assumption. If $T^k v \in U$, then $T^{k+1} v = T(T^k v) \in T(U) \subseteq U$, which completes the induction.

Thus $\{T^k v \mid k \in \mathbb{Z}_{\geq 0}\} \subseteq U$. Since U is a linear subspace, it contains every linear combination of its elements, and therefore contains the span of that set, which is exactly $\langle v \rangle_T$. Hence $\langle v \rangle_T \subseteq U$.

Together, **(a)** and **(b)** say that $\langle v \rangle_T$ is itself a T-invariant subspace containing v and is contained in every other one — that is, it is the smallest such subspace, which is the property used in the proof of Lemma 21.d.8. **Q.E.D.**

Euclidean and Hermitian Spaces (Chapter 22)

Euclidean and Hermitian (Unitary) spaces (Section 22.a)

At last (!) we endow our spaces with a geometric structure. We work here with vector spaces V over $K = \mathbb{R}$ or $K = \mathbb{C}$. Over $\mathbb{R}$ we will have Euclidean spaces. Over $\mathbb{C}$ we will have Hermitian (a.k.a. unitary) spaces.

Definition 22.a.1 (*Scalar Product*): Let V be a vector space over $\mathbb{R}$. A "*scalar product*" (a.k.a. inner product) is a function $V \times V \longrightarrow \mathbb{R}$, denoted by $v, w \mapsto \langle v, w\rangle$ (or (v, w)), that has the following properties:

(a) Linearity in the 1^{st} variable:

$$\langle v_1 + v_2, w\rangle = \langle v_1, w\rangle + \langle v_2, w\rangle \quad \forall v_1, v_2, w \in V$$
$$\langle \alpha v, w\rangle = \alpha\langle v, w\rangle \quad \forall \alpha \in \mathbb{R},\ v, w \in V.$$

(b) Linearity in the 2^{nd} variable:

$$\langle v, w_1 + w_2\rangle = \langle v, w_1\rangle + \langle v, w_2\rangle \quad \forall v, w_1, w_2 \in V$$
$$\langle v, \alpha w\rangle = \alpha\langle v, w\rangle \quad \forall \alpha \in \mathbb{R},\ v, w \in V.$$

(c) Symmetry:

$$\langle v, w\rangle = \langle w, v\rangle \quad \forall v, w \in V.$$

(d) Positivity:

$$\forall 0 \neq v \in V, \quad \langle v, v\rangle > 0.$$

We call $(V, \langle\cdot,\cdot\rangle)$ a "*Euclidean space*", or a space endowed with a scalar product.

Definition 22.a.2 (*Hermitian Product*): Let V be a vector space over $\mathbb{C}$. A "*Hermitian product*" (or inner product) on V is a function $V \times V \longrightarrow \mathbb{C}$, denoted by $v, w \mapsto \langle v, w\rangle$, with the following properties:

(a) Linearity in the 1^{st} variable:

$$\langle v_1 + v_2, w\rangle = \langle v_1, w\rangle + \langle v_2, w\rangle \quad \forall v_1, v_2, w \in V$$
$$\langle \alpha v, w\rangle = \alpha\langle v, w\rangle \quad \forall \alpha \in \mathbb{C},\ v, w \in V.$$

(b) Sesquilinearity (complex anti-linearity) in the 2^{nd} variable:

$$\langle v, w_1 + w_2\rangle = \langle v, w_1\rangle + \langle v, w_2\rangle \quad \forall v, w_1, w_2 \in V$$
$$\langle v, \alpha w\rangle = \overline{\alpha}\langle v, w\rangle \quad \forall \alpha \in \mathbb{C},\ v, w \in V.$$

(c) Hermitian property:

$$\langle v, w\rangle = \overline{\langle w, v\rangle} \quad \forall v, w \in V.$$

(d) Positivity:

$$\forall 0 \neq v \in V, \quad \langle v, v\rangle > 0.$$

We call $(V, \langle\cdot,\cdot\rangle)$ a "*Hermitian*" (or unitary) space.

Remark 22.1: Recall that if $\alpha = a + ib \in \mathbb{C}$, where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$, then $\overline{\alpha} := a - ib$ is the complex conjugate of α. Recall also that $\alpha \cdot \overline{\alpha} = a^2 + b^2 = |\alpha|^2 \geq 0$, with equality if and only if $\alpha = 0$.

The definition of a Hermitian product *"coincides"* with a scalar product if V is over $\mathbb{R}$, and we allow α only in $\mathbb{R}$.

Lemma 22.a.3: Let V be a Euclidean/Hermitian vector space. Then:

(a) $\forall v \in V$, $\langle 0, v\rangle = \langle v, 0\rangle = 0$.
(b) Let $w \in V$. If $\langle v, w\rangle = 0$ for all $v \in V$, then $w = 0$.
(c) Let $w_1, w_2 \in V$. If $\langle v, w_1\rangle = \langle v, w_2\rangle$ for all $v \in V$, then $w_1 = w_2$.

All three parts are left as an exercise.

Definition 22.a.4:

(a) Let V be a Euclidean/Hermitian vector space. For all $v \in V$, define $\|v\| := \sqrt{\langle v, v\rangle} \in \mathbb{R}_{\geq 0}$. The mapping $\|\cdot\|$ is called the "*norm*" on V induced by $\langle \cdot, \cdot \rangle$.
(b) A vector $u \in V$ is called a "*unit vector*" if $\|u\| = 1$. Note that if $0 \neq v \in V$ is any vector, then $\frac{1}{\|v\|}v$ is a unit vector (which is positively proportional to v, so it points in the same direction as v). We call $\frac{v}{\|v\|}$ the "*normalization*" of v.
(c) For $v, w \in V$, we define the "*distance*" between v and w by $d(v, w) := \|v - w\|$.
(d) Two vectors $u, v \in V$ are called "*orthogonal*" (or perpendicular one to the other) if $\langle u, v\rangle = 0$. If this is the case, we sometimes write $u \perp v$.
(e) A subset $S \subseteq V$ is called an "*orthogonal system*" if $0 \notin S$ and for every two distinct elements $u, v \in S$ we have $u \perp v$.
(f) An orthogonal system $S \subseteq V$ is called "*orthonormal*" if for all $u \in S$ we have $\|u\| = 1$.

Theorem 22.a.5 (*Pythagoras' Theorem*): Let V be a Euclidean/Hermitian vector space, and $u, v \in V$ with $u \perp v$. Then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

Proof:

By the definition of the norm and the properties of the inner product, we have:

$$\begin{aligned}\|u+v\|^2 &= \langle u+v, u+v\rangle \\ &= \langle u, u\rangle + \underbrace{\langle u, v\rangle}_{=0} + \underbrace{\langle v, u\rangle}_{=0} + \langle v, v\rangle \\ &= \|u\|^2 + \|v\|^2.\end{aligned}$$

Note that in the Euclidean case, $\langle u, v\rangle = 0$ directly. In the Hermitian case, $\langle v, u\rangle = \overline{\langle u, v\rangle} = \overline{0} = 0$.

Q.E.D.

Exercise 22.1 (*Polarization Identities*):

(a) Let V be a Euclidean space. Show that for all $u, v \in V$ we have

$$\langle u, v\rangle = \frac{1}{2}\left(\|u+v\|^2 - \|u\|^2 - \|v\|^2\right). \quad (*)$$

This shows that $\|\cdot\|$ determines $\langle \cdot, \cdot\rangle$. **Warning:** There exist norms that do **not** come from scalar products.
(b) Find an analogous formula to (*) for a Hermitian vector space.

Examples

(1) The standard scalar product on $\mathbb{R}^n$. Let $\mathbb{R}^n = \mathbb{R}^n_{\text{col}}$. For all $u, v \in \mathbb{R}^n$, define $\langle u, v\rangle = u^\mathsf{T} \cdot v \in \mathcal{M}_{1\times 1}(\mathbb{R}) = \mathbb{R}$.

$$\left\langle \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \right\rangle = \sum_{i=1}^n u_i \cdot v_i$$

$$\left\| \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \right\| = \sqrt{\sum_{i=1}^n v_i^2}.$$

(2) The standard Hermitian inner product on $\mathbb{C}^n$:

$$\langle u, v\rangle := u^\mathsf{T} \cdot \overline{v} = \sum_{k=1}^n u_k \cdot \overline{v}_k \in \mathbb{C}$$

$$\|v\| = \sqrt{\sum_{k=1}^n v_k \cdot \overline{v}_k} = \sqrt{\sum_{k=1}^n |v_k|^2}.$$

(3) If $(V, \langle\cdot,\cdot\rangle)$ is a Euclidean/Hermitian space and $W \subseteq V$ is a subspace, we can restrict $\langle\cdot,\cdot\rangle$ to W and obtain a Euclidean/Hermitian structure on W.

Definition 22.a.6: A matrix $A \in \mathcal{M}_{n\times n}(K)$ (where K is any field) is called "*symmetric*" if $A^\mathsf{T} = A$ (i.e., $a_{ij} = a_{ji}$ for all i, j).

Definition 22.a.7:

(a) Let $A \in \mathcal{M}_{n\times m}(\mathbb{C})$. We write $\overline{A}$ for the matrix whose entry at i, j is $\overline{(A_{ij})}$ (complex conjugation). That is, $(\overline{A})_{ij} := \overline{(A_{ij})}$.

(b) Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$. Define $A^* := \overline{A}^\mathsf{T}$. The matrix A^* is called the "*adjoint matrix*" of A (this has nothing to do with the classical adjoint, $\operatorname{adj}(A)$).

(c) A matrix $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is called "*Hermitian*" if $A^* = A$ ($\iff A^\mathsf{T} = \overline{A}$).

Important example. Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$. Define $\langle\cdot,\cdot\rangle_A : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$\langle v, w\rangle_A = v^\mathsf{T} \cdot A \cdot w.$$

Exercise 22.2 (*Bilinearity of $\langle\cdot,\cdot\rangle_A$*): $\langle\cdot,\cdot\rangle_A$ is bilinear (i.e., linear in both variables).

Assume now that A is symmetric. Then $\langle v, w\rangle_A = \langle w, v\rangle_A$ for all v, w. Indeed,

$$\langle w, v\rangle_A = w^\mathsf{T} \cdot A \cdot v = w^\mathsf{T} A^\mathsf{T} v = (v^\mathsf{T} A \cdot w)^\mathsf{T} = (\langle v, w\rangle_A)^\mathsf{T} = \langle v, w\rangle_A.$$

(Note that this is just a scalar, viewed as a 1×1 matrix).

Is $\langle\cdot,\cdot\rangle_A$ a scalar product? In general, it is not. For example, $A = \left(\begin{smallmatrix} -1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. Then, $\langle \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)\rangle_A = -1$. But for a scalar product, we need $\langle v, v\rangle > 0$ for all $v \neq 0$. So we lack positivity here.

Definition 22.a.8: A symmetric matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is called "*positive definite*" if $v^\mathsf{T} A v > 0$ for all $0 \neq v \in \mathbb{R}^n$.

Exercise 22.3 (*Positive Definiteness Yields a Scalar Product*): If A is positive definite, then $\langle\cdot,\cdot\rangle_A$ is a scalar product.

Claim: If $\langle\cdot,\cdot\rangle_A$ is a scalar product, then A is positive definite. (Proof: a bit later).

Example 22.1:

(a) Let $a_1, \dots, a_n > 0$ (with $a_i \in \mathbb{R}$). Then the diagonal matrix $D = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix}$ is positive definite. Indeed, if $v = (v_1, \dots, v_n)^\mathsf{T}$, then

$$\langle v, v\rangle_D = v^\mathsf{T} \cdot D \cdot v = a_1 v_1^2 + \cdots + a_n v_n^2 \geq 0,$$

with equality if and only if $v_i = 0$ for all i.

(b) Let $A = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right) \in \mathcal{M}_{2\times 2}(\mathbb{R})$. Then A is positive definite if and only if $a > 0$ and $\det(A) > 0$. (Exercise).

The situation for Hermitian products is similar.

Definition 22.a.9: Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be a Hermitian matrix. We say that A is "*positive definite*" if for all $0 \neq v \in \mathbb{C}^n$, we have $v^\mathsf{T} A\overline{v} > 0$. (Note that $v^\mathsf{T} A\overline{v} \in \mathbb{R}$, because $\overline{(v^\mathsf{T} A\overline{v})} = \overline{v}^\mathsf{T}\overline{A}v = (\overline{v}^\mathsf{T}\overline{A}v)^\mathsf{T} = v^\mathsf{T}\overline{A}^\mathsf{T}\overline{v} = v^\mathsf{T} A\overline{v}$. Here, we used $(\cdot)^\mathsf{T} = (\cdot)$ for a scalar/1×1 matrix, and the fact that A is Hermitian. Recall that if $v = (v_1, \dots, v_n)^\mathsf{T} \in \mathbb{C}^n$, then $\overline{v} := (\overline{v}_1, \dots, \overline{v}_n)^\mathsf{T}$).

Given a positive definite complex matrix A, we can define a Hermitian product on $\mathbb{C}^n$ via $\langle v, w\rangle_A = v^\mathsf{T} A\overline{w}$.

Claim: $\langle \cdot, \cdot \rangle_A$ is a Hermitian product if and only if A is positive definite.

Exercise 22.4 (*Positive Definiteness of 2×2 Hermitian Matrices*)**:** $A = \left(\begin{smallmatrix} a & b \\ \overline{b} & d \end{smallmatrix}\right) \in \mathcal{M}_{2\times 2}(\mathbb{C})$ is positive definite if and only if $a \in \mathbb{R}$, $a > 0$, and $\det A \in \mathbb{R}$, $\det A > 0$.

Proof of the Claim:

We will prove it for complex matrices and Hermitian products (the case of symmetric matrices and scalar products is very similar and even simpler — see Exercise 22.5).

Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ and suppose that $\langle v, w\rangle_A := v^\mathsf{T} A\overline{w}$ is a Hermitian product. Then for all $v, w \in \mathbb{C}^n$ we have $\langle v, w\rangle_A = \overline{\langle w, v\rangle_A}$.

$$\implies v^\mathsf{T} A\overline{w} = \overline{w^\mathsf{T} A\overline{v}} \implies v^\mathsf{T} A\overline{w} = \overline{w}^\mathsf{T}\overline{A}v \implies v^\mathsf{T} A\overline{w} = (\overline{w}^\mathsf{T}\overline{A}v)^\mathsf{T} \implies v^\mathsf{T} A\overline{w} = v^\mathsf{T}\overline{A}^\mathsf{T}\overline{w}.$$

Take $v = e_i$, $w = e_j$ (standard basis). Write $A := \begin{pmatrix} a_{11} & \dots & a_{1j} & \dots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \dots & a_{nj} & \dots & a_{nn} \end{pmatrix}$. We have

$$e_i^\mathsf{T} \cdot A \cdot e_j = \begin{pmatrix} - & e_i & - \end{pmatrix} A \begin{pmatrix} | \\ e_j \\ | \end{pmatrix} = \begin{pmatrix} - & e_i & - \end{pmatrix} \begin{pmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{pmatrix} = a_{ij}.$$

Similarly, $e_i^\mathsf{T}\overline{A}^\mathsf{T} e_j = \overline{a}_{ji}$. Thus, $\overline{a}_{ji} = a_{ij}$ for all i, j, which implies $\overline{A}^\mathsf{T} = A$.

The fact that A is positive definite is precisely the condition that $\langle v, v\rangle_A > 0$ for all $v \neq 0$, which is to say $v^\mathsf{T} A\overline{v} > 0$.

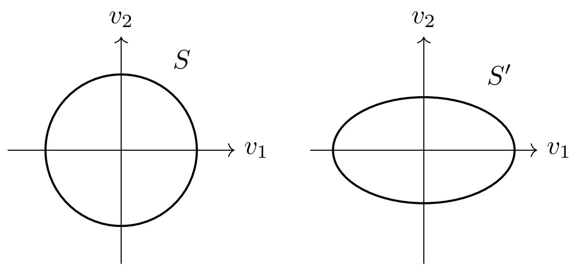
The opposite statement — that a positive definite matrix A makes $\langle \cdot, \cdot \rangle_A$ a Hermitian product — is Exercise 22.6. **Q.E.D.**

Exercise 22.5 (*The Real Case of the Claim*)**:** Carry out the argument above for a real matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ and the form $\langle v, w\rangle_A = v^\mathsf{T} Aw$, thereby proving the Claim on Page 196: if $\langle \cdot, \cdot \rangle_A$ is a scalar product, then A is symmetric and positive definite.

Exercise 22.6 (*Positive Definiteness Yields a Hermitian Product*)**:** Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be a positive definite Hermitian matrix. Show that $\langle v, w\rangle_A := v^\mathsf{T} A\overline{w}$ is a Hermitian product on $\mathbb{C}^n$.

The geometry on V depends on the choice of inner product

Example 22.2: Let $V = \mathbb{R}^2$, $\langle v, w\rangle = v_1w_1 + v_2w_2$ (the standard scalar product on $\mathbb{R}^2$). Let $\langle v, w\rangle' = a_1v_1w_1 + a_2v_2w_2$, where $a_1, a_2 > 0$. Note that $\langle v, w\rangle'$ is $\langle v, w\rangle_A$ for $A = \left(\begin{smallmatrix} a_1 & 0 \\ 0 & a_2\end{smallmatrix}\right)$.
The unit sphere for $\langle\cdot,\cdot\rangle$: $S = \{v \in \mathbb{R}^2 \mid \|v\| = 1\} = \{(v_1, v_2) \mid v_1^2 + v_2^2 = 1\}$.
The unit sphere for $\langle\cdot,\cdot\rangle'$: $S' = \{v \in \mathbb{R}^2 \mid \|v\|' = 1\} = \{(v_1, v_2) \mid a_1v_1^2 + a_2v_2^2 = 1\}$.



Example 22.3: Let $A = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2\end{smallmatrix}\right)$. It is positive definite (why?). Let $\langle\cdot,\cdot\rangle :=$ std. scalar prod, and $\langle\cdot,\cdot\rangle' := \langle\cdot,\cdot\rangle_A$. We have:

$$\left\langle \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}\right\rangle = 1, \quad \left\langle \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}\right\rangle' = \begin{pmatrix}1 & 0\end{pmatrix}\cdot\begin{pmatrix}1 & -1\\ -1 & 2\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1 & -1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = 0.$$

So $\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) \perp' \left(\begin{smallmatrix}1\\1\end{smallmatrix}\right)$ but $\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) \not\perp \left(\begin{smallmatrix}1\\1\end{smallmatrix}\right)$.

A different type of example

Let $C[a,b] =$ space of continuous functions $f : [a,b] \to \mathbb{R}$. It is a vector space over $\mathbb{R}$ with respect to the standard operations. We define a scalar product:

$$\langle f, g\rangle := \int_a^b f(x)\cdot g(x)\,dx.$$

(This is well-defined and finite. Why?)

Exercise 22.7 (*The Integral Pairing is Bilinear and Symmetric*): Show that $\langle\cdot,\cdot\rangle$ is linear in both variables and symmetric.

Let's check positivity: let $0 \neq f \in C[a,b]$. Then $\langle f, f\rangle = \int_a^b f(x)^2\,dx$. Since $f \neq 0$, there exists $x_0 \in (a,b)$ such that $f(x_0) \neq 0$. Since f is continuous, so is f^2. Hence, there exists $\delta > 0$ such that $[x_0-\delta, x_0+\delta] \subseteq [a,b]$ and $f(x) \neq 0$ for all $x \in [x_0-\delta, x_0+\delta]$. This implies $f(x)^2 > 0$ for all $x \in [x_0-\delta, x_0+\delta]$. Therefore,

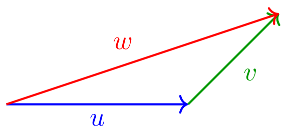
$$\langle f, f\rangle = \int_a^b f(x)^2\,dx \geq \int_{x_0-\delta}^{x_0+\delta} f(x)^2\,dx \geq 2\delta\cdot \min_{x\in[x_0-\delta,x_0+\delta]} f(x)^2 > 0.$$

Remark 22.2: By this scalar product on $C[-\pi,\pi]$, the functions $f(x) = \sin x$ and $g(x) \equiv 1$ are orthogonal ($f \perp g$).

Norms and angles

Definition 22.a.10: Let V be a vector space over K (where $K = \mathbb{R}$ or $\mathbb{C}$). A "*norm*" on V is a function $\|\cdot\| : V \to \mathbb{R}_{\geq 0}$ such that:

(a) $\|u+v\| \leq \|u\| + \|v\|$ for all $u, v \in V$. (Triangle inequality)
(b) $\|\alpha\cdot v\| = |\alpha|\cdot\|v\|$ for all $\alpha \in K$, $v \in V$. (Homogeneity)
(c) If $\|v\| = 0$, then $v = 0$. (Non-degeneracy)



Claim: If $\langle\cdot,\cdot\rangle$ is an inner product on V, then $\|v\| := \sqrt{\langle v,v\rangle}$ is a norm on V. (We'll prove this soon).

Exercise 22.8 (*Examples of Non-Inner-Product Norms*)**:** Define $\left\|\binom{a}{b}\right\|' := \max\{|a|,|b|\}$, and $\left\|\binom{a}{b}\right\|'' := |a|+|b|$ for all $\binom{a}{b} \in \mathbb{R}^2$. Show that $\|\cdot\|'$ and $\|\cdot\|''$ are norms, but they are not induced by any scalar product.
Hint: Recall that if the scalar product $\langle\cdot,\cdot\rangle$ induces $\|\cdot\|$, then $\langle u,v\rangle = \frac{1}{2}(\|u+v\|^2 - \|u\|^2 - \|v\|^2)$.

Theorem 22.a.11 (*Cauchy-Schwarz Inequality*)**:** Let $(V,\langle\cdot,\cdot\rangle)$ be a vector space endowed with an inner product $\langle\cdot,\cdot\rangle$. Then for all $u,v\in V$ we have

$$|\langle u,v\rangle| \le \|u\|\cdot\|v\|,$$

with equality if and only if u and v are proportional (i.e., $u=\alpha\cdot v$, or $v=\alpha\cdot u$ for some $\alpha\in K \iff u,v$ are linearly dependent).

Proof:

If $u=0$, the (in)equality holds, so assume $u\neq 0$. Put $w := v-\lambda\cdot u$, where $\lambda := \frac{\langle v,u\rangle}{\|u\|^2}$. (*)

We have $\langle w,u\rangle = \langle v,u\rangle - \lambda\langle u,u\rangle = 0$. Now:

$$\begin{aligned}
0 \le \|w\|^2 &= \langle v-\lambda u, v-\lambda u\rangle\\
&= \langle v,v\rangle - \overline{\lambda}\langle v,u\rangle - \lambda\langle u,v\rangle + \lambda\overline{\lambda}\langle u,u\rangle\\
&= \|v\|^2 - \overline{\lambda}\cdot\lambda\|u\|^2 - \lambda\cdot\overline{\lambda}\|u\|^2 + \lambda\cdot\overline{\lambda}\|u\|^2 \quad \left(\text{since } \langle v,u\rangle = \lambda\|u\|^2 \text{ and } \langle u,v\rangle = \overline{\langle v,u\rangle} = \overline{\lambda}\|u\|^2\right)\\
&= \|v\|^2 - |\lambda|^2\|u\|^2\\
&= \|v\|^2 - \frac{|\langle v,u\rangle|^2}{\|u\|^4}\cdot\|u\|^2\\
&= \|v\|^2 - \frac{|\langle v,u\rangle|^2}{\|u\|^2}.
\end{aligned}$$

This implies $0 \le \|v\|^2 - \frac{|\langle v,u\rangle|^2}{\|u\|^2}$, which implies $|\langle v,u\rangle| \le \|u\|\cdot\|v\|$.

We have equality if and only if either $u=0$, or $w=0$. But $w=0 \iff v=\lambda u$ where $\lambda = \frac{\langle v,u\rangle}{\|u\|^2}$. But for $u\neq 0$, the condition $v=\lambda u$ implies $\lambda = \frac{\langle v,u\rangle}{\|u\|^2}$. So, equality if and only if either $u=0$ or $v=\lambda u$ for some $\lambda\in K \iff u,v$ are linearly dependent. **Q.E.D.**

Corollary 22.a.12: Let V be a vector space and $\langle\cdot,\cdot\rangle$ an inner product on V. Then $V\ni v\mapsto \|v\| := \sqrt{\langle v,v\rangle}$ is a norm on V.

Proof:

We assume $(V,\langle\cdot,\cdot\rangle)$ is a Hermitian space (the case of Euclidean space is very similar).

Triangle inequality: Let $u,v\in V$.

$$\begin{aligned}
\|u+v\|^2 &= \langle u+v,u+v\rangle = \langle u,u\rangle + \langle u,v\rangle + \langle v,u\rangle + \langle v,v\rangle\\
&= \|u\|^2 + 2\operatorname{Re}\langle u,v\rangle + \|v\|^2. \quad (*)
\end{aligned}$$

Note that for all $z\in\mathbb{C}$ we have $|\operatorname{Re} z| \le |z|$. (Indeed, if $z=x+iy$, $x,y\in\mathbb{R}$, then $|\operatorname{Re} z| = |x| \le \sqrt{x^2+y^2} = |z|$). Applying this to our situation, we have $\operatorname{Re}\langle u,v\rangle \le |\operatorname{Re}\langle u,v\rangle| \le |\langle u,v\rangle|$. Going back to (*), we get:

$$\|u+v\|^2 \leq \|u\|^2 + 2|\langle u,v\rangle| + \|v\|^2 \leq \|u\|^2 + 2\|u\|\cdot\|v\| + \|v\|^2 = (\|u\|+\|v\|)^2,$$

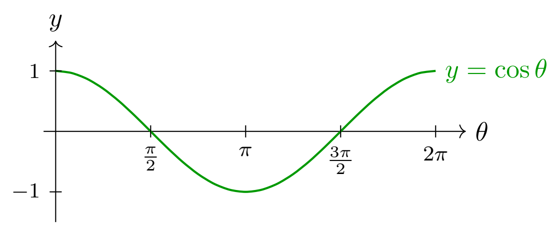
where the second inequality follows from the Cauchy-Schwarz inequality. Since both $\|u+v\|$ and $\|u\|+\|v\|$ are non-negative, the last inequality implies that $\|u+v\| \leq \|u\|+\|v\|$.

The other properties of being a norm are Exercise 22.9. **Q.E.D.**

Exercise 22.9 (*The Remaining Two Norm Axioms*): Complete the proof of Corollary 22.a.12 by verifying the homogeneity $\|\alpha\cdot v\| = |\alpha|\cdot\|v\|$ and the non-degeneracy "$\|v\| = 0 \implies v = 0$" for $\|v\| := \sqrt{\langle v,v\rangle}$.

Definition 22.a.13: Let $(V, \langle\cdot,\cdot\rangle)$ be a Euclidean space. Let $u, v \in V$ be two non-zero vectors. By the Cauchy-Schwarz inequality, $-1 \leq \frac{\langle u,v\rangle}{\|u\|\cdot\|v\|} \leq 1$. We define the "*angle*" between u and v to be the unique $\alpha \in [0,\pi]$ such that:

$$\cos\alpha = \frac{\langle u,v\rangle}{\|u\|\cdot\|v\|} \quad \left(= \left\langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \right\rangle\right).$$



The Gram-Schmidt orthogonalization process.

Theorem 22.a.14: Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space.

(a) If $S \subseteq V$ is an orthogonal system, then S is linearly independent.

(b) If $v_1, \dots, v_n$ is an orthogonal system and $v \in \mathrm{Sp}(v_1,\dots,v_n)$, write $v = a_1v_1 + \dots + a_nv_n$, with $a_j \in K$ (where $K = \mathbb{R}$ or $\mathbb{C}$). Then the coefficients a_j are given by

$$a_j = \frac{\langle v, v_j\rangle}{\langle v_j, v_j\rangle} \quad \forall\, 1 \leq j \leq n.$$

Proof:

(b) Write $v = a_1v_1 + \dots + a_nv_n$. Let $1 \leq j \leq n$. We have:

$$\langle v, v_j\rangle = \sum_{k=1}^{n} a_k \underbrace{\langle v_k, v_j\rangle}_{=0\ \forall k\neq j} = a_j\langle v_j, v_j\rangle,$$

which implies that

$$a_j = \frac{\langle v, v_j\rangle}{\langle v_j, v_j\rangle}.$$

(a) Suppose $\sum_{k=1}^{n} a_kv_k = 0$, where $n \geq 1$, $v_1,\dots,v_n \in S$ are distinct and $a_1,\dots,a_n \in K$. By **(b)**,

$$a_j = \frac{\langle 0, v_j\rangle}{\langle v_j, v_j\rangle} = 0 \quad \forall\, 1 \leq j \leq n.$$

Q.E.D.

Proposition 22.a.15: Let V, W be vector spaces over K $(= \mathbb{R}$ or $\mathbb{C})$ and let $\langle \cdot, \cdot \rangle$ be an inner product on W. Let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V and $\mathcal{C} = (w_1, \ldots, w_m)$ an **orthonormal** basis for W. Let $T : V \to W$ be a linear map. Write $[T]_{\mathcal{C}}^{\mathcal{B}} = (a_{ij})$. Then $a_{ij} = \langle T(v_j), w_i \rangle$ for all i, j. (If $\mathcal{C}$ is only an orthogonal basis, then $a_{ij} = \langle T(v_j), w_i \rangle / \langle w_i, w_i \rangle$). The proof is left as an exercise.

So, having orthogonal/orthonormal bases for our spaces is useful!

Theorem 22.a.16 (*Gram-Schmidt orthogonalization process*): Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over K $(= \mathbb{R}$ or $\mathbb{C})$ and let $n = \dim V$. Let $(v_1, \ldots, v_n)$ be a basis of V. Define a new list of vectors $w_1, \ldots, w_n$ by induction:

$$w_1 := v_1, \quad w_j := v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} \cdot w_i \quad \text{for } j = 2, \ldots, n.$$

Then:

(a) $(w_1, \ldots, w_n)$ is an orthogonal basis for V.

(b) $\left(\frac{w_1}{\|w_1\|}, \ldots, \frac{w_n}{\|w_n\|} \right)$ is an orthonormal basis for V.

(c) $\forall\, 1 \le j \le n$ we have $\mathrm{Sp}(v_1, \ldots, v_j) = \mathrm{Sp}(w_1, \ldots, w_j) = \mathrm{Sp}\left(\frac{w_1}{\|w_1\|}, \ldots, \frac{w_j}{\|w_j\|} \right)$.

Proof:

We claim that for all $1 \le j \le n$, w_j is well defined (i.e., $w_1, \ldots, w_{j-1} \ne 0$, recall that we need to divide by $\langle w_i, w_i \rangle$ in $i = 1, \ldots, j-1$ in order to define w_j), and moreover $(w_1, \ldots, w_j)$ is an orthogonal system with $\mathrm{Sp}(w_1, \ldots, w_j) = \mathrm{Sp}(v_1, \ldots, v_j)$.

We prove this by induction on j.

For $j = 1$, $w_1 = v_1$, so the claim is obvious ($v_1 \ne 0$ because it is a part of a basis).

Let $2 \le j \le n$. Assume the claim holds for $w_1, \ldots, w_{j-1}$. Recall that $w_j = v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} \cdot w_i$. Note that w_j is well defined because $w_i \ne 0$ for all $1 \le i \le j-1$. We claim that $w_j \ne 0$. Indeed, if $w_j = 0$, then

$$v_j = \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} w_i \implies v_j \in \mathrm{Sp}(w_1, \ldots, w_{j-1}) = \mathrm{Sp}(v_1, \ldots, v_{j-1})$$

(by inductive hypothesis), which is impossible because $(v_1, \ldots, v_j)$ are linearly independent.

This shows $w_j \ne 0$. By the inductive hypothesis, $w_1, \ldots, w_{j-1}$ is an orthogonal system. So we have to prove that $w_j \perp w_1, \ldots, w_{j-1}$. Indeed, let $1 \le k \le j-1$. Then

$$\langle w_j, w_k \rangle = \langle v_j, w_k \rangle - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} \cdot \langle w_i, w_k \rangle = \langle v_j, w_k \rangle - \frac{\langle v_j, w_k \rangle}{\langle w_k, w_k \rangle} \cdot \langle w_k, w_k \rangle = 0.$$

This shows $(w_1, \ldots, w_j)$ is an orthogonal system.

To complete the induction, it remains to show that $\mathrm{Sp}(w_1, \ldots, w_j) = \mathrm{Sp}(v_1, \ldots, v_j)$. By the definition of w_j we have

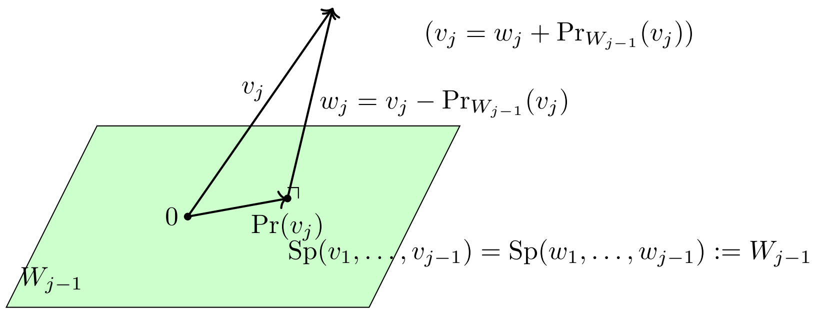
$$w_j \in \mathrm{Sp}(w_1, \ldots, w_{j-1}, v_j) = \mathrm{Sp}(v_1, \ldots, v_{j-1}, v_j) \quad \text{(inductive hypothesis)}.$$

This implies $\mathrm{Sp}(w_1, \ldots, w_{j-1}, w_j) \subseteq \mathrm{Sp}(v_1, \ldots, v_{j-1}, v_j)$. But both these vector spaces have dimension j (because $v_1, \ldots, v_j$ are linearly independent, and also $w_1, \ldots, w_j$ are linearly independent, being an orthogonal system). This implies $\mathrm{Sp}(w_1, \ldots, w_{j-1}, w_j) = \mathrm{Sp}(v_1, \ldots, v_{j-1}, v_j)$.

Q.E.D.

Corollary 22.a.17: Every finite-dimensional inner product space has an orthonormal basis.

Geometric idea behind the Gram-Schmidt process.



Orthogonal projection:

$$\Pr_{W_{j-1}}(v_j) = \sum_{i=1}^{j-1} \frac{\langle v_j, w_i\rangle}{\langle w_i, w_i\rangle} \cdot w_i .$$

We'll see more about this soon.

Solutions to the Exercises

Proof of Lemma 22.a.3:

(a) Homogeneity in the first variable with $\alpha := 0$ gives

$$\langle 0, v\rangle = \langle 0\cdot 0, v\rangle = 0\cdot\langle 0, v\rangle = 0,$$

and homogeneity in the second variable gives $\langle v, 0\rangle = \langle v, 0\cdot 0\rangle = \overline{0}\cdot\langle v, 0\rangle = 0$ (in the Euclidean case simply $0\cdot\langle v, 0\rangle$, since there is no conjugation).

(b) Suppose $\langle v, w\rangle = 0$ for every $v \in V$. Taking in particular $v := w$ gives $\langle w, w\rangle = 0$. If w were non-zero, positivity would force $\langle w, w\rangle > 0$, a contradiction. Hence $w = 0$.

(c) Let $v \in V$ be arbitrary. Additivity in the second variable, together with $\overline{-1} = -1$, gives

$$\langle v, w_1 - w_2\rangle = \langle v, w_1\rangle + \overline{(-1)}\langle v, w_2\rangle = \langle v, w_1\rangle - \langle v, w_2\rangle = 0.$$

Since this holds for every $v \in V$, part **(b)** applied to $w := w_1 - w_2$ gives $w_1 - w_2 = 0$, that is, $w_1 = w_2$.

Q.E.D.

Proof of Exercise 22.1 (a):

Let V be a Euclidean space and $u, v \in V$. Expanding by bilinearity and then using symmetry,

$$\|u+v\|^2 = \langle u+v, u+v\rangle = \langle u,u\rangle + \langle u,v\rangle + \langle v,u\rangle + \langle v,v\rangle = \|u\|^2 + 2\langle u,v\rangle + \|v\|^2.$$

Rearranging gives exactly

$$\langle u, v\rangle = \frac{1}{2}\left(\|u+v\|^2 - \|u\|^2 - \|v\|^2\right).$$

The right-hand side involves nothing but the norm, so two scalar products on V that induce the same norm are equal — this is the sense in which $\|\cdot\|$ determines $\langle\cdot,\cdot\rangle$. The converse fails: not every norm arises this way, as Exercise 22.8 shows. **Q.E.D.**

Solution of Exercise 22.1 (b):

Let V be a Hermitian space. Here $\langle u, v\rangle$ is in general not real, so one formula must recover two real numbers, and we obtain them one at a time.

The real part. Expanding as above, but now with $\langle v, u\rangle = \overline{\langle u, v\rangle}$,

$$\|u+v\|^2 = \|u\|^2 + \langle u,v\rangle + \overline{\langle u,v\rangle} + \|v\|^2 = \|u\|^2 + \|v\|^2 + 2\,\mathrm{Re}\langle u,v\rangle, \tag{22.1}$$

so that $\operatorname{Re}\langle u,v\rangle = \frac{1}{2}\left(\|u+v\|^2 - \|u\|^2 - \|v\|^2\right)$, which is literally the Euclidean formula $(*)$ with $\langle u,v\rangle$ replaced by its real part.

The imaginary part. Apply (22.1) with iv in place of v. Since $\|iv\| = |i|\cdot\|v\| = \|v\|$ and $\langle u,iv\rangle = \bar{i}\langle u,v\rangle = -i\langle u,v\rangle$, we get

$$\|u+iv\|^2 = \|u\|^2 + \|v\|^2 + 2\operatorname{Re}\left(-i\langle u,v\rangle\right).$$

Writing $\langle u,v\rangle = x+iy$ with $x,y\in\mathbb{R}$ gives $-i(x+iy) = y-ix$, whose real part is $y = \operatorname{Im}\langle u,v\rangle$. Hence

$$\operatorname{Im}\langle u,v\rangle = \frac{1}{2}\left(\|u+iv\|^2 - \|u\|^2 - \|v\|^2\right).$$

The formula. Putting the two together,

$$\langle u,v\rangle = \frac{1}{2}\left(\|u+v\|^2 - \|u\|^2 - \|v\|^2\right) + \frac{i}{2}\left(\|u+iv\|^2 - \|u\|^2 - \|v\|^2\right),$$

which is the analogue of $(*)$ asked for. Subtracting the versions of (22.1) for v and for $-v$ (respectively for iv and $-iv$) eliminates $\|u\|^2 + \|v\|^2$ and yields the more symmetric form usually called the polarization identity:

$$\langle u,v\rangle = \frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2\right).$$

Q.E.D.

Proof of Exercise 22.2:

Everything follows from the distributivity and homogeneity of matrix multiplication. For the first variable, using $(v_1+v_2)^\mathsf{T} = v_1^\mathsf{T} + v_2^\mathsf{T}$ and $(\alpha v)^\mathsf{T} = \alpha v^\mathsf{T}$,

$$\langle v_1+v_2, w\rangle_A = (v_1+v_2)^\mathsf{T} Aw = v_1^\mathsf{T} Aw + v_2^\mathsf{T} Aw = \langle v_1,w\rangle_A + \langle v_2,w\rangle_A,$$
$$\langle \alpha v, w\rangle_A = (\alpha v)^\mathsf{T} Aw = \alpha\left(v^\mathsf{T} Aw\right) = \alpha\langle v,w\rangle_A.$$

For the second variable,

$$\langle v, w_1+w_2\rangle_A = v^\mathsf{T} A(w_1+w_2) = v^\mathsf{T} Aw_1 + v^\mathsf{T} Aw_2 = \langle v,w_1\rangle_A + \langle v,w_2\rangle_A,$$
$$\langle v, \alpha w\rangle_A = v^\mathsf{T} A(\alpha w) = \alpha\left(v^\mathsf{T} Aw\right) = \alpha\langle v,w\rangle_A.$$

Note that no hypothesis on A is needed here; symmetry and positivity are what will require one.

Q.E.D.

Proof of Exercise 22.3:

Let $A\in\mathcal{M}_{n\times n}(\mathbb{R})$ be positive definite; by Definition 22.a.8 this includes the requirement that A be symmetric. We check the four axioms of Definition 22.a.1 for $\langle v,w\rangle_A := v^\mathsf{T} Aw$.

Linearity in each of the two variables is Exercise 22.2. Symmetry is the computation carried out in the text above: since $A^\mathsf{T} = A$ and a 1×1 matrix equals its own transpose,

$$\langle w,v\rangle_A = w^\mathsf{T} Av = w^\mathsf{T} A^\mathsf{T} v = \left(v^\mathsf{T} Aw\right)^\mathsf{T} = v^\mathsf{T} Aw = \langle v,w\rangle_A.$$

Positivity is the definition of positive definiteness: $\langle v,v\rangle_A = v^\mathsf{T} Av > 0$ for every $0\neq v\in\mathbb{R}^n$.

Q.E.D.

Proof of Example 22.1 (b):

Let $A = \left(\begin{smallmatrix} a & b \\ b & d\end{smallmatrix}\right)\in\mathcal{M}_{2\times 2}(\mathbb{R})$, which is symmetric, and write $v = \left(\begin{smallmatrix} x \\ y\end{smallmatrix}\right)$. Then

$$Q(v) := v^\mathsf{T} Av = ax^2 + 2bxy + dy^2.$$

Necessity. Suppose A is positive definite. Taking $v = \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$ gives $Q(v) = a > 0$. Taking $v = \left(\begin{smallmatrix}-b/a\\1\end{smallmatrix}\right)$ — which is legitimate now that $a \neq 0$, and is non-zero — gives

$$Q(v) = a \cdot \frac{b^2}{a^2} - 2b \cdot \frac{b}{a} + d = \frac{b^2 - 2b^2 + ad}{a} = \frac{ad - b^2}{a} = \frac{\det(A)}{a} > 0,$$

and since $a > 0$ this forces $\det(A) > 0$.

Sufficiency. Suppose $a > 0$ and $\det(A) = ad - b^2 > 0$. Completing the square,

$$Q(v) = a\left(x + \frac{b}{a}y\right)^2 + \frac{ad - b^2}{a}\,y^2 = a\left(x + \frac{b}{a}y\right)^2 + \frac{\det(A)}{a}\,y^2. \tag{22.2}$$

Both coefficients are positive, so $Q(v) \geq 0$ always, and $Q(v) = 0$ forces first $y = 0$ and then $x + \frac{b}{a}y = x = 0$. Hence $Q(v) > 0$ for every $v \neq 0$, which is positive definiteness. **Q.E.D.**

Proof of Exercise 22.4:

Let $A = \left(\begin{smallmatrix}a & b\\ \bar{b} & d\end{smallmatrix}\right) \in \mathcal{M}_{2\times 2}(\mathbb{C})$ be Hermitian. The two reality assertions are automatic: $A^* = A$ forces $a = \bar{a}$ and $d = \bar{d}$, so $a, d \in \mathbb{R}$, and consequently

$$\det A = ad - b\bar{b} = ad - |b|^2 \in \mathbb{R}.$$

So only the two inequalities are at issue. Write $v = \left(\begin{smallmatrix}x\\y\end{smallmatrix}\right) \in \mathbb{C}^2$ and compute

$$Q(v) := v^{\mathsf{T}} A \bar{v} = a|x|^2 + bx\bar{y} + \bar{b}\,\bar{x}y + d|y|^2 = a|x|^2 + 2\,\mathrm{Re}\,(bx\bar{y}) + d|y|^2,$$

the last step because $\bar{b}\,\bar{x}y = \overline{bx\bar{y}}$, and $z + \bar{z} = 2\,\mathrm{Re}\,z$.

Necessity. If A is positive definite, then $v = \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$ gives $Q(v) = a > 0$, and $v = \left(\begin{smallmatrix}-\bar{b}/a\\1\end{smallmatrix}\right)$ gives, after the computation below, $Q(v) = \det(A)/a > 0$, whence $\det A > 0$.

Sufficiency. Assume $a > 0$ and $\det A > 0$. Completing the square exactly as in the real case, but taking the conjugations into account,

$$a\left|x + \frac{\bar{b}}{a}y\right|^2 = a\left(x + \frac{\bar{b}}{a}y\right)\overline{\left(x + \frac{\bar{b}}{a}y\right)} = a|x|^2 + bx\bar{y} + \bar{b}\,\bar{x}y + \frac{|b|^2}{a}|y|^2,$$

and therefore

$$Q(v) = a\left|x + \frac{\bar{b}}{a}y\right|^2 + \frac{ad - |b|^2}{a}|y|^2 = a\left|x + \frac{\bar{b}}{a}y\right|^2 + \frac{\det(A)}{a}|y|^2. \tag{22.3}$$

Both coefficients are positive real numbers and both squared moduli are non-negative, so $Q(v) \geq 0$; and $Q(v) = 0$ forces $y = 0$ and then $x = 0$. Hence $Q(v) > 0$ for all $v \neq 0$. Reading (22.3) at $x = -\bar{b}y/a$, $y = 1$ also supplies the value $\det(A)/a$ used in the necessity half. **Q.E.D.**

Proof of Exercise 22.5:

Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ and suppose $\langle v, w\rangle_A := v^{\mathsf{T}} A w$ is a scalar product on $\mathbb{R}^n$. Symmetry of the product says $v^{\mathsf{T}} A w = w^{\mathsf{T}} A v$ for all $v, w \in \mathbb{R}^n$. Take $v := e_i$ and $w := e_j$ from the standard basis. As in the complex computation of the text,

$$e_i^{\mathsf{T}} A e_j = a_{ij}, \qquad e_j^{\mathsf{T}} A e_i = a_{ji},$$

since Ae_j is the j-th column of A and left multiplication by e_i^{T} selects its i-th entry. Hence $a_{ij} = a_{ji}$ for all i, j, that is, $A^{\mathsf{T}} = A$: the matrix is symmetric.

Positivity of the scalar product says $\langle v, v\rangle_A = v^{\mathsf{T}} A v > 0$ for every $0 \neq v \in \mathbb{R}^n$, which is precisely the statement that the symmetric matrix A is positive definite (Definition 22.a.8). This proves the Claim on Page 196, and, read together with Exercise 22.3, shows that $\langle\cdot,\cdot\rangle_A$ is a scalar product if and only if A is symmetric and positive definite.

It is indeed simpler than the complex case: no conjugation has to be tracked, and the Hermitian axiom collapses to plain symmetry. **Q.E.D.**

Proof of Exercise 22.6:

Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be Hermitian and positive definite, and put $\langle v, w\rangle_A := v^\mathsf{T} A\overline{w}$. We verify the four axioms of Definition 22.a.2.

Linearity in the first variable. As in Exercise 22.2, $(v_1+v_2)^\mathsf{T} A\overline{w} = v_1^\mathsf{T} A\overline{w} + v_2^\mathsf{T} A\overline{w}$ and $(\alpha v)^\mathsf{T} A\overline{w} = \alpha\left(v^\mathsf{T} A\overline{w}\right)$.

Sesquilinearity in the second variable. Complex conjugation is additive and satisfies $\overline{\alpha w} = \overline{\alpha}\,\overline{w}$, so

$$\langle v, w_1+w_2\rangle_A = v^\mathsf{T} A\left(\overline{w_1}+\overline{w_2}\right) = \langle v, w_1\rangle_A + \langle v, w_2\rangle_A,$$
$$\langle v, \alpha w\rangle_A = v^\mathsf{T} A\left(\overline{\alpha}\,\overline{w}\right) = \overline{\alpha}\left(v^\mathsf{T} A\overline{w}\right) = \overline{\alpha}\langle v, w\rangle_A.$$

The Hermitian property. Using that a 1×1 matrix equals its own transpose, and then $\overline{A}^\mathsf{T} = A$:

$$\overline{\langle w, v\rangle_A} = \overline{w^\mathsf{T} A\overline{v}} = \overline{w}^\mathsf{T}\overline{A} v = \left(\overline{w}^\mathsf{T}\overline{A} v\right)^\mathsf{T} = v^\mathsf{T}\overline{A}^\mathsf{T}\overline{w} = v^\mathsf{T} A\overline{w} = \langle v, w\rangle_A.$$

Positivity. $\langle v, v\rangle_A = v^\mathsf{T} A\overline{v} > 0$ for every $0 \neq v \in \mathbb{C}^n$ is exactly Definition 22.a.9; that this quantity is real at all was checked there.

Together with the half proved in the text, this completes the Claim on Page 197. **Q.E.D.**

Proof of Exercise 22.7:

First, the pairing is well defined and finite — Prof. Biran's *"Why?"*. If $f, g \in C[a,b]$, then $f\cdot g$ is continuous on the closed bounded interval $[a,b]$, hence Riemann integrable there, and its integral is a real number.

Linearity in the first variable. By the linearity of the integral,

$$\langle f_1+f_2, g\rangle = \int_a^b \left(f_1(x)+f_2(x)\right) g(x)\,dx = \int_a^b f_1 g\,dx + \int_a^b f_2 g\,dx = \langle f_1, g\rangle + \langle f_2, g\rangle,$$

and for $\alpha \in \mathbb{R}$,

$$\langle \alpha f, g\rangle = \int_a^b \alpha f(x) g(x)\,dx = \alpha\int_a^b f(x)g(x)\,dx = \alpha\langle f, g\rangle.$$

Symmetry. Pointwise multiplication of real numbers is commutative, so $f(x)g(x) = g(x)f(x)$ for every x, and therefore $\langle f, g\rangle = \langle g, f\rangle$.

Linearity in the second variable now follows from the first variable together with symmetry, without any further computation.

Positivity is the part checked in the text, so $\langle\cdot,\cdot\rangle$ is indeed a scalar product on $C[a,b]$. **Q.E.D.**

Proof of Exercise 22.8:

Write $v = \binom{a}{b}$ and $v' = \binom{a'}{b'}$.

Both are norms. For $\|\cdot\|''$ the three axioms are immediate from the corresponding properties of the absolute value on $\mathbb{R}$:

$$\|v+v'\|'' = |a+a'| + |b+b'| \leq |a| + |a'| + |b| + |b'| = \|v\|'' + \|v'\|'',$$

$\|\alpha v\|'' = |\alpha a| + |\alpha b| = |\alpha|\cdot\|v\|''$, and $\|v\|'' = 0$ forces $|a| = |b| = 0$, so $v = 0$. For $\|\cdot\|'$,

$$\|v+v'\|' = \max\{|a+a'|, |b+b'|\} \leq \max\{|a|+|a'|, |b|+|b'|\} \leq \max\{|a|,|b|\} + \max\{|a'|,|b'|\},$$

since each of $|a|+|a'|$ and $|b|+|b'|$ is bounded by the right-hand side; homogeneity is $\max\{|\alpha a|, |\alpha b|\} = |\alpha|\max\{|a|,|b|\}$, and $\|v\|' = 0$ again forces $v = 0$.

Neither comes from a scalar product. Suppose a norm $\|\cdot\|$ on a real vector space is induced by a scalar product. Expanding as in Exercise 22.1 **(a)** for both $u+v$ and $u-v$ and adding, the cross terms cancel and we obtain the *"parallelogram law"*

$$\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2 \qquad \forall u, v. \tag{22.4}$$

(Equivalently: the formula in the hint would have to define a scalar product, and (22.4) is what its additivity in the first variable amounts to.) So it suffices to exhibit two vectors violating (22.4). Take $u := \binom{1}{0}$ and $v := \binom{0}{1}$, so that $u+v = \binom{1}{1}$ and $u - v = \binom{1}{-1}$.

For $\|\cdot\|'$ we get $\|u\|' = \|v\|' = \|u+v\|' = \|u-v\|' = 1$, so the left-hand side of (22.4) is $1+1=2$ while the right-hand side is $2+2=4$.

For $\|\cdot\|''$ we get $\|u\|'' = \|v\|'' = 1$ and $\|u+v\|'' = \|u-v\|'' = 2$, so the left-hand side is $4+4=8$ while the right-hand side is $2+2=4$.

In both cases (22.4) fails, so neither norm is induced by a scalar product. This is the warning attached to Exercise 22.1 **(a)**: a scalar product determines its norm, but a norm need not come from one. **Q.E.D.**

Proof of Exercise 22.9:

First note that $\|v\| := \sqrt{\langle v, v\rangle}$ makes sense: in the Hermitian case $\langle v, v\rangle = \overline{\langle v, v\rangle}$ by the Hermitian property, so $\langle v, v\rangle$ is real; it is 0 for $v = 0$ by Lemma 22.a.3 **(a)** and positive otherwise, so the square root is defined and lies in $\mathbb{R}_{\geq 0}$.

Homogeneity. For $\alpha \in K$ and $v \in V$,

$$\|\alpha v\|^2 = \langle \alpha v, \alpha v\rangle = \alpha\overline{\alpha}\langle v, v\rangle = |\alpha|^2\|v\|^2,$$

using homogeneity in the first variable and sesquilinearity in the second (in the Euclidean case $\alpha \cdot \alpha = \alpha^2 = |\alpha|^2$). Both $\|\alpha v\|$ and $|\alpha| \cdot \|v\|$ are non-negative, so taking square roots gives $\|\alpha v\| = |\alpha| \cdot \|v\|$.

Non-degeneracy. If $\|v\| = 0$, then $\langle v, v\rangle = 0$. Were $v \neq 0$, positivity would give $\langle v, v\rangle > 0$; hence $v = 0$. **Q.E.D.**

Proof of Proposition 22.a.15:

By the definition of the representation matrix, the j-th column of $[T]^{\mathcal{B}}_{\mathcal{C}} = (a_{ij})$ holds the coordinates of $T(v_j)$ in the basis $\mathcal{C}$, that is,

$$T(v_j) = \sum_{i=1}^{m} a_{ij}\, w_i.$$

Fix $1 \leq k \leq m$ and pair both sides with w_k. Using linearity in the first variable and the orthogonality $\langle w_i, w_k\rangle = 0$ for $i \neq k$,

$$\langle T(v_j), w_k\rangle = \sum_{i=1}^{m} a_{ij}\langle w_i, w_k\rangle = a_{kj}\langle w_k, w_k\rangle.$$

If $\mathcal{C}$ is orthonormal, then $\langle w_k, w_k\rangle = \|w_k\|^2 = 1$ and we get $a_{kj} = \langle T(v_j), w_k\rangle$, which is the claim after renaming k to i. If $\mathcal{C}$ is merely orthogonal, then $\langle w_k, w_k\rangle \neq 0$ (an orthogonal system does not contain 0) and dividing gives

$$a_{ij} = \frac{\langle T(v_j), w_i\rangle}{\langle w_i, w_i\rangle}.$$

This is Theorem 22.a.14 **(b)** applied to the vector $T(v_j) \in \mathrm{Sp}(w_1, \ldots, w_m) = W$. **Q.E.D.**

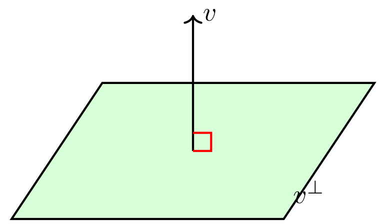
The Orthogonal Complement (Section 22.b)

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$).

Definition 22.b.1 (*The Orthogonal Complement*): Let $\emptyset \neq S \subseteq V$ be a subset. We define the "*orthogonal complement*" of S as:

$$S^\perp := \{v \in V \mid v \perp s \ \forall s \in S\} = \{v \in V \mid \langle v, s\rangle = 0 \ \forall s \in S\}.$$

When $S = \{v\}$, we write $v^\perp := \{v\}^\perp$.



Lemma 22.b.2 (*Properties of the Orthogonal Complement*): Let $\emptyset \neq S, T \subseteq V$. Then:

(a) $S^\perp \subseteq V$ is a linear subspace.
(b) $0^\perp = V$, and $V^\perp = \{0\}$.
(c) $S \cap S^\perp$ is either $\emptyset$ or equal to $\{0\}$.
(d) If $S \subseteq T$, then $S^\perp \supseteq T^\perp$.
(e) $(\mathrm{Sp}(S))^\perp = S^\perp$.
(f) $S \subseteq (S^\perp)^\perp$.

Proof:

(a) We have $0 \in S^\perp$ because $\langle 0, s\rangle = 0$ for all $s \in S$. If $\alpha, \beta \in K$ and $v, w \in S^\perp$, then for all $s \in S$ we have:

$$\langle \alpha v + \beta w, s\rangle = \alpha\langle v, s\rangle + \beta\langle w, s\rangle = 0.$$

This implies that $\alpha v + \beta w \in S^\perp$. Thus, $S^\perp$ is a linear subspace.

(b) Showing $0^\perp = V$ is left as an exercise. Let $v \in V^\perp$. Then $\langle v, w\rangle = 0$ for all $w \in V$. In particular, $\langle v, v\rangle = 0$, which implies $v = 0$. This shows $V^\perp \subseteq \{0\}$. Since it is clear that $\{0\} \subseteq V^\perp$, we conclude $V^\perp = \{0\}$.

(c) If $S \cap S^\perp = \emptyset$, we are done. Otherwise, assume that $s \in S \cap S^\perp$. This implies that $\langle s, s\rangle = 0$, which yields $s = 0$.

(d) (Exercise).

(e) We have $S \subseteq \mathrm{Sp}(S)$. By statement **(d)**, this yields $(\mathrm{Sp}(S))^\perp \subseteq S^\perp$. Next, we will show that $S^\perp \subseteq (\mathrm{Sp}(S))^\perp$. Indeed, let $v \in S^\perp$, and let $u \in \mathrm{Sp}(S)$. We can write $u = \sum_{i=1}^k \alpha_i u_i$ with $\alpha_i \in K$ and $u_i \in S$ for all $1 \le i \le k$. We compute:

$$\langle v, u\rangle = \sum_{i=1}^k \overline{\alpha_i}\underbrace{\langle v, u_i\rangle}_{=0} = 0,$$

where we used the fact that $\langle v, u_i\rangle = 0$ because $v \in S^\perp$ and $u_i \in S$. Thus, $v \perp u$. This holds for all $u \in \mathrm{Sp}(S)$, and therefore $v \in (\mathrm{Sp}(S))^\perp$. This shows that $S^\perp \subseteq (\mathrm{Sp}(S))^\perp$.

(f) Let $s \in S$. Then for all $v \in S^\perp$ we have $\langle v, s\rangle = 0$, which implies $\langle s, v\rangle = 0$. Therefore, $s \perp v$ for all $v \in S^\perp$. Consequently, $s \in (S^\perp)^\perp$.

Q.E.D.

Theorem 22.b.3 (*Orthogonal Complement in Finite Dimensions*): Let V be an inner product space (possibly of infinite dimension), and let $U \subseteq V$ be a finite-dimensional subspace. Then $U^\perp$ is a complement of U in V (i.e., $U + U^\perp = V$ and $U \cap U^\perp = \{0\}$).

Proof:

By Lemma 22.b.2, we already have $U \cap U^\perp = \{0\}$. So, it is enough to show that $U + U^\perp = V$. Put $r := \dim U$ and pick an orthonormal basis $(e_1, \dots, e_r)$ for U. (Note: this is not necessarily the standard basis or anything similar). Let $v \in V$. We introduce the temporary notation $\widetilde{P}_U(v) := \langle v, e_1\rangle e_1 + \cdots + \langle v, e_r\rangle e_r \in U$. (This is the orthogonal projection of v onto U.) We have

$$v = \underbrace{\widetilde{P}_U(v)}_{\in U} + (v - \widetilde{P}_U(v)). \tag{22.5}$$

We claim that $v - \widetilde{P}_U(v) \in U^\perp$. Indeed, for any $1 \le i \le r$,

$$\begin{aligned}\langle v - \widetilde{P}_U(v), e_i\rangle &= \langle v, e_i\rangle - \sum_{k=1}^{r} \langle v, e_k\rangle\langle e_k, e_i\rangle \\ &= \langle v, e_i\rangle - \langle v, e_i\rangle = 0.\end{aligned}$$

This holds for all $1 \le i \le r$, hence $(v - \widetilde{P}_U(v)) \perp \mathrm{Sp}(e_1, \dots, e_r) = U$, meaning $v - \widetilde{P}_U(v) \in U^\perp$. Going back to (22.5), we see that

$$v = \underbrace{\widetilde{P}_U(v)}_{\in U} + \underbrace{(v - \widetilde{P}_U(v))}_{\in U^\perp}.$$

This holds for all $v \in V$, which implies $V = U + U^\perp$. **Q.E.D.**

Remark 22.3: If $U \subseteq V$ is a finite-dimensional subspace, then since $U^\perp$ is a complement of U in V, we have a canonical isomorphism $V \cong U \oplus U^\perp$.

Definition 22.b.4 (*Orthogonal Projection*): Let V be an inner product space and $U \subseteq V$ a finite-dimensional subspace. We have seen that $U^\perp$ is a complement of U in V (which implies that for all $v \in V$, there exist unique vectors $u \in U$ and $w \in U^\perp$ such that $v = u + w$). Define a map $P_U\colon V \to U$ as follows: for all $v \in V$, write $v = u + w$ with $u \in U$ and $w \in U^\perp$ in a unique way. We define $P_U(v) := u$. The map P_U is called the "*orthogonal projection*" on U.

Proposition 22.b.5 (*Properties of Orthogonal Projection*): The map P_U has the following properties:

- **(a)** P_U is a linear map.
- **(b)** $\operatorname{Im} P_U = U$ and $\ker P_U = U^\perp$.
- **(c)** For all $v \in V$, $v - P_U(v) \in U^\perp$.
- **(d)** For all $u \in U$, $P_U(u) = u$.
- **(e)** If $(e_1, \dots, e_r)$ is any orthonormal basis for U, then $P_U(v) = \sum_{i=1}^r \langle v, e_i\rangle \cdot e_i$ (which exactly equals $\widetilde{P}_U(v)$ from the previous proof).

Proof:

Properties **(a)**–**(d)** follow from the general fact that if W_1, W_2 are vector subspaces of W and $W_1 \oplus W_2 = W$, then the map $P_{W_1}\colon W \to W_1$, defined by $P_{W_1}(w) = w_1$, where we write $w = w_1 + w_2$ in a unique way with $w_1 \in W_1, w_2 \in W_2$, is a linear map. Furthermore, $\operatorname{Im} P_{W_1} = W_1$, $\ker P_{W_1} = W_2$, and for all $w \in W$ we have $w - P_{W_1}(w) \in W_2$. (Thus, $w = P_{W_1}(w) + (w - P_{W_1}(w))$ is the unique decomposition of w as $w = w_1 + w_2$). We also have $P_{W_1}(w) = w$ for all $w \in W_1$.

For statement **(e)**, we can write

$$v = \underbrace{\left(\sum_{i=1}^{r} \langle v, e_i\rangle \cdot e_i\right)}_{\text{(I)}} + \underbrace{\left(v - \sum_{i=1}^{r} \langle v, e_i\rangle e_i\right)}_{\text{(II)}}. \tag{22.6}$$

Now, part (I) is exactly $\widetilde{P}_U(v)$ from the previous proof, and we have also seen in that proof that part (II) belongs to $U^\perp$. This implies that (22.6) is the unique decomposition of v with respect to $V \cong U \oplus U^\perp$. Hence, $P_U(v) = \text{(I)}$. **Q.E.D.**

Corollary 22.b.6 (*Bidual Orthogonal Complement*): Let V be an inner product space and let $U \subseteq V$ be a finite-dimensional subspace. Then $(U^\perp)^\perp = U$.

Proof:

By Lemma 22.b.2, we have $U \subseteq (U^\perp)^\perp$. We will show that $(U^\perp)^\perp \subseteq U$ also holds. Indeed, let $v \in (U^\perp)^\perp$. Recall that $U^\perp$ is a complement of U in V. Write $v = u + w$ with $u \in U$ and $w \in U^\perp$. Since $u \in U \subseteq (U^\perp)^\perp$ and also $v \in (U^\perp)^\perp$, we have that $w = v - u \in (U^\perp)^\perp$. But $w \in U^\perp$, hence $w \in U^\perp \cap (U^\perp)^\perp = \{0\}$. This shows that $w = 0$, and therefore, $v = u \in U$. We have proven that $(U^\perp)^\perp \subseteq U$. **Q.E.D.**

Corollary 22.b.7 (*Dimension Formula for the Orthogonal Complement*): If V is a finite-dimensional inner product space and $U \subseteq V$ is a subspace, then $\dim U + \dim U^\perp = \dim V$.

Proof:

Since V is finite-dimensional, so is its subspace U, and Theorem 22.b.3 applies: $U + U^\perp = V$ and $U \cap U^\perp = \{0\}$, so the sum $U + U^\perp$ is direct and equals V. Choosing a basis for U and a basis for $U^\perp$ and concatenating them therefore produces a basis for V, whence

$$\dim U + \dim U^\perp = \dim\left(U \oplus U^\perp\right) = \dim V.$$

Q.E.D.

Orthogonal and Unitary Matrices

Definition 22.b.8 (*Orthogonal and Unitary Matrices*):

(a) A matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is called "*orthogonal*" if its columns form an orthonormal system in $\mathbb{R}^n$, with respect to the standard scalar product of $\mathbb{R}^n$. That is, writing A in terms of its columns,

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix},$$

the condition means that $\langle v_i, v_j\rangle = \delta_{ij}$ for all $1 \le i, j \le n$ (which in particular implies that the columns form an orthonormal basis for $\mathbb{R}^n$). We denote the set of orthogonal matrices of size $n \times n$ by $\mathrm{O}(n) \subseteq \mathcal{M}_{n\times n}(\mathbb{R})$.

(b) A matrix $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is called "*unitary*" if its columns form an orthonormal system in $\mathbb{C}^n$, with respect to the standard Hermitian product on $\mathbb{C}^n$. We denote the set of all unitary matrices of size $n \times n$ by $\mathrm{U}(n) \subseteq \mathcal{M}_{n\times n}(\mathbb{C})$.

Remark 22.4: Why are they called *"orthogonal matrices"* and not *"orthonormal matrices"*? This is simply a historical convention in the terminology!

Lemma 22.b.9 (*Characterization of Orthogonal Matrices*): For a matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$, the following statements are equivalent:

(a) A is an orthogonal matrix.

(b) $A^\mathsf{T} A = I_n$.

(c) $AA^\mathsf{T} = I_n$.

(d) $A^{-1} = A^{\mathsf{T}}$.

Lemma 22.b.10 (*Characterization of Unitary Matrices*): For a matrix $A \in \mathcal{M}_{n\times n}(\mathbb{C})$, the following statements are equivalent:

- **(a)** A is a unitary matrix.
- **(b)** $A^*A = I_n$ (which is equivalent to $\overline{A}^{\mathsf{T}}A = I_n$).
- **(c)** $AA^* = I_n$ (which is equivalent to $A\overline{A}^{\mathsf{T}} = I_n$).
- **(d)** $A^{-1} = A^*$.

Proof of Lemma 22.b.10:

(The proof of Lemma 22.b.9 is extremely similar). Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix},$$

where $v_k \in \mathbb{C}^n_{\text{col}}$ for $1 \le k \le n$. We compute the product:

$$\begin{aligned} A^*A &= \begin{pmatrix} — & \overline{v_1}^{\mathsf{T}} & — \\ & \vdots & \\ — & \overline{v_n}^{\mathsf{T}} & — \end{pmatrix} \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix} \\ &= \left(\overline{v_i}^{\mathsf{T}} \cdot v_j\right)_{1\le i,j\le n} \\ &= \left(\langle v_j, v_i\rangle\right)_{1\le i,j\le n}. \end{aligned} \tag{22.7}$$

Now, A is unitary if and only if $v_1, \dots, v_n$ form an orthonormal system, which is equivalent to saying that the resulting matrix in (22.7) is exactly I_n. This proves **(a)** $\iff$ **(b)**.

For the remaining equivalences, suppose $A^*A = I_n$. Taking determinants and using $\det(A^*) = \det(\overline{A}^{\mathsf{T}}) = \det(\overline{A}) = \overline{\det(A)}$, we obtain $|\det(A)|^2 = 1$, so $\det(A) \neq 0$ and A is invertible. Multiplying $A^*A = I_n$ on the right by A^{-1} gives $A^* = A^{-1}$, which is **(d)**; and then $AA^* = AA^{-1} = I_n$, which is **(c)**. Conversely, **(c)** gives by the same argument $A^* = A^{-1}$ and hence **(b)**, so all four statements are equivalent. **Q.E.D.**

Exercise 22.10 (*The Orthogonal Case of the Characterization*): Carry out the argument above for a real matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ and the standard scalar product on $\mathbb{R}^n$, thereby proving Lemma 22.b.9. (Prof. Biran remarks that it is *“similar”*; the point of writing it out is to see exactly where the complex conjugations of the Hermitian case disappear.)

Corollary 22.b.11 (*Rows of Orthogonal and Unitary Matrices*):

- **(a)** Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$. The matrix $A \in \mathrm{O}(n)$ if and only if $A^{\mathsf{T}} \in \mathrm{O}(n)$, which holds if and only if the rows of A form an orthonormal basis for $\mathbb{R}^n$.
- **(b)** Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$. The matrix $A \in \mathrm{U}(n)$ if and only if $A^* \in \mathrm{U}(n)$, which holds if and only if the rows of A form an orthonormal basis for $\mathbb{C}^n$.

Proof:

- **(a)** By Lemma 22.b.9, $A \in \mathrm{O}(n)$ if and only if $A^{\mathsf{T}}A = I_n$, and equally if and only if $AA^{\mathsf{T}} = I_n$. Applying the same characterization to the matrix A^{T}, whose transpose is A, the condition $A^{\mathsf{T}} \in \mathrm{O}(n)$ reads $(A^{\mathsf{T}})^{\mathsf{T}}A^{\mathsf{T}} = AA^{\mathsf{T}} = I_n$ — the very same equation. Hence $A \in \mathrm{O}(n) \iff A^{\mathsf{T}} \in \mathrm{O}(n)$. Finally, the columns of A^{T} are exactly the rows of A, so $A^{\mathsf{T}} \in \mathrm{O}(n)$ says precisely that the rows of A form an orthonormal system, and an orthonormal system of n vectors in $\mathbb{R}^n$ is a basis by Theorem 22.a.14 **(a)**.

(b) Identical, with Lemma 22.b.10 in place of Lemma 22.b.9 and A^* in place of A^T: the condition $A^* \in \mathrm{U}(n)$ reads $(A^*)^* A^* = AA^* = I_n$, which is one of the equivalent forms of $A \in \mathrm{U}(n)$; and the columns of A^* are the conjugates of the rows of A, which form an orthonormal system exactly when the rows do.

Q.E.D.

QR-Decomposition

Theorem 22.b.12 (*QR-Decomposition for Invertible Matrices*):

(a) Let $A \in \mathrm{GL}_n(\mathbb{R})$. Then there exists $Q \in \mathrm{O}(n)$ and an upper triangular matrix $R \in \mathcal{M}_{n\times n}(\mathbb{R})$ such that $A = Q \cdot R$.

(b) Let $A \in \mathrm{GL}_n(\mathbb{C})$. Then there exists $Q \in \mathrm{U}(n)$ and an upper triangular matrix $R \in \mathcal{M}_{n\times n}(\mathbb{C})$ such that $A = Q \cdot R$.

Proof of Theorem 22.b.12:

We prove parts **(a)** and **(b)** together, writing $K = \mathbb{R}$ or $K = \mathbb{C}$ throughout; the only difference between the two cases is whether the resulting Q is called orthogonal or unitary.

Let $A \in \mathrm{GL}_n(K)$ and write

$$A := \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$$

with $v_k \in K^n_{\mathrm{col}}$. Since $A \in \mathrm{GL}_n(K)$, the vectors $v_1, \dots, v_n$ form a basis for K^n. Apply the Gram-Schmidt orthogonalization process (Theorem 22.a.16) to $v_1, \dots, v_n$ to obtain an orthonormal basis $e_1, \dots, e_n$ for K^n, and recall from part **(c)** of that theorem that

$$\mathrm{Sp}(v_1, \dots, v_j) = \mathrm{Sp}(e_1, \dots, e_j) \qquad \forall\, 1 \le j \le n. \tag{22.8}$$

Fix $1 \le j \le n$. By (22.8) the vector v_j lies in $\mathrm{Sp}(e_1, \dots, e_j)$, and since $(e_1, \dots, e_j)$ is orthonormal, Theorem 22.a.14 **(b)** identifies its coefficients: they are $\langle v_j, e_i\rangle / \langle e_i, e_i\rangle = \langle v_j, e_i\rangle$. So the whole list expands as

$$\begin{aligned} v_1 &= \langle v_1, e_1\rangle e_1, \\ v_2 &= \langle v_2, e_1\rangle e_1 + \langle v_2, e_2\rangle e_2, \\ &\ \ \vdots \\ v_j &= \langle v_j, e_1\rangle e_1 + \dots + \langle v_j, e_j\rangle e_j, \\ &\ \ \vdots \\ v_n &= \langle v_n, e_1\rangle e_1 + \dots + \langle v_n, e_n\rangle e_n, \end{aligned}$$

with no e_i of index $i > j$ ever appearing in v_j. Read column by column, these n identities are exactly one matrix product:

$$\underbrace{\begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} | & & | \\ e_1 & \dots & e_n \\ | & & | \end{pmatrix}}_{Q} \underbrace{\begin{pmatrix} \langle v_1, e_1\rangle & \langle v_2, e_1\rangle & \dots & \langle v_n, e_1\rangle \\ 0 & \langle v_2, e_2\rangle & \dots & \langle v_n, e_2\rangle \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \langle v_n, e_n\rangle \end{pmatrix}}_{R},$$

that is, $R_{ij} = \langle v_j, e_i\rangle$ for $i \le j$ and $R_{ij} = 0$ for $i > j$.

The columns of Q are $e_1, \ldots, e_n$, an orthonormal basis of K^n, so $Q \in \mathrm{O}(n)$ when $K = \mathbb{R}$ and $Q \in \mathrm{U}(n)$ when $K = \mathbb{C}$ (Definition 22.b.8); and R is upper triangular by construction. Hence $A = Q \cdot R$ as required. **Q.E.D.**

Theorem (*Extension of QR-Decomposition for All Matrices*): Let $A \in \mathcal{M}_{n\times n}(K)$, and let $r := \operatorname{rank}(A)$. Then there exist matrices Q and R with $Q \in \mathrm{O}(n)$ if $K = \mathbb{R}$ (and $Q \in \mathrm{U}(n)$ if $K = \mathbb{C}$), and $R = \left(\begin{array}{c|c} C & * \\ \hline 0 & 0 \end{array}\right) \in \mathcal{M}_{n\times n}(K)$, where $C \in \mathcal{M}_{r\times r}(K)$ is an upper triangular matrix, such that $A = Q \cdot R$.

Proof:

Let

$$A := \begin{pmatrix} | & & | \\ v_1 & \ldots & v_n \\ | & & | \end{pmatrix}.$$

Note that $r = \dim \operatorname{Sp}(v_1, \ldots, v_n)$. We may assume $r \geq 1$, because otherwise $r = 0$, meaning $A = 0$, in which case we can take $R = 0$ and Q to be any arbitrary orthogonal/unitary matrix. Note that there exist indices $1 \leq i_1 < \cdots < i_r \leq n$ such that:

(a) $v_{i_1}, \ldots, v_{i_r}$ are linearly independent.

(b) $v_{i_1}, \ldots, v_{i_k}$ is a basis for $\operatorname{Sp}(v_1, v_2, \ldots, v_{i_k})$ (and $k = \dim \operatorname{Sp}(v_1, v_2, \ldots, v_{i_k}) \leq i_k$).

To see this, let $1 \leq i_1$ be the minimal index j such that $v_j \neq 0$. (Since $r \geq 1$, there exists such an index.) Clearly, $\operatorname{Sp}(v_1, \ldots, v_{i_1}) = \operatorname{Sp}(v_{i_1})$. Suppose we have defined $1 \leq i_1 < \cdots < i_k < n$ as above. Consider now the minimal index ℓ, with $i_k < \ell \leq n$ and such that $v_\ell \notin \operatorname{Sp}(v_{i_1}, \ldots, v_{i_k})$. If there is no such ℓ, then $k = r$ and we are done. Otherwise, take $i_{k+1} := \ell$. Continue the construction by induction.

Now, apply the Gram-Schmidt process to the list of vectors $v_{i_1}, \ldots, v_{i_r}$ and obtain an orthonormal system $e_1, \ldots, e_r$ such that $(e_1, \ldots, e_k)$ is a basis for $\operatorname{Sp}(v_{i_1}, \ldots, v_{i_k}) = \operatorname{Sp}(v_1, v_2, \ldots, v_{i_k})$. Extend $e_1, \ldots, e_r$ to an orthonormal basis $(e_1, \ldots, e_r, e_{r+1}, \ldots, e_n)$ of K^n (how? — see Exercise 22.11). Take

$$Q := \begin{pmatrix} | & & | \\ e_1 & \ldots & e_n \\ | & & | \end{pmatrix},$$

and let R be built exactly as in the previous proof, by $R_{ij} := \langle v_j, e_i \rangle$.

The block shape of R is now forced. Every v_j lies in $\operatorname{Sp}(v_1, \ldots, v_j) = \operatorname{Sp}(e_1, \ldots, e_{k(j)})$, where $k(j) \leq r$ counts how many of the indices $i_1 < \cdots < i_r$ do not exceed j; in particular no e_i with $i > r$ ever occurs, so the last $n - r$ rows of R vanish. Among the first r rows, the columns $j = i_1, \ldots, i_r$ carry the upper triangular block $C := \big(\langle v_{i_j}, e_i \rangle\big)_{1 \leq i,j \leq r}$ — upper triangular for the same reason as before — while the remaining columns are the arbitrary entries denoted by $*$. Hence $A = Q \cdot R$ with R of the stated form. **Q.E.D.**

Exercise 22.11 (*Extending an Orthonormal System to an Orthonormal Basis*): Let V be a finite-dimensional inner product space and let $(e_1, \ldots, e_r)$ be an orthonormal system in V with $r \leq n := \dim V$. Show that it can be extended to an orthonormal basis $(e_1, \ldots, e_r, e_{r+1}, \ldots, e_n)$ of V. (This is the step marked *"how?"* in the proof above.)

Exercise 22.12 (*Gram-Schmidt on Linearly Dependent Vectors*): Let V be an inner product space. Suppose $v_1, \ldots, v_m \in V$ are not necessarily linearly independent, but not all of them are 0. What happens if we apply the Gram-Schmidt orthogonalization process?

Suppose that at stage j we already have an orthogonal system $e_1, \ldots, e_k$ that spans $\mathrm{Sp}(v_1, \ldots, v_{i_k})$ and we encounter a vector $v_{i_{k+1}} \in \mathrm{Sp}(v_1, v_2, \ldots, v_{i_k})$. How will the next vector in the Gram-Schmidt process look like?

Solutions to the Exercises

Proof of Lemma 22.b.2 (b), the identity $0^\perp = V$:

Here 0 denotes the subset $\{0\} \subseteq V$, so by Definition 22.b.1

$$0^\perp = \{v \in V \mid \langle v, 0\rangle = 0\}.$$

By Lemma 22.a.3 **(a)** we have $\langle v, 0\rangle = 0$ for **every** $v \in V$, so the defining condition is satisfied by every vector of V and $0^\perp = V$.

The companion statement $V^\perp = \{0\}$, proved in the text, is the mirror image: the larger the set, the more orthogonality conditions, and the smaller its complement. **Q.E.D.**

Proof of Lemma 22.b.2 (d):

Assume $S \subseteq T$ and let $v \in T^\perp$, so that $\langle v, t\rangle = 0$ for every $t \in T$. Every $s \in S$ is in particular an element of T, so $\langle v, s\rangle = 0$ for every $s \in S$, that is, $v \in S^\perp$. Hence $T^\perp \subseteq S^\perp$, which is the assertion $S^\perp \supseteq T^\perp$.

Note that the inclusion reverses: passing to a larger set imposes more conditions on v, so fewer vectors survive. **Q.E.D.**

Proof of Exercise 22.10, that is, of Lemma 22.b.9:

Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \ldots & v_n \\ | & & | \end{pmatrix}, \qquad v_k \in \mathbb{R}^n_{\mathrm{col}},$$

and recall that the standard scalar product on $\mathbb{R}^n$ is $\langle u, v\rangle = u^\mathsf{T} v$.

(a) $\iff$ **(b)**. The rows of A^T are the vectors $v_1^\mathsf{T}, \ldots, v_n^\mathsf{T}$, so

$$A^\mathsf{T} A = \begin{pmatrix} - & v_1^\mathsf{T} & - \\ & \vdots & \\ - & v_n^\mathsf{T} & - \end{pmatrix} \begin{pmatrix} | & & | \\ v_1 & \ldots & v_n \\ | & & | \end{pmatrix} = \left(v_i^\mathsf{T} v_j\right)_{1\le i,j\le n} = \left(\langle v_i, v_j\rangle\right)_{1\le i,j\le n}.$$

By Definition 22.b.8, A is orthogonal exactly when $\langle v_i, v_j\rangle = \delta_{ij}$ for all i, j, which says precisely that the matrix just computed is I_n.

This is where the conjugations of the Hermitian case have gone. There the left factor was $\overline{A}^\mathsf{T}$, so the (i,j) entry came out as $\overline{v_i}^\mathsf{T} v_j = \langle v_j, v_i\rangle$ — the arguments in the opposite order. Over $\mathbb{R}$ there is no conjugation, the scalar product is symmetric, and the entry is $\langle v_i, v_j\rangle$. Since the condition imposed is the symmetric one δ_{ij}, the difference is invisible in the conclusion, but it is real in the computation.

(b) $\implies$ **(d)**. Suppose $A^\mathsf{T} A = I_n$. Taking determinants and using $\det(A^\mathsf{T}) = \det(A)$ gives $\det(A)^2 = 1$, so $\det(A) = \pm 1 \neq 0$ and A is invertible. Multiplying $A^\mathsf{T} A = I_n$ on the right by A^{-1} yields $A^\mathsf{T} = A^{-1}$.

(d) $\implies$ **(c)**. If $A^\mathsf{T} = A^{-1}$, then $AA^\mathsf{T} = AA^{-1} = I_n$.

(c) $\implies$ **(b)**. Suppose $AA^\mathsf{T} = I_n$. Determinants again give $\det(A)^2 = 1$, so A is invertible; multiplying on the left by A^{-1} gives $A^\mathsf{T} = A^{-1}$, and hence $A^\mathsf{T} A = A^{-1} A = I_n$.

The four statements are therefore equivalent. **Q.E.D.**

Proof of Exercise 22.11:

An orthonormal system is in particular an orthogonal system, hence linearly independent by Theorem 22.a.14 **(a)**. So $(e_1,\dots,e_r)$ can be extended to a basis $(e_1,\dots,e_r,u_{r+1},\dots,u_n)$ of V.

Now apply the Gram-Schmidt process (Theorem 22.a.16) to that basis, producing $w_1,\dots,w_n$. The point is that Gram-Schmidt does not disturb an initial segment that is already orthonormal. Indeed $w_1=e_1$, and if $2\le j\le r$ and $w_i=e_i$ for all $i<j$, then

$$w_j = e_j - \sum_{i=1}^{j-1}\frac{\langle e_j,w_i\rangle}{\langle w_i,w_i\rangle}w_i = e_j - \sum_{i=1}^{j-1}\frac{\langle e_j,e_i\rangle}{1}\,e_i = e_j,$$

because $\langle e_j,e_i\rangle=0$ for $i\ne j$ and $\langle e_i,e_i\rangle=1$. By induction $w_i=e_i$ for all $1\le i\le r$.

Therefore $(w_1,\dots,w_n)=(e_1,\dots,e_r,w_{r+1},\dots,w_n)$ is an orthogonal basis of V, and normalizing only the tail (the first r vectors already have norm 1) gives the orthonormal basis

$$\left(e_1,\dots,e_r,\frac{w_{r+1}}{\|w_{r+1}\|},\dots,\frac{w_n}{\|w_n\|}\right)$$

extending the given system.

There is a second route, worth recording because it is the one the geometry suggests: put $U:=\mathrm{Sp}(e_1,\dots,e_r)$. By Corollary 22.b.7, $\dim U^\perp=n-r$; choose an orthonormal basis $(f_{r+1},\dots,f_n)$ of $U^\perp$, which exists by Corollary 22.a.17. Every f_i is orthogonal to every e_j by the definition of $U^\perp$, so the concatenation $(e_1,\dots,e_r,f_{r+1},\dots,f_n)$ is an orthonormal system of n vectors, hence linearly independent, hence a basis of V. **Q.E.D.**

Solution of Exercise 22.12:

The next vector is 0. Suppose that after k successful steps we have an orthogonal system $e_1,\dots,e_k$ with

$$U:=\mathrm{Sp}(e_1,\dots,e_k)=\mathrm{Sp}(v_1,\dots,v_{i_k}),$$

and that the next vector to be processed satisfies $v_{i_{k+1}}\in\mathrm{Sp}(v_1,\dots,v_{i_k})=U$. The Gram-Schmidt formula prescribes

$$w:=v_{i_{k+1}}-\sum_{i=1}^{k}\frac{\langle v_{i_{k+1}},e_i\rangle}{\langle e_i,e_i\rangle}\,e_i.$$

But $(e_1,\dots,e_k)$ is an orthogonal system spanning U, and $v_{i_{k+1}}$ lies in U, so Theorem 22.a.14 **(b)** tells us the coefficients of $v_{i_{k+1}}$ in that system exactly:

$$v_{i_{k+1}}=\sum_{i=1}^{k}\frac{\langle v_{i_{k+1}},e_i\rangle}{\langle e_i,e_i\rangle}\,e_i.$$

The subtracted sum is therefore $v_{i_{k+1}}$ itself — it is the orthogonal projection $P_U(v_{i_{k+1}})$, which fixes U pointwise by Proposition 22.b.5 **(d)** — and consequently

$$w=v_{i_{k+1}}-v_{i_{k+1}}=0.$$

What this means for the process. The construction then stalls: the next step would divide by $\langle w,w\rangle=0$. Geometrically this is exactly right — $v_{i_{k+1}}$ contributes no new direction, so there is nothing left after removing its component along U.

The remedy. Discard every vector that produces 0 and carry on with the next one. The vectors that survive are precisely a maximal linearly independent sublist $v_{i_1},\dots,v_{i_r}$ of $v_1,\dots,v_m$, where $r=\dim\mathrm{Sp}(v_1,\dots,v_m)$, and the output $e_1,\dots,e_r$ is an orthogonal basis of $\mathrm{Sp}(v_1,\dots,v_m)$ satisfying $\mathrm{Sp}(e_1,\dots,e_k)=\mathrm{Sp}(v_1,\dots,v_{i_k})$ for every k.

This is what the proof of the extension of the QR-decomposition on Page 212 arranges in advance: rather than letting the process stall, it selects the indices $i_1<\cdots<i_r$ first and applies Gram-

Schmidt only to $v_{i_1}, \ldots, v_{i_r}$, which are linearly independent by construction.

Q.E.D.

Dual Spaces and Inner Products (Chapter 23)

Dual spaces and inner products (Section 23.a)

Let V be a vector space over K (any field). Recall that we have the dual space $V^* := \mathrm{Hom}(V, K)$, which is the space of linear maps $\varphi\colon V \to K$ (also called "*functionals*"). The dual space V^* is also a vector space over K.

Warm up 23.1: Assume $V = K^n$ and let $\varphi\colon K^n \to K$ be a functional. Then there exists a unique matrix $A = (a_1, \dots, a_n) \in \mathcal{M}_{1\times n}(K)$ such that $\varphi(v) = A \cdot v$ for all $v \in K^n$.

Indeed, let $e_1, \dots, e_n$ be the standard basis for K^n, and define $a_1 := \varphi(e_1), \dots, a_n := \varphi(e_n)$. Let $v = (v_1, \dots, v_n)^\mathsf{T} \in K^n$. Then

$$\varphi(v) = \varphi(v_1 e_1 + \cdots + v_n e_n) = v_1 \varphi(e_1) + \cdots + v_n \varphi(e_n) = v_1 a_1 + \cdots + v_n a_n = A \cdot v.$$

Consider now an inner product space $(V, \langle \cdot, \cdot \rangle)$ over K (where $K = \mathbb{R}$ or $\mathbb{C}$). Let $u \in V$. We define a map $\varphi_u\colon V \to K$ by mapping $v \mapsto \langle v, u \rangle \in K$.

Claim: The map φ_u is linear.

(This is left as an exercise).

Theorem 23.a.1: Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$). Let $\varphi \in V^*$. Then there exists a unique vector $u \in V$ such that $\varphi = \varphi_u$, that is, $\varphi(v) = \langle v, u \rangle$ for all $v \in V$.

In fact, for $K = \mathbb{R}$, the map $\Phi\colon V \to V^*$, given by $\Phi(u) := \varphi_u$, is an isomorphism. For $K = \mathbb{C}$, this map is injective and surjective, but it is only $\mathbb{C}$-antilinear, meaning that $\Phi(u_1 + u_2) = \Phi(u_1) + \Phi(u_2)$ and $\Phi(c \cdot u) = \bar{c} \cdot \Phi(u)$ for all $c \in \mathbb{C}$ and $u \in V$.

So, at least for $K = \mathbb{R}$, a scalar product on V gives us a canonical isomorphism $V \cong V^*$. (However, this isomorphism depends on the scalar product!)

Proof:

First, we prove that for all $\varphi \in V^*$ there exists a unique $u \in V$ such that $\varphi = \varphi_u$. Fix an orthonormal basis $\mathcal{B} := (e_1, \dots, e_n)$ for V, where $n = \dim V$. Let $\varphi \in V^*$, and let $v \in V$. We have $v = \sum_{i=1}^n \langle v, e_i \rangle \cdot e_i$. This implies:

$$\varphi(v) = \varphi\left(\sum_{i=1}^n \langle v, e_i \rangle \cdot e_i\right) = \sum_{i=1}^n \langle v, e_i \rangle \cdot \varphi(e_i) = \left\langle v, \sum_{i=1}^n \overline{\varphi(e_i)} \cdot e_i \right\rangle.$$

(Note that the conjugation is needed here only if $K = \mathbb{C}$). Define $u := \sum_{i=1}^n \overline{\varphi(e_i)} \cdot e_i$. It follows that $\varphi(v) = \langle v, u \rangle$, so $\varphi = \varphi_u$.

Uniqueness of u: Suppose that $\varphi_{u_1} = \varphi_{u_2}$ for $u_1, u_2 \in V$. This implies that $\langle v, u_1 \rangle = \langle v, u_2 \rangle$ for all $v \in V$. By a previous result, this implies that $u_1 = u_2$.

The statements about the linearity of Φ are left as an exercise. **Q.E.D.**

The adjoint map (Section 23.b)

Let V, W be vector spaces over K. Let $T\colon V \to W$ be a linear map. We have seen in Linear Algebra I that T induces a linear map $T^*\colon W^* \to V^*$ defined by $T^*(\varphi) := \varphi \circ T$.

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[dr, "T^*(\varphi)=\varphi\circ T"'] & W \arrow[d, "\varphi"] \\ & K \end{tikzcd}$$

Assume now that $(V, \langle\cdot,\cdot\rangle_V)$ and $(W, \langle\cdot,\cdot\rangle_W)$ are inner product spaces over K (where $K = \mathbb{R}$ or $K = \mathbb{C}$). Assume that $\dim V < \infty$. Given T as above, we can define a map $T' \colon W \to V$ using our *"canonical"* identifications $W \cong W^*$ and $V \cong V^*$. (This is straightforward for $K = \mathbb{R}$, but we have to be careful with $K = \mathbb{C}$.)

$$\begin{tikzcd} W^* \arrow[r, "T^*"] & V^* \arrow[d, bend left, "\Phi_V^{-1}"] \\ W \arrow[u, "\Phi_W"] \arrow[r, dashed, "T'"'] & V \arrow[u, "\Phi_V"'] \end{tikzcd}$$

Here, we define $T' := \Phi_V^{-1} \circ T^* \circ \Phi_W$. (Note that here we are using the fact that $\dim V < \infty$ so that Φ_V^{-1} exists).

What properties does T' have? Note that T' is uniquely defined by $\Phi_V \circ T' = T^* \circ \Phi_W$. We can unpack this relation as follows:

$$\begin{aligned} &\Phi_V \circ T' = T^* \circ \Phi_W \\ &\qquad\iff \forall w \in W, \quad (\Phi_V \circ T')(w) = (T^* \circ \Phi_W)(w) \\ &\qquad\iff \varphi_{T'(w)} = T^*(\varphi_w) \quad (\text{in } V^*) \\ &\qquad\iff \forall v \in V, \quad \varphi_{T'(w)}(v) = (T^*(\varphi_w))(v) \\ &\qquad\iff \langle v, T'(w)\rangle_V = \varphi_w(T(v)) = \langle T(v), w\rangle_W. \end{aligned}$$

In other words,

$$\langle T(v), w\rangle_W = \langle v, T'(w)\rangle_V \quad \forall v \in V, \forall w \in W. \tag{23.1}$$

Claim: The map T' is a linear map.

Proof:

For $K = \mathbb{R}$, the maps Φ_W, Φ_V, and T^* are linear, and therefore T' is also linear. For $K = \mathbb{C}$, the maps Φ_W and Φ_V are additive but only $\mathbb{C}$-antilinear. This implies that Φ_V^{-1} is also additive and $\mathbb{C}$-antilinear (this is an exercise to check!). The map T^* remains linear. Consequently, $T'(w_1 + w_2) = T'(w_1) + T'(w_2)$ (exercise). Now, let $c \in \mathbb{C}$ and $w \in W$. We compute:

$$\begin{aligned} T'(cw) &= (\Phi_V^{-1} \circ T^* \circ \Phi_W)(cw) \\ &= \Phi_V^{-1}(T^*(\bar{c} \cdot \Phi_W(w))) \\ &= \Phi_V^{-1}(\bar{c} \cdot T^*(\Phi_W(w))) \\ &= c \cdot \Phi_V^{-1}(T^*(\Phi_W(w))) \\ &= c \cdot T'(w). \end{aligned}$$

This proves that T' is linear. **Q.E.D.**

Notation 23.1: The linear map $T' \colon W \to V$ is denoted many times by $T^* \colon W \to V$. This makes sense because of the *"canonical"* identifications:

$$T \colon V \to W \quad \rightsquigarrow \quad \begin{tikzcd} W^* \arrow[r, "T^*"] & V^* \\ W \arrow[u, phantom, "\wr\wr"] \arrow[r, "T'"'] & V \arrow[u, phantom, "\wr\wr"] \end{tikzcd}$$

(Here we also assume $\dim W < \infty$ to have the canonical identification on the left). The map $T^*: W \to V$ is called the "*adjoint*" of T.

Basic properties of adjoint maps

Lemma 23.a.2 (*Properties of the Adjoint Map*): Let V, W, U be inner product spaces over K (where $K = \mathbb{R}$ or $\mathbb{C}$), all finite-dimensional. Let $S, T: V \to W$ and $R: W \to U$ be linear maps. Then:

(a) T^* is a linear map.
(b) $(S+T)^* = S^* + T^*$.
(c) $(\lambda T)^* = \overline{\lambda} T^*$ for all $\lambda \in K$.
(d) $(T^*)^* = T$.
(e) $(\mathrm{id}_V)^* = \mathrm{id}_V$.
(f) $(R \circ T)^* = T^* \circ R^*$.

Proof:

(a) This was proved already. (We will use repeatedly that if $\langle x, y_1 \rangle = \langle x, y_2 \rangle$ for all x, then $y_1 = y_2$).

(b) Let $w \in W$ and $v \in V$. We have:
$$\begin{aligned}\langle v, (S+T)^*(w) \rangle &= \langle (S+T)(v), w \rangle \\ &= \langle S(v), w \rangle + \langle T(v), w \rangle \\ &= \langle v, S^*(w) \rangle + \langle v, T^*(w) \rangle \\ &= \langle v, (S^* + T^*)(w) \rangle.\end{aligned}$$
This holds for all $v \in V$, which implies $(S+T)^*(w) = (S^* + T^*)(w)$. Since this holds for all $w \in W$, we conclude that $(S+T)^* = S^* + T^*$.

(c) This is left as an exercise.

(d) Let $v \in V$ and $w \in W$. We have:
$$\begin{aligned}\langle w, (T^*)^*(v) \rangle &= \langle T^*(w), v \rangle \\ &= \overline{\langle v, T^*(w) \rangle} \\ &= \overline{\langle T(v), w \rangle} \\ &= \langle w, T(v) \rangle.\end{aligned}$$
This holds for all $w \in W$, which implies $(T^*)^*(v) = T(v)$. Since this holds for all $v \in V$, we conclude that $(T^*)^* = T$.

(e) This is left as an exercise.

(f) Let $v \in V$ and $u \in U$. We have:
$$\begin{aligned}\langle v, (R \circ T)^*(u) \rangle &= \langle (R \circ T)(v), u \rangle \\ &= \langle R(T(v)), u \rangle \\ &= \langle T(v), R^*(u) \rangle \\ &= \langle v, T^*(R^*(u)) \rangle \\ &= \langle v, (T^* \circ R^*)(u) \rangle.\end{aligned}$$
This holds for all $v \in V$, which implies $(R \circ T)^*(u) = (T^* \circ R^*)(u)$. Since this holds for all $u \in U$, we conclude that $(R \circ T)^* = T^* \circ R^*$.

Q.E.D.

Lemma 23.a.3 (*Orthogonal Relations of Adjoint Maps*): Let V, W be finite-dimensional inner product spaces. Let $T\colon V \to W$ and let $T^*\colon W \to V$ be the adjoint of T. Then:

(a) $\ker(T^*) = (\operatorname{Im} T)^\perp$.
(b) $\operatorname{Im} T^* = (\ker T)^\perp$.
(c) $\ker T = (\operatorname{Im} T^*)^\perp$.
(d) $\operatorname{Im} T = (\ker T^*)^\perp$.

Proof:

(a) Let $w \in W$. Then:

$$\begin{aligned} w \in \ker(T^*) &\iff T^*(w) = 0 \\ &\iff \forall v \in V, \quad \langle v, T^*(w)\rangle = 0 \\ &\iff \forall v \in V, \quad \langle T(v), w\rangle = 0 \\ &\iff w \perp T(v) \quad \forall v \in V \\ &\iff w \perp \operatorname{Im}(T) \\ &\iff w \in (\operatorname{Im} T)^\perp. \end{aligned}$$

(b) Use statement **(d)** with T^* instead of T:

$$\operatorname{Im} T^* = (\ker (T^*)^*)^\perp = (\ker T)^\perp.$$

(c) Apply the orthogonal complement $(-)^\perp$ to statement **(b)**:

$$(\operatorname{Im} T^*)^\perp = ((\ker T)^\perp)^\perp = \ker T.$$

(d) Apply the orthogonal complement $(-)^\perp$ to both sides of **(a)**:

$$(\ker T^*)^\perp = ((\operatorname{Im} T)^\perp)^\perp = \operatorname{Im} T.$$

Q.E.D.

Matrix representation of adjoint maps

Proposition 23.a.4 (*Matrix Representation of the Adjoint*): Let V, W be finite-dimensional inner product spaces and $T\colon V \to W$ a linear map. Let $\mathcal{B} := (e_1, \dots, e_n)$ be an orthonormal basis for V, and $\mathcal{C} := (f_1, \dots, f_m)$ an orthonormal basis for W. Then:

$$[T^*]^{\mathcal{C}}_{\mathcal{B}} = \left(\overline{[T]^{\mathcal{B}}_{\mathcal{C}}}\right)^{\mathsf{T}} = \left([T]^{\mathcal{B}}_{\mathcal{C}}\right)^* \qquad (\text{“}\textit{adjoint matrix}\text{”})$$

Proof:

Write $A := (a_{ij})_{\substack{1\le i\le m\\ 1\le j\le n}} := [T]^{\mathcal{B}}_{\mathcal{C}}$ and $B := (b_{ij})_{\substack{1\le i\le n\\ 1\le j\le m}} := [T^*]^{\mathcal{C}}_{\mathcal{B}}$. We have:

$$b_{ij} = \langle T^*(f_j), e_i\rangle = \overline{\langle e_i, T^*(f_j)\rangle} = \overline{\langle T(e_i), f_j\rangle} = \overline{a_{ji}}.$$

Therefore, $B = (\overline{A})^{\mathsf{T}} = A^*$. Q.E.D.

Corollary 23.a.5 (*Adjoint of a Matrix Map*): Let $A \in \mathcal{M}_{m\times n}(K)$ (where $K = \mathbb{R}$ or $\mathbb{C}$), and let $T_A\colon K^n \to K^m$ be the linear map defined by $T_A(v) = A \cdot v$. Endow K^n and K^m with their standard inner products. Then $(T_A)^* = T_{A^*}$.

Proposition 23.a.6 (*Properties of the Adjoint Matrix*): Let $K = \mathbb{R}, \mathbb{C}$, and let $A, B \in \mathcal{M}_{m\times n}(K)$ and $C \in \mathcal{M}_{n\times p}(K)$. Then:

(a) $\overline{(A+B)}^{\mathsf{T}} = \overline{A}^{\mathsf{T}} + \overline{B}^{\mathsf{T}}$.
(b) $\overline{(\lambda A)}^{\mathsf{T}} = \overline{\lambda}\,\overline{A}^{\mathsf{T}}$ for all $\lambda \in K$.

(c) $\overline{(\overline{A}^{\mathsf{T}})}^{\mathsf{T}} = A$.
(d) $\overline{I_n}^{\mathsf{T}} = I_n$.
(e) $\overline{(A \cdot C)}^{\mathsf{T}} = \overline{C}^{\mathsf{T}} \cdot \overline{A}^{\mathsf{T}}$.

Proof:

This is left as an exercise. **Q.E.D.**

Exercise Solutions

Proof of Linearity of φ_u (on Page 216):

Let $v_1, v_2 \in V$ and $c \in K$. We have:

$$\varphi_u(v_1 + v_2) = \langle v_1 + v_2, u\rangle = \langle v_1, u\rangle + \langle v_2, u\rangle = \varphi_u(v_1) + \varphi_u(v_2),$$

and

$$\varphi_u(cv_1) = \langle cv_1, u\rangle = c\langle v_1, u\rangle = c\varphi_u(v_1).$$

Thus, φ_u is a linear map. **Q.E.D.**

Proof of Theorem 23.a.1:

For $K = \mathbb{R}$, we have $\Phi(u) = \varphi_u$. Then:

$$\Phi(u_1 + u_2)(v) = \langle v, u_1 + u_2\rangle = \langle v, u_1\rangle + \langle v, u_2\rangle = \Phi(u_1)(v) + \Phi(u_2)(v),$$

and

$$\Phi(cu)(v) = \langle v, cu\rangle = c\langle v, u\rangle = c\Phi(u)(v).$$

(Here we used the fact that for $K = \mathbb{R}$, $c = \overline{c}$). Thus Φ is linear.

For $K = \mathbb{C}$, the additivity follows identically. For scalar multiplication:

$$\Phi(cu)(v) = \langle v, cu\rangle = \overline{c}\langle v, u\rangle = \overline{c}\Phi(u)(v).$$

Thus $\Phi(cu) = \overline{c}\Phi(u)$, making it $\mathbb{C}$-antilinear.

For the inverse Φ^{-1}: since Φ is additive, taking $\varphi_1 = \Phi(u_1)$ and $\varphi_2 = \Phi(u_2)$ gives $\Phi^{-1}(\varphi_1 + \varphi_2) = \Phi^{-1}(\varphi_1) + \Phi^{-1}(\varphi_2)$, so Φ^{-1} is additive. For scalar multiplication, let $\varphi = \Phi(u)$ and $\lambda \in \mathbb{C}$. Then $\overline{\lambda}\varphi = \overline{\lambda}\Phi(u) = \Phi(\lambda u)$. Applying Φ^{-1} to both sides gives $\Phi^{-1}(\overline{\lambda}\varphi) = \lambda u = \lambda\Phi^{-1}(\varphi)$. Setting $c = \overline{\lambda}$ (so $\lambda = \overline{c}$) gives $\Phi^{-1}(c\varphi) = \overline{c}\Phi^{-1}(\varphi)$, proving Φ^{-1} is $\mathbb{C}$-antilinear. **Q.E.D.**

Proof of Additivity of T' (on Page 217):

We have:

$$\begin{aligned}
T'(w_1 + w_2) &= (\Phi_V^{-1} \circ T^* \circ \Phi_W)(w_1 + w_2) \\
&= (\Phi_V^{-1} \circ T^*)(\Phi_W(w_1) + \Phi_W(w_2)) && \text{(since } \Phi_W \text{ is additive)} \\
&= \Phi_V^{-1}(T^*(\Phi_W(w_1)) + T^*(\Phi_W(w_2))) && \text{(since } T^* \text{ is linear)} \\
&= \Phi_V^{-1}(T^*(\Phi_W(w_1))) + \Phi_V^{-1}(T^*(\Phi_W(w_2))) && \text{(since } \Phi_V^{-1} \text{ is additive)} \\
&= T'(w_1) + T'(w_2).
\end{aligned}$$

Q.E.D.

Proof of Lemma 23.a.2 (c):

Let $v \in V$ and $w \in W$. We have:

$$\langle v, (\lambda T)^*(w)\rangle = \langle (\lambda T)(v), w\rangle = \langle \lambda T(v), w\rangle = \lambda\langle T(v), w\rangle = \lambda\langle v, T^*(w)\rangle = \langle v, \overline{\lambda}T^*(w)\rangle.$$

This holds for all $v \in V$, so $(\lambda T)^*(w) = \overline{\lambda} T^*(w)$. Thus $(\lambda T)^* = \overline{\lambda} T^*$. **Q.E.D.**

Proof of Lemma 23.a.2 (e):

Let $v, w \in V$. We have:

$$\langle v, (\mathrm{id}_V)^*(w) \rangle = \langle \mathrm{id}_V(v), w \rangle = \langle v, w \rangle.$$

This holds for all $v \in V$, so $(\mathrm{id}_V)^*(w) = w = \mathrm{id}_V(w)$. Thus $(\mathrm{id}_V)^* = \mathrm{id}_V$. **Q.E.D.**

Proof of Proposition 23.a.6:

These statements follow directly from the properties of matrix transpose and complex conjugation:

(a) $\overline{(A+B)}^{\mathsf{T}} = (\overline{A} + \overline{B})^{\mathsf{T}} = \overline{A}^{\mathsf{T}} + \overline{B}^{\mathsf{T}}$.

(b) $\overline{(\lambda A)}^{\mathsf{T}} = (\overline{\lambda A})^{\mathsf{T}} = \overline{\lambda A}^{\mathsf{T}}$.

(c) $\overline{(\overline{A}^{\mathsf{T}})}^{\mathsf{T}} = (\overline{\overline{A}}^{\mathsf{T}})^{\mathsf{T}} = (A^{\mathsf{T}})^{\mathsf{T}} = A$.

(d) $\overline{I_n}^{\mathsf{T}} = I_n^{\mathsf{T}} = I_n$ (since I_n is a real matrix).

(e) $\overline{(A \cdot C)}^{\mathsf{T}} = (\overline{A} \cdot \overline{C})^{\mathsf{T}} = \overline{C}^{\mathsf{T}} \cdot \overline{A}^{\mathsf{T}}$.

Q.E.D.

Spectral Theorems (Chapter 24)

AI-Note: The professor left a note to *"explain the name"* of the Spectral Theorems. In linear algebra, the "*spectrum*" of a linear operator is its set of eigenvalues. The Spectral Theorems provide conditions under which an operator can be completely described by its spectrum—specifically, when there exists a basis for the entire space consisting solely of the operator's eigenvectors (making the operator diagonalizable).

Definition 24.a.1: Let V be an inner product space and $T\colon V \to V$ an endomorphism. We say that T is "*orthogonally diagonalizable*" if there exists an orthonormal basis for V consisting of eigenvectors of T.

Lemma 24.a.2: If T is orthogonally diagonalizable, then $T \circ T^* = T^* \circ T$.

Proof:

Let $\mathcal{B}$ be an orthonormal basis for V such that all the vectors in $\mathcal{B}$ are eigenvectors of T. This implies that the representation matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a diagonal matrix.

Since $\mathcal{B}$ is an orthonormal basis, we also know that the matrix of the adjoint map is the conjugate transpose of the matrix of the map, meaning $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$. Since the conjugate transpose of a diagonal matrix is still a diagonal matrix, $[T^*]_{\mathcal{B}}^{\mathcal{B}}$ is also a diagonal matrix.

Recall that if M and N are diagonal matrices of the form

$$\begin{pmatrix} * & & 0 \\ & \ddots & \\ 0 & & * \end{pmatrix},$$

then they must commute, meaning $M \cdot N = N \cdot M$. Therefore, we have:

$$[T]_{\mathcal{B}}^{\mathcal{B}} \cdot [T^*]_{\mathcal{B}}^{\mathcal{B}} = [T^*]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}}.$$

By the properties of representation matrices, this implies that

$$[T \circ T^*]_{\mathcal{B}}^{\mathcal{B}} = [T^* \circ T]_{\mathcal{B}}^{\mathcal{B}},$$

and therefore, $T \circ T^* = T^* \circ T$. **Q.E.D.**

Definition 24.a.3: Let V be an inner product space. An endomorphism $T\colon V \to V$ is called "*normal*" if $T \circ T^* = T^* \circ T$.

Similarly, a matrix $A \in \mathcal{M}_{n\times n}(K)$ (where $K = \mathbb{R}$ or $\mathbb{C}$) is called "*normal*" if $A \cdot \overline{A}^{\mathsf{T}} = \overline{A}^{\mathsf{T}} \cdot A$. (Note that A is a normal matrix if and only if the map $T_A\colon K^n \to K^n$ is a normal endomorphism, provided we endow K^n with its standard inner product).

We have seen in the previous lemma (Lemma 24.a.2) that if $T\colon V \to V$ is orthogonally diagonalizable, then T is normal.

Question 24.1: Is the converse statement true?

Over $K = \mathbb{R}$, the answer is NO! For example, consider the matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The linear map $T_A\colon \mathbb{R}^2 \to \mathbb{R}^2$ is a clockwise rotation by $90°$, so there are no eigenvalues (in $\mathbb{R}$) and no eigenvectors. Thus, it cannot be diagonalizable. However, $\overline{A}^{\mathsf{T}} = A^{\mathsf{T}} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and we can easily check that $A \cdot A^{\mathsf{T}} = A^{\mathsf{T}} \cdot A = I_2$, meaning A is indeed normal.

What about over $\mathbb{C}$?

Theorem 24.a.4 (*Spectral Theorem over $\mathbb{C}$*): Let V be a finite-dimensional inner product space over $\mathbb{C}$, and let $T\colon V \to V$ be a linear map. Then T is orthogonally diagonalizable if and only if T is normal.

For the proof, we need a lemma.

Lemma 24.a.5 (*Properties of Normal Endomorphisms*): Let V be a finite-dimensional inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$). Let $T\colon V \to V$ be a normal endomorphism. Then:

(a) For all $v \in V$, $\|Tv\| = \|T^*v\|$.

(b) For all $\lambda \in K$, the endomorphism $T - \lambda\,\mathrm{id}_V$ is also normal.

(c) If v is an eigenvector of T with eigenvalue λ, then v is also an eigenvector of T^* with eigenvalue $\overline{\lambda}$.

(d) Let v_1, v_2 be eigenvectors of T with eigenvalues λ_1, λ_2 respectively. If $\lambda_1 \neq \lambda_2$, then $\langle v_1, v_2\rangle = 0$ (i.e., $v_1 \perp v_2$).

Proof:

(a) We compute the squared norm:
$$\|Tv\|^2 = \langle Tv, Tv\rangle = \langle v, T^*Tv\rangle = \langle v, TT^*v\rangle = \langle T^*v, T^*v\rangle = \|T^*v\|^2,$$
where we used the definition of the adjoint and the fact that T is normal ($T \circ T^* = T^* \circ T$).

(b) We expand the compositions:
$$\begin{aligned}(T - \lambda\,\mathrm{id}_V) \circ (T - \lambda\,\mathrm{id}_V)^* &= (T - \lambda\,\mathrm{id}_V) \circ (T^* - \overline{\lambda}\,\mathrm{id}_V) \\ &= T \circ T^* - \overline{\lambda}T - \lambda T^* + \lambda\overline{\lambda}\,\mathrm{id}_V\,. \quad (*)\end{aligned}$$
On the other hand:
$$\begin{aligned}(T - \lambda\,\mathrm{id}_V)^* \circ (T - \lambda\,\mathrm{id}_V) &= (T^* - \overline{\lambda}\,\mathrm{id}_V) \circ (T - \lambda\,\mathrm{id}_V) \\ &= T^* \circ T - \lambda T^* - \overline{\lambda}T + \overline{\lambda}\lambda\,\mathrm{id}_V\,. \quad (**)\end{aligned}$$
Because $T \circ T^* = T^* \circ T$, the expressions $(*)$ and $(**)$ are identically equal.

(c) If v is an eigenvector of T with eigenvalue λ, then $(T - \lambda\,\mathrm{id}_V)v = 0$. By statement **(b)**, $T - \lambda\,\mathrm{id}_V$ is normal, hence by statement **(a)**, we have:
$$\|(T^* - \overline{\lambda}\,\mathrm{id}_V)v\| = \|(T - \lambda\,\mathrm{id}_V)v\| = \|0\| = 0.$$
This implies that $(T^* - \overline{\lambda}\,\mathrm{id}_V)v = 0$, and therefore $T^*v = \overline{\lambda}v$.

(d) We compute the inner product in two ways:
$$\lambda_1\langle v_1, v_2\rangle = \langle \lambda_1 v_1, v_2\rangle = \langle Tv_1, v_2\rangle = \langle v_1, T^*v_2\rangle.$$
By statement **(c)**, $T^*v_2 = \overline{\lambda_2}v_2$. Thus:
$$\langle v_1, T^*v_2\rangle = \langle v_1, \overline{\lambda_2}v_2\rangle = \lambda_2\langle v_1, v_2\rangle.$$
Equating the two results gives $\lambda_1\langle v_1, v_2\rangle = \lambda_2\langle v_1, v_2\rangle$, which implies $(\lambda_1 - \lambda_2)\langle v_1, v_2\rangle = 0$. Since $\lambda_1 \neq \lambda_2$, we must have $\langle v_1, v_2\rangle = 0$.

Q.E.D.

Proof of the Spectral Theorem over $\mathbb{C}$ (Theorem 24.a.4):

We only need to prove the direction *"T is normal $\implies$ T is orthogonally diagonalizable"*, as the reverse direction was already established in Lemma 24.a.2.

By a previous theorem, T is trigonalizable (here we explicitly use the fact that $K = \mathbb{C}$ and the Fundamental Theorem of Algebra). That is, there exists a basis $\mathcal{C}$ for V such that $[T]_{\mathcal{C}}^{\mathcal{C}}$ is an upper triangular matrix. By applying the Gram-Schmidt orthogonalization process to the basis $\mathcal{C}$, we obtain a new basis $\mathcal{B} := (v_1, \ldots, v_n)$ which is orthonormal, and such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is still upper triangular. (This is left as an exercise to check. The main point is that if $\mathcal{C} = (u_1, \dots, u_n)$, then Gram-Schmidt gives an orthonormal basis $\mathcal{B} = (v_1, \dots, v_n)$ with $\mathrm{Sp}(v_1, \dots, v_k) = \mathrm{Sp}(u_1, \dots, u_k)$ for all $1 \le k \le n$).

We claim that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is actually diagonal. To see this, let us write the matrix explicitly as $[T]_{\mathcal{B}}^{\mathcal{B}} = (a_{ij})_{1 \le i \le n, 1 \le j \le n}$:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}.$$

Since $\mathcal{B}$ is orthonormal, the norm of Tv_1 is simply the norm of the first column of the matrix, which means $\|Tv_1\|^2 = |a_{11}|^2$. We also know that $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$, which means taking the conjugate transpose:

$$\left(\overline{[T]_{\mathcal{B}}^{\mathcal{B}}}\right)^{\mathsf{T}} = \begin{pmatrix} \overline{a_{11}} & 0 & \dots & 0 \\ \overline{a_{12}} & \overline{a_{22}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \overline{a_{1n}} & \overline{a_{2n}} & \dots & \overline{a_{nn}} \end{pmatrix}.$$

Thus, the first column of $[T^*]_{\mathcal{B}}^{\mathcal{B}}$ is the conjugate transpose of the first row of $[T]_{\mathcal{B}}^{\mathcal{B}}$. Therefore, we have:

$$\|T^* v_1\|^2 = \sum_{k=1}^{n} |a_{1k}|^2 = |a_{11}|^2 + |a_{12}|^2 + \dots + |a_{1n}|^2.$$

As T is normal, Lemma 24.a.5 **(a)** tells us that $\|Tv_1\| = \|T^* v_1\|$. This implies that

$$|a_{11}|^2 = |a_{11}|^2 + |a_{12}|^2 + \dots + |a_{1n}|^2,$$

hence $a_{12} = \dots = a_{1n} = 0$. This shows that the matrix now looks like:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & * & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & * & \dots & * \end{pmatrix}.$$

Now, consider v_2. We have $\|Tv_2\|^2 = |a_{22}|^2$ and $\|T^* v_2\|^2 = |a_{22}|^2 + |a_{23}|^2 + \dots + |a_{2n}|^2$. As $\|Tv_2\| = \|T^* v_2\|$, we similarly obtain $a_{23} = \dots = a_{2n} = 0$.

We can continue this process by induction, and show that for all $1 \le k < n$ we have $a_{k,k+1} = \dots = a_{k,n} = 0$. Hence, $[T]_{\mathcal{B}}^{\mathcal{B}}$ is diagonal. Since $\mathcal{B}$ is an orthonormal basis that diagonalizes T, it consists of eigenvectors, proving that T is orthogonally diagonalizable. **Q.E.D.**

Remark 24.1: In the proof above, we used $K = \mathbb{C}$ **only** at one place: when we stated that T is trigonalizable. Therefore, we immediately obtain the following corollary for the real case.

Corollary 24.a.6: Let V be a finite-dimensional inner product space over $\mathbb{R}$, and let $T\colon V \to V$ be an endomorphism which is normal. If T is trigonalizable (which is true if and only if the characteristic polynomial $P_T(x)$ factors as a product of linear factors over $\mathbb{R}$), then T is orthogonally diagonalizable.

The Spectral Theorem over $\mathbb{R}$

We've seen that over $\mathbb{R}$, the statement *"$T \in \mathrm{End}(V)$ is normal $\implies$ T is orthogonally diagonalizable"* does not hold in general. We need a stronger condition.

Lemma 24.a.7: Let V be a finite-dimensional inner product space over $\mathbb{R}$, and let $T \in \operatorname{End}(V)$ be orthogonally diagonalizable. Then $T^* = T$.

Proof:

Let $\mathcal{B}$ be an orthonormal basis such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a diagonal matrix. Since we are over $\mathbb{R}$, the conjugate transpose is simply the standard transpose. Then:

$$[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^{\mathsf{T}} = [T]_{\mathcal{B}}^{\mathcal{B}},$$

where the first equality holds because we are over $\mathbb{R}$, and the second equality holds because $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a diagonal matrix. Because their representation matrices with respect to the same basis are equal, it follows that $T^* = T$. **Q.E.D.**

Definition 24.a.8: Let V be an inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$), and $T \in \operatorname{End}(V)$. T is called "*self adjoint*" if $T^* = T$.

Similarly, a matrix $A \in \mathcal{M}_{n\times n}(K)$ is called "*self adjoint*" if $A^* = A$. (Note that over $K = \mathbb{R}$, a self adjoint matrix is precisely a symmetric matrix, $A^{\mathsf{T}} = A$).

Exercise 24.1 (*Matrix Representation of Self-Adjoint Maps*): Show that $T\colon V \to V$ is self adjoint if and only if for every orthonormal basis $\mathcal{B}$ of V, the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is self adjoint.

Theorem 24.a.9 (*Spectral Theorem over $\mathbb{R}$*): Let V be a Euclidean space (i.e., an inner product space over $\mathbb{R}$) of finite dimension. Then T is orthogonally diagonalizable if and only if T is self adjoint.

For the proof, we need the following lemma:

Lemma 24.a.10: Let V be an inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$), and let $T \in \operatorname{End}(V)$ be self adjoint. Then:

- **(a)** All eigenvalues of T are real.
- **(b)** The characteristic polynomial $P_T(x)$ factors into a product of linear factors over K.

Proof:

- **(a)** If $K = \mathbb{R}$, there is nothing to show. So assume $K = \mathbb{C}$. Let $\lambda \in \mathbb{C}$ be an eigenvalue and $v \in V$ a corresponding non-zero eigenvector. We've seen in Lemma 24.a.5 **(c)** that v is also an eigenvector of T^* with eigenvalue $\overline{\lambda}$. But $T^* = T$. This implies:

  $$\lambda v = Tv = T^* v = \overline{\lambda} v.$$

  Since $v \neq 0$, we must have $\lambda = \overline{\lambda}$, which implies $\lambda \in \mathbb{R}$.
- **(b)** For $K = \mathbb{C}$, this follows immediately from the Fundamental Theorem of Algebra. So assume $K = \mathbb{R}$. Let $\mathcal{B}$ be an orthonormal basis for V and consider the matrix $A := [T]_{\mathcal{B}}^{\mathcal{B}} \in \mathcal{M}_{n\times n}(\mathbb{R})$. The statement would follow if we show that the characteristic polynomial $P_A(x)$ factors as a product of linear terms with real coefficients.

  Now, $T^* = T$, hence $A = [T]_{\mathcal{B}}^{\mathcal{B}} = [T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^* = A^* = A^{\mathsf{T}}$. Consider $\mathbb{C}^n$ endowed with its standard inner product, and view A as a matrix in $\mathcal{M}_{n\times n}(\mathbb{C})$. Let $T_A\colon \mathbb{C}^n \to \mathbb{C}^n$ be the corresponding linear map. The standard basis $\mathcal{E}_n = (e_1, \ldots, e_n)$ is orthonormal, and $[T_A]_{\mathcal{E}_n}^{\mathcal{E}_n} = A$.

  We know that $P_{T_A}(x) = P_A(x)$. When viewed as a polynomial over $\mathbb{C}$ (i.e., in $\mathbb{C}[x]$), it factors as a product of linear terms:

  $$P_A(x) = \prod_{k=1}^{n} (x - \lambda_k) \in \mathbb{C}[x], \quad (*) \tag{24.1}$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of T_A. But T_A is self adjoint (even when viewed over $\mathbb{C}$ since $A^\mathsf{T} = A = A^*$), so by part **(a)**, all the eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$. Therefore, the factorization $(*)$ actually holds in $\mathbb{R}[x]$.

Q.E.D.

Proof of the Spectral Theorem over $\mathbb{R}$ (Theorem 24.a.9):

The forward direction was proven in Lemma 24.a.7. For the reverse direction, if T is self adjoint, then by Lemma 24.a.10, its characteristic polynomial splits over $\mathbb{R}$. This implies T is trigonalizable over $\mathbb{R}$. Furthermore, since $T = T^*$, T trivially commutes with its adjoint, making it normal. By Corollary 24.a.6, a normal map that is trigonalizable over $\mathbb{R}$ is orthogonally diagonalizable.

Q.E.D.

Spectral Theorems for Matrices

Let $K = \mathbb{R}$ or $\mathbb{C}$. Recall that a matrix $A \in \mathcal{M}_{n\times n}(K)$ is called normal if $A \cdot A^* = A^* \cdot A$.

Example 24.1 (*Classes of Normal Matrices*):

(a) For $K = \mathbb{R}$: Symmetric matrices and orthogonal matrices $\mathrm{O}(n)$ are normal.

(b) For $K = \mathbb{C}$: Self adjoint matrices and unitary matrices $\mathrm{U}(n)$ are normal.

Remark 24.2: There exist symmetric matrices in $\mathcal{M}_{n\times n}(\mathbb{C})$ that are not normal. For example, let $A = \left(\begin{smallmatrix} i & i \\ i & 1 \end{smallmatrix}\right)$. We have $A^\mathsf{T} = A$, so it is symmetric. However, $A^* = \left(\begin{smallmatrix} -i & -i \\ -i & 1 \end{smallmatrix}\right)$. A quick calculation shows that $A \cdot A^* \neq A^* \cdot A$, so A is not normal.

Theorem 24.a.11 (*Matrix Spectral Theorems*):

(a) If $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is normal, then there exists a unitary matrix $U \in \mathrm{U}(n)$ such that $U^{-1}AU$ is diagonal. (Note that $U^{-1} = U^*$).

(b) If $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is a symmetric matrix, then there exists an orthogonal matrix $O \in \mathrm{O}(n)$ such that $O^{-1}AO$ is diagonal. (Note that $O^{-1} = O^\mathsf{T}$).

Proof Sketch:

The proof is left as an exercise, based on the following key points translating between maps and matrices:

(a) $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is normal if and only if $T_A \colon \mathbb{C}^n \to \mathbb{C}^n$ is normal (when we endow $\mathbb{C}^n$ with its standard inner product).

(b) If $\mathcal{B}$ is an orthonormal basis for $\mathbb{C}^n$, then the transition matrix $U := [\mathrm{id}_{\mathbb{C}^n}]^{\mathcal{B}}_{\mathcal{E}_n}$ is unitary.

(c) $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is symmetric if and only if $T_A \colon \mathbb{R}^n \to \mathbb{R}^n$ is self adjoint, when we endow $\mathbb{R}^n$ with its standard inner product.

(d) If $\mathcal{B}$ is an orthonormal basis for $\mathbb{R}^n$, then the transition matrix $O := [\mathrm{id}_{\mathbb{R}^n}]^{\mathcal{B}}_{\mathcal{E}_n}$ is orthogonal.

Q.E.D.

Exercise Solutions

Proof of Gram-Schmidt Preserving Upper Triangularity (on Page 223):

Let $\mathcal{C} := (u_1, \ldots, u_n)$ be a basis for V such that the representation matrix $[T]^{\mathcal{C}}_{\mathcal{C}}$ is upper triangular. This means that for any $1 \leq k \leq n$, the vector $T(u_k)$ is a linear combination of $u_1, \ldots, u_k$. In other words, $T(u_k) \in \mathrm{Sp}(u_1, \ldots, u_k)$. Consequently, the subspace $W_k := \mathrm{Sp}(u_1, \ldots, u_k)$ is T-invariant, meaning $T(W_k) \subseteq W_k$.

Applying the Gram-Schmidt orthogonalization process to the basis $\mathcal{C}$ produces an orthonormal basis $\mathcal{B} := (v_1, \ldots, v_n)$ with the property that for every $1 \leq k \leq n$, the spans are preserved: $\mathrm{Sp}(v_1, \ldots, v_k) = \mathrm{Sp}(u_1, \ldots, u_k) = W_k$.

Now, consider the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$. The k-th column of this matrix consists of the coordinates of $T(v_k)$ with respect to the basis $\mathcal{B}$. Since $v_k \in W_k$ and W_k is T-invariant, we must have $T(v_k) \in W_k = \mathrm{Sp}(v_1, \ldots, v_k)$. This means $T(v_k)$ can be written strictly as a linear combination of $v_1, \ldots, v_k$, meaning all coefficients for vectors v_j with $j > k$ are exactly zero. Therefore, $[T]_{\mathcal{B}}^{\mathcal{B}}$ is an upper triangular matrix. **Q.E.D.**

Proof of Matrix Self Adjointness Equivalence (on Page 225):

Let $T \in \mathrm{End}(V)$. First, suppose T is self adjoint, so $T = T^*$. Let $\mathcal{B}$ be an arbitrary orthonormal basis for V. By the properties of representation matrices for adjoint maps in orthonormal bases, we have $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$. Since $T = T^*$, we substitute to get $[T]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$, which means the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is self adjoint.

Conversely, suppose that for any orthonormal basis $\mathcal{B}$ of V, the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is self adjoint, i.e., $[T]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$. Again, using the identity $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$, we see that $[T]_{\mathcal{B}}^{\mathcal{B}} = [T^*]_{\mathcal{B}}^{\mathcal{B}}$. Because the map sending an endomorphism to its representation matrix with respect to a fixed basis is an isomorphism, this equality of matrices implies the equality of the maps, hence $T = T^*$, meaning T is self adjoint. **Q.E.D.**

Proof of Theorem 24.a.11:

(a) Let $A \in \mathcal{M}_{n \times n}(\mathbb{C})$ be a normal matrix. By point (a), the linear map $T_A : \mathbb{C}^n \to \mathbb{C}^n$ is a normal endomorphism with respect to the standard inner product. By the Spectral Theorem over $\mathbb{C}$ (Theorem 24.a.4), T_A is orthogonally diagonalizable. Thus, there exists an orthonormal basis $\mathcal{B} := (v_1, \ldots, v_n)$ for $\mathbb{C}^n$ consisting of eigenvectors of T_A.

The representation matrix of T_A with respect to $\mathcal{B}$, which is $[T_A]_{\mathcal{B}}^{\mathcal{B}}$, is a diagonal matrix D. Let $\mathcal{E}_n$ be the standard basis for $\mathbb{C}^n$. The transition matrix from $\mathcal{B}$ to $\mathcal{E}_n$ is $U := [\mathrm{id}_{\mathbb{C}^n}]_{\mathcal{E}_n}^{\mathcal{B}}$, whose columns are precisely the vectors $v_1, \ldots, v_n$. Since $\mathcal{B}$ is an orthonormal basis, point **(b)** guarantees that U is a unitary matrix.

Using the change of basis formula, we have:

$$D = [T_A]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_{\mathbb{C}^n}]_{\mathcal{B}}^{\mathcal{E}_n} \cdot [T_A]_{\mathcal{E}_n}^{\mathcal{E}_n} \cdot [\mathrm{id}_{\mathbb{C}^n}]_{\mathcal{E}_n}^{\mathcal{B}} = U^{-1} \cdot A \cdot U.$$

Thus, $U^{-1}AU$ is diagonal.

(b) Let $A \in \mathcal{M}_{n \times n}(\mathbb{R})$ be a symmetric matrix. By point (c), the linear map $T_A : \mathbb{R}^n \to \mathbb{R}^n$ is self adjoint. By the Spectral Theorem over $\mathbb{R}$ (Theorem 24.a.9), T_A is orthogonally diagonalizable. Thus, there exists an orthonormal basis $\mathcal{B}$ for $\mathbb{R}^n$ consisting of eigenvectors of T_A, making $[T_A]_{\mathcal{B}}^{\mathcal{B}}$ a diagonal matrix D.

Let $O := [\mathrm{id}_{\mathbb{R}^n}]_{\mathcal{E}_n}^{\mathcal{B}}$ be the transition matrix. Since $\mathcal{B}$ is an orthonormal basis in $\mathbb{R}^n$, point (d) guarantees that O is an orthogonal matrix. By the change of basis formula:

$$D = [T_A]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_{\mathbb{R}^n}]_{\mathcal{B}}^{\mathcal{E}_n} \cdot [T_A]_{\mathcal{E}_n}^{\mathcal{E}_n} \cdot [\mathrm{id}_{\mathbb{R}^n}]_{\mathcal{E}_n}^{\mathcal{B}} = O^{-1} \cdot A \cdot O.$$

Thus, $O^{-1}AO$ is diagonal.

Q.E.D.

Orthogonal and Unitary Maps / Isometries (Chapter 25)

Linear Isometries (Section 25.a)

Definition 25.a.1 (*Linear Isometry*): Let V, W be inner product spaces over K (where $K = \mathbb{R}$ or $\mathbb{C}$). A map $T\colon V \to W$ is called a "*linear isometry*" if T is linear and for all $v_1, v_2 \in V$,

$$\langle Tv_1, Tv_2\rangle_W = \langle v_1, v_2\rangle_V.$$

(In other words, T preserves the inner product).

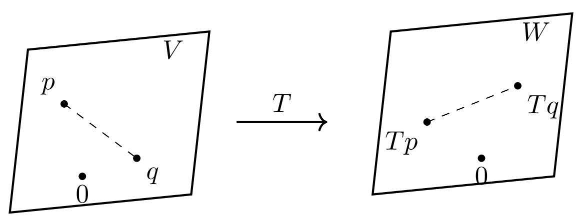
Remark 25.1: If $T\colon V \to W$ is a linear isometry, then:

(a) For all $v \in V$, $\|Tv\| = \|v\|$.

(b) For all $p, q \in V$, the distance is preserved:

$$\operatorname{dist}_W(Tp, Tq) = \|Tp - Tq\|_W = \|p - q\|_V = \operatorname{dist}_V(p, q).$$

(c) T is injective. Indeed, if $v \in \ker T$, then $0 = \|Tv\| = \|v\|$, which implies $v = 0$. So, $\ker T = \{0\}$.



From now on, we will concentrate on the case where $W = V$, meaning $T \in \operatorname{End}(V)$. If such a map T is an isometry, we also call it an "*orthogonal endomorphism*" if $K = \mathbb{R}$, or a "*unitary endomorphism*" if $K = \mathbb{C}$.

Theorem 25.a.2 (*Characterization of Isometries*): Let V be a finite-dimensional inner product space over $\mathbb{R}$ or $\mathbb{C}$, and let $T \in \operatorname{End}(V)$. The following statements are equivalent:

(a) T is an isometry.

(b) For all $v \in V$, $\|Tv\| = \|v\|$.

(c) For every orthonormal system $e_1, \ldots, e_k$ of V, the list of vectors $Te_1, \ldots, Te_k$ is also an orthonormal system.

(d) There exists an orthonormal **basis** $\mathcal{B} := (e_1, \ldots, e_n)$ of V such that $Te_1, \ldots, Te_n$ is also an orthonormal basis.

(e) $T \circ T^* = \mathrm{id}_V$.

(f) $T^* \circ T = \mathrm{id}_V$.

(g) $T^* = T^{-1}$.

(h) T^* is an isometry.

Proof:

We will prove some of the equivalences here; the rest are left as exercises (and for the exercise sheet).

(b) $\Longrightarrow$ (a): Using the polarization identity for the real case (on Page 195), we have:

$$\begin{aligned}\langle Tv, Tw\rangle &= \frac{1}{2}\left(\|Tv+Tw\|^2 - \|Tv\|^2 - \|Tw\|^2\right)\\ &= \frac{1}{2}\left(\|T(v+w)\|^2 - \|Tv\|^2 - \|Tw\|^2\right)\\ &= \frac{1}{2}\left(\|v+w\|^2 - \|v\|^2 - \|w\|^2\right)\\ &= \langle v, w\rangle.\end{aligned}$$

(a) $\Longrightarrow$ (c): Let $e_1, \ldots, e_k$ be an orthonormal system. Then, for all $1 \le i, j \le k$, we have:

$$\langle Te_i, Te_j\rangle = \langle e_i, e_j\rangle = \delta_{ij}.$$

This implies that $Te_1, \ldots, Te_k$ is also an orthonormal system.

The remaining implications, in particular **(d) $\Longrightarrow$ (e)**, are established in the exercise solutions at the end of this chapter. **Q.E.D.**

Orthogonal and Unitary Matrices (Section 25.b)

Back to matrices. Recall that:

- $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is orthogonal if $A^\intercal A = I \iff AA^\intercal = I \iff A^{-1} = A^\intercal$. The set of such matrices is $\mathrm{O}(n)$.
- $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is unitary if $A^*A = I \iff AA^* = I \iff A^{-1} = A^*$. The set of such matrices is $\mathrm{U}(n)$.

Proposition 25.a.3: Let V be an n-dimensional inner product space over $\mathbb{R}$ (resp. $\mathbb{C}$). Let $\mathcal{B}$ be an orthonormal basis for V, and let $T \in \mathrm{End}(V)$. Then, T is orthogonal (resp. unitary) if and only if $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathrm{O}(n)$ (resp. $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathrm{U}(n)$).

Proof:

This is left as an exercise. **Q.E.D.**

Note that for all $A \in \mathrm{O}(n)$, we have $\det(A) = \pm 1$. Indeed, by Theorems 20.c.1 and 20.b.5,

$$1 = \det(I) = \det(AA^\intercal) = \det(A)\cdot\det(A^\intercal) = \det(A)\cdot\det(A) = \det(A)^2.$$

Similarly, for all $A \in \mathrm{U}(n)$, we have $|\det(A)| = 1$. Indeed, by the same two results,

$$1 = \det(AA^*) = \det(A)\cdot\det(\overline{A}^\intercal) = \det(A)\cdot\det(\overline{A}) = \det(A)\cdot\overline{\det(A)} = |\det(A)|^2.$$

(Note that $\det(A) \in \mathbb{C}$).

Definition 25.a.4 (*Special Orthogonal and Unitary Groups*): We define the following subsets:

$$\begin{aligned}\mathrm{SO}(n) &:= \{A \in \mathrm{O}(n) \mid \det(A) = 1\} \subseteq \mathrm{O}(n) \subseteq \mathrm{GL}_n(\mathbb{R}),\\ \mathrm{SU}(n) &:= \{A \in \mathrm{U}(n) \mid \det(A) = 1\} \subseteq \mathrm{U}(n) \subseteq \mathrm{GL}_n(\mathbb{C}).\end{aligned}$$

Proposition 25.a.5 (*Group Properties*): Let G be any of the sets $\mathrm{O}(n), \mathrm{SO}(n), \mathrm{U}(n)$, or $\mathrm{SU}(n)$. Then:

(a) $I_n \in G$.

(b) For all $A, B \in G$, $A \cdot B \in G$.

(c) For all $A \in G$, $A^{-1} \in G$.

Proof:

This is left as an exercise. **Q.E.D.**

Classification of the Elements of O(2), O(3), etc. (Section 25.c)

Lemma 25.a.6: Let $A \in \mathrm{O}(n)$. If λ is a real eigenvalue of A, then $\lambda = \pm 1$.

Proof:

Suppose $Av = \lambda v$, where $0 \neq v \in \mathbb{R}^n$. Since the standard basis of $\mathbb{R}^n$ is orthonormal, Proposition 25.a.3 shows that T_A is an isometry, and therefore, it preserves the standard scalar product. Then:

$$\langle v, v\rangle = \langle Av, Av\rangle = \langle \lambda v, \lambda v\rangle = \lambda^2 \langle v, v\rangle.$$

As $v \neq 0$, we have $\langle v, v\rangle \neq 0$, which implies $\lambda^2 = 1$, and therefore $\lambda = \pm 1$. **Q.E.D.**

Proposition 25.a.7 (*Classification of* $\mathrm{O}(2)$): Let $A \in \mathrm{O}(2)$.

(a) If $\det A = 1$, then there exists $\theta \in [0, 2\pi)$ such that

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} := R_\theta.$$

The map $T_A \colon \mathbb{R}^2 \to \mathbb{R}^2$ is an anti-clockwise rotation by an angle θ.

(b) If $\det A = -1$, then there exists a basis $\mathcal{B} := (v_1, v_2)$ for $\mathbb{R}^2$ such that $T_A \colon \mathbb{R}^2 \to \mathbb{R}^2$ has the following representation matrix in the basis $\mathcal{B}$:

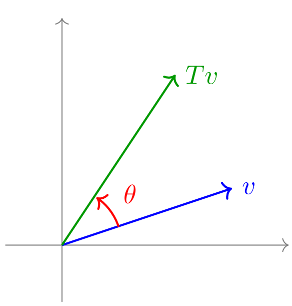
$$[T_A]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{25.1}$$

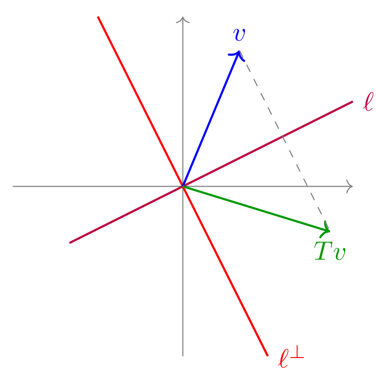
In other words, T_A performs a reflection with respect to the line $\ell := \mathrm{Sp}(v_1)$. Moreover, there exists $\theta \in [0, 2\pi)$ such that

$$A = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} = R_\theta \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The basis $\mathcal{B}$ for which (25.1) holds is given by

$$v_1 = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) \end{pmatrix}, \quad v_2 = \begin{pmatrix} \cos\left(\frac{\theta}{2} + \frac{\pi}{2}\right) \\ \sin\left(\frac{\theta}{2} + \frac{\pi}{2}\right) \end{pmatrix}.$$





Proof:

Note that if $v \in \mathbb{R}^2$ has $\|v\| = 1$, then there exists $\theta \in [0, 2\pi)$ such that $v = \left(\begin{smallmatrix} \cos\theta \\ \sin\theta \end{smallmatrix}\right)$.

Now, let

$$A = \begin{pmatrix} | & | \\ u_1 & u_2 \\ | & | \end{pmatrix} \in \mathrm{O}(2).$$

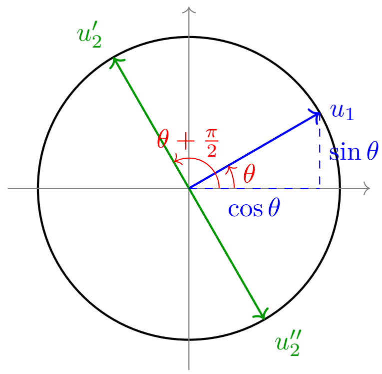
Since A is orthogonal, its columns form an orthonormal basis by Definition 22.b.8. Thus, $\|u_1\| = 1$, which implies $u_1 = \left(\begin{smallmatrix}\cos\theta \\ \sin\theta\end{smallmatrix}\right)$ for some $\theta \in [0, 2\pi)$. We also have $u_1 \perp u_2$ and $\|u_2\| = 1$.

By Corollary 22.b.7, the orthogonal complement of u_1 is a line, namely:

$$\begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix}^{\perp} = \mathrm{Sp}\begin{pmatrix}\cos(\theta+\frac{\pi}{2}) \\ \sin(\theta+\frac{\pi}{2})\end{pmatrix} = \mathrm{Sp}\begin{pmatrix}-\sin\theta \\ \cos\theta\end{pmatrix}.$$

Within the space $\mathrm{Sp}\left(\begin{smallmatrix}-\sin\theta \\ \cos\theta\end{smallmatrix}\right)$, there are precisely two elements with norm 1, namely:

$$u_2' = \begin{pmatrix}-\sin\theta \\ \cos\theta\end{pmatrix}, \quad u_2'' = \begin{pmatrix}\sin\theta \\ -\cos\theta\end{pmatrix}.$$



If $u_2 = u_2'$, then $A = \left(\begin{smallmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{smallmatrix}\right) = R_\theta$, and $\det A = 1$.

If $u_2 = u_2''$, then $A = \left(\begin{smallmatrix}\cos\theta & \sin\theta \\ \sin\theta & -\cos\theta\end{smallmatrix}\right) = R_\theta \cdot \left(\begin{smallmatrix}1 & 0 \\ 0 & -1\end{smallmatrix}\right)$, and $\det A = -1$.

In the second case, a direct calculation shows that $v_1 = \left(\begin{smallmatrix}\cos(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2})\end{smallmatrix}\right)$ and $v_2 = \left(\begin{smallmatrix}\cos(\frac{\theta}{2}+\frac{\pi}{2}) \\ \sin(\frac{\theta}{2}+\frac{\pi}{2})\end{smallmatrix}\right)$ are eigenvectors of A for the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = -1$, respectively. **Q.E.D.**

Summary 25.1: $\mathrm{SO}(2) =$ rotations.
$\mathrm{O}(2) \setminus \mathrm{SO}(2) =$ reflections.

Corollary 25.a.8: Let V be a 2-dimensional Euclidean space and $T\colon V \to V$ a linear isometry. Then $\det(T) = \pm 1$. Moreover, there exists an orthonormal basis $\mathcal{B}$ for V such that:

(a) If $\det(T) = 1$, then $[T]_{\mathcal{B}}^{\mathcal{B}} = R_\theta$.

(b) If $\det(T) = -1$, then $[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}$.

Proposition 25.a.9 (*Classification of* $\mathrm{SO}(3)$): Let $A \in \mathrm{SO}(3)$. Then 1 is an eigenvalue of A. Moreover, there exists an orthonormal basis $\mathcal{B} := (v_1, v_2, v_3)$ and $\theta \in [0, 2\pi)$ such that

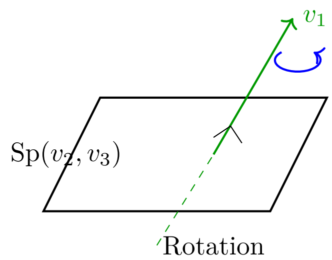
$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix}1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta\end{pmatrix} = \begin{pmatrix}1 & 0 \\ 0 & R_\theta\end{pmatrix}.$$

So, T_A is a rotation with angle θ around the axis $\mathrm{Sp}(v_1)$. Moreover:

(a) If -1 is an eigenvalue of A, then $\theta = \pi$, hence

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix}1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1\end{pmatrix}.$$

(b) If -1 is **not** an eigenvalue of A (which is true if and only if 1 is the only real eigenvalue), then $\theta \neq \pi$.



Proof:

The characteristic polynomial $P_A(x)$ of A is a polynomial of degree 3 with real coefficients. By a theorem from analysis (the Intermediate Value Theorem), every polynomial of odd degree with real coefficients has a real zero, so there exists $\lambda \in \mathbb{R}$ with $P_A(\lambda) = 0$. This implies that A has at least one real eigenvalue. By Lemma 25.a.6, $\lambda = \pm 1$.

Let $v_1 \in \mathbb{R}^3$ be an eigenvector of λ. By normalizing it if necessary, we may assume $\|v_1\| = 1$. Consider the subspace

$$L := v_1^\perp = \{v \in \mathbb{R}^3 \mid v \perp v_1\}.$$

Since A is orthogonal, we have $A \cdot v \in L$ for all $v \in L$ (i.e., $T_A(L) \subseteq L$, meaning L is invariant under T_A). Indeed, if $v \in L$, this implies $0 = \langle v, v_1 \rangle$. Because $A \in \mathrm{O}(3)$, the map T_A preserves the scalar product by Proposition 25.a.3, and therefore,

$$0 = \langle v, v_1 \rangle = \langle Av, Av_1 \rangle = \langle Av, \pm v_1 \rangle = \pm \langle Av, v_1 \rangle,$$

which implies that $Av \in L$.

By Theorem 22.b.3, we also have $\mathbb{R}^3 = \mathrm{Sp}(v_1) \oplus L$, hence $\dim L = 2$ by Corollary 22.b.7. Now, $T_A|_L \colon L \to L$ is a linear isometry. Let $\mathcal{L} = (w_1, w_2)$ be an orthonormal basis for L. By Proposition 25.a.3, the representation matrix is $Q := [T_A|_L]_{\mathcal{L}}^{\mathcal{L}} \in \mathrm{O}(2)$.

Case (a): $\lambda = 1$. Take the basis $\mathcal{B} := (v_1, w_1, w_2)$. Then

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & \multicolumn{2}{c}{Q} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix}.$$

Now, by Corollary 20.c.8 and Proposition 20.c.4, $1 = \det A = \det(T_A) = 1 \cdot \det(Q)$, which implies $\det Q = 1$. So, $Q \in \mathrm{SO}(2)$. By Proposition 25.a.7, there exists $\theta \in [0, 2\pi)$ such that $Q = R_\theta$. Thus,

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & R_\theta \end{pmatrix}.$$

Case (b): $\lambda = -1$. By the same argument as in case **(a)**, we get that $1 = \det A = (-1) \cdot \det(T_A|_L)$, which implies $\det(T_A|_L) = -1$. By Corollary 25.a.8, there exists an orthonormal basis $\mathcal{L} = (w_1, w_2)$ for L such that $[T_A|_L]_{\mathcal{L}}^{\mathcal{L}} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. Take $\mathcal{B}' := (v_1, w_1, w_2)$. Then

$$[T_A]_{\mathcal{B}'}^{\mathcal{B}'} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

And for the reordered basis $\mathcal{B} := (w_1, v_1, w_2)$, we have

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & R_\pi \end{pmatrix}.$$

Note that in case **(b)** too, 1 is an eigenvalue of A (corresponding to the eigenvector w_1).

Q.E.D.

Proposition 25.a.10 (*Classification of* $\mathrm{O}(3) \setminus \mathrm{SO}(3)$): Let $A \in \mathrm{O}(3) \setminus \mathrm{SO}(3)$. Then there exists an orthonormal basis $\mathcal{B} := (v_1, v_2, v_3)$ for $\mathbb{R}^3$ such that

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & R_\theta \end{pmatrix}$$

for some $\theta \in [0, 2\pi)$. (So T_A does a rotation by θ around $\mathrm{Sp}(v_1)$, followed by a reflection with respect to $\mathrm{Sp}(v_2, v_3)$).

Proof:

Let $A' := -A = (-I) \cdot A$. Since $-I \in \mathrm{O}(3)$, Proposition 25.a.5 gives $A' \in \mathrm{O}(3)$, and by Theorem 20.b.5, $\det A' = \det(-I) \cdot \det A = (-1)^3 \cdot \det A = (-1) \cdot (-1) = 1$. This implies $A' \in \mathrm{SO}(3)$. By Proposition 25.a.9, there exists an orthonormal basis $\mathcal{B}$ for $\mathbb{R}^3$, and $\theta' \in [0, 2\pi)$, such that $[T_{A'}]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & R_{\theta'} \end{smallmatrix}\right)$. This implies:

$$\begin{aligned} [T_A]_{\mathcal{B}}^{\mathcal{B}} &= [-\mathrm{id}_{\mathbb{R}^3}]_{\mathcal{B}}^{\mathcal{B}} \cdot [T_{A'}]_{\mathcal{B}}^{\mathcal{B}} \\ &= -I \cdot \begin{pmatrix} 1 & 0 \\ 0 & R_{\theta'} \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & -R_{\theta'} \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & R_\theta \end{pmatrix}, \end{aligned}$$

where $\theta = (\theta' + \pi) \pmod{2\pi}$. (Note that $-R_{\theta'} = R_\pi \circ R_{\theta'} = R_{\theta'+\pi}$). **Q.E.D.**

Exercise Solutions

Proof of Theorem 25.a.2 (d) $\implies$ (e):

AI-Note: In the handwritten notes, this proof is labeled as proving **(d)** $\implies$ **(e)**. However, the first part actually proves **(d)** $\implies$ **(f)** (which is $T^* \circ T = \mathrm{id}_V$), from which **(e)** is then deduced. We have clarified the steps here.

Assume there exists an orthonormal basis $\mathcal{B} = (e_1, \ldots, e_n)$ of V such that $Te_1, \ldots, Te_n$ is also an orthonormal basis. Let $1 \le j \le n$. Then, for all $1 \le i \le n$, the defining property (23.1) of the adjoint map gives

$$\langle e_i, e_j \rangle = \delta_{ij} = \langle Te_i, Te_j \rangle = \langle e_i, T^*Te_j \rangle. \tag{25.2}$$

Since $(e_1, \ldots, e_n)$ is a basis for V, the equality (25.2) extends by linearity in the first argument to

$$\langle v, e_j \rangle = \langle v, T^*Te_j \rangle \quad \text{for all } v \in V,$$

and therefore, $T^*Te_j = e_j$. This last equality holds for all $1 \le j \le n$, so the linear map $T^* \circ T$ agrees with id_V on the basis $\mathcal{B}$. Thus, $T^* \circ T = \mathrm{id}_V$, which proves **(d)** $\implies$ **(f)**.

Finally, $T^* \circ T = \mathrm{id}_V$ shows that T is injective, and since V is finite-dimensional, T is an isomorphism by Corollary 15.b.10. Consequently, $T^* = T^* \circ T \circ T^{-1} = T^{-1}$, and therefore, $T \circ T^* = \mathrm{id}_V$, which proves **(e)**. **Q.E.D.**

Proof of Proposition 25.a.3:

Let $\mathcal{B} = (e_1, \ldots, e_n)$ be an orthonormal basis for V. First, suppose T is an orthogonal (resp. unitary) endomorphism. By Theorem 25.a.2, $T \circ T^* = \mathrm{id}_V$ and $T^* \circ T = \mathrm{id}_V$. Taking the representation matrix with respect to $\mathcal{B}$, we have:

$$[T^* \circ T]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I_n.$$

By the properties of representation matrices, this splits into $[T^*]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} = I_n$. Since $\mathcal{B}$ is an orthonormal basis, Proposition 23.a.4 gives $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$. Thus, $\left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^* \cdot [T]_{\mathcal{B}}^{\mathcal{B}} = I_n$, which by Lemma 22.b.9 (resp. Lemma 22.b.10) means that the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is orthogonal (resp. unitary).

Conversely, assume that $A := [T]_{\mathcal{B}}^{\mathcal{B}}$ is orthogonal (resp. unitary). This means $A^*A = I_n$. Again because $\mathcal{B}$ is an orthonormal basis, Proposition 23.a.4 yields $A^* = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^* = [T^*]_{\mathcal{B}}^{\mathcal{B}}$. Substituting this back yields $[T^*]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} = I_n$, and therefore $[T^* \circ T]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}}$. This implies the maps are equal, so $T^* \circ T = \mathrm{id}_V$. By Theorem 25.a.2, this guarantees T is an isometry. **Q.E.D.**

Proof of Proposition 25.a.5:

We will prove this generically for $G = \mathrm{U}(n)$ and $G = \mathrm{SU}(n)$ (the real cases are identical, replacing the conjugate transpose * with the transpose $^\mathsf{T}$).

(a) The identity matrix I_n satisfies $I_n^* \cdot I_n = I_n \cdot I_n = I_n$, so $I_n \in \mathrm{U}(n)$. Furthermore, $\det(I_n) = 1$, so $I_n \in \mathrm{SU}(n)$.

(b) Let $A, B \in \mathrm{U}(n)$. We evaluate $(AB)^*(AB) = (B^*A^*)(AB) = B^*(A^*A)B$. Since $A \in \mathrm{U}(n)$, $A^*A = I_n$, so this reduces to $B^*I_nB = B^*B$. Since $B \in \mathrm{U}(n)$, $B^*B = I_n$. Thus, $AB \in \mathrm{U}(n)$. If additionally $A, B \in \mathrm{SU}(n)$, then by Theorem 20.b.5, $\det(AB) = \det(A)\det(B) = 1 \cdot 1 = 1$, so $AB \in \mathrm{SU}(n)$.

(c) Let $A \in \mathrm{U}(n)$. This implies $A^*A = I_n$, meaning $A^{-1} = A^*$. We check if A^{-1} is unitary: $(A^{-1})^*A^{-1} = (A^*)^*A^* = AA^* = I_n$. Thus, $A^{-1} \in \mathrm{U}(n)$. If additionally $A \in \mathrm{SU}(n)$, then by Corollary 20.b.6, $\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{1} = 1$, so $A^{-1} \in \mathrm{SU}(n)$.

Q.E.D.

Singular Value Decomposition (Chapter 26)

Theorem 26.a.1 (*Singular Value Decomposition*): Let $K = \mathbb{R}$ or $\mathbb{C}$, and let $A \in \mathcal{M}_{m\times n}(K)$. Then there exist $P \in \mathrm{O}(m)$ (if $K = \mathbb{R}$) or $P \in \mathrm{U}(m)$ (if $K = \mathbb{C}$), and $Q \in \mathrm{O}(n)$ (if $K = \mathbb{R}$) or $Q \in \mathrm{U}(n)$ (if $K = \mathbb{C}$), and a *"diagonal"* matrix

$$D = \begin{pmatrix} \begin{matrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_k \end{matrix} \\ \hline 0 \dots 0 \end{pmatrix} \in \mathcal{M}_{m\times n}(\mathbb{R})$$

with $\lambda_1, \dots, \lambda_k > 0$, where $k = \operatorname{rank} A$, such that $A = PDQ^{-1}$.

Remark 26.1: Recall that we previously defined the equivalence of matrices. Here, the factorization $A = PDQ^{-1}$ demonstrates that we have an orthogonal (or unitary) equivalence.

Proof:

We will prove only a special case: we assume $\operatorname{rank} A = n$ (which is true if and only if T_A is injective. Hence, we also assume $n \leq m$).

Define $B := A^* \cdot A \in \mathcal{M}_{n\times n}(K)$. We claim that the linear map $T_B \colon K^n \to K^n$ is injective. Indeed, for all $0 \neq v \in K^n$, we have:

$$\langle T_B v, v\rangle = \langle Bv, v\rangle = \langle A^*Av, v\rangle = \langle Av, Av\rangle > 0. \tag{26.1}$$

This implies that $T_B v \neq 0$. Therefore, $\ker T_B = \{0\}$, hence T_B is injective.

Note also that if $\eta \in K$ is an eigenvalue of B, then $\eta \in \mathbb{R}$ (even if $K = \mathbb{C}$) and $\eta > 0$. Indeed, B is self adjoint ($B^* = (A^*A)^* = A^*(A^*)^* = A^*A = B$), hence $\eta \in \mathbb{R}$. To see that $\eta > 0$, let v be an eigenvector for η, so that $v \neq 0$ and $Bv = \eta v$. Substituting this into (26.1) gives

$$0 < \langle Bv, v\rangle = \langle \eta v, v\rangle = \eta \cdot \langle v, v\rangle,$$

where the last step uses linearity in the first variable together with $\eta \in \mathbb{R}$. Since $v \neq 0$, positivity of the inner product gives $\langle v, v\rangle > 0$, and therefore, $\eta > 0$.

By a previous theorem, B is orthogonally diagonalizable, hence there exists $Q \in \mathrm{O}(n)$ (or $\mathrm{U}(n)$) such that

$$Q^{-1}A^*AQ = \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix},$$

where $\eta_j \in \mathbb{R}$, and $\eta_j > 0$ for all j.

Let $\lambda_j := +\sqrt{\eta_j} > 0$ for $j = 1, \dots, n$. Define

$$C := AQ \cdot \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} \in \mathcal{M}_{m\times n}(K).$$

We have:

$$C^*C = \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} \underbrace{Q^{-1}A^*AQ}_{= \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix}} \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix}$$

$$= I_n.$$

It follows that the columns of C form an orthonormal system of vectors in K^m. To see this explicitly, let $c_1, \dots, c_n$ be the columns of C. The identity $C^*C = I_n$ means:

$$C^*C = \begin{pmatrix} - & \overline{c_1}^\mathsf{T} & - \\ & \vdots & \\ - & \overline{c_n}^\mathsf{T} & - \end{pmatrix} \begin{pmatrix} | & & | \\ c_1 & \dots & c_n \\ | & & | \end{pmatrix} = I_n.$$

Extend $c_1, \dots, c_n$ to an orthonormal basis $c_1, \dots, c_n, f_{n+1}, \dots, f_m$ of K^m and define:

$$P := \begin{pmatrix} | & & | & | & & | \\ c_1 & \dots & c_n & f_{n+1} & \dots & f_m \\ | & & | & | & & | \end{pmatrix} \in \mathcal{M}_{m\times m}(K).$$

Clearly, $P \in \mathrm{O}(m)$ (or $\mathrm{U}(m)$).

We have:

$$AQ \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} = C = P \cdot \begin{pmatrix} \begin{matrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{matrix} \\ \hline 0 \end{pmatrix} \begin{matrix} \left.\vphantom{\begin{matrix} 1 \\ \ddots \\ 1 \end{matrix}}\right\} n \\ \}\, m-n \end{matrix}$$

Multiplying from the right by the diagonal matrix of λ_j's and then by Q^{-1}, we obtain:

$$A = P \cdot \begin{pmatrix} \begin{matrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{matrix} \\ \hline 0 \end{pmatrix} \cdot Q^{-1}.$$

This proves the theorem (under the assumption that $\operatorname{rank} A = n$).

For the general case, see the script, chapter 7.5. **Q.E.D.**

Bilinear and Quadratic Forms (Chapter 27)

Definition 27.a.1 (*Bilinear Form*): Let V be a vector space over K (a general field). A function $B\colon V \times V \to K$ is called a "*bilinear form*" if the following two conditions are satisfied:

(a) For all $w \in V$, the map $B(-, w)\colon V \to K$, $v \mapsto B(v, w)$ is linear.

(b) For all $v \in V$, the map $B(v, -)\colon V \to K$, $w \mapsto B(v, w)$ is linear.

Lemma 27.a.2 (*Matrix Representation of a Bilinear Form*): Let V be a vector space over K and $\mathcal{C} := (e_1, \ldots, e_n)$ a basis for V. Let $B\colon V \times V \to K$ be a bilinear form. Then there exists a matrix $[B]_\mathcal{C} \in \mathcal{M}_{n\times n}(K)$ such that

$$B\left(\sum_{i=1}^n c_i e_i, \sum_{j=1}^n d_j e_j\right) = (c_1, \ldots, c_n) \cdot [B]_\mathcal{C} \cdot (d_1, \ldots, d_n)^\mathsf{T} \quad \forall c_i, d_j \in K.$$

The matrix $[B]_\mathcal{C}$ is called the "*matrix representation*" of B in the basis $\mathcal{C}$.

Proof:

Define $[B]_\mathcal{C} := (a_{ij})$, where $a_{ij} := B(e_i, e_j)$. Then we compute:

$$\begin{aligned} B\left(\sum_{i=1}^n c_i e_i, \sum_{j=1}^n d_j e_j\right) &= \sum_{i=1}^n c_i B\left(e_i, \sum_{j=1}^n d_j e_j\right) \\ &= \sum_{i=1}^n c_i \left(\sum_{j=1}^n d_j B(e_i, e_j)\right) \\ &= \sum_{i=1}^n \sum_{j=1}^n c_i d_j B(e_i, e_j) \\ &= (c_1, \ldots, c_n) \cdot [B]_\mathcal{C} \cdot \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix}. \end{aligned}$$

Q.E.D.

We saw that also for $T \in \mathrm{End}(V)$ and $\mathcal{C}$ a basis for V, we have a matrix representation $[T]_\mathcal{C}^\mathcal{C} \in \mathcal{M}_{n\times n}(K)$, and that if $\mathcal{D}$ is another basis for V, then

$$[T]_\mathcal{D}^\mathcal{D} = [\mathrm{id}_V]_\mathcal{D}^\mathcal{C} \cdot [T]_\mathcal{C}^\mathcal{C} \cdot [\mathrm{id}_V]_\mathcal{C}^\mathcal{D}, \tag{27.1}$$

where $[\mathrm{id}_V]_\mathcal{C}^\mathcal{D} = ([\mathrm{id}_V]_\mathcal{D}^\mathcal{C})^{-1}$ is the transition matrix from basis $\mathcal{D}$ to $\mathcal{C}$.

Question 27.1: What happens for bilinear forms?

Lemma 27.a.3 (*Change of Basis for Bilinear Forms*): Let V be a finite-dimensional vector space over K and $\mathcal{C}, \mathcal{D}$ two bases of V. Let $B\colon V \times V \to K$ be a bilinear form. Then

$$[B]_\mathcal{D} = \left([\mathrm{id}_V]_\mathcal{C}^\mathcal{D}\right)^\mathsf{T} \cdot [B]_\mathcal{C} \cdot [\mathrm{id}_V]_\mathcal{C}^\mathcal{D}.$$

(Note that this is quite different from (27.1)).

Proof:

Let $\mathcal{C} := (v_1, \ldots, v_n)$ and $\mathcal{D} := (w_1, \ldots, w_n)$. We have $([B]_\mathcal{C})_{j,k} = B(v_j, v_k)$, and $([B]_\mathcal{D})_{j,k} = B(w_j, w_k)$. Recall that column j in $[\mathrm{id}_V]^\mathcal{D}_\mathcal{C}$ contains the coordinates of w_j in the basis $\mathcal{C}$:

$$w_j = \sum_{p=1}^{n} \left([\mathrm{id}_V]^\mathcal{D}_\mathcal{C}\right)_{p,j} \cdot v_p.$$

This implies:

$$\begin{aligned}
([B]_\mathcal{D})_{j,k} &= B(w_j, w_k) \\
&= B\left(\sum_{p=1}^{n} \left([\mathrm{id}_V]^\mathcal{D}_\mathcal{C}\right)_{p,j} v_p, \sum_{q=1}^{n} \left([\mathrm{id}_V]^\mathcal{D}_\mathcal{C}\right)_{q,k} v_q\right) \\
&= \sum_{p,q=1}^{n} \left([\mathrm{id}_V]^\mathcal{D}_\mathcal{C}\right)_{p,j} \cdot B(v_p, v_q) \cdot \left([\mathrm{id}_V]^\mathcal{D}_\mathcal{C}\right)_{q,k} \\
&= \left(\left([\mathrm{id}_V]^\mathcal{D}_\mathcal{C}\right)^\mathsf{T} \cdot [B]_\mathcal{C} \cdot [\mathrm{id}_V]^\mathcal{D}_\mathcal{C}\right)_{j,k}.
\end{aligned}$$

Q.E.D.

Definition 27.a.4 (*Symmetric, Positive Definite, and Quadratic Forms*):

(a) A bilinear form $B \colon V \times V \to K$ is called "*symmetric*" if $B(v, w) = B(w, v)$ for all $v, w \in V$.

(b) Assume $K = \mathbb{R}$. A bilinear form $B \colon V \times V \to \mathbb{R}$ is called "*positive definite*" if $B(v, v) > 0$ for all $0 \neq v \in V$.

(c) For a bilinear form $B \colon V \times V \to K$, we define $q_B \colon V \to K$ by $q_B(v) := B(v, v)$. We call q_B the "*quadratic form*" associated to B. Sometimes we just denote it by q.

Example 27.1 (*Standard Bilinear and Quadratic Forms*):

(a) Let $(V, \langle \cdot, \cdot \rangle)$ be a Euclidean vector space. Then $B(v, w) := \langle v, w \rangle$ is a symmetric, positive definite bilinear form. Its associated quadratic form is $q(v) = \|v\|^2$.

(b) Let $B \in \mathcal{M}_{n\times n}(K)$. Then $B \colon K^n \times K^n \to K$ defined by $B(v, w) := v^\mathsf{T} B w$ is a bilinear form. If $B^\mathsf{T} = B$ (i.e., B is a symmetric matrix), then B is a symmetric bilinear form (left as an exercise). The associated quadratic form is $q(v) = v^\mathsf{T} B v$.

Remark 27.1: Note that $B \in \mathcal{M}_{n\times n}(K)$ and $B^\mathsf{T} \in \mathcal{M}_{n\times n}(K)$ give the same quadratic form $q \colon K^n \to K$. Indeed, for all $v \in K^n$, we have:

$$q_B(v) = v^\mathsf{T} B v = (v^\mathsf{T} B v)^\mathsf{T} = v^\mathsf{T} B^\mathsf{T} v = q_{B^\mathsf{T}}(v),$$

where we used the fact that $v^\mathsf{T} B v$ is a scalar, and thus equals its own transpose.

Assume now that $2 \neq 0$ in K (i.e., $1 + 1 \neq 0$). Then $\frac{1}{2}(B + B^\mathsf{T})$ also gives the same quadratic form as B.

Conclusion 27.1: If $2 \neq 0$ in K, then for the discussion on quadratic forms $q \colon K^n \to K$, we can assume they all come from symmetric matrices (or symmetric bilinear forms).

Theorem (*Polarization Identity*): Assume $2 \neq 0$ in K. (Note that $2 \neq 0 \implies 4 \neq 0$. Why? This is left as an exercise). If B is symmetric and defines $q := q_B$, then for all $v, w \in V$ we have:

(a) $B(v, w) = \frac{1}{4} q(v + w) - \frac{1}{4} q(v - w)$.

(b) $B(v, w) = \frac{1}{2}(q(v + w) - q(v) - q(w))$.

Theorem 27.a.5 (*Sylvester's Law of Inertia*): Let $B \colon V \times V \to \mathbb{R}$ be a **symmetric** bilinear form on a vector space V over $\mathbb{R}$, of dimension n. Then there exists a basis $\mathcal{C}$ for V such that

$$[B]_{\mathcal{C}} = \begin{pmatrix} I_k & 0 & 0 \\ 0 & -I_\ell & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

for some $k, \ell \in \mathbb{Z}_{\geq 0}$ with $k + \ell \leq n$. (The zero block in the bottom right has size $n-(k+\ell)$).
Moreover, k and ℓ do **not** depend on the basis $\mathcal{C}$ (they depend only on B). In fact:

$$\begin{aligned} k &= \max\{\dim W \mid W \subseteq V \text{ is a subspace and } B|_{W\times W} \text{ is positive definite}\}, \\ \ell &= \max\{\dim U \mid U \subseteq V \text{ is a subspace and } -B|_{U\times U} \text{ is positive definite}\}, \\ n-(k+\ell) &= \max\{\dim Z \mid Z \subseteq V \text{ is a subspace and } B|_{Z\times Z} \equiv 0\}. \end{aligned}$$

Remark 27.2:

(a) In the basis $\mathcal{C}$, the quadratic form q associated to B looks like

$$q(x_1, \ldots, x_n) = x_1^2 + \cdots + x_k^2 - x_{k+1}^2 - \cdots - x_{k+\ell}^2.$$

(b) k is called the "*positive index*" of B, ℓ is the "*negative index*", and $n-(k+\ell)$ is the "*nullity*". The value $\sigma(B) := \operatorname{sgn}(B) := k-\ell$ is called the "*signature*" of B. (This is sometimes referred to as type $(k, \ell, 0)$).

Proof:

Choose any basis $\mathcal{C}'$ for V and let $A := [B]_{\mathcal{C}'} \in \mathcal{M}_{n\times n}(\mathbb{R})$. Note that A is a symmetric matrix (since B is symmetric). This implies that A is orthogonally diagonalizable, i.e., there exists $P \in \mathrm{O}(n)$ such that $P^{-1}AP$ is a diagonal matrix. But $P^{-1} = P^{\mathsf{T}}$, so $P^{\mathsf{T}}AP$ is diagonal.

Let $\mathcal{C}''$ be the (unique) basis of V such that $[\mathrm{id}_V]^{\mathcal{C}'}_{\mathcal{C}''} = P$. (That is, if $\mathcal{C}' = (v_1', \ldots, v_n')$ and $\mathcal{C}'' = (v_1'', \ldots, v_n'')$, then $v_j'' = \sum_{i=1}^n P_{ij} v_i'$ for all j). We have

$$[B]_{\mathcal{C}''} = P^{\mathsf{T}}AP = \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix},$$

where $\eta_j \in \mathbb{R}$ for all j.

By changing the order of the elements in the basis $\mathcal{C}''$, we may assume $\eta_1, \ldots, \eta_k > 0$, $\eta_{k+1}, \ldots, \eta_{k+\ell} < 0$, and $\eta_{k+\ell+1} = \cdots = \eta_n = 0$ for some $k, \ell \in \mathbb{Z}_{\geq 0}$ with $k+\ell \leq n$.

We can write:

$$\begin{aligned} &\eta_i = \lambda_i^2, \text{ with } \lambda_i > 0, \quad \forall 1 \leq i \leq k \quad (\text{so } B(v_i'', v_i'') = \lambda_i^2), \\ &\eta_i = -\lambda_i^2, \text{ with } \lambda_i > 0, \quad \forall k+1 \leq i \leq k+\ell \quad (\text{so } B(v_i'', v_i'') = -\lambda_i^2), \\ &\eta_i = 0, \quad \forall k+\ell+1 \leq i \leq n \quad (\text{so } B(v_i'', v_i'') = 0). \end{aligned}$$

Define $\mathcal{C} := (v_1, \ldots, v_k, v_{k+1}, \ldots, v_{k+\ell}, v_{k+\ell+1}, \ldots, v_n)$, where

$$v_i = \frac{1}{\lambda_i} v_i'' \quad \forall 1 \leq i \leq k+\ell, \qquad v_i = v_i'' \quad \forall k+\ell+1 \leq i \leq n.$$

In other words:

$$\mathcal{C} = \left(\frac{1}{\lambda_1} v_1'', \ldots, \frac{1}{\lambda_k} v_k'', \frac{1}{\lambda_{k+1}} v_{k+1}'', \ldots, \frac{1}{\lambda_{k+\ell}} v_{k+\ell}'', v_{k+\ell+1}'', \ldots, v_n'' \right).$$

Note that $v_i = c_i v_i''$ for some scalars $c_i \neq 0$, for all i. It is left as an exercise to show that $\mathcal{C}$ is indeed a basis.

We have $([B]_{\mathcal{C}})_{i,j} = B(v_i, v_j) = B(c_i v_i'', c_j v_j'') = 0$ for all $i \neq j$. And for $i = j$:

- if $1 \leq i \leq k$, then $([B]_{\mathcal{C}})_{i,i} = B(v_i, v_i) = B\left(\frac{1}{\lambda_i} v_i'', \frac{1}{\lambda_i} v_i''\right) = \frac{1}{\lambda_i^2}\lambda_i^2 = 1$,

- if $k+1 \le i \le k+\ell$, then $([B]_{\mathcal{C}})_{i,i} = B(v_i, v_i) = B\left(\frac{1}{\lambda_i}v_i'', \frac{1}{\lambda_i}v_i''\right) = \frac{1}{\lambda_i^2}(-\lambda_i^2) = -1$,
- if $k+\ell+1 \le i \le n$, then $([B]_{\mathcal{C}})_{i,i} = B(v_i, v_i) = B(v_i'', v_i'') = 0$.

This proves the existence of the basis $\mathcal{C}$ claimed in the theorem.

We will prove now that k is independent of $\mathcal{C}$. Put

$$k' := \max\{\dim W \mid W \subseteq V \text{ is a subspace such that } B|_{W\times W} \text{ is positive definite}\}.$$

We will show $k = k'$. Indeed, $k \le k'$, because $B|_{U\times U}$ is positive definite on $U = \mathrm{Sp}(v_1, \dots, v_k)$. This is because:

$$B(a_1v_1 + \cdots + a_kv_k, a_1v_1 + \cdots + a_kv_k) \tag{27.2}$$

$$= (a_1, \dots, a_k, 0, \dots, 0) \cdot \begin{pmatrix} I_k & 0 & 0 \\ 0 & -I_\ell & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} \tag{27.3}$$

$$= a_1^2 + \cdots + a_k^2 \tag{27.4}$$

$$> 0, \tag{27.5}$$

if not all $a_i = 0$.

Suppose by contradiction that $k' \ge k+1$. This implies there exists $W \subseteq V$ of dimension $\ge k+1$ such that $B|_{W\times W}$ is positive definite. Take $X := \mathrm{Sp}(v_{k+1}, \dots, v_n)$. Then, by the dimension formula:

$$\dim(W \cap X) = \dim W + \dim X - \dim(W+X) \ge (k+1) + (n-k) - \dim(W+X) \ge n+1-n = 1.$$

This implies $W \cap X \ne \{0\}$. Take $0 \ne w \in W \cap X$. We can write $w = a_{k+1}v_{k+1} + \cdots + a_nv_n$. Then $B(w,w) = -a_{k+1}^2 - \cdots - a_{k+\ell}^2 \le 0$. But, by the assumption that W is positive definite, $B(w,w) > 0$. This is a contradiction, which implies $k' \le k$. This proves $k = k'$.

The proofs for ℓ and $n-(k+\ell)$ are similar. **Q.E.D.**

Theorem 27.a.6 (*Sylvester's Law for Euclidean Spaces*): Let V be a finite-dimensional Euclidean space of dimension n. Let $B\colon V \times V \to \mathbb{R}$ be a symmetric bilinear form. Then there exists an **orthonormal** basis $\mathcal{C}$ for V in which

$$[B]_{\mathcal{C}} = \begin{pmatrix} \eta_1 & & & & & 0 \\ & \ddots & & & & \\ & & \eta_k & & & \\ & & & \eta_{k+1} & & \\ & & & & \ddots & \\ 0 & & & & & \eta_{k+\ell} \end{pmatrix},$$

where $k+\ell \le n$, with $\eta_1, \dots, \eta_k > 0$ and $\eta_{k+1}, \dots, \eta_{k+\ell} < 0$. As before, k, ℓ are the positive and negative indices of B. Moreover, the scalars $\eta_1, \dots, \eta_k, \eta_{k+1}, \dots, \eta_{k+\ell}, 0, \dots, 0$ are independent of $\mathcal{C}$. They depend only on B (and on the scalar product on V). They are the eigenvalues of $[B]_{\mathcal{D}}$ for every **orthonormal** basis $\mathcal{D}$ of V (independent of $\mathcal{D}$ and on whether or not $[B]_{\mathcal{D}}$ is diagonal).

Remark 27.3: This theorem has a geometric meaning. If $q\colon \mathbb{R}^n \to \mathbb{R}$ is a quadratic form $q(x_1, \dots, x_n) = \sum a_{ij}x_ix_j$ for some $a_{ij} \in \mathbb{R}$, then the set

$$Q = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid q(x_1, \dots, x_n) = C\} \quad (\text{for some constant } C \in \mathbb{R})$$

is called a "*quadric*". (Examples include an ellipse/ellipsoid, parabola/paraboloid, hyperbola/hyperboloid). The theorem tells us that there exists a linear isometry $T\colon \mathbb{R}^n \to \mathbb{R}^n$ such that

$$T(Q) = \left\{ (y_1, \ldots, y_n) \in \mathbb{R}^n \;\middle|\; \underbrace{\eta_1 y_1^2 + \cdots + \eta_k y_k^2}_{\text{positive coeffs.}} + \underbrace{\eta_{k+1} y_{k+1}^2 + \cdots + \eta_{k+\ell} y_{k+\ell}^2}_{\text{negative coeffs.}} = C \right\}.$$

Proof Outline:

Pick any orthonormal basis $\mathcal{C}'$ for V (similar to the previous proof, only now we take $\mathcal{C}'$ to be orthonormal). Define $A := [B]_{\mathcal{C}'}$, and proceed exactly as in the previous proof, but instead of defining $\mathcal{C}$ with the scaling factors λ_i, simply take $\mathcal{C} = \mathcal{C}''$.

To see why $\eta_1, \ldots, \eta_{k+\ell}, 0, \ldots, 0$ depend only on B etc., note that if $\mathcal{D}$ and $\mathcal{D}'$ are two orthonormal bases for V, then $[\mathrm{id}_V]^{\mathcal{D}}_{\mathcal{D}'} \in \mathrm{O}(n)$ (left as an exercise). Put $Q := [\mathrm{id}_V]^{\mathcal{D}}_{\mathcal{D}'}$. We have

$$[B]_{\mathcal{D}} = Q^{\mathsf{T}} \cdot [B]_{\mathcal{D}'} \cdot Q = Q^{-1}[B]_{\mathcal{D}'} Q,$$

since $Q \in \mathrm{O}(n)$. Hence, the matrices $[B]_{\mathcal{D}}$ and $[B]_{\mathcal{D}'}$ are similar, which means their eigenvalues are exactly the same. **Q.E.D.**

Theorem 27.a.7 (*Diagonalization of Symmetric Bilinear Forms*): Let V be a finite-dimensional vector space over a field K of characteristic 0. Let $B\colon V \times V \to K$ be a symmetric bilinear form. Then there exists a basis $\mathcal{C}$ of V such that $[B]_{\mathcal{C}}$ is a diagonal matrix.

Proof:

We proceed by induction on $n := \dim V$. We need to show there exists a basis $\mathcal{B} = (v_1, \ldots, v_n)$ such that $B(v_i, v_j) = 0$ for all $i \neq j$.

If $B = 0$, or if $n = 1$, there is nothing to prove.

Let $n \geq 2$. Assume the statement of the theorem holds for all vector spaces of dimension $\leq n-1$. Let V be an n-dimensional vector space. If $B(v,v) = 0$ for all $v \in V$, then the polarization formula shows $B = 0$ and we are done. So assume that there exists $v_1 \in V$ such that $B(v_1, v_1) \neq 0$.

Put $W := \mathrm{Sp}(v_1)$. The subspace W is 1-dimensional (because $v_1 \neq 0$). Define the orthogonal complement with respect to B as:

$$W^{\perp_B} := \{w \in V \mid B(v_1, w) = 0\}.$$

It is left as an exercise to show that $W^{\perp_B} \subseteq V$ is a subspace.

We claim that $W \cap W^{\perp_B} = \{0\}$. Indeed, if $w \in W \cap W^{\perp_B}$, then $w = c \cdot v_1$ and $B(v_1, w) = 0$. But $B(v_1, w) = B(v_1, c \cdot v_1) = c \cdot B(v_1, v_1)$. Since $B(v_1, v_1) \neq 0$, this implies $c = 0$, which means $w = 0$.

We claim that $V = W + W^{\perp_B}$. Indeed, if $v \in V$, then the vector

$$u := v - \frac{B(v_1, v)}{B(v_1, v_1)} \cdot v_1$$

belongs to $W^{\perp_B}$. To see this, we calculate:

$$B(v_1, u) = B(v_1, v) - \frac{B(v_1, v)}{B(v_1, v_1)} \cdot B(v_1, v_1) = 0.$$

This implies $u \in W^{\perp_B}$. So we can write v as:

$$v = \underbrace{\frac{B(v_1, v)}{B(v_1, v_1)} \cdot v_1}_{\in W} + \underbrace{u}_{\in W^{\perp_B}} \in W + W^{\perp_B}.$$

It follows that $V = W \oplus W^{\perp_B}$. This implies $\dim W^{\perp_B} = n - 1$.

Now, $B|_{W^{\perp_B} \times W^{\perp_B}}$ is a symmetric bilinear form on an $(n-1)$-dimensional vector space ($W^{\perp_B}$). By the induction hypothesis, there exists a basis $\mathcal{C}' = (v_2, \dots, v_n)$ for $W^{\perp_B}$ such that $B(v_i, v_j) = 0$ for all $i \neq j$ with $i, j \geq 2$.

Take $\mathcal{C} := (v_1, \dots, v_n)$. Since $v_1 \in W$ and $v_2, \dots, v_n \in W^{\perp_B}$, we obtain that $B(v_i, v_j) = 0$ for all $i \neq j$. **Q.E.D.**

Question 27.2: What happens over $\mathbb{C}$?

Theorem 27.a.8 (*Complex Symmetric Bilinear Forms*): Let $B\colon V \times V \to \mathbb{C}$ be a symmetric bilinear form on a vector space V over $\mathbb{C}$, of dimension n. Then there exists a basis $\mathcal{C}$ of V such that

$$[B]_{\mathcal{C}} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

for some $0 \leq r \leq n$. (The zero block in the bottom right has size $n-r$).

The number r is independent of the basis $\mathcal{C}$ — it depends only on B. In fact,

$$n - r = \max\{\dim W \mid W \subseteq V \text{ is a subspace such that } B|_{W \times W} \equiv 0\}.$$

The number r is sometimes called the "*rank*" of B.

Proof:

By the previous theorem (Theorem 27.a.7), there exists a basis $\mathcal{C}' = (v_1', \dots, v_n')$ of V such that

$$[B]_{\mathcal{C}'} = \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix},$$

where $0 \leq r \leq n$ and $\eta_j \in \mathbb{C}$ for all $1 \leq j \leq r$, and $\eta_j = 0$ for $r < j \leq n$.

Choose $\lambda_j \in \mathbb{C}$ such that $\lambda_j^2 = \eta_j$ for $j = 1, \dots, r$. (This is always possible over $\mathbb{C}$). Define $v_j := \frac{1}{\lambda_j} v_j'$ for $j = 1, \dots, r$, and $v_q := v_q'$ for all $r < q \leq n$.

Take $\mathcal{C} := (v_1, \dots, v_n)$. Then $B(v_i, v_j) = 0$ for all $i \neq j$. Furthermore, for $1 \leq j \leq r$, we have:

$$B(v_j, v_j) = B\left(\frac{1}{\lambda_j} v_j', \frac{1}{\lambda_j} v_j'\right) = \frac{1}{\lambda_j^2} \cdot \eta_j = 1,$$

and for all $r+1 \leq j \leq n$, we have $B(v_j, v_j) = 0$. **Q.E.D.**

More on Positive Definite Bilinear Forms

Recall: A symmetric bilinear form $B\colon V \times V \to \mathbb{R}$ is called "*positive definite*" if $B(v,v) > 0$ for all $0 \neq v \in V$.

Theorem 27.a.9 (*Sylvester's Criterion*): Let V be a finite-dimensional vector space over $\mathbb{R}$ and $B\colon V \times V \to \mathbb{R}$ a symmetric bilinear form. Then the following conditions are equivalent:

(a) B is positive definite.

(b) There exists a basis $\mathcal{C} := (e_1, \dots, e_n)$ for V such that the principal minors

$$\det(B(e_j, e_k))_{j,k=1,\dots,\ell}$$

for all $1 \leq \ell \leq n$, are positive.

(c) Like **(b)**, but instead of "there exists a basis...", it holds for "all bases...".

Proof:

(b) $\implies$ **(a)**: We proceed by induction on $n = \dim V$.

For $n = 1$, the assumption in **(b)** says $B(e_1, e_1) > 0$. This implies that for all $0 \neq c \in \mathbb{R}$, $B(ce_1, ce_1) = c^2 B(e_1, e_1) > 0$.

Suppose the implication holds for all spaces of dimension $\leq n$. Let V be an $(n+1)$-dimensional vector space, and $B\colon V \times V \to \mathbb{R}$ a symmetric bilinear form. Let $\mathcal{C} := (e_1, \ldots, e_{n+1})$ be a basis for V such that all the principal minors of $[B]_{\mathcal{C}}$ are positive.

Let $W := \mathrm{Sp}(e_1, \ldots, e_n)$ with the basis $\mathcal{C}_W := (e_1, \ldots, e_n)$. The principal minors of $[B|_{W \times W}]_{\mathcal{C}_W}$ are exactly the same as the first n principal minors of $[B]_{\mathcal{C}}$. Since these are positive, the induction hypothesis implies that $B|_{W \times W}$ is positive definite.

This implies that if we define $\langle w, w' \rangle_B := B(w, w')$, then $\langle \cdot, \cdot \rangle_B$ is a valid scalar product on W.

Consider the linear functional $L\colon W \to \mathbb{R}$ defined by $L(w) = B(w, e_{n+1})$. By a previous result (Riesz Representation Theorem, Theorem 23.a.1), there exists $w' \in W$ such that $L(w) = \langle w, w' \rangle_B$ for all $w \in W$. This implies:

$$B(w, w') = B(w, e_{n+1}) \quad \forall w \in W. \tag{27.6}$$

Define a new basis for V, $\mathcal{C}' = (e_1, \ldots, e_n, e_{n+1} - w')$. (It is left as an exercise to show this is a basis. Hint: $w' \in W$). By (27.6), we have $B(e_j, e_{n+1} - w') = 0$ for all $1 \leq j \leq n$.

This implies that the matrix representation in the new basis is block diagonal:

$$[B]_{\mathcal{C}'} = \begin{pmatrix} C & 0 \\ 0 & d \end{pmatrix} = S^{\mathsf{T}} [B]_{\mathcal{C}} S,$$

where $d = B(e_{n+1} - w', e_{n+1} - w')$, $S = [\mathrm{id}_V]^{\mathcal{C}}_{\mathcal{C}'}$, and $C = [B|_{W \times W}]_{\mathcal{C}_W}$.

Taking the determinant, we get:

$$\det([B]_{\mathcal{C}'}) = \det(C) \cdot d = \det(S)^2 \cdot \det([B]_{\mathcal{C}}).$$

We know $\det(C) > 0$ (it is a principal minor), $\det(S)^2 > 0$ (since S is invertible), and $\det([B]_{\mathcal{C}}) > 0$ (by assumption). This implies that $d > 0$.

Let $v \in V$. We can write $v = w + \alpha \cdot (e_{n+1} - w')$, with $w \in W$ and $\alpha \in \mathbb{R}$. Recall that $\langle w, e_{n+1} - w' \rangle_B = 0$. This implies:

$$B(v, v) = B(w, w) + \alpha^2 B(e_{n+1} - w', e_{n+1} - w') = B(w, w) + \alpha^2 d > 0,$$

whenever either $w \neq 0$ or $\alpha \neq 0$. This implies that B is positive definite. The proof of "**(b)** $\implies$ **(a)**" follows now by induction.

(a) $\implies$ **(b)**: Left as an exercise. Similar ideas apply. **Q.E.D.**

Exercise Solutions

Proof that $2 \neq 0 \implies 4 \neq 0$ (on Page 238):

Assume $2 \neq 0$ in a field K. This means $1 + 1 \neq 0$. If $4 = 0$, then $2 + 2 = 0$, which means $2 \cdot 2 = 0$. Since K is a field, it has no zero divisors. Thus, $2 \cdot 2 = 0$ implies $2 = 0$, which contradicts our assumption. Therefore, $4 \neq 0$. **Q.E.D.**

Proof that $W^{\perp_B}$ is a subspace (on Page 241):

Let $w_1, w_2 \in W^{\perp_B}$ and $\alpha, \beta \in K$. We need to show that $\alpha w_1 + \beta w_2 \in W^{\perp_B}$. By definition of $W^{\perp_B}$, we have $B(v_1, w_1) = 0$ and $B(v_1, w_2) = 0$. Using the linearity of B in the second argument, we compute:

$$B(v_1, \alpha w_1 + \beta w_2) = \alpha B(v_1, w_1) + \beta B(v_1, w_2) = \alpha \cdot 0 + \beta \cdot 0 = 0.$$

Thus, $\alpha w_1 + \beta w_2 \in W^{\perp_B}$, proving it is a subspace. **Q.E.D.**

Proof that $\mathcal{C}$ is a basis (on Page 238):

We are given a basis $\mathcal{C}'' := (v_1'', \dots, v_n'')$. The new set $\mathcal{C} := (v_1, \dots, v_n)$ is formed by scaling each vector in $\mathcal{C}''$ by a non-zero scalar c_i (where $c_i = 1/\lambda_i$ or $c_i = 1$). Since scaling basis vectors by non-zero scalars preserves linear independence and the span, $\mathcal{C}$ remains a basis for V. **Q.E.D.**

Proof that $\mathcal{C}'$ is a basis (on Page 242):

We are given the basis $\mathcal{C} = (e_1, \dots, e_n, e_{n+1})$. We define $\mathcal{C}' := (e_1, \dots, e_n, e_{n+1} - w')$, where $w' \in W = \operatorname{Sp}(e_1, \dots, e_n)$. We can write $w' = \sum_{i=1}^n \alpha_i e_i$. Suppose a linear combination of $\mathcal{C}'$ equals zero:

$$\sum_{i=1}^n c_i e_i + c_{n+1}(e_{n+1} - w') = 0 \implies \sum_{i=1}^n (c_i - c_{n+1}\alpha_i)e_i + c_{n+1}e_{n+1} = 0.$$

Because $\mathcal{C}$ is a basis, all coefficients must be zero. In particular, $c_{n+1} = 0$. Substituting this back into the other coefficients yields $c_i = 0$ for all $1 \le i \le n$. Thus, the vectors in $\mathcal{C}'$ are linearly independent and form a basis. **Q.E.D.**

Proof of Theorem 27.a.9 (a) $\implies$ (b):

Assume B is positive definite. Let $\mathcal{C} = (e_1, \dots, e_n)$ be any basis for V. For any $1 \le \ell \le n$, consider the subspace $W_\ell = \operatorname{Sp}(e_1, \dots, e_\ell)$. Since B is positive definite on V, its restriction $B|_{W_\ell \times W_\ell}$ is also positive definite. The matrix representation of this restriction in the basis $(e_1, \dots, e_\ell)$ is exactly the $\ell \times \ell$ principal submatrix of $[B]_{\mathcal{C}}$. A positive definite matrix has strictly positive eigenvalues (as shown in the proof of Sylvester's Law of Inertia), and its determinant is the product of these eigenvalues. Therefore, the determinant of every principal submatrix (the principal minors) must be strictly positive. **Q.E.D.**

The Jordan Normal Form (Chapter 28)

Introduction and Motivation (Section 28.a)

Let V be a finite-dimensional vector space over K, and let $T\colon V \to V$ be a linear map.

The easiest situation to understand is when T is diagonalizable. That is, there exists a basis $\mathcal{B}$ for V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of T.

Using the basis $\mathcal{B}$, it is also easy to calculate powers of T. For example,

$$[T^k]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & \lambda_n^k \end{pmatrix} \quad \forall k \in \mathbb{Z}_{\geq 1}.$$

But **not all** $T \in \operatorname{End}(V)$ are diagonalizable! Why? There are two possible reasons:

(a) The characteristic polynomial $P_T(x)$ does not split as a product of linear factors. This is equivalent to saying that T is not (even) trigonalizable (hence, all the more so not diagonalizable!). In other words, $P_T(x)$ doesn't have enough zeroes in K (or even no zeroes at all).

For example, consider $T_A\colon \mathbb{R}^2 \to \mathbb{R}^2$, where $A = R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$, with $\theta \neq \pi k$ for $k \in \mathbb{Z}$. The map T_A has **no** eigenvalues (in $\mathbb{R}$). It is a rotation, but not by 0° or 180°.

Sometimes we can overcome this "problem" by enlarging (extending) K. For example, $K = \mathbb{R}$ can be extended to $K = \mathbb{C}$. Then the above matrix A has two eigenvalues when viewed as $A \in \mathcal{M}_{2\times 2}(\mathbb{C})$, namely $\lambda_1 = e^{i\theta}$ and $\lambda_2 = e^{-i\theta}$ (Exercise: check this).

(b) T is trigonalizable, or equivalently $P_T(x)$ splits as a product of linear factors, but for some eigenvalues λ of T, the geometric multiplicity is strictly less than the algebraic multiplicity:

$$m_g(\lambda) < m_a(\lambda).$$

Let's try to understand case **(b)** better.

Definition 28.a.1 (*Jordan Matrix*): Let $\lambda \in K$ and $n \in \mathbb{Z}_{\geq 1}$. Define:

$$J_{\lambda,n} = \begin{pmatrix} \lambda & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{pmatrix} \in \mathcal{M}_{n\times n}(K).$$

This matrix has λ on the main diagonal, 1's on the diagonal immediately above the main diagonal, and 0's everywhere else.

For $n = 1$, $J_{\lambda,1} = (\lambda)$. For $n = 2$, $J_{\lambda,2} = \left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right)$. For $n = 3$, $J_{\lambda,3} = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$, etc.

The matrix $J_{\lambda,n}$ is called the "*n-dimensional Jordan matrix with eigenvalue λ*".

Lemma 28.a.2 (*Properties of the Jordan Matrix*):

(a) λ is the only eigenvalue of $J_{\lambda,n}$.

(b) $P_{J_{\lambda,n}}(x) = (\lambda - x)^n$, and $m_a(J_{\lambda,n};\lambda) = n$.
(c) $m_g(J_{\lambda,n};\lambda) = 1$. In fact, $\mathrm{Eig}_{J_{\lambda,n}}(\lambda) = \mathrm{Sp}(e_1)$, where $e_1 = (1,0,\dots,0)^\mathsf{T}$.

Proof:

(a) The characteristic polynomial is the determinant of an upper triangular matrix:

$$P_{J_{\lambda,n}}(x) = \det\begin{pmatrix} \lambda - x & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda - x \end{pmatrix} = (\lambda - x)^n.$$

The only root of $P_{J_{\lambda,n}}(x)$ is λ, so λ is the only eigenvalue of $J_{\lambda,n}$.

(b) From part **(a)**, the root λ has multiplicity n, so $m_a(J_{\lambda,n};\lambda) = n$.

(c) To find $\mathrm{Eig}_{J_{\lambda,n}}(\lambda)$, we need to solve the equation $(J_{\lambda,n} - \lambda I_n)x = 0$:

$$(J_{\lambda,n} - \lambda I_n)\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Subtracting λI_n clears the main diagonal and leaves only the superdiagonal of 1's, so this reads

$$\begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

This immediately implies the system of equations:

$$\begin{cases} x_2 = 0 \\ x_3 = 0 \\ \vdots \\ x_n = 0 \\ 0 = 0 \end{cases}$$

The only free variable is x_1. Therefore, $\mathrm{Eig}_{J_{\lambda,n}}(\lambda) = \mathrm{Sp}\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \mathrm{Sp}(e_1)$, which means the geometric multiplicity is 1.

Q.E.D.

We've seen that the statement *"$P_T(x)$ splits as a product of linear factors"* does **not** imply that *"T is diagonalizable"*, and $J_{\lambda,n}$ is a clear counterexample.

Question 28.1: Is $J_{\lambda,n}$ the *"only reason"* for this non-implication?

Theorem 28.a.3 (*Jordan Normal Form*): Let V be an n-dimensional vector space over K and $T \in \mathrm{End}(V)$ such that $P_T(x)$ splits as a product of linear factors in $K[x]$ (e.g., if $K = \mathbb{C}$, this always happens). Then there exists a basis $\mathcal{B}$ for V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} J_{\lambda_1,n_1} & & & \\ & J_{\lambda_2,n_2} & & \\ & & \ddots & \\ & & & J_{\lambda_k,n_k} \end{pmatrix},$$

where $k \geq 1$, $\lambda_1, \ldots, \lambda_k \in K$, $n_1, \ldots, n_k \geq 1$, and $n_1 + \cdots + n_k = n$.

Moreover, up to permutation, the blocks $J_{\lambda_1,n_1}, \ldots, J_{\lambda_k,n_k}$ are unique.

Remark 28.1: The basis $\mathcal{B}$ is called a "*Jordan basis*" for T. The resulting block diagonal matrix is called the "*Jordan Normal Form*" (JNF) of T. A similar theorem holds for matrices $A \in \mathcal{M}_{n\times n}(K)$: there exists an invertible matrix P such that $P^{-1}AP$ is in Jordan Normal Form.

Example 28.1 (*Jordan Normal Form Block Structure*): Consider a matrix $A \in \mathcal{M}_{13\times 13}(K)$ in Jordan Normal Form:

$$A = \begin{pmatrix}
\lambda_1 & 1 & 0 & & & & & & & & & & \\
0 & \lambda_1 & 1 & & & & & & & & & & \\
0 & 0 & \lambda_1 & & & & & & & & & & \\
& & & \lambda_1 & 1 & 0 & & & & & & & \\
& & & 0 & \lambda_1 & 1 & & & & & & & \\
& & & 0 & 0 & \lambda_1 & & & & & & & \\
& & & & & & \lambda_2 & 1 & & & & & \\
& & & & & & 0 & \lambda_2 & & & & & \\
& & & & & & & & \lambda_2 & & & & \\
& & & & & & & & & \lambda_2 & & & \\
& & & & & & & & & & \lambda_3 & 1 & 0 \\
& & & & & & & & & & 0 & \lambda_3 & 1 \\
& & & & & & & & & & 0 & 0 & \lambda_3
\end{pmatrix}$$

Here, $n = 13$. Let's assume the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ are distinct.

- **Characteristic polynomial:** $P_A(x) = (\lambda_1 - x)^6 \cdot (\lambda_2 - x)^4 \cdot (\lambda_3 - x)^3$.
- **Algebraic multiplicity:** $m_a(\lambda_1) = 6$, $m_a(\lambda_2) = 4$, $m_a(\lambda_3) = 3$.
- **Geometric multiplicity:** $m_g(\lambda)$ is exactly the number of blocks with eigenvalue λ. So, $m_g(\lambda_1) = 2$, $m_g(\lambda_2) = 3$, and $m_g(\lambda_3) = 1$.
- **Minimal polynomial:** $\mu_A(x) = (x-\lambda_1)^{\ell_1} \cdot (x-\lambda_2)^{\ell_2} \cdot (x-\lambda_3)^{\ell_3}$, where $1 \leq \ell_i \leq m_a(\lambda_i)$. We will see that ℓ_i is precisely the size of the **biggest** block with eigenvalue λ_i. Thus,
$$\mu_A(x) = (x-\lambda_1)^3 \cdot (x-\lambda_2)^2 \cdot (x-\lambda_3)^3.$$

Powers of Jordan Matrices and Minimal Polynomials (Section 28.b)

Let's go back to the Jordan matrices $J_{\lambda,n}$. Recall that $J_{\lambda,n} \in \mathcal{M}_{n\times n}(K)$ for $n \in \mathbb{Z}_{\geq 1}$ and $\lambda \in K$. We can write $J_{\lambda,n} = \lambda \cdot I_n + N$, where

$$N = \begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix} = J_{0,n}.$$

Lemma 28.a.4 (*Powers of the Nilpotent Matrix N and Jordan Blocks*):

(a) The powers of N shift the diagonal of 1's further up:

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & & 0 \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & \ddots & 0 \\ 0 & & & & 0 \end{pmatrix}, \quad N^3 = \begin{pmatrix} 0 & 0 & 0 & 1 & \dots \\ & \ddots & \ddots & \ddots & \ddots \\ & & \ddots & \ddots & \ddots \end{pmatrix}, \quad \dots,$$

$$N^{n-1} = \begin{pmatrix} 0 & \dots & 0 & 1 \\ \vdots & & & 0 \\ \vdots & & & \vdots \\ 0 & \dots & \dots & 0 \end{pmatrix}.$$

Furthermore, $N^n = 0$, and $N^k = 0$ for all $k \geq n$. In other words, N^k has 1's in the k-th *"diagonal"* above the *"central diagonal"* and 0's everywhere else.

(b) For all $k \geq 1$, we have:

$$J_{\lambda,n}^k = \begin{pmatrix} \lambda^k & \binom{k}{1}\lambda^{k-1} & \binom{k}{2}\lambda^{k-2} & \dots \\ & \lambda^k & \binom{k}{1}\lambda^{k-1} & \dots \\ & & \ddots & \dots \\ & & & \binom{k}{1}\lambda^{k-1} \\ 0 & & & \lambda^k \end{pmatrix}.$$

Written differently, we have $J_{\lambda,n}^k = \sum_{j=0}^k \binom{k}{j}\lambda^{k-j}N^j$ for all $k \geq 1$.

Proof:

(a) This can be shown by direct calculation (exercise). Alternatively, denote by $e_1, \dots, e_n$ the standard basis for K^n. Then the linear map $T_N \colon K^n \to K^n$ acts as a shift operator:

$$T_N e_i = e_{i-1} \quad \forall 2 \leq i \leq n, \quad \text{and} \quad T_N e_1 = 0.$$

In other words:

$$e_n \xrightarrow{T_N} e_{n-1} \xrightarrow{T_N} \dots \xrightarrow{T_N} e_1 \xrightarrow{T_N} 0.$$

The map $T_{N^2} (= T_N \circ T_N = T_N^2)$ applies the shift twice:

$$e_n \xrightarrow{T_N^2} e_{n-2} \xrightarrow{T_N^2} e_{n-4} \dots \to 0$$

$$e_{n-1} \xrightarrow{T_N^2} e_{n-3} \xrightarrow{T_N^2} \dots \to 0$$

By induction on k, it follows that for all $k \geq 1$:

$$T_{N^k}(e_j) = \begin{cases} e_{j-k} & \text{if } k+1 \leq j \leq n \\ 0 & \text{if } j \leq k \end{cases}.$$

Since $N^k = [T_{N^k}]_{\mathcal{E}_n}^{\mathcal{E}_n}$, the matrix form follows immediately.

(b) Since the scalar matrix λI_n and the matrix N commute, we can use the standard binomial theorem:

$$J_{\lambda,n}^k = (\lambda I_n + N)^k = \sum_{j=0}^k \binom{k}{j}(\lambda I_n)^{k-j}N^j = \sum_{j=0}^k \binom{k}{j}\lambda^{k-j}N^j.$$

Using the structure of N^j from part **(a)**, this yields the upper triangular matrix with binomial coefficients along the diagonals.

Q.E.D.

Corollary 28.a.5 (*Multiplicities and the Minimal Polynomial from JNF*): Suppose $T \in \mathrm{End}(V)$ admits a matrix representation in Jordan Normal Form in some basis $\mathcal{B}$:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} J_{\lambda_1,n_1} & & \\ & \ddots & \\ & & J_{\lambda_k,n_k} \end{pmatrix}. \tag{28.1}$$

Then for every eigenvalue λ of T, the geometric multiplicity $m_g(\lambda)$ equals the number of blocks in (28.1) labeled with λ (i.e., blocks of the type $J_{\lambda,\dots}$). Note that blocks of size 1×1 also count. In other words,

$$m_g(\lambda) = \#\{j \mid 1 \le j \le k, \lambda_j = \lambda\}.$$

Furthermore, the minimal polynomial $\mu_T(x)$ is given by:

$$\mu_T(x) = \prod_{\lambda \in \mathrm{Spec}(T)} (x-\lambda)^{s(\lambda)},$$

where $s(\lambda)$ is the size of the largest block labeled by λ, i.e., $s(\lambda) = \max\{n_j \mid 1 \le j \le k, \lambda_j = \lambda\}$.

Proof:

Geometric Multiplicity: Suppose that $A \in \mathcal{M}_{n\times n}(K)$ has the block diagonal shape $A = \left(\begin{smallmatrix} B & 0 \\ 0 & C \end{smallmatrix}\right)$ with $B \in \mathcal{M}_{\ell\times\ell}(K)$ and $C \in \mathcal{M}_{m\times m}(K)$. Let $0 \ne u = \left(\begin{smallmatrix} v \\ w \end{smallmatrix}\right)$ with $v \in K^\ell$ and $w \in K^m$. Then u is an eigenvector of A for the eigenvalue λ if and only if $Bv = \lambda v$ and $Cw = \lambda w$.

Moreover, the eigenspace decomposes as $\mathrm{Eig}_A(\lambda) = \mathrm{Eig}_B(\lambda) \oplus \mathrm{Eig}_C(\lambda)$. In particular, $m_g(A,\lambda) = m_g(B,\lambda) + m_g(C,\lambda)$. (We use the convention here that if λ is not an eigenvalue of a matrix D, then $m_g(D,\lambda) = 0$).

Now recall from Lemma 28.a.2 that for a single Jordan block, $m_g(J_{\lambda,r};\lambda') = \begin{cases} 1 & \text{if } \lambda' = \lambda \\ 0 & \text{if } \lambda' \ne \lambda \end{cases}$. Applying this additively to $A := [T]_{\mathcal{B}}^{\mathcal{B}}$, we obtain:

$$m_g(T,\lambda) = m_g(A,\lambda) = \sum_{i=1}^{k} m_g(J_{\lambda_i,n_i};\lambda) = \#\{i \mid \lambda_i = \lambda\}.$$

Minimal Polynomial: Let $A := \left(\begin{smallmatrix} B & 0 \\ 0 & C \end{smallmatrix}\right)$ as above. We have $A^k = \begin{pmatrix} B^k & 0 \\ 0 & C^k \end{pmatrix}$ (check!). This implies that for every polynomial $q(x) \in K[x]$, we have $q(A) = \begin{pmatrix} q(B) & 0 \\ 0 & q(C) \end{pmatrix}$.

Consider a single Jordan matrix $J_{\lambda,n}$. For every $q(x) \in K[x]$, we have:

$$q(J_{\lambda,n}) = \begin{pmatrix} q(\lambda) & & * \\ & \ddots & \\ 0 & & q(\lambda) \end{pmatrix} \quad \text{(exercise)}.$$

Note the following key properties:

(a) If $q(\lambda) \ne 0$, then $q(J_{\lambda,n})$ is an upper triangular matrix with non-zero diagonal entries, hence it is invertible.

(b) If $q(x) = (x-\lambda)^s$, then $q(J_{\lambda,n}) = (J_{\lambda,n} - \lambda I_n)^s = N^s$. So, for $q(x) = (x-\lambda)^s$, $q(J_{\lambda,n}) = 0$ if and only if $s \ge n$.

(c) Now, take a general polynomial $q(x) \in K[x]$ of positive degree (i.e., $q(x) \ne \text{const.}$) and let λ be a zero of $q(x)$. We can write $q(x) = (x-\lambda)^s \cdot p(x)$, where $s \ge 1$ and $p(\lambda) \ne 0$. This implies $q(J_{\lambda,n}) = N^s \cdot p(J_{\lambda,n})$. Since $p(\lambda) \ne 0$, $p(J_{\lambda,n})$ is invertible by **(a)**. Therefore, $q(J_{\lambda,n}) = 0$ if and only if $N^s = 0$, which occurs if and only if $s \ge n$.

Let A be a matrix in Jordan Normal Form, and let $q(x) \in K[x]$ be a polynomial of positive degree such that $q(A) = 0$. We must have $q(\lambda_j) = 0$ for all $1 \leq j \leq k$, because if $q(\lambda_j) \neq 0$, then $q(J_{\lambda_j, n_j})$ would be invertible (and thus non-zero) by **(a)**, which would imply that $q(A) \neq 0$. This proves that $\lambda_1, \ldots, \lambda_k$ are all zeroes of $q(x)$.

Let $\lambda \in \{\lambda_1, \ldots, \lambda_k\}$. Write $q(x) = (x - \lambda)^s \cdot p(x)$ where $p(\lambda) \neq 0$. By **(c)**, we must have $s \geq n_j$ for all j with $\lambda_j = \lambda$. That is, $s \geq \max\{n_j \mid \lambda_j = \lambda\} = s(\lambda)$. This implies that any annihilating polynomial $q(x)$ must be of the form:

$$q(x) = \prod_{\lambda \in \{\lambda_1, \ldots, \lambda_k\}} (x - \lambda)^{s(\lambda)} \cdot r(x) \quad \text{for some } r(x) \in K[x]. \tag{28.2}$$

On the other hand, we claim that the specific polynomial $\tilde{q}(x) := \prod_{\lambda \in \{\lambda_1, \ldots, \lambda_k\}} (x - \lambda)^{s(\lambda)}$ satisfies $\tilde{q}(A) = 0$. If we prove this, then together with (28.2), it establishes that $\tilde{q}(x)$ is the monic polynomial of least degree annihilating A, meaning $\mu_T(x) = \tilde{q}(x)$.

Indeed, let $A := \text{blockdiag}(J_{\lambda_1, n_1}, \ldots, J_{\lambda_k, n_k})$. We have:

$$\tilde{q}(A) = \text{blockdiag}(\tilde{q}(J_{\lambda_1, n_1}), \ldots, \tilde{q}(J_{\lambda_k, n_k})). \tag{28.3}$$

Let $1 \leq j \leq k$, and consider the block $\tilde{q}(J_{\lambda_j, n_j})$. We can write $\tilde{q}(x) = (x - \lambda_j)^{s(\lambda_j)} \cdot p_j(x)$ where $p_j(x) \in K[x]$. By the definition of $s(\lambda_j)$, we have $s(\lambda_j) \geq n_j$. By property **(c)** above, $\tilde{q}(J_{\lambda_j, n_j}) = 0$. This holds for all $1 \leq j \leq k$, hence the entire block diagonal matrix in (28.3) is zero. This proves the last claim, and hence concludes the proof of the corollary. **Q.E.D.**